

*Introduction
to
Geophysical
Fluid Dynamics*

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Contents

I	Fundamentals of Geophysical Fluid Dynamics	1
1	Equations of Motion	3
1.1	Time Derivatives for Fluids	3
1.2	The Mass Continuity Equation	8
1.3	The Momentum Equation	13
1.4	The Equation of State	17
1.5	The Thermodynamic Equation	20
1.6	Sound Waves	34
1.7	Compressible and Incompressible Flow	35
1.8	* More Thermodynamics of Liquids	39
1.9	The Energy Budget	46
1.10	An Introduction to Non-Dimensionalization and Scaling	50
2	A Shallow Fluid with Rotation and Stratification	59
2.1	Equations in a Rotating Frame	59
2.2	Equations of Motion in Spherical Co-ordinates	64
2.3	Cartesian Approximations: The Tangent Plane	75
2.4	The Boussinesq Equations	78
2.5	The Anelastic Equations	85
2.6	Changing Vertical Co-ordinate	90
2.7	Scaling for Hydrostatic Balance	97
2.8	Static Instability and the Parcel Method	102
2.9	Gravity waves	110
2.10	* Acoustic-gravity waves in an ideal gas	112

3	Boundary Layers	123
3.1	The Ekman Layer	124
3.2	* The Near Surface Layer	137
4	Vorticity and Potential Vorticity	139
4.1	Vorticity and Circulation	139
4.2	The Vorticity Equation	142
4.3	Vorticity and Circulation Theorems	145
4.4	Vorticity in a Rotating Frame	154
4.5	Potential Vorticity Conservation	157
4.6	Potential Vorticity in Approximate Models	164
4.7	Potential Vorticity on Isentropic Surfaces	165
4.8	The Impermeability of Isentropes to Potential Vorticity	166
5	Shallow Water Systems	175
5.1	Dynamics of a Single Layer	176
5.2	Multi-Layer Shallow Water Equations	184
5.3	Conservation properties	190
5.4	Rotating Shallow Water Flow	193
5.5	Potential Vorticity, revisited	195
5.6	Shallow Water Waves	197
5.7	Scaling Properties	202
5.8	The Planetary Geostrophic Equations	207
5.9	The Quasi-Geostrophic equations	210
5.10	The Two-layer and Multi-layer Quasi-Geostrophic Systems	215
5.11	Rossby Waves	218
5.12	Geostrophic Adjustment	223
5.13	†Frontal Geostrophic Dynamics	232
5.14	†Higher Order Quasi-Geostrophic Dynamics	232
6	Simplified Equations for Ocean and Atmosphere	239
6.1	Elementary scaling in the momentum equation	240
6.2	Planetary Geostrophic Equations	245
6.3	The Quasi-Geostrophic System	250
6.4	Two-Level Quasi-Geostrophic System	257
6.5	Energetics of Quasi-Geostrophy	259
6.6	* Quasi-geostrophy, Revisited	262
6.7	Quasi-geostrophy in Isentropic Co-ordinates	265
6.8	Rossby Waves in Stratified Quasi-Geostrophic Flow	266
6.9	Available Potential Energy	269

II	Waves, Instabilities and Turbulence	273
7	Barotropic and Baroclinic Instability	275
7.1	Kelvin-Helmholtz Instability	276
7.2	Instability of Parallel Shear Flow	278
7.3	Necessary Conditions for Instability	289
7.4	Bounds on the Growth Rate	293
7.5	Baroclinic Instability	296
7.6	Linearized Quasi-Geostrophic Equations	298
7.7	The Eady Problem	301
7.8	Two-Layer Baroclinic Instability	311
7.9	* Beta, shear and stratification in a continuous model	320
7.10	Energetics and Diagnostics of Baroclinic Instability	328
7.11	Physical Interpretation of Baroclinic Instability	333
8	Wave-Mean Flow Interaction	339
8.1	Quasi-geostrophic Preliminaries	340
8.2	The Eliassen-Palm Flux	342
8.3	Properties and Consequences of Wave Activity	346
8.4	The Transformed Eulerian Mean	349
8.5	The TEM, momentum fluxes, and form drag	363
8.6	Examples of Wave-Meanflow Interaction	366
8.7	Necessary Conditions for Instability	366
8.8	* Necessary Conditions for Instability: Pseudoenergy	370

Preface

These notes contain the material for a graduate course in geophysical fluid dynamics, such as AOS 571. The notes contain more detail than is necessary to go through on the first reading. In particular, starred sections may be skimmed over or omitted. Some of the material on the shallow water equations, although not particularly advanced, is inessential for the later work, but on the other hand it also provides a gentle and straightforward introduction to many topics that appear complicated in their continuously stratified form. Finally, depending on the details of the course, the order in which chapter 5 and chapter 4 are covered could be reversed with just a little effort, or one could interleave them.

The notes are not finished and contain errors. Use at your own risk!

Please send comments and corrections to me at gkv@princeton.edu

Part I

**FUNDAMENTALS OF
GEOPHYSICAL FLUID
DYNAMICS**

Are you sitting comfortably? Then I'll Begin.

Julia Lang, *Listen With Mother*, BBC radio program, 1950–1982.

CHAPTER ONE

Equations of Motion

THIS CHAPTER establishes the fundamental governing equations of motion for a fluid, with particular reference to the fluids of the earth's atmosphere and ocean. Our approach in many places is quite informal, and more extensive treatments on some of the topics can be found in the references at the end of the chapter.¹ In particular, our treatment of viscosity and pressure is very brief. On the other hand, we have a somewhat more extensive treatment of thermodynamics and the equation of state than is common in many fluid texts.

1.1 TIME DERIVATIVES FOR FLUIDS

The equations of motion of fluid mechanics differs from those of rigid-body mechanics because fluids form a continuum and because they flow and deform. Thus, even though both solid and fluid media are governed by the same relatively simple physical laws (Newton's laws and the laws of thermodynamics) the expression of these laws differs between the two. To determine the equations of motion for fluids we must first clearly establish what the time derivative of some property of fluid actually means, and that is the subject of this section.

1.1.1 Field and material viewpoints

In solid-body mechanics one is normally concerned with the position and momentum of identifiable objects — the angular velocity of a spinning top or the motions of the planets around the sun are two well-worn examples. The position and velocity of a particular object, x_i , is then computed as a function of time by formulating equations of the general form

$$\frac{dx_i}{dt} = F(\{x_i\}, t) \quad (1.1)$$

where $\{x_i\}$ is the set of positions and velocities of all the interacting objects and the operator F on the right-hand side is formulated using Newton's laws of motion. For example, two massive point objects interacting via their gravitational field obey

$$\frac{d\mathbf{r}_i}{dt} = \mathbf{v}_i, \quad \frac{d\mathbf{v}_i}{dt} = -\frac{Gm_j}{(\mathbf{r}_i - \mathbf{r}_j)^2} \hat{\mathbf{r}}_{i,j} \quad i = 1, 2; \quad j = 3 - i \quad (1.2)$$

We thereby predict the positions, $\mathbf{r}_i$ and velocities, $\mathbf{v}_i$ of the objects given their masses, m_i and the gravitational constant G , and where $\hat{\mathbf{r}}_{i,j}$ is a unit vector directed between the objects. This is essentially a 'material' point of view (sometimes called a 'Lagrangian' point of view), because we are concerned with identifiable pieces of material, and it is the natural one for this type of mechanics.

In fluid dynamics such a viewpoint would lead us to try to describe the fluid in terms of the positions and momenta of marked fluid elements, each identified by some label (which might simply be their position at an initial time). This procedure is perfectly acceptable in principle, and if followed it would lead to equations similar to (1.1) which would provide a complete and closed description of the fluid dynamics. However, from a practical point of view it is much more than we need: we don't necessarily *want* to know the position and velocity of each marked fluid element, and such a description would likely be extremely complicated. Instead, for most problems we would like to know what the values of velocity, density and so on are at *fixed points* in space as time passes. (A weather forecast we might care about tells us how warm it will be where we live, and we don't particularly care where the fluid parcel that was here today will be tomorrow.) Since the fluid is a continuum, this knowledge is equivalent to knowing how the fields of the dynamical variables evolve in space and time, and this is often known as the 'field' or 'Eulerian' viewpoint. Thus, whereas in the material or Lagrangian view we consider the time evolution of identifiable fluid elements, in the field or Eulerian view we consider time evolution of the fluid field from a particular frame of reference;² that is, we seek evolution equations of the general form:

$$\frac{\partial}{\partial t} \phi(x, y, z, t) = F \quad (1.3)$$

In this equation the field $\phi(x, y, z, t)$ is a dynamical variable (e.g., velocity, density, temperature) which gives the value at any point in space-time, and F is some operator to be determined from Newton's laws of motion and appropriate thermodynamic laws.

Although the field viewpoint will turn out to be the most practically useful, the material description is invaluable both in deriving the equations and in the subsequent insight it frequently provides. This is because the important quantities from a fundamental point of view are often those which are associated with a given fluid element: it is these which directly enter Newton's laws of motion and the thermodynamic equations. It is thus important to have a relationship between the rate of change of quantities associated with a given fluid element and the local rate of change of a field. The material or advective derivative provides this relationship.

1.1.2 The material derivative of a fluid element

The material derivative of a fluid property is the rate of change of the fluid property of a particular fluid element. (By fluid element we mean an infinitesimal, indivisible, piece of fluid.) It is also known as the 'substantive derivative' (the derivative associated with a parcel of fluid substance), the 'advective derivative' (because the fluid property is being advected), the 'convective derivative' (convection is a slightly old-fashioned name for advection, still used in engineering), or the 'Lagrangian derivative' (after Lagrange, even if erroneously). It is simply the total time derivative of a property of a piece of fluid.

Let us suppose that a fluid is characterized by a (given) velocity field $\mathbf{v}(\mathbf{x}, t)$, which determines its velocity throughout. Let us also suppose that it has another property ϕ (temperature, for example), and let us seek an expression for the rate of change of ϕ of a fluid element. Since ϕ is changing in time and in space we use the chain rule:

$$\begin{aligned}\delta\phi &= \frac{\partial\phi}{\partial t}\delta t + \frac{\partial\phi}{\partial x}\delta x + \frac{\partial\phi}{\partial y}\delta y + \frac{\partial\phi}{\partial z}\delta z \\ &= \frac{\partial\phi}{\partial t}\delta t + \delta\mathbf{x} \cdot \nabla\phi\end{aligned}\tag{1.4}$$

This is true in general for any δt , δx , etc. Thus the total time derivative is

$$\frac{d\phi}{dt} = \frac{\partial\phi}{\partial t} + \frac{d\mathbf{x}}{dt} \cdot \nabla\phi.\tag{1.5}$$

Now, the material derivative is specifically the rate of change of a property of a given fluid element. The second term on the right hand side of (1.5) thus takes a simple form, because the time derivative of position of a fluid element is nothing but its velocity. That is

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}\tag{1.6}$$

and in Cartesian co-ordinates

$$\frac{d\mathbf{x}}{dt} = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} + \frac{dz}{dt}\mathbf{k} = u\mathbf{i} + v\mathbf{j} + w\mathbf{k}, \quad (1.7)$$

where u is the component of velocity in the direction of increasing x , and similarly for v and y , and w and z . Hence, (1.5) may be written

$$\frac{d\phi}{dt} = \frac{\partial\phi}{\partial t} + \mathbf{v} \cdot \nabla\phi \quad (1.8)$$

Since the expression is so common, and to emphasize that it is the total derivative of a material fluid element, we denote the material derivative by the special notation

$$\boxed{\frac{D\phi}{Dt} = \frac{\partial\phi}{\partial t} + (\mathbf{v} \cdot \nabla)\phi} . \quad (1.9)$$

This expression relates the total time derivative d/dt of a property of a fluid element to the partial time ($\partial/\partial t$) and spatial derivatives ($\partial/\partial x$, etc.) derivatives for a field. Although strictly unnecessary, the brackets in the last term of this equation is helpful in reminding us that $(\mathbf{v} \cdot \nabla)$ is an operator acting on ϕ ; they will be particularly useful when the material derivative acts on a vector field.

Material derivative of vector field

The material derivative is not restricted in its action to a scalar field. For a vector field $\mathbf{b}$ it is given by

$$\frac{D\mathbf{b}}{Dt} \equiv \frac{\partial\mathbf{b}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{b}. \quad (1.10)$$

In Cartesian co-ordinates this is

$$\frac{D\mathbf{b}}{Dt} = \frac{\partial\mathbf{b}}{\partial t} + u\frac{\partial\mathbf{b}}{\partial x} + v\frac{\partial\mathbf{b}}{\partial y} + w\frac{\partial\mathbf{b}}{\partial z}, \quad (1.11)$$

and for a particular component of $\mathbf{b}$,

$$\frac{Db_x}{Dt} = \frac{\partial b_x}{\partial t} + u\frac{\partial b_x}{\partial x} + v\frac{\partial b_x}{\partial y} + w\frac{\partial b_x}{\partial z}, \quad (1.12)$$

or, in Cartesian tensor notation,

$$\frac{Db_i}{Dt} = \frac{\partial b_i}{\partial t} + v_j \frac{\partial b_i}{\partial x_j} = \frac{\partial b_i}{\partial t} + v_j \partial_j b_i. \quad (1.13)$$

where the subscripts denote the Cartesian components and repeated indices are summed over. In co-ordinate systems other than Cartesian the advective derivative of a vector is not simply the sum of the advective derivative of its components, because the co-ordinate vectors change direction with position; this will be important when we deal with spherical co-ordinates (for example, see problem 2.6). Finally, we note that the advective derivative of the position of a fluid element, $\mathbf{r}$ say, is its velocity, and this may easily be checked by explicitly evaluating $D\mathbf{r}/Dt$.

1.1.3 Material derivative of a finite volume

Frequently, conservation laws arise that will require us to consider the material derivative of the integral of a fluid property over a volume of fluid. First we calculate how the volume of a material element changes through advection. Let the volume of a certain amount of fluid be $\int dV$. This fluid volume is deformed and advected by the fluid motion, and there is no particular reason why its volume should be preserved. In general its volume may change as a result of the movement of each element of its bounding material surface; the volume will change if the velocity has a non-zero normal component at its surface. Thus,

$$\frac{D}{Dt} \int_V dV = \int_S \mathbf{v} \cdot d\mathbf{S} \quad (1.14)$$

where the subscript V indicates that the integral is a definite integral over some finite volume V , although its limits may be functions of time. The integral on the right-hand side is over the closed surface, S , bounding the volume. Although intuitively apparent (to some), this expression may be derived more formally using Leibnitz's formula for the rate of change of an integral whose limits are changing (problem 1.2). Using the divergence theorem on the right-hand side, (1.14) becomes

$$\frac{D}{Dt} \int_V dV = \int_V \nabla \cdot \mathbf{v} dV. \quad (1.15)$$

The rate of change of the volume of an infinitesimal fluid element of volume ΔV is then obtained by taking the limit of this expression as the volume tends to zero, giving

$$\lim_{\Delta V \rightarrow 0} \frac{1}{\Delta V} \frac{D\Delta V}{Dt} = \nabla \cdot \mathbf{v}. \quad (1.16)$$

We will write such expressions informally as

$$\frac{D\Delta V}{Dt} = \Delta V \nabla \cdot \mathbf{v}, \quad (1.17)$$

with the limit implied.

Consider now the material derivative of a property, ϕ , of an infinitesimal fluid element; that is, ϕ is the amount per unit volume of ϕ -stuff — it might, for example, be mass density or density of a dye, per unit volume, in a fluid, and ϕ might even be a vector. Then

$$\frac{D}{Dt}(\phi \Delta V) = \phi \frac{D\Delta V}{Dt} + \Delta V \frac{D\phi}{Dt}. \quad (1.18)$$

Using (1.17) this becomes

$$\frac{D}{Dt}(\phi \Delta V) = \Delta V \left(\phi \nabla \cdot \mathbf{v} + \frac{D\phi}{Dt} \right), \quad (1.19)$$

and the analogous result for a finite fluid volume is just

$$\frac{D}{Dt} \int_V \phi \, dV = \int_V \left(\phi \nabla \cdot \mathbf{v} + \frac{D\phi}{Dt} \right) dV \quad (1.20)$$

This expression is to be contrasted with the Eulerian derivative, for which the volume remains fixed in space. Then, when the limits of integration are fixed, we have

$$\frac{d}{dt} \int_V \phi \, dV = \int_V \frac{\partial \phi}{\partial t} dV, \quad (1.21)$$

Now consider the material derivative of the integral of fluid property χ multiplied by the mass density, ρ . This arises, for example, in the derivation of the momentum equation, where the momentum of a fluid element is $\rho \mathbf{v} \Delta V$. In general the material derivative of $\chi \rho \Delta V$ is given by

$$\frac{D}{Dt} (\chi \rho \Delta V) = \rho \Delta V \frac{D\chi}{Dt} + \chi \frac{D}{Dt} (\rho \Delta V) \quad (1.22)$$

But $\rho \Delta V$ is just the total mass of the fluid element, and this is constant — because that is how a fluid element is defined. Thus,

$$\frac{D}{Dt} (\chi \rho \Delta V) = \rho \Delta V \frac{D\chi}{Dt}, \quad (1.23)$$

with the analogous result for a finite volume:

$$\frac{D}{Dt} \int_V \chi \rho \, dV = \int_V \rho \frac{D\chi}{Dt} dV. \quad (1.24)$$

This expression may also be derived more formally using Leibnitz's formula for the material derivative of an integral, and the result also holds when χ is a vector. Again, the result is quite different from the corresponding Eulerian expression:

$$\frac{d}{dt} \int_V \chi \rho \, dV = \int_V \frac{\partial}{\partial t} (\chi \rho) dV. \quad (1.25)$$

1.2 THE MASS CONTINUITY EQUATION

In classical mechanics mass is absolutely conserved, and transformations between mass and energy that occur in relativity theory do not occur. In solid-body mechanics conservation of mass is implicit, and there is no need for any equation to describe its effects. However, in fluid mechanics fluid flows into and away from regions, and fluid density may change, and an equation which explicitly accounts for the flow of mass is one of the 'equations of motion' of the fluid.

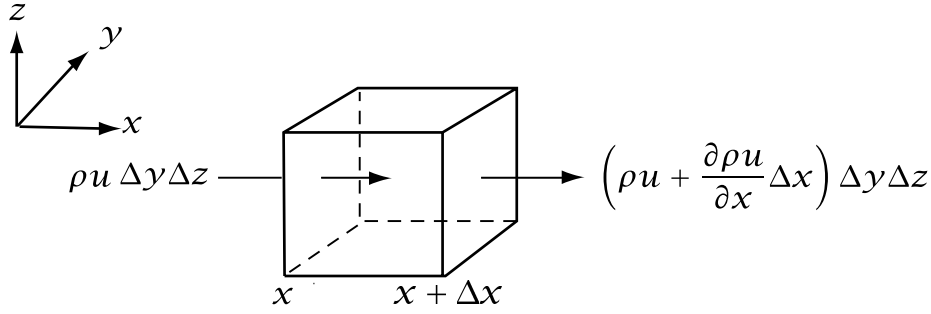


Figure 1.1 Mass conservation in a cubic Eulerian control volume.

1.2.1 A field derivation

We will first derive the mass conservation equation from an Eulerian point of view; that is to say, our reference frame is fixed in space and the fluid flows through it. First we give an elementary derivation in Cartesian co-ordinates (refer to Fig. 1.1).

Consider a rectangular, infinitesimal control volume $\delta V = \delta x \delta y \delta z$ fixed in space. Fluid moves into or out of the volume through its surface S . Consider the y - z plane, of area $\delta A = \delta y \delta z$ at co-ordinate x . The fluid flow through this plane, into the control volume, is

$$(\rho u)_x \times \text{Area} = (\rho u)_x \delta y \delta z \quad (1.26)$$

where u is the component of velocity in the x -direction, and the subscript x here denotes the co-ordinate of the argument. A small distance to the right the flow out of the control volume is

$$(\rho u)_{x+\delta x} \delta y \delta z \quad (1.27)$$

So the accumulation of fluid within the control volume, due to motion in the x -direction only

$$\delta y \delta z [(\rho u)_x - (\rho u)_{x+\delta x}] = -\frac{\partial(\rho u)}{\partial x} \delta x \delta y \delta z \quad (1.28)$$

To this must be added the effects of motion in the y - and z -directions, namely

$$-\left[\frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} \right] \delta x \delta y \delta z \quad (1.29)$$

This accumulation of fluid must be accompanied by a corresponding increase of fluid mass within the control volume. This is

$$\frac{\partial}{\partial t} (\text{Density} \times \text{Volume}) = \delta x \delta y \delta z \frac{\partial \rho}{\partial t}, \quad (1.30)$$

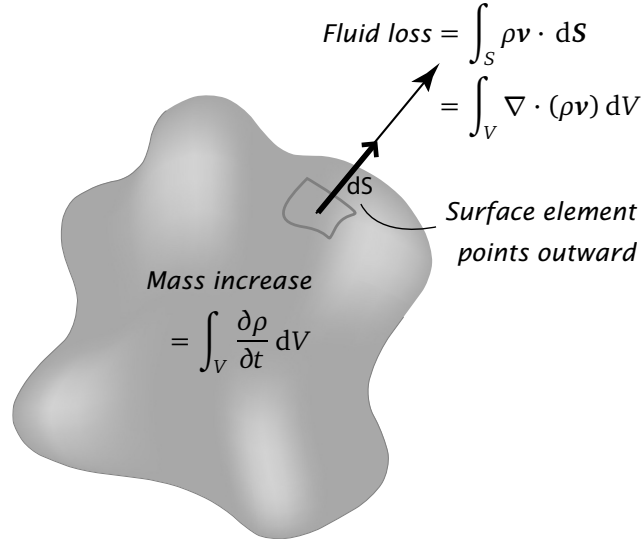


Figure 1.2 Mass conservation in an arbitrary Eulerian control volume V bounded by a surface S . The mass gain, $\int_V (\partial \rho / \partial t) dV$ is equal to the mass flowing into the volume, $-\int_S (\rho \mathbf{v}) \cdot d\mathbf{S} = -\int_V \nabla \cdot (\rho \mathbf{v}) dV$.

since the volume is constant. And thus from (1.28), (1.29) and (1.30),

$$\delta x \delta y \delta z \left[\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} \right] = 0 \quad (1.31)$$

Because the control volume is arbitrary the quantity in square brackets is zero and we have the *mass continuity equation*:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (1.32)$$

Vector derivation

Consider an arbitrary control volume V bounded by a surface S , fixed in space, with by convention the direction of $\mathbf{S}$ being toward the outside of V , as in Fig. 1.2. The rate of fluid loss due to flow through the closed surface S is then given by

$$\text{Fluid loss} = \int_S \rho \mathbf{v} \cdot d\mathbf{S} = \int_V \nabla \cdot (\rho \mathbf{v}) dV \quad (1.33)$$

using the divergence theorem. This must be balanced by a change in the mass M of the fluid within the control volume, which, since its volume is fixed, implies a density change. That is

$$\text{Fluid loss} = -\frac{dM}{dt} = -\frac{d}{dt} \int_V \rho dV = -\int_V \frac{\partial \rho}{\partial t} dV. \quad (1.34)$$

Equating (1.33) and (1.34) yields

$$\int_V \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) \right] dV = 0 \quad (1.35)$$

Again, because the volume is arbitrary, the integrand must vanish and we recover (1.32).

1.2.2 Mass continuity via the material derivative

We now derive the mass continuity equation (1.32) from a material perspective. This is the most fundamental approach of all since the principle of mass conservation states simply that the mass of a given element of fluid is, by definition of the element, constant. Thus, consider a small mass of fluid of density ρ and volume ΔV . Then conservation of mass may be represented by

$$\frac{D}{Dt}(\rho \Delta V) = 0 \quad (1.36)$$

Both the density and the volume of the parcel may change, so

$$\Delta V \frac{D\rho}{Dt} + \rho \frac{D\Delta V}{Dt} = 0 \quad (1.37)$$

The fluid volume changes by the motion of each element of the bounding surface. Thus, using (1.15) in (1.37) gives

$$\frac{D}{Dt}(\rho \Delta V) = \Delta V \left(\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} \right) = 0. \quad (1.38)$$

Sine the volume element is arbitrary, we must have

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0 \quad (1.39)$$

After expansion of the first term this becomes identical to (1.32).

The above result may be derived more formally by re-writing (1.36) as the integral expression

$$\frac{D}{Dt} \int_V \rho dV = 0. \quad (1.40)$$

Expanding the derivative using (1.20) gives

$$\frac{D}{Dt} \int_V \rho dV = \int_V \left(\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} \right) dV = 0 \quad (1.41)$$

Various Material and Field Derivatives

The material derivative of a scalar and vector fields are given by

$$\frac{D\phi}{Dt} = \frac{\partial\phi}{\partial t} + \mathbf{v} \cdot \nabla\phi \quad \text{and} \quad \frac{D\mathbf{b}}{Dt} = \frac{\partial\mathbf{b}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{b}. \quad (\text{D.1})$$

Various material derivatives of integrals are:

$$\frac{D}{Dt} \int_V \phi \, dV = \int_V \left(\frac{D\phi}{Dt} + \phi \nabla \cdot \mathbf{v} \right) dV = \int_V \left(\frac{\partial\phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}) \right) dV, \quad (\text{D.2})$$

$$\frac{D}{Dt} \int_V dV = \int_V \nabla \cdot \mathbf{v} \, dV, \quad (\text{D.3})$$

$$\frac{D}{Dt} \int_V \rho \phi \, dV = \int_V \rho \frac{D\phi}{Dt} \, dV. \quad (\text{D.4})$$

These formulae also hold if ϕ is a vector. The Eulerian, or field, derivative of an intergral is:

$$\frac{d}{dt} \int_V \phi \, dV = \int_V \frac{\partial\phi}{\partial t} \, dV, \quad (\text{D.5})$$

so that

$$\frac{d}{dt} \int_V dV = 0 \quad \text{and} \quad \frac{d}{dt} \int_V \rho \phi \, dV = \int_V \frac{\partial\rho\phi}{\partial t} \, dV. \quad (\text{D.6})$$

Because the volume over which the integral is taken is arbitrary the integrand itself must vanish and we recover (1.39). Summarizing, equivalent partial differential equation representing conservation of mass are

$$\left[\begin{array}{l} \frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0 \\ \frac{\partial\rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \end{array} \right]. \quad (1.42)$$

1.2.3 A general continuity equation

The derivation of continuity equation for a general scalar property of a fluid is similar to that for density, except that there may be an external source or sink, or

a means of transferring the property from one location to another than by fluid motion, for example by diffusion. If ϕ is the amount of some property of the fluid per unit volume, and if the net effect per unit volume of all nonconservative processes is denoted by $Q[\phi]$, then the general continuity equation may be written:

$$\frac{D}{Dt}(\phi \Delta V) = Q[\phi] \Delta V \quad (1.43)$$

Expanding the left hand side and using (1.17) we obtain

$$\frac{D\phi}{Dt} + \phi \nabla \cdot \mathbf{v} = Q[\phi] \quad (1.44)$$

or equivalently

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}) = Q[\phi]. \quad (1.45)$$

If we are interested in a tracer that is normally measured per unit mass of fluid (useful when considering thermodynamic quantities) then the conservation equation would be written

$$\frac{D}{Dt}(\xi \rho \Delta V) = Q[\xi] \rho \Delta V, \quad (1.46)$$

where ξ is the tracer concentration per unit fluid mass $Q[\xi]$ is the corresponding source terms. we write the source term for ξ , per unit mass, as $\dot{\xi}$. Then, since $\rho \Delta V$ is constant we obtain

$$\frac{D\xi}{Dt} = Q[\xi]. \quad (1.47)$$

The source term $Q[\xi]$ here is evidently equal to the rate of change of ξ of a fluid element. When this is so it is common to write the source term simply as $\dot{\xi}$, so that

$$\frac{D\xi}{Dt} = \dot{\xi}. \quad (1.48)$$

A tracer obeying (1.48) with $\dot{\xi} = 0$ is said to be ‘materially conserved’. If $\dot{\xi} \neq 0$, then ξ is ‘semi-materially conserved’, meaning that the tracer is materially conserved in the absence of explicit source terms.

1.3 THE MOMENTUM EQUATION

The momentum equation is a partial differential equation that describes how the velocity or momentum of a fluid responds to imposed forces. We will derive it using material methods and initially without paying any attention to the form of the momentum sources. Then we will examine the influences of momentum sources, informally deducing the forms of the pressure, gravitational and viscous forces.

1.3.1 Advection

Let $\mathbf{m}(x, y, z, t)$ be the momentum-density field (momentum per unit volume) of the fluid. Thus, $\mathbf{m} = \rho \mathbf{v}$ and the total momentum of a volume of fluid is given by the volume integral $\int_V \mathbf{m} dV$ and the momentum of an elemental fluid mass is given by $\mathbf{m} \Delta V$. Now, for a fluid the rate of change of a momentum of an identifiable fluid mass is given by the material derivative, and by Newton's second law this is equal to the force acting on it. Thus,

$$\frac{D}{Dt} \int_V \rho \mathbf{v} dV = \int_V \mathbf{F} dV \quad (1.49)$$

Now, because the mass of a fluid element is constant [or using (1.24) with χ replaced by the fluid velocity)] we have

$$\frac{D}{Dt} \int_V \rho \mathbf{v} dV = \int_V \rho \frac{D\mathbf{v}}{Dt} dV, \quad (1.50)$$

and (1.49) becomes

$$\int_V \left(\rho \frac{D\mathbf{v}}{Dt} - \mathbf{F} \right) dV = 0. \quad (1.51)$$

Because the volume is arbitrary the integrand itself must vanish and we obtain

$$\rho \frac{D\mathbf{v}}{Dt} = \mathbf{F}. \quad (1.52)$$

Expanding the material derivative, this may be written

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = \frac{\mathbf{F}}{\rho} \quad (1.53)$$

We have thus obtained an expression for how a fluid accelerates if subject to known forces. These forces are however not all external to the fluid itself; a stress arises from the direct contact between one fluid parcel and another, giving rise to pressure and viscous forces, sometimes referred to as *contact* forces. We treat both of these very informally and intuitively.

1.3.2 The pressure force

Within or at the boundary of a fluid the pressure is the normal force per unit area due to the collective action of molecular motion. Thus

$$d\mathbf{F}_p = -p d\mathbf{S}. \quad (1.54)$$

If we grant ourselves this intuitive notion, it is a simple matter to assess the influence of pressure on a fluid, for the pressure force *on* a volume of fluid is the integral of the pressure over the its boundary and so

$$\mathbf{F}_p = - \int_S p d\mathbf{S} \quad (1.55)$$

The minus sign arises because the pressure force is directed inward, whereas $\mathbf{S}$ is a vector normal to the surface and directed outward. Applying a relative of the divergence theorem gives

$$\mathbf{F}_p = - \int_V \nabla p \, dV \quad (1.56)$$

and so the pressure force per unit volume is just $-\nabla p$. Inserting this into (1.53) we obtain

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{\nabla p}{\rho} + \mathbf{F}' \quad (1.57)$$

where $\mathbf{F}'$ now includes only viscous and body forces, per unit mass.

1.3.3 Viscosity and diffusion

Viscosity, like pressure, is a force due to the internal motion of molecules. The effects of viscosity are apparent in many situations: the flow of treacle or volcanic lava are obvious examples. In many large scale phenomena, for example the flow of air in the atmosphere, viscosity is to a first approximation negligible. However, for a constant density fluid viscosity is the *only* way that energy may be removed from the fluid, so that if energy is being added in some way viscosity must ultimately become important if the fluid is to reach an equilibrium where energy input equals energy dissipation. When tea is stirred in a cup, viscosity is the mechanism whereby the fluid eventually stops spinning after we have removed our spoon.

It is somewhat difficult to rigorously derive a formulation for the effects of viscosity in a fluid. A number of textbooks³ show that, for most Newtonian fluids, the viscous force per unit volume can be represented by the term $\mu \nabla^2 \mathbf{v}$, where μ is a coefficient of diffusivity. With this term, the momentum equation becomes,

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{v} \quad (1.58)$$

where $\nu \equiv \mu / \rho$ is the *kinematic viscosity*. Although not exactly true, this expression holds to a very good approximation for most liquids and gases. For gases dimensional arguments suggest that the magnitude of ν should be given by

$$\nu \sim \langle \text{mean free path} \times \text{mean molecular velocity} \rangle \quad (1.59)$$

which for a typical molecular velocity of $300 \, \text{m s}^{-1}$ and a mean free path of $7 \times 10^{-8} \, \text{m}$ gives the not unreasonable estimate of $2.1 \times 10^{-5} \, \text{m}^2 \text{s}^{-1}$, well within a factor of two of the experimental value (table 1.1). Interestingly, the kinematic viscosity is *less* for water and mercury than it is for air.

Table 1.1 Experimental values of viscosity for air, water and mercury at room temperature and pressure.

	μ (kg m ⁻¹ s ⁻¹)	ν (m ² s ⁻¹)
Air	1.8 10 ⁻⁵	1.5 10 ⁻⁵
Water	1.1 10 ⁻³	1.1 10 ⁻⁶
Mercury	1.6 10 ⁻³	1.2 10 ⁻⁷

1.3.4 Hydrostatic balance

The vertical component — meaning the component parallel to the gravitational force — of the momentum equation is

$$\frac{Dw}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g \quad (1.60)$$

where w is the vertical component of the velocity. If the fluid is static the gravitational term is balanced by the pressure term and we have

$$\frac{\partial p}{\partial z} = -\rho g. \quad (1.61)$$

which is known as *hydrostatic balance*. In this case, it is clear that the pressure at a point is given by the weight of the fluid above it, provided $p = 0$ at the top of the fluid. It might also appear that (1.61) would be a good *approximation* to (1.60) provided vertical accelerations, Dw/Dt , are small compared to gravity, which is nearly always the case in the atmosphere and atmosphere. While this statement is true if we need only a reasonably accurate representation of the pressure at a point or in a column, the satisfaction of this condition is not sufficient to ensure that (1.61) provides an accurate enough pressure to determine the *horizontal* pressure gradients responsible for producing motion. We return to this point in section 2.7.

Momentum and Mass Continuity Equations

The *mass continuity* equation is

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad \text{or} \quad \frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0. \quad (\text{MM.1})$$

If density is constant this reduces to $\nabla \cdot \mathbf{v} = 0$.

The *momentum equation* is

$$\frac{D\mathbf{v}}{Dt} = -\frac{\nabla p}{\rho} + \nu \nabla^2 \mathbf{v} + \mathbf{F}, \quad (\text{MM.2})$$

where $\mathbf{F}$ represents the effects of body forces such as gravity. If density is constant, or pressure is a function of density alone (e.g., $p = C\rho^\gamma$), then the above equations form a complete system.

The equations describing fluid motion, including the thermodynamic equation and equation of state as needed, are called the *Euler equations* if viscosity is not included and the *Navier-Stokes equations* if viscosity is included.⁴

1.4 THE EQUATION OF STATE

In three dimensions the momentum and continuity equations provide four equations, but contain five unknowns — three components of velocity, density and pressure. Obviously other equations are needed, and an *equation of state* relates the various thermodynamic variables to each other.⁵ Conventionally, ‘the’ equation of state is an expression that relates temperature, pressure, composition (salinity), and density, since these are all readily measurable quantities, and we may write

$$p = p(\rho, T, S). \quad (1.62)$$

For an ideal gas we have

$$p = \rho RT, \quad (1.63)$$

where R is the gas constant and T is temperature. (R is related to the universal gas constant R_u by $R = R_u/m$ where m is the mean molecular weight of the constituents of the gas. Also, $R = nk$ where k is Boltzmann’s constant and n is the number of molecules per unit mass. For dry air, $R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$.) Air has virtually constant composition, except for variations in water vapour content. A measure of this is the water vapour mixing ratio, $w = \rho_w / \rho_d$ where ρ_w and ρ_d are the densities of water vapour and dry air, respectively, and in the atmosphere w

varies between 0 and 0.03. This variation makes the gas constant in the equation of state a weak function of water vapour mixing ratio, namely, $p = \rho R_{\text{eff}} T$ where $R_{\text{eff}} = R_d(1 + wR_v/R_d)/(1 + w)$ where R_d and R_v are the gas constants of dry air and water vapour. Since $w \sim 0.01$ the dependence is quite weak and is often ignored, especially in theoretical studies.⁶

For a liquid such as seawater no such analytic expression is easily derivable, and semi-empirical equations are usually resorted to. For pure water in a laboratory setting a reasonable approximation of the equation of state is

$$\rho = \rho_0[1 - \beta_T(T - T_0)] \quad (1.64)$$

where ρ_0 and T_0 are a constant reference density and temperature, and β_T is a thermal expansion coefficient. Even for pure water, this equation is only valid for small variations around a reference state — note, for example, that pure water generally expands when heated, but that it contracts as its temperature rises from 0°C to 4°C, so that β_T is certainly not constant. In the ocean both salinity and pressure affect the density substantially, and an accurate equation of equation of state is a complicated affair. Let us write it in the general form

$$\alpha = \alpha(T, S, p) \quad (1.65)$$

where $\alpha = 1/\rho$ is the specific volume and S the salinity. Small variations around a reference value can be obtained from

$$\begin{aligned} d\alpha &= \left(\frac{\partial \alpha}{\partial T} \right)_{S,p} dT + \left(\frac{\partial \alpha}{\partial S} \right)_{T,p} dS + \left(\frac{\partial \alpha}{\partial p} \right)_{T,S} dp \\ &= \alpha(\beta_T dT - \beta_S dS - \beta_p dp) \end{aligned} \quad (1.66)$$

where the second line serves to define the thermal expansion coefficient β_T , the saline contraction coefficient β_S , and the compressibility coefficient β_p (equal to α divided by the bulk modulus). These are in general not constants, but for small variations around a reference state they may be treated as such and we have

$$\alpha = \alpha_0[1 + \beta_T(T - T_0) - \beta_S(S - S_0) - \beta_p(p - p_0)]. \quad (1.67)$$

Typical values of these parameters, with variations typically encountered through the ocean, are: $\beta_T \approx 2(\pm 1.5) \times 10^{-4} \text{ K}^{-1}$ (values increase with both temperature and pressure), $\beta_S \approx 7.6(\pm 0.2) \times 10^{-4} \text{ ppt}^{-1}$ (ppt = parts per thousand), $\beta_p \approx 4.1(\pm 0.5) \times 10^{-10} \text{ Pa}^{-1}$. Since the variations around the mean density are small (1.67) can also be written

$$\rho = \rho_0([1 - \beta_T(T - T_0) + \beta_S(S - S_0) + \beta_p(p - p_0)]). \quad (1.68)$$

A linear equation of state for seawater is *not* accurate enough for quantitative oceanography; the β parameters in (1.67) themselves vary with pressure, temperature and (more weakly) salinity so introducing nonlinearities to the equation.

The most important of these are captured by an equation of state of the form

$$\alpha = \alpha_0 [1 + (\beta_T + \gamma^* p)(T - T_0) + \frac{1}{2} \beta_T^* (T - T_0)^2 - \beta_S (S - S_0) - \beta_p (p - p_0)]. \quad (1.69)$$

The starred constants β_T^* and γ^* capture the leading nonlinearities: γ^* is the *thermobaric* parameter and β_T^* is the second thermal expansion coefficient. Even this expression has quantitative deficiencies and more complicated semi-empirical formulae are often used.⁷ Nonetheless, (1.69) is a very useful expression both as an aid to understanding and for quantitative calculations where the variations about a mean state are not large. More discussion is to be found in section 1.8.2.

The equation of state might also (and sometimes very usefully) be written as an expression for the internal energy of a body as a function of the extensive quantities mass, entropy, and chemical composition (extensive meaning that their value depends of the amount of fluid present, as opposed to an intensive quantity such as temperature). For our purposes it is more convenient to divide all of these by the mass of fluid present, so expressing the internal energy per unit volume, I , as a function of the specific volume (or inverse density) $\alpha = \rho^{-1}$, the specific entropy η , and, if appropriate, the salinity S . Thus,

$$I = I(\alpha, \eta, S). \quad (1.70)$$

or $\eta = \eta(I, \alpha, S)$. The equations (1.62) and (1.70) are both *bona fide* equations of state, and given the form of one the other may be derived using the various thermodynamic relations we discuss in next section. Indeed, the thermodynamic state of a body is fully specified by a knowledge of any two of p, ρ, T, η and I , plus its composition.

Clearly, the equation of state introduces, in general, a sixth unknown, temperature, and we will have to introduce another physical principle — the first law of thermodynamics or the principle of energy conservation — to obtain a complete set of equations. However, note that if the equation of state were such that it linked only density and pressure, without introducing another variable, then the equations would be complete. For an important class of fluids this is indeed the case, the simplest cases of all being constant density fluids for which the equation of state is just $\rho = \text{constant}$. Another instance is adiabatic flow in an ideal gas, for which $p\rho^{-\gamma} = \text{constant}$ where $\gamma = c_p/c_v$. A fluid for which the density is a function of pressure alone is called a *barotropic fluid*; otherwise, it is a *baroclinic fluid*. Barotropic fluids are sometimes called *homentropic*, because their internal energy may be written as a function of density alone (and not entropy; see problem 1.13), and equations of state of the form $p = C\rho^\gamma$ are sometimes called polytropic.

1.5 THE THERMODYNAMIC EQUATION

The thermodynamic equation embodies the principle of the conservation of energy, and in fluids in which the equation of state involves temperature (and not just density and pressure) then the thermodynamic equation is necessary to obtain a closed system of equations.

1.5.1 Basic equations

Conservation of energy states that the internal energy of a body may change because of work done by or on it, or because of a heat input, or because of a change in its chemical composition. We write this as

$$dI = dQ - dW + dC \quad (1.71)$$

where dW is the work done by the body, dQ is the heat input to the body, and dC accounts for the change in internal energy caused by a change in its chemical composition (e.g., its salinity). This is the first law of thermodynamics. It is applicable to a definite fluid mass, so we can regard dI as the change in internal energy per unit mass, and similarly for the other quantities. Our goal is to express each term in terms of state variables — the pressure, volume, temperature and so forth and, ultimately, to then obtain a single prognostic equation for a fluid parcel that represents the conservation of energy. First we consider how the three terms on the right-hand-side may separately vary.

Heat Input: The heat input dQ is not the differential of any quantity, and we cannot unambiguously define the heat content of a body as a function of its state. However, the second law of thermodynamics provides a relationship between the heat input and the change in the entropy of a body, namely that in an (infinitesimal) quasi-static or reversible process, with constant composition,

$$Td\eta = dQ \quad (1.72)$$

where η is the *entropy* of the body (remember we use η for entropy and S for salinity). The entropy is a function of the state of a body and it is, by definition, an adiabatic invariant. Entropy is an extensive quantity, meaning that if we double the amount of material then we double the entropy. Here we will be dealing with the amount of a quantity per unit mass, so that η is the specific entropy.

Work done: The work done by a body is equal to the pressure times the change in its volume. That is

$$dW = p d\alpha \quad (1.73)$$

where $\alpha = 1/\rho$ is the specific volume of the fluid and p is the pressure.

Composition: Seawater is a solution of various ions in water — chloride ($\approx 1.9\%$ by weight) sodium (1%), sulfate (0.26%), magnesium (0.13%) and so on, with a total average concentration of about 35‰ (ppt, or parts per thousand). The ratio of the fractions of the various salts is more-or-less constant throughout the ocean, and the concentration of these salts may be parameterized by a single measure, the *salinity*, S . We write

$$dC = \mu dS \quad (1.74)$$

where μ is the *chemical potential* of the solution. Salinity is a compositional invariant; that is, the salinity of a parcel of fluid is conserved unless there are explicit sources and sinks, such as precipitation and evaporation at the surface and molecular diffusion. Salinity is important in determining the density of seawater, but the effects of evaporation and precipitation on changing the internal energy of a fluid parcel are in fact small. In the atmosphere the composition of a parcel of air primarily varies according to the amount of water vapour in it, but this is not a chemical effect. Rather, when condensation or evaporation occurs, heat is released (or required) which provides an entropy source in (1.72).

Collecting equations (1.71) – (1.74) together we have

$$\boxed{dI = T d\eta - p d\alpha + \mu dS} . \quad (1.75)$$

We refer to this (often with $dS = 0$) as *the fundamental thermodynamic relation*. Much of classical thermodynamics follows from this, along with the observation that the macroscopic state of a body in thermodynamic equilibrium is fully specified by a knowledge of its entropy as a function of its internal energy and the specific volume α ; that is, by an equation of state.

1.5.2 * Some thermodynamic relations

From (1.75) it follows that

$$T = \left(\frac{\partial I}{\partial \eta} \right)_{\alpha, S}, \quad p = - \left(\frac{\partial I}{\partial \alpha} \right)_{\eta, S}, \quad \mu = \left(\frac{\partial I}{\partial S} \right)_{\eta, \alpha} . \quad (1.76)$$

These may be regarded as the defining relations for these variables. It can be shown that the pressure and temperature defined this way are indeed related to the internal motion of motion of the molecules that constitute the fluid. Note that if we write

$$d\eta = \frac{1}{T} dI + \frac{p}{T} d\alpha - \frac{\mu}{T} dS, \quad (1.77)$$

it is also clear that

$$p = T \left(\frac{\partial \eta}{\partial \alpha} \right)_{I,S}, \quad T^{-1} = \left(\frac{\partial \eta}{\partial I} \right)_{\alpha,S}, \quad \mu = -T \left(\frac{\partial \eta}{\partial S} \right)_{I,\alpha}. \quad (1.78)$$

In the following derivations, we will unless noted suppose that the composition of a fluid parcel is fixed, and drop the suffix S on partial derivatives unless ambiguity might arise.

Because the right-hand-side of (1.75) is equal to an exact differential, the second derivatives are independent of the order of differentiation. That is,

$$\frac{\partial^2 I}{\partial \eta \partial \alpha} = \frac{\partial^2 I}{\partial \alpha \partial \eta} \quad (1.79)$$

and therefore, using (1.76)

$$\left(\frac{\partial T}{\partial \alpha} \right)_\eta = - \left(\frac{\partial p}{\partial \eta} \right)_\alpha. \quad (1.80)$$

This is one of the *Maxwell relations*, which are a collection of four similar relationships which follow directly from the fundamental thermodynamic relation (1.75) and simple relations between second derivatives. A couple of others will be useful.

Define the *enthalpy* of a fluid by

$$h \equiv I + p\alpha \quad (1.81)$$

then, for a parcel of constant composition, (1.75) becomes

$$dh = T d\eta + \alpha dp. \quad (1.82)$$

But h is a function only of η and p so that in general

$$dh = \left(\frac{\partial h}{\partial \eta} \right)_p d\eta + \left(\frac{\partial h}{\partial p} \right)_\eta dp. \quad (1.83)$$

Comparing the last two equations we have

$$T = \left(\frac{\partial h}{\partial \eta} \right)_p \quad \text{and} \quad \alpha = \left(\frac{\partial h}{\partial p} \right)_\eta. \quad (1.84)$$

Noting that

$$\frac{\partial^2 h}{\partial \eta \partial p} = \frac{\partial^2 h}{\partial p \partial \eta} \quad (1.85)$$

we evidently must have

$$\left(\frac{\partial T}{\partial p} \right)_\eta = \left(\frac{\partial \alpha}{\partial \eta} \right)_p. \quad (1.86)$$

Maxwell's Relations

The four Maxwell equations are:

$$\begin{aligned} \left(\frac{\partial T}{\partial \alpha} \right)_\eta &= - \left(\frac{\partial p}{\partial \eta} \right)_\alpha, & \left(\frac{\partial T}{\partial p} \right)_\eta &= \left(\frac{\partial \alpha}{\partial \eta} \right)_p, \\ \left(\frac{\partial \eta}{\partial p} \right)_T &= - \left(\frac{\partial \alpha}{\partial T} \right)_p, & \left(\frac{\partial \eta}{\partial \alpha} \right)_T &= \left(\frac{\partial p}{\partial T} \right)_\alpha. \end{aligned} \quad (\text{M.1})$$

These imply:

$$\frac{\partial(T, \eta)}{\partial(p, \alpha)} \equiv \left(\frac{\partial T}{\partial p} \right)_\eta \left(\frac{\partial \eta}{\partial \alpha} \right)_T - \left(\frac{\partial T}{\partial \alpha} \right)_\eta \left(\frac{\partial \eta}{\partial p} \right)_T = 0. \quad (\text{M.2})$$

We can also write

$$dI = T d\eta - p d\alpha = d(T\eta) - \eta dT - d(p\alpha) + \alpha dp, \quad (1.87)$$

or

$$dG = -\eta dT + \alpha dp, \quad (1.88)$$

where $G \equiv I - T\eta + p\alpha$ is called the ‘Gibbs free energy’. Now, formally, we have

$$dG = \left(\frac{\partial G}{\partial T} \right)_p dT + \left(\frac{\partial G}{\partial p} \right)_T dp. \quad (1.89)$$

Comparing the last two equations we see that $\eta = -(\partial G / \partial T)_p$ and $\alpha = (\partial G / \partial p)_T$. Furthermore, because

$$\frac{\partial^2 G}{\partial p \partial T} = \frac{\partial^2 G}{\partial T \partial p} \quad (1.90)$$

we have our third Maxwell equation,

$$\left(\frac{\partial \eta}{\partial p} \right)_T = - \left(\frac{\partial \alpha}{\partial T} \right)_p. \quad (1.91)$$

The fourth Maxwell equation, whose derivation is left to the reader, is

$$\left(\frac{\partial \eta}{\partial \alpha} \right)_T = \left(\frac{\partial p}{\partial T} \right)_\alpha, \quad (1.92)$$

and all four Maxwell equations are summarized in the box at the top of the page.

1.5.3 * Internal energy and specific heats

We can obtain some useful relations between the internal energy and specific heat capacities, and some useful estimates of their values, by some simple manipulations of the fundamental thermodynamic relation. Assuming that the composition of the fluid is constant (1.75) is

$$T d\eta = dI + p d\alpha, \quad (1.93)$$

so that

$$T d\eta = \left(\frac{\partial I}{\partial T} \right)_{\alpha} dT + \left[\left(\frac{\partial I}{\partial \alpha} \right)_T + p \right] d\alpha. \quad (1.94)$$

From this, we see that the heat capacity at constant volume (or constant α) c_v is given by

$$c_v \equiv T \left(\frac{\partial \eta}{\partial T} \right)_{\alpha} = \left(\frac{\partial I}{\partial T} \right)_{\alpha}. \quad (1.95)$$

Similarly, using (1.82) we have

$$\begin{aligned} T d\eta &= dh - \alpha dp \\ &= \left(\frac{\partial h}{\partial T} \right)_p dT + \left[\left(\frac{\partial h}{\partial p} \right)_T - \alpha \right] dp. \end{aligned} \quad (1.96)$$

The heat capacity at constant pressure, c_p , is then given by

$$c_p \equiv T \left(\frac{\partial \eta}{\partial T} \right)_p = \left(\frac{\partial h}{\partial T} \right)_p. \quad (1.97)$$

For later use, we define the ratios $\gamma = c_p/c_v$ and $\kappa = R/c_p$.

For an ideal gas $h = I + RT = T(c_v + R)$. But $c_p = (\partial h/\partial T)_p$, and hence $c_p = c_v + R$ [and $(\gamma - 1)/\gamma = \kappa$]. For an ideal gas the internal energy is dependent only on temperature, and statistical mechanics tells us it is equal to $kT/2$ per molecule, or $RT/2$ per unit mass, for each excited degree of freedom, where k is the Boltzmann constant and R the gas constant. The diatomic molecules N_2 and O_2 that comprise most of our atmosphere have two rotational and three translational degrees of freedom, so that $I \approx 5RT/2$, and so $c_v \approx 5R/2$ and $c_p \approx 7R/2$, both being constants. These are in fact very good approximations to the measured values for the earth's atmosphere, and give $c_p \approx 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$. The internal energy is simply $c_v T$ and the enthalpy is $c_p T$. For a liquid, especially one containing dissolved salts such as seawater, no such simple relations are possible: the heat capacities are functions of the state of the fluid, and the internal energy is a function of pressure (or density) as well as temperature.

1.5.4 Equations for fluids

The various thermodynamic relations above apply to identifiable bodies or systems; thus, the heat input affects the fluid parcel to which it is applied, and we can apply the material derivative to the above thermodynamic relations to obtain equations of motion for a moving fluid. But in doing so we make two assumptions:

- (i) That locally the fluid is in thermodynamic equilibrium. This means that, although the thermodynamic quantities like temperature, pressure and density vary in space and time, locally they are related by the thermodynamic relations such as the equation of state and Maxwell's relations.
- (ii) That macroscopic fluid motions are reversible and so not entropy producing. Thus, the diabatic term dQ represents the entropy sources — such effects as viscous dissipation of energy, radiation, and conduction — whereas the macroscopic fluid motion itself is adiabatic.

The first point requires that the temperature variation on the macroscopic scales must be slow enough that there can exist a volume that is small compared to the scale of macroscopic variations, so that temperature is effectively constant within in, but that is also sufficiently large to contain enough molecules so that macroscopic variables such as temperature have a proper meaning. Accepting these assumptions, the expression

$$T d\eta = dQ \quad (1.98)$$

implies that we may write

$$T \frac{D\eta}{Dt} = \dot{Q}, \quad (1.99)$$

where $\dot{Q}$ is the total rate of heat input per unit mass. This is a thermodynamic equation of motion of the fluid.

For seawater a full specification of its thermodynamic state requires a knowledge of the salinity S , and this is determined by the conservation equation

$$\frac{DS}{Dt} = \dot{S}, \quad (1.100)$$

where $\dot{S}$ represents effects of evaporation and precipitation at the ocean surface, and molecular diffusion. (We will write source terms for a materially advected property ϕ as $\dot{\phi}$, so that $D\phi/Dt = \dot{\phi}$. In the equation of motion for a quantity χ that is not necessarily of this form we denote source terms as $Q[\chi]$.) Somewhat analogously, for atmosphere the thermodynamics involve water vapour whose evolution is given by the conservation of water vapour mixing ratio

$$\frac{Dw}{Dt} = \dot{w} \quad (1.101)$$

where $\dot{w}$ represents the effects of condensation and evaporation. Salt has an important effect on the density of seawater, whereas the effect of water vapour on the density of air is slight.

Equation (1.99) is not a *useful* equation unless the entropy can be related to the other fluid variables, temperature, pressure and density. This can be done using an equation of state and the thermodynamic relations we have derived, and is the subject of the following sections. An ideal gas is the simplest case with which to start.

1.5.5 Thermodynamic equation for an ideal gas

For a fluid parcel of constant composition the fundamental thermodynamic relation is

$$dQ = dI + p d\alpha \quad (1.102)$$

For an ideal gas the internal energy is a function of temperature only and $dI = c_v dT$ (problems 1.9 and 1.11), so that

$$dQ = c_v dT + p d\alpha. \quad (1.103)$$

or

$$dQ = c_p dT - \alpha dp, \quad (1.104)$$

using (1.63) and $c_p - c_v = R$. For any fluid, (1.102) implies

$$\frac{dI}{dt} + p \frac{D\alpha}{Dt} = \dot{Q}, \quad (1.105)$$

where $\dot{Q}$ is the heating rate per unit mass. Similarly, (1.103) and (1.104) imply

$$c_v \frac{DT}{Dt} + p \frac{D\alpha}{Dt} = \dot{Q}, \quad (1.106)$$

and

$$c_p \frac{DT}{Dt} - \frac{RT}{p} \frac{Dp}{Dt} = \dot{Q}, \quad (1.107)$$

where these two equations hold only for an ideal gas. Although (1.106) and (1.107) are equations in the state variables p , T and/or α , time derivatives act on two variables and this is not convenient for many purposes. Using the mass continuity equation, (1.106) may be written

$$c_v \frac{DT}{Dt} + p\alpha \nabla \cdot \mathbf{v} = \dot{Q}. \quad (1.108)$$

Using the ideal gas equation we may eliminate T in favor of p and α , giving the equivalent equation

$$\frac{Dp}{Dt} + \gamma p \nabla \cdot \mathbf{v} = \dot{Q} \frac{\rho R}{c_v}. \quad (1.109)$$

Potential temperature and potential density

When a fluid parcel changes pressure adiabatically, it will expand or contract and, using (1.104), its temperature change is determined by

$$c_p dT = \alpha dp. \quad (1.110)$$

We define the *potential temperature* to be the temperature that a fluid would have if moved adiabatically to some reference pressure (often taken to be the 1000 hPa, which is close to the pressure at the earth's surface). Thus, in adiabatic flow the potential temperature of a fluid parcel is conserved, essentially by definition, and $D\theta/Dt = 0$. To relate θ to the other thermodynamic variables we use (1.104) and the ideal gas equation to write the thermodynamic equation as

$$d\eta = c_p d \ln T - R d \ln p. \quad (1.111)$$

The definition of potential temperature then implies that

$$c_p d \ln \theta = c_p d \ln T - R d \ln p, \quad (1.112)$$

and this is satisfied by

$$\theta = \left(\frac{p_{\text{ref}}}{p} \right)^\kappa T \quad (1.113)$$

where p_{ref} is a reference pressure and $\kappa = R/c_p$.

Note that

$$d\eta = c_p \frac{d\theta}{\theta} \quad (1.114)$$

and, if c_p is constant,

$$\eta = c_p \ln \theta. \quad (1.115)$$

Equation (1.114) is in fact a general expression for potential temperature of a fluid parcel of constant composition (see section 1.8.1), but (1.115) applies only if c_p is constant, as in, to a very good approximation, the earth's atmosphere.

Using (1.112), the thermodynamic equation in the presence of heating is simply

$$\boxed{c_p \frac{D\theta}{Dt} = \frac{\theta}{T} \dot{Q}} \quad (1.116)$$

with θ given by (1.113). Equations (1.108), (1.109) and (1.116) are all equivalent forms of the thermodynamic equation for an ideal gas.

The *potential density*, ρ_θ , is the density that a fluid parcel would have if moved adiabatically and at constant composition to a reference pressure, p_{ref} . If the equation of state is written as

$$\rho = f(p, T) \quad (1.117)$$

then the potential density is just

$$\rho_\theta = f(p_{\text{ref}}, \theta). \quad (1.118)$$

For an ideal gas we therefore have

$$\rho_\theta = \frac{p_{\text{ref}}}{\theta R}; \quad (1.119)$$

that is, potential density is proportional to the inverse of potential temperature.

Finally, for later use we note that for small variations around a reference state manipulation of the ideal gas equation gives

$$\frac{\delta \theta}{\theta} = \frac{\delta T}{T} - \kappa \frac{\delta p}{p} = \frac{1}{\gamma} \frac{\delta p}{p} - \frac{\delta \rho}{\rho}. \quad (1.120)$$

1.5.6 * Thermodynamic equation for liquids

For a liquid such as seawater no simple exact equation of state exists. Thus, although (1.116) holds at constant salinity for a liquid by virtue of the definition of potential temperature, but an accurate expression relating potential temperature to the other thermodynamic variables is nonlinear, complicated and, to most eyes, uninformative. Yet for both theoretical and modelling work a thermodynamic equation is needed to represent energy conservation, and an equation of state needed to close the system, and one of two approaches is thus generally taken: For most theoretical work and for idealized models a simple analytic but approximate equation of state is used, but in situations where more accuracy is called for, such as quantitative modelling or observational work, an accurate but complex semi-empirical equation of state is used. This section outlines how relatively simple thermodynamic equations may be derived that are adequate in many circumstances, and which illustrate the principles used in deriving more complicated equations.

Thermodynamic equation using pressure and density

If we regard η as a function of pressure and density (and salinity if appropriate) we obtain

$$\begin{aligned} T d\eta &= T \left(\frac{\partial \eta}{\partial \rho} \right)_{p,S} d\rho + T \left(\frac{\partial \eta}{\partial p} \right)_{\rho,S} dp + T \left(\frac{\partial \eta}{\partial S} \right)_{\rho,p} dS \\ &= T \left(\frac{\partial \eta}{\partial \rho} \right)_{p,S} d\rho - T \left(\frac{\partial \eta}{\partial \rho} \right)_{p,S} \left(\frac{\partial \rho}{\partial p} \right)_{\eta,S} dp + T \left(\frac{\partial \eta}{\partial S} \right)_{\rho,p} dS. \end{aligned} \quad (1.121)$$

From this, and using with (1.99) and (1.100), we obtain

$$T \left(\frac{\partial \eta}{\partial \rho} \right)_{p,S} \frac{D\rho}{Dt} - T \left(\frac{\partial \eta}{\partial \rho} \right)_{p,S} \left(\frac{\partial \rho}{\partial p} \right)_{\eta,S} \frac{Dp}{Dt} = Q - T \left(\frac{\partial \eta}{\partial S} \right)_{\rho,p} Q_S \quad (1.122)$$

But $(\partial p / \partial \rho)_{\eta,S} = c_s^2$ where c_s is the speed of sound (see section 1.6). This is a measurable quantity in a fluid, and often nearly constant, and so useful to keep in an equation. Then the thermodynamic equation may be written in the form

$$\boxed{\frac{D\rho}{Dt} - \frac{1}{c_s^2} \frac{Dp}{Dt} = Q[\rho]} \quad (1.123)$$

where $Q[\rho] = (\partial \rho / \partial \eta)_{p,S} \dot{Q} / T - (\partial \rho / \partial S)_{\rho,p} Q[S]$ appropriately represents the effects of all diabatic and salinity source terms. This form of the thermodynamic equation is valid for both liquids and gases.

Approximations: The speed of sound in a fluid is related to its compressibility — the less compressible the fluid, the greater the sound speed. In a fluid it is often sufficiently high that the second term in (1.123) can be neglected, and the thermodynamic equation takes the simple form:

$$\frac{D\rho}{Dt} = Q[\rho]. \quad (1.124)$$

This equation is a very good approximation for many laboratory fluids. Note that this equation expresses the principle of conservation of energy for a liquid. It is a very different equation from the mass conservation equation, which for compressible fluids is also an evolution equation for density.

In the ocean the enormous pressures resulting from columns of seawater kilometers deep mean that the second term in (1.123) is not negligible, and a better approximation results if we suppose that the pressure is given by the weight of the fluid above it — the hydrostatic approximation. In this case $dp = -\rho g dz$ and (1.123) becomes

$$\frac{D\rho}{Dt} + \frac{\rho g}{c_s^2} \frac{Dz}{Dt} = Q[\rho]. \quad (1.125)$$

In the second term the height field varies much more than the density field, so a good approximation is to replace ρ by a constant, ρ_0 , just in this term. Taking the speed of sound also to be constant gives

$$\frac{D}{Dt} \left[\rho + \frac{\rho_0 z}{H_\rho} \right] = Q[\rho] \quad (1.126)$$

where

$$H_\rho = c_s^2 / g \quad (1.127)$$

is the *density scale height* of the ocean. In water, $c_s \approx 1500 \text{ m s}^{-1}$ so that $H_\rho \approx 200 \text{ km}$. The quantity in square brackets in (1.126) is (in this approximation) the *potential density*, as it is the density that a parcel would have if moved adiabatically and with constant composition to the reference height $z = 0$. The density scale height as defined here is due to the mean compressibility (i.e., the change in density with pressure) of seawater and, because sound speed varies only slightly in the ocean, this is nearly a constant. The adiabatic lapse rate of density is the rate at which the density of a parcel changes when undergoing an adiabatic displacement. From (1.126) it is approximately

$$-\left(\frac{\partial \rho}{\partial z}\right)_\eta \approx \frac{\rho_0 g}{c_s^2} \approx 5 \text{ (kg m}^{-3}\text{) / km} \quad (1.128)$$

so that if a parcel is moved adiabatically from the surface to the deep ocean (5 km) its density will increase by about 25 kg m^{-3} , a fractional density increase of about 1/40 or 2.5%.

Finally, we remark that the expression (1.125) is as valid in the gaseous atmosphere as the liquid ocean, but (1.126) is not so, because of the $\mathcal{O}(1)$ density variations with height in the atmosphere. A simple expression is nevertheless available in the atmospheric case, as in section 1.5.5.

Thermodynamic equation using pressure and temperature

Taking entropy to be a function of pressure and temperature (and salinity if appropriate) we have

$$\begin{aligned} T d\eta &= T \left(\frac{\partial \eta}{\partial T}\right)_{p,S} dT + T \left(\frac{\partial \eta}{\partial p}\right)_{T,S} dp + T \left(\frac{\partial \eta}{\partial S}\right)_{T,p} dS \\ &= c_p dT + T \left(\frac{\partial \eta}{\partial p}\right)_{T,S} dp + T \left(\frac{\partial \eta}{\partial S}\right)_{T,p} dS. \end{aligned} \quad (1.129)$$

For a moving fluid, and using (1.99) and (1.100), this implies,

$$\frac{DT}{Dt} + \frac{T}{c_p} \left(\frac{\partial \eta}{\partial p}\right)_{T,S} \frac{Dp}{Dt} = Q[T]. \quad (1.130)$$

where $Q[T] = \dot{Q}/c_p - T c_p^{-1} Q[S](\partial \eta / \partial S)$ includes the effects of the entropy and saline source terms. Assuming salt to be conserved, substitute the Maxwell

relation (1.91) in the form

$$\left(\frac{\partial \eta}{\partial p} \right)_T = \frac{1}{\rho^2} \left(\frac{\partial \rho}{\partial T} \right)_p \quad (1.131)$$

to give

$$\frac{DT}{Dt} + \frac{T}{c_p \rho^2} \left(\frac{\partial \rho}{\partial T} \right)_p \frac{Dp}{Dt} = Q[T], \quad (1.132a)$$

or, equivalently,

$$\frac{DT}{Dt} - \frac{T}{c_p} \left(\frac{\partial \alpha}{\partial T} \right)_p \frac{Dp}{Dt} = Q[T]. \quad (1.132b)$$

The density and temperature are related through a measurable coefficient of thermal expansion β_T where

$$\left(\frac{\partial \rho}{\partial T} \right)_p = -\beta_T \rho \quad (1.133)$$

Equation (1.132) then becomes

$$\boxed{\frac{DT}{Dt} - \frac{\beta_T T}{c_p \rho} \frac{Dp}{Dt} = Q[T]}. \quad (1.134)$$

This is form of the thermodynamic equation is valid for both liquids and gases, and in an ideal gas $\beta_T = 1/T$.

Approximations: Liquids are characterized by a small coefficient of thermal expansion, and it is sometimes acceptable in laboratory fluids to neglect the second term on the left-hand-side of (1.134). We then obtain an equation analogous to (1.124), namely

$$\frac{DT}{Dt} = Q[T]. \quad (1.135)$$

This approximation relies on the smallness of the coefficient of thermal expansion, which is related to the high speed of sound in liquids. A better approximation is to again suppose that the pressure in (1.134) varies according only to the weight of the fluid above it. Then $dp = -\rho g dz$ and (1.134) becomes

$$\frac{1}{T} \frac{DT}{Dt} + \frac{\beta_T g}{c_p} \frac{Dz}{Dt} = \frac{Q[T]}{T}. \quad (1.136)$$

For small variations of T , and if β_T is nearly constant, this simplifies to

$$\frac{D}{Dt} \left(T + \frac{T_0 z}{H_T} \right) = Q[T] \quad (1.137)$$

where

$$H_T = c_p / (\beta_T g) \quad (1.138)$$

is the *temperature scale height* of the fluid. The quantity $T + T_0 z / H_T$ is (in this approximation) the *potential temperature*, θ , so called because it is the temperature that a fluid at a depth z would have if moved adiabatically to a reference depth, here taken as $z = 0$ — the temperature changing because of the work done by or on the fluid parcel as it expands or is compressed. That is,

$$\theta \approx T + \frac{\beta_T g T_0}{c_p} z \quad (1.139)$$

In seawater, however, the expansion coefficient β_T and c_p are functions of pressure and (1.139) is not good enough for quantitative calculations. With the approximate values for the ocean of $\beta_T \approx 2 \times 10^{-4} \text{ K}^{-1}$ and $c_p \approx 4 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$ we obtain $H_T \approx 2000 \text{ km}$.

The adiabatic lapse rate is rate at which the temperature of a parcel changes in the vertical when undergoing an adiabatic displacement. From (1.136) it is

$$\Gamma_{\text{ad}} = - \left(\frac{\partial T}{\partial z} \right)_\eta = \frac{T g \beta_T}{c_p}. \quad (1.140)$$

In general it is a function of temperature, salinity and pressure, but it is a calculable quantity if β_T is known. With the oceanic values above, it is approximately 0.15 K km^{-1} . Again this is not accurate enough for quantitative oceanography because the expansion coefficient is a function of pressure. Nor is it a good measure of stability, because of the effects of salt.

It is interesting that the scale heights given by (1.127) and (1.138) differ so much. As mentioned, the first is due to the compressibility of seawater [and so related to c_s^2 , or β_p in (1.68)] whereas the second is due to the change of density with temperature [β_T in (1.68)], and is the distance over which the the difference between temperature and potential temperature changes by an amount equal to the temperature itself (i.e., by about 273 K). The two heights differ so much because the value of thermal expansion coefficient is not directly tied to the compressibility — for example, fresh water at 4° C has a zero thermal expansion, and so would have an infinite temperature scale height, but its compressibility differs little from water at 20° C.⁸ (See also problem 1.16.)

In the atmosphere the ideal gas relationship gives $\beta_T = 1/T$ and so

$$\Gamma_{\text{ad}} = \frac{g}{c_p} \quad (1.141)$$

which is approximately 10 K km^{-1} . The only approximation involved in deriving this is the use of the hydrostatic relationship.

Forms of the Thermodynamic Equation

General form

For a parcel of constant composition the thermodynamic equation is

$$T \frac{D\eta}{Dt} = \dot{Q} \quad \text{or} \quad c_p \frac{D \ln \theta}{Dt} = \frac{1}{T} \dot{Q} \quad (\text{T.1})$$

where η is the entropy, θ is the potential temperature, $c_p \ln \theta = \eta$ and $\dot{Q}$ is the heating rate. Applying the first law of thermodynamics $T d\eta = dI + p d\alpha$ gives:

$$\frac{DI}{Dt} + p \frac{D\alpha}{Dt} = \dot{Q} \quad \text{or} \quad \frac{DI}{Dt} + RT \nabla \cdot \mathbf{v} = \dot{Q} \quad (\text{T.2})$$

where I is the internal energy.

Ideal gas

For an ideal gas $dI = c_v dT$ and the (adiabatic) thermodynamic equation may be written in the following equivalent, exact, forms:

$$\begin{aligned} c_p \frac{DT}{Dt} - \alpha \frac{Dp}{Dt} &= 0, & \frac{Dp}{Dt} + \gamma p \nabla \cdot \mathbf{v} &= 0, \\ c_v \frac{DT}{Dt} + p \alpha \nabla \cdot \mathbf{v} &= 0, & \frac{D\theta}{Dt} &= 0, \end{aligned} \quad (\text{T.3})$$

where $\theta = T(p_{\text{ref}}/p)^\kappa$. The two expressions on the second line are usually the most useful in modelling and theoretical work.

Liquids

For liquids we may usefully write the (adiabatic) thermodynamic equation as a conservation equation for potential temperature θ or potential density ρ_{pot} and represent these in terms of other variables. For example:

$$\frac{D\theta}{Dt} = 0, \quad \theta \approx \begin{cases} T & \text{(approximately)} \\ T + (\beta_T g z / c_p) & \text{(with some thermal expansion),} \end{cases} \quad (\text{T.4a})$$

$$\frac{D\rho_{\text{pot}}}{Dt} = 0, \quad \rho_{\text{pot}} \approx \begin{cases} \rho & \text{(very approximately)} \\ \rho + (\rho_0 g z / c_s^2) & \text{(with some compression).} \end{cases} \quad (\text{T.4b})$$

Unlike (T.3) these are not equivalent forms. More accurate semi-empirical expressions that may also include saline effects are often used for quantitative applications.

Thermodynamic equation using density and temperature

In this section we leave details and approximations to the reader. Taking entropy to be a function of density and temperature (and salinity if appropriate) we have

$$\begin{aligned} T d\eta &= T \left(\frac{\partial \eta}{\partial T} \right)_{\alpha, S} dT + T \left(\frac{\partial \eta}{\partial \alpha} \right)_{T, S} d\alpha + T \left(\frac{\partial \eta}{\partial S} \right)_{T, \alpha} dS \\ &= c_v dT + T \left(\frac{\partial \eta}{\partial \alpha} \right)_{T, S} d\alpha + T \left(\frac{\partial \eta}{\partial S} \right)_{T, \alpha} dS. \end{aligned} \quad (1.142)$$

For a moving fluid this implies,

$$\frac{DT}{Dt} + \frac{T}{c_v} \left(\frac{\partial \eta}{\partial \alpha} \right)_{T, S} \frac{D\alpha}{Dt} = Q_2[T]. \quad (1.143)$$

where $Q_2[T] = (c_p/c_v)Q[T]$. In an incompressible fluid the second term in (1.143) is small, and $c_p \approx c_v$.

The thermodynamic equations for a fluid are summarized on page 33. Also, note that for ideal gas (1.123) and (1.134) are exactly equivalent to (1.108) or (1.109) (problem 1.8).

1.6 SOUND WAVES

Full of sound and fury, signifying nothing.
Shakespeare, *Macbeth*.

We now consider, rather briefly, one of the most common phenomena in fluid dynamics yet one which is relatively unimportant for geophysical fluid dynamics — sound waves. Sound itself is not a meteorologically or oceanographically important phenomenon, except in a few special cases, for the pressure disturbance produced by sound waves is a tiny fraction of the ambient pressure and too small to be of importance for the circulation. For example, the ambient surface pressure in the atmosphere is about 10^5 Nm^{-2} and variations due to large-scale weather phenomena are about 10^3 Nm^{-2} , often larger, whereas sound waves of 60 dB produce pressure variations of about 10 Nm^{-2} .

The smallness of the disturbance produced by sound waves justifies a linearization of the equations of motion about a spatially uniform basic state (denoted by a subscript 0) that is a time-independent solution to the equations of motion. Thus, we write $\mathbf{v} = \mathbf{v}_0 + \mathbf{v}'$, $\rho = \rho_0 + \rho'$ and so on, substitute in the equations of motion, and neglect terms involving products of primed quantities,

as these are assumed small. By choice of our reference frame we will simplify matters further by setting $\mathbf{v}_0 = 0$.

The linearized momentum and mass conservation equations are then

$$\rho_0 \frac{\partial \mathbf{v}'}{\partial t} = -\nabla p' \quad (1.144)$$

$$\rho_0 \frac{\partial \rho'}{\partial t} = -\rho_0 \nabla \cdot \mathbf{v}' \quad (1.145)$$

Notice that we are not eliminating the nonlinear term in the momentum equation via a comparison of the magnitude of $\mathbf{v}' \cdot \nabla \mathbf{v}'$ with that of $\mathbf{v}_0 \cdot \nabla \mathbf{v}'$, since the latter is zero. Rather, we neglect the nonlinear term by assuming it is smaller than the local time-derivative term $\partial \mathbf{v}' / \partial t$, and that the *dominant balance* in the equation lies between this term and the pressure term on the right-hand-side. The ensuing linear equations do not in themselves determine the magnitude of the disturbance, and the linear approximation must be checked *a posteriori* (problem 1.4).

Now, sound waves are largely adiabatic. Thus,

$$\frac{dp}{dt} = \left. \frac{\partial p}{\partial \rho} \right|_{\eta} \frac{d\rho}{dt}, \quad (1.146)$$

where $(\partial p / \partial \rho)_{\eta}$ is the derivative at constant entropy, whose particular form is given by the equation of state for the fluid at hand. Then, from (1.144) – (1.146) we obtain a single equation for pressure,

$$\frac{\partial p'}{\partial t} = c_s^2 \nabla^2 p', \quad (1.147)$$

where $c_s^2 = (\partial p / \partial \rho)_{\eta}$. Eq. (1.147) is the classical wave equation; solutions propagate at a speed c_s which thus may be identified as the speed of sound. In an ideal gas manipulation of the equation of state leads to $p = C\rho^{\gamma}$ for adiabatic compressions, whence $c_s^2 = \gamma p / \rho = \gamma RT$ where $\gamma = c_p / c_v$. Values of γ typically range from 5/3 for a monatomic gas to 7/5 for a diatomic gas. For air, which is almost entirely diatomic, this leads to $c_s \approx 350 \text{ m s}^{-1}$ at 300 K. In seawater no such theoretical approximation is easily available but measurements show that $c_s \approx 1500 \text{ m s}^{-1}$.

1.7 COMPRESSIBLE AND INCOMPRESSIBLE FLOW

There are many important circumstances in fluid dynamics when the dynamics are completely specified by the momentum equation and a form of the mass conservation equation. This arises in general when the equation of state does not

introduce temperature, and the simplest such equation of state would be that density be constant. Although there are probably no fluids of exactly constant density, in many cases the density of a fluid will vary so little that it is a very good approximation to consider the density effectively constant in the mass conservation equation. The motion is then said to be incompressible. For example, in the earth's oceans the density varies by less than 5%, even though the pressure at the ocean bottom, a few kilometers below the surface, is several hundred times the atmospheric pressure at the surface.

Let us first note how the mass conservation equation simplifies when density is strictly constant, and then consider under what conditions this is a good approximation.

1.7.1 Constant density fluids

If a fluid is strictly of constant density, the mass continuity equation (MM.1) simplifies easily by neglecting all derivatives of density yielding

$$\boxed{\nabla \cdot \mathbf{v} = 0} . \quad (1.148)$$

The *prognostic* equation (1.42) has become a *diagnostic* equation (1.148), or a constraint to be satisfied by the velocity at each instant of the fluid motion. A consequence of this equation is that the volume of each material fluid element remains constant. To see this recall the expression for the conservation of mass in the form

$$\frac{D}{Dt}(\rho \Delta V) = 0 \quad (1.149)$$

If ρ is constant this reduces to an expression for volume conservation, namely $D\Delta V/Dt = 0$, whence (1.148) is recovered because $D\Delta V/Dt = \Delta V \nabla \cdot \mathbf{v}$. The absolute constancy of density is an idealization which is rarely, if ever, realized in the natural world. However, there are many cases in which it is a very good approximation to use (1.148) in place of the full continuity equation. These depend not only on the physical nature of the fluid but also on the flow itself, as we shall now discuss.

1.7.2 Incompressible flows

An incompressible fluid is one in which the density of a given fluid element does not change.⁹ Thus, in the mass continuity equation

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0, \quad (1.150)$$

the material derivative of density is neglected and we obtain,

$$\nabla \cdot \mathbf{v} = 0. \quad (1.151)$$

In reality no fluid is truly incompressible and for (1.151) to approximately hold we just require that that

$$|D\rho/Dt| \ll |\rho \nabla \cdot \mathbf{v}|. \quad (1.152)$$

Our working definition of incompressibility, then, is that *in an incompressible fluid density changes (from whatever cause) are so small that they have a negligible affect on the mass balance*, and (1.150) may be replaced by (1.151). We do not need to assume that the densities of differing fluid elements are similar to each other, just that the density of any particular fluid element is nearly constant, but in the ocean (and in most liquids) it is in fact the case that variations in density are everywhere small compared to the mean density. The atmosphere is not incompressible and (1.151) does not in general hold there. Note also that the fact that $\nabla \cdot \mathbf{v} = 0$ does *not* imply that we may independently use $D\rho/Dt = 0$. Indeed for a liquid with equation of state $\rho = \rho_0(1 - \beta_T(T - T_0))$ and thermodynamic equation $c_p DT/Dt = \dot{Q}$ we obtain

$$\frac{D\rho}{Dt} = -\frac{\beta_T \rho_0}{c_p} \dot{Q}. \quad (1.153)$$

Some conditions for incompressibility

The condition that density is largely unaffected by pressure gives a necessary condition for the legitimate use of (1.151), as follows. First assume adiabatic flow, and omit the gravitational term. Then

$$\frac{dp}{dt} = \left(\frac{\partial p}{\partial \rho} \right)_\eta \frac{d\rho}{dt} = c_s^2 \frac{d\rho}{dt} \quad (1.154)$$

so that the density and pressure variations of a fluid parcel are related by

$$\delta p \sim c_s^2 \delta \rho. \quad (1.155)$$

From the momentum equation we estimate

$$\frac{U^2}{L} \sim \frac{1}{L} \frac{\delta p}{\rho_0}, \quad (1.156)$$

where U and L are typical velocities and lengths and where ρ_0 is a representative value of the density. Using (1.155) and (1.156) gives $U^2 \sim c_s^2 \delta \rho / \rho_0$. The incompressibility condition $\delta \rho / \rho_0 \ll 1$ then becomes

$$\frac{U^2}{c_s^2} \ll 1. \quad (1.157)$$

That is, for a flow to be incompressible the fluid velocities must be less than the speed of sound, or the Mach number, $M \equiv U/c_s$, must be small.

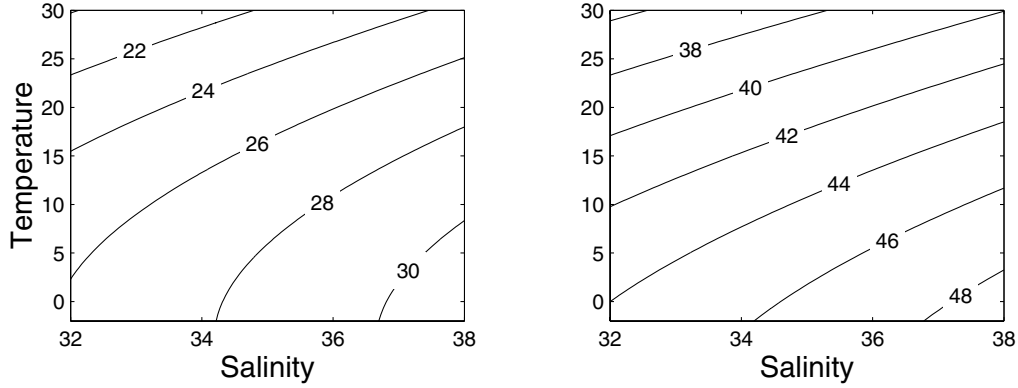


Figure 1.3 A temperature-salinity diagram for seawater, calculated using an empirical equation of state. Contours are (density - 1000) kg m⁻³, and the temperature is potential temperature, which in the deep ocean may be less than *in situ* temperature a degree or so (see Fig. 1.4). Left panel: at sea-level ($p = 10^5$ Pa = 1000 mb). Right panel: at $p = 4 \times 10^7$ Pa, a depth of about 4 km. Note that in both cases the contours are slightly convex.

In the earth's atmosphere it is apparent that density changes significantly with height. Assuming hydrostatic balance and an ideal gas, then

$$\frac{1}{\rho} \frac{\partial p}{\partial z} = -g, \quad (1.158)$$

and if (for simplicity) we assume that atmosphere is isothermal then

$$\frac{\partial p}{\partial z} = \left(\frac{\partial p}{\partial \rho} \right)_T \frac{\partial \rho}{\partial z} = RT_0 \frac{\partial \rho}{\partial z}. \quad (1.159)$$

Using (1.158) and (1.159) gives

$$\rho = \rho_0 \exp[-z/H_\rho], \quad (1.160)$$

where $H_\rho = RT_0/g$ is the (density) *scale height* of the atmosphere. It is easy to see that density changes are negligible only if we concern ourselves with motion less than the scale height, so this is another necessary condition for incompressibility.

In the atmosphere, although the Mach number is small for most flows, vertical displacements often exceed the scale height and in those cases the flow cannot be considered incompressible. In the ocean density changes from all causes are small and the ocean may be considered an incompressible fluid. We return to this in the next chapter when we consider the Boussinesq equations.

1.8 * MORE THERMODYNAMICS OF LIQUIDS

1.8.1 Potential temperature, potential density and entropy

For an ideal gas we were able to derive a single prognostic equation for a single variable, potential temperature. Since potential temperature is in turn simply related to the temperature and pressure, this is a useful prognostic equation. Can we achieve something similar with a more general equation of state, with non-constant coefficients of expansion?

Potential temperature

The potential temperature is defined as the temperature that a parcel would have if moved adiabatically to a given reference pressure p_{ref} , often taken as 10^5 Pa (or or 1000 hPa, or 1000 mb, approximately the pressure at the sea-surface). Thus it may be calculated, at least in principal, through an integral of the form

$$\theta(S, T, p; p_{\text{ref}}) = T + \int_p^{p_{\text{ref}}} \Gamma'_{\text{ad}}(S, T, p) dp \quad (1.161)$$

where $\Gamma'_{\text{ad}} = (\partial T / \partial p)_{\eta}$.

The potential temperature of a fluid parcel is directly related to its entropy, provided its composition does not change. We already demonstrated this for an ideal gas, and to see it explicitly in the general case let us first write the equation of state in the form

$$\eta = \eta(S, T, p). \quad (1.162)$$

Now, by definition of potential temperature we have

$$\eta = \eta(S, \theta; p_{\text{ref}}), \quad (1.163)$$

and similarly

$$\theta = \theta(\eta, S; p_{\text{ref}}). \quad (1.164)$$

For a parcel of constant salinity, changes in entropy are caused only by changes in potential temperature so that

$$d\eta = \frac{\partial \eta(S, \theta; p_{\text{ref}})}{\partial \theta} d\theta. \quad (1.165)$$

Now, if we express entropy as a function of temperature and pressure then

$$\begin{aligned} T d\eta &= T \left(\frac{\partial \eta}{\partial T} \right)_p dT + T \left(\frac{\partial \eta}{\partial p} \right)_T dp \\ &= c_p dT - T \left(\frac{\partial \alpha}{\partial T} \right)_p dp. \end{aligned} \quad (1.166)$$

using one of the Maxwell relations. Suppose a fluid parcel moves adiabatically, then $d\eta = 0$ and, by (1.165), $d\theta = 0$. That is, the potential temperature at each point along its trajectory is constant and $\theta = \theta(\eta)$. How do we evaluate this function? Simply note that the temperature at the reference pressure p_{ref} is the potential temperature, so that directly from (1.166)

$$\boxed{d\eta = c_p(p_{\text{ref}}, \theta) \frac{d\theta}{\theta}}, \quad (1.167)$$

and $d\eta/d\theta = c_p(p_{\text{ref}}, \theta)/\theta$. If c_p is constant this integrates to

$$\eta = c_p \ln \theta + \text{constant} \quad (1.168)$$

as for an ideal gas (1.115).

Since potential temperature is conserved in adiabatic motion, the thermodynamic equation can be written

$$c_p \frac{D\theta}{Dt} = \frac{\theta}{T} \dot{Q}. \quad (1.169)$$

where the right-hand-side represents heating. (If salinity is changing, then the right-hand side should also include any saline source terms and saline diffusion. However, such terms usually have a very small effect.) This equation is equivalent to (1.123) or (1.134), although it is only useful if θ can be simply related to the other state variables. In principle this can be done using (1.161), and in practice empirical relationships have been derived that express potential temperature in terms of the local temperature, pressure and salinity, and density in terms of potential temperature, salinity and pressure (see section 1.8.2 for more discussion).

Potential density

Potential density, ρ_θ , is defined as the density that a parcel would have if moved adiabatically and with fixed composition to a given reference pressure p_{ref} often, but not always, taken as 1 bar. If the equation of state is of the form

$$\rho = \rho(S, T, p) \quad (1.170)$$

then by definition we have

$$\rho_\theta = \rho(S, \theta; p_{\text{ref}}). \quad (1.171)$$

For a parcel moving adiabatically (i.e., fixed salinity and entropy or potential temperature) its potential density is therefore conserved.

For an ideal gas we have

$$\rho = \frac{p}{RT} \quad (1.172)$$

and so

$$\rho_\theta = \frac{p_{\text{ref}}}{R\theta}. \quad (1.173)$$

Since p_{ref} and R are constants, potential density is just the inverse of potential temperature and provides no new information. But in the oceans potential density accounts for the effect of salinity on density and so is a much better measure of the static stability of a column of water than density itself.

From (1.126) an approximate expression for the potential density in the ocean is

$$\rho_\theta = \left(\rho - \frac{\rho_0 g z}{c_s^2} \right). \quad (1.174)$$

Although this may suffice for theoretical or some modelling work, the vertical gradient of potential temperature in the ocean is often close to zero and a still more accurate, generally semi-empirical, expression is needed to determine stability properties.

Because density is so nearly constant in the ocean, it is common to subtract the amount 1000 kg m^{-3} before quoting its value. Depending on whether the density is calculated *in situ* or is the potential density the results are called σ_t ('sigma-tee') or σ_θ ('sigma-theta') respectively. Thus,

$$\sigma_t = \rho(p, T, S) - 1000 \quad (1.175a)$$

$$\sigma_\theta = \rho(p_{\text{ref}}, \theta, S) - 1000. \quad (1.175b)$$

If the potential density is referenced to a particular level, this is denoted by a subscript on the σ . Thus, σ_2 is the potential density referenced to 200 bars of pressure, or about 2 kilometers depth.

1.8.2 * More About Seawater

We now consider, rather didactically, some of the properties of the equation of state for seawater, noting in particular those nonlinearities that, although small, give it somewhat peculiar properties. We use a prototypical equation of state, (1.69) that, although not highly accurate except for small variations around a reference state, does capture the essential nonlinearities.¹⁰ That equation of state is

$$\alpha = \alpha_0 [1 + (\beta_T + \gamma^* p)(T - T_0) + \frac{1}{2} \beta_T^* (T - T_0)^2 - \beta_S (S - S_0) - \beta_p (p - p_0)]. \quad (1.176)$$

where we may take $p_0 = 0$ and $\beta_p = \alpha_0 / c_0^2$, where c_0 is a reference sound speed. The starred parameters are associated with the nonlinear terms: β_T^* is the second expansion coefficient and γ^* is the 'thermobaric parameter' which determines the extent to which the thermal expansion of water depends on temperature. An equation of this form is useful because its coefficients can, in principle, be measured in the field or in the laboratory, and approximate values are given in table

Parameter	Description	Value
ρ_0	Reference Density	1.027 kg m^{-3}
α_0	Reference Specific Volume	$9.738 \times 10^{-4} \text{ m}^3 \text{ kg}^{-1}$
T_0	Reference temperature	283 K
S_0	Reference salinity	35 psu $\approx 35\text{‰}$
c_0	Reference sound speed	1490 m s^{-1}
β_T	First thermal expansion coefficient	$1.67 \times 10^{-4} \text{ K}^{-1}$
β_T^*	Second thermal expansion coefficient	$1.00 \times 10^{-5} \text{ K}^{-2}$
β_S	Haline contraction coefficient	$0.78 \times 10^{-3} \text{ psu}^{-1}$
β_p	Inverse bulk modulus $\approx \alpha_0 / c_0^2$	$4.39 \times 10^{-10} \text{ m s}^2 \text{ kg}^{-1}$
γ	Thermobaric parameter	$1.09 \times 10^{-8} \text{ Pa}^{-1}$
c_{p0}	Specific heat capacity at constant pressure	$3986 \text{ J kg}^{-1} \text{ K}^{-1}$

Table 1.2 Various thermodynamic and equation-of-state parameters for seawater. Specifically, these parameters may be used in the approximate equations of state (1.69) and (1.188).

1.2. However, it may not be the most useful form for modelling or observational work, because T is not materially conserved. Let us use this equation to deduce various thermodynamic quantities of interest, and also transform it to a more useful form for modelling.

Potential temperature of seawater

It would be useful to express (1.176) in terms of materially conserved variables, and in particular in terms of potential temperature rather than temperature. By definition the potential temperature is obtained by integrating the adiabatic lapse rate from the in situ pressure to the reference pressure (zero); that is

$$\theta - T = \int_z^{z(p=0)} \left(\frac{\partial T}{\partial z} \right)_\eta dz = \int_p^0 \left(\frac{\partial T}{\partial p} \right)_\eta dp \quad (1.177)$$

Using (1.166), the adiabatic lapse rate is

$$\left(\frac{\partial T}{\partial p} \right)_\eta = \frac{T}{c_p} \left(\frac{\partial \alpha}{\partial T} \right)_{p,S} \quad (1.178)$$

$$= \frac{T}{c_p} \alpha_0 [\beta_T (1 + \gamma^* p) + \beta_T^* (T - T_0)] \quad (1.179)$$

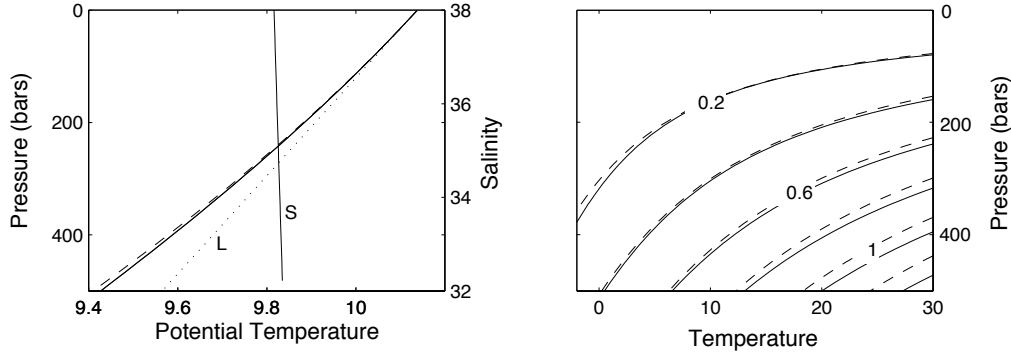


Figure 1.4 Examples of variation of potential temperature of seawater with pressure, temperature and salinity. Left panel: the sloping lines show potential temperature as a function of pressure at fixed salinity ($S = 35$ psu) and temperature (13.36°C , the value of potential temperature at the surface). The solid line is computed using an accurate, complicated, empirical equation of state (Bryden 1973), the almost-coincident dashed line uses the simpler expression (1.186) and the dotted line (labelled L) uses the even simpler linear expression (1.187). The near vertical solid line, labelled S, shows the variation of potential temperature with salinity at fixed temperature and pressure. Right panel: Contours of the difference between temperature and potential temperature, $(T - \theta)$ in the pressure-temperature plane, again for $S = 35$ psu. The solid lines use the accurate empirical formula, and the dashed lines use (1.186). The simpler equation can be improved locally, but not globally, by tuning the coefficients. (100 bars of pressure (10^7 Pa or 10 MPa) is approximately 1 km depth.)

Now, c_p satisfies

$$c_p = T \left(\frac{\partial \eta}{\partial T} \right)_p \quad (1.180)$$

so that

$$\begin{aligned} \left(\frac{\partial c_p}{\partial p} \right)_{T,S} &= T \left(\frac{\partial}{\partial T} \left(\frac{\partial \eta}{\partial p} \right)_T \right)_p \\ &= T \frac{\partial^2 \alpha}{\partial T^2} \end{aligned} \quad (1.181)$$

using the Maxwell relation (1.91). Thus, for our equation of state, we have

$$\left(\frac{\partial c_p}{\partial p} \right)_{T,S} = -T \alpha_0 \beta_T^*, \quad (1.182)$$

and therefore

$$c_p = c_{p0}(T, S) - p T \alpha_0 \beta_T^*. \quad (1.183)$$

The first term cannot be determined solely from the equation of state; in fact for seawater specific heat varies very little with temperature (of order one part in a

thousand for a 10 K temperature variation). It varies more with salinity, changing by about $-5 \text{ J kg}^{-1} \text{ K}^{-1}$ per part-per-thousand change in salinity. Thus we take

$$c_{p0}(T, S) = c_{p1} + c_{p2}(S - S_0) \quad (1.184)$$

where c_{p1} and c_{p2} are constants that may be experimentally determined.

Using (1.183) and (1.179) in (1.177) gives,

$$\theta = T \exp \left\{ -\frac{\alpha_0 \beta_T p}{c_{p0}} \left[1 + \frac{1}{2} \gamma^* p + \frac{\beta_T^*}{\beta_T} (T - T_0) \right] \right\} \quad (1.185)$$

The exponent itself is small, the second and third terms in square brackets are small compared to unity, and the deviations of both T and θ from T_0 are also presumed small. Taking advantage of all of this enables the expression to be rewritten as

$$T' \approx \frac{T_0 \alpha_0 \beta_T}{c_{p0}} p \left(1 + \frac{1}{2} \gamma^* p + T_0 \frac{\alpha_0 \beta_T^*}{c_{p0}} p \right) + \theta' \left(1 + T_0 \frac{\alpha_0 \beta_T^*}{c_{p0}} p \right) \quad (1.186)$$

where $T' = T - T_0$ and $\theta' = \theta - T_0$. These last two equations are relationships between T , θ and p analogous to (1.113) for an ideal gas. For a linear equation of state we have simply

$$T \approx \frac{T_0 \alpha_0 \beta_T}{c_{p0}} p + \theta \quad (1.187)$$

and this is a useful expression for calculating approximate differences between temperature and potential temperature. Making use of the hydrostatic approximation reveals that this is essentially the same as (1.139). Note that the potential temperature is related to temperature via the thermal expansion coefficient and not, as one might perhaps have expected, the compressibility coefficient or bulk modulus. An example of the *actual* difference between temperature and potential temperature is sketched in Fig. 1.4.

Given (1.186) our equation of state may be written to good approximation as

$$\alpha = \alpha_0 \left[1 - a_1(p) + \beta_T (1 + \gamma'^* p) \theta' + \frac{1}{2} \beta_T^* \theta'^2 - \beta_S (S - S_0) \right] \quad (1.188)$$

where $\gamma'^* = \gamma^* + 2T_0 \beta_T^* \alpha_0 / c_{p0} \approx \gamma^*$ and $a_1(p) = \alpha_0 p / c_0^2$ where c_0 is a reference value of the speed of sound. Given this, it is in principle straightforward, although in practice rather tedious, to compute various thermodynamic quantities of interest. Most of these are for the reader to do, or not, but a calculation of the buoyancy frequency is of interest.

Buoyancy frequency for seawater

This is given by

$$N^2 = -\frac{g}{\rho} \left(\frac{\partial \rho_{\text{pot}}}{\partial z} \right)_{\text{env}} = \frac{g}{\alpha} \left(\frac{\partial \alpha_{\text{pot}}}{\partial z} \right)_{\text{env}} = -\frac{g^2}{\alpha^2} \left(\frac{\partial \alpha_{\text{pot}}}{\partial p} \right)_{\text{env}} \quad (1.189)$$

where α_{pot} and ρ_{pot} are the potential inverse density (or potential specific volume) and potential density — the values of the respective property that a parcel has if taken adiabatically to a reference pressure p_{ref} . The subscript ‘env’ indicates that the gradient is that occurring in the environment, and the last equality uses hydrostasy. From (1.188) we have

$$\begin{aligned}\alpha_{\text{pot}} &= \alpha(\theta, S; p_{\text{ref}}) \\ &= \alpha_0 \left[1 - \frac{\alpha_0}{c_0^2} p_{\text{ref}} + \beta_T (1 + \gamma^* p_{\text{ref}}) \theta' + \frac{1}{2} \beta_T^* \theta'^2 - \beta_S (S - S_0) \right]\end{aligned}\quad (1.190)$$

Using this and (1.189) we obtain an expression for the buoyancy frequency, namely

$$N^2 = -\frac{g^2}{\alpha^2} \alpha_0 \left[\beta_T \left(1 + \gamma p_{\text{ref}} + \frac{\beta_T^*}{\beta_T} \theta \right) \left(\frac{\partial \theta}{\partial p} \right)_{\text{env}} - \beta_S \left(\frac{\partial S}{\partial p} \right)_{\text{env}} \right]. \quad (1.191)$$

Having written this we note that for an accurate determination of the buoyancy frequency the reference pressure p_{ref} must be the pressure itself; that is, the pressure where the buoyancy frequency is to be calculated.

Another way to derive (1.191) is as follows: The sound speed is given by

$$c^{-2} = -\frac{1}{\alpha^2} \left(\frac{\partial \alpha}{\partial p} \right)_{\theta, S} = \frac{1}{\alpha^2} \left(\frac{\alpha_0^2}{c_0^2} - \gamma \alpha_1 \theta \right) \quad (1.192)$$

and, using (1.189) [see also (2.224)] the square of the buoyancy frequency may be written

$$N^2 = \frac{g}{\alpha} \left(\frac{\partial \alpha}{\partial z} \right)_{\text{env}} - \frac{g^2}{c^2} = -\frac{g^2}{\alpha^2} \left[\left(\frac{\partial \alpha}{\partial p} \right)_{\text{env}} + \frac{\alpha^2}{c^2} \right] \quad (1.193)$$

Using (1.188), (1.192) and (1.193) we recover (1.191), although now with p explicitly in place of p_{ref} . The variation of density with potential temperature and salinity is illustrated in Fig. 1.3.

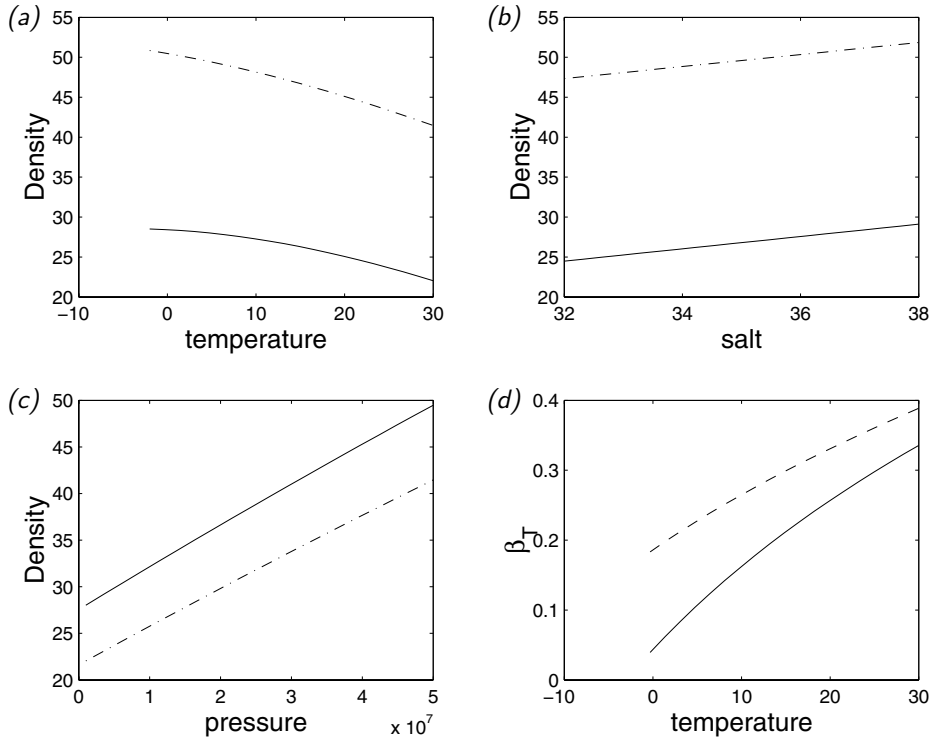


Figure 1.5 Examples of the variation of density of seawater (minus 1000 kg m^{-3}) with: (a) potential temperature; (b) salt; and (c) pressure, for seawater. In (a) salinity is fixed at $S = 35$ psu and the two curves are for sea-level pressure (solid curve) and $p = 5 \times 10^7$ Pa, which is about 5 km depth. In (b) the solid curve is for $T = 13^\circ\text{C}$ and sea-level, and the dashed curve is for $T = 5^\circ\text{C}$ and $p = 5 \times 10^7$ Pa, so corresponding to cold deep water. In (c) the salinity is fixed at $S = 35$ psu and $T = 13^\circ\text{C}$ (solid curve) and $T = 30^\circ\text{C}$ (dashed curve). The lower right panel, (d), shows the thermal expansion coefficient $\beta_T = -\rho_0^{-1}(\partial\rho/\partial T)_{p,S} \text{ K}^{-1}$, multiplied by 10^3 , for each of the two curves in panel (a).

1.9 THE ENERGY BUDGET

The total energy of a fluid includes the kinetic, potential and internal energies. Both fluid flow and pressure forces will in general move energy from place to place, but we nevertheless expect energy to be conserved in an enclosed volume. Let us now consider what form energy conservation takes in a fluid.

1.9.1 Constant density fluid

For a constant density fluid the momentum equation and the mass continuity equation $\nabla \cdot \mathbf{v} = 0$, are sufficient to completely determine the evolution of a

system. The momentum equation is

$$\frac{D\mathbf{v}}{Dt} = -\nabla(\phi + \Phi) + \nu\nabla^2\mathbf{v}, \quad (1.194)$$

where $\phi = p/\rho_0$ and Φ is the potential for any conservative force (e.g., gz for a uniform gravitational field).

We can rewrite the advective term on the left-hand-side using the identity,

$$(\mathbf{v} \cdot \nabla)\mathbf{v} = -\mathbf{v} \times \boldsymbol{\omega} + \nabla(\tfrac{1}{2}\mathbf{v}^2) \quad (1.195)$$

where $\boldsymbol{\omega} \equiv \nabla \times \mathbf{v}$ is the *vorticity*, discussed more in later chapters. Then, omitting viscosity, have

$$\frac{\partial\mathbf{v}}{\partial t} + \boldsymbol{\omega} \times \mathbf{v} = -\nabla B, \quad (1.196)$$

where $B = (\phi + \Phi + \mathbf{v}^2/2)$ is the *Bernoulli function* for constant density flow. Consider for a moment steady flows ($\partial\mathbf{v}/\partial t = 0$). Streamlines are, by definition, parallel to $\mathbf{v}$ everywhere, and the vector $\mathbf{v} \times \boldsymbol{\omega}$ is everywhere orthogonal to the streamlines, so that taking the dot product of the steady version of (1.196) with $\mathbf{v}$ gives $\mathbf{v} \cdot \nabla B = 0$. That is, for steady flows the Bernoulli function is constant along a streamline, and $DB/Dt = 0$.

Reverting back to the time-varying case, take the dot product with $\mathbf{v}$ and include the density to yield

$$\tfrac{1}{2} \frac{\partial\rho_0\mathbf{v}^2}{\partial t} + \rho_0\mathbf{v} \cdot (\boldsymbol{\omega} \times \mathbf{v}) = -\rho_0\mathbf{v} \cdot \nabla B \quad (1.197)$$

The second term on the left-hand-side vanishes identically. Defining the kinetic energy density K , or energy per unit volume, by $K = \rho_0\mathbf{v}^2/2$, (1.197) becomes an expression for the rate of change of K ,

$$\frac{\partial K}{\partial t} + \nabla \cdot (\rho_0\mathbf{v}B) = 0. \quad (1.198)$$

Because Φ is time-independent this may be written

$$\frac{\partial}{\partial t} [\rho_0(\tfrac{1}{2}\mathbf{v}^2 + \Phi)] + \nabla \cdot [\rho_0\mathbf{v}(\tfrac{1}{2}\mathbf{v}^2 + \Phi + \phi)] = 0. \quad (1.199)$$

or

$$\frac{\partial E}{\partial t} + \nabla \cdot [\mathbf{v}(E + p)] = 0. \quad (1.200)$$

where $E = K + \rho_0\Phi$ is the total energy density (i.e., the total energy per unit volume). This has the form of a general conservation equation in which a local change in a quantity is balanced by the divergence of its flux. However, the energy flux, $\mathbf{v}(\rho_0\mathbf{v}^2/2 + \rho_0\Phi + \rho_0\phi)$, is *not* simply the velocity times the energy

density $\rho_0(\mathbf{v}^2/2 + \Phi)$; there is an additional term, $\mathbf{v}p$, that represents the energy transfer occurring when work is done by the fluid against the pressure force.

Now consider a volume through which there is no mass flux, for example a domain bounded by rigid walls. The rate of change of energy within that volume is then given by the integral of (1.197)

$$\frac{dK}{dt} \equiv \frac{d}{dt} \int_V K dV = - \int_V \nabla \cdot (\rho_0 \mathbf{v} B) dV = - \int_S \rho_0 B \mathbf{v} \cdot d\mathbf{S} = 0, \quad (1.201)$$

using the divergence theorem. Thus, the total energy within the volume is conserved. Note that for a constant density fluid the gravitational potential energy, $\mathcal{P}$, is given by

$$\mathcal{P} = \int_V \rho_0 g z dV, \quad (1.202)$$

which is a constant, not affected by a re-arrangement of fluids. Thus, in a constant density fluid there is no exchange between kinetic energy and potential energy and the kinetic energy itself is conserved.

1.9.2 Variable density fluids

We start with the momentum equation

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p - \rho \nabla \Phi, \quad (1.203)$$

and take its dot product with $\mathbf{v}$ to obtain an equation for the evolution of kinetic energy,

$$\begin{aligned} \frac{1}{2} \rho \frac{D\mathbf{v}^2}{Dt} &= -\mathbf{v} \cdot \nabla p - \rho \mathbf{v} \cdot \nabla \Phi \\ &= -\nabla \cdot (p\mathbf{v}) + p \nabla \cdot \mathbf{v} - \rho \mathbf{v} \cdot \nabla \Phi. \end{aligned} \quad (1.204)$$

From (1.93) the internal energy equation for flow with no heat sources is

$$\begin{aligned} \rho \frac{DI}{Dt} &= -\rho p \frac{D\alpha}{Dt} \\ &= -p \nabla \cdot \mathbf{v} \end{aligned} \quad (1.205)$$

where the second line follows by use of the continuity equation. Finally, and somewhat trivially, the potential energy density obeys

$$\rho \frac{D\Phi}{Dt} = \rho \mathbf{v} \cdot \nabla \Phi. \quad (1.206)$$

Adding (1.204), (1.205) and (1.206) we obtain

$$\rho \frac{D}{Dt} \left(\frac{1}{2} \mathbf{v}^2 + I + \Phi \right) = -\nabla \cdot (p\mathbf{v}), \quad (1.207)$$

or, expanding the material derivative and using the mass conservation equation,

$$\frac{\partial}{\partial t} [\rho (\frac{1}{2} \mathbf{v}^2 + I + \Phi)] + \nabla \cdot [\rho \mathbf{v} (\frac{1}{2} \mathbf{v}^2 + I + \Phi + p/\rho)] = 0. \quad (1.208)$$

or

$$\boxed{\frac{\partial E}{\partial t} + \nabla \cdot [\mathbf{v}(E + p)] = 0.} \quad (1.209)$$

where $E = \rho(\mathbf{v}^2/2 + I + \Phi)$ is the total energy per unit volume, or the total energy density, of the fluid. This is the energy equation for an unforced, inviscid and adiabatic, compressible fluid. Just as for the constant density case, the energy flux contains the term $p\mathbf{v}$ that represents the work done against pressure and, again, the second term vanishes when integrated over a closed domain with rigid boundaries, implying that the total energy is conserved. However, now there can now be an exchange of energy between kinetic, potential and internal components. The quantity $\sigma = I + p\alpha + \Phi = h + \Phi$ is sometimes referred to as the static energy, or the dry static energy. However, it is not a component of the globally conserved total energy; the conserved energy contains only the quantity $I + \Phi$ plus the kinetic energy, and it is only the *flux* of static energy that affects the energetics. For an ideal gas we have $\sigma = c_v T + RT + \Phi = c_p T + \Phi$, and if the gas is also in a uniform gravitational field then $\sigma = c_p T + gz$.

Bernoulli's theorem

For steady flow $\partial/\partial t = 0$ and $\nabla \cdot \rho \mathbf{v} = 0$ so that (1.208) may be written $\mathbf{v} \cdot \nabla B = 0$ where B is the Bernoulli function given by

$$B = (\frac{1}{2} \mathbf{v}^2 + I + \Phi + p/\rho) = (\frac{1}{2} \mathbf{v}^2 + h + \Phi). \quad (1.210)$$

This means that

$$\frac{DB}{Dt} = 0, \quad (1.211)$$

and, for steady flow, the Bernoulli function is a constant along streamline. For an ideal gas in a constant gravitation field $B = \mathbf{v}^2/2 + c_p T + gz$.

For adiabatic flow we also have $D\theta/Dt = 0$. Thus, steady flow is both along surfaces of constant θ and along surfaces of constant B , and the vector

$$\mathbf{l} = \nabla \theta \times \nabla B \quad (1.212)$$

is parallel to streamlines.¹¹ An analogue of this result in unsteady flow is given in section 4.8.

1.9.3 Viscous Effects

We might expect that viscosity will always act to reduce the kinetic energy of a flow, and we will demonstrate this for the constant density case. Retaining the viscous term in (1.194), the energy equation becomes

$$\frac{d\hat{E}}{dt} \equiv \frac{d}{dt} \int_V E dV = \mu \int_V \mathbf{v} \cdot \nabla^2 \mathbf{v} dV. \quad (1.213)$$

The right hand side is negative definite. To see this we use the vector identity

$$\nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{v}) - \nabla^2 \mathbf{v}, \quad (1.214)$$

and because $\nabla \cdot \mathbf{v} = 0$ we have $\nabla^2 \mathbf{v} = -\nabla \times \boldsymbol{\omega}$, where $\boldsymbol{\omega} \equiv \nabla \times \mathbf{v}$. Thus,

$$\frac{d\hat{E}}{dt} = -\mu \int_V \mathbf{v} \cdot (\nabla \times \boldsymbol{\omega}) dV = -\mu \int_V \boldsymbol{\omega} \cdot (\nabla \times \mathbf{v}) dV = -\mu \int_V \boldsymbol{\omega}^2 dV, \quad (1.215)$$

after integrating by parts, providing $\mathbf{v} \times \boldsymbol{\omega}$ vanishes at the boundary. Thus, viscosity acts to extract kinetic energy from the flow. The loss of kinetic energy reappears as an irreversible warming of the fluid ('Joule heating'), and the total energy of the fluid is conserved, but this effect plays no role in a constant density fluid. The effect is normally locally small, at least in the earth's ocean and atmosphere, although it is sometimes included in comprehensive General Circulation Models.

1.10 AN INTRODUCTION TO NON-DIMENSIONALIZATION AND SCALING

The units we use to measure length, velocity and so on are irrelevant to the dynamics, and not necessarily the most appropriate units for a given problem. Rather, it is convenient to express the equations of motion, so far as is possible, in 'nondimensional' variables, by which we mean expressing every variable (such as velocity) as the ratio of its value to some reference value. For velocity the reference could, for example, be the speed of light — but this would not be very helpful for fluid dynamical problems in the earth's atmosphere or ocean! Rather, we should choose the reference value as a natural one for a given flow, in order that, so far as possible, the nondimensional variables are order-unity quantities, and doing this is called *scaling the equations*. Evidently, there is no reference velocity that is universally appropriate, and much of the art of fluid dynamics lies in choosing sensible scaling factors for the problem at hand. Non-dimensionalization plays an important role throughout this book, and we introduce it here with a simple, important, example.

1.10.1 The Reynolds number

To be specific, consider the constant-density momentum equation in Cartesian coordinates. If a typical velocity is U and a typical length is L (implying a timescale $T \sim L/U$) then the approximate sizes of the various terms in the equation of motion are given by.

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla \phi + \nu \nabla^2 \mathbf{v}$$

$$\frac{U}{T} \quad \frac{U^2}{L} \quad \sim \quad \frac{P}{\rho L} \quad \nu \frac{U}{L^2}. \quad (1.216)$$

The ratio of the inertial terms to the viscous terms is $(U^2/L) : (\nu U/L^2) = UL/\nu$, and this is the *Reynolds number*.¹²

More formally, we can nondimensionalize the momentum equation by writing

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{U}, \quad \hat{\mathbf{x}} = \frac{\mathbf{x}}{L}, \quad \hat{t} = \frac{t}{T}, \quad \hat{\phi} = \frac{\phi}{\Phi}, \quad (1.217)$$

where the terms with hats on are *nondimensional* values of the variables and the capitalized quantities are known as *scaling values*. If there are well-defined length and velocity scales in the problem, and we choose these scales to perform the nondimensionalization, then the nondimensional quantities are of order unity. Because there are no other external forces in this problem, appropriate scaling values for time and pressure are

$$T = \frac{L}{U} \quad \Phi = U^2. \quad (1.218)$$

Substituting (1.217) and (1.218) into the momentum equation we obtain

$$\frac{U^2}{L} \left(\frac{\partial \hat{\mathbf{v}}}{\partial \hat{t}} + (\hat{\mathbf{v}} \cdot \nabla) \hat{\mathbf{v}} \right) = -\frac{U^2}{L} \nabla \hat{\phi} + \frac{\nu U}{L^2} \nabla^2 \mathbf{v}, \quad (1.219)$$

or

$$\frac{\partial \hat{\mathbf{v}}}{\partial \hat{t}} + (\hat{\mathbf{v}} \cdot \nabla) \hat{\mathbf{v}} = -\nabla \hat{\phi} + \frac{1}{\text{Re}} \nabla^2 \hat{\mathbf{v}}, \quad (1.220)$$

where

$$\text{Re} \equiv \frac{UL}{\nu} \quad (1.221)$$

is, again, the Reynolds number. If we have chosen our length and velocity scales sensibly — that is, if we have scaled them properly — each variable in (1.220) is order unity, with the viscous term being multiplied by $1/\text{Re}$. There are three important conclusions:

- (i) The ratio of the importance of the inertial terms to the viscous terms is given by the *Reynolds number*, defined by (1.221).

- (ii) The Reynolds number is the only non-dimensional parameter explicitly appearing in the momentum equation, in the absence of other forces (such as gravity and rotation). Hence its value, along with the boundary conditions, controls the behaviour of the system.
- (iii) More generally, by scaling the equations of motion appropriately the parameters determining the behaviour of the system become explicit. Scaling the equations is just intelligent nondimensionalization.

Notes

1 See references under 'further reading' below for general material. Some aspects of this chapter and the next draw on notes prepared over the years for a graduate class at Princeton University and originally written by Steve Garner, Isaac Held and Yoshio Kurihara.

2 Joseph-Louis Lagrange (1736–1813) was a Franco-Italian, born and raised in Turin who then lived and worked mainly in Germany and France. He made notable contributions in analysis, number theory and mechanics and was recognized as one of the greatest mathematicians of the 18th century. He laid the foundations of the calculus of variations (to wit, the 'Lagrange multiplier') and first formulated the principle of least action, and his monumental treatise *Mécanique Analytique* (1788) provides a unified analytic framework (it contains no diagrams, a feature virtually emulated in Whitaker's *Analytical Dynamics*, 1927) for all Newtonian mechanics.

Leonard Euler (1707–1783), a Swiss mathematician who lived and worked for extended periods in Berlin and St. Petersburg, made important contributions in many areas of mathematics and mechanics, including the analytical treatment of algebra, the theory of equations, calculus, number theory and classical mechanics. He was the first to establish the form of the equations of motion of fluid mechanics, writing down both the field description of fluids *and* what we now call the material or advective (and sometimes the Lagrangian!) derivative. Euler also went out of his way to help Lagrange early in his (Lagrange's) career.

Truesdell (1954) (p. 30) points out that 'Eulerian' and 'Lagrangian' co-ordinates, especially the latter, are inappropriate eponyms. The so-called Eulerian description was introduced by d'Alembert in 1749 and generalized by Euler in 1752, and the so-called Lagrangian description was introduced by Euler in 1759. The modern confusion evidently stems from a monograph by Dirichlet in 1860 that credits Euler in 1757 and Lagrange in 1788 for the respective methods — see Truesdell for the full references. To avoid incorrect personal attributions, one may refer to the two points of view as the 'field' or 'spatial' view, and the 'material' view.

3 For example Batchelor (1967).

4 Claude-Louis-Marie-Henri Navier (1785–1836) was a French civil engineer, professor at the École Polytechnique and later at the École des Ponts et Chaussée.

He was a expert in road and bridge building (he developed the theory of suspension bridges) and, relatedly, he made lasting theoretical contributions to the theory of elasticity, being the first to publish a set of general equations for the dynamics of an elastic solid. In fluid mechanics, he laid down the now-called *Navier-Stokes equations*, including the viscous terms, in 1822.

George Gabriel Stokes (1819–1903). Irish born (in Skreen, County Sligo), he held the Lucasian chair of Mathematics at Cambridge from 1849 until his retirement, a chair once held by Isaac Newton. As well as having a role in the development of fluid mechanics, especially through his considerations of viscous effects, Stokes worked on the dynamics of elasticity, fluorescence, the wave theory of light, and was (rather ill-advisedly in hindsight) a proponent of the idea of an ether permeating all space.

- 5 Our treatment of the equation of state, and thermodynamics in general, is not a comprehensive review of the subject; rather, it is a refresher for those with some prior exposure to the subject, and an introduction to the role of thermodynamics in geofluids. Those with no prior exposure to thermodynamics are advised to consult one of many standard texts in the subject.
- 6 R_d and R_v are related by the molecular weights of water and dry air, M_v and M_d , so that $\alpha \equiv R_d/R_v = M_v/M_d = 0.62$. Rather than allow the gas constant to vary, meteorologists sometimes incorporate the variation in humidity into the temperature, so that instead of $p = \rho R_{\text{eff}} T$ we use $p = \rho R_d T_v$, so defining the ‘virtual temperature’, T_v . It is easy to show that $T_v \approx [1 + w(\alpha^{-1} - 1)]T$. Atmospheric GCMs often use a virtual temperature.
- 7 The Unesco (1981) equation of state of seawater is an empirical equation that fits measurements to an accuracy of order 10^{-5} (see Fofonoff 1985). For nearly all purposes simpler and more easily computable formulae are adequate, for example, Mellor (1991), Wright (1997), Jackett & McDougall (1995) and McDougall *et al.* (2002). These authors provide formulae for the density (or potential density) in terms of temperature, salinity and pressure. The formulae of Wright and McDougall *et al.* are both very accurate and easy to compute, and are of the form:

$$\rho = \frac{p + p_0}{\lambda + \alpha_0(p + p_0)}$$

where α_0 , p_0 and λ are expressed as polynomials in potential temperature and salinity. They are constructed using the Gibbs function of Feistel & Hagen (1995), and are used for Fig. 1.5 and Fig. 1.3. Bryden (1973) provides polynomial formulae for potential temperature of seawater in terms of temperature, salinity and pressure and this is used in Fig. 1.4. de Szoeke (2003) provides a review of many of the salient issues. In numerical ocean models potential temperature and salinity are the usual prognostic thermodynamic variables and an empirical equation of state is used.

- 8 I am grateful for comments from Trevor McDougall and Roland de Szoeke on many aspects of geofluid thermodynamics.
- 9 Definitions vary slightly from source to source. Some take it to mean that density is unaffected by pressure, but this alone is insufficient to guarantee that

the mass conservation equation can be approximated by $\nabla \cdot \mathbf{v} = 0$.

- 10 Following de Szoëke (2003), where a more complete discussion is to be found.
- 11 These results, usually known as Bernoulli's theorem, were developed by Daniel and Johannes Bernoulli and so should be known as Bernoullis' theorem.
- 12 Osborne Reynolds (1842-1912) was an Irish born (Belfast) physicist who was professor of engineering at Manchester University from 1868-1905. His early work was in electricity and magnetism, but he is now most famous for his work in hydrodynamics. The 'Reynolds number,' which determines the ratio of inertial to viscous forces, and the 'Reynolds stress,' which is the stress on the mean flow due to the fluctuating components, are both named after him. He was also one of the first scientists to think about the concept of group velocity.

Further Reading

There are numerous books on hydrodynamics an early one being:

Lamb, H. 1932. *Hydrodynamics*, 6th edn.

This is a classic text in the subject, although its notation is now too dated to make it really useful as an introduction. Another very well-known text is:

Batchelor, G. K. 1967. *An Introduction to Fluid Dynamics*.

This mainly treats incompressible flow. It is rather heavy going for the true beginner, but nevertheless is a very good reference for the fundamentals. Two other useful references are

Tritton, D. J. 1988. *Physical Fluid Dynamics*, 2nd edn.

and

Kundu, P. and I. M. Cohen 1990. *Fluid Mechanics*.

Both are introductions written at the advanced undergraduate/beginning graduate level, and are easier-going than Batchelor. Kundu's book has more material on geophysical fluid dynamics.

A number of textbooks exist on classical thermodynamics. Two of the clearest (in my opinion) are:

Reif, F., 1965. *Fundamentals of Statistical and Thermal Physics*,

and

Landau, L. and E. M. Lifshitz, 1987 *Statistical Physics, Part I*.

An introduction to thermodynamic effects in fluids, with an emphasis on fundamental properties, is provided by:

Salmon, R, 1998, *Lectures on Geophysical Fluid Dynamics*.

Problems

It is by the solution of problems that the investigator tests the temper of his steel; he finds new methods and new outlooks, and gains a wider and freer horizon.

David Hilbert (1862-1943).

- 1.1 For an infinitesimal volume, informally show that

$$\frac{D}{Dt}(\rho\phi\Delta V) = \rho\Delta V \frac{D\phi}{Dt}, \quad (\text{P1.1})$$

and hence deduce that

$$\frac{D}{Dt} \int_V \rho\phi \, dV = \int_V \rho \frac{D\phi}{Dt} \, dV. \quad (\text{P1.2})$$

where ϕ is an arbitrary, differentiable, property of the fluid.

- 1.2 Show that the derivative of an integral is given by

$$\frac{d}{dt} \int_{x_1(t)}^{x_2(t)} \phi(x, t) \, dx = \int_{x_1}^{x_2} \frac{\partial \phi}{\partial t} \, dx + \frac{dx_2}{dt} \phi(x_2, t) - \frac{dx_1}{dt} \phi(x_1, t). \quad (\text{P1.3})$$

By generalizing to three-dimensions show that the material derivative of an integral of a fluid property is given by

$$\begin{aligned} \frac{D}{Dt} \int_V \phi(\mathbf{x}, t) \, dV &= \int_V \frac{\partial \phi}{\partial t} \, dV + \int_S \phi \mathbf{v} \cdot d\mathbf{S} \\ &= \int_V \left[\frac{\partial \phi}{\partial t} + \nabla \cdot (\mathbf{v}\phi) \right] \, dV, \end{aligned} \quad (\text{P1.4})$$

where the surface integral ($\int_S$) is over the surface bounding the volume V . Hence deduce that

$$\frac{D}{Dt} \int_V \rho\phi \, dV = \int_V \rho \frac{D\phi}{Dt} \, dV. \quad (\text{P1.5})$$

- 1.3 By invoking Galilean invariance we can often choose, without loss of generality, the basic state for problems in sound waves to be such that $u_0 \equiv 0$. The perturbation velocity is then certainly larger than the basic state velocity. How can we then justify ignoring the nonlinear term in the perturbation equation, as the term $u' \partial u' / \partial x$ is certainly no smaller than the linear term $u_0 \partial u' / \partial x$?
- 1.4 For sound waves in air of amplitude 60 dB, show that the linear approximation to the equations of motion is a good one. For example, in a one-dimensional problem, show that the term $u' \partial u' / \partial x$ is much smaller than $\partial u' / \partial t$. What amplitude of sound wave is required for the nonlinear terms to become important? Is this achieved at a rock concert?
- 1.5 Using the observed value of molecular diffusion of heat in water, estimate how long it would take for a temperature perturbation to mix from the top of the ocean to the bottom, assuming that molecular diffusion alone is responsible. Comment on whether you think the real ocean has reached equilibrium after the last ice age (which ended about 20,000 years ago).

- 1.6 Consider the following flow:

$$\begin{aligned} u &= \Gamma z \\ v &= V \sin[k(x - ct)] \end{aligned}$$

where Γ , V , k and c are positive constants. (This is similar to the flow in the mid-latitude troposphere — an eastward flow increasing with height, with a transverse wave superimposed.) Suppose that $\Gamma z > c$ for the region of interest. Consider particles located along the $y = 0$ axis at $t = 0$, and compute their position at some later time t . Compare this with the *streamfunction* for the flow at the same time. (*Hint*: Show that the meridional particle displacement is $\eta = \psi / (u - c)$, where ψ is the streamfunction and u and c are parameters.)

1.7 Consider the two-dimensional flow

$$\begin{aligned} u &= A(y) \sin \omega t \\ v &= A(y) \cos \omega t. \end{aligned}$$

The time-mean of this at a fixed point flow is zero. If A is independent of y , then fluid parcels move clockwise in a circle. What is its radius? If A does depend on y , find an *approximate* expression for the average drift of a particle,

$$\lim_{t \rightarrow \infty} \frac{\mathbf{x}(\mathbf{a}, t)}{t}$$

where $\mathbf{a}$ is a particle label and A is suitably 'small'. Be precise about what small means.

Partial solution: Write $A(y) = A_0 + ay$ and solve

$$\frac{dy}{dt} = (A_0 + ay) \cos \omega t \quad (\text{P1.6})$$

as a polynomial function of $\sin \omega t$. Then solve

$$\frac{dx}{dt} = (A_0 + ay) \sin \omega t$$

using the solution for y . The rectification of the standing oscillation by the oscillating trajectory now appears, because $\langle \sin \omega t \sin \omega t \rangle \neq 0$. The small parameter is the relative variation of A across a particle displacement, namely

$$\frac{\delta A}{A} = \frac{(aA_0/b)}{A_0} = \frac{a}{b}.$$

That is, the shear is small compared to the frequency.

- 1.8 (a) Show that for an ideal gas (1.123) is equivalent to (1.109). You may use the Maxwell relation $(\partial \alpha / \partial \eta)_p = (\partial T / \partial p)_\eta$.
- (b) Show that for an ideal gas (1.134) is equivalent to (1.108). You may use the results of part (a).
- 1.9 Show that it follows directly from the equation of state, $P = RT/\alpha$, that the internal energy of an ideal gas is a function of temperature only.

Solution: From the first law of thermodynamics [c.f. (1.94)]

$$d\eta = \frac{1}{T} \left(\frac{\partial I}{\partial T} \right)_\alpha dT + \left[\frac{1}{T} \left(\frac{\partial I}{\partial \alpha} \right)_T + \frac{R}{\alpha} \right] d\alpha. \quad (\text{P1.7})$$

But, mathematically,

$$d\eta = \left(\frac{\partial \eta}{\partial T} \right)_{\alpha} dT + \left(\frac{\partial \eta}{\partial \alpha} \right)_T d\alpha. \quad (\text{P1.8})$$

Thus,

$$\begin{aligned} \left(\frac{\partial \eta}{\partial T} \right)_{\alpha} &= \frac{1}{T} \left(\frac{\partial I}{\partial T} \right)_{\alpha} \\ \left(\frac{\partial \eta}{\partial \alpha} \right)_T &= \frac{1}{T} \left(\frac{\partial I}{\partial \alpha} \right)_T + \frac{R}{\alpha}. \end{aligned} \quad (\text{P1.9})$$

Noting that $\partial^2 \eta / (\partial \alpha \partial T) = \partial^2 \eta / (\partial T \partial \alpha)$ we obtain

$$\frac{\partial^2 I}{\partial \alpha \partial T} = \frac{\partial^2 I}{\partial T \partial \alpha} - \left(\frac{\partial I}{\partial \alpha} \right)_T. \quad (\text{P1.10})$$

Thus, $(\partial I / \partial \alpha)_T = 0$. Because, in general, the internal energy may be considered *either* a function of either temperature and density *or* temperature and pressure, this proves that for an ideal gas the internal energy is a function only of temperature, not density or pressure.

- 1.10 Show that it follows directly from the equation of state $P = RT/\alpha$, that for an ideal gas the heat capacity at constant volume, c_v , is, at most, a function of temperature.
- 1.11 Show that for an ideal gas

$$Td\eta = c_v dT + p d\alpha. \quad (\text{P1.11})$$

and that its internal energy is given by $I = \int c_v dT$.

Solution: Let us regard η as a function of T and α , where α is the specific volume $1/\rho$. Then

$$\begin{aligned} Td\eta &= T \left(\frac{\partial \eta}{\partial T} \right)_{\alpha} dT + T \left(\frac{\partial \eta}{\partial \alpha} \right)_T d\alpha \\ &= c_v dT + T \left(\frac{\partial \eta}{\partial \alpha} \right)_T d\alpha \end{aligned} \quad (\text{P1.12})$$

by definition of c_v . For an ideal gas the internal energy is a function of temperature alone (problem 1.9), so that using (1.78) the pressure of a fluid $p = T (\partial \eta / \partial \alpha)_T = T (\partial \eta / \partial \alpha)_T$ and (P1.12) becomes

$$Td\eta = c_v dT + p d\alpha \quad (\text{P1.13})$$

But, *in general*, the fundamental thermodynamic relation is

$$Td\eta = dI + p d\alpha. \quad (\text{P1.14})$$

The terms on the right hand side of (P1.13) are identifiable as the change in the internal energy and the work done on a fluid, and so $dI = c_v dT$. The heat capacity need not necessarily be constant, although it very nearly is, but it must be a function of temperature only.

- 1.12 (a) Beginning with the expression for potential temperature for an ideal gas, $\theta = T(p_{\text{ref}}/p)^\kappa$, where $\kappa = R/c_p$, show that

$$d\theta = \frac{\theta}{T}(dT - \alpha dp) \quad (\text{P1.15})$$

and therefore that the first law of thermodynamics may be written as

$$dQ = Td\eta = c_p \frac{T}{\theta} d\theta. \quad (\text{P1.16})$$

- (b) Show that (P1.16) is generally true, without making the ideal gas approximation.
- 1.13 A barotropic fluid is one for which the gradient of pressure is everywhere parallel to the gradient of density. Show that this condition is equivalent to stating that the internal energy is a function only of the density, and not the entropy. (It is for this reason that barotropic fluids are sometimes called homentropic fluids.)
- 1.14 Consider an ocean at rest with known vertical profiles of potential temperature and salinity, $\theta(z)$ and $S(z)$. Suppose we also know the equation of state in the form $\rho = \rho(\theta, S, p)$. Obtain an expression for the buoyancy frequency. Check your expression by substituting the equation of state for an ideal gas and recovering a known expression for the buoyancy frequency.
- 1.15 $\diamond$ Obtain an expression, in terms of temperature and pressure, for the potential temperature of a van der Waals gas, with equation of state $(p + a/\alpha^2)(\alpha - b) = RT$, where a and b are constants. Show that it reduces to the expression for an ideal gas in the limit $a \rightarrow 0$, $b \rightarrow 0$.
- 1.16 $\diamond$ In seawater, why does potential temperature differ from in situ temperature? Is it because of the compressibility (with pressure) or the thermal expansion of water? A good answer will include, *inter alia*, a physical explanation of the difference in scale heights of (1.127) and (1.138) and a discussion of the lapse rate and scale heights of water that is between 0° and 4°C .

If a body is moving in any direction, there is a force, arising from the earth's rotation, which always deflects it to the right in the northern hemisphere, and to the left in the southern hemisphere.

William Ferrel, *The influence of the Earth's rotation upon the relative motion of bodies near its surface*, 1858.

CHAPTER TWO

A Shallow Fluid with Rotation and Stratification

THE ATMOSPHERE AND OCEAN are shallow layers of fluid on a sphere, in the sense that their thickness or depth is much less than their horizontal extent. Furthermore, their motion is strongly influenced by two effects: rotation and stratification, the latter meaning that there is a mean vertical gradient of (potential) density that is often large compared with its horizontal gradient. Here we consider how the equations of motion are affected by these effects. First, we consider some elementary effects of rotation on a fluid and derive the Coriolis and centrifugal forces, and then we write down the equations of motion appropriate for motion on a sphere. Then we discuss some approximations to the equations of motion that are appropriate for large-scale flow in the ocean and atmosphere, in particular the hydrostatic approximation, and how the presence of strong stratification can be used to further simplify the governing equations.

2.1 THE EQUATIONS OF MOTION IN A ROTATING FRAME OF REFERENCE

Newton's second law of motion, that the acceleration on a body is proportional to the imposed force divided by the body's mass, applies in so-called inertial frames of reference. The earth rotates with a period of about 24 hours and the surface

of the earth manifestly is not, in that sense, an inertial frame. Nevertheless, because the surface of the earth is moving at velocities of up to a few hundreds of meters per second it is very convenient to describe the flow relative to the earth's surface, rather than in some inertial frame. This necessitates recasting the equations into a form that is appropriate for a rotating frame of reference, and that is the subject of this section.

2.1.1 Rate of change of a vector

Consider first a vector $\mathbf{C}$ of constant length rotating relative to an inertial frame at a constant angular velocity $\boldsymbol{\Omega}$. Then, in a frame rotating with that same angular velocity it appears stationary and constant. If in small interval of time δt it the vector $\mathbf{C}$ rotates through a small angle $\delta\lambda$ then the change in $\mathbf{C}$, as perceived in the inertial frame, is given by (see Fig. 2.1)

$$\delta\mathbf{C} = |\mathbf{C}| \cos\theta \delta\lambda \mathbf{m}, \quad (2.1)$$

where the vector $\mathbf{m}$ is the unit vector in the direction of change of $\mathbf{C}$, which is perpendicular to both $\mathbf{C}$ and $\boldsymbol{\Omega}$. But the rate of change of the angle λ is just, by definition, the angular velocity so that $\delta\lambda = |\boldsymbol{\Omega}| \delta t$ and

$$\delta\mathbf{C} = |\mathbf{C}| |\boldsymbol{\Omega}| \sin\hat{\theta} \mathbf{m} \delta t = \boldsymbol{\Omega} \times \mathbf{C} \delta t. \quad (2.2)$$

using the definition of the vector cross product, where $\hat{\theta} = (\pi/2 - \theta)$ is the angle between $\boldsymbol{\Omega}$ and $\mathbf{C}$. Thus

$$\left(\frac{d\mathbf{C}}{dt} \right)_I = \boldsymbol{\Omega} \times \mathbf{C} \quad (2.3)$$

where the left hand side is the rate of change of $\mathbf{C}$ as perceived in the inertial frame.

To generalize slightly, now consider a vector $\mathbf{B}$ that changes in the inertial frame. In a small time δt the change in $\mathbf{B}$ as seen in the rotating frame will differ from the change seen in the inertial frame; that is,

$$(\delta\mathbf{B})_I = (\delta\mathbf{B})_R + (\delta\mathbf{B})_{\text{rot}} \quad (2.4)$$

where the terms are, respectively, the change seen in the inertial frame, the change due to the vector itself changing as measured in the rotating frame, and the change due to the rotation itself. Using (2.2) $(\delta\mathbf{B})_{\text{rot}} = \boldsymbol{\Omega} \times \mathbf{B} \delta t$, and so the rates of change of the vector $\mathbf{B}$ in the inertial and rotating frames are related by

$$\boxed{\left(\frac{d\mathbf{B}}{dt} \right)_I = \left(\frac{d\mathbf{B}}{dt} \right)_R + \boldsymbol{\Omega} \times \mathbf{B}}. \quad (2.5)$$

This relation applies to a vector $\mathbf{B}$ that, as measured at any one time, is the same in both inertial and rotating frames.

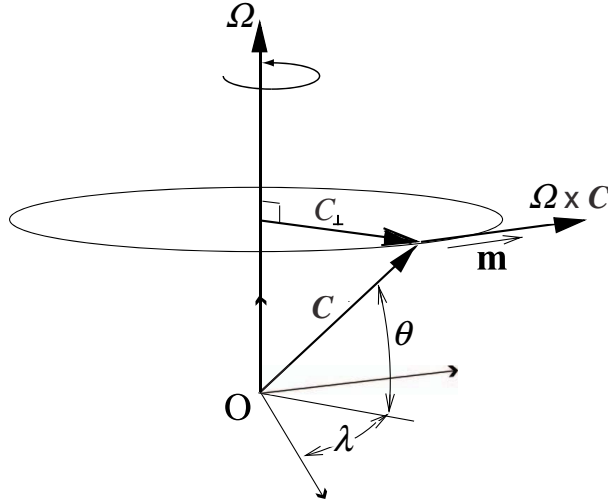


Figure 2.1 A vector C rotating at an angular velocity Ω . It appears to be a constant vector in the rotating frame, whereas in the inertial frame it evolves according to $(dC/dt)_I = \Omega \times C$.

2.1.2 Velocity and acceleration in a rotating frame

The velocity of a body is not measured to be the same in the inertial and rotating frames, so care must be taken when applying (2.5) to velocity. First apply (2.5) to r , the position of a particle to obtain

$$\left(\frac{dr}{dt} \right)_I = \left(\frac{dr}{dt} \right)_R + \Omega \times r \quad (2.6)$$

or

$$v_I = v_R + \Omega \times r. \quad (2.7)$$

We refer to v_R and v_I as the relative and inertial velocity, respectively. Eq. (2.7) relates the velocity in the rotating frame to that in the inertial frame. Apply (2.5) again, this time to the velocity v_R to give

$$\left(\frac{dv_R}{dt} \right)_I = \left(\frac{dv_R}{dt} \right)_R + \Omega \times v_R \quad (2.8)$$

or, using (2.7)

$$\left(\frac{d}{dt} (v_I - \Omega \times r) \right)_I = \left(\frac{dv_R}{dt} \right)_R + \Omega \times v_R, \quad (2.9)$$

or

$$\left(\frac{dv_I}{dt} \right)_I = \left(\frac{dv_R}{dt} \right)_R + \Omega \times v_R + \frac{d\Omega}{dt} \times r + \Omega \times \left(\frac{dr}{dt} \right)_I. \quad (2.10)$$

Then, noting that

$$\left(\frac{dr}{dt} \right)_I = \left(\frac{dr}{dt} \right)_R + \Omega \times r = (v_R + \Omega \times r), \quad (2.11)$$

and assuming that the rate of rotation is constant, (2.10) becomes

$$\left(\frac{d\mathbf{v}_R}{dt} \right)_R = \left(\frac{d\mathbf{v}_I}{dt} \right)_I - 2\boldsymbol{\Omega} \times \mathbf{v}_R - \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}). \quad (2.12)$$

This equation may be interpreted as follows. The first term on the left-hand side is the rate of change of the relative velocity as measure in the rotating frame. The first term on the right-hand side is the rate of change of the inertial velocity as measured in the inertial frame (or, loosely, the inertial acceleration). Thus, by Newton's second law, it is equal to force on a fluid parcel divided by its mass. The second term and third terms on the right-hand side (including the minus signs) are the 'Coriolis force' and the 'centrifugal force' per unit mass. Neither of these are true forces — they may be described as quasi-forces (i.e., 'as if' forces); that is, when a body is observed from a rotating frame it seems to behave as if unseen forces are present that affect its motion.¹

Centrifugal force

Noting that $\boldsymbol{\Omega} \times \mathbf{r} = \boldsymbol{\Omega} \times \mathbf{r}_\perp$, where $\mathbf{r}_\perp$ is the perpendicular distance from the axis of rotation (see Fig. 2.1, and substitute $\mathbf{r}$ for $\mathbf{C}$), and using the vector identity $\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_\perp) = (\boldsymbol{\Omega} \cdot \mathbf{r}_\perp)\boldsymbol{\Omega} - (\boldsymbol{\Omega} \cdot \boldsymbol{\Omega})\mathbf{r}_\perp$ then, since $\boldsymbol{\Omega}$ is perpendicular to $\mathbf{r}_\perp$, the centrifugal force per unit mass is just given by

$$\mathbf{F}_{ce} = -\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) = \Omega^2 \mathbf{r}_\perp. \quad (2.13)$$

This may usefully be written as the gradient of a scalar potential,

$$\mathbf{F}_{ce} = -\nabla \Phi_{ce}. \quad (2.14)$$

where $\Phi_{ce} = -(\Omega^2 r_\perp^2)/2 = -(\boldsymbol{\Omega} \times \mathbf{r}_\perp)^2/2$.

Coriolis force

The effects of the Coriolis force play a central role in much of geophysical fluid dynamics and will be considered extensively later on. For now, we just note three basic properties:

- (i) There is zero Coriolis force on bodies that are stationary in the rotating frame.
- (ii) The Coriolis force acts to deflect moving bodies at right angles to their direction of travel.
- (iii) The Coriolis force does no work on a body, a consequence of the fact that $\mathbf{v}_R \cdot (\boldsymbol{\Omega} \times \mathbf{v}_R) = 0$.

2.1.3 Momentum equation in a rotating frame

Since (2.12) simply relates the accelerations of a particle in the inertial and rotating frames, then in the rotating frame of reference Euler's equation may be written

$$\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} = -\frac{1}{\rho}\nabla p - \nabla\Phi. \quad (2.15)$$

We have dropped the subscript R ; henceforth, unless noted, all velocities without a subscript will be considered to be relative to the rotating frame.

2.1.4 Mass and tracer conservation in a rotating frame

Let ϕ be a scalar field that, in the inertial frame, obeys

$$\frac{D\phi}{Dt} + \phi \nabla \cdot \mathbf{v}_I = 0 \quad (2.16)$$

Now, observers in both the rotating and inertial frame measure the same value of ϕ . Further, $D\phi/Dt$ is simply the rate of change of ϕ associated with a material parcel, and therefore is reference frame invariant. Thus,

$$\left(\frac{D\phi}{Dt} \right)_R = \left(\frac{D\phi}{Dt} \right)_I \quad (2.17)$$

where $(D\phi/Dt)_R = (\partial\phi/\partial t)_R + \mathbf{v}_R \cdot \nabla\phi$ and $(D\phi/Dt)_I = (\partial\phi/\partial t)_I + \mathbf{v}_I \cdot \nabla\phi$ and the local temporal derivatives $(\partial\phi/\partial t)_R$ and $(\partial\phi/\partial t)_I$ are evaluated at fixed locations in the rotating and inertial frames, respectively.

Further, since $\mathbf{v} = \mathbf{v}_I - \boldsymbol{\Omega} \times \mathbf{r}$, we have that

$$\nabla \cdot \mathbf{v}_I = \nabla \cdot (\mathbf{v}_I - \boldsymbol{\Omega} \times \mathbf{r}) = \nabla \cdot \mathbf{v}_R \quad (2.18)$$

since $\nabla \cdot (\boldsymbol{\Omega} \times \mathbf{r}) = 0$. Thus, using (2.17) and (2.18), (2.16) is equivalent to

$$\frac{D\phi}{Dt} + \phi \nabla \cdot \mathbf{v} = 0 \quad (2.19)$$

where all observables are measured in the *rotating* frame. Thus, the equation for the evolution of a scalar whose measured value is the same in rotating and inertial frames is unaltered by the presence of rotation. In particular, the mass conservation equation is unaltered by the presence of rotation.

Although we have taken (2.17) as true *a priori*, the individual components of the material derivative differ in the rotating and inertial frames. In particular

$$\left(\frac{\partial\phi}{\partial t} \right)_I = \left(\frac{\partial\phi}{\partial t} \right)_R - (\boldsymbol{\Omega} \times \mathbf{r}) \cdot \nabla\phi \quad (2.20)$$

because $\boldsymbol{\Omega} \times \mathbf{r}$ is the velocity, in the inertial frame, of a uniformly rotating body. Similarly,

$$\mathbf{v}_I \cdot \nabla \phi = (\mathbf{v}_R + \boldsymbol{\Omega} \times \mathbf{r}) \cdot \nabla \phi. \quad (2.21)$$

Thus, adding these equations,

$$\left(\frac{D\phi}{Dt} \right)_I = \left(\frac{D\phi}{Dt} \right)_R. \quad (2.22)$$

This may be taken as a pedestrian confirmation of (2.17).

2.2 EQUATIONS OF MOTION IN SPHERICAL CO-ORDINATES

We now derive the equations of motion in a form that is useful when describing fluid dynamics on the surface of a nearly spherical, rotating planet, and we perform some simple manipulations on these equations. Before that we discuss a seemingly obvious matter, the appropriateness of spherical co-ordinates.

2.2.1 The centrifugal force and spherical co-ordinates

The centrifugal force is a potential force, like gravity, and so we may therefore define an ‘effective gravity’ equal to the sum of the true gravity and the centrifugal force. The true gravitational force is directed toward the center of the earth, except possibly for tiny effects due to the earth’s inhomogeneity, but the line of action of the effective gravity will in general differ slightly from this, and therefore have a component in the ‘horizontal’ plane, that is the perpendicular to the radial direction. The magnitude of the centrifugal force is $\Omega^2 r_\perp$, and so the effective gravity is given by

$$\mathbf{g} = \mathbf{g}_{\text{grav}} + \Omega^2 \mathbf{r}_\perp \quad (2.23)$$

where $\mathbf{g}_{\text{grav}}$ is the gravitational force due to the gravitational attraction of the earth and $r_\perp = r \cos \vartheta$. Both gravity and centrifugal force are potential forces and therefore we may define the *geopotential*, Φ , such that

$$\mathbf{g} = -\nabla \Phi \quad (2.24)$$

Surfaces of constant Φ are not quite spherical because $r_\perp$, and hence the centrifugal force, vary with latitude.

The components of the centrifugal force parallel and perpendicular to the radial direction $\Omega^2 r \cos^2 \vartheta$ and $\Omega^2 r \cos \vartheta \sin \vartheta$. Newtonian gravity is much larger than either of these, and at the earth’s surface the ratio of centrifugal to gravitational terms is approximately, and no more than,

$$\alpha \approx \frac{\Omega^2 a}{g} \approx \frac{(7.27 \times 10^{-5})^2 \times 6.4 \times 10^6}{10} \approx 3 \times 10^{-3} \quad (2.25)$$

The angle between $\mathbf{g}$ and the line to the center of the earth is given by a similar expression and so is also small, typically around 3×10^{-3} radians. However, the horizontal component of the centrifugal force is still large compared to the Coriolis force, their ratio in mid-latitudes being given by

$$\frac{\text{Horizontal centrifugal force}}{\text{Coriolis force}} \approx \frac{\Omega^2 a \cos \vartheta \sin \vartheta}{2\Omega u} \approx \frac{\Omega a}{4|u|} \approx 10, \quad (2.26)$$

using $u = 10 \text{ m s}^{-1}$. The centrifugal term therefore dominates the Coriolis term, and is largely balanced by a pressure force. Thus, if we adhered to true spherical co-ordinates, both the horizontal and radial components of the momentum equation would be dominated by a static balance between a pressure gradient and gravity or centrifugal terms. Although in principle there is nothing wrong with writing the equations this way, it obscures the dynamical balances involving the Coriolis force and pressure that determine the large-scale horizontal flow. A way around this is to use the direction of the geopotential force to *define* the vertical direction, and then to regard the surfaces of constant Φ as being true spheres.² The horizontal component of apparent gravity is then identically zero, and we have traded a potentially large dynamical error for a small geometric error. (If the angle between apparent gravity and a natural direction of the co-ordinate system were not small then more heroic measures would be called for.) The geopotential Φ is then a function of the vertical co-ordinate alone, and in fact for many purposes we can just take $\Phi = gz$. In fact, over time, the earth has developed an equatorial bulge to partly compensate for the centrifugal force, so that the effective gravity does in fact act in a direction virtually normal to the earth's surface, and the surface of the earth is an oblate spheroid of nearly constant geopotential and, because the oblateness is very small (the equatorial radius is only about 20 km more than the radius at the pole) using spherical co-ordinates is a very good approximation. The direction normal to geopotential surfaces, the local vertical, is the direction of increasing r in spherical coordinates, in this approximation.

2.2.2 Some identities in spherical co-ordinates

The location of a point is given by the co-ordinates (λ, ϑ, r) where λ is the angular distance eastward (i.e., longitude), ϑ is angular distance poleward (i.e., latitude) and r is the radial distance from the center of the earth. (In many fields co-latitude is used as a spherical co-ordinate, but meteorology and oceanography use latitude.) If a is the radius of the earth, then we also define $z = r - a$. At a given location we may also define the Cartesian increments $(\delta x, \delta y, \delta z) = (r \cos \vartheta \delta \lambda, r \delta \vartheta, \delta r)$.

For a scalar quantity ϕ the material derivative in spherical co-ordinates is

$$\frac{D\phi}{Dt} = \frac{\partial \phi}{\partial t} + \frac{u}{r \cos \vartheta} \frac{\partial \phi}{\partial \lambda} + \frac{v}{r} \frac{\partial \phi}{\partial \vartheta} + w \frac{\partial \phi}{\partial r}, \quad (2.27)$$

where the velocity components corresponding to the co-ordinates λ, ϑ, r are

$$(u, v, w) \equiv \left(r \cos \vartheta \frac{D\lambda}{Dt}, r \frac{D\vartheta}{Dt}, \frac{Dr}{Dt} \right). \quad (2.28)$$

That is, u is the zonal velocity, v is the meridional velocity and w the vertical velocity. If we define $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ to be the unit vectors in the direction of increasing (λ, ϑ, r) then

$$\mathbf{v} = \mathbf{i}u + \mathbf{j}v + \mathbf{k}w. \quad (2.29)$$

Note also that $Dr/Dt = Dz/Dt$.

The divergence of a vector $\mathbf{B} = \mathbf{i}B_\lambda + \mathbf{j}B_\vartheta + \mathbf{k}B_r$ is

$$\nabla \cdot \mathbf{B} = \frac{1}{\cos \vartheta} \left[\frac{1}{r} \frac{\partial B_\lambda}{\partial \lambda} + \frac{1}{r} \frac{\partial}{\partial \vartheta} (B_\vartheta \cos \vartheta) + \frac{\cos \vartheta}{r^2} \frac{\partial}{\partial r} (r^2 B_r) \right]. \quad (2.30)$$

The vector gradient of a scalar is:

$$\nabla \phi = \mathbf{i} \frac{1}{r \cos \vartheta} \frac{\partial \phi}{\partial \lambda} + \mathbf{j} \frac{1}{r} \frac{\partial \phi}{\partial \vartheta} + \mathbf{k} \frac{\partial \phi}{\partial r} \quad (2.31)$$

The Laplacian of a scalar is:

$$\nabla^2 \phi \equiv \nabla \cdot \nabla \phi = \frac{1}{r^2 \cos \vartheta} \left[\frac{1}{\cos \vartheta} \frac{\partial^2 \phi}{\partial \lambda^2} + \frac{\partial}{\partial \vartheta} \left(\cos \vartheta \frac{\partial \phi}{\partial \vartheta} \right) + \cos \vartheta \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) \right]. \quad (2.32)$$

The curl of a vector is:

$$\text{curl } \mathbf{B} = \nabla \times \mathbf{B} = \frac{1}{r^2 \cos \vartheta} \begin{vmatrix} \mathbf{i} r \cos \vartheta & \mathbf{j} r & \mathbf{k} \\ \partial / \partial \lambda & \partial / \partial \vartheta & \partial / \partial r \\ B_\lambda r \cos \vartheta & B_\vartheta r & B_r \end{vmatrix} \quad (2.33)$$

The vector Laplacian $\nabla^2 \mathbf{B}$ (used for example when calculating viscous terms in the momentum equation) may be obtained from the vector identity:

$$\nabla^2 \mathbf{B} = \nabla(\nabla \cdot \mathbf{B}) - \nabla \times (\nabla \times \mathbf{B}) \quad (2.34)$$

Only in Cartesian coordinates does this take the simple form,

$$\nabla^2 \mathbf{B} = \frac{\partial^2 \mathbf{B}}{\partial x^2} + \frac{\partial^2 \mathbf{B}}{\partial y^2} + \frac{\partial^2 \mathbf{B}}{\partial z^2}. \quad (2.35)$$

The expansion in spherical co-ordinates is rather uninformative and rarely needed.

Rate of change of unit vectors

In spherical co-ordinates the unit vectors that define the co-ordinate system on the sphere are $\mathbf{i}$, the unit vector pointing eastward, parallel to a line of latitude; $\mathbf{j}$ is the unit vector pointing polewards, parallel to a meridian; and $\mathbf{k}$, the unit vector pointing radially outward. The directions these vectors change with location, and in fact this is the case in nearly all coordinate systems, with the notable exception of the Cartesian one, and thus their material derivative is not zero. One way to evaluate this is to consider geometrically how the co-ordinate axes change with position, and this is set as a problem, 2.6. We will approach the problem a little differently, by first obtaining the effective rotation rate Ω_{flow} , relative to the earth, of a unit vector as it moves with the flow, and then applying (2.3). Specifically, let the fluid velocity be $\mathbf{v} = (u, v, w)$. The meridional component, v , produces a displacement $r\delta\vartheta = v\delta t$, and this give rise a local effective vector rotation rate around the local zonal axis of $-(v/r)\mathbf{i}$, the minus sign arising because a displacement in the direction of the north pole is produced by negative rotational displacement around the $\mathbf{i}$ axis. Similarly, the zonal component, u , produces a displacement $\delta\lambda r \cos \vartheta = u\delta t$ and so an effective rotation rate, but now about the earth's rotation axis, of $u/(r \cos \vartheta)$. Now, a rotation around the earth's rotation axis may be written as (see Fig. 2.2)

$$\Omega = \Omega(\mathbf{j} \cos \vartheta + \mathbf{k} \sin \vartheta). \quad (2.36)$$

If the scalar rotation rate is not Ω but is $u/(r \cos \vartheta)$, then the vector rotation rate is

$$\frac{u}{r \cos \vartheta}(\mathbf{j} \cos \vartheta + \mathbf{k} \sin \vartheta) = \mathbf{j} \frac{u}{r} + \mathbf{k} \frac{u \tan \vartheta}{r}. \quad (2.37)$$

Thus, the total rotation rate of a vector that moves with the flow is

$$\Omega_{\text{flow}} = -\mathbf{i} \frac{v}{r} + \mathbf{j} \frac{u}{r} + \mathbf{k} \frac{u \tan \vartheta}{r} \quad (2.38)$$

Then, applying (2.3), we have

$$\frac{D\mathbf{i}}{Dt} = \Omega_{\text{flow}} \times \mathbf{i} = \frac{u}{r \cos \vartheta}(\mathbf{j} \sin \vartheta - \mathbf{k} \cos \vartheta) \quad (2.39a)$$

$$\frac{D\mathbf{j}}{Dt} = \Omega_{\text{flow}} \times \mathbf{j} = -\mathbf{i} \frac{u}{r} \tan \vartheta - \mathbf{k} \frac{v}{r} \quad (2.39b)$$

$$\frac{D\mathbf{k}}{Dt} = \Omega_{\text{flow}} \times \mathbf{k} = \mathbf{i} \frac{u}{r} + \mathbf{j} \frac{v}{r}. \quad (2.39c)$$

2.2.3 Equations of motion

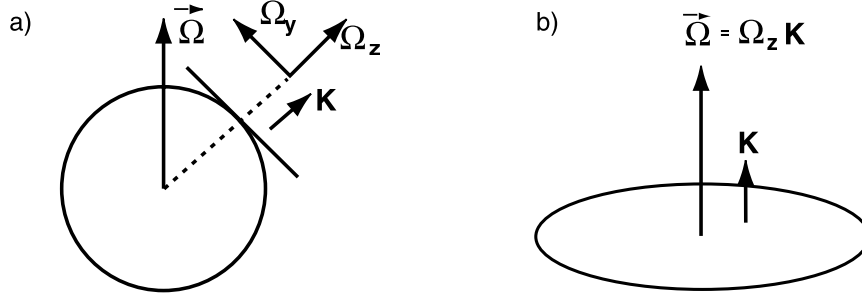


Figure 2.2 a) On the sphere the rotation vector $\vec{\Omega}$ can be decomposed into two components, one in the local vertical and one in the local horizontal, pointing toward the pole. That is, $\vec{\Omega} = \Omega_y \mathbf{j} + \Omega_z \mathbf{k}$ where $\Omega_y = \Omega \cos \vartheta$ and $\Omega_z = \Omega \sin \vartheta$. In geophysical fluid dynamics, the rotation vector in the local vertical is often the more important component in the horizontal momentum equations. On a rotating disk, (b), the rotation vector $\vec{\Omega}$ is parallel to the local vertical $\mathbf{k}$.

Mass Conservation and Thermodynamic Equation

The mass conservation equation is:

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0 \quad (2.40)$$

which in spherical co-ordinates is

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \frac{u}{r \cos \vartheta} \frac{\partial \rho}{\partial \lambda} + \frac{v}{r} \frac{\partial \rho}{\partial \vartheta} + w \frac{\partial \rho}{\partial r} \\ + \frac{\rho}{r \cos \vartheta} \left[\frac{\partial u}{\partial \lambda} + \frac{\partial}{\partial \vartheta} (v \cos \vartheta) + \frac{1}{r} \frac{\partial}{\partial r} (wr^2 \cos \vartheta) \right] = 0 \end{aligned} \quad (2.41)$$

or equivalently,

$$\frac{\partial \rho}{\partial t} + \frac{1}{r \cos \vartheta} \frac{\partial (u\rho)}{\partial \lambda} + \frac{1}{r \cos \vartheta} \frac{\partial}{\partial \vartheta} (v\rho \cos \vartheta) + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 w \rho) = 0. \quad (2.42)$$

The thermodynamic equation is a tracer advection equation. Thus, using (2.27), its adiabatic form is

$$\frac{D\theta}{Dt} = \frac{\partial \theta}{\partial t} + \frac{u}{r \cos \vartheta} \frac{\partial \theta}{\partial \lambda} + \frac{v}{r} \frac{\partial \theta}{\partial \vartheta} + w \frac{\partial \theta}{\partial r} = 0, \quad (2.43)$$

and similarly for tracers such as water vapour or salt.

Momentum Equation

Recall that inviscid momentum equation is:

$$\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} = -\frac{1}{\rho}\nabla p - \nabla\Phi. \quad (2.44)$$

where Φ is the geopotential. In spherical coordinates the directions of the coordinate axes change with position and so the component expansion of (2.44) is

$$\frac{D\mathbf{v}}{Dt} = \frac{Du}{Dt}\mathbf{i} + \frac{Dv}{Dt}\mathbf{j} + \frac{Dw}{Dt}\mathbf{k} + u\frac{D\mathbf{i}}{Dt} + v\frac{D\mathbf{j}}{Dt} + w\frac{D\mathbf{k}}{Dt} \quad (2.45a)$$

$$= \frac{Du}{Dt}\mathbf{i} + \frac{Dv}{Dt}\mathbf{j} + \frac{Dw}{Dt}\mathbf{k} + \boldsymbol{\Omega}_{\text{flow}} \times \mathbf{v} \quad (2.45b)$$

using (2.39). Using either (2.45a) and the expressions for the rates of change of the unit vectors given in (2.39), or (2.45b) and the expression for $\boldsymbol{\Omega}_{\text{flow}}$ given in (2.38), this becomes

$$\begin{aligned} \frac{D\mathbf{v}}{Dt} = & \mathbf{i} \left(\frac{Du}{Dt} - \frac{uv \tan \vartheta}{r} + \frac{uw}{r} \right) + \mathbf{j} \left(\frac{Dv}{Dt} + \frac{u^2 \tan \vartheta}{r} + \frac{vw}{r} \right) \\ & + \mathbf{k} \left(\frac{Dw}{Dt} - \frac{u^2 + v^2}{r} \right). \end{aligned} \quad (2.46)$$

Using the definition of a vector cross product the Coriolis term is:

$$\begin{aligned} 2\boldsymbol{\Omega} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2\Omega \cos \vartheta & 2\Omega \sin \vartheta \\ u & v & w \end{vmatrix} \\ &= \mathbf{i} (2\Omega w \cos \vartheta - 2\Omega v \sin \vartheta) + \mathbf{j} 2\Omega u \sin \vartheta - \mathbf{k} 2\Omega u \cos \vartheta. \end{aligned} \quad (2.47)$$

Using (2.46) and (2.47), and the gradient operator given by (2.31), the momentum equation (2.44) becomes:

$$\frac{Du}{Dt} - \left(2\Omega + \frac{u}{r \cos \vartheta} \right) (v \sin \vartheta - w \cos \vartheta) = -\frac{1}{\rho r \cos \vartheta} \frac{\partial p}{\partial \lambda}, \quad (2.48a)$$

$$\frac{Dv}{Dt} + \frac{wv}{r} + \left(2\Omega + \frac{u}{r \cos \vartheta} \right) u \sin \vartheta = -\frac{1}{\rho r} \frac{\partial p}{\partial \vartheta}, \quad (2.48b)$$

$$\frac{Dw}{Dt} - \frac{u^2 + v^2}{r} - 2\Omega u \cos \vartheta = -\frac{1}{\rho} \frac{\partial p}{\partial r} - g. \quad (2.48c)$$

The terms involving Ω are called Coriolis terms, and the quadratic terms on the left-hand sides involving $1/r$ are often called metric terms.

2.2.4 The primitive equations

The so-called *primitive equations* of motion are simplifications of the above equations frequently used in atmospheric and oceanic modelling.³ Three related approximations are involved; these are:

1. The *hydrostatic approximation*. In the vertical momentum equation the gravitational term is assumed to be exactly balanced by the pressure gradient term, so that

$$\frac{\partial p}{\partial z} = -\rho g. \quad (2.49)$$

The advection of vertical velocity, the Coriolis terms, and the metric term $(u^2 + v^2)/r$ are all neglected.

2. The *shallow-fluid approximation*. We write $r = a + z$ where the constant a is the radius of the earth and z increases in the radial direction. The coordinate r is then replaced by a except where it is used as the differentiating argument. Thus, for example,

$$\frac{1}{r^2} \frac{\partial(r^2 w)}{\partial r} \rightarrow \frac{\partial w}{\partial z}. \quad (2.50)$$

3. The *traditional approximation*. Terms in the horizontal momentum equations involving the Coriolis parameter multiplied by the vertical velocity, and the still smaller metric terms uw/r and vw/r , are neglected.

The second and third of these approximations should be taken, or not, together, the underlying reason being that they both relate to the presumed small aspect ratio of the motion, so the approximations succeed or fail together. If we make one approximation but not the other then we are being asymptotically inconsistent, and angular momentum and energy conservation are not assured [see section 2.2.7]. The hydrostatic approximation also depends on the small aspect ratio of the flow but in a slightly different way. For large-scale flow in the terrestrial atmosphere and ocean all three approximations are in fact all very accurate approximations. We defer a more complete treatment until section 2.7, in part because a treatment of the hydrostatic approximation is done most easily in the context of the Boussinesq equations.

Making these approximations, the momentum equations become

$$\frac{Du}{Dt} - 2\Omega \sin \vartheta v - \frac{uv}{a} \tan \vartheta = -\frac{1}{a\rho \cos \vartheta} \frac{\partial p}{\partial \lambda}, \quad (2.51a)$$

$$\frac{Dv}{Dt} + 2\Omega \sin \vartheta u + \frac{u^2 \tan \vartheta}{a} = -\frac{1}{\rho a} \frac{\partial p}{\partial \vartheta}, \quad (2.51b)$$

$$0 = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g. \quad (2.51c)$$

where

$$\frac{D}{Dt} = \left(\frac{\partial}{\partial t} + \frac{u}{a \cos \vartheta} \frac{\partial}{\partial \lambda} + \frac{v}{a} \frac{\partial}{\partial \vartheta} + w \frac{\partial}{\partial z} \right). \quad (2.52)$$

We note the ubiquity of the factor $2\Omega \sin \vartheta$, and take the opportunity to define the *Coriolis parameter*, $f \equiv 2\Omega \sin \vartheta$.

The corresponding mass conservation equation for a shallow fluid layer is:

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \frac{u}{a \cos \vartheta} \frac{\partial \rho}{\partial \lambda} + \frac{v}{a} \frac{\partial \rho}{\partial \vartheta} + w \frac{\partial \rho}{\partial z} \\ + \rho \left[\frac{1}{a \cos \vartheta} \frac{\partial u}{\partial \lambda} + \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (v \cos \vartheta) + \frac{\partial w}{\partial z} \right] = 0 \end{aligned} \quad (2.53)$$

or equivalently,

$$\frac{\partial \rho}{\partial t} + \frac{1}{a \cos \vartheta} \frac{\partial (u\rho)}{\partial \lambda} + \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (v\rho \cos \vartheta) + \frac{\partial (w\rho)}{\partial z} = 0. \quad (2.54)$$

2.2.5 Primitive equations in vector form

The primitive equations may be written in a compact vector form provided we make a slight reinterpretation of the material derivative of the co-ordinate axes. Let $\mathbf{u} = u\mathbf{i} + v\mathbf{j} + 0\mathbf{k}$ be the horizontal velocity. The primitive equations (2.51a) and (2.51b) may be written as

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\frac{1}{\rho} \nabla_z p \quad (2.55)$$

where $\mathbf{f} = f\mathbf{k} = 2\Omega \sin \vartheta \mathbf{k}$ and $\nabla_z p = -[(a\rho \cos \vartheta)^{-1} \partial p / \partial \lambda, (\rho a)^{-1} \partial p / \partial \vartheta]$. In this equation the material derivative of the horizontal velocity is given by

$$\frac{D\mathbf{u}}{Dt} = \mathbf{i} \frac{Du}{Dt} + \mathbf{j} \frac{Dv}{Dt} + u \frac{D\mathbf{i}}{Dt} + v \frac{D\mathbf{j}}{Dt}, \quad (2.56)$$

where

$$\begin{aligned} \frac{D\mathbf{i}}{Dt} &= \tilde{\boldsymbol{\Omega}}_{\text{flow}} \times \mathbf{i} = \mathbf{j} \frac{u \tan \vartheta}{a}, \\ \frac{D\mathbf{j}}{Dt} &= \tilde{\boldsymbol{\Omega}}_{\text{flow}} \times \mathbf{j} = -\mathbf{i} \frac{u \tan \vartheta}{a}, \end{aligned} \quad (2.57)$$

where $\tilde{\boldsymbol{\Omega}}_{\text{flow}} = \mathbf{k} u \tan \vartheta / a$ [c.f., (2.38).] Note that the advection of the horizontal wind $\mathbf{u}$ is still by the three-dimensional velocity $\mathbf{v}$.

The vertical momentum equation is the hydrostatic equation, (2.51c), and the mass conservation equation is

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0 \quad \text{or} \quad \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (2.58)$$

where D/Dt on a scalar is given by (2.52), and the second expression is written out in full in (2.54).

2.2.6 The vector invariant form of the momentum equation

The ‘vector invariant’ form of the momentum equation is so-called because it appears to take the same form in all co-ordinate systems — there is no advective derivative of the co-ordinate system to worry about. After a little manipulation, the three dimensional momentum equation may be written:

$$\frac{\partial \mathbf{v}}{\partial t} + (2\boldsymbol{\Omega} + \boldsymbol{\omega}) \times \mathbf{v} = -\nabla B. \quad (2.59)$$

where $B = \phi + \mathbf{v}^2/2 + \Phi$ is the Bernoulli function and $\boldsymbol{\omega}$ is the relative vorticity, $\boldsymbol{\omega} = \nabla \times \mathbf{v}$. In spherical coordinates the vorticity is:

$$\boldsymbol{\omega} = \nabla \times \mathbf{v} = \frac{1}{r^2 \cos \vartheta} \begin{vmatrix} \mathbf{i} r \cos \vartheta & \mathbf{j} r & \mathbf{k} \\ \partial/\partial \lambda & \partial/\partial \vartheta & \partial/\partial r \\ ur \cos \vartheta & rv & w \end{vmatrix} \quad (2.60)$$

$$= \mathbf{i} \frac{1}{r} \left(\frac{\partial w}{\partial \vartheta} - \frac{\partial(rv)}{\partial r} \right) - \mathbf{j} \frac{1}{r \cos \vartheta} \left(\frac{\partial w}{\partial \lambda} - \frac{\partial}{\partial r}(ur \cos \vartheta) \right) + \mathbf{k} \frac{1}{r \cos \vartheta} \left(\frac{\partial v}{\partial \lambda} - \frac{\partial}{\partial \vartheta}(u \cos \vartheta) \right). \quad (2.61)$$

Making the traditional approximation, and considering the horizontal and vertical components of the momentum equation separately, gives

$$\frac{\partial \mathbf{u}}{\partial t} + (2\boldsymbol{\Omega} + \mathbf{k}\zeta) \times \mathbf{u} + w \frac{\partial \mathbf{u}}{\partial z} = -\nabla_z B_h \quad (2.62)$$

where $\mathbf{u} = (u, v, 0)$, $\boldsymbol{\Omega} = \mathbf{k}2\Omega \sin \vartheta$, $B_h = \Phi + \mathbf{u}^2/2$, ∇_z is the horizontal gradient operator (the gradient at a constant value of z), and using (2.61), ζ is given by

$$\zeta = \frac{1}{a \cos \vartheta} \frac{\partial v}{\partial \lambda} - \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta}(u \cos \vartheta) = \frac{1}{a \cos \vartheta} \frac{\partial v}{\partial \lambda} - \frac{1}{a} \frac{\partial u}{\partial \vartheta} + \frac{u}{a} \tan \vartheta. \quad (2.63)$$

The separate components of the momentum equation are then found to be:

$$\frac{\partial u}{\partial t} - (f + \zeta)v + w \frac{\partial u}{\partial z} = -\frac{1}{a} \frac{\partial B_h}{\partial \lambda}, \quad (2.64)$$

and

$$\frac{\partial v}{\partial t} + (f + \zeta)u + w \frac{\partial v}{\partial z} = -\frac{1}{a} \frac{\partial B_h}{\partial \vartheta}, \quad (2.65)$$

where $f = 2\Omega \sin \vartheta$.

2.2.7 Angular Momentum

The zonal momentum equation can be usefully expressed as a statement about axial angular momentum, i.e., angular momentum about the rotation axis. The zonal angular momentum per unit mass is the component of angular momentum in the direction of the axis of rotation and it is given by, without making any shallow atmosphere approximation,

$$m = (u + \Omega r \cos \vartheta) r \cos \vartheta. \quad (2.66)$$

The evolution equation for this quantity follows from the zonal momentum equation and has the simple form

$$\frac{Dm}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial \lambda} \quad (2.67)$$

where the material derivative is

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \frac{u}{r \cos \vartheta} \frac{\partial}{\partial \lambda} + \frac{v}{r} \frac{\partial}{\partial \vartheta} + w \frac{\partial}{\partial r}. \quad (2.68)$$

Using the mass continuity equation, this can be written as

$$\frac{D\rho m}{Dt} + \rho m \nabla \cdot \mathbf{v} = -\frac{\partial p}{\partial \lambda} \quad (2.69)$$

or

$$\frac{\partial \rho m}{\partial t} + \frac{1}{r \cos \vartheta} \frac{\partial (\rho u m)}{\partial \lambda} + \frac{1}{r \cos \vartheta} \frac{\partial}{\partial \vartheta} (\rho v m \cos \vartheta) + \frac{\partial}{\partial z} (\rho m w) = -\frac{\partial p}{\partial \lambda}. \quad (2.70)$$

Integrating over longitude the pressure term on the right-hand side vanishes and we obtain

$$\frac{\partial \rho m}{\partial t} + \frac{1}{r \cos \vartheta} \frac{\partial}{\partial \vartheta} (v m \cos \vartheta) + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 m w) = 0. \quad (2.71)$$

This is evidently an angular momentum conservation equation.

If the fluid is confined to a shallow layer near the surface of a sphere, then we may replace r , the radial co-ordinate, by a , the radius of the sphere, in the definition of m , so that $\tilde{m} = (u + \Omega a \cos \vartheta) a \cos \vartheta$. Then (2.67) is replaced by

$$\frac{D\tilde{m}}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial \lambda} \quad (2.72)$$

where now

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \frac{u}{a \cos \vartheta} \frac{\partial}{\partial \lambda} + \frac{v}{a} \frac{\partial}{\partial \vartheta} + w \frac{\partial}{\partial z}. \quad (2.73)$$

Using mass continuity this may be written as

$$\frac{\partial \rho \tilde{m}}{\partial t} + \frac{u}{a \cos \vartheta} \frac{\partial \tilde{m}}{\partial \lambda} + \frac{v}{a} \frac{\partial \tilde{m}}{\partial \vartheta} + w \frac{\partial \tilde{m}}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial \lambda}. \quad (2.74)$$

As before we may integrate around a latitude circle to obtain

$$\frac{\partial \rho \tilde{m}}{\partial t} + \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (v \tilde{m} \cos \vartheta) + \frac{\partial}{\partial z} (\tilde{m} w) = 0. \quad (2.75)$$

which is the appropriate angular momentum conservation equation for a shallow atmosphere.

** From angular momentum to the spherical component equations*

A somewhat indirect way to derive the three components of the momentum equation in spherical polar co-ordinates is to *begin* with (2.67) and the principle of conservation of energy. That is, we take the equations for conservation of angular momentum and energy as true *a priori* and demand that the forms of the momentum equation be constructed to satisfy this. Expanding the material derivative, noting that that $Dr/Dt = w$ and $D \cos \vartheta / Dt = -(v/r) \sin \vartheta$, immediately gives (2.48a). Multiplication by u then yields

$$u \frac{Du}{Dt} - 2\Omega uv \sin \vartheta + 2\Omega uw \cos \vartheta - \frac{u^2 v \tan \vartheta}{r} + \frac{u^2 w}{r} = -\frac{u}{\rho r \cos \vartheta} \frac{\partial p}{\partial \lambda}. \quad (2.76)$$

Now suppose that the meridional and vertical momentum equations are of the form

$$\frac{Dv}{Dt} + \text{Coriolis and metric terms} = -\frac{1}{\rho r} \frac{\partial p}{\partial \vartheta} \quad (2.77a)$$

$$\frac{Dw}{Dt} + \text{Coriolis and metric terms} = -\frac{1}{\rho} \frac{\partial p}{\partial r}. \quad (2.77b)$$

but that we do not know what form the Coriolis and metric terms take. Then, construct the kinetic energy equation by multiplying these equations by v and w . Now, the metric terms must vanish when we sum the resulting equations, so that (2.77a) must contain the Coriolis term $2\Omega u \sin \vartheta$ as well as the metric term $u^2 \tan \vartheta / r$, and (2.77b) must contain a $-2\Omega u \cos \vartheta$ as well as the metric term u^2 / r . But if (2.77b) contains the term u^2 / r it must also contain the term v^2 / r by isotropy, and therefore (2.77a) must also contain the term vw / r . In this way, (2.48) is precisely reproduced, although the skeptic might argue that the uniqueness of the form has not been proven.

A particular advantage of this approach arises in determining the appropriate momentum equations that conserve angular momentum and energy in the

shallow-fluid approximation. We begin with (2.72) and expand to obtain (2.51a). Multiplying by u gives

$$u \frac{Du}{Dt} - 2\Omega uv \sin \vartheta - \frac{u^2 v \tan \vartheta}{a} = -\frac{u}{\rho a \cos \vartheta} \frac{\partial p}{\partial \lambda}. \quad (2.78)$$

Evidently, the meridional momentum equation must contain the Coriolis term $2\Omega u \sin \vartheta$ and the metric term $u^2 \tan \vartheta / a$, but the vertical momentum equation must have neither of the metric terms appearing in (2.48c). Thus we deduce the following equations:

$$\frac{Du}{Dt} - \left(2\Omega \sin \vartheta + \frac{u \tan \vartheta}{a} \right) v = -\frac{1}{\rho a \cos \vartheta} \frac{\partial p}{\partial \lambda} \quad (2.79a)$$

$$\frac{Dv}{Dt} + \left(2\Omega \sin \vartheta + \frac{u \tan \vartheta}{a} \right) v = -\frac{1}{\rho a} \frac{\partial p}{\partial \vartheta} \quad (2.79b)$$

$$\frac{Dw}{Dt} = -\frac{1}{\rho} \frac{\partial \rho}{\partial r} - g. \quad (2.79c)$$

This equation set, when used in conjunction with the thermodynamic and mass continuity equations, conserves appropriate forms of angular momentum and energy. In the hydrostatic approximation the material derivative of w in (2.79c) is *additionally* neglected. Thus, the hydrostatic approximation is mathematically and physically consistent with the shallow-fluid approximation, but it is an additional approximation with slightly different requirements that one may choose, rather than being required, to make. From an asymptotic perspective, the difference lies in the small parameter necessary for either approximation to hold, namely

$$\text{Shallow fluid and traditional approximations: } \gamma \equiv \frac{H}{a} \ll 1 \quad (2.80a)$$

$$\text{Small aspect ratio for hydrostatic approximation: } \alpha \equiv \frac{H}{L} \ll 1. \quad (2.80b)$$

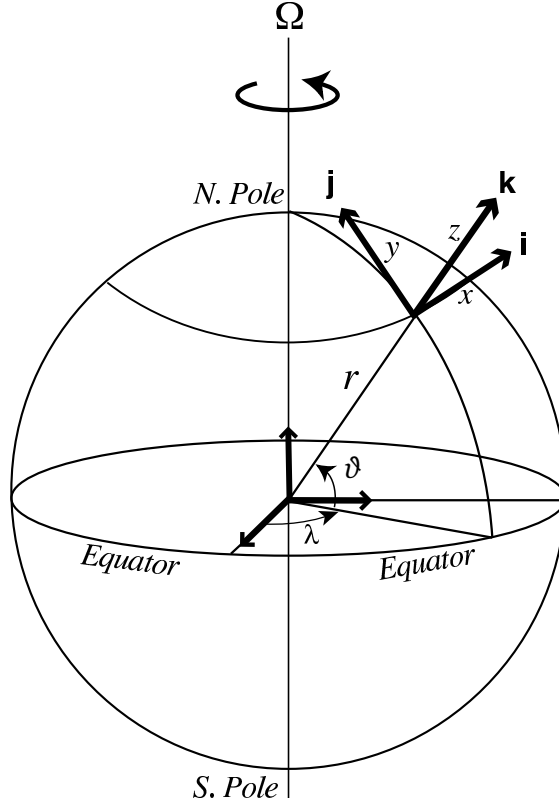
where L is the horizontal scale of the motion and a is the radius of the earth. For hemispheric or global scale phenomena $L \sim a$ and the two approximations coincide. (Requirement (2.80b) for the hydrostatic approximation is derived in section 2.7.)

2.3 CARTESIAN APPROXIMATIONS: THE TANGENT PLANE

2.3.1 The f-plane

Although the rotation of the earth is central for many dynamical phenomena, the sphericity of the earth is not always so. This is especially true for phenomena

Figure 2.3 The spherical coordinate system. The orthogonal unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ point in the direction of increasing longitude λ , latitude ϑ , and altitude z . Locally, one may apply a Cartesian system with variables x , y and z measuring distances along $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$.



on a scale somewhat smaller than global where the use of spherical coordinates becomes awkward, and it is more convenient to use a locally Cartesian representation of the equations. Referring to Fig. 2.2 we will define a plane tangent to the surface of the earth at a latitude ϑ_0 , and then use a Cartesian coordinate system (x, y, z) to describe motion on that plane. For small excursions on the plane, $(x, y, z) \approx (a\lambda \cos \vartheta_0, a(\vartheta - \vartheta_0), z)$. Consistently, the velocity is $\mathbf{v} = (u, v, w)$, so that u, v and w are the components of the velocity *in the tangent plane*. These are approximately in the east-west, north-south and vertical directions, respectively.

The momentum equations for flow in this plane are then

$$\frac{\partial u}{\partial t} + (\mathbf{v} \cdot \nabla)u + 2\Omega_y w - 2\Omega_z v = -\frac{1}{\rho} \frac{\partial p}{\partial x}, \quad (2.81a)$$

$$\frac{\partial v}{\partial t} + (\mathbf{v} \cdot \nabla)v + 2\Omega_z u = -\frac{1}{\rho} \frac{\partial p}{\partial y}, \quad (2.81b)$$

$$\frac{\partial w}{\partial t} + (\mathbf{v} \cdot \nabla)w + 2(\Omega_x v - \Omega_y u) = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g, \quad (2.81c)$$

where the rotation vector $\boldsymbol{\Omega} = \Omega_x \mathbf{i} + \Omega_y \mathbf{j} + \Omega_z \mathbf{k}$ and $\Omega_x = 0$, $\Omega_y = \Omega \cos \vartheta_0$ and $\Omega_z = \Omega \sin \vartheta_0$. If we make the traditional approximation, and so ignore the

components of $\boldsymbol{\Omega}$ not in the direction of the local vertical, then

$$\frac{Du}{Dt} - f_0 v = -\frac{1}{\rho} \frac{\partial p}{\partial x}, \quad (2.82a)$$

$$\frac{Dv}{Dt} + f_0 u = -\frac{1}{\rho} \frac{\partial p}{\partial y}, \quad (2.82b)$$

$$\frac{Dw}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - \rho g. \quad (2.82c)$$

where $f_0 = 2\Omega_z \sin \vartheta_0$. Defining the horizontal velocity vector $\mathbf{u} = (u, v, 0)$, the first two equations may also be written as

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f}_0 \times \mathbf{u} = -\frac{1}{\rho} \nabla_z p, \quad (2.83)$$

where $D\mathbf{u}/Dt = \partial\mathbf{u}/\partial t + \mathbf{v} \cdot \nabla\mathbf{u}$. Eq. (2.83) may also be written in the form

$$\frac{\partial\mathbf{u}}{\partial t} + (\mathbf{f}_0 + \mathbf{k}\zeta) \times \mathbf{u} = -\frac{1}{\rho} \nabla_z B \quad (2.84)$$

where B is the Bernoulli function, $\mathbf{f}_0 = 2\Omega \sin \vartheta_0 \mathbf{k} = f_0 \mathbf{k}$, $\zeta = \partial v / \partial x - \partial u / \partial y$, and $\mathbf{k}$ is the direction perpendicular to the plane (it does not change its orientation with latitude). These equations are, evidently, exactly the same as the momentum equations in a system in which the rotation vector is aligned with the local vertical, as illustrated in the right panel in Fig. 2.2. They will describe flow on the surface of a rotating sphere to a good approximation provided the flow is of limited latitudinal extent so that the effects of sphericity are unimportant. This is known as the *f-plane* approximation since the Coriolis parameter is a constant.

We may in addition make the hydrostatic approximation, in which case (2.82c) becomes the familiar $\partial p / \partial z = -\rho g$. However, there is no requirement that it be made: equations (2.82a), (2.82b) and (2.82c) are in themselves a consistent Cartesian equation set.

2.3.2 The beta-plane approximation

The magnitude of the vertical component of rotation varies with latitude, and this has important dynamical consequences. We can approximate this effect by allowing the effective rotation vector to vary. Thus, noting that, for small variations in latitude,

$$f = 2\Omega \sin \vartheta \approx 2\Omega \sin \vartheta_0 + 2\Omega \cos \vartheta_0 (\vartheta - \vartheta_0), \quad (2.85)$$

then on the tangent plane the Coriolis parameter will vary as

$$\boxed{f = f_0 + \beta y}, \quad (2.86)$$

where $f_0 = 2\Omega \sin \vartheta$ and $\beta = \partial f / \partial y = (2\Omega \cos \vartheta_0) / a$. This important approximation is known as the *beta-plane*, or *β -plane*, approximation. It turns out that it captures the most important *dynamical* effects of sphericity, without the complicating *geometric* effects, which are not essential to describe many phenomena. The momentum equations (2.82a), (2.82b) and (2.82c) (or its hydrostatic counterpart) are unaltered, save that f_0 is replaced by $f_0 + \beta y$ to represent a varying Coriolis parameter. Thus, sphericity combined with rotation is dynamically equivalent to a *differentially rotating* system. For future reference, we write down the horizontal momentum equations:

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\frac{1}{\rho} \nabla_z p, \quad (2.87)$$

or

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{f} + \mathbf{k}\zeta) \times \mathbf{u} = -\frac{1}{\rho} \nabla_z B \quad (2.88)$$

where $\mathbf{f} = (f_0 + \beta y)\hat{\mathbf{k}}$, and the hydrostatic relationship

$$\frac{1}{\rho} \frac{\partial p}{\partial z} = -\rho g. \quad (2.89)$$

In component form the horizontal momentum equations are

$$\frac{Du}{Dt} - fv = -\frac{1}{\rho} \frac{\partial p}{\partial x}, \quad (2.90a)$$

$$\frac{Dv}{Dt} + fu = -\frac{1}{\rho} \frac{\partial p}{\partial y}, \quad (2.90b)$$

The mass conservation and thermodynamic equations in the β -plane approximation are the same as the usual Cartesian forms of those equations.

2.4 EQUATIONS FOR A STRATIFIED OCEAN AND THE BOUSSINESQ APPROXIMATION

The density variations in the ocean are quite small compared to the mean density, and it is useful to exploit this to derive somewhat simpler but still quite accurate equations of motion — the *Boussinesq equations*.

2.4.1 Variation of density in the ocean

The variation of density in the ocean are due to three effects: the compression of water by pressure (which we denote as $\Delta\rho_p$), the thermal expansion of water if its temperature changes ($\Delta\rho_T$), and the haline contraction if its salinity changes

($\Delta\rho_S$). How big are these? An appropriate equation of state to approximately evaluate these effects is the linear one [c.f., (1.188)]:

$$\rho = \rho_0 \left(1 - \beta_T(T - T_0) + \beta_S(S - S_0) + \frac{p}{\rho_0 c_s^2} \right) \quad (2.91)$$

where $\beta_T \approx 2 \times 10^{-4} \text{ K}^{-1}$, $\beta_S \approx 10^{-3} \text{ psu}^{-1}$ and $c_s \approx 1500 \text{ m s}^{-1}$. The three effects are then:

Pressure compressibility: We have $\Delta\rho_p \approx \Delta p / c^2 \approx \rho_0 g H / c^2$ where H is the depth and the pressure change is quite accurately evaluated using the hydrostatic approximation. Thus,

$$\frac{|\Delta\rho_p|}{\rho_0} \ll 1 \quad \text{if} \quad \frac{gH}{c^2} \ll 1, \quad (2.92)$$

or if $H \ll c^2 / g$. The quantity $c^2 / g \approx 200 \text{ km}$ is the density scale height of the ocean. Thus, the pressure at the bottom of the ocean (say $H = 10 \text{ km}$ in the deep trenches), enormous as it is, is insufficient to compress the water enough to make a significant change in its density. Changes in density due to dynamical variations of pressure are small if the Mach number is small, and this is also the case.

Thermal expansion: We have $\Delta\rho_T \approx -\beta_T \rho_0 \Delta T$ and therefore

$$\frac{|\Delta\rho_T|}{\rho_0} \ll 1 \quad \text{if} \quad \beta_T \Delta T \ll 1. \quad (2.93)$$

For $\Delta T = 20 \text{ K}$, $\beta_T \Delta T \approx 4 \times 10^{-3}$, and evidently we would require temperature differences of order $\beta_T^{-1} \approx 5000 \text{ K}$ to obtain order one variations in density.

Saline contraction: We have $\Delta\rho_S \approx \beta_S \rho_0 \Delta S$ and therefore

$$\frac{|\Delta\rho_S|}{\rho_0} \ll 1 \quad \text{if} \quad \beta_S \Delta S \ll 1. \quad (2.94)$$

Since changes in salinity in the ocean rarely exceed 5 psu, the fractional change in the density of seawater is correspondingly very small.

Evidently, fractional density changes in the ocean are very small.

2.4.2 The Boussinesq equations

The *Boussinesq equations* are a set of equations that exploit the smallness of density variations in many liquids.⁴ To set notation we write

$$\begin{aligned}\rho &= \rho_0 + \delta\rho(x, y, z, t) \\ &= \rho_0 + \hat{\rho}(z) + \rho'(x, y, z, t) \\ &= \tilde{\rho}(z) + \rho'(x, y, z, t)\end{aligned}\tag{2.95}$$

where ρ_0 is a constant and we assume that

$$|\hat{\rho}|, |\rho'|, |\delta\rho| \ll \rho_0.\tag{2.96}$$

We need not assume that $|\rho'| \ll |\hat{\rho}|$, but this is often the case in the ocean. Associated with the reference density is a reference pressure that is defined to be in hydrostatic balance. That is,

$$\begin{aligned}p &= p_0(z) + \delta p(x, y, z, t) \\ &= \tilde{p}(z) + p'(x, y, z, t)\end{aligned}\tag{2.97}$$

where

$$\frac{dp_0}{dz} \equiv -g\rho_0,\tag{2.98}$$

and

$$\frac{d\tilde{p}}{dz} \equiv -g\tilde{\rho}.\tag{2.99}$$

Note that $\nabla_z p = \nabla_z p' = \nabla_z \delta p$ and that $p_0 \approx \tilde{p}$ if $|\hat{\rho}| \ll \rho_0$.

Momentum equations

The momentum equation can be written as

$$(\rho_0 + \delta\rho) \left(\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} \right) = -\nabla p - g(\rho_0 + \delta\rho)\mathbf{k}\tag{2.100}$$

which, without approximation, is

$$(\rho_0 + \delta\rho) \left(\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} \right) = -\nabla \delta p - g\delta\rho\mathbf{k}.\tag{2.101}$$

If density variations are small this becomes

$$\boxed{\left(\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} \right) = -\nabla \phi + b\mathbf{k}}.\tag{2.102}$$

where $\phi = \delta p / \rho_0$ and $b = -g\delta\rho / \rho_0$ is the *buoyancy*. Note that we should not and do not neglect the term $g\delta\rho$, for there is no reason to believe it to be small. Eq. (2.102) is the momentum equation in the *Boussinesq approximation*, and it is common to say that this approximation ignores all variations of density of a fluid in the momentum equation, except when associated with the gravitational term; that is, in the term $\rho\mathbf{g}$.

For most large-scale motions in the ocean the *deviation* pressure and density fields are also approximately in hydrostatic balance, and in that case the vertical component of (2.102) becomes

$$\frac{\partial\phi}{\partial z} = b. \quad (2.103)$$

However, there may be regions of strong convective activity where the hydrostatic approximation is poor, and it is important to note that the Boussinesq approximation does not necessarily imply hydrostatic balance. Furthermore, note that a condition for (2.103) to hold is that vertical accelerations are small *compared to* $g\delta\rho / \rho_0$, and not compared to the acceleration due to gravity itself (also see section 2.7).

Mass Conservation

The unapproximated mass conservation equation is

$$\frac{D\delta\rho}{Dt} + (\rho_0 + \delta\rho)\nabla \cdot \mathbf{v} = 0. \quad (2.104)$$

Provided that time scales advectively — that is to say that D/Dt scales in the same way as $\mathbf{v} \cdot \nabla$ — then we may approximate this equation by

$$\boxed{\nabla \cdot \mathbf{v} = 0}. \quad (2.105)$$

which is the same as that for a constant density fluid. Note that this *absolutely does not* allow one to go back and use (2.104) to say that $D\delta\rho/Dt = 0$. The evolution of density is given by the thermodynamic equation in conjunction with an equation of state, and this should not be confused with the mass conservation equation. Note too that in eliminating the time-derivative of density we eliminate the possibility of sound waves.

Thermodynamic equation and equation of state

The Boussinesq equations are closed by the addition of an equation of state, a thermodynamic equation and, as appropriate, a salinity equation. Because the fluid is incompressible, there can be no conversion from kinetic and potential energy into internal energy (see section 1.9). Thus, to ensure that the energetics

Summary of Boussinesq Equations

The simplest form is, for an adiabatic fluid:

$$\text{Momentum equations: } \frac{D\mathbf{v}}{Dt} + \mathbf{f} \times \mathbf{v} = -\nabla\phi + b\mathbf{k} \quad (\text{B.1})$$

$$\text{Mass conservation: } \nabla \cdot \mathbf{v} = 0 \quad (\text{B.2})$$

$$\text{Buoyancy equation: } \frac{Db}{Dt} = 0 \quad (\text{B.3})$$

The above equations form a complete set. A more complex form replaces the buoyancy equation by:

$$\text{Thermodynamic equation: } \frac{D\theta}{Dt} = 0 \quad (\text{B.4})$$

$$\text{Salinity equation (as appropriate): } \frac{DS}{Dt} = 0 \quad (\text{B.5})$$

$$\text{Equation of state: } \rho = \rho(\theta, \phi, S) \quad (\text{B.6})$$

$$\text{Buoyancy definition: } b = -\frac{g\delta\rho}{\rho_0} \quad (\text{B.7})$$

be consistent we should neglect the divergence of velocity in the thermodynamic equation. Writing the thermodynamic equation in terms of temperature and density we have

$$c_v \frac{DT}{Dt} + T \left(\frac{\partial \eta}{\partial \alpha} \right)_T \frac{D\alpha}{Dt} = \dot{Q} \quad (2.106)$$

and neglecting the second term gives $c_v DT/Dt = \dot{Q}$. If the density variations are proportional to density perturbations we straightforwardly obtain

$$\boxed{\frac{Db}{Dt} = \dot{b}}, \quad (2.107)$$

where $\dot{b}$ is the buoyancy source due to heating and is proportional to $\dot{Q}$. The momentum equation (2.102), mass continuity equation (2.105) and thermodynamic equation (2.107) then form a closed set.

* *More thermodynamic details*

Another approach to the thermodynamics is write the (adiabatic) thermodynamic equation as

$$\frac{D\rho}{Dt} - \frac{1}{c^2} \frac{Dp}{Dt} = 0 \quad (2.108)$$

where c is the speed of sound. The most severe approximation is just to neglect the the second term on the left hand side, and the thermodynamic equation becomes again

$$\frac{Db}{Dt} = 0. \quad (2.109)$$

A more accurate approach retains the compressibility of the fluid in (2.108) that is due to the hydrostatic pressure. To a good approximation the sound speed is constant and we can evaluate the pressure variations using the hydrostatic approximation, giving

$$\frac{D\rho_{\text{pot}}}{Dt} = 0 \quad (2.110)$$

where $\rho_{\text{pot}} = \rho_0 + \delta\rho - p_0(z)/c_s^2 = \rho_0 + \delta\rho + \rho_0 g z / c_s^2$ is, in this approximation, the potential density [c.f., (1.126)]. If we define the *potential buoyancy*, the buoyancy based on potential density, as

$$b_\sigma \equiv -\frac{g}{\rho_0} \left(\delta\rho - \frac{p_0(z)}{c_s^2} \right) = -\frac{g}{\rho_0} \left(\delta\rho + \frac{\rho_0 z}{H_\rho} \right) \quad (2.111)$$

where $H_\rho = c_s^2/g$ then the thermodynamic equation may be written

$$\frac{Db_\sigma}{Dt} = 0. \quad (2.112)$$

A convenient way to write down the full equations arises if we rewrite the momentum equation to use potential buoyancy directly. To do this, define a modified pressure such that

$$\phi^* = \phi - \frac{gz^2}{2H_\rho} \quad (2.113)$$

Then, $-\nabla\phi^* + b_\sigma\mathbf{k} = -\nabla\phi + b\mathbf{k}$ and the momentum equation becomes

$$\frac{D\mathbf{v}}{Dt} = -\nabla\phi^* + b_\sigma\mathbf{k}. \quad (2.114)$$

The equations (2.114), (2.112) and (2.105) form a complete set. They have the same form as, but are slightly more accurate than the set (2.102), (2.107) and (2.105). In practice the difference arises mainly in the interpretation of the terms and possibly in the boundary conditions. Use of the Boussinesq equations either of these forms seems to have avoided the use of a full equation of state. In reality,

this makes its presence felt in any diabatic source terms that would appear right-hand side of (2.107) or (2.112) and in the boundary conditions on b or b_σ .

It is often the case that we do need to use an accurate equation of state. In that case, it is simplest to write (from section 1.5.6) the thermodynamic equation as

$$\frac{D\theta}{Dt} = 0 \quad (2.115)$$

where θ is the potential temperature. Thus, (2.107) may be replaced by (2.115), a salinity evolution equation $DS/Dt = 0$ if needed, and an equation of state of the general form

$$\rho = \rho(\theta, S, \phi), \quad (2.116)$$

an example of which is (1.188), along with the definition of buoyancy $b = -g\delta\rho/\rho_0$. Using an accurate equation of state and the Boussinesq approximation is the procedure used in many comprehensive ocean general circulation models. The complication of this procedure is hidden in the equation of state, and the energetic properties of such a system are not easily determined, or necessarily consistent. The Boussinesq equations, which with the hydrostatic and traditional approximations are often considered to be the oceanic primitive equations, are summarized in the table on page 82.

** Mean stratification and the buoyancy frequency*

The processes that cause density to vary in the vertical often differ from those that cause it to vary in the horizontal. For this reason it is sometimes useful to write $\rho = \rho_0 + \hat{\rho}(z) + \rho'(x, y, z, t)$ and define $\hat{b}(z) \equiv -g\hat{\rho}/\rho_0$ and $b' \equiv -g\rho'/\rho_0$. Using the hydrostatic equation to evaluate pressure, the thermodynamic equation (2.108) becomes, to a good approximation,

$$\boxed{\frac{Db'}{Dt} + N^2 w = 0}, \quad (2.117)$$

where

$$N^2(z) = \left(\frac{d\hat{b}}{dz} - \frac{g^2}{c_s^2} \right) = -\frac{g}{\rho_0} \left(\frac{d\hat{\rho}}{dz} + \frac{g}{c_s^2} \right) \quad (2.118)$$

is the square of the buoyancy frequency.

2.4.3 Energetics of the Boussinesq system

In a uniform gravitational field but with no other forcing or dissipation, we write the Boussinesq equations as

$$\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} = b\mathbf{k} - \nabla\phi \quad (2.119a)$$

$$\nabla \cdot \mathbf{v} = 0 \quad (2.119b)$$

$$\frac{Db}{Dt} = 0 \quad (2.119c)$$

From (2.119a) and (2.119b) the kinetic energy density evolution is given by

$$\frac{1}{2} \frac{Dv^2}{Dt} = bw - \nabla \cdot (\phi \mathbf{v}) \quad (2.120)$$

where the constant reference density ρ_0 is omitted. Let us now define the potential Φ such that $\nabla\Phi = -\mathbf{k}$ (so $\Phi = -z$). Then

$$\frac{D\Phi}{Dt} = \nabla \cdot (\mathbf{v}\Phi) = -w \quad (2.121)$$

and using this and (2.119c) gives

$$\frac{D}{Dt}(b\Phi) = -wb. \quad (2.122)$$

Adding this to (2.120) and expanding the material derivative gives

$$\frac{\partial}{\partial t} \left(\frac{1}{2} v^2 + b\Phi \right) + \nabla \cdot \left[\mathbf{v} \left(\frac{1}{2} v^2 + b\Phi + \phi \right) \right] = 0. \quad (2.123)$$

This constitutes an energy equation for the Boussinesq system, and may be compared to (1.208). (See also problems 2.12 and 2.13.) The energy density (divided by ρ_0) is just $v^2/2 + b\Phi$. What does the second term represent? Its integral, multiplied by ρ_0 , is the potential energy of the flow minus that of the basic state, or $\int g(\rho - \rho_0)z \, dz$. If there were a heating term on the right-hand side of (2.119c) this would directly provide a source of potential energy, rather than internal energy as in the compressible system.

2.5 EQUATIONS FOR A STRATIFIED ATMOSPHERE AND THE ANELASTIC APPROXIMATION

In the atmosphere the density varies significantly, especially in the vertical. However deviations of both ρ and p from a statically balanced state are often quite small, and the relative vertical variation of potential temperature is also small. We can usefully exploit these observations to give a somewhat simplified set of equations, useful both for theoretical and numerical analysis because sound waves are eliminated by way of an ‘anelastic’ approximation.⁵

To begin we set

$$\rho = \tilde{\rho} + \delta\rho(x, y, z, t) \quad (2.124a)$$

$$p = \tilde{p}(z) + \delta p(x, y, z, t) \quad (2.124b)$$

where we assume that $|\delta\rho| \ll |\tilde{\rho}|$ and we define $\tilde{p}$ such that

$$\frac{\partial \tilde{p}}{\partial z} \equiv -g\tilde{\rho}(z). \quad (2.125)$$

The notation is similar to that for the Boussinesq case except that, importantly, our basic state of density is now a (given) function of vertical co-ordinate. Rather like with the Boussinesq case, the idea is to ignore dynamic variations of density (i.e., $\delta\rho$) except where associated with gravity. First recall a couple of ideal gas relationships involving potential temperature, θ and entropy s (divided by c_p , so $s \equiv \log \theta$), namely

$$s \equiv \log \theta = \log T - \frac{R}{c_p} \log p \quad (2.126)$$

$$= \frac{1}{\gamma} \log p - \log \rho \quad (2.127)$$

where $\gamma = c_p/c_v$, implying

$$\delta s = \frac{1}{\gamma} \frac{\delta p}{p} - \frac{\delta \rho}{\rho} \approx \frac{1}{\gamma} \frac{\delta p}{\tilde{p}} - \frac{\delta \rho}{\tilde{\rho}} \quad (2.128)$$

Further, if $\tilde{s} \equiv \gamma^{-1} \log \tilde{p} - \log \tilde{\rho}$ then

$$\begin{aligned} \frac{d\tilde{s}}{dz} &= \frac{1}{\gamma\tilde{p}} \frac{d\tilde{p}}{dz} - \frac{1}{\tilde{\rho}} \frac{d\tilde{\rho}}{dz} \\ &= -\frac{g\tilde{\rho}}{\gamma\tilde{p}} - \frac{1}{\tilde{\rho}} \frac{d\tilde{\rho}}{dz}. \end{aligned} \quad (2.129)$$

In the atmosphere, the left-hand-side is, typically, much smaller than either of the terms on the right-hand side.

2.5.1 The Momentum equation

The exact inviscid horizontal momentum equation is

$$(\tilde{\rho} + \rho') \frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_z \delta p. \quad (2.130)$$

Neglecting ρ' where it appears with $\tilde{\rho}$ leads to

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_h \phi, \quad (2.131)$$

where $\phi = \delta p / \bar{\rho}$, which is obviously similar to the corresponding equation in the Boussinesq approximation.

The vertical component of the inviscid momentum equation is, without approximation,

$$\begin{aligned} (\bar{\rho} + \delta\rho) \frac{Dw}{Dt} &= -\frac{\partial \tilde{p}}{\partial z} - \frac{\partial \delta p}{\partial z} - g\bar{\rho} - g\delta\rho \\ &= -\frac{\partial \delta p}{\partial z} - g\delta\rho. \end{aligned} \quad (2.132)$$

Neglecting $\delta\rho$ on the left-hand-side we obtain

$$\begin{aligned} \frac{Dw}{Dt} &= -\frac{1}{\bar{\rho}} \frac{\partial \delta p}{\partial z} - g \frac{\delta\rho}{\bar{\rho}} \\ &= -\frac{\partial}{\partial z} \left(\frac{\delta p}{\bar{\rho}} \right) - \frac{\delta p}{\bar{\rho}^2} \frac{\partial \bar{\rho}}{\partial z} - g \frac{\delta\rho}{\bar{\rho}} \end{aligned} \quad (2.133)$$

This is not a useful form for a gaseous atmosphere, since the variation of the mean density cannot be ignored. However, we may eliminate $\delta\rho$ in favour of δs using (2.128) to give

$$\frac{Dw}{Dt} = g\delta s - \frac{\partial}{\partial z} \left(\frac{\delta p}{\bar{\rho}} \right) - \frac{g}{\gamma} \frac{\delta p}{\bar{p}} - \frac{\delta p}{\bar{\rho}^2} \frac{\partial \bar{\rho}}{\partial z} \quad (2.134)$$

and using (2.129) gives

$$\frac{Dw}{Dt} = g\delta s - \frac{\partial}{\partial z} \left(\frac{\delta p}{\bar{\rho}} \right) + \frac{d\tilde{s}}{dz} \frac{\delta p}{\bar{\rho}}. \quad (2.135)$$

What have these manipulations gained us? Two things:

- (i) The gravitational term now involves δs rather than $\delta\rho$ which enables a more direct connection with the thermodynamic equation.
- (ii) The potential temperature scale height (~ 100 km) in the atmosphere is much larger than the density scale height (~ 10 km), and so the last term in (2.135) is small.

The second item thus suggests that we choose our reference state to be one of constant potential temperature (see also problem 2.17). The term $d\tilde{s}/dz$ then vanishes and the vertical momentum equation becomes

$$\boxed{\frac{Dw}{Dt} = g\delta s - \frac{\partial \phi}{\partial z}}. \quad (2.136)$$

where $\phi = \delta p / \tilde{\rho}$, $\delta s = \delta \theta / \tilde{\theta}$ and $\tilde{\theta} = \theta_0$, a constant. If we define a buoyancy by $b_a = g \delta s = g \delta \theta / \tilde{\theta}$, then (2.131) and (2.136) have the same form as the Boussinesq momentum equations, but with different definitions of b and ϕ .

Note that although we have defined the reference components of the pressure and density field ($\tilde{p}$ and $\tilde{\rho}$) to be hydrostatically related, it is not necessary to assume that the dynamical fields δp and $\delta \rho$ are in hydrostatic balance.

2.5.2 Mass conservation

Using (2.124b) the mass conservation equation may be written, without approximation, as

$$\frac{\partial \delta \rho}{\partial t} + \nabla \cdot ((\tilde{\rho} + \delta \rho) \mathbf{v}) = 0. \quad (2.137)$$

We neglect $\delta \rho$ where it appears with $\tilde{\rho}$ in the divergence term. Further, the local time derivative will be small if time itself is scaled advectively (i.e., $T \sim L/U$ and sound waves do not dominate), giving

$$\nabla \cdot \mathbf{u} + \frac{1}{\tilde{\rho}} \frac{\partial}{\partial z} (\tilde{\rho} w) = 0 \quad (2.138)$$

It is here that the eponymous ‘anelastic approximation’ arises: the elastic compressibility of the fluid is neglected, and this serves to eliminate sound waves. For reference, in spherical coordinates the equation is

$$\frac{1}{a \cos \vartheta} \frac{\partial u}{\partial \lambda} + \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (v \cos \vartheta) + \frac{1}{\tilde{\rho}} \frac{\partial (w \tilde{\rho})}{\partial z} = 0. \quad (2.139)$$

In the Boussinesq approximation the term $\tilde{\rho}$ becomes ρ_0 , a constant, and so cancels from the last term.

Thermodynamic equation

The thermodynamic equation for an ideal gas may be written

$$\frac{D \ln \theta}{Dt} = \frac{\dot{Q}}{T c_p} \quad (2.140)$$

where $s = \ln \theta$. In the anelastic equations, $\theta = \tilde{\theta} + \delta \theta$ where $\tilde{\theta}$ is constant, and the thermodynamic equation is, with $\dot{Q} = 0$,

$$\frac{D \delta s}{Dt} = 0. \quad (2.141)$$

Summarizing, the complete set of anelastic equations, with rotation but with no dissipation or diabatic terms, is

$$\boxed{\begin{aligned}\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} &= \mathbf{k}b_a - \nabla\phi \\ \frac{Db_a}{Dt} &= 0 \\ \nabla \cdot (\tilde{\rho}\mathbf{v}) &= 0\end{aligned}} \quad (2.142)$$

where $b_a = g\delta s = g\delta\theta/\tilde{\theta}$. The main difference between the anelastic and Boussinesq sets of equations is in the mass continuity equation, and when $\tilde{\rho} = \rho_0 = \text{constant}$ the two equation sets are formally identical. However, whereas the Boussinesq approximation is a very good one for ocean dynamics, the anelastic approximation is much less so for large-scale atmosphere flow: the constancy of the reference potential temperature state is not a particularly good approximation for large-scale flow, and so the deviations in density from its reference profile are not especially small leading to inaccuracies in the momentum equation. Nevertheless, the anelastic equations have been used very productively in limited area ‘large-eddy-simulations’ where one does not wish to make the hydrostatic approximation but where sound waves are unimportant.⁶ They also provide a good jumping-off point for the still simpler models that will be considered in the next chapter.

2.5.3 * Energetics of the anelastic equations

Conservation of energy follows in much the same way as for the Boussinesq equations, except that $\tilde{\rho}$ enters. Take the dot product of (2.142a) with $\tilde{\rho}\mathbf{v}$ to obtain

$$\tilde{\rho} \frac{D}{Dt} \left(\frac{1}{2} \mathbf{v}^2 \right) = -\nabla \cdot (\phi \tilde{\rho} \mathbf{v}) + b_a \tilde{\rho} w \quad (2.143)$$

Now, define a potential $\Phi(z)$ such that $\nabla\Phi = -\mathbf{k}$, and so

$$\tilde{\rho} \frac{D\Phi}{Dt} = -w\tilde{\rho}. \quad (2.144)$$

Thus, with the thermodynamic equation (2.142b),

$$\tilde{\rho} \frac{D(b_a\Phi)}{Dt} = -wb_a\tilde{\rho}. \quad (2.145)$$

Combining this with (2.143) gives

$$\tilde{\rho} \frac{D}{Dt} \left(\frac{1}{2} \mathbf{v}^2 + b_a\Phi \right) = -\nabla \cdot (\phi \tilde{\rho} \mathbf{v}) \quad (2.146)$$

or, expanding the material derivative,

$$\frac{\partial}{\partial t} [\tilde{\rho} (\frac{1}{2} \mathbf{v}^2 + b_a \Phi)] + \nabla \cdot [\tilde{\rho} \mathbf{v} (\frac{1}{2} \mathbf{v}^2 + b_a \Phi + \phi)] = 0. \quad (2.147)$$

This equation has the form

$$\frac{\partial E}{\partial t} + \nabla \cdot [\mathbf{v}(E + \tilde{\rho} \phi)] = 0 \quad (2.148)$$

where $E = \tilde{\rho}(\mathbf{v}^2/2 + b_a \Phi)$ is the energy density of the flow. The integral of this over a closed domain evidently vanishes. The term $b_a \Phi$ is the potential energy of the system. However, as in the Boussinesq equations the internal energy does not explicitly enter and interpretation of the energetics requires care.

2.6 CHANGING VERTICAL CO-ORDINATE

Although using z as a vertical co-ordinate seems and is natural in our Cartesian worldview, it is not the only choice nor in many circumstances is it the most useful choice. Any variable that has a one-to-one correspondence with z in the vertical, so any variable that varies monotonically with z , could be used, and we will consider two possibilities, pressure and entropy. In the atmosphere pressure almost always falls monotonically with height, and using it instead of z provides a useful simplification of the mass conservation and geostrophic relations, as well as a more direct connection with observations, which are often taken at fixed values of pressure. (In the ocean pressure is almost the same as height, because density is almost constant.) Entropy seems an exotic vertical co-ordinate, but in practice this means using potential temperature in the atmosphere or buoyancy (density) in the ocean; such co-ordinate systems are generically called *isentropic co-ordinates*, and sometimes *isopycnal co-ordinates* if density is used. They are useful because in an adiabatic process entropy is conserved, and the thermodynamic equation then becomes a statement for the conservation of the mass of fluid with a given value of potential temperature or density. The flow of both the atmosphere and ocean is largely along isentropic surfaces, and the realization of this, most easily expressed in isentropic co-ordinates, is of great help in understanding large-scale flows.

2.6.1 Pressure co-ordinates

The primitive equations of motion for an ideal gas can be written,

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\frac{1}{\rho} \nabla p, \quad (2.149a)$$

$$\frac{\partial p}{\partial z} = -\rho g, \quad (2.149b)$$

$$\frac{D\theta}{Dt} = 0, \quad (2.149c)$$

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0 \quad (2.149d)$$

where $p = \rho RT$ and $\theta = T (p_s / p)^{R/c_p}$. These equations can be put into a form very similar to the Boussinesq equations by transforming from Cartesian [i.e., (x, y, z)] to *pressure coordinates*, (x, y, p) . The analog of the vertical velocity is $\omega \equiv Dp/Dt$, and the advective derivative itself is given by

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla_p + \omega \frac{\partial}{\partial p}. \quad (2.150)$$

The horizontal and time derivatives are taken at constant pressure. However, x and y are still purely horizontal coordinates, and $\mathbf{u} = u\mathbf{i} + v\mathbf{j}$ is still a strictly horizontal velocity, perpendicular to the vertical (z) axis. The operator D/Dt is of course the same in pressure or height co-ordinates because it is simply the total derivative of some property of a fluid parcel. However, the individual terms comprising it in general differ between height and pressure co-ordinates.

To obtain an expression for the pressure force, first consider a general vertical coordinate, ξ . Then the chain rule gives

$$\left. \frac{\partial}{\partial x} \right|_{\xi} = \left. \frac{\partial}{\partial x} \right|_z + \left. \frac{\partial z}{\partial x} \right|_{\xi} \left. \frac{\partial}{\partial z} \right|_{\xi} \quad (2.151)$$

Now let $\xi = p$ and apply the relationship to p itself to give

$$0 = \left. \frac{\partial p}{\partial x} \right|_z + \left. \frac{\partial z}{\partial x} \right|_p \left. \frac{\partial p}{\partial z} \right|_p \quad (2.152)$$

which, using the hydrostatic relationship, gives

$$\left. \frac{\partial p}{\partial x} \right|_z = \rho \left. \frac{\partial \Phi}{\partial x} \right|_p \quad (2.153)$$

where $\Phi = gz$ is the *geopotential*. Thus, the horizontal pressure force in the momentum equations is

$$\frac{1}{\rho} \nabla_z p = \nabla_p \Phi \quad (2.154)$$

where the subscripts on the gradient operator indicate that the horizontal derivatives are taken at constant z or constant p .

Mass continuity

The mass conservation equation simplifies attractively in pressure co-ordinates, if the hydrostatic approximation is used. Recall that the mass conservation equation can be derived from the Lagrangian form

$$\frac{D}{Dt}\rho\delta V = 0 \quad (2.155)$$

where $\delta V = \delta x\delta y\delta z$ is a volume element. But by the hydrostatic relationship $\rho\delta z = (1/g)\delta p$ and thus

$$\frac{D}{Dt}(\delta x\delta y\delta p) = 0. \quad (2.156)$$

This is completely analogous to the expression for the Lagrangian conservation of volume in an incompressible fluid, (1.17). Thus, without further ado, we write the mass conservation in pressure co-ordinates as

$$\nabla_p \cdot \mathbf{u} + \frac{\partial \omega}{\partial p} = 0 \quad (2.157)$$

where the horizontal derivative is taken at constant pressure. (See also problem 2.19.) The primitive equations in pressure co-ordinates are thus:

$$\begin{aligned} \frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} &= -\nabla_p \Phi \\ \frac{\partial \Phi}{\partial p} &= -\alpha \\ \frac{D\theta}{Dt} &= 0 \\ \nabla_p \cdot \mathbf{u} + \frac{\partial \omega}{\partial p} &= 0 \end{aligned}$$

(2.158)

where $\alpha = 1/\rho$ and D/Dt is given by (2.150). The equation set is completed with the addition of the ideal gas equation and the definition of potential temperature. These are not quite isomorphic to the Boussinesq equations, because the hydrostatic equation is $\partial\phi/\partial p = -\alpha = -(\theta R/p_s)(p/p_s)^\gamma$ and not $\partial\phi/\partial p = -\theta$.

The main practical difficulty with these equations is the lower boundary condition. Using

$$w \equiv \frac{Dz}{Dt} = \frac{\partial z}{\partial t} + \mathbf{u} \cdot \nabla_p z + \omega \frac{\partial z}{\partial p}, \quad (2.159)$$

and (2.158), the boundary condition of $w = 0$ at $z = z_s$ becomes

$$\frac{\partial \Phi}{\partial t} + \mathbf{u} \cdot \nabla_p \Phi - \alpha \omega = 0 \quad (2.160)$$

at $p(x, y, z_s, t)$. In theoretical studies, it is common to assume that the lower boundary is in fact a constant pressure surface and simply assume that $\omega = 0$, or sometimes the condition $\omega = -\alpha^{-1} \partial \Phi / \partial t$ is used. For realistic studies (with general circulation models, say) the fact that the level $z = 0$ is not a coordinate surface must be properly accounted for. For this reason, and especially if the lower boundary is uneven because of the presence of topography, so-called *sigma coordinates* are sometimes used, in which the vertical coordinate is chosen so that the lower boundary is a co-ordinate surface. Sigma co-ordinates may use height itself as a measure of displacement (typical in oceanic applications) or use pressure (typical in atmospheric applications). In the latter case the vertical co-ordinate is $\sigma = p/p_s$ where $p_s(x, y, t)$ is the surface pressure. The difficulty of applying (2.160) is replaced by a prognostic equation for the surface pressure, which is derived from the mass conservation equation (problem 20).

The thermal wind equation takes a simple form in pressure co-ordinates. The geostrophic wind satisfies

$$\mathbf{f} \times \mathbf{u}_g = -\nabla_p \Phi \quad (2.161)$$

Taking the vertical derivative (i.e., derivative with respect to pressure) and using the hydrostatic relationship gives

$$\mathbf{f} \times \frac{\partial \mathbf{u}_g}{\partial p} = \nabla_p \alpha = \frac{R}{p} \nabla_p T \quad (2.162)$$

where the last equality follows using the ideal gas equation and because the horizontal derivative is at constant pressure. Thus,

$$\frac{\partial \mathbf{u}_g}{\partial \ln p} = -\frac{R}{f} \mathbf{k} \times \nabla_p T = -\frac{R}{f} \nabla_p^\perp T \quad (2.163)$$

Log-pressure co-ordinates

A variant of pressure co-ordinates is *log-pressure* co-ordinates, in which the vertical co-ordinate is $Z = -H \ln(p/p^*)$ where p^* is a reference pressure (say 1000 mb) and H a constant (for example the scale height RT_s/g) so that Z has units of length. The ‘vertical velocity’ for the system is now

$$W \equiv \frac{DZ}{Dt} \quad (2.164)$$

and the advective derivative is now

$$\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla_p + W \frac{\partial}{\partial Z}, \quad (2.165)$$

It is straightforward to show (problem 2.21) that the primitive equations of motion in this co-ordinate set are:

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla\Phi \quad (2.166a)$$

$$\frac{\partial\Phi}{\partial Z} = \frac{RT}{H} \quad (2.166b)$$

$$\frac{D\theta}{Dt} = 0 \quad (2.166c)$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial W}{\partial Z} - \frac{W}{H} = 0. \quad (2.166d)$$

2.6.2 Isentropic co-ordinates

We now consider potential temperature or potential density as a vertical co-ordinate. Although a seemingly odd thing to do, this is perfectly acceptable if these variables have a monotonic profile, and for adiabatic flow in particular the resulting equations of motion have an attractive form. Thus, for example, we use potential temperature θ as the vertical co-ordinate, where the thermodynamic equation is $D\theta/Dt = \dot{\theta}$, and transform the equations from (x, y, z) co-ordinates to (x, y, θ) co-ordinates. The material derivative in this co-ordinate system is

$$\begin{aligned} \frac{D}{Dt} &= \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} \Big|_{\theta} + v \frac{\partial}{\partial y} \Big|_{\theta} + \frac{D\theta}{Dt} \frac{\partial}{\partial \theta} \\ &= \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla_{\theta} + \dot{\theta} \frac{\partial}{\partial \theta} \end{aligned} \quad (2.167)$$

where the last term is zero for adiabatic flow, and the two-dimensional velocity $\mathbf{u} \equiv (u, v)$ is parallel to the isentropes.

A hydrostatic Boussinesq fluid

In the Boussinesq equations (the first three equations on page 82) the buoyancy is the relevant thermodynamic variable. With hydrostatic balance the horizontal and vertical momentum equations are, in height co-ordinates,

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla\phi, \quad b = \phi_z \quad (2.168)$$

where we use the symbol b for the buoyancy, the variable analogous to the potential temperature θ of an ideal gas. The adiabatic thermodynamic equation is

$$\frac{Db}{Dt} = 0, \quad (2.169)$$

and because $b = -g\delta\rho/\rho_0$, isentropic co-ordinates are the same as isopycnal co-ordinates.

Using (2.151) the horizontal pressure gradient may be transformed to isentropic co-ordinates

$$\left. \frac{\partial\phi}{\partial x} \right|_z = \left. \frac{\partial\phi}{\partial x} \right|_b - \left. \frac{\partial z}{\partial x} \right|_b \frac{\partial\phi}{\partial z} = \left. \frac{\partial\phi}{\partial x} \right|_b - b \left. \frac{\partial z}{\partial x} \right|_b = \left. \frac{\partial M}{\partial x} \right|_b \quad (2.170)$$

where

$$M \equiv \phi - zb. \quad (2.171)$$

Thus, the horizontal momentum equation becomes

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_b M. \quad (2.172)$$

where the material derivative is given by (2.167). Using (2.171) the hydrostatic equation becomes

$$\frac{\partial M}{\partial b} = -z. \quad (2.173)$$

The mass continuity equation may be derived by noting that the mass element may be written as

$$\delta m = \frac{\partial z}{\partial b} \delta b \delta x \delta y. \quad (2.174)$$

The mass continuity equation, $Dm/Dt = 0$, becomes

$$\frac{D}{Dt} \frac{\partial z}{\partial b} + \frac{\partial z}{\partial b} \nabla_3 \cdot \mathbf{v} = 0 \quad (2.175)$$

where $\nabla_3 \cdot \mathbf{v}$ is the three dimensional derivative of the velocity in isentropic co-ordinates. Thus,

$$\frac{D\sigma}{Dt} + \sigma \nabla_b \cdot \mathbf{u} = -\sigma \frac{\partial \dot{b}}{\partial b} \quad (2.176)$$

where $\sigma \equiv \partial z / \partial b$ is a measure of the thickness between two isentropic surfaces. Equations (2.172), (2.173) and (2.176) comprise a closed set.

A hydrostatic ideal gas

Deriving the equations of motion for this system requires a little more work than in the Boussinesq case but the idea is the same. For an ideal gas in hydrostatic balance we have, using (1.120),

$$\frac{\delta\theta}{\theta} = \frac{\delta T}{T} - \kappa \frac{\delta p}{p} = \frac{\delta T}{T} + \frac{\delta\Phi}{c_p T} = \frac{1}{c_p T} \delta M \quad (2.177)$$

where $M \equiv c_p T + \Phi$ is the ‘Montgomery potential’, and also the dry static energy. (We use some of the same symbols as in the Boussinesq case to facilitate comparison, but their meanings are slightly different.) From this

$$\frac{\partial M}{\partial \theta} = \Pi \quad (2.178)$$

where $\Pi \equiv c_p T / \theta = c_p (p / p_r)^{R/c_p}$ is the ‘Exner function.’ Equation (2.178) represents the hydrostatic relation in isentropic co-ordinates. Note also that $M = \theta \Pi + \Phi$.

To obtain the an appropriate form for the horizontal pressure gradient first note that, in the usual height co-ordinates, it is given by

$$\frac{1}{\rho} \nabla_z p = \theta \nabla_z \Pi \quad (2.179)$$

where $\Pi = c_p T / \theta$. Using (2.151) gives

$$\theta \nabla_z \Pi = \theta \nabla_\theta \Pi - \frac{\theta}{g} \frac{\partial \Pi}{\partial z} \nabla_\theta \Phi. \quad (2.180)$$

Then, using the definition of Π and the hydrostatic approximation to help evaluate the vertical derivative, we obtain

$$\frac{1}{\rho} \nabla_z p = c_p \nabla_\theta T + \nabla_\theta \Phi = \nabla_\theta M. \quad (2.181)$$

Thus, the horizontal momentum equation is

$$\frac{D\mathbf{v}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_\theta M. \quad (2.182)$$

As in the Boussinesq case the mass continuity equation may be derived by noting that the mass element may be written as

$$\delta m = \frac{\partial p}{\partial \theta} \delta \theta \delta x \delta y. \quad (2.183)$$

The mass continuity equation, $Dm/Dt = 0$, becomes

$$\frac{D}{Dt} \frac{\partial p}{\partial \theta} + \frac{\partial p}{\partial \theta} \nabla_3 \cdot \mathbf{v} = 0 \quad (2.184)$$

or

$$\frac{D\sigma}{Dt} + \sigma \nabla_\theta \cdot \mathbf{u} = -\sigma \frac{\partial \dot{\theta}}{\partial \theta} \quad (2.185)$$

where now $\sigma \equiv \partial p / \partial \theta$ is a measure of the (pressure) thickness between two isentropic surfaces. Equations (2.178), (2.182) and (2.185) form a closed set, analogous to (2.173), (2.172) and (2.176).

Analogy to shallow water equations

The equations of motion in isentropic co-ordinates have an obvious analogy with the shallow water equations, and we may think of the shallow water equations to be a finite-difference representation of the primitive equations written in isentropic co-ordinates, or think of the latter as the continuous limit of the shallow water equations as the number of layers increases. For example, consider a two-isentropic-level representation of (2.178), (2.182) and (2.185), in which the lower boundary is an isentrope. A natural finite differencing gives

$$-M_1 = -z_0 \Delta \theta_0 \quad (2.186a)$$

$$M_1 - M_2 = -z_1 \Delta \theta_1 \quad (2.186b)$$

and the momentum equations for each layer become

$$\frac{D\mathbf{u}_1}{Dt} + \mathbf{f} \times \mathbf{u}_1 = -\Delta \theta_0 \nabla z_0 \quad (2.187a)$$

$$\frac{D\mathbf{u}_2}{Dt} + \mathbf{f} \times \mathbf{u}_2 = -\Delta \theta_0 \nabla z_0 - \Delta \theta_1 \nabla z_1 \quad (2.187b)$$

which, together with the mass continuity equation for each level, are just like the two-layer shallow water equations (5.53).

2.7 SCALING FOR HYDROSTATIC BALANCE

Consider the relative sizes of terms in (2.81c),

$$\frac{W}{T} + \frac{UW}{L} + \frac{W^2}{H} + \Omega U \sim \frac{1}{\rho} \frac{\partial p}{\partial z} - g \quad (2.188)$$

For most large-scale motion in the atmosphere and ocean the terms on the right-hand side are orders of magnitude larger than those on the left, and therefore must be approximately equal. Explicitly, suppose $W \sim 1 \text{ cm s}^{-1}$, $L \sim 10^5 \text{ m}$, $H \sim 10^3 \text{ m}$, $U \sim 10 \text{ m s}^{-1}$, $T = L/U$. Then by substituting into (2.188) it would seem that the pressure term is the only one which could balance the gravitational term, and we appear to be led to the following vertical momentum equation,

$$\frac{\partial p}{\partial z} = -\rho g. \quad (2.189)$$

otherwise known as *hydrostatic balance*.

However, this reasoning is misleading! Let us suppose that the density is constant, ρ_0 , and then we can write the pressure as

$$p(x, y, z, t) = p_0(z) + p'(x, y, z, t) \quad (2.190)$$

where

$$\frac{\partial p_0}{\partial z} \equiv -\rho_0 g. \quad (2.191)$$

That is, p_0 and ρ_0 are in hydrostatic balance. The vertical momentum equation becomes, without approximation,

$$\frac{Dw}{Dt} = -\frac{1}{\rho_0} \frac{\partial p'}{\partial z} + \nu \nabla^2 w. \quad (2.192)$$

Thus, for constant density fluids, the gravitational term has no dynamical effect: there is no buoyancy force, and the pressure term in the horizontal momentum equations can be replaced by p' . Hydrostatic balance, and in particular (2.191), is certainly not an appropriate vertical momentum equation in this case. If the fluid is stratified, we should therefore subtract off the hydrostatic pressure associated with the mean density before we can determine whether hydrostasy is a useful dynamical approximation. This is automatic in the Boussinesq equations, where the vertical momentum equation is

$$\frac{Dw}{Dt} = -\frac{\partial \phi}{\partial z} + b. \quad (2.193)$$

and the hydrostatic balance of the basic state is already subtracted out. In the more general equation,

$$\frac{Dw}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g. \quad (2.194)$$

it is apparent that we should compare the advective term on the left-hand side with the pressure variations arising from horizontal flow.

2.7.1 The aspect ratio

In a non-rotating Boussinesq fluid the horizontal momentum equation $D\mathbf{u}/Dt = -\nabla\phi$ implies the scaling

$$\phi \sim U^2 \quad (2.195)$$

If we now use mass conservation to scale vertical velocity, so that

$$\frac{W}{H} \sim \frac{U}{L} \quad (2.196)$$

then the advective terms in vertical momentum equation all scale as

$$\frac{Dw}{Dt} \sim \frac{UW}{L} = \frac{U^2 H}{L^2} \quad (2.197)$$

Using (2.195) and (2.197) the ratio of the advective term to the pressure gradient term in the vertical momentum equations then scales as

$$\frac{|Dw/Dt|}{|(1/\rho)\partial\phi/\partial z|} \sim \frac{U^2 H/L^2}{U^2/H} \sim \left(\frac{H}{L}\right)^2. \quad (2.198)$$

Thus, the condition for hydrostasy is:

$$\alpha^2 \equiv \left(\frac{H}{L} \right)^2 \ll 1, \quad (2.199)$$

in which case the advective term in the vertical momentum is small. Thus, hydrostasy is an *aspect ratio approximation*; it holds when the aspect ratio $\alpha \equiv H/L$ is small.

Effects of rotation

In the presence of rapid rotation the scaling for pressure differs: geostrophic-balance scaling $\phi \sim \rho fUL$ holds and we obtain

$$\frac{|Dw/Dt|}{|(1/\rho)\partial p/\partial z|} \sim \frac{WH}{fL^2} = Ro \frac{WH}{UL}. \quad (2.200)$$

Furthermore, in low Rossby number flow the horizontal flow is near non-divergent, so that $W \sim UH/L$ is an overestimate of the magnitude of the vertical velocity. Let us thus write $W \sim \epsilon UH/L$ where $\epsilon \ll 1$. (Later on we'll see that ϵ is in fact related to the Rossby number.) Using this in (2.200) gives

$$\frac{|Dw/Dt|}{|(1/\rho)\partial p/\partial z|} \sim \epsilon Ro \left(\frac{H}{L} \right)^2, \quad (2.201)$$

which is evidently very small for the large-scale flow. In particular, because $\epsilon Ro \ll 1$, rotation tends to weaken further the conditions needed for hydrostasy; that is, a rapidly rotating fluid is more likely to be in hydrostatic balance than a non-rotating fluid, other conditions being equal.

** Effects of stratification*

The above results say little about the dynamics that might give rise to hydrostatic or non-hydrostatic flow. Our intuition suggests that hydrostatic balance might be questionable in small-scale convective activity where the vertical velocity is high, and most applicable in highly stratified flow for then vertical velocity is diminished. But in highly stratified flow the estimate of W from (2.197) may too restrictive, so here we give another estimate of the vertical velocity that takes explicit account of known stratification and use it to derive a slightly weaker condition for hydrostasy.

To obtain an estimate of the vertical velocity, we use the Boussinesq approximation with a vertical momentum equation:

$$\frac{Dw}{Dt} = -\frac{\partial \phi}{\partial z} + b. \quad (2.202)$$

For hydrostatic balance we demand

$$\frac{UW}{L} \ll \Delta b \quad (2.203)$$

where Δb is the scaling magnitude for b . It is only the horizontal variations of b that matter, so let us assume that the magnitude for w is given by the thermodynamic equation written in the form

$$\frac{Db'}{Dt} + N^2 w = 0, \quad (2.204)$$

where the mean stratification (giving N^2) is given — determined, for example, by the larger scale circulation. Then,

$$W \sim \frac{U \Delta b}{L N^2} \quad (2.205)$$

This implies that $W/H = (U/L) \Delta b / (H N^2)$, and, assuming that $\Delta b/H \ll N^2$, that $W/H \ll U/L$ and horizontal advection dominates vertical advection of the buoyancy anomaly. Using (2.205), the condition for hydrostatic balance, (2.203), becomes

$$\boxed{\frac{U^2}{L^2 N^2} \ll 1}. \quad (2.206)$$

Since the buoyancy frequency N is a measure of stratification (the higher the frequency, the more stratified the fluid), (2.206) formalizes our intuitive expectation that the more stratified a fluid the more vertical motion is suppressed and the more likely hydrostatic balance is to hold.

The *Froude number* may be defined by

$$F \equiv \frac{U}{NH}. \quad (2.207)$$

Then

$$\frac{U^2}{L^2 N^2} = F^2 \frac{H^2}{L^2}, \quad (2.208)$$

and using (2.206) the hydrostatic condition is

$$\boxed{F^2 \alpha^2 \ll 1}, \quad (2.209)$$

where $\alpha = H/L$ is the aspect ratio of the motion. Thus, for a given Froude number a small aspect ratio will favor hydrostasy. This derivation differs from that leading to (2.199) in its use of the thermodynamic equation, rather than the mass conservation equation, to give an estimate of the vertical velocity, and this may be appropriate in regions of convective activity.

Why bother with any of this scaling? Why not just say that hydrostatic balance holds when $|Dw/Dt| \ll |b|$? One reason is that don't really have a good idea of the value of W from direct measurements, and it may change significantly in different oceanic and atmospheric parameter regimes. On the other hand the Froude number and the aspect ratio are familiar nondimensional parameters with a wide applicability in other contexts, and which we can control in a laboratory setting or estimate more easily in the ocean or atmosphere. Still, as in most scaling theory, deciding which parameters are given and which should be derived is often a *choice*, rather than being set *a priori*.

Hydrostatic balance in waves

If the motion is predominantly wavelike, rather than advective, then the advective derivative scales like a frequency: $D/Dt \sim \omega$. Using the vertical momentum equation we then require

$$\omega w \ll b, \quad (2.210)$$

and the thermodynamic equation has the scaling

$$\omega b \sim N^2 w. \quad (2.211)$$

These together demand that

$$\omega^2 \ll N^2 \quad (2.212)$$

for hydrostatic balance to hold. Buoyancy oscillations have $\omega \approx N$ and are essentially non-hydrostatic.

Oceanic applicability

For the large scale ocean circulation, let $N \sim 10^{-2} \text{ s}^{-1}$, $U \sim 0.1 \text{ m s}^{-1}$ and $H \sim 1 \text{ km}$. Then

$$F = \frac{U}{NH} \sim 10^{-2} \ll 1 \quad (2.213)$$

Thus, $F^2 \gamma^2 \ll 1$ even for unit aspect-ratio motion. For gyre scale flow $L \sim 10^6 \text{ m}$ and $F^2 \gamma^2 \sim 10^{-10}$ and hydrostatic balance is a very good approximation indeed.

For intense convection, for example in the Labrador Sea, the hydrostatic approximation may be less appropriate. The intense descending plumes may have an aspect ratio (H/L) of one or greater and the stratification is very weak: the hydrostatic condition is then simply a requirement that stratification is sufficiently strong that the Froude number is small. Representative orders of magnitude are $U \sim W \sim 0.1 \text{ m s}^{-1}$, $H \sim 1 \text{ km}$ and $N \sim 10^{-3} \text{ s}^{-1}$ – 10^{-4} s^{-1} . For these values F ranges between 0.1 and 1, and at the upper end of this range hydrostatic balance is violated.

Atmospheric applicability

Similar considerations apply to the atmosphere, although the details differ. Over much of the troposphere $N \sim 10^{-2} \text{ s}^{-1}$ (in fact this is the ‘standard atmosphere’ value) so that with $U = 10 \text{ m s}^{-1}$ and $H = 1 \text{ km}$ for large-scale flow $F \sim 1$. Hydrostasy is then maintained because the aspect ratio H/L is much less than unity. For smaller scale atmospheric motion associated with fronts and, especially, convection, there can be little expectation that hydrostatic balance will be a good approximation.

2.8 STATIC INSTABILITY AND THE PARCEL METHOD

Considering the pressure and gravitational forces on a displaced parcel in a stratified fluid is the simplest and most direct way to see how oscillations — gravity waves — are produced in a fluid. Consider a fluid at rest in a constant gravitational field, and therefore in hydrostatic balance. Suppose that a small parcel of the fluid is adiabatically displaced upwards by the small distance δz , without altering the overall pressure field — that is, the fluid parcel instantly assumes the pressure of its environment. If after the displacement the parcel is lighter than its environment, it will accelerate upwards, because the upward pressure gradient force is now greater downwards gravity force on the parcel — that is, the parcel is *buoyant*, a consequence of Archimedes’ principle, and the fluid *statically unstable*. If on the other hand the fluid parcel finds itself heavier than its surroundings, the downward gravitational force will be greater than the upward pressure force and the fluid will sink back towards its original position and an oscillatory motion will develop. The equilibrium is *stable*. Using such simple ‘parcel’ arguments we will now develop criteria for the stability of the environmental profile.

2.8.1 An density-conserving fluid

Consider first the simple case of an incompressible fluid in which the density of the displaced parcel is conserved. If the mean profile is $\tilde{\rho}(z)$ parcel displacements are stable $\partial \tilde{\rho} / \partial z < 0$ and unstable if $\partial \tilde{\rho} / \partial z > 0$. The upward force (per unit volume) on the displaced parcel is

$$F = -g\delta\rho = -g(\rho - \rho_0) = g \frac{\partial \rho_0}{\partial z} \delta z. \quad (2.214)$$

and thus

$$\rho_0(z) \frac{\partial^2 \delta z}{\partial t^2} = g \frac{\partial \rho_0}{\partial z} \delta z \quad (2.215)$$

or

$$\frac{\partial^2 \delta z}{\partial t^2} = (g/\rho_0) \frac{\partial \tilde{\rho}}{\partial z} \delta z = -N^2 \delta z \quad (2.216)$$

where $N^2 = -(g/\rho_0)\partial\tilde{\rho}/\partial z$ is the *Brunt-Väisälä frequency*, or the *buoyancy frequency* — the frequency at which the fluid parcel oscillates.

2.8.2 A compressible fluid

Consider next the slightly more general case of an adiabatic displacement in which the density and pressure displacements are related by [see (1.123)]

$$\delta\rho = \frac{1}{c_s^2}\delta p. \quad (2.217)$$

But if the fluid is hydrostatic $\delta p = \rho g \delta z$ so that

$$\delta\rho = -\frac{1}{c_s^2}\rho g \delta z. \quad (2.218)$$

Thus, if a parcel is displaced adiabatically, its density changes according to (c.f., (1.125))

$$\left(\frac{\partial\rho}{\partial z}\right)_\eta = -\frac{\rho g}{c_s^2} \quad (2.219)$$

The parcel's density decreases with increasing height, because it is compressed less because pressure decreases as z increases.

The stability or otherwise of the column of fluid is determined by whether a parcel displaced a small distance is more or less dense than its surroundings. If a parcel is displaced a distance δz upwards then the density difference between it and its new surroundings is

$$\delta\rho = \left[\left(\frac{\partial\rho}{\partial z}\right)_\eta - \left(\frac{\partial\rho}{\partial z}\right)_{\text{env}} \right] \delta z \quad (2.220)$$

where $(\partial\rho/\partial z)_{\text{env}}$ is the environmental gradient. If the parcel is less dense than its surroundings it is unstable; thus, the condition for stability is

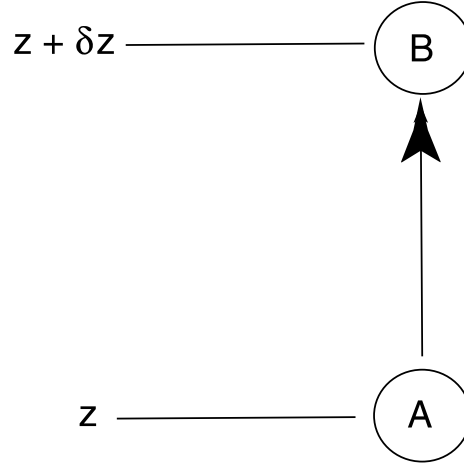
$$\text{Stability :} \quad -\left(\frac{\partial\rho}{\partial z}\right)_{\text{env}} > \frac{\rho g}{c_s^2} \quad (2.221a)$$

$$\text{Instability :} \quad -\left(\frac{\partial\rho}{\partial z}\right)_{\text{env}} < \frac{\rho g}{c_s^2}. \quad (2.221b)$$

Assume now that the environmental profile is in fact stable. Then the restoring force per unit mass is given by g times (2.220). Since the displacement is δz , the parcel will obey

$$\begin{aligned} \rho \frac{\partial^2 \delta z}{\partial t^2} &= -g \left[\left(\frac{\partial\rho}{\partial z}\right)_\eta - \left(\frac{\partial\rho}{\partial z}\right)_{\text{env}} \right] \delta z \\ &= g \left[\frac{\rho g}{c_s^2} + \left(\frac{\partial\rho}{\partial z}\right)_{\text{env}} \right] \delta z \end{aligned} \quad (2.222)$$

Figure 2.4 A parcel A is adiabatically displaced upward from level z to $z + \delta z$. If its density at $z + \delta z$ is greater than that of its surroundings (e.g., parcel B), the buoyancy force is negative and the displacement is stable. But if $\rho_A(z + \delta z) < \rho_B(z + \delta z)$ the parcel is less dense than its surroundings, and its upward displacement will continue; the displacement is unstable.



Since the right-hand-side is negative, we write

$$\rho \frac{\partial^2 \delta z}{\partial t^2} = -N^2 \delta z \quad (2.223)$$

where

$$N^2 = -g \left[\frac{g}{c_s^2} + \frac{1}{\rho} \left(\frac{\partial \rho}{\partial z} \right)_{\text{env}} \right]. \quad (2.224)$$

This expression holds for both liquids and gases, but for ideal gases a more convenient form will be derived in the next section. A good approximation for liquids is to use a reference value ρ_0 for the undifferentiated density in the denominator. Equation (2.223) is the equation for a simple harmonic oscillator, and the quantity N is known the *Brunt-Väisälä frequency*, or the *buoyancy frequency*. Typical values in the upper ocean where the density is changing most rapidly (the pycnocline — ‘pycno’ for density, ‘cline’ for changing) are about 0.01 s^{-1} , falling to 0.001 s^{-1} in the more homogeneous abyssal ocean. These correspond to periods ($2\pi/N$) of about 10 and 100 minutes, respectively. In the troposphere (the lowest several kilometers of the atmosphere, below the tropopause) N is also about 0.01 s^{-1} .

* *Using potential density*

Consider again a parcel that is displaced adiabatically a vertical distance δz from A (level z , say) to B (level $z + \delta z$; refer again to Fig. 2.4). The parcel preserves its potential density during the displacement which, the reader will recall, is the density that a parcel would have if moved adiabatically to some reference pressure. Thus, if we use the level $z + \delta z$ as the potential density reference level, then the *in situ* density of the parcel at $z + \delta z$ is equal to its potential density,

which in turn is just the potential density of the environment at z . The difference in *in situ* density between the parcel and the environment at $z + \delta z$, namely $\delta\rho$, is thus equal the difference between the potential density of the environment at z and at $z + \delta z$. Thus,

$$\delta\rho = -\frac{\partial\rho_\theta}{\partial z}\delta z. \quad (2.225)$$

If the right-hand-side is positive, the parcel is heavier than its surroundings and the displacement is stable. Thus, the condition for stability is just:

$$\text{Stability :} \quad \frac{\partial\rho_\theta}{\partial z} < 0 \quad (2.226a)$$

$$\text{Instability :} \quad \frac{\partial\rho_\theta}{\partial z} > 0 \quad (2.226b)$$

which looks the same as (2.221) except for the absence of quantity g/c_s^2 , and the gradient is that of the locally referenced environmental potential density. The equation of motion of the fluid parcel is determined by

$$\begin{aligned} \frac{\partial^2\delta z}{\partial t^2} &= \frac{g}{\rho} \left(\frac{\partial\rho_\theta}{\partial z} \right) \delta z \\ &= -N^2\delta z \end{aligned} \quad (2.227)$$

where

$$N^2 = -\frac{g}{\rho_\theta} \left(\frac{\partial\rho_\theta}{\partial z} \right)_{\text{env}}. \quad (2.228)$$

In this formula, we must use a *local* reference level for potential density. In the atmosphere $\rho_\theta = p_{\text{ref}}/(\theta R)$ and no difficulties arise — the buoyancy frequency is just

$$N^2 = \frac{g}{\theta} \left(\frac{\partial\theta}{\partial z} \right)_{\text{env}}, \quad (2.229)$$

which is easily computed from the temperature and pressure. In contrast, for seawater, no simple, accurate, analytic expression is available for computing static stability. If the ocean had no salt, then the potential density referenced to the surface would in fact be a measure of the stability of a fluid column.⁷ However, in the presence of salinity, the surface-referenced potential density is not necessarily even a measure of the sign of stability, because the coefficients of compressibility β_T and β_S vary in different ways with pressure. To see this, suppose two small neighbouring parcels (one just above the other, say) at the surface have the same potential density, but different salinities and temperatures. Displace them both adiabatically to the deep ocean. Although their potential densities are still equal, we can say little about their actual densities, and hence their stability relative to each other, without doing a detailed calculation because they will each have been compressed by different amounts.

* *Cabbeling*

Cabbeling is an instability that arises because of the nonlinear equation of state of seawater. From Fig. 1.3 we see that the contours are slightly convex, bowing upward, especially in the plot at sea level. Suppose we mix two parcels of water, each with the same density ($\sigma_\theta = 28$, say), but with different initial values of temperature and salinity. Then the resulting parcel of water will have a temperature and a salinity equal to the average of the two parcels, but its density will be *higher* than either of the two original parcels. In the appropriate circumstances such mixing may thus lead to a convective instability, and may be an important source of ‘bottom water’ formation in the Weddell Sea, off Antarctica.⁸

2.8.3 An ideal gas

The buoyancy frequency for an ideal gas is given by (2.229), but we briefly re-derive it directly, and then consider the effects of moisture.

A dry atmosphere

In a dry atmosphere potential density is conserved in adiabatic flow. Again we displace a parcel from level z to $z + \delta z$. The potential temperature difference between it and its surroundings is then

$$\delta\theta = - \left(\frac{\partial\theta}{\partial z} \right)_{\text{env}} \delta z. \quad (2.230)$$

where the subscript ‘env’ denotes the environmental gradient. Using the definition of potential temperature and the ideal gas equation we have

$$\theta = T \left(\frac{p_r}{p} \right)^{R/c_p} = \frac{P}{R\rho} \left(\frac{p_r}{p} \right)^{R/c_p}. \quad (2.231)$$

Since the parcel is at the same pressure as its surroundings, the density difference between the parcel and its surroundings is determined by

$$\frac{\delta\rho}{\rho} = -\frac{\delta\theta}{\theta} = \frac{1}{\theta} \frac{\partial\theta}{\partial z} \delta z \quad (2.232)$$

Thus, the parcel experiences a buoyancy force equal to $(g\rho/\theta)\delta\theta/\delta z$ and therefore obeys

$$\rho \frac{\partial^2 \delta z}{\partial t^2} = -\frac{\rho g}{\theta} \frac{\partial\theta}{\partial z} \delta z \quad (2.233)$$

or

$$\frac{\partial^2 \delta z}{\partial t^2} = -N^2 \delta z, \quad (2.234)$$

where

$$N^2 = \frac{g}{\theta} \frac{\partial \theta}{\partial z}. \quad (2.235)$$

Thus, for positive N^2 , a displaced parcel will oscillate around its initial position with a frequency N and period $2\pi/N$. For the troposphere the value of N is similar to that in the upper ocean, namely 10^{-2} s^{-1} , with a corresponding period of about 10 minutes. If N^2 is negative then the displacement is unstable, and convection will occur.

The (negative of the) rate of change of temperature in the vertical is known as the *lapse rate*, and the rate corresponding to $\partial\theta/\partial z = 0$ is the *adiabatic lapse rate*. Using $\theta = T(p_0/p)^{R/c_p}$ and $\partial p/\partial z = -\rho g$ we easily find

$$\frac{\partial T}{\partial z} = \frac{T}{\theta} \frac{\partial \theta}{\partial z} - \frac{g}{c_p}, \quad (2.236)$$

so that the adiabatic lapse rate is g/c_p , as in (1.141). This is often denoted Γ_d , where the subscript d , for dry, indicates that the effects of moisture are not included. It is about 10 K km^{-1} . The conditions for static stability are thus:

$$\text{Stability :} \quad \frac{\partial \theta}{\partial z} > 0; \quad -\frac{\partial T}{\partial z} < \Gamma_d \equiv \frac{g}{c_p}, \quad (2.237a)$$

$$\text{Instability :} \quad \frac{\partial \theta}{\partial z} < 0; \quad -\frac{\partial T}{\partial z} > \Gamma_d \equiv \frac{g}{c_p}. \quad (2.237b)$$

It turns out that the atmosphere is in fact generally stable by the criteria (2.237): the observed lapse rate, corresponding to an observed buoyancy frequency of about 10^{-2} s^{-1} , is often about 7 K km^{-1} , compared to a dry adiabatic lapse rate of about 10 K km^{-1} . Why the discrepancy? One reason, particularly important in the tropics, is that the atmosphere contains water vapour as we now see the effects of this.

* Effects of water vapour on an ideal gas

A given volume of air can contain only so much water vapour before becoming saturated, at which stage the water vapour condenses. The warmer a parcel of air is, the more water vapour it can hold — indeed, the partial pressure of the water vapour at saturation, e_s , is given as function of temperature by the Clausius-Clapeyron equation

$$\frac{de_s}{dT} = \frac{L_c e_s}{R_v T^2} \quad (2.238)$$

where L_c is the latent heat of condensation or vapourization and R_v is the gas constant for water vapour. If a parcel is forced to rise adiabatically it will cool, and

at some height (known as the ‘lifting condensation level’) the parcel will become saturated and any further ascent will cause the water vapour to condense. This releases latent heat, which causes the parcel’s temperature, and buoyancy, to increase. The parcel thus rises further, causing more water vapour to condense, and so on, and the consequence of this is that an environmental profile that is stable if dry may be unstable if saturated. Let us now derive an expression for the lapse rate of a saturated parcel that is ascending adiabatically apart from the affects of condensing water vapour.

Let w denote the mass of water vapour per unit mass of dry air, the mixing ratio, and let w_s be the saturation mixing ratio. ($w_s = \alpha e_s / (p - e_s) \approx \alpha e_s / p$ where $\alpha = 0.62$, the ratio of the mass of a water molecule to one of dry air.) The diabatic heating associated with condensation is then given by

$$Q_{\text{cond}} = -L_c \frac{Dw_s}{Dt} \quad (2.239)$$

so that the thermodynamic equation is

$$c_p \frac{D \ln \theta}{Dt} = -\frac{L_c}{T} \frac{Dw_s}{Dt}, \quad (2.240)$$

or, in terms of p and T

$$c_p \frac{D \ln T}{Dt} - R \frac{D \ln P}{Dt} = -\frac{L_c}{T} \frac{Dw_s}{Dt}. \quad (2.241)$$

If a parcel is ascending, then this becomes

$$\frac{d \ln T}{dz} - \frac{R}{c_p} \frac{d \ln p}{dz} = -\frac{L_c}{T c_p} \frac{dw_s}{dz} \quad (2.242)$$

and, using the hydrostatic relationship and the fact that w_s is a function of T and, more weakly, p ,

$$\frac{dT}{dz} + \frac{g}{c_p} = -\frac{L_c}{c_p} \left[\left(\frac{\partial w_s}{\partial T} \right)_p \frac{dT}{dz} - \left(\frac{\partial w_s}{\partial p} \right)_T \rho g \right]. \quad (2.243)$$

Solving for dT/dz , the lapse rate of an ascending saturated parcel is given by

$$\Gamma_s = -\frac{dT}{dz} = \frac{g}{c_p} \frac{1 - \rho L_c (\partial w_s / \partial p)_T}{1 + (L_c / c_p) (\partial w_s / \partial T)_p}. \quad (2.244)$$

This is variously called the *pseudoadiabatic* or *moist adiabatic* or *saturated adiabatic* lapse rate. Since g/c_p is the dry adiabatic lapse rate Γ_d , $\Gamma_s < \Gamma_d$. Values of Γ_s can be explicitly calculated using the Clausius-Clapeyron relationship and are typically around 6 K km^{-1} in the earth’s atmosphere. Since dw_s/dT is an increasing function of T , Γ_s decreases with increasing temperature, and can be as low as

3.5 K km^{-1} . For a saturated parcel, the stability conditions analogous to (2.237) are

$$\text{Stability :} \quad -\frac{\partial T}{\partial z} < \Gamma_s, \quad (2.245a)$$

$$\text{Instability :} \quad -\frac{\partial T}{\partial z} > \Gamma_s. \quad (2.245b)$$

The observed environmental profile in convecting situations is often a combination of the dry adiabatic and moist adiabatic profiles: if a column of air is heated from below a parcel may eventually become sufficiently warm that it is unstable; if so, it will rise and cool following a dry adiabat Γ_d until it becomes saturated at the ‘lifting condensation level’, above which it will rise following a saturation adiabat, Γ_s . Such convection will proceed until the atmospheric column is stable and, in low latitudes, the lapse rate of the atmosphere is largely determined by such convective processes.

** Equivalent potential temperature*

Suppose that all the moisture in a parcel of air condenses, and that all the heat released goes into heating the parcel. The *equivalent potential temperature*, θ_{eq} is the potential temperature that the parcel then achieves. We may obtain an approximate analytic expression for it by noting that the first law of thermodynamics, $dQ = Td\eta$, then implies, by definition of potential temperature,

$$-L_c dw = c_p T d \ln \theta \quad (2.246)$$

where dw is the change in water vapour mixing ratio, so that a reduction of w via condensation leads to heating. Integrating gives, by definition of equivalent potential temperature,

$$-\int_w^0 \frac{L_c w}{c_p T} dw = \int_{\theta}^{\theta_{\text{eq}}} d \ln \theta, \quad (2.247)$$

and so, if T and L_c are assumed constant,

$$\theta_{\text{eq}} = \theta \exp \left(\frac{L_c w}{c_p T} \right). \quad (2.248)$$

The equivalent potential temperature so defined is approximately conserved during condensation, the approximation arising going from (2.247) to (2.248). It is a useful expression for diagnostic purposes, and in constructing theories of convection, but it is not accurate enough to use as a prognostic variable in a numerical model that has pretensions to reality. The ‘equivalent temperature’ may be defined in terms of the equivalent potential temperature by

$$T_{\text{eq}} = \theta_{\text{eq}} \left(\frac{p}{p_{\text{ref}}} \right)^{\kappa}. \quad (2.249)$$

2.9 GRAVITY WAVES

The parcel approach to oscillations and stability, while simple and direct, seems divorced from the fluid-dynamical equations of motion and, as a consequence, makes it hard to include other effects such as rotation, or explore the effects of possible differences between the hydrostatic and non-hydrostatic cases. To remedy this, use the equations of motion to analyze the motion resulting from a small disturbance.

2.9.1 Gravity waves and convection in a Boussinesq fluid

Assume the flow is a function only of x and z , that N^2 is constant, and ignore background rotation. Then linearizing about this stratified, resting, background state gives

$$\frac{\partial u}{\partial t} = -\frac{\partial \phi}{\partial x}, \quad (2.250)$$

$$\frac{\partial w}{\partial t} = -\frac{\partial \phi}{\partial z} + b, \quad (2.251)$$

$$\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0, \quad (2.252)$$

$$\frac{\partial b}{\partial t} + wN^2 = 0, \quad (2.253)$$

and a little algebra gives the single equation for w ,

$$\left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} \right) \frac{\partial^2}{\partial t^2} + N^2 \frac{\partial^2}{\partial x^2} \right] w = 0. \quad (2.254)$$

Seeking solutions of the form $w = \text{Re } W \exp[i(kx + mz - \omega t)]$ (where Re denotes the real part) yields the important dispersion relationship for gravity waves,

$$\omega^2 = \frac{k^2 N^2}{k^2 + m^2}. \quad (2.255)$$

The frequency (see Fig. 2.5) is thus always less than N , approaching N for small horizontal scales, $k \gg m$. If we explicitly neglect pressure perturbations, as in the parcel argument, then the following two equations,

$$\frac{\partial w}{\partial t} = b \quad (2.256)$$

$$\frac{\partial b}{\partial t} + wN^2 = 0, \quad (2.257)$$

form a closed set, and give $\omega^2 = N^2$.

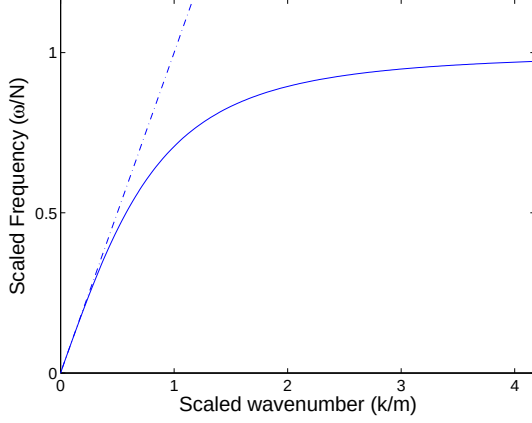


Figure 2.5 Scaled frequency, ω/N , plotted as a function of scaled horizontal wavenumber, k/m , using the full dispersion relation of (2.255) (solid line, asymptoting to unit value for large k/m) and with the hydrostatic dispersion relation (2.263) (dashed line, tending to ∞ for large k/m).

If $\partial\rho_0/\partial z > 0$ and $N^2 < 0$ then we expect the background state to be unstable. Indeed, (2.255) then gives

$$\sigma = \frac{\pm k\tilde{N}}{(k^2 + m^2)^{1/2}} \quad (2.258)$$

where $\tilde{N}^2 = (g/\rho_0)\partial\rho_0/\partial z$ and the disturbance grows exponentially according to $\exp(\sigma t)$. Most convective activity in the ocean and atmosphere is, ultimately, related to an instability of this form, although of course there are many complicating issues — water vapour in the atmosphere, salt in the ocean, the effects of rotation and so forth.

Hydrostatic gravity waves and convection

Suppose that we make the hydrostatic assumption. The linearized two-dimensional equations of motion become

$$\frac{\partial u}{\partial t} = -\frac{\partial \phi}{\partial x} \quad (2.259)$$

$$0 = -\frac{\partial \phi}{\partial z} + b \quad (2.260)$$

$$\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0 \quad (2.261)$$

$$\frac{\partial b}{\partial t} + wN^2 = 0 \quad (2.262)$$

Then a little algebra gives the dispersion relation,

$$\omega^2 = \frac{k^2 N^2}{m^2}. \quad (2.263)$$

The frequency and, if N^2 is negative the growth rate, is unbounded for as $k/m \rightarrow \infty$, and the hydrostatic approximation thus has quite unphysical behaviour for small horizontal scales (see also problem 2.10).⁹

2.10 * ACOUSTIC-GRAVITY WAVES IN AN IDEAL GAS

We now consider wave motion in a stratified, compressible fluid such as the earth's atmosphere. The complete problem is complicated and uninformative; we will specialize to the case of an isothermal, stationary atmosphere and ignore the effects of rotation and sphericity, but otherwise we will make few approximations. In this section we will denote the unperturbed state with a subscript 0 and the perturbed state with a prime ($'$). Because it is at rest, the basic state is in hydrostatic balance,

$$\frac{\partial p_0}{\partial z} = -\rho_0 g \quad (2.264)$$

Ignoring variations in the y -direction for algebraic simplicity, the linearized equations of motion are:

$$\text{u-momentum:} \quad \rho_0 \frac{\partial u'}{\partial t} = -\frac{\partial p'}{\partial x} \quad (2.265a)$$

$$\text{w-momentum:} \quad \rho_0 \frac{\partial w'}{\partial t} = -\frac{\partial p'}{\partial z} - \rho' g \quad (2.265b)$$

$$\text{mass conservation:} \quad \frac{\partial \rho'}{\partial t} + w' \frac{\partial \rho_0}{\partial z} = -\rho_0 \left(\frac{\partial u'}{\partial x} + \frac{\partial w'}{\partial z} \right) \quad (2.265c)$$

$$\text{thermodynamic:} \quad \frac{\partial \theta'}{\partial t} + w' \frac{\partial \theta_0}{\partial z} = 0 \quad (2.265d)$$

$$\text{equation of state:} \quad \frac{\theta'}{\theta_0} + \frac{\rho'}{\rho_0} = \frac{1}{\gamma} \frac{p'}{p_0}. \quad (2.265e)$$

For an isothermal basic state we have $p_0 = \rho_0 R T_0$ where T_0 is a constant, so that $\rho_0 = \rho_s e^{-z/H}$ and $p_0 = p_s e^{-z/H}$ where $H = R T_0 / g$. Further, using $\theta = T(p_s / p)^\kappa$ where $\kappa = R / c_p$, we have that $\theta_0 = T_0 e^{\kappa z / H}$ and so $N^2 = \kappa g / H$. Using these then, after a little algebra, the set of equations (2.265) becomes

$$\left(\frac{\partial^2}{\partial t^2} - c^2 \frac{\partial^2}{\partial x^2} \right) u' = c^2 \left(\frac{\partial}{\partial z} - \frac{1}{\gamma H} \right) \frac{\partial}{\partial z} w' \quad (2.266a)$$

and

$$\left(\frac{\partial^2}{\partial t^2} - c^2 \left[\frac{\partial^2}{\partial x^2} - \frac{1}{H} \frac{\partial}{\partial z} \right] \right) w' = c^2 \left(\frac{\partial}{\partial z} - \frac{\kappa}{H} \right) \frac{\partial u'}{\partial x}, \quad (2.266b)$$

where $c^2 = \partial(\partial p / \rho)_\eta = \gamma R T_0$ is the square of the speed of sound. These two equations in turn combine to give

$$\frac{\partial^4 w'}{\partial t^4} - c^2 \frac{\partial^2}{\partial t^2} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} - \frac{1}{H} \frac{\partial}{\partial z} \right) w' - c^2 \frac{\kappa g}{H} \frac{\partial^2 w'}{\partial x^2} = 0. \quad (2.267)$$

If we set $w' = W(x, z, t)e^{z/(2H)}$ (so that $W = (\rho_0 / \rho_s)^{1/2} w$) then the term with the single z -derivative is eliminated, giving

$$\frac{\partial^4 W}{\partial t^4} - c^2 \frac{\partial^2}{\partial t^2} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} - \frac{1}{4H^2} \right) W - c^2 \frac{\kappa g}{H} \frac{\partial^2 W}{\partial x^2} = 0. \quad (2.268)$$

Although superficially complicated, this equation has constant coefficients and we may seek wavelike solutions of the form

$$W = \text{Re } \tilde{W} e^{i(kx + mz - \omega t)}, \quad (2.269)$$

where $\tilde{W}$ is the complex amplitude, and this leads to the dispersion relation for acoustic-gravity waves, namely

$$\omega^4 - c^2 \omega^2 \left(k^2 + m^2 + \frac{1}{4H^2} \right) + c^2 N^2 k^2 = 0, \quad (2.270)$$

with solution

$$\omega^2 = \frac{1}{2} c^2 K^2 \left[1 \pm \left(1 - \frac{4N^2 k^2}{c^2 K^4} \right)^{1/2} \right] \quad (2.271)$$

where $K^2 = k^2 + m^2 + 1/(4H^2)$. (The factor $[1 - 4N^2 k^2 / (c^2 K^4)]$ is always positive — see problem 2.22.) For an isothermal, ideal-gas, atmosphere $4N^2 H^2 / c^2 \approx 0.8$ and so this may be written

$$\frac{\omega^2}{N^2} \approx 2.5 \hat{K}^2 \left[1 \pm \left(1 - \frac{0.8 \hat{k}^2}{\hat{K}^4} \right)^{1/2} \right] \quad (2.272)$$

where $\hat{K}^2 = \hat{k}^2 + \hat{m}^2 + 1/4$, and $(\hat{k}, \hat{m}) = (kH, mH)$.

2.10.1 Interpretation

Acoustic and gravity waves

There are two branches of roots in (2.271), corresponding to acoustic waves (using the plus sign in the dispersion relation) and internal gravity waves (using the minus sign). These (and the Lamb wave, described below) are plotted in Fig.

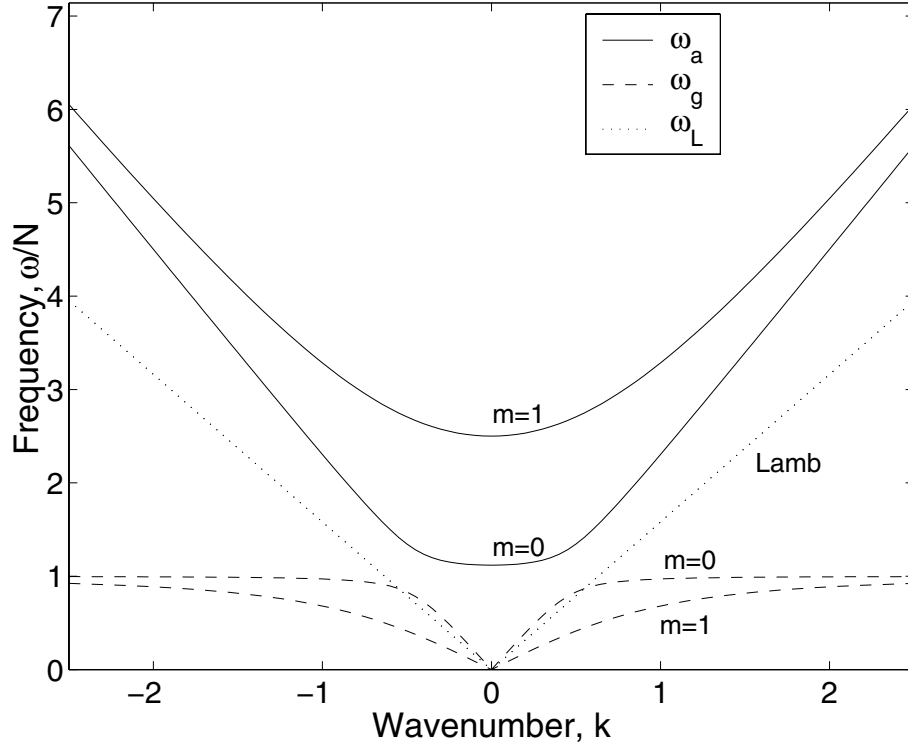


Figure 2.6 Dispersion diagram for acoustic gravity waves in an isothermal atmosphere, calculated using (2.272). The frequency is given in units of the buoyancy frequency N , and the wavenumbers are non-dimensionalized by the inverse of the scale height, H . The solid curves indicate acoustic waves, whose frequency is always higher than that of the corresponding Lamb wave at the same wavenumber (i.e. ck), and of the base acoustic frequency $\approx 1.12N$. The dashed curves indicate internal gravity waves, whose frequency asymptotes to N at small horizontal scales.

2.6. If $4N^2k^2/c^2K^4 \ll 1$ then the two sets of waves are well separated. From (2.271) this is satisfied when

$$\frac{4\kappa}{\gamma}(kH)^2 \approx 0.8(kH)^2 \ll \left[(kH)^2 + (mH)^2 + \frac{1}{4}\right]^2; \quad (2.273)$$

that is, when *either* $mH \gg 1$ or $kH \gg 1$. The two roots of the dispersion relation are then

$$\omega_a^2 \approx c^2K^2 = c^2\left(k^2 + m^2 + \frac{1}{4H^2}\right) \quad (2.274)$$

and

$$\omega_g^2 \approx \frac{N^2k^2}{k^2 + m^2 + 1/(4H^2)}. \quad (2.275)$$

corresponding to acoustic and gravity waves, respectively. The acoustic waves owe their existence to the presence of compressibility in the fluid, and they have

no counterpart in the Boussinesq system. On the other hand, the internal gravity waves are just modified forms of those found in the Boussinesq system, and if we take the limit $(kH, mH) \rightarrow \infty$ then the gravity wave branch reduces to $\omega_g^2 = N^2 k^2 / (k^2 + m^2)$, which is the dispersion relationship for gravity waves in the Boussinesq approximation. We may consider this to be the limit of infinite scale height or (equivalently) the case in which wavelengths of the internal waves are sufficiently small that the fluid is essentially incompressible.

Vertical structure

Recall that $w' = W(x, z, t)e^{z/(2H)}$ and, by inspection of (2.266), u' has the same vertical structure. That is,

$$w' \propto e^{z/(2H)}, \quad u' \propto e^{z/(2H)}, \quad (2.276)$$

the amplitude of the velocity field internal waves increasing with height. The pressure and density perturbation amplitudes fall off with height, varying like

$$p' \propto e^{-z/(2H)}, \quad \rho' \propto e^{-z/(2H)}. \quad (2.277)$$

The kinetic energy of the perturbation, $\rho_0(u'^2 + w'^2)$ is *constant* with height, because $\rho_0 = \rho_s e^{-z/H}$.

Hydrostatic approximation and Lamb waves

Equations (2.266) also admit to a solution with $w' = 0$. We then have

$$\left(\frac{\partial^2}{\partial t^2} - c^2 \frac{\partial^2}{\partial x^2} \right) u' = 0 \quad \text{and} \quad \left(\frac{\partial}{\partial z} - \frac{\kappa}{H} \right) \frac{\partial u'}{\partial x} = 0 \quad (2.278)$$

and these have solutions of the form

$$u' = \text{Re} \tilde{U} e^{\kappa z/H} e^{i(kx - \omega t)} \quad (2.279)$$

where $\tilde{U}$ is a constant, complex, amplitude and

$$\omega = ck \quad (2.280)$$

These are horizontally propagating sound waves, known as *Lamb waves* after the hydrodynamicist Horace Lamb. Their velocity perturbation amplitude increases with height, now varying as

$$u' \propto e^{\kappa z/H} \approx e^{2z/(7H)} \quad (2.281)$$

but their pressure perturbation falls off as

$$p' \propto e^{(\kappa-1)z/H} \approx e^{-5z/(7H)}. \quad (2.282)$$

Their kinetic energy density $\rho_0 u^2$ varies as

$$K.E. \propto e^{-z/H+2\kappa z/H} = e^{(2R-c_p)z/(c_p H)} = e^{(R-c_v)z/(c_p H)} \approx e^{-3z/(7H)} \quad (2.283)$$

for an ideal gas. ($c_v = nR/2$ where n is the number of excited degrees of freedom, 5 for a diatomic molecule.) In this sense Lamb waves are trapped at the surface and are an example of edge waves or surface waves.

Consider now the slightly more general case, in which we make the hydrostatic approximation *ab initio*, but do not restrict the perturbation to have $w' = 0$. The linearized equations are identical to (2.265), except that (2.265b) is replaced by

$$\frac{\partial p'}{\partial z} = -\rho' g \quad (2.284)$$

It is left as a problem for the reader to show that the dispersion relation is now

$$\omega^2 = \frac{N^2 k^2}{m^2 + 1/(4H^2)}. \quad (2.285)$$

This may be compared with the corresponding Boussinesq result (2.263). Again, the frequency increases without bound as the horizontal wavelength diminishes.

Notes

- 1 If (2.12) is written, as is common, with the terms $+2\mathbf{\Omega} \times \mathbf{v}_r$ and $+\mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{r})$ on the left-hand-side then these terms should really be referred to as the Coriolis and centrifugal *accelerations*, but this distinction is not made uniformly in the literature.
- 2 Phillips (1973). Stommel & Moore (1989) and Gill (1982) delve into the issue of the shape of the earth in a little more detail, and Gill shows that an appropriate co-ordinate system to use is a slight distortion from oblate spheroidal co-ordinates. To a very good approximation, the equations in this co-ordinate system are the same as those written in spherical co-ordinates. Note the typographic errors in Gill's equations (4.12.11) and (4.12.12).
- 3 Phillips (1966). See White (2003) for a derivation and review of the primitive equations and the use of spherical co-ordinates. Associated with increasing computer power there is a tendency for comprehensive numerical models to use non-hydrostatic equations of motion that do not make the shallow-fluid or traditional approximations.
- 4 The Boussinesq approximation is named after Boussinesq (1903), although similar approximations were used earlier by Oberbeck (1879, 1888). Spiegel & Veronis (1960) give a physically based derivation, and Mihaljan (1962) provides a systematic asymptotic derivation of the equations, expanding the Navier-Stokes equations in two small parameters, proportional to the coefficient of volume expansion and the square of the thermal diffusivity.

- 5 Various versions of anelastic equations exist — for example Batchelor (1953), ?, Gough (1969), ?, and ?, although not all have potential vorticity and energy conservation laws (???). The system we derive is most similar to that of ? and unpublished notes by J. S. A. Green, who also looks into the effects of a basic state with nonconstant θ .
- 6 A numerical model that includes sound waves must take very small timesteps in order to maintain numerical stability, in particular to satisfy the CFL criterion, or else use an implicit time-stepping scheme. This latter procedure effectively lets the numerics do the filtering of the sound waves, and is increasingly favoured by many numerical modellers. If we make the hydrostatic approximation then all sound waves except those that propagate horizontally are eliminated, and there is little need to make the anelastic approximation. Most atmospheric general circulation models follow this route, although many are now becoming nonhydrostatic.
- 7 For suppose we move parcel A upwards to position B , but reference all potential densities to the surface. Then the difference in parcel A 's potential density and that of its surroundings will always be of the same sign as the difference between its density and the surroundings. To see this, suppose that after it is displaced to the level of B , parcel A is less dense than a parcel at B . Then, because they are both at the same pressure, parcel A must be warmer. Thus, parcel A must have a higher potential temperature also (because the relationship between T and θ is monotonic at constant pressure) and therefore it also has a lower potential density. However, the buoyancy frequency would not be accurately given by (2.227) if the reference level were the surface, because the difference in potential density is not *equal* to the difference in density.
- 8 Foster (1972).
- 9 Many numerical models of the large-scale circulation in the atmosphere and ocean do make the hydrostatic approximation. In these models convection must be *parameterized*; otherwise, convection would simply occur at the smallest scale available, namely the size of the numerical grid, and this type of unphysical behaviour should be avoided. For numerical studies of convection the hydrostatic approximation is thus inadequate. Of course in non-hydrostatic models convection must also be parameterized if the horizontal resolution of the model is too coarse to properly resolve the convective scales. An alternative to such parameterization schemes is suggested by problem 2.10.

Further Reading

Cushman-Roisin, B., 1994. *An Introduction to Geophysical Fluid Dynamics*.

This compact book provides a clear introduction to a variety of topics in GFD.

Gill, A. E. 1982. *Atmosphere-Ocean Dynamics*.

A rich, textured textbook and reference, especially strong on equatorial dynamics and gravity wave motion.

Holton, J. R. 1992. *An Introduction to Dynamical Meteorology*.

A well-known textbook at the upper-division undergraduate/beginning graduate level. It contains much more meteorology than is in this book.

Pedlosky, J., 1987. *Geophysical Fluid Dynamics*.

A primary reference, especially for flow at low Rossby number. Although the book requires some effort, there is a handsome pay-off for those who study it closely.

White (2002) provides a clear and thorough summary of the equations of motion for meteorology, including the non-hydrostatic and primitive equations.

Problems

2.1 For an ideal gas in hydrostatic balance, show that:

- (a) The integral of the potential plus internal energy from the surface to the top is equal to its enthalpy;
- (b) $d\sigma/dz = c_p(T/\theta)d\theta/dz$, where $\sigma = I + p\alpha + \Phi$ is the dry static energy;
- (c) The following expressions for the pressure gradient force are all equal (even without hydrostatic balance):

$$-\frac{1}{\rho}\nabla p = -\theta\nabla\Pi = -\frac{c_s^2}{\rho\theta}\nabla(\rho\theta). \quad (\text{P2.1})$$

where $\Pi = c_p T/\theta$ is the Exner function.

2.2 Show that, without approximation, the unforced, inviscid momentum equation may be written in the forms

$$\frac{D\mathbf{v}}{Dt} = T\nabla\eta - \nabla(p\alpha + I) \quad (\text{P2.2})$$

and

$$\frac{\partial\mathbf{v}}{\partial t} + \boldsymbol{\omega} \times \mathbf{v} = T\nabla\eta - \nabla B \quad (\text{P2.3})$$

where $\boldsymbol{\omega} = \nabla \times \mathbf{v}$, η is the specific entropy ($d\eta = c_p d \ln \theta$) and $B = I + \mathbf{v}^2/2 + p\alpha$ where I is the internal energy per unit mass.

Hint: First show that $T\nabla\eta = \nabla I + p\nabla\rho^{-1}$, and note also the vector identity $\mathbf{v} \times (\nabla \times \mathbf{v}) = \frac{1}{2}\nabla(\mathbf{v} \cdot \mathbf{v}) - (\mathbf{v} \cdot \nabla)\mathbf{v}$.

2.3 Consider two-dimensional fluid flow in a rotating frame of reference on the f -plane. Linearize the equations about a state of rest.

- (a) Ignore the pressure term and determine the general solution to the resulting equations. Show that the speed of fluid parcels is constant. Show that the trajectory of the fluid parcels is a circle with radius $|U|/f$, where $|U|$ is the fluid speed.
- (b) What is the period of oscillation of a fluid parcel? Why is this not the same as the period of rotation of the frame of reference?

2.4 For rapidly rotating, constant density flow, with the rotation vector parallel to gravity, show that the velocity is approximately independent of the vertical co-ordinate.

- 2.5 A fluid at rest evidently satisfies the hydrostatic relation, which says that the pressure at the surface is given by the weight of the fluid above it. Now consider a *deep* atmosphere on a spherical planet. A unit cross-sectional area at the planet's surface supports a column of fluid whose cross-section increases with height, because the total area of the atmosphere increases with distance away from the center of the planet. Is the pressure at the surface still given by the hydrostatic relation, or is it greater than this because of the increased mass in the column? If it is still given by the hydrostatic relation, then the pressure at the surface, integrated over the entire area of the planet, is less than the total weight of the fluid. Resolve this paradox. But if the pressure at the surface is greater than that implied by hydrostatic balance, explain how the hydrostatic relation fails.
- 2.6 By considering how the direction of the coordinate axes change with position, show that in spherical coordinates: $D\mathbf{i}/Dt = u\partial\mathbf{i}/\partial x = (u/r \cos\vartheta)(\mathbf{j} \sin\vartheta - \mathbf{k} \cos\vartheta)$; $D\mathbf{j}/Dt = u\partial\mathbf{j}/\partial x + v\partial\mathbf{j}/\partial y = -\mathbf{i}(u/r) \tan\vartheta - \mathbf{k}v/a$; and $D\mathbf{k}/Dt = u\partial\mathbf{k}/\partial x + v\partial\mathbf{k}/\partial y = \mathbf{i}(u/r) + \mathbf{j}(v/r)$. Then, using

$$\frac{D\mathbf{v}}{Dt} = \mathbf{i}\frac{Du}{Dt} + \mathbf{j}\frac{Dv}{Dt} + \mathbf{k}\frac{Dw}{Dt} + u\frac{D\mathbf{i}}{Dt} + v\frac{D\mathbf{j}}{Dt} + w\frac{D\mathbf{k}}{Dt}, \quad (\text{P2.4})$$

show that

$$\begin{aligned} \frac{D\mathbf{v}}{Dt} = & \mathbf{i} \left(\frac{Du}{Dt} - \frac{uv \tan\vartheta}{r} + \frac{uw}{r} \right) + \mathbf{j} \left(\frac{Dv}{Dt} + \frac{u^2 \tan\vartheta}{r} + \frac{vw}{r} \right) \\ & + \mathbf{k} \left(\frac{Dw}{Dt} - \frac{u^2 + v^2}{r} \right). \end{aligned} \quad (\text{P2.5})$$

- 2.7 At what latitude is the angle between the direction of Newtonian gravity (due solely to the mass of the earth) and that of apparent gravity (Newtonian gravity plus centrifugal terms) the largest? At what latitudes, if any, is this angle zero?
- 2.8 $\diamond$ Write the momentum equations in true spherical co-ordinates, including the centrifugal and gravitational terms. Show that for reasonable values of the wind, the dominant balance in the meridional component of this equation involve a balance between centrifugal and pressure gradient terms. Can this balance be subtracted out of the equations in a sensible way, so leaving a useful horizontal momentum equation that involves the Coriolis and acceleration terms? If so, obtain a closed set of equations for the flow this way. Discuss the pros and cons of this approach versus the geometric approximation discussed in section 2.2.1.
- 2.9 Consider an ocean at rest with known vertical profiles of potential temperature and salinity, $\theta(z)$ and $S(z)$. Suppose we also know the equation of state in the form $\rho = \rho(\theta, S, p)$. Obtain an expression for the buoyancy frequency. Check your expression by substituting the equation of state for an ideal gas and recovering a known expression for the buoyancy frequency.

2.10 Convection and its parameterization.

- (i) Consider a Boussinesq system in which the vertical momentum equation is modified by the parameter α to read

$$\alpha \frac{Dw}{Dt} = -\frac{\partial \phi}{\partial z} + b, \quad (\text{P2.6})$$

and the other equations are unchanged. (If $\alpha = 0$ the system is hydrostatic, and if $\alpha = 1$ the system is the original one.) Linearise these equations about a state of rest and of constant stratification (as in section 2.9.1) and obtain the dispersion relation for the system, and plot it for various values of α , including 0 and 1. Show that, for $\alpha > 1$ the system approaches the limiting frequency of N more rapidly than with $\alpha = 1$.

- (ii) $\diamond$ Argue that if $N^2 < 0$, convection in a system with $\alpha > 1$ generally occurs at a larger scale than with $\alpha = 1$. Show this explicitly by adding some diffusion or friction to the right-hand sides of the equations of motion and obtaining the dispersion relation. You may do this approximately, or by some other method.
- 2.11 (a) The *geopotential height* is the height of a given pressure level. Show that in an atmosphere with a uniform lapse rate (i.e., $dT/dz = \Gamma = \text{constant}$) the geopotential height at a pressure p is given by

$$z = \frac{T_0}{\Gamma} \left[1 - \left(\frac{p_0}{p} \right)^{-\Gamma/g} \right] \quad (\text{P2.7})$$

where T_0 is the temperature at $z = 0$.

- (b) In an isothermal atmosphere, obtain an expression for the geopotential height as function of pressure, and show that this is consistent with the expression (P2.7) in the appropriate limit.
- 2.12 Consider the Boussinesq equations in the form $D\mathbf{v}/Dt = \mathbf{k}b + \nu \nabla^2 \mathbf{v}$, $\nabla \cdot \mathbf{v} = 0$, $Db/Dt = Q + \kappa \nabla^2 b$. Obtain an energy equation similar to (2.123) but now with the terms on the right-hand side that represent viscous and diabatic effects. Give a brief interpretation of these terms.
- 2.13 $\diamond$ Consider the Boussinesq equations in the form

$$\frac{D\mathbf{v}}{Dt} = -\nabla\phi + b\mathbf{k}, \quad \nabla \cdot \mathbf{v} = 0, \quad \frac{D\theta}{Dt} = 0, \quad (\text{P2.8})$$

where θ is the potential temperature and b is the buoyancy. If $\theta = \theta(b) + f(z)$, where f is an arbitrary function of z , show that the system has an appropriate energy conservation statement. Is this true for the more general equations of state, $\theta = \theta(b, z)$ and $\theta = \theta(b, \phi)$? *Hint:* Show that a function ϕ^* exists where $-\nabla\phi^* + \theta\mathbf{k} = -\nabla\phi + b\mathbf{k}$ everywhere.

- 2.14 Consider a rapidly rotating (i.e., in near geostrophic balance) Boussinesq fluid on the f -plane.
- (a) Show that the pressure divided by the density scales as $\phi \sim fUL$.
- (b) Show that the horizontal divergence of the geostrophic wind vanishes. Thus, argue that the scaling $W \sim UH/L$ is an *overestimate* for the magnitude of the vertical velocity. (Optional extra: obtain a scaling estimate for the magnitude of vertical velocity in rapidly rotating flow.)

- (c) Using these results, or otherwise, discuss whether hydrostatic balance is more or less likely to hold in a rotating flow than in non-rotating flow.
- 2.15 Using either the anelastic equations in height co-ordinates, or the equations of motion in pressure co-ordinates, show that for rapidly rotating flow the vertical shear of horizontal velocity is proportional to the horizontal temperature gradient. Estimate the size of the zonal wind 5 km above the surface in the midlatitude atmosphere in summer and winter using (approximate) values for the meridional temperature gradient in the atmosphere. What is the shear corresponding to the pole-equator temperature gradient in the ocean?
- 2.16 Using approximate but realistic values for the observed stratification, what is the buoyancy period for (a) the mid-latitude troposphere, (ii) the oceanic thermocline, (c) the oceanic abyss (the deep ocean below the thermocline)?
- 2.17 Consider a dry, hydrostatic, ideal-gas atmosphere whose lapse rate is one of constant potential temperature. What is its vertical extent? That is, at what height does the density vanish? Is this a problem for the anelastic approximation discussed in the text?
- 2.18 Show that for an ideal gas, the expressions (2.224), (2.228), (2.235) are all equivalent, and express N^2 terms of the temperature lapse rate, $\partial T / \partial z$.
- 2.19 Begin with the mass conservation in the height-co-ordinates, namely

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0, \quad (\text{P2.9})$$

Transform this into pressure co-ordinates using the chain rule (or otherwise) and derive the mass conservation equation in the form

$$\nabla_p \cdot \mathbf{u} + \frac{\partial \omega}{\partial p} = 0. \quad (\text{P2.10})$$

- 2.20 Starting with the primitive equations in pressure coordinates, derive the form of the primitive equations of motion in sigma-pressure coordinates. In particular, show that the prognostic equation for surface pressure is,

$$\frac{\partial p_s}{\partial t} + \nabla \cdot (p_s \mathbf{u}) + p_s \frac{\partial \sigma}{\partial \sigma} = 0 \quad (\text{P2.11})$$

and that hydrostatic balance may be written:

$$\frac{\partial \Phi}{\partial \sigma} = -\frac{RT}{\sigma}. \quad (\text{P2.12})$$

- 2.21 Starting with the primitive equations in pressure coordinates, derive the form of the primitive equations of motion in log-pressure coordinates in which $Z = -H \ln(p/p_r)$ is the vertical coordinate. Here, H is a reference height (e.g., a scale height RT_r/g where T_r is a typical or an average temperature) and p_r is

a reference pressure (e.g., 1000 mb). In particular, show that if the 'vertical velocity' is $W = DZ/Dt$ then $W = -H\omega/p$ and that

$$\frac{\partial \omega}{\partial p} = -\frac{\partial}{\partial p} \left(\frac{pW}{H} \right) = \frac{\partial W}{\partial Z} - \frac{W}{H}. \quad (\text{P2.13})$$

and obtain the mass conservation equation (2.166d). Show that this can be written in the form

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{1}{\rho_s} \frac{\partial}{\partial Z} (\rho_s W) = 0 \quad (\text{P2.14})$$

where $\rho_s = \rho_r \exp(-Z/H)$ where ρ_r is a constant.

- 2.22 (a) Prove that the argument of the square root in (2.271) is always positive.
Solution: The largest value or the argument occurs when $m = 0$ and $k^2 = 1/(4H^2)$. The argument is then $1 - 4H^2 N^2 / c^2$. But $c^2 = \gamma RT_0 = \gamma gH$ and $N^2 = g\kappa/H$ so that $4N^2 H^2 / c^2 = 4\kappa/\gamma \approx 0.8$.
- (b) $\diamond$ This argument seems to depend on the parameters in the ideal gas equation of state. Is it more general than this? Is a natural system possible for which the argument is negative, and if so what physical interpretation could one ascribe to the situation?
- 2.23 Consider an ideal-gas isothermal atmosphere that obeys the hydrostatic equations of motion. Show that small deviations around a resting state, as described by (2.265), but with (2.265b) replaced by (2.284), give rise to internal gravity waves with a dispersion relation given by (2.285). Show also that the Lamb wave is present in this system.

CHAPTER THREE

Boundary Layers

THE VALUES OF PHYSICAL PROPERTIES IN THE ‘INTERIOR’ OF A FLUID are generally set by the physics and dynamics pertaining in that interior. The values at the boundary, however, are often set by different physical processes, which may occur on a faster timescale than the interior processes and which thereby in effect provide a ‘boundary condition’ that the fluid must satisfy. The consequent transition region between the interior properties and the boundary values is called a ‘boundary layer,’ and often it is much thinner than the interior region and different physical balances may hold. There are many different types of boundary layer: for example, the molecular viscosity of a fluid leads ‘no-slip’ boundary condition which says that the tangential velocity of a fluid at a solid boundary is strictly zero. But millimeters away the velocity might be determined by other forces, resulting in a quite thin boundary layer, sometimes called the Blasius boundary layer. It may be a region of quite strong fluxes of momentum, necessary to decelerate the fluid from its interior value to zero. In the atmosphere and ocean, boundary layers are the means whereby both heat and momentum are transferred from one to the other, and are therefore crucially important in the dynamics of both systems. There are many different boundary layers at the interface, from the millimeter thick Blasius layer through a logarithmic boundary layer to the planetary Ekman layer, and our discussion is quite selective. We are mainly concerned with quasi-vertical boundary layers along horizontal boundaries, such as the land or ocean surface.

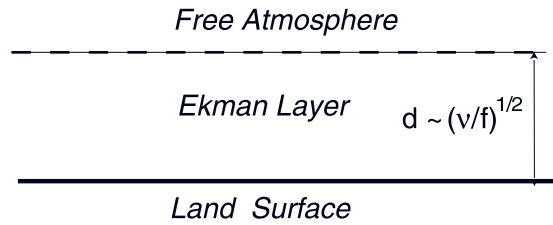


Figure 3.1 The Ekman layer in the atmosphere. . .

Boundary layers nearly always involve one or both of viscosity and diffusion. This is because these appear in the terms of highest differential order in the equation of motion, and so these terms are responsible for the number and type of boundary conditions that the equations must satisfy. In their absence, the full fluid boundary conditions cannot generally be satisfied. In the atmosphere and ocean, the direct effects of molecular viscosity and diffusion are nearly always locally negligible at distances more than a few millimeters away from a solid boundary. But it is inconceivable that the entire boundary layer between the free atmosphere (say) and the ground is only a few millimeters thick; it is more likely that there is a turbulent boundary layer involving eddies of multiple sizes; fully understanding this is a research topic of extreme difficulty, and there is no proper solution to this problem. In rapidly rotating fluids, foremost among the turbulent boundary layers is the *Ekman layer*, and this is the main topic of this short chapter.

3.1 THE EKMAN LAYER

The Ekman layer is a boundary layer in rotating fluids that connects a geostrophically balanced interior flow with the surface. Such a layer exists in rotating tanks in the laboratory, in rotating stars and the giant planets, and of course, in the ocean and atmosphere of the earth where it may be tens of meters to a kilometer or so thick. It is boundary layer that serves to transfer the stress at the surface to the ‘free’ interior flow. By ‘free’ we mean that friction and diffusion are not important at lowest order, so that the motion is largely inviscid and adiabatic.

If the boundary layer is turbulent, we immediately run into a virtually insurmountable problem. That is, we do not know how to parameterize the transports (of heat and momentum say) in terms of the mean fields themselves, or in particular in terms of their gradients. A common (and time honoured) ‘engineering’ device is to suppose that the transport is related to the mean gradient by way of some empirical ‘eddy viscosity’ or ‘eddy diffusion,’ and indeed we shall follow that route in section 3.1.4. But because of the inherent uncertainties in any such approach, we cannot rely too much on the details of any analysis. For example,

any property that relies on the detailed form of an eddy momentum flux parameterization (e.g., an eddy viscosity) should not be trusted as a representation of reality. It is for this reason that we shall first look for more general properties of the boundary layer.

To aid the analysis let us assume:

1. The Ekman layer is Boussinesq. This is a very good assumption for the ocean. For the atmosphere it is not bad if we are only dealing with the lower kilometer of the atmosphere.
2. The nonlinear and time dependent terms in the equations of motion are negligible, and geostrophic balance holds in the horizontal away from the boundary layer. Hydrostatic balance holds in the vertical.
3. Where we need an explicit form for friction, we shall assume that it can be parameterized by a Newtonian viscosity, which may be an eddy viscosity, the vertical component of which is dominant. Thus, $\mathbf{F} = \rho_0 A \partial^2 \mathbf{u} / \partial z^2$ where A is constant.

3.1.1 Equations of motion and scaling

Frictional-geostrophic balance in the horizontal momentum equation is:

$$\mathbf{f} \times \mathbf{u} = -\nabla\phi + \frac{1}{\rho_0} \frac{\partial \boldsymbol{\tau}}{\partial z}, \quad (3.1)$$

or, with an explicit eddy viscosity

$$\mathbf{f} \times \mathbf{u} = -\nabla\phi + A \frac{\partial^2 \mathbf{u}}{\partial z^2}. \quad (3.2)$$

The equation set is completed by adding hydrostatic balance,

$$\frac{\partial \phi}{\partial z} = b \quad (3.3)$$

and mass conservation

$$\nabla \cdot \mathbf{v} = 0 \quad (3.4)$$

By assumption, the friction term in (3.2) is small. That is, away from a boundary layer geostrophic balance holds. Since we are making a Boussinesq assumption, we will incorporate the density into the stress by defining a 'kinematic stress' $\tilde{\boldsymbol{\tau}} \equiv \boldsymbol{\tau} / \rho_0$.

How thick is the Ekman layer? To obtain a scaling estimate, non-dimensionalize the equations by setting

$$(u, v) = U(\hat{u}, \hat{v}), \quad (x, y) = L(\hat{x}, \hat{y}), \quad f = f_0 \hat{f}, \quad z = H\hat{z}, \quad \phi = \Phi \hat{\phi}, \quad (3.5)$$

Here, f_0 and the terms in capital letters are constant representative values of the variables in lower case, and a hat over a variable (e.g., $\hat{z}$) indicates it is non-dimensional. H is a scaling for the height, and at this stage we will suppose it to be some height scale in the free atmosphere or ocean, not the height of the Ekman layer itself. Geostrophic balance suggests that $\Phi = f_0 UL$.

The resulting non-dimensional equations are

$$\hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\nabla \hat{\phi} + \left(\frac{A}{f_0 H^2} \right) \frac{\partial^2 \hat{\mathbf{u}}}{\partial \hat{z}^2} \quad (3.6)$$

or

$$\hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\hat{\nabla} \hat{\phi} + Ek \frac{\partial^2 \hat{\mathbf{u}}}{\partial \hat{z}^2} \quad (3.7)$$

The parameter $Ek = (A / f_0 H^2)$ is the *Ekman number*, and it determines the importance of frictional terms in the geostrophic balance equation. If $Ek \ll 1$ then the friction is small. However, since it multiplies the term of largest differential order, it cannot be neglected *a priori*, since it will in general be important near boundaries, no matter how small Ek is. If $Ek \geq 1$ friction is important everywhere. This is less interesting from a mathematical perspective and, more importantly, it is in fact generally not the case that Ek is large. Since the true value of A is not well known, the underlying reason for this subtlety; in part it is because A is determined by the flow field itself, rather than being an intrinsic property of the fluid. Near a surface the flow may be turbulent and hence A , and thence Ek , is large and so friction important. Away from a surface and the boundary, we do not expect such intense turbulence and so A may be smaller. This reasoning seems a little circular ('away from the boundary layer friction is small because we are away from the boundary layer') but not specious.

Suppose we know a value for A , and that it is constant, then how thick is the Ekman layer? Returning to the dimensional equations, the frictional term will balance the Coriolis force if the vertical scale is small enough. That is, if

$$\mathbf{f} \times \mathbf{u} \sim A \frac{\partial^2 \mathbf{u}}{\partial z^2}, \quad (3.8)$$

then

$$f_0 \sim \frac{A}{\delta^2}. \quad (3.9)$$

or

$$\delta \sim \left(\frac{A}{f_0} \right)^{1/2}. \quad (3.10)$$

where δ is the vertical scale of the Ekman layer.

If we define $\delta = (A / f_0)^{1/2}$ then the Ekman number may be written

$$Ek = \left(\frac{\delta}{H} \right)^2 \quad (3.11)$$

Thus, frictional effects are small in the free fluid if the depth of the frictional boundary layer given by (3.10) is small compared to the depth of the fluid layer itself. If $f = 10^{-4} \text{ s}^{-1}$ and $A = 10^{-1} \text{ m}^2 / \text{ s}$ then $\delta \sim 30 \text{ m}$. If the vertical scale H of motions in the atmosphere is 3 km then $Ek = 10^{-3}$. Let us invert this calculation for the atmosphere. Observations indicate that deviations from geostrophic balance are very weak above 1 km. Taking this as the height of the Ekman layer we obtain $A = 10^2 \text{ m}^2 \text{ s}^{-1}$, a value which applies of course (if at all) only in the Ekman layer itself.

3.1.2 Elementary boundary layer analysis

If the Ekman number is small, then since the Ekman layer is a boundary layer it is natural to apply boundary layer techniques, just as is done when looking at western boundary layers in the ocean (see chapter ??), or the thermocline (section ??). For definiteness, suppose there is a solid surface at $z = 0$, above which lies the fluid. Sufficiently far away from the boundary the velocity field is known, say $\mathbf{u}_i(x, y)$ where the subscript indicates flow in the *interior*. The corresponding interior pressure is given by geostrophic balance. We then write the velocity field and the pressure field as the sum of the interior geostrophic part plus a boundary layer correction:

$$\hat{\mathbf{u}} = \hat{\mathbf{u}}_g + \hat{\mathbf{u}}_e, \quad \hat{\phi} = \hat{\phi}_g + \hat{\phi}_e, \quad \hat{\mathbf{b}} = \hat{\mathbf{b}}_g + \hat{\mathbf{b}}_e \quad (3.12)$$

where the Ekman layer corrections, denoted with a subscript e , are negligible away from the boundary layer. That is, in the interior, geostrophic balance

$$\mathbf{f} \times \mathbf{u}_g = -\nabla \phi_g. \quad (3.13)$$

pertains. Now, in the boundary layer we anticipate that the vertical scales will be much smaller than H , and so the scaling of the previous section is not appropriate for the boundary layer correction. Rather, we *stretch* the vertical co-ordinate near the point of failure $z = 0$ by defining a new co-ordinate $\hat{\zeta}$ such that

$$\hat{z} = \epsilon \hat{\zeta} \quad (3.14)$$

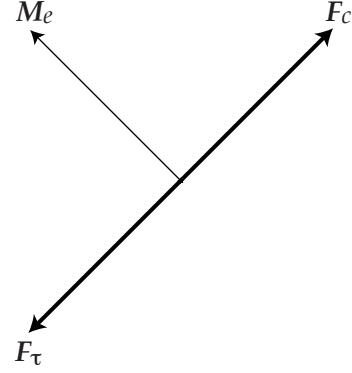
where ϵ is a small parameter, to be determined by the requirement that $\hat{\zeta} = O(1)$ in the boundary layer. Then $\hat{\mathbf{u}}_e = \hat{\mathbf{u}}_e(\hat{\mathbf{x}}, \hat{\zeta})$ and (3.7) becomes

$$\hat{\mathbf{f}} \times \hat{\mathbf{u}}_e = -\nabla \hat{\phi}_e + Ek \frac{\partial^2 \hat{\mathbf{u}}_i}{\partial \hat{z}^2} + \frac{Ek}{\epsilon^2} \frac{\partial^2 \hat{\mathbf{u}}_e}{\partial \hat{\zeta}^2} \quad (3.15)$$

and the hydrostatic equation becomes

$$\frac{\partial \hat{\phi}_e}{\partial \hat{z}} = 0. \quad (3.16)$$

Figure 3.2 Vertically integrated balance of within a bottom Ekman layer. M_e is the boundary layer transport, F_τ the integrated frictional force, proportional to the surface stress $\tau = A\partial\mathbf{u}/\partial z$, and F_C the Coriolis force on the ageostrophic transport. The surface stress is balanced only by the Coriolis force on the ageostrophic transport, implying that this transport is at right angles to the surface stress.



But since $\hat{\phi}_e$ is negligible in the interior ($\hat{\phi}(\hat{\mathbf{x}}, \zeta) \rightarrow 0$ as $\zeta \rightarrow \infty$) ϕ_e must be zero everywhere. This is an important result: *there is no boundary layer in the pressure field*. Note that this is a much stronger result than saying that pressure is continuous, which is nearly always true in fluids; rather, it is a special result about Ekman layers.

Given (3.16), and $Ek \ll 1$, (3.15) becomes

$$\hat{\mathbf{f}} \times \hat{\mathbf{u}}_e = \frac{Ek}{\epsilon^2} \frac{\partial^2 \hat{\mathbf{u}}_e}{\partial \hat{\zeta}^2}. \quad (3.17)$$

This tells us that the dominant force balance in the Ekman layer is between the Coriolis force and the friction. If we demand a nontrivial balance and that the terms on the right-hand-side are $O(1)$, then

$$\epsilon = E^{1/2} = \left(\frac{A}{f_0 H^2} \right)^{1/2} \quad (3.18)$$

The corresponding dimensional thickness of the Ekman layer is then given by

$$\delta = \epsilon H = \left(\frac{A}{f_0} \right)^{1/2}. \quad (3.19)$$

as before.

3.1.3 Integral properties of the Ekman layer

What can we deduce about the Ekman layer without specifying the detailed form of the frictional term? Reverting to dimensional notation, we recall frictional-geostrophic balance,

$$\mathbf{f} \times \mathbf{u} = -\nabla\phi + \frac{\partial \tilde{\tau}}{\partial z} \quad (3.20)$$

where $\tilde{\tau}$ is zero at the edge of the Ekman layer, and in the Ekman layer itself

$$\mathbf{f} \times \mathbf{u}_e = \frac{\partial \tilde{\tau}}{\partial z}. \quad (3.21)$$

Consider either a top or bottom Ekman layer, and integrate over its thickness. From (3.21) we obtain

$$\mathbf{f} \times \mathbf{M}_e = \tilde{\tau}_t - \tilde{\tau}_b \quad (3.22)$$

where

$$\mathbf{M}_e = \int_{\text{Ek}} \mathbf{u}_e \, dz \quad (3.23)$$

is the ageostrophic transport in the Ekman layer, and where $\tilde{\tau}_t$ and $\tilde{\tau}_b$ is the stress at the top and the bottom of the layer. The former (latter) will be zero in a bottom (top) Ekman layer. Taking the curl of (3.22) gives the simple but important results:

Top Ekman Layer:

$$\mathbf{M}_e = -\frac{1}{f} \mathbf{k} \times \tilde{\tau}_t \quad (3.24)$$

Bottom Ekman Layer:

$$\mathbf{M}_e = \frac{1}{f} \mathbf{k} \times \tilde{\tau}_b \quad (3.25)$$

The transport in the Ekman layer is at right-angles to the stress at the surface. This has a simple physical explanation: integrated over the depth of the Ekman layer the surface stress must be balanced by the Coriolis force, which in turn acts at right angles to the mass transport (see Fig. 3.2).

This result is particularly important in the top Ekman layer in the ocean. The stress at the top of the Ekman layer is primarily due to the wind, and can be regarded as given independent of the interior flow, and at the bottom of the Ekman layer the stress is zero. If f is positive, as in the Northern hemisphere, then an Ekman transport is induced 90° to the right of the direction of the wind stress (see Fig. 3.3). This has innumerable important consequences — for example, in inducing coastal upwelling when, as is not uncommon, the wind blows parallel to the coast.

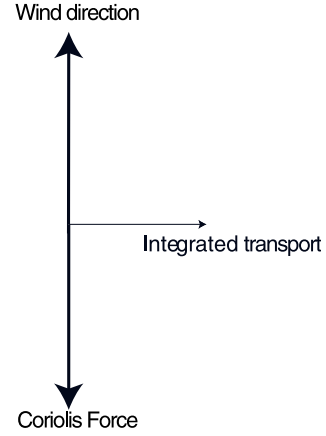
Finally, we obtain an expression for the vertical velocity induced by an Ekman layer. The mass conservation equation is

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0. \quad (3.26)$$

Integrating this over an Ekman layer gives

$$\nabla \cdot \mathbf{M}_t = -(w_t - w_b) \quad (3.27)$$

Figure 3.3 Vertically integrated balance of forces within an oceanic Ekman layer. The frictional force is given by the surface stress, and so is in the direction of the wind. For it to be in balance with the Coriolis force, the direction of the transport in the Ekman layer must be at right angles to the wind.



where $\mathbf{M}_t$ is the total (Ekman plus geostrophic) transport in the Ekman layer,

$$\mathbf{M}_t = \int_{\text{Ek}} \mathbf{u} \, dz = \int_{\text{Ek}} (\mathbf{u}_g + \mathbf{u}_e) \, dz = \mathbf{M}_g + \mathbf{M}_e, \quad (3.28)$$

and w_t and w_b are the vertical velocities at the top and bottom of the Ekman layer; the former (latter) is zero in a top (bottom) Ekman layer. From (3.22)

$$\mathbf{k} \times (\mathbf{M}_t - \mathbf{M}_g) = \frac{1}{f} (\tilde{\tau}_t - \tilde{\tau}_b). \quad (3.29)$$

Taking the curl of this (i.e., cross-differentiating) gives

$$\nabla \cdot (\mathbf{M}_t - \mathbf{M}_g) = \text{curl}_z [(\tilde{\tau}_t - \tilde{\tau}_b) / f] \quad (3.30)$$

Using (3.27) we obtain, for top and bottom Ekman layers separately,

$$w_b = \text{curl}_z \frac{\tilde{\tau}_t}{f} + \nabla \cdot \mathbf{M}_g \quad (3.31)$$

$$w_t = \text{curl}_z \frac{\tilde{\tau}_b}{f} - \nabla \cdot \mathbf{M}_g \quad (3.32)$$

where $\nabla \cdot \mathbf{M}_g = -\beta \mathbf{M}_g / f$ is the divergence of the geostrophic transport in the Ekman layer, which is often small compared to the other terms in these equations. These are perhaps the most used results in Ekman layer theory, and they say that friction induces a vertical velocity proportional to the curl of the stress at the surface. In addition to the fundamental importance of the results, numerical models sometimes do not have the vertical resolution to explicitly resolve an Ekman layer, and (3.31) and (3.32) provide a means of *parameterizing* the Ekman layer in terms of resolved or known fields.

3.1.4 Explicit solutions

We now assume an explicit form for the frictional terms in the momentum equations, namely

$$\mathbf{f} \times \mathbf{u} = -\nabla\phi + A \frac{\partial^2 \mathbf{u}}{\partial z^2} \quad (3.33)$$

or

$$\mathbf{f} \times (\mathbf{u} - \mathbf{u}_g) = A \frac{\partial^2 \mathbf{u}}{\partial z^2} \quad (3.34)$$

where

$$f(u_g, v_g) = \left(-\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y} \right). \quad (3.35)$$

We also assume there are no horizontal gradients of temperature, so that, via thermal wind, $\partial u_g / \partial z = \partial v_g / \partial z = 0$. It is now simply a matter of solving these equations, with appropriate boundary conditions.

Solution for a bottom Ekman layer

Appropriate boundary conditions are:

$$z = 0 : \quad u = 0, \quad v = 0 \quad (\text{the no slip condition}) \quad (3.36)$$

and

$$z \rightarrow \infty : \quad u = u_g, \quad v = v_g \quad (\text{a geostrophic interior}). \quad (3.37)$$

Let us assume solutions of the form

$$u = u_g + A_0 e^{\alpha z}, \quad (3.38)$$

$$v = v_g + B_0 e^{\alpha z} \quad (3.39)$$

where A_0 and B_0 are constants. Substituting into (3.34) gives two homogeneous algebraic equations

$$A_0 f - B_0 A \alpha^2 = 0 \quad (3.40)$$

$$A_0 A \alpha^2 - B_0 f = 0 \quad (3.41)$$

and thus the solvability condition

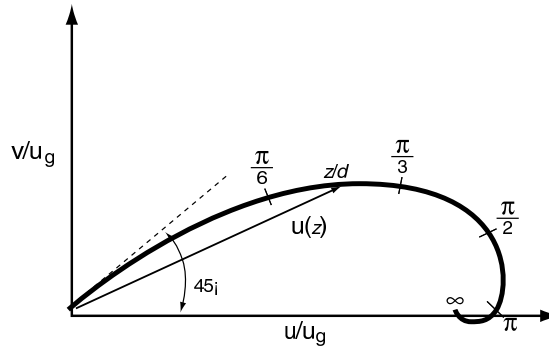
$$\alpha^4 = -\frac{A^2}{f^2}. \quad (3.42)$$

or $\alpha = \pm(1 \pm i)\sqrt{f/2A}$. Now using the boundary conditions we obtain the solution

$$u = u_g - e^{-z/d} [u_g \cos(z/d) + v_g \sin(z/d)] \quad (3.43a)$$

$$v = v_g + e^{-z/d} [u_g \sin(z/d) - v_g \cos(z/d)] \quad (3.43b)$$

Figure 3.4 The idealised Ekman layer solution in the lower atmosphere. The figure is a 'hodograph' of the wind components: the arrows show the velocity vectors at various heights, and the curve traces out the continuous variation of the velocity. The values on the curve are of the nondimensional variable αz , where $\alpha = (f/2A)^{1/2}$, and v_g is here chosen to be zero.



where $d = \sqrt{2A/f}$. This is, within a constant factor, the depth of the Ekman layer obtained from our earlier scaling considerations. The solution decays exponentially from the surface with this e-folding scale, so that the height d may be taken as the top of the Ekman layer. Note that the boundary layer correction depends on the interior flow, since the boundary layer serves to bring the flow to zero at the surface.

Suppose that the pressure force is directed in the y -direction (northward), so that the geostrophic current is eastward. Then the solution, the now-famous *Ekman spiral*, is plotted in Fig. 3.4. The wind falls to zero at the surface, and that its direction just above the surface is northeastward; that is, it is rotated by 45° to the right of its direction in the free atmosphere. Although this result is independent of the value of the frictional coefficients, it is dependent on the form of the friction chosen, a result that was appreciated by Ekman himself. The force balance in the Ekman layer is between the Coriolis force, the frictional or turbulent stress, and the pressure force. At the surface the Coriolis force is zero, and the balance is entirely between the northward pressure force and the southwards stress force.

Mass transport and force balance

The cross-isobaric flow is given by (for $v_g = 0$)

$$V = \int_0^\infty v \, dz = \int_0^\infty u_g e^{-z/d} \sin(z/d) \, dz = \frac{u_g d}{2} \quad (3.44)$$

For positive f , this is to the left of the geostrophic flow—that is, down the pressure gradient. In the general case ($v_g \neq 0$) we easily obtain

$$V = \int_0^\infty (v - v_g) \, dz = \frac{d}{2} (u_g - v_g) \quad (3.45)$$

Similarly, the additional zonal transport produced by frictional effects are, for $v_g = 0$,

$$U = \int_0^\infty (u - u_g) dz = - \int_0^\infty e^{-z/d} \sin(z/d) dz = -\frac{u_g d}{2} \quad (3.46)$$

and in the general case

$$U = \int_0^\infty (u - u_g) dz = -\frac{d}{2}(u_g + v_g) \quad (3.47)$$

So the total transport caused by frictional forces is

$$\mathbf{M}_e = \frac{d}{2} [-\mathbf{i}(u_g + v_g) + \mathbf{j}(u_g - v_g)] . \quad (3.48)$$

The total stress at the bottom surface $z = 0$ induced by frictional forces is

$$\tilde{\tau}_b = A \frac{\partial \mathbf{u}}{\partial z} = \frac{A}{d} [\mathbf{i}(u_g - v_g) + \mathbf{j}(u_g + v_g)] . \quad (3.49)$$

Thus, using (3.48), (3.49) and $d^2 = 2A/f_0$, the total frictionally induced transport is related to the stress at the surface by

$$\mathbf{M}_e = \frac{\mathbf{k} \times \tilde{\tau}_b}{f} \quad (3.50)$$

which is perpendicular to the frictional stress. Consistent with our previous more general analysis [c.f., (3.25)], this relationship is not dependent upon A , nor on the form of the frictional (turbulent flux) parameterization at all.

Explicit solution for the upper ocean

The wind provides a stress on the upper ocean, and the Ekman layer serves to communicate this to the oceanic interior. Appropriate boundary conditions are thus:

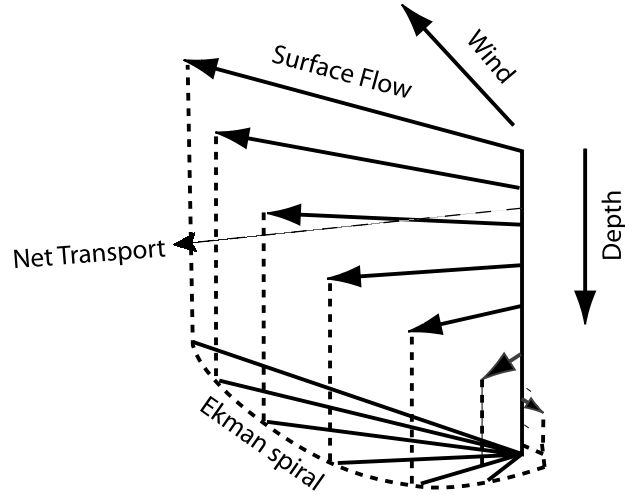
$$z = 0 : A \frac{\partial u}{\partial z} = \tau^x, \quad A \frac{\partial v}{\partial z} = \tau^y \quad (\text{the no slip condition}) \quad (3.51)$$

$$z \rightarrow -\infty : u \rightarrow u_g, \quad v \rightarrow v_g \quad (\text{a geostrophic interior}) \quad (3.52)$$

where τ is the given wind stress at the surface. Solutions to these differential equations are found as before, and are

$$u = u_g + \frac{\sqrt{2}}{fd} e^{z/d} [\tau^x \cos(z/d - \pi/4) - \tau^y \sin(z/d - \pi/4)] , \quad (3.53)$$

Figure 3.5 An idealized Ekman spiral in a southern-hemisphere ocean, driven by an imposed wind-stress. A northern hemisphere spiral would be the reflection of this about the vertical axis. Such a clean spiral is rarely, if ever, observed in the real ocean. The net transport is at right angles to the wind, independent of the detailed form of the friction. The angle of the surface flow is 45° to the wind only for a Newtonian viscosity.



and

$$v = v_g + \frac{\sqrt{2}}{fd} e^{z/d} [\tau^x \sin(z/d - \pi/4) - \tau^y \cos(z/d - \pi/4)] . \quad (3.54)$$

Note that the boundary layer correction depends only on the imposed surface stress, and not the interior flow itself. This is a consequence of the type of boundary conditions chosen, for in the absence of an imposed stress the boundary layer correction is zero—the interior flow already satisfies the gradient boundary condition at the top surface. Just as in the bottom boundary layer the velocity vectors of the solution trace a diminishing spiral as they descend into the interior (Fig. 3.5, which is drawn for the southern hemisphere).

Mass flux

The mass flux induced by the surface stress is obtained by integrating (3.53) and (3.54) from the surface to $-\infty$. We explicitly find

$$U = \int_{-\infty}^0 (u - u_g) dz = \frac{1}{f} \tau^y \quad (3.55)$$

and

$$V = \int_{-\infty}^0 (v - v_g) dz = -\frac{1}{f} \tau^x \quad (3.56)$$

or

$$\mathbf{f} \times \mathbf{M}_e = \boldsymbol{\tau}_s \quad (3.57)$$

indicating that the ageostrophic mass transport perpendicular to the wind-stress, as we obtained previously from more general considerations.

We note finally that the flow right at the surface is 45° to the right of (in the Northern hemisphere) the imposed surface wind. Suppose for simplicity that the geostrophic flow is zero, and that the surface wind is northward. Then $\tau^x = 0$ and the solutions immediately give

$$\begin{aligned} u(0) &= (\tau^y / fd) \cos(\pi/4) \\ v(0) &= (\tau^y / fd) \sin(\pi/4). \end{aligned} \quad (3.58)$$

Therefore the magnitudes of the flow in the x - and y - directions are equal to each other, and the flow is 45° to the right of the wind. However, this result is dependent on the form of the frictional parameterization chosen.

Vertical velocity at the edge of the Ekman layer

We recall that, at the edge of either a top or bottom Ekman layer, the vertical velocity induced by frictional effects is given by

$$w = \text{curl}_z \frac{\tilde{\tau}}{f} \quad (3.59)$$

In the top Ekman layer of the ocean the stress does not, to a very good approximation, depend on the interior geostrophic flow in the ocean. Then the vertical velocity at the base of the Ekman layer is proportional to the curl of the wind-stress. (It is usually, but not always, the case that the wind-stress varies more rapidly than the Coriolis parameter, so that f can be extracted from the *curl* operator in (3.59). Indeed, in the quasi-geostrophic approximation, f must be taken as constant there.)

In the bottom Ekman layer the surface stress is a function of the interior geostrophic flow. To obtain the form we return to the explicit solution and hold the Coriolis parameter fixed. Using either (3.48) (or (3.49)) we obtain

$$W = -\nabla \cdot \mathbf{M} = \text{curl}_z \tilde{\tau}_b / f = V_x - U_y = \frac{d}{2} \zeta_g$$

where ζ_g is the vorticity of the geostrophic flow. Thus, the vertical velocity at the top of the Ekman layer is:

- Proportional to the geostrophic vorticity in the free fluid;
- Proportional to the Ekman layer height $\sqrt{2A/f}$;
- Proportional to the magnitude of the cross-isobaric flow.

The induced vertical velocity at the edge of the Ekman layer is, so far as the large-scale circulation of either atmosphere or ocean, the most important aspect of the Ekman layer. Boundary layer fluxes are communicated to the free interior fluid quite efficiently via a secondary circulation, and this is one of the most notable differences between rotating and non-rotating fluids (Fig. 3.6).

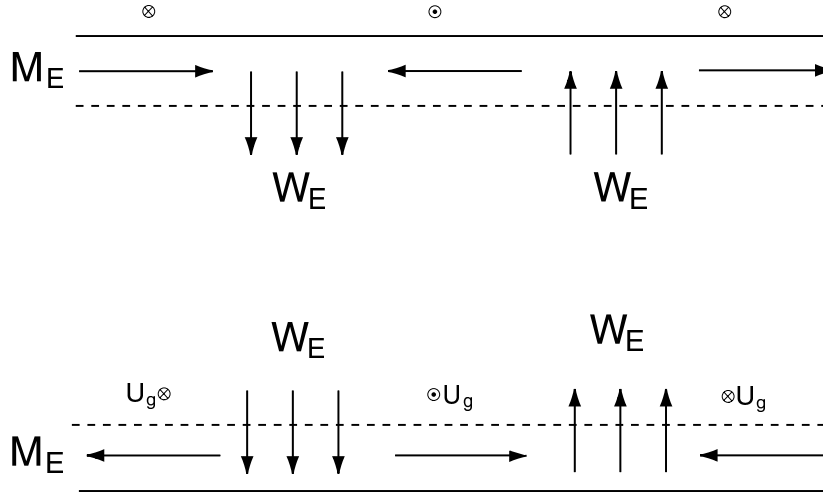


Figure 3.6 Upper and lower Ekman layers. The upper Ekman layer in the ocean is primarily driven by the imposed wind stress, τ . The lower Ekman layer in the atmosphere or ocean results from an interior geostrophic velocity, and acts as a drag on the flow. The upper part of figure shows the resulting vertical Ekman ‘pumping’ velocities for the given wind stress, and the lower part the Ekman pumping interior for a given velocity field. Only in unrealistic circumstances ($\beta = 0$ and barotropic flow) are the upper and lower Ekman layers necessarily aligned as in the figure.

3.1.5 Frictional parameterization

Suppose that the free atmosphere is described by the barotropic quasi-geostrophic vorticity equation,

$$\frac{D\zeta_g}{Dt} = f \frac{\partial w}{\partial z} \quad (3.60)$$

Then, integrating from the top of the Ekman layer to the tropopause (say at a height H where $w = 0$), then

$$\frac{D\zeta_g}{Dt} = \frac{f[w(H) - w(d)]}{H - d} \quad (3.61)$$

$$= -\frac{fd}{H - d} \zeta_g \quad (3.62)$$

This equation shows that the Ekman layer acts as a *linear drag* on the interior flow, with a ‘drag coefficient’ r given by fd/H (assuming $H \gg d$), and associated timescale τ_{Ek} given by

$$\tau_{Ek} = \frac{H}{fd} = \frac{H}{\sqrt{2fA}} \quad (3.63)$$

In the oceanic case the corresponding vorticity equation for the interior flow is

$$\frac{D\zeta_g}{Dt} = \frac{1}{H} \text{curl}_z \tau_s \quad (3.64)$$

where τ_s is the surface stress. The surface stress thus acts as if it were a body force on the interior flow. Interestingly, neither the Coriolis parameter nor the depth of the Ekman layer explicitly appear in this formula.

The secondary circulation induced by the Ekman layer is a much more efficient way of communicating surface stresses to the interior of the fluid than turbulent mixing alone. Note that of course we are assuming some level of turbulent mixing by using an eddy mixing coefficient A . But suppose that eddy mixing were the sole mechanism of transferring stress from the surface to the free atmosphere. Then the timescale of spindown of the free atmosphere is given by using

$$\frac{D\zeta_g}{Dt} = A \frac{\partial^2 \zeta}{\partial z^2} \quad (3.65)$$

implying a turbulent spindown time, τ_t of

$$\tau_t \sim \frac{H^2}{A} \quad (3.66)$$

where H is the depth over which we require a spin-down. Then

$$\frac{\tau_{\text{Ek}}}{\tau_t} = \frac{(H/fd)}{(H^2/A)} = \frac{d}{2H}, \quad (3.67)$$

using $d = \sqrt{2A/f}$. Thus, provided that the Ekman layer depth is shallower than the thickness of the free atmosphere, a secondary Ekman circulation is proportionally more efficient than simple turbulent diffusion in providing a frictional spin-down to that free atmosphere. Similar considerations apply to the way a surface stress is communicated to the oceanic interior.

3.2 * THE NEAR SURFACE LAYER

This section may be omitted in the final version.

3.2.1 The logarithmic boundary layer

Very close to the surface, especially in the atmosphere, it seems unlikely that rotational effects are directly important: the Rossby number is too high. Further, it also seems likely that the concept of a constant eddy diffusivity is wrong, since an eddy diffusivity implies a mixing length which is surely not constant. However, there may be vigorous mixing taking place so that the stress is constant throughout this layer. Absent other dynamical scales, there is no reason to assume any characteristic size of the eddies other than the distance from the surface itself. The size of the eddies, l , may then determine the mixing length in a coefficient of eddy viscosity (i.e., $A \propto l \times V$).

Thus, let us explicitly make two assumptions:

1. The stress is constant throughout the mixed layer.
2. The mixing can be parameterized by the use of an eddy mixing coefficient, the size of which is proportional to the distance from the surface. S

The magnitude of the stress is given by

$$\tau = \rho_0 \left[\overline{(u'w')^2} + \overline{(v'w')^2} \right]^{1/2}. \quad (3.68)$$

Since the stress is assumed constant, we can define the constant 'frictional velocity' u^* by

$$u^{*2} \equiv \frac{\tau}{\rho_0}. \quad (3.69)$$

Now using the second assumption the stress is given by

$$\tau = \rho_0 A \frac{\partial u}{\partial z} \quad (3.70)$$

where the eddy mixing coefficient is

$$A = u^* k z \quad (3.71)$$

where k is a (supposedly) universal constant (von Karman's constant) and u^* is the frictional velocity.

Using (3.69), (3.70) and (3.71) we obtain

$$\frac{\partial u}{\partial z} = \frac{u^*}{Az} \quad (3.72)$$

which integrates to give

$$u = \frac{u^*}{k} \ln(z/z_0). \quad (3.73)$$

where z_0 is a constant of integration. This is the classical logarithmic velocity law. In the atmosphere it might exist below an Ekman layer, perhaps in the lowest 100 m or so of a stably-stratified atmosphere, but plainly, just from inspection of its mathematical form. it cannot extend all the way to the surface.

Problems

- 3.1 Consider a wind stress imposed by a mesoscale cyclonic storm (in the atmosphere) given by

$$\boldsymbol{\tau} = -Ae^{-(r/\lambda)^2} (\mathbf{y}\mathbf{i} + \mathbf{x}\mathbf{j}) \quad (\text{P3.1})$$

where $r^2 = x^2 + y^2$, and A and λ are constants. Also assume constant Coriolis gradient $\beta = \partial f / \partial y$ and constant ocean depth H . Find (a) the Ekman transport, (b) the vertical velocity $w_e(x, y, z)$ below the Ekman layer, (c) the northward velocity $v(x, y, z)$ below the Ekman layer and (d) indicate how you would find the westward velocity $u(x, y, z)$ below the Ekman layer.

All real fluid motions are rotational.
Clifford Truesdell, *The Kinematics of Vorticity*, 1954

CHAPTER FOUR

Vorticity and Potential Vorticity

VORTICITY AND POTENTIAL VORTICITY both play a central role in geophysical fluid dynamics — indeed, we shall find that the large scale circulation of the ocean and atmosphere is in large-part governed by the evolution of the latter. In this chapter we define these quantities and deduce some of their dynamical properties. Along the way we will come across *Kelvin’s circulation theorem*, one of the most fundamental conservation laws in all of fluid mechanics, and we will find that the conservation of potential vorticity is intimately tied to this.

4.1 VORTICITY AND CIRCULATION

4.1.1 Preliminaries

Vorticity is defined to be the curl of velocity, and is normally denoted by the symbol ω . Thus

$$\omega \equiv \nabla \times \mathbf{v}. \quad (4.1)$$

Circulation is defined to be the integral of velocity around a closed fluid loop. That is,

$$C = \oint \mathbf{v} \cdot d\mathbf{l}, \quad (4.2)$$

which by Stokes' theorem can be written as

$$C = \int_S \boldsymbol{\omega} \cdot d\mathbf{S}. \quad (4.3)$$

where S is any surface bounded by the loop. That is, the circulation around some path is equal to the integral of the normal component of vorticity over *any* surface bounded by that path. The circulation is not a field like vorticity and velocity. Rather, we think of the circulation around a particular material line of finite length, and so its value generally depends on the path chosen. If δS is an infinitesimal surface element whose normal points in the direction of the unit vector $\hat{\mathbf{n}}$, then

$$\hat{\mathbf{n}} \cdot (\nabla \times \mathbf{v}) = \frac{1}{\delta S} \oint_{\delta l} \mathbf{v} \cdot d\mathbf{l} \quad (4.4)$$

where the line integral is around the infinitesimal area. Thus at a point the component of vorticity in the direction of $\mathbf{n}$ is proportional to the circulation around the surrounding infinitesimal fluid element, divided by the elemental area bounded by the path of the integral. A heuristic test for the presence of vorticity is then to imagine a small paddle-wheel in the flow: the paddle wheel acts as a 'circulation-meter' and so rotates if the vorticity is non-zero.

4.1.2 Simple axi-symmetric examples

Consider axi-symmetric motion in two dimensions, so that the flow is confined to a plane. We use cylindrical co-ordinates (r, ϕ, z) where z is the direction perpendicular to the plane. Then

$$\begin{aligned} u_z &= u_r = 0 \\ u_\phi &\neq 0. \end{aligned} \quad (4.5)$$

Rigid Body Motion

The velocity distribution is given by

$$u_\phi = \Omega r \quad (4.6)$$

and associated with this is the vorticity

$$\boldsymbol{\omega} = \nabla \times \mathbf{v} = \omega_z \mathbf{k}, \quad (4.7)$$

where

$$\omega_z = \frac{1}{r} \frac{\partial}{\partial r} (r u_\phi) = \frac{1}{r} \frac{\partial}{\partial r} (r^2 \Omega) = 2\Omega. \quad (4.8)$$

The vorticity of a fluid in solid body rotation is thus twice the angular velocity of the fluid, and is pointed in a direction orthogonal to the plane of rotation.

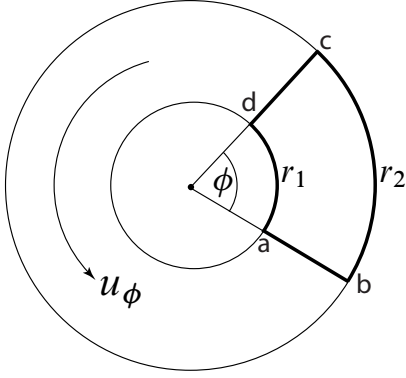


Figure 4.1 Evaluation of circulation in the axis-symmetric vr vortex. The circulation around path $a-b-c-d-a$ is zero. This result does not depend on the size of the path, and so the circulation around any infinitesimal path not enclosing the origin is zero. Thus the vorticity is zero everywhere except at the origin.

The ' vr ' vortex

This vortex is so-called because the tangential velocity distribution is such that the product vr is constant. That is:

$$u_\phi = \frac{K}{r} \quad (4.9)$$

where K is a constant determining the vortex strength. Evaluating the z -component of vorticity gives

$$\omega_z = \frac{1}{r} \frac{\partial}{\partial r} (ru_\phi) = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{K}{r} \right) = 0 \quad (4.10)$$

except where $r = 0$, at which the expression is singular and the vorticity is infinite. Obviously the paddle wheel rotates when placed at the vortex center, but, less obviously, does not if placed elsewhere.

We can also obtain the result by calculating the circulation $\oint \mathbf{u} \cdot d\mathbf{l}$ around an appropriate contour. If the contour is a circle that encloses the origin, the circulation is given by

$$C_o = \oint \frac{K}{r} r d\phi = 2\pi K \quad (4.11)$$

This does not depend on the radius, and so it is true if the radius is infinitesimal, provided the contour encloses the origin. Since the vorticity is the circulation divided by the area, the vorticity at the origin must be infinite. Consider now a contour that does not enclose the origin, for example the contour $a-b-c-d-a$ in Fig. 4.1. Over the segments $a-b$ and $c-d$ the velocity is orthogonal to the contour, and so the contribution is zero. Over $b-c$ and $d-a$ we have

$$C_{bc} = \frac{K}{r_2} \theta r_2 = K\phi, \quad C_{da} = -\frac{K}{r_1} \theta r_1 = -K\phi \quad (4.12)$$

The net circulation around the contour $C_{abcd a}$ is zero. If we shrink this contour to an infinitesimal size then within this contour, by Stokes' theorem, the vorticity is zero. Thus the vorticity is everywhere zero, except at the origin.

4.2 THE VORTICITY EQUATION

Using the vector identity,

$$\mathbf{v} \times (\nabla \times \mathbf{v}) = \frac{1}{2} \nabla(\mathbf{v} \cdot \mathbf{v}) - (\mathbf{v} \cdot \nabla) \mathbf{v}, \quad (4.13)$$

we can write the momentum equation as

$$\frac{\partial \mathbf{v}}{\partial t} + \boldsymbol{\omega} \times \mathbf{v} = -\frac{1}{\rho} \nabla p - \frac{1}{2} \nabla v^2 + \nu \nabla^2 \mathbf{v}. \quad (4.14)$$

Taking the curl of this gives the vorticity equation

$$\frac{\partial \boldsymbol{\omega}}{\partial t} + \nabla \times (\boldsymbol{\omega} \times \mathbf{v}) = \frac{1}{\rho^2} (\nabla \rho \times \nabla p), \quad (4.15)$$

where here, and in most of the rest of this chapter, we ignore viscosity.

Now, the vector identity

$$\nabla \times (\mathbf{a} \times \mathbf{b}) = (\mathbf{b} \cdot \nabla) \mathbf{a} - (\mathbf{a} \cdot \nabla) \mathbf{b} + \mathbf{a} \nabla \cdot \mathbf{b} - \mathbf{b} \nabla \cdot \mathbf{a}, \quad (4.16)$$

implies that the second term on the left hand side of (4.15) may be written

$$\nabla \times (\boldsymbol{\omega} \times \mathbf{v}) = (\mathbf{v} \cdot \nabla) \boldsymbol{\omega} - (\boldsymbol{\omega} \cdot \nabla) \mathbf{v} + \boldsymbol{\omega} \nabla \cdot \mathbf{v} - \mathbf{v} \nabla \cdot \boldsymbol{\omega}. \quad (4.17)$$

Because vorticity is the curl of velocity, its divergence vanishes, whence we obtain

$$\frac{\partial \boldsymbol{\omega}}{\partial t} + (\mathbf{v} \cdot \nabla) \boldsymbol{\omega} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{v} - \boldsymbol{\omega} \nabla \cdot \mathbf{v} + \frac{1}{\rho^2} (\nabla \rho \times \nabla p). \quad (4.18)$$

The divergence term may be eliminated with the aid of the mass-conservation equation

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0 \quad (4.19)$$

to give

$$\boxed{\frac{D\tilde{\boldsymbol{\omega}}}{Dt} = (\tilde{\boldsymbol{\omega}} \cdot \nabla) \mathbf{v} + \frac{1}{\rho^3} (\nabla \rho \times \nabla p)} \quad (4.20)$$

where $\tilde{\boldsymbol{\omega}} = \boldsymbol{\omega} / \rho$.

The last term on the right-hand side of (4.18), as well as the term on the right-hand side of (4.20), is called the *baroclinic* term, the *non-homentropic* term, or the *solenoidal* term. The solenoidal vector is defined by

$$\mathbf{S} = \frac{1}{\rho^2} \nabla \rho \times \nabla p = -\nabla \alpha \times \nabla p \quad (4.21)$$

A solenoid is a tube directed perpendicular to both $\nabla \alpha$ and ∇p , with elements of length proportional to $\nabla p \times \nabla \alpha$. If the isolines of p and α are parallel to each

other, then solenoids do not exist. This occurs when the density is a function only of pressure for then

$$\nabla \rho \times \nabla p = \nabla \rho \times \nabla \rho \frac{dp}{d\rho} = 0. \quad (4.22)$$

The solenoidal vector may also be written

$$\mathbf{S} = -\nabla \eta \times \nabla T. \quad (4.23)$$

(This follows most easily by first writing the momentum equation in the form $\partial \mathbf{v} / \partial t + \boldsymbol{\omega} \times \mathbf{v} = T \nabla \eta - \nabla B$, and taking its curl; see problem 2.2.) Evidently, then, solenoidal term vanishes if: (i) isolines of pressure and density are parallel; (ii) isolines of temperature and entropy are parallel; (iii) density or entropy or temperature or pressure are constant. A *barotropic* fluid has by definition $p = p(\rho)$ and therefore no solenoids. A *baroclinic* fluid is one for which ∇p is not parallel to $\nabla \rho$.

For a barotropic fluid the vorticity equation takes the simple and elegant form,

$$\frac{D\tilde{\boldsymbol{\omega}}}{Dt} = (\tilde{\boldsymbol{\omega}} \cdot \nabla) \mathbf{v}. \quad (4.24)$$

If the fluid is also incompressible, meaning that $\nabla \cdot \mathbf{v} = 0$, then we have the even simpler form,

$$\frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{v}. \quad (4.25)$$

The terms on the right-hand-side of (4.24) or (4.25) are conventionally divided into ‘stretching’ and ‘tipping’ (or ‘tilting’) terms, and we return to these in in section ??.

An integral conservation property

Consider a single Cartesian component in (4.18). Then

$$\begin{aligned} \frac{\partial \omega_x}{\partial t} &= -\mathbf{v} \cdot \nabla \omega_x - \omega_x \nabla \cdot \mathbf{v} + (\boldsymbol{\omega} \cdot \nabla) v_x + S_x \\ &= -\nabla \cdot (\mathbf{v} \omega_x) + \nabla \cdot (\boldsymbol{\omega} v_x) + S_x \end{aligned} \quad (4.26)$$

where S_x is the (x -component of) the solenoidal term. This may be written as

$$\frac{\partial \omega_x}{\partial t} + \nabla \cdot (\mathbf{v} \omega_x - \boldsymbol{\omega} v_x) = S_x, \quad (4.27)$$

and this implies the Cartesian tensor form of the vorticity equation, namely

$$\frac{\partial \omega_i}{\partial t} + \frac{\partial}{\partial x_j} (v_j \omega_i - v_i \omega_j) = S_i \quad (4.28)$$

with summation over repeated indices. The tendency of vorticity is given by the solenoidal term plus the divergence of a vector field, and thus if the former vanishes the volume integrated vorticity can only be altered by boundary effects. In both atmosphere and ocean the solenoidal term is important, but we will see in section 4.5 that a useful conservation law for a scalar quantity can still be obtained.

4.2.1 Two-dimensional flow

In two-dimensional flow the fluid is confined to a surface, and independent of the third dimension normal to that surface. Let us initially stay in a Cartesian geometry, and then two-dimensional flow is flow on a plane, and the velocity normal to the plane, and the rate of change of any quantity normal to that plane, are zero. Let the normal direction be the z -direction and then the velocity in the plane, denoted by $\mathbf{u}$, is

$$\mathbf{v} = \mathbf{u} = u\mathbf{i} + v\mathbf{j}, \quad w = 0. \quad (4.29)$$

Only one component of vorticity non-zero and this is given by

$$\boldsymbol{\omega} = \mathbf{k} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right). \quad (4.30)$$

That is, in two-dimensional flow the vorticity is perpendicular to the velocity. We let $\zeta \equiv \omega_z = \boldsymbol{\omega} \cdot \mathbf{k}$. Both the stretching and tilting terms vanish in two-dimensional flow, and the two-dimensional vorticity equation becomes, for incompressible flow,

$$\frac{D\zeta}{Dt} = 0, \quad (4.31)$$

where $D\zeta/Dt = \partial\zeta/\partial t + \mathbf{u} \cdot \nabla\zeta$. That is, in two-dimensional flow vorticity is conserved following the fluid elements; each material parcel of fluid keeps its value of vorticity even as it is being advected around. Furthermore, specification of the vorticity completely determines the flow field. To see this, we use the incompressibility condition to define a streamfunction ψ such that

$$u = -\frac{\partial\psi}{\partial y}, \quad v = \frac{\partial\psi}{\partial x}, \quad (4.32)$$

and

$$\zeta = \nabla^2\psi. \quad (4.33)$$

Given the vorticity, the Poisson equation (4.33) can be solved for the streamfunction and the velocity fields obtained through (4.32), and this process is called ‘inverting the vorticity’.

Numerical integration of (4.31) is then a process of time-stepping plus inversion. The vorticity equation may then be written as an advection equation for vorticity,

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta = 0 \quad (4.34)$$

in conjunction with (4.32) and (4.33). The vorticity is stepped forward one time-step using a finite-difference representation of (4.34), and the vorticity inverted to obtain a velocity using (4.33) and (4.32). (The notion that complete or nearly complete information about the flow may be obtained by inverting one field plays an important role in geophysical fluid dynamics, as we will see later on.)

In the presence of rotation the two-dimensional vorticity equation becomes

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla (\zeta + f) = 0 \quad (4.35)$$

where $f = 2\boldsymbol{\Omega} \cdot \mathbf{k}$. This may be written

$$\frac{D}{Dt} (\zeta + f) = 0, \quad (4.36)$$

implying that the absolute vorticity, $\zeta_a \equiv \zeta + f$, is materially conserved. If f is a constant, then (4.35) reduces to (4.34), and background rotation plays no role. If f varies linearly with y , so that $f = f_0 + \beta y$, then (4.35) becomes

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta + \beta v = 0 \quad (4.37)$$

which is known as the β -plane vorticity equation.

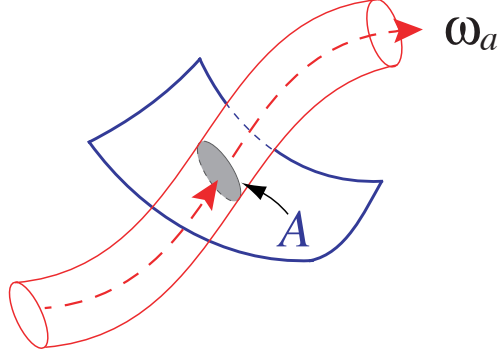
Finally, we note that two-dimensional flow is not restricted to a single, Cartesian plane, and we may certainly envision two-dimensional flow on the surface of a sphere. In this case, the velocity normal to the spherical surface (the ‘vertical velocity’) vanishes. However, aside from their expression in spherical coordinates the equations of motion are largely unchanged, and in particular vorticity (absolute vorticity if the sphere is rotating) is conserved on parcels as they move over the spherical surface.

4.3 VORTICITY AND CIRCULATION THEOREMS

4.3.1 The ‘frozen-in’ property of vorticity

Let us first consider some simple topological properties of the vorticity field and its evolution. We define a *vortex-line* to be a line drawn through the fluid which is everywhere in the direction of the local vorticity. This definition is analogous to that of a streamline, which is everywhere in the direction of the local velocity. A *vortex-tube* is formed by the collection of vortex lines passing through a closed

Figure 4.2 A vortex tube passing through a material sheet. The circulation is the integral of the velocity around the boundary of A , and is equal to the integral of the normal component of vorticity over A .



curve (Fig. 4.2). A *material-line* is just a line that connects material fluid elements. Suppose we draw a vortex line through the fluid; such a line obviously connects fluid elements and therefore defines a co-incident material line. As the fluid moves the material line deforms, and the vortex line also evolves in a manner determined by the equations of motion. A remarkable property of vorticity is that, for an unforced and inviscid barotropic fluid, the flow evolution is such that a vortex line remains co-incident with the same material line with which it was initially associated. Put another way, a vortex line always contains the same material elements — the vorticity is ‘frozen’ to the material fluid.

To prove this we consider how an infinitesimal material line element $\delta \mathbf{l}$ evolves, $\delta \mathbf{l}$ being the infinitesimal material element connecting $\mathbf{l}$ with $\mathbf{l} + \delta \mathbf{l}$. The rate of change of $\delta \mathbf{l}$ following the flow is given by

$$\frac{D\delta \mathbf{l}}{Dt} = \frac{1}{\delta t} (\delta \mathbf{l}(t + \delta t) - \delta \mathbf{l}(t)), \quad (4.38)$$

which follows from the definition of the material derivative in the limit $\delta t \rightarrow 0$. From the Taylor expansion of $\delta \mathbf{l}(t)$ and the definition of velocity it is also apparent that

$$\begin{aligned} \delta \mathbf{l}(t + \delta t) &= \mathbf{l}(t) + \delta \mathbf{l}(t) + (\mathbf{v} + \delta \mathbf{v})\delta t - (\mathbf{l}(t) + \mathbf{v}\delta t) \\ &= \delta \mathbf{l} + \delta \mathbf{v}\delta t, \end{aligned} \quad (4.39)$$

as illustrated in Fig. 4.3. Substituting into (4.38) gives

$$\frac{D\delta \mathbf{l}}{Dt} = \delta \mathbf{v} \quad (4.40)$$

But since $\delta \mathbf{v} = (\delta \mathbf{l} \cdot \nabla) \mathbf{v}$ we have that

$$\frac{D\delta \mathbf{l}}{Dt} = (\delta \mathbf{l} \cdot \nabla) \mathbf{v} \quad (4.41)$$

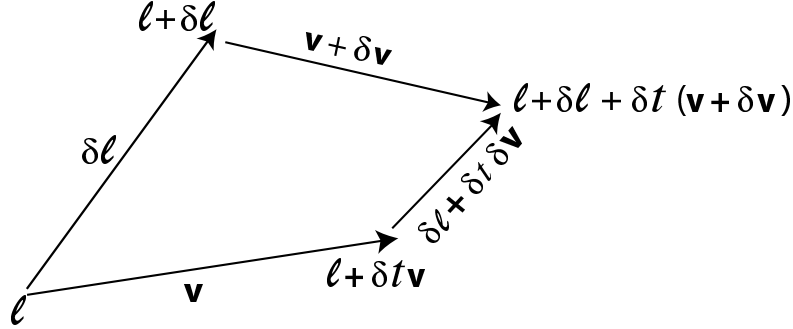


Figure 4.3 Evolution of an infinitesimal material line δl from time t to time $t + \delta t$. It can be seen from the diagram that $D\delta l/Dt = \delta v$.

Comparing this with (4.20), we see that vorticity evolves in the same way as a line element. To see what this means, at some initial time we can define an infinitesimal material line element parallel to the vorticity at that location, that is,

$$\delta l(\mathbf{x}, t) = A\boldsymbol{\omega}(\mathbf{x}, t) \quad (4.42)$$

where A is a (dimensional) constant. Then, for all subsequent times, the magnitude of the vorticity of that fluid element (wherever that particular element may be in the fluid) remains proportional to its length, and is in that direction, i.e.,

$$\boldsymbol{\omega}(\mathbf{x}', t') = A^{-1}\delta l(\mathbf{x}', t'). \quad (4.43)$$

To see the same result in a slightly different way note that a vortex line element is determined by the condition $\delta l = A\boldsymbol{\omega}$ or, because A is just an arbitrary scaling factor, $\boldsymbol{\omega} \times \delta l = 0$. Now, for any line element we have that

$$\frac{D}{Dt}(\boldsymbol{\omega} \times \delta l) = \frac{D\boldsymbol{\omega}}{Dt} \times \delta l - \frac{D\delta l}{Dt} \times \boldsymbol{\omega}. \quad (4.44)$$

We also have that

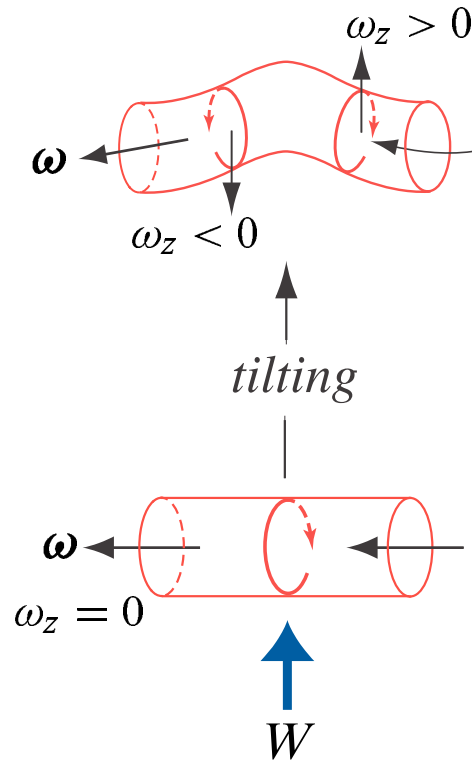
$$\frac{D\delta l}{Dt} = \delta v = \delta l \cdot \nabla v \quad (4.45a)$$

and

$$\frac{D\boldsymbol{\omega}}{Dt} = \boldsymbol{\omega} \cdot \nabla v. \quad (4.45b)$$

If the line element is initially a vortex line element then, at $t = 0$, $\delta l = A\boldsymbol{\omega}$ and, using (4.45), the right hand side of (4.44) vanishes. Thus, the tendency of $\boldsymbol{\omega} \times \delta l$ is zero, and the vortex line continues to be a material line.

Figure 4.4 The tilting of vorticity. Suppose that vorticity, ω is initially directed horizontally, as in the lower figure, so that ω_z , its vertical component, is zero. The material lines, and therefore the vortex lines also, are tilted by the positive vertical velocity W , so creating a vertically oriented vorticity. This mechanism is important in creating vertical vorticity in the atmospheric boundary layer (and, one may show, the β -effect in large-scale flow).



Stretching and tilting

The frozen-in property of vorticity allows us to interpret the terms on the right-hand side of (4.24) or (4.25) in terms of ‘stretching’ and ‘tipping’ (or ‘tilting’). Consider a single Cartesian component of (4.25),

$$\frac{D\omega_x}{Dt} = \omega_x \frac{\partial u}{\partial x} + \omega_y \frac{\partial u}{\partial y} + \omega_z \frac{\partial u}{\partial z}. \quad (4.46)$$

The second and third terms of this are the tilting or tipping terms because they involve changes in the orientation of the vorticity vector. They tell us that vorticity in x -direction may be generated from vorticity in the y - and z -directions if the advection acts to tilt the material lines. Because vorticity is tied to these lines, vorticity oriented in one direction becomes oriented in another, and in Fig. 4.4 we see how vorticity in the z -direction may be created by the tilting of vortex lines that initially have a purely horizontal orientation.

The first term on the right-hand-side of (4.46) is the stretching term, and it acts to intensify the x -component of vorticity if the velocity is increasing in the x -direction — that is, if the material lines are being ‘stretched’ (Fig. 4.5). This effect again arises because a vortex line is tied to a material line, and therefore vorticity is amplified if the material line aligned with it is stretched, and in proportion to

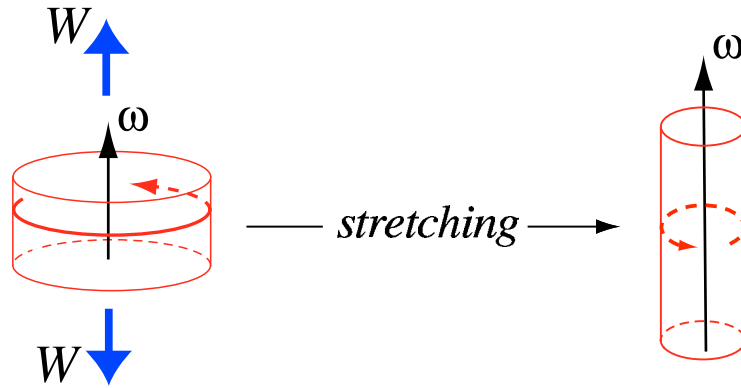


Figure 4.5 Stretching of material lines distorts the cylinder of fluid as shown. Vorticity is tied to material lines, and so is amplified in the direction of the stretching. However, because the volume of fluid is conserved, the end surfaces shrink, the material lines through the cylinder ends converge, and the integral of vorticity over a material surface (the circulation) remains constant, as discussed in section 4.3.2.

that stretching. This effect leads to the amplification of vorticity in tornadoes, to give one example.

If the fluid is incompressible the volume of a mass of material fluid remains fixed, so that stretching in one direction is accompanied by convergence in another, and this leads to the conservation of circulation, as discussed in the next section.

4.3.2 Kelvin's Circulation Theorem

Kelvin's circulation theorem states that under certain circumstances the circulation around a material fluid parcel is conserved; that is, the circulation is conserved 'following the flow'. The primary restrictions are that body forces are conservative (i.e., they are representable as potential forces, and therefore that the flow be inviscid), and that the fluid is barotropic [i.e., $p = p(\rho)$]. Of these, the latter is the more restrictive for geophysical fluids.

The circulation in the theorem is defined with respect to the inertial frame of reference. That is, the velocity in (4.50) is the velocity relative to the inertial frame. We will give a straightforward proof, beginning with the inviscid momentum equation:

$$\frac{D\mathbf{v}}{Dt} = -\frac{1}{\rho}\nabla p - \nabla\Phi, \quad (4.47)$$

where $\nabla\Phi$ represents the conservative body forces on the system. Applying the

material derivative to the circulation, (4.2), gives

$$\begin{aligned}
 \frac{DC}{Dt} &= \frac{D}{Dt} \oint \mathbf{v} \cdot d\mathbf{r} = \oint \left(\frac{D\mathbf{v}}{Dt} \cdot d\mathbf{r} + \mathbf{v} \cdot d\mathbf{v} \right) \\
 &= \oint \left[\left(-\frac{1}{\rho} \nabla p - \nabla \Phi \right) \cdot d\mathbf{r} + \mathbf{v} \cdot d\mathbf{v} \right] \\
 &= \oint -\frac{1}{\rho} \nabla p \cdot d\mathbf{r}
 \end{aligned} \tag{4.48}$$

using the momentum equation and $D(\delta l)/Dt = \delta \mathbf{v}$. The line integration is over a closed, material, circuit. The second and third terms on the second line vanish separately, because they are exact differentials integrated around a closed loop. The term on the last line vanishes if density is constant or, more generally, if pressure is a function of density alone in which case ∇p is parallel to $\nabla \rho$. To see this, note that

$$\oint \frac{1}{\rho} \nabla p \cdot d\mathbf{r} = \int_S \nabla \times \left(\frac{\nabla p}{\rho} \right) \cdot d\mathbf{A} = \int_A \frac{-\nabla \rho \times \nabla p}{\rho^2} \cdot d\mathbf{A}, \tag{4.49}$$

using Stokes's theorem where A is any surface bounded by the path of the line integral, and this evidently vanishes identically if p is a function of ρ alone. The last term is the integral of the solenoidal vector, and if it is zero (4.48) becomes

$$\boxed{\frac{D}{Dt} \oint \mathbf{v} \cdot d\mathbf{r} = 0} . \tag{4.50}$$

This is Kelvin's circulation theorem. In words, the circulation around a material loop is invariant for a barotropic fluid that is subject only to conservative forces. Using Stokes' theorem, the circulation theorem may also be written

$$\boxed{\frac{D}{Dt} \int \boldsymbol{\omega} \cdot d\mathbf{S} = 0} . \tag{4.51}$$

That is, the area-integral of the normal component of vorticity across any material surface is constant, under the same conditions. This form is both natural and useful, and it arises because of the way vorticity is tied to material fluid elements, as we now see.

Stretching and circulation

We now informally consider how vortex stretching and mass conservation work together to give the circulation theorem. Let the fluid be incompressible so that the volume of a fluid mass is constant, and consider a surface normal to a vortex

tube, that is the plane of the surface perpendicular to the direction of the vorticity, as in Fig. 4.5). Let the volume of a small material box around the surface be δV , the length of the material lines be δl and the surface area be δA . Then

$$\delta V = \delta l \delta A. \quad (4.52)$$

Because of the frozen-in property, vorticity passing through the surface is proportional to the length of the material lines. That is $\omega \propto \delta l$, and

$$\delta V \propto \omega \delta A. \quad (4.53)$$

The right-hand side is just the circulation around the surface. Now, if the corresponding material tube is stretched δl increases, but the volume, δV , remains constant by mass conservation. Thus, the circulation given by the right-hand side of (4.53) also remains constant. In other words, because of the frozen-in property vorticity itself is amplified by the stretching, but the vortex lines get closer together in such a way that the product $\omega \delta S$ remains constant and circulation is conserved.

Baroclinic flow and the solenoidal term

In baroclinic flow, the circulation is not generally conserved. and we have

$$\frac{DC}{Dt} = - \oint \frac{\nabla p}{\rho} \cdot d\mathbf{l} = - \oint \frac{dp}{\rho} \quad (4.54)$$

and this is sometimes called Bjerknes's circulation theorem.¹ Noting the fundamental thermodynamic relation $T d\eta = dI + p d\alpha$ we have

$$\alpha dp = d(p\alpha) - T d\eta + dI \quad (4.55)$$

so that the solenoidal term on the right-hand side of (4.54) may be written as

$$S_o = - \oint \alpha dp = \oint T d\eta = - \oint \eta dT = -R \oint T d \log p \quad (4.56)$$

where the last equality holds only for an ideal gas. Using Stokes's theorem this also be written

$$S_o = - \int_A \nabla \alpha \times \nabla p \cdot d\mathbf{A} = - \int_A \left(\frac{\partial \alpha}{\partial T} \right)_p \nabla \alpha \times \nabla p \cdot d\mathbf{A} = \int_A \nabla T \times \nabla \eta \cdot d\mathbf{A}. \quad (4.57)$$

The rate of change of the circulation across a surface depends on the existence of such terms — see Fig. 4.6 and problem 4.6.

However, even if the solenoidal vector is in general non-zero, circulation is conserved if the material path is in a surface of constant entropy, η , and if $D\eta/Dt = 0$. In this case the solenoidal term vanishes and, because $D\eta/Dt = 0$, entropy remains constant on that same material loop as it evolves. This result gives rise to the conservation of potential vorticity, discussed in section 4.5

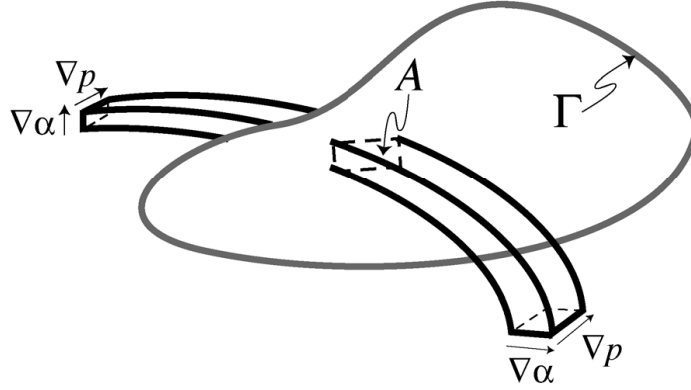


Figure 4.6 A schematic of the solenoids and the circulation theorem. The circulation over a surface is the integral of the vortex lines passing through it. Solenoids are tubes perpendicular to both $\nabla\alpha$ and ∇p , and they have a non-zero cross-sectional area if isolines of α and p do not coincide. The rate of change of circulation over a material surface is given by the sum of all the solenoidal areas crossing the surface. If $\nabla\alpha \times \nabla p = 0$ there are no solenoids.

Circulation in a rotating frame

In a rotating frame of reference, the appropriate form of the circulation theorem is obtained by using

$$\frac{D}{Dt} \oint (\mathbf{v}_r + \boldsymbol{\Omega} \times \mathbf{r}) \cdot d\mathbf{r} = \oint \left[\left(\frac{D\mathbf{v}_r}{Dt} + \boldsymbol{\Omega} \times \mathbf{v}_r \right) \cdot d\mathbf{r} + (\mathbf{v}_r + \boldsymbol{\Omega} \times \mathbf{r}) \cdot d\mathbf{v}_r \right], \quad (4.58)$$

where for emphasis we explicitly use the subscript r to denote the rotating frame. But $\oint \mathbf{v}_r \cdot d\mathbf{v}_r = 0$ and, integrating by parts,

$$\begin{aligned} \oint (\boldsymbol{\Omega} \times \mathbf{r}) \cdot d\mathbf{v}_r &= \oint \left\{ d[(\boldsymbol{\Omega} \times \mathbf{r}) \cdot \mathbf{v}_r] - (\boldsymbol{\Omega} \times d\mathbf{r}) \cdot \mathbf{v}_r \right\} \\ &= \oint \left\{ d[(\boldsymbol{\Omega} \times \mathbf{r}) \cdot \mathbf{v}_r] + (\boldsymbol{\Omega} \times \mathbf{v}_r) \cdot d\mathbf{r} \right\}. \end{aligned} \quad (4.59)$$

The first term on the right-hand side is zero and so (4.58) becomes

$$\begin{aligned} \frac{D}{Dt} \oint (\mathbf{v}_r + \boldsymbol{\Omega} \times \mathbf{r}) \cdot d\mathbf{r} &= \oint \left(\frac{D\mathbf{v}_r}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v}_r \right) \cdot d\mathbf{r} \\ &= - \oint \frac{dp}{\rho}, \end{aligned} \quad (4.60)$$

using the momentum equation. The term on the last line vanishes if the fluid is barotropic, and if so the circulation theorem is:

$$\frac{D}{Dt} \oint (\mathbf{v}_r + \boldsymbol{\Omega} \times \mathbf{r}) \cdot d\mathbf{r} = 0. \quad (4.61)$$

Using Stokes's theorem this may be written,

$$\frac{D}{Dt} \int (\boldsymbol{\omega}_r + 2\boldsymbol{\Omega}) \cdot d\mathbf{S} = 0. \quad (4.62)$$

having used $\nabla \times (\boldsymbol{\Omega} \times \mathbf{r}) = 2\boldsymbol{\Omega}$, and where $\boldsymbol{\omega}_r$ is the relative vorticity, or the vorticity of a fluid parcel as measured in the rotating frame.²

4.3.3 The circulation theorem for hydrostatic flow

A version of Kelvin's circulation theorem holds for hydrostatic flow. For simplicity we restrict attention to the f -plane, and start with the hydrostatic momentum equations,

$$\frac{D\mathbf{u}_r}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{u}_r = -\frac{1}{\rho} \nabla_z p \quad (4.63a)$$

$$0 = -\frac{1}{\rho} \frac{\partial p}{\partial z} - \nabla \Phi, \quad (4.63b)$$

where now $\Phi = gz$ is the gravitational potential and the rotation vector is vertical so that $\boldsymbol{\Omega} = \Omega \mathbf{k}$. The advecting field is three-dimensional, and in particular we still have $D\delta\mathbf{r}/Dt = \delta\mathbf{v} = (\delta\mathbf{r} \cdot \nabla)\mathbf{v}$. Thus, using (4.63) we have

$$\begin{aligned} \frac{D}{Dt} \oint (\mathbf{u}_r + \boldsymbol{\Omega} \times \mathbf{r}) \cdot d\mathbf{r} &= \oint \left[\left(\frac{D\mathbf{u}_r}{Dt} + \boldsymbol{\Omega} \times \mathbf{v}_r \right) \cdot d\mathbf{r} + (\mathbf{u}_r + \boldsymbol{\Omega} \times \mathbf{r}) \cdot d\mathbf{v}_r \right] \\ &= \oint \left(\frac{D\mathbf{u}_r}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{u}_r \right) \cdot d\mathbf{r} \\ &= \oint \left(-\frac{1}{\rho} \nabla p - \nabla \Phi \right) \cdot d\mathbf{r}, \end{aligned} \quad (4.64)$$

as with (4.60), having used $\boldsymbol{\Omega} \times \mathbf{v}_r = \boldsymbol{\Omega} \times \mathbf{u}_r$. The gradient operator ∇ is three-dimensional. The last term on the right-hand side vanishes because it is the integral of the gradient of a potential around a closed path. Thus, the circulation theorem is

$$\frac{D}{Dt} \oint (\mathbf{u}_r + \boldsymbol{\Omega} \times \mathbf{r}) \cdot d\mathbf{r} = 0 \quad (4.65)$$

for a barotropic fluid. Using Stokes's theorem we have the equivalent form

$$\frac{D}{Dt} \int (\boldsymbol{\omega}_{hy} + 2\boldsymbol{\Omega}) \cdot d\mathbf{S} = 0, \quad (4.66)$$

where the subscript 'hy' denotes hydrostatic and, in Cartesian co-ordinates,

$$\boldsymbol{\omega}_{hy} = \nabla \times \mathbf{u} = -\mathbf{i} \frac{\partial v}{\partial z} + \mathbf{j} \frac{\partial u}{\partial z} + \mathbf{k} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right). \quad (4.67)$$

4.4 VORTICITY IN A ROTATING FRAME

Perhaps easiest way to derive the vorticity equation appropriate for a rotating reference frame is to begin with the momentum equation in the form

$$\frac{\partial \mathbf{v}_r}{\partial t} + (2\boldsymbol{\Omega} + \boldsymbol{\omega}_r) \times \mathbf{v}_r = -\frac{1}{\rho} \nabla p + \nabla \left(\Phi - \frac{1}{2} \mathbf{v}_r^2 \right), \quad (4.68)$$

where the potential Φ contains the gravitational and centrifugal forces. Take the curl of this and use the identity (4.16), which here implies

$$\nabla \times [(2\boldsymbol{\Omega} + \boldsymbol{\omega}_r) \times \mathbf{v}_r] = (2\boldsymbol{\Omega} + \boldsymbol{\omega}_r) \nabla \cdot \mathbf{v}_r + (\mathbf{v}_r \cdot \nabla)(2\boldsymbol{\Omega} + \boldsymbol{\omega}_r) - [(2\boldsymbol{\Omega} + \boldsymbol{\omega}_r) \cdot \nabla] \mathbf{v}_r, \quad (4.69)$$

(noting that $\nabla \cdot (2\boldsymbol{\Omega} + \boldsymbol{\omega}) = 0$), to give the vorticity equation

$$\frac{D\boldsymbol{\omega}_r}{Dt} = [(2\boldsymbol{\Omega} + \boldsymbol{\omega}_r) \cdot \nabla] \mathbf{v} - (2\boldsymbol{\Omega} + \boldsymbol{\omega}_r) \nabla \cdot \mathbf{v}_r + \frac{1}{\rho^2} (\nabla \rho \times \nabla p). \quad (4.70)$$

Note that because $\boldsymbol{\Omega}$ is a constant, $D\boldsymbol{\omega}_r/Dt = D\boldsymbol{\omega}_a/Dt$ where $\boldsymbol{\omega}_a = \boldsymbol{\Omega} + \boldsymbol{\omega}_r$ is the absolute vorticity. We see that the only difference between the vorticity equation in the rotating and inertial frames of reference is in the presence of the solid-body vorticity $2\boldsymbol{\Omega}$ in the right-hand-side of this equation.

As before, the divergent term may be folded in to the material derivative using the mass continuity equation to give

$$\frac{D}{Dt} \left(\frac{\boldsymbol{\omega}_a}{\rho} \right) = \frac{1}{\rho} (\boldsymbol{\omega}_r + 2\boldsymbol{\Omega}) \cdot \nabla \mathbf{v}_r + \frac{1}{\rho^3} (\nabla \rho \times \nabla p), \quad (4.71)$$

but now it is the absolute vorticity that appears on the left-hand-side. If ρ is constant $\boldsymbol{\omega}_a$ may be replaced by $\boldsymbol{\omega}_r$.

4.4.1 The circulation theorem and vortex tilting

What are the implications of the circulation theorem on a rotating, spherical planet? Let us define relative circulation over some material loop as

$$C_r \equiv \oint \mathbf{v}_r \cdot d\mathbf{l}, \quad (4.72)$$

and because $\mathbf{v}_r = \mathbf{v}_a - 2\boldsymbol{\Omega} \times \mathbf{r}$ we have

$$C_r = C_a - \int 2\boldsymbol{\Omega} \cdot d\mathbf{S} = C_a - 2\boldsymbol{\Omega} A_{\perp} \quad (4.73)$$

where C_a is the total or absolute circulation and $A_{\perp}$ is the area enclosed by the projection of the material circuit onto the plane normal to the rotation vector;

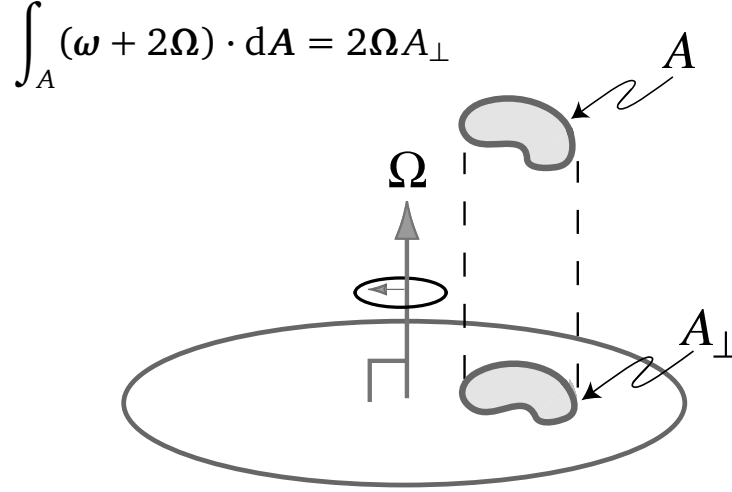


Figure 4.7 The projection of a material circuit on to the equatorial plane. If a fluid element moves poleward, keeping its orientation to the local vertical fixed (e.g., it stays horizontal) then the area of its projection on to the equatorial plane increases. If its total (absolute) circulation is to be maintained, then the vertical component of the relative vorticity must diminish. That is, $\int_A (\omega + 2\Omega) \cdot d\mathbf{A} = \int_A (\zeta + f) dA = \text{constant}$. Thus, the β term in $D(\zeta + f)/Dt = D\zeta/Dt + \beta v = 0$ ultimately arises from the *tilting* of a parcel relative to the axis of rotation as it moves meridionally.

that is, onto the equatorial plane (see Fig. 4.7). If the solenoidal term is zero then the circulation theorem, (4.62), may be written as

$$\frac{D}{Dt}(C_r + 2\Omega A_\perp) = 0 \quad (4.74)$$

This equation tells us that the relative circulation around a circuit will change if the orientation of the plane changes; that is, if the area of its projection on to the equatorial plane changes. In large scale dynamics the most common cause of this is when a fluid parcel changes its latitude. For example, consider the two-dimensional flow of an infinitesimal, horizontal, homentropic fluid surface at a latitude ϑ with area δS , so that the projection of its area on the equatorial plane is $\delta S \sin \vartheta$. If the fluid surface moves, but remains horizontal, then directly from (4.74) the relative vorticity changes as

$$\frac{D\zeta_r}{Dt} = -2\Omega \frac{D}{Dt} \sin \vartheta = -v_r \frac{2\Omega \cos \vartheta}{a}. \quad (4.75)$$

The means by which the relative vorticity of a parcel changes by virtue of its latitudinal displacement is known as the *beta effect*, or the β effect. It is a manifestation of the tilting term in the vorticity equation, and it is often the most

important means by which relative vorticity does change. We define

$$\beta \equiv \frac{df}{dy} = \frac{2\Omega}{a} \cos \vartheta, \quad (4.76)$$

so that in this instance $D\zeta_r/Dt = -\beta v_r$. The β term directly arises in the full vorticity equation, as we now see.

4.4.2 The vertical vorticity equation

The most important, although not the largest, component of the vorticity equation — at least for large-scale dynamics — is the vertical component, because this contains much of the information about the horizontal flow. We can obtain an explicit expression for it by taking the vertical component of (4.70), although care must be taken because the unit vectors ($\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$) are functions of position (see problem 2.6.) An alternative derivation that results in a useful form begins with the horizontal momentum equations in the form

$$\frac{\partial u}{\partial t} - v(\zeta + f) + w \frac{\partial u}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial x} - \frac{1}{2} \frac{\partial}{\partial x} (u^2 + v^2) + F_x \quad (4.77a)$$

$$\frac{\partial v}{\partial t} + u(\zeta + f) + w \frac{\partial v}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial y} - \frac{1}{2} \frac{\partial}{\partial y} (u^2 + v^2) + F_y. \quad (4.77b)$$

$$(4.77c)$$

where in this section we again drop the subscript r on variables measured in the rotating frame. Cross-differentiating gives, after a little algebra,

$$\begin{aligned} \frac{D}{Dt}(\zeta + f) = & -(\zeta + f) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \left(\frac{\partial v}{\partial z} \frac{\partial w}{\partial x} - \frac{\partial u}{\partial z} \frac{\partial w}{\partial y} \right) \\ & + \left(\frac{\partial \rho}{\partial x} \frac{\partial p}{\partial y} - \frac{\partial \rho}{\partial y} \frac{\partial p}{\partial x} \right) + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right). \end{aligned} \quad (4.78)$$

The brave reader may verify that this equation holds as written in spherical coordinates, with $\partial/\partial x = (r \cos \vartheta)^{-1} \partial/\partial \lambda$ and $\partial/\partial y = r^{-1} \partial/\partial \vartheta$. We interpret the various terms as follows:

$D\zeta/Dt = \partial\zeta/\partial t + \mathbf{v} \cdot \nabla\zeta$: The material derivative of the vertical component of the vorticity.

$Df/Dt = v\partial f/\partial y = v\beta$: The β -effect. The vorticity is affected by the meridional motion of the fluid, so that, apart from the terms on the right-hand-side, $(\zeta + f)$ is conserved on parcels. Because the Coriolis parameter changes with latitude this is like saying that the system has differential rotation. This effect is precisely that due to the change in orientation of fluid surfaces with latitude, as given above in section 4.4.1 and Fig. 4.7.

$-(\zeta + f)(\partial u / \partial x + \partial v / \partial y)$: The divergence term, which gives rise to vortex stretching. In an incompressible fluid this may be written $(\zeta + f)\partial w / \partial z$, so that vorticity is amplified if the vertical velocity increases with height, so stretching the material lines and the vorticity.

$[(\partial v / \partial z)(\partial w / \partial x) - (\partial u / \partial z)(\partial w / \partial y)]$: The tilting term, whereby a vertical component of vorticity may be generated by a vertical velocity acting on a horizontal vorticity. See Fig. 4.4.

$[(\partial \rho / \partial x)(\partial p / \partial y) - (\partial \rho / \partial y)(\partial p / \partial x)]$: The non-homentropic or baroclinic term, arising when isosurfaces of pressure and density are not parallel.

$(\partial F_y / \partial x - \partial F_x / \partial y)$: The friction term. If the only frictional effect is that of molecular viscosity then this is $\nu \nabla^2 \zeta$.

In inviscid two-dimensional incompressible flow, all of the terms on the right-hand-side vanish and we have the simple equation

$$\frac{D(\zeta + f)}{Dt} = 0. \quad (4.79)$$

For inviscid shallow water flow, we can show that (see chapter 5)

$$\frac{D(\zeta + f)}{Dt} = -(\zeta + f) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right). \quad (4.80)$$

The vanishing of the tilting term is the only aspect which is perhaps not immediately apparent.

4.5 POTENTIAL VORTICITY CONSERVATION

Although Kelvin's circulation theorem is a general statement about vorticity conservation, in its original form it is not always a very *useful* statement for two reasons. First, it is not an immediate statement about a *field*, such as vorticity itself. Second, it is not satisfied for baroclinic flow, such as is found in the atmosphere and ocean. (Of course non-conservative forces such as viscosity also lead to circulation non-conservation, but this applies to virtually all conservation laws and does not diminish them.) It turns out that it is possible to derive a beautiful conservation law that overcomes both of these failings and one, furthermore, that is extraordinarily useful in geophysical fluid dynamics. This is the conservation of *potential vorticity* introduced first by Rossby and then in a more general form by Ertel.³ The idea is that we can use a scalar field that is being advected by the flow to keep track of, or to take care of, the evolution of fluid elements. For a baroclinic fluid this scalar field must be chosen in a special way (it must be a function of the density and pressure alone), but the restriction to barotropic

fluid can then be avoided. Then using the scalar evolution equation in conjunction with the vorticity equation gives us a scalar conservation equation. In the next few subsections we derive the equation for potential vorticity conservation in several different ways. We do not treat rotation explicitly until section 4.5.5, and so the vorticity in what follows should be regarded as the absolute vorticity.

4.5.1 PV conservation from the circulation theorem

Barotropic fluids

Let us begin with the simple case of a barotropic fluid. For an infinitesimal volume we write Kelvin's theorem as:

$$\frac{D}{Dt} [(\boldsymbol{\omega}_a \cdot \mathbf{n}) \delta A] = 0 \quad (4.81)$$

where $\mathbf{n}$ is a unit vector normal to an infinitesimal surface δA (see Fig. 4.8). The volume is bounded by two isosurfaces of values χ and $\chi + \delta\chi$, where χ is any conserved tracer satisfying

$$\frac{D\chi}{Dt} = 0. \quad (4.82)$$

Since $\mathbf{n} = \nabla\chi/|\nabla\chi|$ and the infinitesimal volume $\delta V = \delta h \delta A$, where δh is the separation between the two surfaces, we have that

$$\boldsymbol{\omega}_a \cdot \mathbf{n} \delta A = \boldsymbol{\omega}_a \cdot \frac{\nabla\chi}{|\nabla\chi|} \frac{\delta V}{\delta h}. \quad (4.83)$$

Now, the separation between the two surfaces, δh may be obtained from

$$\delta\chi = \delta\mathbf{r} \cdot \nabla\chi = \delta h |\nabla\chi|, \quad (4.84)$$

so that $|\nabla\chi| = \delta\chi/\delta h$. Thus we have

$$\frac{D}{Dt} \left[\frac{(\boldsymbol{\omega}_a \cdot \nabla\chi) \delta V}{\delta\chi} \right] = 0. \quad (4.85)$$

Furthermore, χ is conserved on material elements and so $\delta\chi$ may be taken out of the differentiation. The mass of the volume element $\rho \delta V$ is also conserved, so that (4.85) becomes (4.81) becomes

$$\frac{\rho \delta V}{\delta\chi} \frac{D}{Dt} \left(\frac{\boldsymbol{\omega}_a}{\rho} \cdot \nabla\chi \right) = 0 \quad (4.86)$$

or

$$\frac{D}{Dt} (\tilde{\boldsymbol{\omega}}_a \cdot \nabla\chi) = 0 \quad (4.87)$$

where $\tilde{\boldsymbol{\omega}}_a = \boldsymbol{\omega}_a / \rho$.

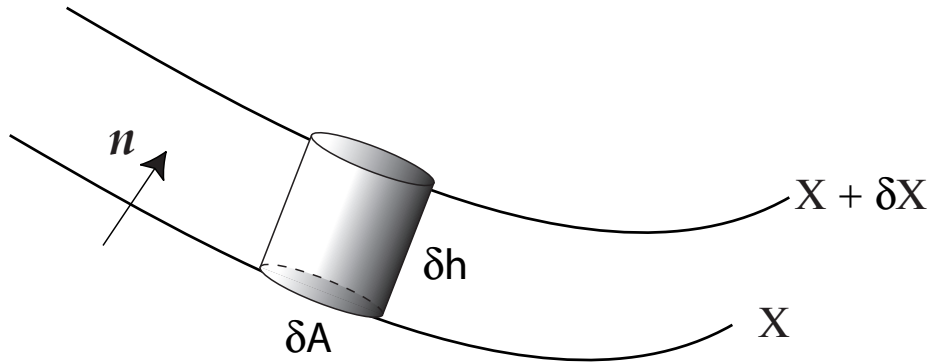
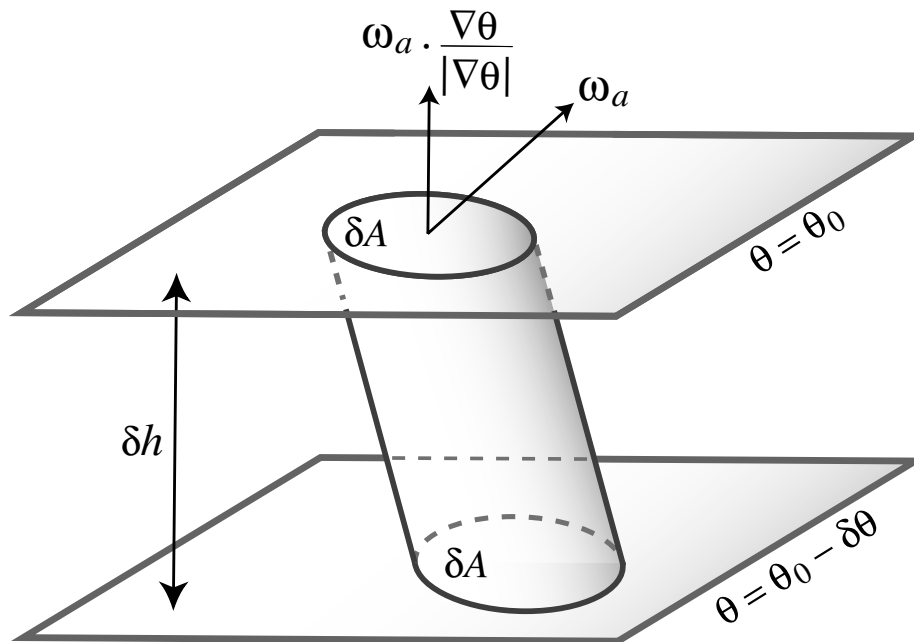


Figure 4.8 An infinitesimal fluid element, bounded by two isosurfaces of the conserved tracer χ .



$$\text{Mass: } \rho \delta A \delta h = \text{constant}$$

$$\text{Entropy: } |\nabla\theta| \delta h = \text{constant}$$

Figure 4.9 Geometry of potential vorticity conservation. The circulation around isentropic sheets is conserved. (Expand caption).

The general case

For a baroclinic fluid the above derivation fails simply because the statement of the conservation of circulation, (4.81) is not, in general, true: there are solenoidal terms on the right-hand-side. How may these be annihilated? Simply by choosing the circuit around which we evaluate the circulation be such that the solenoidal term is identically zero. Since $\mathbf{S} = -\nabla\alpha \times \nabla p = -\nabla\eta \times \nabla T$ this occurs if the values of any of p, ρ, η, T are constant on that circuit; that is, if $\chi = p, \rho, \eta$ or T . But of course the derivation still demands that χ be a materially conserved quantity, which usually restricts its choice to η (or potential temperature), or to ρ itself if the thermodynamic equation is $D\rho/Dt = 0$. Thus, the conservation of potential vorticity for inviscid, adiabatic flow is

$$\frac{D}{Dt} (\tilde{\omega}_a \cdot \nabla\theta) = 0 \quad (4.88)$$

where $D\theta/Dt = 0$. For diabatic flow source terms appear on the right-hand side, and we derive these later on.

4.5.2 PV conservation from the frozen-in property

Potential vorticity conservation may be thought of as being consequence of the frozen-in property of vorticity. First consider the case in which the solenoidal term vanishes from the outset.

Barotropic fluids

If χ is a materially conserved tracer, and so obeying (4.82), then difference in χ between two infinitesimally close fluid elements is also conserved and

$$\frac{D}{Dt} (\chi_1 - \chi_2) = \frac{D\delta\chi}{Dt} = 0. \quad (4.89)$$

But $\delta\chi = \nabla\chi \cdot \delta\mathbf{l}$ where $\delta\mathbf{l}$ is the infinitesimal vector connecting the two fluid elements. Thus

$$\frac{D}{Dt} (\nabla\chi \cdot \delta\mathbf{l}) = 0 \quad (4.90)$$

But since the line element and the vorticity (divided by density) obey the same equation, we can replace the line element by vorticity (divided by density) in (4.90) to obtain again

$$\frac{D}{Dt} \left(\frac{\nabla\chi \cdot \omega_a}{\rho} \right) = 0. \quad (4.91)$$

That is, the potential vorticity, $Q = (\tilde{\omega}_a \cdot \nabla\chi)$ is a material invariant, where χ is any scalar quantity that satisfies $D\chi/Dt = 0$.

Baroclinic fluids

In baroclinic fluids we cannot casually substitute the vorticity for that of a line element in (4.90) because of the presence of the solenoidal term, and in any case a little more care would not be amiss. From (4.90) we obtain

$$\delta \mathbf{l} \cdot \frac{D \nabla \chi}{Dt} + \nabla \chi \cdot \frac{D \delta \mathbf{l}}{Dt} = 0 \quad (4.92)$$

or, using (4.41),

$$\delta \mathbf{l} \cdot \frac{D \nabla \chi}{Dt} + \nabla \chi \cdot [(\delta \mathbf{l} \cdot \nabla) \mathbf{v}] = 0. \quad (4.93)$$

Now, let us choose $\delta \mathbf{l}$ to correspond to a vortex line, so that at the initial time $\delta \mathbf{l} = \epsilon \tilde{\boldsymbol{\omega}}_a$. (Note that in this case the association of $\delta \mathbf{l}$ with a vortex line can only be made instantaneously, and we cannot set $D \delta \mathbf{l} / Dt \propto D \boldsymbol{\omega}_a / Dt$.) Then,

$$\tilde{\boldsymbol{\omega}}_a \cdot \frac{D \nabla \chi}{Dt} + \nabla \chi \cdot [(\boldsymbol{\omega}_a \cdot \nabla) \mathbf{v}] = 0, \quad (4.94)$$

or, using the vorticity equation (4.20),

$$\tilde{\boldsymbol{\omega}}_a \cdot \frac{D \nabla \chi}{Dt} + \nabla \chi \cdot \left[\frac{D \tilde{\boldsymbol{\omega}}_a}{Dt} - \frac{1}{\rho^3} \nabla \rho \times \nabla p \right] = 0. \quad (4.95)$$

This may be written

$$\frac{D}{Dt} \tilde{\boldsymbol{\omega}}_a \cdot \nabla \chi = \frac{1}{\rho^3} \nabla \chi \cdot (\nabla \rho \times \nabla p). \quad (4.96)$$

The term on the right hand side is in general non-zero for an arbitrary choice of scalar, but it will evidently vanish if ∇p , $\nabla \rho$ and $\nabla \chi$ are coplanar. If χ is any function of p and ρ this will be satisfied, but χ must also be a materially conserved scalar. If, as for an ideal gas, $\rho = \rho(\eta, p)$ (or $\eta = \eta(p, \rho)$) where η is the entropy (which is materially conserved), and if χ is a function of entropy η alone, then χ satisfies both conditions. Explicitly, the solenoidal term vanishes because

$$\nabla \chi \cdot (\nabla \rho \times \nabla p) = \frac{d\chi}{d\eta} \nabla \eta \cdot \left[\left(\frac{\partial \rho}{\partial p} \nabla p + \frac{\partial \rho}{\partial \eta} \nabla \eta \right) \times \nabla p \right] = 0. \quad (4.97)$$

Thus, provided χ satisfies

$$\frac{D\chi}{Dt} = 0 \quad \text{and} \quad \chi = \chi(p, \rho), \quad (4.98)$$

then (4.96) becomes

$$\frac{D}{Dt} \left(\frac{\boldsymbol{\omega}_a \cdot \nabla \chi}{\rho} \right) = 0. \quad (4.99)$$

The natural choice for χ is potential temperature, whence

$$\boxed{\frac{D}{Dt} \left(\frac{\boldsymbol{\omega}_a \cdot \nabla \theta}{\rho} \right) = 0}. \quad (4.100)$$

The presence of a density term in the denominator is not necessary for incompressible flows (i.e., if $\nabla \cdot \mathbf{v} = 0$).

4.5.3 * An algebraic derivation

Finally, we give a straightforward algebraic derivation of potential vorticity conservation. We will take the opportunity to include frictional and diabatic processes, although these may also be included in the derivations above.⁴ We begin with the frictional vorticity equation in the form

$$\frac{D\tilde{\boldsymbol{\omega}}_a}{Dt} = (\tilde{\boldsymbol{\omega}}_a \cdot \nabla) \mathbf{v} + \frac{1}{\rho^3} (\nabla \rho \times \nabla p) + \frac{1}{\rho} (\nabla \times \mathbf{F}). \quad (4.101)$$

where $\mathbf{F}$ represents any nonconservative force term on the right-hand-side of the momentum equation (i.e., $D\mathbf{v}/Dt = -\rho^{-1}\nabla p + \mathbf{F}$). We have also the equation for our materially conserved scalar χ ,

$$\frac{D\chi}{Dt} = \dot{\chi} \quad (4.102)$$

where $\dot{\chi}$ represents any sources and sinks of χ . Now

$$(\tilde{\boldsymbol{\omega}}_a \cdot \nabla) \frac{D\chi}{Dt} = \tilde{\boldsymbol{\omega}}_a \cdot \frac{D\nabla\chi}{Dt} + [(\tilde{\boldsymbol{\omega}}_a \cdot \nabla) \mathbf{v}] \cdot \nabla\chi. \quad (4.103)$$

which may be obtained just by expanding the left-hand side. Thus, using (4.102),

$$\tilde{\boldsymbol{\omega}}_a \cdot \frac{D\nabla\chi}{Dt} = (\tilde{\boldsymbol{\omega}}_a \cdot \nabla) \dot{\chi} - [(\tilde{\boldsymbol{\omega}}_a \cdot \nabla) \mathbf{v}] \cdot \nabla\chi. \quad (4.104)$$

Now take the dot product of (4.101) with $\nabla\chi$:

$$\nabla\chi \cdot \frac{D\tilde{\boldsymbol{\omega}}_a}{Dt} = \nabla\chi \cdot [(\tilde{\boldsymbol{\omega}}_a \cdot \nabla) \mathbf{v}] + \nabla\chi \cdot \left[\frac{1}{\rho^3} (\nabla \rho \times \nabla p) \right] + \nabla\chi \cdot \left[\frac{1}{\rho} (\nabla \times \mathbf{F}) \right]. \quad (4.105)$$

The sum of the last two equations yields

$$\frac{D}{Dt} (\tilde{\boldsymbol{\omega}}_a \cdot \nabla\chi) = \tilde{\boldsymbol{\omega}}_a \cdot \nabla\dot{\chi} + \nabla\chi \cdot \left[\frac{1}{\rho^3} (\nabla \rho \times \nabla p) \right] + \frac{\nabla\chi}{\rho} \cdot (\nabla \times \mathbf{F}). \quad (4.106)$$

This equation reprises (4.96), but with the addition of frictional and diabatic terms. As before, the solenoidal term is annihilated if we choose $\chi = \theta(p, \rho)$,

so giving the evolution equation for potential vorticity in the presence of forcing and diabatic terms, namely:

$$\boxed{\frac{D}{Dt} (\tilde{\omega}_a \cdot \nabla \theta) = \tilde{\omega}_a \cdot \nabla \dot{\theta} + \frac{\nabla \theta}{\rho} \cdot (\nabla \times \mathbf{F})} . \quad (4.107)$$

4.5.4 Effects of salinity

For seawater the equation of state may be written as

$$\theta = \theta(\rho, p, S) \quad (4.108)$$

where θ is potential temperature and S is salinity. In the absence of diabatic terms (which include saline diffusion) potential temperature is a materially conserved quantity. However, because of the presence of salinity, θ cannot be used to annihilate the solenoidal term; that is

$$\nabla \theta \cdot (\nabla \rho \times \nabla p) = \left(\frac{\partial \theta}{\partial S} \right)_{p,\rho} \nabla S \cdot (\nabla \rho \times \nabla p) \neq 0. \quad (4.109)$$

Strictly speaking then, *there is no potential vorticity conservation principle for seawater*. However, such a blunt statement rather overemphasizes the importance of salinity, and the nonconservation of potential vorticity because of this effect is rather small.

4.5.5 Effects of rotation

In a rotating frame the potential vorticity conservation equation is obtained simply by replacing ω_a by $\omega + 2\Omega$, where Ω is the rotation rate of the rotating frame. The operator D/Dt is reference frame invariant, and so may be evaluated using the usual formula with velocities measured in the rotating frame.

The conservation of potential vorticity has profound consequences. The non-conservative terms are often small, and then large-scale flow in the ocean atmosphere are characterized by conservation of potential vorticity. This is a very powerful constraint on the flow, and it indeed turns out that potential vorticity is a much more useful quantity for baroclinic, or nonhomentropic fluids than for barotropic fluids, because the required use of a special conserved scalar imparts additional information. A large fraction of the remainder of this book explores, in one way or another, the consequences of (4.100).

4.6 POTENTIAL VORTICITY IN APPROXIMATE MODELS

Appropriate approximate forms of potential vorticity are conserved by the respective approximate models — Boussinesq, hydrostatic and so on, and we consider a few such cases here.

4.6.1 The Boussinesq equations

The Boussinesq equations are incompressible (that is, the volume of a fluid element is conserved) and the equation for vorticity itself is isomorphic to that for a line element. However the Boussinesq equations are not barotropic, $\nabla\rho$ is not parallel to ∇p , and although the pressure gradient term $\nabla\phi$ disappears on taking its curl (or equivalently disappears on integration around a closed path) the buoyancy term $\mathbf{k}b$ does not, and it is this that prevents Kelvin's circulation from holding. However, taking the dot product of the vorticity equation with ∇b is sufficient to annihilate this effect, because $\nabla b \cdot \nabla \times (\mathbf{k}b) = 0$. Thus, if the thermodynamic equation is

$$\frac{Db}{Dt} = 0. \quad (4.110)$$

then potential vorticity conservation is $DQ/Dt = 0$ where

$$Q = (\boldsymbol{\omega} + 2\boldsymbol{\Omega}) \cdot \nabla b. \quad (4.111)$$

Expanding (4.111) in Cartesian coordinates we obtain,

$$Q = (v_x - u_y)b_z + (w_y - v_z)b_x + (u_z - w_x)b_y + fb_z. \quad (4.112)$$

4.6.2 The hydrostatic equations

Making the hydrostatic approximation has no effect on Kelvin's circulation theorem or, in a baroclinic fluid, Bjerknes's theorem. Thus, in a baroclinic hydrostatic fluid we have

$$\frac{D}{Dt} \int (\boldsymbol{\omega}_{\text{hy}} + 2\boldsymbol{\Omega}) \cdot d\mathbf{S} = - \int \nabla\alpha \times \nabla p \cdot d\mathbf{S} \quad (4.113)$$

where, from (4.67) $\boldsymbol{\omega}_{\text{hy}} = \nabla \times \mathbf{u} = -i v_z + j u_z + \mathbf{k}(v_x - u_y)$, but the gradient operator and material derivative are fully three-dimensional. Derivation of potential vorticity conservation then proceeds, as in section 4.5.1, by choosing the circuit over which the circulation is calculated to be such that the right-hand-side vanishes; that is, to be such that the solenoidal term is annihilated. Precisely as before, this occurs if the circuit is barotropic and without further ado we write

$$\frac{DQ_{\text{hy}}}{Dt} = \frac{D}{Dt} \left(\frac{(\boldsymbol{\omega}_h + 2\boldsymbol{\Omega}) \cdot \nabla\theta}{\rho} \right) = 0. \quad (4.114)$$

Expanding this gives in Cartesian co-ordinates

$$Q_{\text{hy}} = \frac{1}{\rho} [(v_x - u_y)\theta_z - v_z\theta_x + u_z\theta_y + 2\Omega\theta_z]. \quad (4.115)$$

In spherical co-ordinates the hydrostatic approximation is usually accompanied by the traditional approximation and the expanded expression for a conserved potential vorticity is more complicated. It can still be derived from Kelvin's theorem, but this is left as a problem for the reader (problem 4.4).

4.7 POTENTIAL VORTICITY ON ISENTROPIC SURFACES

If we begin with the primitive equations in isentropic co-ordinates then potential vorticity conservation follows quite simply. Cross differentiating the horizontal momentum equations (2.172) gives the vorticity equation [c.f. (5.71)]

$$\frac{D}{Dt}(\zeta + f) + (\zeta + f)\nabla_\theta \cdot \mathbf{u} = 0. \quad (4.116)$$

where $D/Dt = \partial/\partial t + \mathbf{u} \cdot \nabla_\theta$. The thermodynamic equation is

$$\frac{D\sigma}{Dt} + \sigma\nabla \cdot \mathbf{u} = 0 \quad (4.117)$$

where $\sigma = \partial z/\partial b$ (Boussinesq) or $\partial p/\partial \theta$ (ideal gas) or is the thickness of an isopycnal layer. Eliminating the divergence between these last two equations gives

$$\frac{D}{Dt} \left(\frac{\zeta + f}{\sigma} \right) = 0 \quad (4.118)$$

The derivation, and the result, are precisely the same as with the shallow water equations (section 5.3.1).⁵

4.7.1 A connection with height co-ordinates

The hydrostatic potential vorticity written in height co-ordinates may be transformed into a form that reveals its intimate connection with isentropic surfaces. Let us make the Boussinesq approximation for which the potential vorticity is

$$Q_{\text{hy}} = (v_x - u_y)b_z - v_z b_x + u_z b_y. \quad (4.119)$$

where b is the buoyancy. We can write this as

$$Q_{\text{hy}} = b_z \left[\left(v_x - v_z \frac{b_x}{b_z} \right) - \left(u_y - u_z \frac{b_y}{b_z} \right) \right]. \quad (4.120)$$

But the terms in the inner brackets are just the horizontal velocity derivatives at constant b . To see this, note that

$$\begin{aligned} \left(\frac{\partial v}{\partial x} \right)_b &= \left(\frac{\partial v}{\partial x} \right)_z + \frac{\partial v}{\partial z} \left(\frac{\partial z}{\partial x} \right)_b \\ &= \left(\frac{\partial v}{\partial x} \right)_z - \frac{\partial v}{\partial z} \left(\frac{\partial b}{\partial x} \right)_z / \frac{\partial b}{\partial z}, \end{aligned} \quad (4.121)$$

with a similar expression for $(\partial u / \partial y)_b$. (These relationships follow from standard rules of partial differentiation. Derivatives with respect to z are taken at constant x and y .) Thus, we obtain

$$Q_{hy} = \frac{\partial b}{\partial z} \left[\left(\frac{\partial v}{\partial x} \right)_b - \left(\frac{\partial u}{\partial y} \right)_b \right] = \frac{\partial b}{\partial z} \zeta_b \quad (4.122)$$

Thus, potential vorticity is simply the horizontal vorticity evaluated on a surface of constant buoyancy, multiplied by the vertical derivative of buoyancy, a measure of static stability. An entirely analogous derivation, with an almost identical result, proceeds for the ideal gas equations, with potential temperature replacing buoyancy.

4.8 THE IMPERMEABILITY OF ISENTROPES TO POTENTIAL VORTICITY

A kinematical result is a result valid forever.

Clifford Truesdell, *The Kinematics of Vorticity*, 1954.

An interesting property of isentropic surfaces is that they are ‘impermeable’ to potential vorticity, meaning that the mass integral of potential vorticity ($\int Q \rho dV$) over a volume bounded by an isentropic surface remains constant, even in the presence of diabatic sources, provided the surfaces do not intersect a non-isentropic surface like the ground.⁶ This may seem surprising, especially because unlike most conservation laws the result does not require adiabatic flow, and for that reason it leads to interesting dynamical interpretations of a number of phenomena. However, at the same time impermeability is a consequence of the definition of potential vorticity rather than the equations of motion, and in that sense it is a kinematic property.

To derive the result we define $s \equiv \rho Q = \nabla \cdot (\theta \omega)$ and integrate over some volume V to give

$$I = \int_V s dV = \int_V \nabla \cdot (\theta \omega) dV = \int_S \theta \omega \cdot dS, \quad (4.123)$$

using the divergence theorem, where S is the surface surrounding the volume V . If this is an isentropic surface then we have

$$I = \theta \int_S \omega \cdot dS = \theta \int_V \nabla \cdot \omega dV = 0, \quad (4.124)$$

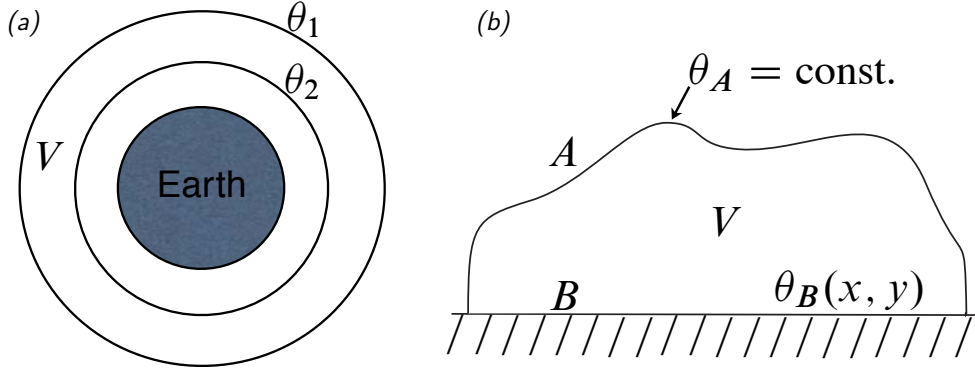


Figure 4.10 (a) Two isentropic surfaces that do not intersect the ground. The integral of PV substance over the volume between them, V , is zero, even if there is heating and the contours move. (b) An isentropic surface, A , intersects the ground, B , so enclosing a volume V . The rate of change of PV substance over the volume is given by an integral over B .

again using the divergence theorem. That is, over a volume wholly enclosed by a single isentropic surface the integral of $s = \rho Q$ vanishes. The quantity s may be called ‘potential vorticity substance’, or ‘PV substance’, being the total amount of potential vorticity that a fluid element has per unit volume — that is $Q \delta(\text{mass}) / \delta(\text{volume})$. If the volume is bounded by more than one isentropic surface neither of which intersect the surface, for example by concentric spheres of different radii as in Fig. 4.10a, the result still holds.

Suppose now that fluid volume is enclosed by an isentrope that intersects the ground, as in Fig. 4.10b. Let A denote the isentropic surface, B denote the ground, θ_A the constant value of θ on the isentrope, and $\theta_B(x, y, t)$ the non-constant value of θ on the ground. The integral of s over the volume is then

$$\begin{aligned}
 s &= \int_V \nabla \cdot (\theta \boldsymbol{\omega}) dV = \theta_A \int_A \boldsymbol{\omega} \cdot d\mathbf{S} + \int_B \theta_B \boldsymbol{\omega} \cdot d\mathbf{S} \\
 &= \theta_A \int_{A+B} \boldsymbol{\omega} \cdot d\mathbf{S} + \int_B (\theta_B - \theta_A) \boldsymbol{\omega} \cdot d\mathbf{S} \\
 &= \int_B (\theta_B - \theta_A) \boldsymbol{\omega} \cdot d\mathbf{S}.
 \end{aligned} \tag{4.125}$$

The first term on the second line vanishes after using the divergence theorem. Thus, the value of s , and so its rate of change, is a function *only of an integral over the surface B* , and the PV flux there must be calculated using the full equations of motion. However, we do not need to be concerned with a flux of PV substance through the isentropic surface. Put another way, the total amount of PV substance in a volume can change only when isentropes enclosing it intersect a boundary such as the earth’s surface.

4.8.1 Interpretation and application

Motion of the isentropic surface

How can the above results hold in the presence of heating? The isentropic surfaces must move in such a way that the total amount of PV substance contained between them nevertheless stays fixed, and we now demonstrate this explicitly. The potential vorticity equation may be written

$$\frac{\partial Q}{\partial t} + \mathbf{v} \cdot \nabla Q = S_Q, \quad (4.126)$$

where, from (4.107), $S_Q = (\boldsymbol{\omega} / \rho) \cdot \nabla \dot{\theta} + \nabla \theta \cdot (\nabla \times \mathbf{F}) / \rho$. Using mass continuity this may be written as

$$\frac{\partial s}{\partial t} + \nabla \cdot \mathbf{J} = 0 \quad (4.127)$$

where

$$\mathbf{J} \equiv \rho \mathbf{v} Q + \mathbf{N} \quad (4.128)$$

and $\nabla \cdot \mathbf{N} = -\rho S_Q$. Written this way, the quantity $\mathbf{J} / (\rho Q)$ is a notional velocity, $\mathbf{v}_Q$ say, and s satisfies

$$\frac{\partial s}{\partial t} + \nabla \cdot (\mathbf{v}_Q s) = 0, \quad (4.129)$$

and so s evolves as if it were being fluxed by the velocity $\mathbf{v}_Q$. A chemical tracer χ , where χ is the amount of tracer per unit volume, obeys a similar equation, to wit

$$\frac{\partial \chi}{\partial t} + \nabla \cdot (\chi \mathbf{v}) = 0, \quad (4.130)$$

However, whereas (4.130) implies that $D(\chi / \rho) / Dt = 0$, (4.129) does not imply that $\partial Q / \partial t + \mathbf{v}_Q \cdot \nabla Q = 0$ because $\partial \rho / \partial t + \nabla \cdot (\rho \mathbf{v}_Q) \neq 0$.

Now, the impermeability result tells us that there can be no notional velocity across an isentropic surface. How this is satisfied by the equations of motion? We write the right-hand side of (4.126) as

$$\rho S_Q = \nabla \cdot (\dot{\theta} \boldsymbol{\omega} + \theta \nabla \times \mathbf{F}) = \nabla \cdot (\dot{\theta} \boldsymbol{\omega} + \mathbf{F} \times \nabla \theta). \quad (4.131)$$

Thus, the $\mathbf{J}$ vector is

$$\mathbf{J} = \rho \mathbf{v} Q - \dot{\theta} \boldsymbol{\omega} - \mathbf{F} \times \nabla \theta \quad (4.132a)$$

$$= \rho Q (\mathbf{v}_\perp + \mathbf{v}_\parallel) - \dot{\theta} \boldsymbol{\omega}_\parallel - \mathbf{F} \times \nabla \theta, \quad (4.132b)$$

where

$$\mathbf{v}_\parallel = \mathbf{v} - \frac{\mathbf{v} \cdot \nabla \theta}{|\nabla \theta|^2} \nabla \theta, \quad \mathbf{v}_\perp = -\frac{\partial \theta / \partial t}{|\nabla \theta|^2} \nabla \theta, \quad (4.133a)$$

$$\boldsymbol{\omega}_\parallel = \boldsymbol{\omega} - \frac{\boldsymbol{\omega} \cdot \nabla \theta}{|\nabla \theta|^2} \nabla \theta = \boldsymbol{\omega} - \frac{Q}{|\nabla \theta|^2} \nabla \theta. \quad (4.133b)$$

The subscripts ‘ $\perp$ ’ and ‘ $\parallel$ ’ denote components perpendicular and parallel to the local isentropic surface, and $\mathbf{v}_\perp$ is the velocity of the isentropic surface normal to itself. Equation (4.132) may be verified by using (4.133) and $D\theta/Dt = \dot{\theta}$.

The ‘parallel’ terms in (4.132) are all vectors parallel to the local isentropic surface, and therefore do not lead to any flux of PV substance across that surface. Furthermore, the term $\rho Q \mathbf{v}_\perp$ is ρQ multiplied by the normal velocity of the surface. That is to say, the notional velocity associated with the flux normal to the isentropic surface is equal to the normal velocity of the isentropic surface itself, and so it too provides no flux of PV substance across that surface (even though there may well be a mass flux across the surface). Put simply, the isentropic surface always moves in such a way as to ensure that there is no flux of PV substance across it. In our proof of the impermeability result in the previous section we used the fact that the potential vorticity (times density) is the divergence of something. In the demonstration above we used the fact that the terms *forcing* it are the divergence of something.

** Dynamical choices of PV flux and a connection to Bernoulli's theorem*

If we add a non-divergent vector to the flux, $\mathbf{J}$, then it has no effect on the evolution of s . This gauge invariance means that the notional velocity, $\mathbf{v}_Q = \mathbf{J}/(\rho Q)$ is similarly non-unique, although it does not mean that there are not dynamical choices for it that are more appropriate in given circumstances. To explore this, let us obtain a general expression for $\mathbf{J}$ by starting with the definition of s , so that

$$\begin{aligned} \frac{\partial s}{\partial t} &= \nabla \theta \cdot \frac{\partial \boldsymbol{\omega}}{\partial t} + \boldsymbol{\omega} \cdot \nabla \frac{\partial \theta}{\partial t} \\ &= \nabla \theta \cdot \nabla \times \frac{\partial \mathbf{v}}{\partial t} + \nabla \cdot \left(\boldsymbol{\omega} \frac{\partial \theta}{\partial t} \right) = -\nabla \cdot \mathbf{J}' \end{aligned} \quad (4.134)$$

where

$$\mathbf{J}' = \nabla \theta \times \frac{\partial \mathbf{v}}{\partial t} - \frac{\partial \theta}{\partial t} \boldsymbol{\omega} + \nabla \phi \times \nabla \chi. \quad (4.135)$$

The last term in this expression is an arbitrary divergence-free vector. If we choose $\phi = \theta$ and $\chi = B$, where B is the Bernoulli function given by $B = I + \mathbf{v}^2/2 + p/\rho$ where I is the internal energy per unit mass, then

$$\mathbf{J}' = \nabla \theta \times \left(\nabla B + \frac{\partial \mathbf{v}}{\partial t} \right) - \boldsymbol{\omega}(\dot{\theta} - \mathbf{v} \cdot \nabla \theta), \quad (4.136)$$

having used the thermodynamic equation $D\theta/Dt = \dot{\theta}$. Now, the momentum equation may be written, without approximation, in the form (see problems 2.1 and 2.2)

$$\frac{\partial \mathbf{v}}{\partial t} = -\boldsymbol{\omega} \times \mathbf{v} + T \nabla \eta + \mathbf{F} - \nabla B \quad (4.137)$$

where η is the specific entropy ($d\eta = c_p d \ln \theta$). Using (4.136) and (4.137) gives

$$\mathbf{J}' = \rho Q \mathbf{v} - \dot{\theta} \boldsymbol{\omega} + \nabla \theta \times \mathbf{F}. \quad (4.138)$$

which is the same as (4.132a). Furthermore, using (4.135) for steady flow,

$$\mathbf{J} = \nabla \theta \times \nabla B. \quad (4.139)$$

That is, the flux of potential vorticity (in this gauge) is aligned with the intersection of θ - and B -surfaces. For steady *inviscid and adiabatic* flow the Bernoulli function is constant along streamlines, that is surfaces of constant Bernoulli function are aligned with streamlines, and, because θ is materially conserved, streamlines are formed at intersecting θ - and B -surfaces, as in (1.212). In the presence of forcing, this property is replaced by (4.139), that the flux of PV substance is along such intersections.

This choice of gauge leading to (4.138) is physical in that it reduces to the true advective flux $\mathbf{v} \rho Q$ for unforced, adiabatic flow, but it is not a unique choice, nor mandated by the dynamics. Choosing $\chi = 0$ leads to

$$\mathbf{J}_1 = \rho Q \mathbf{v} - \dot{\theta} \boldsymbol{\omega} + \nabla \theta \times (\mathbf{F} - \nabla B) \quad (4.140)$$

and using (4.135) this vanishes for steady flow, a potentially useful property.

Remarks

The impermeability result has a number of consequences, some obvious with hindsight, and it also provides a useful point of view and diagnostic tool.⁷ Here, we will just remark:

- ★ There can be no net transport of potential vorticity across an isentropic surface, and the total amount of potential vorticity in a volume wholly enclosed by isentropic surfaces is zero.
- ★ Thus, and trivially, the amount of potential vorticity contained between two isentropes isolated from the earth's surface in the northern hemisphere is the negative of the corresponding amount in the southern hemisphere.
- ★ Potential vorticity flux lines (i.e., lines everywhere parallel to $\mathbf{J}$) can either close in on themselves or begin and end at boundaries (e.g., the ground, the ocean surface). However, $\mathbf{J}$ may change its character. Thus, for example, at the base of the oceanic mixed layer $\mathbf{J}$ may change from being a diabatic flux above to an adiabatic advective flux below. There may be a similar change in character at the atmospheric tropopause.
- ★ The flux vector $\mathbf{J}$ is defined only to within the curl of a vector. Thus the vector $\mathbf{J}' = \mathbf{J} + \nabla \times \mathbf{A}$, where $\mathbf{A}$ is an arbitrary vector, is as valid as is $\mathbf{J}$ in the above derivations and diagnostics.

- ★ There can be no net turbulent ‘mixing’ of potential vorticity across an isentropic surface, in the absence of compensating diabatic source terms. Relatedly, one need only consider the along-isentrope fluxes of potential vorticity in wave-mean flow problems, considered in chapter 8.

Notes

- 1 Bjerknes (1898). His theorem is sometimes stated to be that the evolution of circulation around a circuit is determined by the number of solenoids passing through any surface bounded by that circuit, but the meaning is that of (4.54). Vilhelm Bjerknes (1862–1951) was a physicist and hydrodynamicist, who in 1917 moved to the University of Bergen as founding head of the Bergen Geophysical Institute. Here, late in his career, he did what is probably his most influential work in meteorology, setting up and contributing to the ‘Bergen School of Meteorology’. Among other things he and his colleagues were the first to consider, as a practical proposition, the use of numerical methods — initial data in conjunction with the fluid equations of motion — to forecast the state of the atmosphere, based on an earlier paper describing how that task might be done (Bjerknes 1904). Inaccurate initial velocity fields compounded with the shear complexity of the effort ultimately defeated them, but the effort was continued (also unsuccessfully) by L. F. Richardson (Richardson 1922), before J. Charney, R. Fjortoft and J. Von Neumann eventually made the first successful numerical forecast (Charney *et al.* 1950). Their success can be attributed to the use of a simplified, filtered, set of equations and the use of an electronic computer.

Vilhelm’s son, Jacob Bjerknes (1897–1975) was a leading player in the Bergen school. He was responsible for the now-famous frontal model of cyclones (Bjerknes 1919) and discussed the role of cyclones in the general circulation of the atmosphere. In collaboration with Halvor Solberg and Tor Bergeron the model led to a prescient picture of the life-cycle of extra-tropical cyclones (see chapter ??), in which a wave grows initially on the polar front (akin to baroclinic instability with the meridional temperature gradient compressed to a front, but that theory was not yet developed), develops into a mature cyclone, occludes and decays. In 1939 Bjerknes moved to the U.S. and, largely because of WWII, stayed, joining UCLA and heading its Dept. of Meteorology after its formation in 1945. He developed an interest in air-sea interactions, and notably proposed the essential mechanism governing El Niño, a feedback between sea-surface temperatures and the strength of the trade winds (Bjerknes 1969). [See also Friedman (1989), Cressman (1996), articles in Shapiro & Gronas (1999), and a memoir by Arnt Eliassen available from <http://www.nap.edu/readingroom/books/biomems/jbjerknes.html>.]

- 2 This result is sometimes attributed to ?, although it was evidently known to ?.
- 3 The expression ‘potential vorticity’ was introduced by Rossby (1940). In this paper, Rossby shows that in a homogeneous layer of fluid the quantity $q = (\zeta + f)/h$ is a material invariant in adiabatic flow — that is, $Dq/Dt = 0$.

Here, h is the thickness of the fluid column and ζ is computed using horizontal derivatives taken at constant density or potential temperature, for example $\zeta = (\partial v / \partial x)_\theta - (\partial u / \partial y)_\theta$. Rossby introduces the expression as follows: '*This quantity, which may be called the potential vorticity, represents the vorticity the air column would have if it were brought, isopycnally or isentropically, to a standard latitude (f_0) and stretched or shrunk vertically to a standard depth D_0 or weight Δ_0 .*' (Rossby's italics.) That is,

$$\text{Potential Vorticity} = \zeta_0 = \left(\frac{\zeta + f}{h} \right) D_0 - f_0, \quad (4.141)$$

which follows from his eq. (11), and this is the sense he uses it in that paper. However, potential vorticity has come to mean the quantity $(\zeta + f)/h$, which of course does not have the dimensions of vorticity. We use it in this latter, now conventional, sense throughout this book. Ironically, quasi-geostrophic potential vorticity as usually defined is closer to Rossby's definition and does have the dimensions of vorticity.

The expression for potential vorticity in a continuously stratified fluid was given by Ertel (1942*b*), and its relationship to circulation was given by Ertel (1942*a*). It is now commonly known as the *Ertel potential vorticity*. As noted above, Rossby (1940) had previously discovered the potential vorticity invariant for the shallow water equations, and in that paper we also find the Fermat-like comment 'It is possible to derive corresponding results for an atmosphere in which the potential temperature varies continuously with elevation.... The generalized treatment will be presented in another place.'(!) It is not clear whether Rossby and Ertel's derivations were independent: Charney (in Lindzen *et al.* 1990, p35) suggests they were, and Cressman (1996) remarks that the origin of the concept of potential vorticity is a 'delicate one that has aroused some passion in private correspondences'. Later, Ertel and Rossby did publish a short joint paper, Ertel & Rossby (1949), describing their conservation theorem.

- 4 Pedlosky (1987) provides a similar derivation. Truesdell (1951, 1954) and Obukhov (1962) were early explorers of the consequences of heating and friction on potential vorticity.
- 5 Potential vorticity evaluated on an isentropic surface is sometimes referred to as 'isentropic potential vorticity.' This expression should be eschewed, because it implies that it is somehow a different quantity than normal potential vorticity.
- 6 Haynes & McIntyre (1987, 1990). See also Danielsen (1990), ? [who obtained the result in (4.139)], ? and ?.
- 7 See for example McIntyre & Norton (1990) and Marshall & Nurser (1992). The latter use $\mathbf{J}$ vectors to study the creation and transport of potential vorticity in the oceanic thermocline.

Further Reading

Truesdell, C. 1954. *The Kinematics of Vorticity*.

A marvellous book, written in Truesdell's inimitable style, on many aspects of vorticity and with many historical references.

Batchelor, G. K. 1967. *An Introduction to Fluid Dynamics*.

This contains an extensive discussion of vorticity and vortices.

Salmon, R. S. 1998. *Geophysical Fluid Dynamics*.

Chapter 4 contains a brief discussion of potential vorticity, and chapter 7 a longer discussion of Hamiltonian fluid dynamics, in which the particle relabeling symmetry that gives rise to potential vorticity conservation is discussed.

Problems

- 4.1 For the νr vortex, choose a contour of arbitrary shape (e.g., a square) with segments neither parallel nor perpendicular to the radius, and not enclosing the origin. Show explicitly that the circulation around the contour is zero. (This problem is a little perverse.)
- 4.2 $\diamond$ Here we show how vortex stretching may be balanced by viscosity, leading to a steady state. Suppose there is an incompressible swirling flow given in cylindrical co-ordinates (r, ϕ, z) :

$$\mathbf{v} = (v_r, v_\phi, v_z) = (-\frac{1}{2}\alpha r, v_\phi, \alpha z) \quad (\text{P4.1})$$

Show that this satisfies the mass conservation equation. Show too that vorticity is only non-zero in the vertical direction. Show that the vertical component of the vorticity equation contains only the stretching term and that in steady state it is

$$-\frac{1}{2}\alpha r \frac{\partial \zeta}{\partial r} = \zeta \alpha + \nu \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \zeta}{\partial r} \right). \quad (\text{P4.2})$$

Show that this may be integrated to $\zeta = \zeta_0 \exp(-\alpha r^2/4\nu)$. Thus deduce that there is a rotational core of thickness $r_o = 2(\nu/\alpha)^{1/2}$, and that the radial velocity field is given by

$$v_\phi(r) = -\frac{1}{r} \frac{2\nu}{\alpha} \zeta_0 \exp\left[-\frac{\alpha r^2}{4\nu}\right] + \frac{A}{r} \quad (\text{P4.3})$$

where $A = 2\nu\zeta_0/\alpha$. What is the swirling velocity field? (Batchelor 1967)

- 4.3 (a) Beginning with the three-dimensional vorticity equation in a rotating frame of reference [either (4.70) or (4.71)] show that in Cartesian co-ordinates the evolution of the vertical component of vorticity is given by

$$\begin{aligned} \frac{D\zeta}{Dt} = & -(\zeta + f) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \\ & + \left(\frac{\partial v}{\partial z} \frac{\partial w}{\partial x} - \frac{\partial u}{\partial z} \frac{\partial w}{\partial y} \right) + \left(\frac{\partial \rho}{\partial x} \frac{\partial p}{\partial y} - \frac{\partial \rho}{\partial y} \frac{\partial p}{\partial x} \right) \end{aligned} \quad (\text{P4.4})$$

where 'vertical' means the direction parallel to the rotation vector.

- (b) $\diamond$ Repeat this derivation, but in now spherical co-ordinates where vertical means the radial direction, and discuss the differences between the resulting equation and (P4.4). In particular, show carefully how the β -term arises, and that it may be interpreted as a tilting term in the vorticity equation.

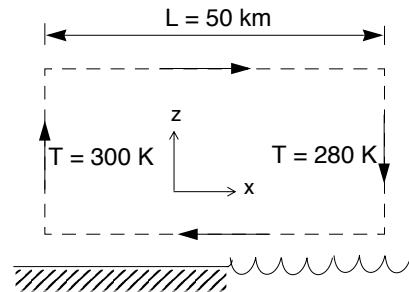
- 4.4 $\diamond$ Making use of Kelvin's circulation theorem obtain an expression for the potential vorticity that is conserved following the flow (for an adiabatic and unforced fluid) and that is appropriate for the hydrostatic primitive equations on a spherical planet. Express this in terms of the components of the spherical co-ordinate system.
- 4.5 $\diamond$ In pressure co-ordinates for hydrostatic flow on the f-plane, the horizontal momentum equation takes the form

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla\phi \quad (\text{P4.5})$$

On taking the curl of this, there appears to be no baroclinic term. Show that Kelvin's circulation theorem is nevertheless not in general satisfied, even for unforced, adiabatic flow. By appropriately choosing a path on which to evaluate the circulation obtain an expression for potential vorticity, in this co-ordinate system, that is conserved following the flow.

4.6 *Solenoids and sea-breezes.*

A land-sea temperature contrast of 20 K forces a sea breeze in the surface "mixed layer" (potential temperature nearly uniform with height), as illustrated schematically. The layer extends through the lowest 10% of the mass of the atmosphere.



- (a) In the absence of dissipation and diffusion, at what rate does the circulation change on a material circuit indicated? You may assume the horizontal flow is isobaric, and express your answer in m/s per hour.
- (b) Suppose the sea breeze is equilibrated by a nonlinear surface drag of the form

$$\frac{dV}{dt} = -\frac{V^2}{L_F} \quad (\text{P4.6})$$

with $L_F = (3 \text{ m s}^{-1})(3600 \text{ s})$. What is the steady speed of the horizontal wind in the case $L = 50 \text{ km}$?

- (c) Suppose that the width of the circulation is determined by a horizontal thermal diffusion of the form

$$\frac{d\theta}{dt} = \kappa_H \frac{\partial^2 \theta}{\partial x^2} \quad (\text{P4.7})$$

Provide an estimate of κ_H that is consistent with $L = 50 \text{ km}$. Comment on whether you think the extent of real sea breezes is determined this way.

Another advantage of a mathematical statement is that it is so definite that it might be definitely wrong. . . Some verbal statements have not this merit.

L. F. Richardson (1881–1953).

CHAPTER FIVE

Shallow Water Systems

Conventionally, ‘the’ shallow water equations describe a thin layer of constant density fluid in hydrostatic balance, rotating or not, bounded from below by a rigid surface and from above by a free surface, above which we suppose is another fluid of negligible inertia. This set-up can be generalized to multiple layers of immiscible fluids lying one on top of each other, forming a ‘stacked shallow water’ system, and this class of systems is the subject of this chapter.

Such systems are useful approximations of the real fluid system when the horizontal length-scale of the motion is much larger than the fluid depth, the same condition that ensures hydrostatic balance. They are one of the simplest useful models in geophysical fluid dynamics because they allow one to consider the effects of rotation in a simple framework when the full effects of stratification are not needed. However, the usefulness of the shallow water equations is more than pedagogical — there is a close physical and mathematical analogy between them and a description of the ocean or atmosphere written in isopycnal or isentropic co-ordinates, with a meaning beyond a coincidental similarity in the equations.

This chapter could also serve as a minicourse in geophysical fluid dynamics for those more interested in the methodology than the phenomenology.

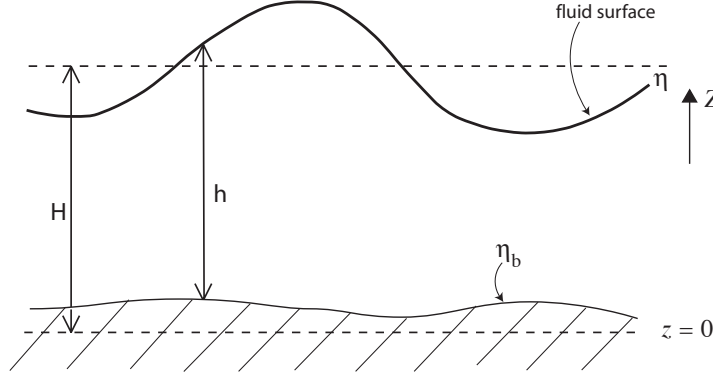


Figure 5.1 A shallow water system. H is the mean thickness of the fluid, $h(x, y)$ the thickness of a water column, $\eta(x, y)$ the height of the free surface and η_b is the height of the lower, rigid, surface above some arbitrary origin. By convention z is the vertical co-ordinate, and x and y the horizontal co-ordinates.

5.1 DYNAMICS OF A SINGLE LAYER

Shallow water dynamics apply, by definition, to a fluid layer of constant density in which the horizontal scale of the flow is much greater than the depth of the water. The fluid motion is then fully determined by the momentum and mass continuity equations, and because of the assumed small aspect ratio the hydrostatic approximation is well satisfied, and we invoke this from the outset.

Consider, then, fluid in a container above which is another fluid of negligible density (and therefore negligible inertia) relative to the fluid of interest, as illustrated in Fig. 5.1. As usual, our notation is that $\mathbf{v} = u\mathbf{i} + v\mathbf{j} + w\mathbf{k}$ is the three dimensional velocity and $\mathbf{u} = u\mathbf{i} + v\mathbf{j}$ is the horizontal velocity. $h(x, y)$ is thickness of the liquid column, H is its mean height, and η is the height of the free surface. In a flat-bottomed container $\eta = h$, whereas in general $h = \eta - \eta_b$, where η_b is the height of the floor of the container.

5.1.1 Momentum equations

The vertical momentum equation is the hydrostatic equation,

$$\frac{\partial p}{\partial z} = -\rho g \quad (5.1)$$

Since density is assumed constant we integrate this to

$$p(x, y, z) = -\rho g z + p_o \quad (5.2)$$

At the top of the fluid, $z = \eta$, the pressure is determined by the weight of the overlying fluid and this is assumed negligible. Thus, $p = 0$ at $z = \eta$ giving

$$p(x, y, z) = \rho g(h(x, y) - z) \quad (5.3)$$

The consequence of this is that the horizontal gradient of pressure is independent of height. That is

$$\nabla_z p = \rho g \nabla_z \eta \quad (5.4)$$

where

$$\nabla_z = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} \quad (5.5)$$

is the gradient operator at constant z . (In the rest of this chapter we will drop the subscript z unless that causes ambiguity. The three-dimensional gradient operator will be denoted ∇_3 . We will also mostly use Cartesian co-ordinates, but the shallow water equations may certainly be applied over a spherical planet — indeed, ‘Laplace’s tidal equations’ are essentially the shallow water equations on a sphere.) The horizontal momentum equations therefore become

$$\frac{D\mathbf{u}}{Dt} = -\frac{1}{\rho} \nabla p = -g \nabla \eta \quad (5.6)$$

The right-hand side of this equation is independent of the vertical co-ordinate z . Thus, if the flow is initially independent of z , it must stay so. (This z -independence is unrelated to that arising from the rapid rotation necessary for the Taylor-Proudman effect.) The velocities u and v are functions only of x, y and t and the horizontal momentum equation is therefore

$$\frac{D\mathbf{u}}{Dt} = \frac{\partial \mathbf{u}}{\partial t} + u \frac{\partial \mathbf{u}}{\partial x} + v \frac{\partial \mathbf{u}}{\partial y} = -g \nabla \eta. \quad (5.7)$$

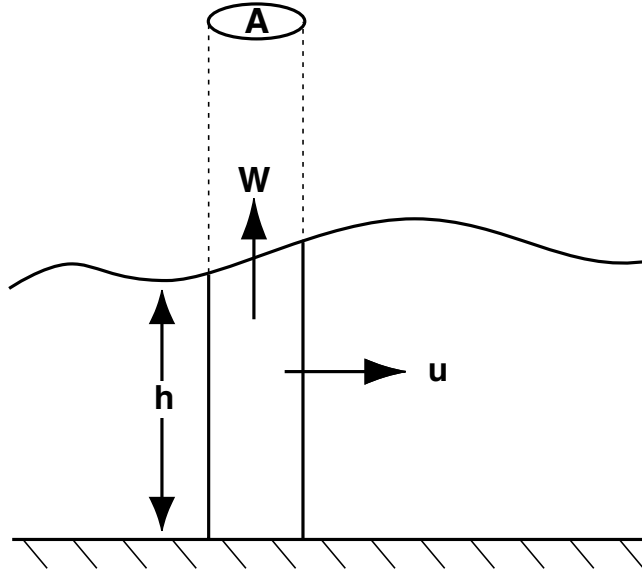
That the horizontal velocity is independent of z is a consequence of the hydrostatic equation, which ensures that the horizontal pressure gradient is independent of height. (Another starting point would be to take this independence of the horizontal motion with height as the *definition* of shallow water flow.) In real physical situations this is not exactly satisfied — for example, friction at the bottom may induce a vertical dependence of the flow in a boundary layer.

In the presence of rotation (5.8) easily generalizes to

$$\boxed{\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -g \nabla \eta} \quad (5.8)$$

where $\mathbf{f} = f\mathbf{k}$. Just as with the primitive equations, f may be constant or may vary with latitude. In the former case, $f = 2\Omega$ whereas on a spherical planet $f = 2\Omega \sin \vartheta$ and on the β -plane $f = f_0 + \beta y$.

Figure 5.2 The mass budget for a column of area A in a shallow water system. The fluid leaving the column is $\oint \rho \mathbf{h} \mathbf{u} \cdot \mathbf{n} dl$ where $\mathbf{n}$ is the unit vector normal to the boundary of the fluid column. There is a non-zero vertical velocity at the top of the column if the mass convergence into the column is non-zero.



5.1.2 Mass conservation equation

From first principles

The mass contained in a fluid column of height h and cross-sectional area A is given by $\int_A \rho h dA$ (Fig. 5.2). If there is a net flux of fluid across the column boundary (by advection) then this must be balanced by a net increase in the mass in A , and therefore a net increase in the height of the water column. The mass convergence into the column is given by

$$F_m = \text{Mass flux in} = - \int_S \rho \mathbf{u} \cdot d\mathbf{S} \quad (5.9)$$

where S is the area of the vertical boundary of the column. The surface area of the column is comprised of elements of area $h \mathbf{n} \delta l$, where δl is a line element circumscribing the column and $\mathbf{n}$ is a unit vector perpendicular to the boundary, pointing outwards. Thus (5.9) becomes

$$F_m = - \oint \rho \mathbf{h} \mathbf{u} \cdot \mathbf{n} dl \quad (5.10)$$

Using the divergence theorem in two-dimensions, (5.10) simplifies to

$$F_m = - \int_A \nabla \cdot (\rho \mathbf{u} h) dA. \quad (5.11)$$

where the integral is over the cross-sectional area of the fluid column (looking down from above). This is balanced by the local increase in height of the water

column, given by

$$F_m = \frac{d}{dt} \int \rho dV = \frac{d}{dt} \int_A \rho h dA = \int_A \rho \frac{\partial h}{\partial t} dA \quad (5.12)$$

The balance between (5.11) and (5.12) thus leads to

$$\int_A \left[\frac{\partial}{\partial t} h + \nabla \cdot (\mathbf{u}h) \right] dA = 0 \quad (5.13)$$

Because the area is arbitrary the integrand itself must vanish, whence,

$$\boxed{\frac{\partial h}{\partial t} + \nabla \cdot (\mathbf{u}h) = 0}, \quad (5.14)$$

or equivalently

$$\boxed{\frac{Dh}{Dt} + h \nabla_h \cdot \mathbf{u} = 0}. \quad (5.15)$$

This derivation holds whether or not the lower surface is flat. If it is, then $h = \eta$, and if not $h = \eta - \eta_b$.

From the 3D mass conservation equation

Since the fluid is incompressible, the three-dimensional mass continuity equation is just $\nabla \cdot \mathbf{v} = 0$. Writing this out in component form

$$\frac{\partial w}{\partial z} = - \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = -\nabla \cdot \mathbf{u} \quad (5.16)$$

Integrate this from 0 to h , noting that the right-hand side is independent of z , to give

$$w(\eta) - w(\eta_b) = -h \nabla \cdot \mathbf{u}. \quad (5.17)$$

If the bottom of the container is flat the vertical velocity is zero there. At the top the vertical velocity is the material derivative of the position of a particular fluid element. But the position of the fluid at the top is just η , and therefore (see Fig. 5.2)

$$w(\eta) = \frac{D\eta}{Dt}. \quad (5.18)$$

At the bottom of the fluid we have similarly

$$w(\eta_b) = \frac{D\eta_b}{Dt}. \quad (5.19)$$

The Shallow Water Equations

For a single-layer fluid, and including the Coriolis term, the inviscid shallow water equations are:

Momentum:

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -g\nabla\eta. \quad (\text{SW.1})$$

Mass Conservation:

$$\frac{Dh}{Dt} + h\nabla \cdot \mathbf{u} = 0 \quad \text{or} \quad \frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{u}) = 0. \quad (\text{SW.2})$$

where $\mathbf{u}(x, y, t)$ is the horizontal velocity and $h(x, y, t) = \eta(x, y, t) - \eta_b(x, y)$ is the total fluid thickness, η is the height of the upper free surface and η_b is the height of the lower surface (the bottom topography). The material derivative is

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla = \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y}, \quad (\text{SW.3})$$

with the rightmost expression holding in Cartesian co-ordinates.

(where, absent earthquakes and similar, $\partial\eta_b/\partial t = 0$). Using the last two equations, (5.17) becomes

$$\frac{D}{Dt}(\eta - \eta_b) + h\nabla \cdot \mathbf{u} = 0 \quad (5.20)$$

or

$$\frac{Dh}{Dt} + h\nabla_h \cdot \mathbf{u} = 0. \quad (5.21)$$

as in (5.15).

5.1.3 A rigid lid

The case where the *upper* surface is held flat by the imposition of a rigid lid is sometimes of interest. The ocean suggests one such example, for here the bathymetry at the bottom of the ocean provides much larger variations in fluid thickness than do the small variations in the height of the ocean surface. Suppose then the upper surface is at a constant height H then, from (5.14) with $\partial h/\partial t = 0$ and $h = h_b = H - \eta_b$ the mass conservation equation becomes

$$\nabla_h \cdot (h\mathbf{u}_b) = 0. \quad (5.22)$$

Note that this allows us to define an incompressible *mass-transport* velocity $\mathbf{U} = h_b \mathbf{u}$.

Although the upper surface is flat, the pressure there is no longer constant because a force must be provided by the rigid lid to keep the surface flat. The horizontal momentum equation is

$$\frac{D\mathbf{u}}{Dt} = -\frac{1}{\rho_0} \nabla p_{\text{lid}} \quad (5.23)$$

where p_{lid} is the pressure at the lid, and the complete equations of motion are then (5.22) and (5.23).¹ If the lower surface is flat, the two-dimensional flow itself is divergence-free, and the equations reduce to the two-dimensional incompressible Euler equations.

5.1.4 Stretching and the vertical velocity

Because the horizontal velocity is depth independent, the vertical velocity plays no role in advection. It is certainly not zero for then the free surface would be unable to move up or down, but because of the vertical independence of the horizontal flow it does have a simple vertical structure. Write the mass conservation equation as

$$\frac{\partial w}{\partial z} = -\nabla \cdot \mathbf{u} \quad (5.24)$$

and integrate this upwards from the bottom to give

$$w = w_b - (\nabla_h \cdot \mathbf{u})(z - h_b) \quad (5.25)$$

or

$$\frac{Dz}{Dt} = \frac{Dh_b}{Dt} - (\nabla \cdot \mathbf{u})(z - h_b) \quad (5.26)$$

Evidently, the vertical velocity is a linear function of height. At the upper surface $w = D\eta/Dt$ so that here (5.26) is

$$\frac{D\eta}{Dt} = \frac{Dh_b}{Dt} - (\nabla \cdot \mathbf{u})(\eta - h_b), \quad (5.27)$$

Eliminating the divergence term from the last two equations gives

$$\frac{D}{Dt}(z - h_b) = \frac{z - h_b}{\eta - h_b} \frac{D}{Dt}(\eta - \eta_b) \quad (5.28)$$

which gives

$$\frac{D}{Dt} \left[\frac{z - h_b}{\eta - h_b} \right] = \frac{D}{Dt} \left[\frac{z - h_b}{h} \right] = 0. \quad (5.29)$$

This kinematic property of the shallow water system equation, which we might perhaps have guessed, means that the fluid stretches uniformly in a column; the ratio of the height of a fluid parcel above the floor to the total depth of the column is fixed.

5.1.5 Analogy with Compressible Flow

The shallow water equations (5.8) and (5.14) are directly analogous to the compressible gas dynamic equations in two dimensions. These are

$$\frac{D\mathbf{u}}{Dt} = -\frac{1}{\rho}\nabla p \quad (5.30)$$

and

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\mathbf{u}\rho) = 0 \quad (5.31)$$

along with an equation of state which we take to be $p = f(\rho)$. The mass conservation equations (5.14) and (5.31) are identical, with the replacement $\rho \Leftrightarrow h$. If $p = C\rho^\gamma$, then (5.30) becomes

$$\frac{D\mathbf{u}}{Dt} = -\frac{1}{\rho} \frac{dp}{d\rho} \nabla \rho = -C\gamma \rho^{\gamma-2} \nabla \rho \quad (5.32)$$

If $\gamma = 2$ then the momentum equations (5.8) and (5.32) become equivalent, with $\rho \Leftrightarrow h$ and $C\gamma \Leftrightarrow g$. In an ideal gas $\gamma = c_p/c_v$ and values typically are in fact less than two (e.g., 7/5); however, if the equations are linearized, then the analogy is exact for all values of γ , for then (5.32) becomes

$$\frac{\partial \mathbf{v}'}{\partial t} = -\frac{1}{\rho_0} c_s^2 \nabla \rho' \quad (5.33)$$

where $c_s^2 = dp/d\rho$, and this is analogous to the linearized shallow water momentum equation,

$$\frac{\partial \mathbf{u}'}{\partial t} = -\frac{1}{H} (gH) \nabla h' \quad (5.34)$$

where H is the mean fluid height. Equations (5.34) and (5.33) are formally identical, with the identifications $\rho \Leftrightarrow h$ and $\rho_0 \Leftrightarrow H$, where H is the mean fluid height. In (5.33) the constant c_s^2 is the sound speed, and in (5.34) this becomes gH , indicating that we may expect the shallow water equations to have wave-like solutions travelling at a speed $\sqrt{gH}$, as we see in section 5.6.

5.1.6 Reduced gravity equations

Consider now a single shallow moving layer of fluid overlying a deep, quiescent fluid layer. This configuration is often used as a model of the upper ocean: the upper layer represents flow in perhaps the upper few hundred meters of the ocean, the lower layer the near-stagnant abyss. If we turn the model upside-down we have a model, perhaps slightly less realistic, of the atmosphere: the lower layer represents motion in the troposphere above which lies an inactive stratosphere. The equations of motion are virtually the same in both cases.

5.1.7 Pressure gradient in the active layer

We'll derive the equations for the oceanic case (active layer on top) in two cases, which differ slightly in the assumption made about the upper surface.

I Free upper surface

The pressure in the upper layer is given by integrating the hydrostatic equation down from the upper surface. Thus, at a height z in the upper layer

$$p_1(z) = g\rho_1(\eta_0 - z) \quad (5.35)$$

where η_0 is the height of the upper surface. Hence, everywhere in the upper layer,

$$\frac{1}{\rho_1} \nabla p_1 = -g \nabla \eta_0. \quad (5.36)$$

and the momentum equation is

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -g \nabla \eta_0. \quad (5.37)$$

In the lower layer the the pressure is also given by the weight of the fluid above it. Thus, at some level z in the lower layer,

$$p_2(z) = \rho_1 g(\eta_0 - \eta_1) + \rho_2 g(\eta_1 - z). \quad (5.38)$$

But if this layer is motionless the horizontal pressure gradient in it is zero and therefore, to within a dynamically unimportant constant,

$$\rho_1 g \eta_0 = -\rho_1 g' \eta_1 \quad (5.39)$$

where $g' = g(\rho_2 - \rho_1) / \rho_1$ is the *reduced gravity*. Finally, the equations are completed by the usual mass conservation equation,

$$\frac{Dh}{Dt} + h \nabla \cdot \mathbf{u} = 0 \quad (5.40)$$

where $h = \eta_0 - \eta_1$. Equations (5.37), (5.40) and (5.39) are a complete set.

Because $g \gg g'$, (5.39) shows that surface displacements are *much smaller* than the displacements at the interior interface. We see this in the real ocean where the interior isopycnal displacements may be several tens of meters but variations in the height of ocean surface are of order centimeters.

II The rigid lid approximation

The smallness of the upper surface displacement suggests that we will make little error if we impose a *rigid lid* at the top of the fluid. Displacements are no longer allowed, but the lid will in general impart a pressure force to the fluid. Suppose that this is $P(x, y, t)$, then the horizontal pressure gradient in the upper layer is simply

$$\nabla p_1 = \nabla P. \quad (5.41)$$

The pressure in the lower layer is again given by hydrostasy, and is

$$\begin{aligned} p_2 &= -\rho_1 g \eta_1 + \rho_2 g(\eta_1 - z) + P \\ &= \rho_1 g h - \rho_2 g(h + z) + \rho_1 g h + P, \end{aligned} \quad (5.42)$$

so that

$$\nabla p_2 = -g(\rho_2 - \rho_1)\nabla h + \nabla P \quad (5.43)$$

Then if $\nabla p_2 = 0$ we have

$$g(\rho_2 - \rho_1)\nabla h = \nabla P \quad (5.44)$$

and the momentum equation for the upper layer is just

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -g'_1 \nabla h. \quad (5.45)$$

where $g' = g(\rho_2 - \rho_1) / \rho_1$. These equations differ from the shallow water equations only in the use of a reduced gravity g' in place of g itself. It is the *density difference* between the two layers that is important: if we take a ‘conventional’ shallow water system, with the moving layer on the bottom, but we suppose that overlying it is a stationary fluid of finite density, then we would easily find that the fluid equations for the moving layer are the same as if the fluid on top had zero inertia except that g would be replaced by an appropriate reduced gravity (problem 5.1).

5.2 MULTI-LAYER SHALLOW WATER EQUATIONS

We now consider the dynamics of multiple layers of fluid stacked on top of each other. This is a crude representation of continuous stratification, but it turns out to be a powerful model of many geophysically interesting phenomena as well as being physically realizable in the laboratory. The pressure is continuous across the interface, but the density jumps discontinuously and this allows the horizontal velocity to have a corresponding discontinuity. The set up is illustrated in Fig. 5.3.

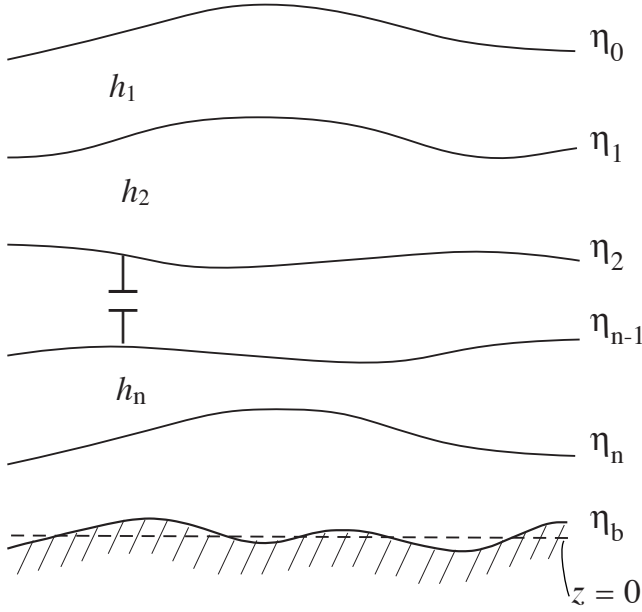


Figure 5.3 The multi-layer shallow water system. The layers are numbered from the top down. The co-ordinates of the interfaces are denoted η_i , and the layer thicknesses h_i .

In each layer pressure is given by the hydrostatic approximation, and so anywhere in the interior we can find the pressure by integrating down from the top. Thus, at a height z in the first layer we have

$$p_1 = \rho_1 g(\eta_0 - z), \quad (5.46)$$

and in the second layer,

$$p_2 = \rho_1 g(\eta_0 - \eta_1) + \rho_2 g(\eta_1 - z) = \rho_1 g\eta_0 + \rho_2 g'\eta_1 - \rho_2 g z \quad (5.47)$$

where $g'_1 = g(\rho_2 - \rho_1)/\rho_1$, and so on. The term involving z is irrelevant for the dynamics, because only the horizontal derivative enters the equation of motion. Omitting this term, for the n 'th layer the 'dynamical pressure' is given by the sum from the top down:

$$p_n = \rho_1 \sum_{i=0}^{n-1} g'_i \eta_i \quad (5.48)$$

where $g'_i = g(\rho_{i+1} - \rho_i)/\rho_1$ and $\rho_0 = 0$. This may be expressed in terms of the layer thicknesses by summing from the bottom up:

$$\eta_n = \eta_b + \sum_{i=n+1}^{i=N} h_i. \quad (5.49)$$

The momentum equation for each layer may then be written, in general,

$$\frac{D\mathbf{u}_n}{Dt} + \mathbf{f} \times \mathbf{u}_n = \frac{1}{\rho_n} \nabla p_n \quad (5.50)$$

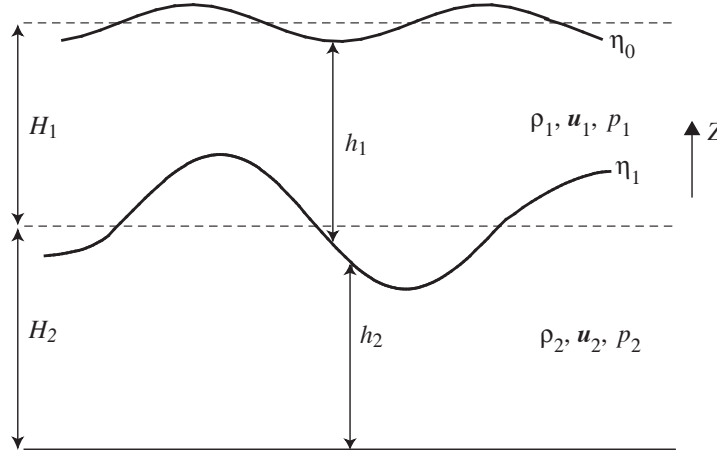


Figure 5.4 The two layer shallow water system. A fluid of density ρ_1 lies over a denser fluid of density ρ_2 . In the reduced gravity case the lower layer may be arbitrarily thick and is assumed stationary and so has no horizontal pressure gradient. In the 'rigid-lid' approximation the top surface displacement is neglected, but there is then a non-zero pressure gradient induced by the lid.

where the pressure is given by (5.48) and in terms of the layer depths using (5.50). If we make the Boussinesq approximation then ρ_n on the right-hand side of (5.50) is replaced by ρ_1 .

Finally, the mass conservation equation for each layer has the same form as the single-layer case, and is

$$\frac{Dh_n}{Dt} + h_n \nabla \cdot \mathbf{u}_n = 0. \quad (5.51)$$

The two- and three-layer cases

The two-layer model is the simplest model to capture the effects of stratification. Evaluating the pressures using (5.48) and (5.49) we find:

$$p_1 = \rho_1 g \eta_0 = \rho_1 g (h_1 + h_2 + h_3) \quad (5.52a)$$

$$p_2 = \rho_1 [g \eta_0 + g'_1 \eta_1] = \rho_1 [g (h_1 + h_2 + \eta_b) + \rho_1 g'_1 (h_2 + h_b)] \quad (5.52b)$$

The momentum equations for the two layers are then

$$\frac{D\mathbf{u}_1}{Dt} + \mathbf{f} \times \mathbf{u}_1 = -g \nabla \eta_0 = -g \nabla (h_1 + h_2 + h_b). \quad (5.53a)$$

and in the bottom layer

$$\begin{aligned} \frac{D\mathbf{u}_2}{Dt} + \mathbf{f} \times \mathbf{u}_2 &= -\frac{\rho_1}{\rho_2} (g\nabla\eta_0 + g'_1\nabla\eta_1) \\ &= -\frac{\rho_1}{\rho_2} [g\nabla(\eta_b + h_1 + h_2) + g'_1\nabla(h_2 + h_b)] \end{aligned} \quad (5.53b)$$

In the Boussinesq approximation ρ_1/ρ_2 is replaced by unity.

In a three layer model the dynamical pressures are found to be

$$p_1 = \rho_1 gh \quad (5.54a)$$

$$p_2 = \rho_1 [gh + g'_1(h_2 + h_3 + h_b)] \quad (5.54b)$$

$$p_3 = \rho_1 [gh + g'_1(h_2 + h_3 + h_b) + g'_2(h_3 + h_b)] \quad (5.54c)$$

where $h = \eta_0 = h_b + h_1 + h_2 + h_3$ and $g'_2 = g(\rho_3 - \rho_2)/\rho_1$. More layers can obviously be added in a systematic fashion.

The mass conservation equation for each laer of a multi-layer model is

$$\frac{\partial h_n}{\partial t} + \nabla \cdot (\mathbf{u}_n h_n) = 0. \quad (5.55)$$

5.2.1 Reduced-gravity multi-layer equation

As with a single active layer, we may envision multiple layers of fluid overlying a deeper stationary layer. This is a useful model of the stratified upper ocean overlying a nearly stationary and nearly unstratified abyss. Indeed we use such a model to study the ‘ventilated thermocline’ in chapter ?? and a detailed treatment may be found there. If we suppose there is a lid at the top, then the model is almost the same as that of the previous section. However, now the horizontal pressure gradient in the lowest model layer is zero, and so we may obtain the pressures in all the active layers by integrating the hydrostatic equation upwards from this layer. Suppose we have N moving layers, then the reader may verify that the dynamic pressure in the n ’th layer is given by

$$p_n = - \sum_{i=n}^{i=N} \rho_1 g'_i \eta_i \quad (5.56)$$

where as before $g'_i = g(\rho_{i+1} - \rho_i)/\rho_1$. If we have a lid at the top, so that $\eta_0 = 0$, then the interface displacements are related to the layer thicknesses by

$$\eta_n = - \sum_{i=1}^{i=n} h_i. \quad (5.57)$$

From these expressions the momentum equation in each layer is easily constructed.

5.2.2 Thermal wind

The slope of an interfacial surface is directly related to the difference in pressure gradient on either side and so, by geostrophic balance, to the shear of the flow. This is the shallow water analog of the thermal wind relation. To see this, consider the interface, η , between two layers labelled 1 and 2. The pressure in two layers is given by the hydrostatic relation and so,

$$p_1 = A(x, y) - \rho_1 g z \quad (\text{at some } z \text{ in layer 1}) \quad (5.58a)$$

$$\begin{aligned} p_2 &= A(x, y) - \rho_1 g \eta + \rho_2 g (\eta - z) \\ &= A(x, y) + \rho_1 g'_1 \eta - \rho_2 g z \quad (\text{at some } z \text{ in layer 2}) \end{aligned} \quad (5.58b)$$

where $A(x, y)$ is the pressure at where $z = 0$, but we don't need to specify where this is. Thus we find

$$\frac{1}{\rho_1} \nabla(p_1 - p_2) = -g'_1 \nabla \eta. \quad (5.59)$$

If the flow is geostrophically balanced and Boussinesq then, in each layer, the velocity obeys

$$f \mathbf{u}_i = \frac{1}{\rho_1} \mathbf{k} \times \nabla p_i. \quad (5.60)$$

Using (5.59) then gives

$$f(\mathbf{u}_1 - \mathbf{u}_2) = -\mathbf{k} \times g'_1 \nabla \eta, \quad (5.61)$$

or in general

$$f(\mathbf{u}_n - \mathbf{u}_{n+1}) = \mathbf{k} \times g'_n \nabla \eta. \quad (5.62)$$

This is the thermal wind equation for the shallow water system. It applies at any interface, and it implies *the shear is proportional to the interface slope*.

Suppose that we represent the atmosphere by two layers of fluid then a meridionally decreasing temperature may be represented by an interface that slopes upward toward the pole, and in the Northern hemisphere $\partial \eta / \partial y > 0$. Then, in the northern hemisphere where f is positive we have

$$u_1 - u_2 = \frac{g_1}{f} \frac{\partial \eta}{\partial y} > 0, \quad (5.63)$$

and the temperature gradient is associated with a positive shear. (See problem 5.2.)

5.2.3 Form Drag

When the interface between two layers varies with position — that is, when it is wavy — the layers exert a pressure force on each other. Similarly, if the bottom is

not flat then the topography and the bottom layer can exert forces on each other. This is known as *form drag*, and it is an important means whereby momentum can be added to or extracted from a flow.² Consider a layer confined between two interfaces, $\eta_1(x, y)$ and $\eta_2(x, y)$. Then over some zonal interval L the average zonal pressure force on that fluid layer is given by

$$F_p = -\frac{1}{L} \int_{x_1}^{x_2} \int_{\eta_2}^{\eta_1} \frac{\partial p}{\partial x} dx dz. \quad (5.64)$$

Integrating by parts first in z and then in x , and noting that by hydrostasy $\partial p / \partial z$ does not depend on horizontal position within the layer, we obtain

$$\begin{aligned} F_p &= -\frac{1}{L} \int_{x_1}^{x_2} \left[\frac{\partial p}{\partial x} z \right]_{\eta_2}^{\eta_1} dx \\ &= -\overline{\eta_1 \frac{\partial p_1}{\partial x}} + \overline{\eta_2 \frac{\partial p_2}{\partial x}} = \overline{+p_1 \frac{\partial \eta_1}{\partial x}} - \overline{p_2 \frac{\partial \eta_2}{\partial x}}, \end{aligned} \quad (5.65)$$

where p_1 is the pressure at η_1 , and similarly for p_2 , and to obtain the second line we suppose that the integral is around a closed path, such as a circle of latitude, and the average is denoted with an overbar. These terms represent the transfer of momentum from one layer to the next, and at a particular interface, i , we may define the form drag, τ_i , by

$$\tau_i \equiv \overline{p_i \frac{\partial \eta_i}{\partial x}} = -\overline{\eta_i \frac{\partial p_i}{\partial x}}. \quad (5.66)$$

The form drag is a stress, and as the layer depth shrinks to zero its vertical derivative, $\partial \tau / \partial z$, is the force on the fluid. It is a particularly important mechanism for the vertical transfer of momentum and its ultimate removal in an eddying fluid, and it one of the the main mechanisms wherby the wind stress at the top of the ocean is communicated to the ocean bottom. At the fluid bottom the form drag is $\overline{p \eta_{bx}}$, where η_b is the bottom topography, and this is proportional to the momentum exchange with the solid earth. This is a significant mechanism for the ultimate removal of momentum in the ocean, especially in the Antactic Circumpolar Current where it is likely to be much larger than bottom (Ekman) drag arising from small scale turbulence and friction. In the two layer, flat-bottomed case the only form drag occurring as that at the interface, and the momentum transfer between the layers is just $\overline{p_1 \partial \eta_1 / \partial x}$ or $-\overline{\eta_1 \partial p / \partial x}$; then, the force on each layer due to the other is equal and opposite, as we would expect from momentum conservation.

For flows in geostrophic balance, the form drag is closely related to the meridional heat flux. The pressure gradient and velocity are related by $\rho f v' = \partial p' / \partial x$ and the interfacial displacement is proportional to the temperature perturbation, b' (in fact one may show that $\eta' = -b' / \partial \bar{b} / \partial z$). Thus $-\overline{\eta' \partial p' / \partial x} \propto \overline{v' b'}$, a correspondence that will re-occur when we consider the *Eliassen-Palm flux* in chapter 8.

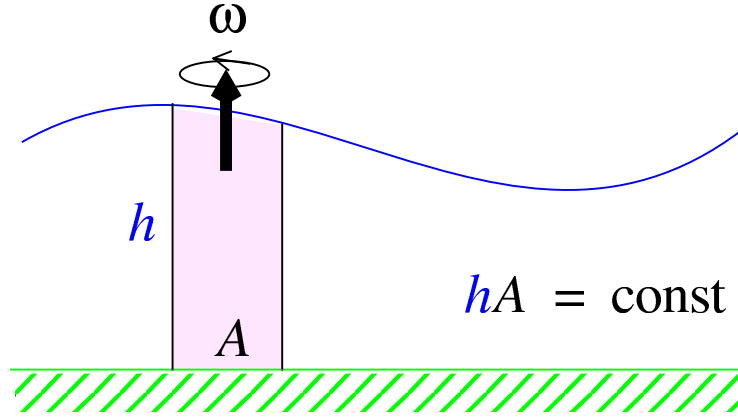


Figure 5.5 The mass of a column of fluid, hA , is conserved in the shallow water system. Furthermore, the vorticity is tied to material lines so that ζA is also a material invariant, where $\zeta = \boldsymbol{\omega} \cdot \mathbf{k}$ is the vertical component of the vorticity. From this, ζ/h must be materially conserved; that is $D(\zeta/h)/Dt = 0$, which is the conservation of potential vorticity in a shallow water system. In a rotating system this readily generalizes to $D[(\zeta + f)/h]/Dt = 0$.

5.3 CONSERVATION PROPERTIES

5.3.1 A Lagrangian invariant: potential vorticity

Because of the presence of a free surface vorticity is not longer conserved on parcels in the shallow water equations, whereas it is for 2D incompressible flow. However, there is a conserved quantity related to vorticity. Making use of the vector identity

$$(\mathbf{u} \cdot \nabla)\mathbf{u} = \frac{1}{2}\nabla(\mathbf{u} \cdot \mathbf{u}) - \mathbf{u} \times (\nabla \times \mathbf{u}) \quad (5.67)$$

we write the momentum equation (5.8) as

$$\frac{\partial \mathbf{u}}{\partial t} + \boldsymbol{\omega} \times \mathbf{u} = -g\nabla(h + \frac{1}{2}\mathbf{u}^2) \quad (5.68)$$

where $\boldsymbol{\omega} = (\partial v / \partial x - \partial u / \partial y)\mathbf{k}$ is the two-dimensional vorticity (see also chapter 4). To obtain an evolution equation for the vorticity we take the curl of (5.68), making use of the vector identity

$$\begin{aligned} \nabla \times \boldsymbol{\omega} \times \mathbf{u} &= (\mathbf{u} \cdot \nabla)\boldsymbol{\omega} - (\boldsymbol{\omega} \cdot \nabla)\mathbf{u} + \boldsymbol{\omega} \nabla \cdot \mathbf{u} - \mathbf{u} \nabla \cdot \boldsymbol{\omega} \\ &= (\mathbf{u} \cdot \nabla)\boldsymbol{\omega} + \boldsymbol{\omega} \nabla \cdot \mathbf{u}. \end{aligned} \quad (5.69)$$

Here, $\nabla \cdot \boldsymbol{\omega}$ is the divergence of a curl and therefore zero, and $(\boldsymbol{\omega} \cdot \nabla)\mathbf{u} = 0$ because $\boldsymbol{\omega}$ is perpendicular to any variation in $\mathbf{u}$. Thus, the curl of (5.68) becomes

$$\frac{\partial \boldsymbol{\omega}}{\partial t} + (\mathbf{u} \cdot \nabla)\boldsymbol{\omega} = -\boldsymbol{\omega} \nabla \cdot \mathbf{u} \quad (5.70)$$

or, defining ζ by letting $\omega = \mathbf{k}\zeta$,

$$\frac{\partial \zeta}{\partial t} + (\mathbf{u} \cdot \nabla)\zeta = -\zeta \nabla \cdot \mathbf{u}. \quad (5.71)$$

However, from the mass conservation equation (1.19), the right-hand side is

$$-\zeta \nabla \cdot \mathbf{u} = \frac{\zeta}{h} \frac{Dh}{Dt}, \quad (5.72)$$

and hence (5.71) is

$$\frac{\partial \zeta}{\partial t} + (\mathbf{u} \cdot \nabla)\zeta = \frac{\zeta}{h} \frac{Dh}{Dt}, \quad (5.73)$$

which simplifies to

$$\frac{D}{Dt} \left(\frac{\zeta}{h} \right) = 0 \quad (5.74)$$

The important quantity ζ/h , often denoted Q , is known as the *potential vorticity*, and (5.74) is known as the potential vorticity equation. We re-derive this conservation law in a more general way in section 5.5

In two-dimensional non-divergent flow, the right hand side of (5.71) is zero and we are lead to conservation of vorticity on parcels. There is no vortex stretching. In the shallow water equations vorticity only has one component, so there can be no twisting terms in the vorticity equation. However, because of variations in free surface height there can be vortex stretching. As height increases, vortex tubes are stretched and vorticity is proportionally enhanced, and the quantity ζ/h is conserved. It is a Lagrangian invariant, being conserved on parcels.

Not only is potential vorticity, Q , a Lagrangian invariant, but so is any function of Q . To see this, multiply (5.74) by $F'(Q)$ where F denotes any differentiable function, and the prime denotes a derivative. Then we obtain

$$F'(Q) \frac{DQ}{Dt} = \frac{D}{Dt} F(Q) = 0 \quad (5.75)$$

Since F is arbitrary, there are an infinite number of Lagrangian invariants corresponding to different choices of F .

5.3.2 Integral Invariants: Energy and Vorticity

Vortical Conservation Relations

Consider the integral

$$C = \int_A \zeta \, dA = \int_A qh \, dA, \quad (5.76)$$

where the integral is over a surface 'A', the cross-sectional area of a column of height h . Taking the material derivative of this gives

$$\frac{DC}{Dt} = \int \frac{DQ}{Dt} h dA + \int Q \frac{D}{Dt} (h dA). \quad (5.77)$$

The first term is zero, by (5.73); the second term is just the derivative of the volume of a column of fluid and it too is zero, by mass conservation. Thus,

$$\frac{DC}{Dt} = \frac{D}{Dt} \int \zeta dA = 0 \quad (5.78)$$

Thus, the integral of the vorticity over a some cross-section area of the fluid is unchanging, although both the vorticity and area of the fluid may individually change. This is Kelvin's circulation theorem applied to the shallow water system, and using Stokes' theorem, it may be written

$$\frac{DC}{Dt} = \frac{D}{Dt} \oint \mathbf{u} \cdot d\mathbf{l} \quad (5.79)$$

where the line integral is around the boundary of 'A'.

A slight generalization of (5.79) is possible. Consider the integral $I = \int F(Q) h dA$ where again F is any differentiable function of its argument. It is clear that

$$\frac{D}{Dt} \int F(Q) h dA = 0. \quad (5.80)$$

If the area of integration in (5.68) or (5.80) is the whole domain (enclosed by frictionless walls, for example) then it is clear that the integral of $hF(Q)$ is a constant, including as a special case the integral of ζ .

Energy

Since we have made various simplifications in deriving the shallow water system, it is not self-evident that energy should be conserved, or indeed what form the energy takes. The kinetic energy density, that is the kinetic energy per unit area, is $\rho h \mathbf{u}^2 / 2$. The potential energy density of the fluid is

$$P.E. = \int_0^h \rho_0 g z dz = \frac{1}{2} \rho_0 g h^2 \quad (5.81)$$

The factor ρ_0 appears in both kinetic and potential energies and, because it is a constant, we will omit it.

Using the mass conservation equation (5.15) we obtain an equation for the evolution of potential energy density:

$$\frac{D}{Dt} \frac{gh^2}{2} + gh^2 \nabla \cdot \mathbf{u} = 0 \quad (5.82a)$$

or

$$\frac{\partial}{\partial t} \frac{gh^2}{2} + \nabla \cdot \left(\mathbf{u} \frac{gh^2}{2} \right) + \frac{gh^2}{2} \nabla \cdot \mathbf{u} = 0. \quad (5.82b)$$

From the momentum and mass continuity equations we obtain an equation for the evolution of kinetic energy density, namely

$$\frac{D}{Dt} \frac{hu^2}{2} + \frac{u^2 h}{2} \nabla \cdot \mathbf{u} = -g\mathbf{u} \cdot \nabla \frac{h^2}{2} \quad (5.83a)$$

or

$$\frac{\partial}{\partial t} \frac{hu^2}{2} + \nabla \cdot \left(\mathbf{u} \frac{hu^2}{2} \right) + g\mathbf{u} \cdot \nabla \frac{h^2}{2} = 0. \quad (5.83b)$$

Adding (5.82b) and (5.83b) we obtain

$$\frac{\partial}{\partial t} \frac{1}{2} (hu^2 + gh^2) + \nabla \cdot \left[\frac{1}{2} \mathbf{u} (gh^2 + hu^2 + gh^2) \right] = 0 \quad (5.84)$$

or

$$\frac{\partial E}{\partial t} + \nabla \cdot [\mathbf{u} (E + \frac{1}{2} gh^2)] = 0 \quad (5.85)$$

where $E = K.E. + P.E. = (hu^2 + gh^2)/2$ is the density of the total energy. If the fluid is confined to a domain bounded by rigid walls, on which the normal component of velocity vanishes, then on integrating (5.84) over that area and using Gauss's theorem, the total energy is seen to be conserved; that is

$$\frac{dE}{dt} = \frac{d}{dt} \frac{1}{2} \int_A hu^2 + gh^2 dA = 0. \quad (5.86)$$

Such an energy principle also holds in the case with bottom topography, but this is left as an exercise for the reader (problem 5.10).

Just as we found in the case for a compressible fluid, the energy flux in (5.85), namely $\mathbf{u}(gh^2 + hu^2 + gh^2)/2$, is not just the energy density multiplied by the velocity but it contains an additional term $g\mathbf{u}h^2/2$, and this represents the energy transfer occurring when the fluid does work against the pressure force (see also problem 5.3).

5.4 ROTATING SHALLOW WATER FLOW

5.4.1 Vorticity and potential vorticity equations

In a rotating frame of reference, the shallow water momentum equation may be written

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -g\nabla h \quad (5.87)$$

Using the vector identity

$$(\mathbf{u} \cdot \nabla) \mathbf{u} = \frac{1}{2} \nabla(\mathbf{u} \cdot \mathbf{u}) - \mathbf{u} \times (\nabla \times \mathbf{u}) \quad (5.88)$$

this may be written in vector invariant form as

$$\frac{\partial \mathbf{u}}{\partial t} + (\boldsymbol{\omega} + \mathbf{f}) \times \mathbf{u} = -\nabla(gh + \frac{1}{2} \mathbf{u}^2) \quad (5.89)$$

where $\boldsymbol{\omega} \equiv \nabla \times \mathbf{u} = (\partial v / \partial x - \partial u / \partial y) \mathbf{k} = \zeta \mathbf{k}$ and (as before) $\mathbf{f} = f \mathbf{k}$.

To get the vorticity equation we take the curl of (5.89). The term on the right hand side vanishes and the second term on the left hand side is

$$\nabla \times (\boldsymbol{\omega}_a \times \mathbf{u}) = (\mathbf{u} \cdot \nabla) \boldsymbol{\omega}_a - (\boldsymbol{\omega}_a \cdot \nabla) \mathbf{u} \quad (5.90)$$

where $\boldsymbol{\omega}_a = \boldsymbol{\omega} + \mathbf{f}$ is the total or absolute vorticity of the fluid. This vector points in the z -direction, and may be written

$$\mathbf{k} \cdot \nabla \times (\boldsymbol{\omega}_a \times \mathbf{u}) = (\mathbf{u} \cdot \nabla) \zeta - \zeta \frac{\partial w}{\partial z} = (\mathbf{u} \cdot \nabla)(\zeta + f) + (\zeta + f) \nabla \cdot \mathbf{u}. \quad (5.91)$$

Thus, the vorticity equation, which is the curl of (5.89) is

$$\frac{\partial \zeta}{\partial t} + (\mathbf{u} \cdot \nabla)(\zeta + f) = -(\zeta + f) \nabla \cdot \mathbf{u}. \quad (5.92)$$

This is, of course, the same as the shallow water vorticity equation in a non-rotating frame, save that ζ is replaced by $\zeta + f$. This is consistent with the notion that f is the vorticity that the fluid has by virtue of the background rotation. Thus, (5.92) is simply the equation of motion for the total vorticity.

The potential vorticity equation is obtained by combining (5.92) with the mass conservation equation,

$$\frac{Dh}{Dt} + h \nabla \cdot \mathbf{u} = 0 \quad (5.93)$$

From (5.92) and (5.93)

$$\frac{1}{\zeta + f} \frac{D}{Dt} (\zeta + f) = \frac{1}{h} \frac{Dh}{Dt} \quad (5.94)$$

whence

$$\boxed{\frac{D}{Dt} \left(\frac{\zeta + f}{h} \right) = 0}. \quad (5.95)$$

That is, $Q \equiv (\zeta + f) / h$, the potential vorticity in a rotating shallow system, is a material invariant.

5.5 POTENTIAL VORTICITY, REVISITED

5.5.1 Relation to Kelvin's theorem

Kelvin's circulation theorem tells us that the integral of the circulation around a material curve is conserved, in the absence of non-conservative forces and baroclinic (solenoidal) terms. By Stokes theorem this is the same as the integral of vorticity over a surface bounded by that material line, so that if we consider an infinitesimal fluid element

$$\frac{D}{Dt}(\boldsymbol{\omega}_3 \cdot \delta \mathbf{S}) = 0 \quad (5.96)$$

where $\boldsymbol{\omega}_3$ is the curl of the three-dimensional velocity and $\delta \mathbf{S} = \hat{\mathbf{n}} \delta S$ is an arbitrary infinitesimal vector surface element, pointing in the direction normal to the surface. If we separate the vorticity and surface element into vertical and horizontal components we can write (5.96) as

$$\frac{D}{Dt} [(\zeta + f) \delta A + \boldsymbol{\omega}_h \cdot \delta \mathbf{S}_h] = 0 \quad (5.97)$$

where $\boldsymbol{\omega}_h$ and $\delta \mathbf{S}_h$ are the horizontally-directed components of the vorticity and the surface element, and $\delta A = \mathbf{k} \delta \mathbf{S}$ is the area of a horizontal cross-section of a fluid column. In Cartesian form the horizontal component of the vorticity is

$$\boldsymbol{\omega}_h = \mathbf{i} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) - \mathbf{j} \left(\frac{\partial w}{\partial x} - \frac{\partial v}{\partial z} \right) = \mathbf{i} \frac{\partial w}{\partial y} - \mathbf{j} \frac{\partial w}{\partial x} \quad (5.98)$$

where vertical derivatives of the horizontal velocity are zero by virtue of the nature of the shallow water system. Now, the vertical velocity in the shallow water system is smaller than the horizontal velocity by the order of the aspect ratio, the ratio of the fluid depth to the horizontal scale of motion. Furthermore, the size of the horizontally-directed surface element is also an aspect-ratio smaller than the vertically-directed component. That is,

$$|\boldsymbol{\omega}_h| \sim \alpha |\zeta| \quad (5.99)$$

and

$$|\delta \mathbf{S}_h| \sim \alpha |\delta A| \quad (5.100)$$

where α is the aspect ratio of the system. Thus $\boldsymbol{\omega}_h \cdot \delta \mathbf{S}_h$ is an aspect-number squared smaller than the term $\zeta \delta A$ and (consistent with the hydrostatic approximation) it should therefore be neglected. Kelvin's circulation theorem then becomes simply

$$\frac{D}{Dt} [(\zeta + f) \delta A] = 0 \quad (5.101)$$

or

$$\frac{D}{Dt} \left[\frac{(\zeta + f)}{h} h \delta A \right] = 0 \quad (5.102)$$

where h is the depth of the fluid column. But $h\delta A$ is the volume of the fluid column, and this is constant. Thus, as before,

$$\frac{D}{Dt} \left(\frac{\zeta + f}{h} \right) = 0 \quad (5.103)$$

We note again that horizontal velocities are independent of the vertical coordinate so that the term $w\partial/\partial z$ in the material derivative has no effect or, equivalently, the advection term is purely horizontal. Potential vorticity conservation thus follows as an immediate consequence of Kelvin's circulation theorem — an important, general result.

5.5.2 * An alternative derivation

The general form of potential vorticity conservation in a fluid is

$$\frac{D}{Dt} \left(\frac{\boldsymbol{\omega} \cdot \nabla \theta}{\rho} \right) = 0 \quad (5.104)$$

where θ is a materially-conserved scalar. In the flat-bottomed shallow water system, an appropriate scalar is the ratio z/h , where h is the local thickness of the fluid column because, from (5.29),

$$\frac{D}{Dt} \left(\frac{z}{h} \right) = 0. \quad (5.105)$$

if $h_b = 0$ (for simplicity). With this choice of scalar, potential vorticity conservation becomes

$$\frac{D}{Dt} \left[\boldsymbol{\omega} \cdot \nabla \left(\frac{z}{h} \right) \right] = 0 \quad (5.106)$$

where $\boldsymbol{\omega}$ and D/Dt are fully three dimensional. Expanding the dot product gives

$$\frac{D}{Dt} \left[\frac{\zeta + f}{h} - \frac{z}{h^2} \boldsymbol{\omega}_h \cdot \nabla_h h \right] = 0. \quad (5.107)$$

For an order unity Rossby number, the ratio of the size of the two terms in this equation is

$$\frac{|\zeta|}{|(z/h)\boldsymbol{\omega}_h \cdot \nabla_h h|} \sim \frac{[U/L]}{[WH/L^2]} = \frac{UL}{WH} = \alpha^2 \ll 1. \quad (5.108)$$

Thus, the second term in (5.107) is an aspect-ratio squared smaller than the first, and should be neglected, whereupon (5.103) is recovered.

5.6 SHALLOW WATER WAVES

5.6.1 Non-rotating

To first understand the basic properties of wave motion in shallow water, we linearize the equations around a mean height H , and a state of rest. Thus let

$$h(x, y, t) = H + h'(x, y, t), \quad (5.109a)$$

$$\mathbf{u}(x, y, t) = \mathbf{u}'(x, y, t). \quad (5.109b)$$

The mass conservation equation becomes

$$\frac{\partial h'}{\partial t} + (H + h')\nabla \cdot \mathbf{u} + \mathbf{u}' \cdot \nabla h' = 0, \quad (5.110)$$

and neglecting squares of small quantities this becomes

$$\frac{\partial h'}{\partial t} + H\nabla \cdot \mathbf{u}' = 0. \quad (5.111)$$

The advective terms drop out of the momentum equation, since we are linearizing around a state of rest, yielding

$$\frac{\partial \mathbf{u}'}{\partial t} = -g\nabla h' \quad (5.112)$$

For the remainder of the section we will drop the primes on the linearized fields.

Eliminating velocity by differentiating (5.111) with respect to time and taking the divergence of (5.112) leads to

$$\frac{\partial^2 h}{\partial t^2} - gH\nabla^2 h = 0 \quad (5.113)$$

which may be recognized as a wave equation. We can find the dispersion relationship for this by substituting the trial solution

$$h = \text{Re } h_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \quad (5.114)$$

where h_0 is a complex constant, $\mathbf{k} = \mathbf{i}k + \mathbf{j}l$ is the horizontal wavenumber, and Re indicates that the real part of the solution should be taken. If for simplicity we restrict attention for the moment to the one-dimensional problem, with no variation in the y -direction, then substituting into (5.113) leads to the dispersion relationship

$$\omega = \pm ck, \quad (5.115)$$

where $c = \sqrt{gH}$. That is, the wave speed is proportional to the square root of the mean fluid depth and is independent of the wavenumber — that is, the waves

are dispersionless. The general solution is a superposition of all such waves, with the amplitudes of each wave (or Fourier component) being determined by the Fourier decomposition of the initial conditions.

Because the waves are dispersionless, the general solution can be written

$$h(x, t) = \frac{1}{2} [F(x - ct) + F(x + ct)] \quad (5.116)$$

where $F(x)$ is the height field at $t = 0$. From this, it is easy to see that the shape of an initial disturbance is preserved as it propagates both to the right and to the left at speed c . (See also problem 5.7.)

5.6.2 Rotating, constant Coriolis parameter

Linearizing the rotating, flat-bottomed shallow water equations about a state of rest we obtain

$$\frac{\partial u}{\partial t} - f_0 v = -g \frac{\partial h}{\partial x}, \quad (5.117a)$$

$$\frac{\partial v}{\partial t} + f_0 u = -g \frac{\partial h}{\partial x}, \quad (5.117b)$$

$$\frac{\partial h}{\partial t} + H(u_x + v_y) = 0. \quad (5.117c)$$

We nondimensionalize these equations with length- and time-scales L and T respectively, so that $U = L/T$, and nondimensionalize the surface height deviation by the depth of the fluid, H . The equations become

$$\frac{\partial \hat{u}}{\partial \hat{t}} - \hat{f}_0 \hat{v} = -\hat{c}^2 \frac{\partial \hat{h}}{\partial \hat{x}}, \quad (5.118a)$$

$$\frac{\partial \hat{v}}{\partial \hat{t}} + \hat{f}_0 \hat{u} = -\hat{c}^2 \frac{\partial \hat{h}}{\partial \hat{y}}, \quad (5.118b)$$

$$\frac{\partial \hat{h}}{\partial \hat{t}} + (\hat{u}_{\hat{x}} + \hat{v}_{\hat{y}}) = 0. \quad (5.118c)$$

All the variables with hats on are nondimensional, and $\hat{f}_0 = f_0 T$ and $\hat{c} = \sqrt{gH}/U$ is the nondimensional speed of nonrotating shallow-water waves. (It is also the inverse of the Froude number $U/\sqrt{gH}$.) To obtain a dispersion relationship we substitute

$$(\hat{u}, \hat{v}, \hat{h}) = (u_0, v_0, h_0) e^{i(\hat{k} \cdot \hat{x} - \hat{\omega} \hat{t})} \quad (5.119)$$

into (5.118), giving

$$\begin{pmatrix} i\hat{\omega} & -\hat{f}_0 & i\hat{c}^2 \hat{k}_x \\ \hat{f}_0 & i\hat{\omega} & i\hat{c}^2 \hat{k}_y \\ i\hat{k}_x & i\hat{k}_y & i\hat{\omega} \end{pmatrix} \begin{pmatrix} u_0 \\ v_0 \\ h_0 \end{pmatrix} = 0 \quad (5.120)$$

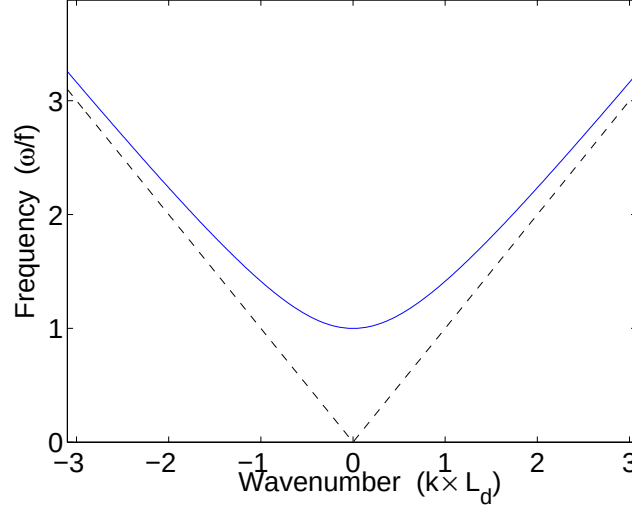


Figure 5.6 Dispersion relation for Poincaré waves (solid line) and nonrotating shallow water waves (dashed line). The frequency is normalized by the Coriolis frequency f and the wavenumber by the (inverse) deformation radius $\sqrt{gH}/f$. The frequency is always greater than f . For long waves (small wavenumbers) the frequency is approximately f , as for inertial oscillations. For short waves (high wavenumbers) the frequency asymptotes to that of nonrotating shallow water waves, namely (in dimensional form) $\sqrt{gH}k$.

where $\hat{k}$ and $\hat{\omega}$ are nondimensional wavenumber and frequency, respectively. This homogeneous equation has nontrivial solutions only if the determinant of the matrix vanishes. This gives

$$\hat{\omega}(\hat{\omega}^2 - \hat{f}_0^2 - \hat{c}^2 \hat{k}^2) = 0. \quad (5.121)$$

There are two classes of solution to this equation. The first is simply $\hat{\omega} = 0$, time-independent flow corresponding to geostrophic balance in (5.117). (Because geostrophic balance gives a divergence-free velocity field for constant Coriolis parameter the equations are satisfied by a time-independent solution.) The second set of solutions is given by

$$\hat{\omega}^2 = \hat{f}_0^2 + \hat{c}^2(\hat{k}_x^2 + \hat{k}_y^2). \quad (5.122)$$

If we make the choice $L = f_0^{-1}$ and $L = L_d = \sqrt{gH}/f_0$ then this is

$$\hat{\omega}^2 = 1 + (\hat{k}_x^2 + \hat{k}_y^2). \quad (5.123)$$

and in dimensional form it is

$$\boxed{\omega^2 = f_0^2 + gH(k_x^2 + k_y^2)}. \quad (5.124)$$

The corresponding waves are known as *Poincaré*³ waves, and the dispersion relationship is illustrated in Fig. 5.6. A few things to note are:

- (i) The frequency is always greater than the Coriolis frequency f_0 .
- (ii) The short waves limit: If

$$k^2 \gg \frac{f_0^2}{gH} \quad (5.125)$$

then the dispersion relationship reduces to that of the nonrotating case (5.115). This condition is equivalent to requiring that the wavelength be much shorter than the deformation radius, L_d . Specifically, if $\lambda = 2\pi/k$ is the wavelength, the condition is

$$\lambda^2 \ll L_d^2 (2\pi)^2 \quad (5.126)$$

The numerical factor of $(2\pi)^2$ is more than an order of magnitude, so care must be taken when deciding if the condition is satisfied in particular cases. Furthermore, the wavelength must still be longer than the depth of the fluid, else the shallow water condition is not met.

- (iii) The long wave limit: If

$$k^2 \ll \frac{f_0^2}{gH}, \quad (5.127)$$

that is if the wavelength is much longer than the deformation radius L_d , then the dispersion relationship is

$$\omega = f_0. \quad (5.128)$$

These are known as *inertial oscillations*. The equations of motion giving rise to them are

$$\frac{\partial u}{\partial t} - f_0 v = 0, \quad \frac{\partial v}{\partial t} + f_0 u = 0, \quad (5.129)$$

which are equivalent to Lagrangian equations for free particles in a rotating frame, namely

$$\frac{d^2 x}{dt^2} - f_0 v = 0, \quad \frac{d^2 y}{dt^2} + f_0 u = 0. \quad (5.130)$$

In this limit fluid parcels are thus undergoing free, inertial, oscillations in the rotating co-ordinate system, unrestrained by pressure forces (see problem 5.9).

5.6.3 Kelvin waves

The Kelvin wave is a particular type of shallow water wave that exists in the presence of both rotation and a lateral boundary.⁴ Suppose there is a solid boundary, a wall, at $y = 0$, then clearly harmonic solutions in the y -direction are not allowable, as these would not satisfy the condition of no-normal flow at the boundary. Do any wavelike solutions exist? The affirmative answer to this question was provided by Kelvin⁵ and the associated waves are now eponymously known as *Kelvin waves*.

We begin with the linearized shallow water equations

$$\frac{\partial u}{\partial t} - f_0 v = -g \frac{\partial \eta}{\partial x} \quad (5.131a)$$

$$\frac{\partial v}{\partial t} + f_0 u = -g \frac{\partial \eta}{\partial y} \quad (5.131b)$$

$$\frac{\partial \eta}{\partial t} + H \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0 \quad (5.131c)$$

The fact that $v = 0$ at $x = 0$ suggests that we look for solutions $v = 0$ everywhere, whence the equations become

$$\frac{\partial u}{\partial t} = -g \frac{\partial \eta}{\partial x} \quad (5.132a)$$

$$f_0 u = -g \frac{\partial \eta}{\partial y} \quad (5.132b)$$

$$\frac{\partial \eta}{\partial t} + \frac{\partial u}{\partial x} = 0 \quad (5.132c)$$

Equations (5.132a) and (5.132c) lead to the standard wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad (5.133)$$

where $c = \sqrt{gH}$, the usual wave speed of shallow water waves. The solution of (5.133) is

$$u = F_1(x + ct, y) + F_2(x - ct, y), \quad (5.134)$$

with corresponding surface displacement

$$\eta = \sqrt{H/g} \left[-F_1(x + ct, y) + F_2(x - ct, y) \right]. \quad (5.135)$$

The solution represents the superposition of two waves, one (F_1) travelling in the negative x -direction, and the other in the positive x -direction. To obtain the y -dependence of these functions we use (5.132b). This gives

$$\frac{\partial F_1}{\partial y} = \frac{f}{\sqrt{gH}} F_1, \quad \frac{\partial F_2}{\partial y} = -\frac{f}{\sqrt{gH}} F_2 \quad (5.136)$$

with solutions

$$F_1 = F(x + ct)e^{y/l_d} \quad F_2 = G(x - ct)e^{-y/l_d} \quad (5.137)$$

where $L_d = \sqrt{gH}/f_0$ is the radius of deformation. The solution F_1 grows exponentially away from the wall, and so fails to satisfy the condition of boundedness at infinity. It must be thus eliminated, leaving the general solution

$$u = e^{-yu/l_d}G(x - ct), \quad v = 0 \quad (5.138a)$$

$$\eta = \sqrt{H/g}e^{-yu/l_d}G(x - ct). \quad (5.138b)$$

The solution thus decays exponentially away from the boundary. If f_0 is positive, as in the Northern hemisphere, the boundary is to the right of an observer moving with the wave. Now, if f is constant, as we have assumed, we could have obtained a solution on a meridional wall, in which case we would find that the wave again moves such that the wall is to the right of the wave direction. (This is obvious once it is realized that f -plane dynamics are isotropic in x and y .) Thus, in the northern hemisphere the wave moves anticlockwise round a basin, and conversely in the Southern hemisphere. Thus, in both Northern and Southern hemispheres, the wave moves cyclonically round an enclosed basin.

5.7 SCALING PROPERTIES OF THE ROTATING SHALLOW WATER EQUATIONS

I have no satisfaction in formulas unless I feel their numerical magnitude.

William Thomson, Lord Kelvin (1824–1907).

Scaling and non-dimensionalization lie at the heart of fluid dynamics in general, and geophysical fluid dynamics in particular. The idea is that we *choose* the order of magnitude of various variables and parameters (the length scale, the magnitude of velocity, the timescale and so on), either on an *a priori* basis or using observations or some other physical theory. We then attempt to derive a consistent set of equations that is simpler than the original set but that consistently describes motion of the chosen scale. The set is simpler because, if we are lucky, our choice of scales will tell us that some terms in the equations are negligibly small and we can drop them. Asymptotics is the theory that has been developed to tell us systematically which terms we can drop and which we should keep. The combined approach — scaling plus asymptotics — has proven enormously useful, and it is fair to say that much progress in geophysical fluid dynamics has been intimately linked to this. However, it is good to keep in mind a few salient points:

- ★ Scaling is a choice. For example, we may choose a length-scale of say 10000 km if we are studying the global atmospheric circulation, and any resulting equations we derive will only be valid for that length-scale. Any interactions that the motion has with different length-scales is then likely to be inaccurately represented or neglected in the approximations.
- ★ This choice has, hopefully, a basis in physical reality but this is not given *a priori*. If the choice is made on an observational basis, then presumably there is some assurance of a degree of self consistency and physical consistency. However, by doing this we have not explained the existence of such scales of motion.
- ★ Even given the scaled nondimensional equations, an asymptotic (or similar) analysis to derive simplified equations of motion can only guarantee self-consistency, rather than uniqueness, especially at higher order.

We will denote scales by capital letters, and nondimensionalize the variables using these scales. Thus, if L is a typical length scale of the motion we wish to describe, we let

$$(x, y) = L \times (\hat{x}, \hat{y}) \quad (5.139)$$

where (x, y) are the dimensional variables and $(\hat{x}, \hat{y})$ are the nondimensional variables which, by supposition, are $\mathcal{O}(1)$. (We indicate orders of magnitude with the ' $\sim$ ' symbol, as in $x \sim L$ which means the same to us as $x = \mathcal{O}(L)$.)

We will first perform the scaling with values typical of large-scale (but not too large) flow in the atmosphere and ocean. The scales are illustrated in table (5.1). Remember that neither the atmosphere nor ocean really obeys the shallow water system of equations; we are using this as model, as a precursor to more detailed and realistic calculations later on. However, given that, we choose values for the variables to have geophysical relevance. Note too that we do not put down a value for a scale for the height deviation. This is because its value is tied to that of the more easily observed velocity scale.

Now consider the size of various terms in the momentum equation

$$\frac{\partial u}{\partial t} + \mathbf{u} \cdot \nabla u - f v = -g \frac{\partial \eta}{\partial x} \quad (5.140a)$$

$$\frac{U}{T} \quad \frac{U^2}{L} \quad f U \sim g \frac{\mathcal{H}}{L} \quad (5.140b)$$

where $\mathcal{H}$ is the amplitude of the *variations* in the surface displacement, and similarly for the v -momentum equation. We must now first pick a minimal set of scales for the phenomenon we wish to explore (as say in table 5.1), without making choices that are inconsistent with the equation itself, and then choose how to make the terms in the equation balance.

Variable	Meaning	Symbol	Atmos. value	Ocean value
(x, y)	Horizontal length	L	10^6 m	10^5 m
t	Timescale	T	1 day (10^5 s)	10 days (10^6 s)
(u, v)	Horizontal velocity	U	10 m/s	0.1 m/s

Table 5.1 Scales of large-scale flow in atmosphere and ocean. The choices given are representative of large-scale eddy motion in both systems.

5.7.1 Small Rossby number scaling

The ratio of the advective term to the rotational term in the momentum equation (5.140) is $U^2/L : fU = U/fL$, or the *Rossby number*.⁶ Using the values from table 5.1 we find that $Ro \approx 0.1$ for the atmosphere and $Ro \approx 0.01$ for the ocean, small in both cases. Suppose that we are interested in motion that has the advective timescale $T = L/U$. Then the local time derivative scales like the advective term, namely as U^2/L , and both of these terms are order Rossby number smaller than the rotation term. Then, we either have a state of no motion with

$$-fv = 0 \quad (5.141)$$

or else the Coriolis force it is balanced by the pressure force, and the dominant balance is

$$-fv = -g \frac{\partial \eta}{\partial x} \quad (5.142)$$

or *geostrophic balance* (see Fig. 5.7). If we make this nontrivial choice, then the equation informs us that η scales according to

$$\eta \sim \mathcal{H} = \frac{fUL}{g} \quad (5.143)$$

We can also write $\mathcal{H}$ as

$$\mathcal{H} = Ro \frac{f^2 L^2}{g} = Ro H \frac{L^2}{L_d^2}. \quad (5.144)$$

where $L_d = \sqrt{gH}/f$ is the *deformation radius*, (so-called for reasons we discuss later) and H is the mean depth of the fluid. The variations in fluid height thus scale as

$$\frac{\eta}{H} \sim Ro \frac{L^2}{L_d^2}, \quad (5.145)$$

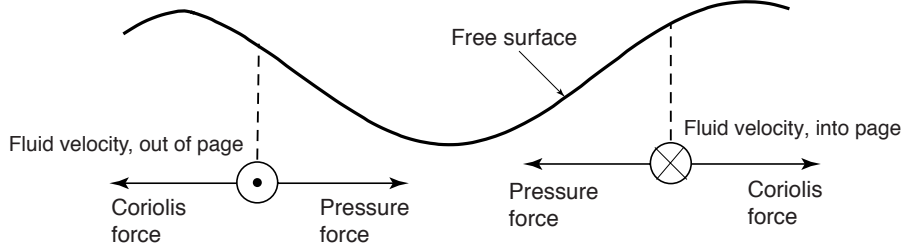


Figure 5.7 Geostrophic flow in a shallow water system, with a positive value of the Coriolis parameter f , as in the Northern hemisphere. The pressure force is directed down the gradient of the height field, and this can be balanced by the Coriolis force if the fluid velocity is at right angles to it. If f were negative, the geostrophic flow would be reversed.

and the total height of the fluid may be written as

$$\eta = H \left(1 + Ro \frac{L^2}{L_d^2} \hat{\eta} \right), \quad (5.146)$$

where $\hat{\eta}$ is the $\mathcal{O}(1)$ nondimensional value of the surface height deviation.

The momentum equation

Nondimensionalizing all the variables, that is, writing $(u, v) = U(\hat{u}, \hat{v})$, $(x, y) = L(\hat{x}, \hat{y})$, $t = T\hat{t}$ where $T = L/U$, and substituting into the momentum equation we obtain the nondimensional momentum equation

$$Ro \left[\frac{\partial \hat{\mathbf{u}}}{\partial \hat{t}} + (\hat{\mathbf{u}} \cdot \nabla) \hat{\mathbf{u}} \right] + \hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\nabla \hat{\eta} \quad (5.147)$$

where $\hat{\mathbf{f}} = \mathbf{k}\hat{f} = \mathbf{k}f/f_0$, where f_0 is a representative value of the Coriolis parameter. If f is a constant, then $\hat{f} = 1$, but it is useful to keep it in the equations to indicate the presence of Coriolis parameter. All the variables in this equation will be supposed to be of order unity, and the Rossby number (a parameter here, not a variable) multiplying the local time derivative and the advective terms indicates the smallness of those terms. The dominant balance in this equation is between the last two terms, since these are the only order-unity terms. Thus, we can expect,

$$\hat{\mathbf{f}} \times \hat{\mathbf{u}} \approx -\nabla \hat{\eta} \quad (5.148)$$

or in dimensional form

$$f(\mathbf{k} \times \mathbf{u}) \approx -g\nabla\eta. \quad (5.149)$$

This is, of course, *geostrophic balance*, and we may define the geostrophic velocity by

$$f(\mathbf{k} \times \mathbf{u}_g) = -g\nabla\eta. \quad (5.150)$$

The mass continuity (height) equation

The (dimensional) mass continuity equation can be written

$$\frac{1}{H} \frac{\partial \eta}{\partial t} + \frac{1}{H} \mathbf{u} \cdot \nabla \eta + \left(1 + \frac{\eta}{H}\right) \nabla \cdot \mathbf{u} = 0 \quad (5.151)$$

Using (5.146), and assuming that time scales advectively and $T = LU$ this becomes

$$Ro \left(\frac{L}{L_d}\right)^2 \frac{D\hat{\eta}}{Dt} + H \left(1 + Ro \left(\frac{L}{L_d}\right)^2 \eta\right) \hat{\nabla} \cdot \mathbf{v} = 0. \quad (5.152)$$

or

$$FRo \left(\frac{\partial \hat{\eta}}{\partial \hat{t}} + \hat{\mathbf{u}} \cdot \nabla \hat{\eta}\right) + (1 + FRo\hat{\eta}) \nabla \cdot \hat{\mathbf{u}} = 0 \quad (5.153)$$

where the parameter F is given by

$$F \equiv \frac{f^2 L^2}{gH} = \frac{L^2}{L_d^2} \quad (5.154)$$

Equations (5.147) and (5.153) are the nondimensional versions of the full equations of motion, and if we leave it there we have gained little. However, some terms in the equations of motion are evidently small and may be eliminated, but this should be done systematically. Given that we have a small parameter — the Rossby number — one way to do this is to expand all the variables in an asymptotic series in that small parameter. Because of the presence of a second nondimensional parameter, F , the way this is done will depend on the choice we make for the size of F_r . Sections 5.9 and 5.8 describe the two most useful ways of proceeding, each of which gives rise to a commonly used set of equations.

A note on Froude and Burger numbers

The Froude number is usually defined as the ratio of a the fluid particle speed to the wave speed. For a shallow water fluid this gives

$$Fr \equiv \frac{U}{\sqrt{gH}}. \quad (5.155)$$

In terms of the deformation radius this is

$$Fr = \frac{U}{f_0 L_d} = Ro \frac{L}{L_d}. \quad (5.156)$$

The Burger number⁷ is a useful measure of scale of motion of the fluid, relative to the deformation radius, and may be defined by

$$Bu \equiv \left(\frac{L_d}{L} \right)^2 \quad (5.157)$$

where L_d is the deformation radius and L the horizontal scale of the fluid motion. Equivalently, this is

$$Bu = \frac{gH}{f_0^2 L^2}, \quad (5.158)$$

or, in terms of the Rossby and Froude numbers,

$$Bu = \left(\frac{Ro}{Fr} \right)^2 \quad (5.159)$$

Sometimes it is more convenient to work with the inverse of the Burger number

$$F \equiv Bu^{-1} = \frac{f_0^2 L^2}{gH} = \left(\frac{Fr}{Ro} \right)^2 = \left(\frac{L}{L_d} \right)^2 \quad (5.160)$$

which is like the square of a Froude number but uses the rotational speed fL instead of U in the numerator.

5.8 THE PLANETARY GEOSTROPHIC EQUATIONS

The planetary geostrophic equations are perhaps the simplest ‘geostrophic equations’ that can be derived for rapidly rotating fluids with small Rossby number. We’ll derive equations these with two levels of rigour.

5.8.1 Informal derivation

To informally derive the planetary geostrophic equations, we first note that the advection term in the momentum equation (5.147) is order Rossby number smaller than the Coriolis and pressure terms (the term in square brackets is multiplied by Ro), and therefore we neglect it. The momentum equation straightforwardly becomes

$$\hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\nabla \hat{\eta}. \quad (5.161)$$

The mass conservation equation (5.153), contains two nondimensional parameters, Ro and the inverse Burger number F and we must make a choice as to the relationship of these two numbers. We will choose

$$FRo = \mathcal{O}(1) \quad (5.162)$$

which in turn implies

$$L^2 \gg L_d^2. \quad (5.163)$$

That is to say, we suppose that the scales of motion are much larger than the deformation scale. Given this choice, all the terms in the mass conservation equation are of roughly the same size (they are of the same asymptotic order), and we retain all terms in (5.153). In dimensional form, the shallow water planetary geostrophic equations are

$$\left. \begin{aligned} \mathbf{f} \times \mathbf{u} &= -g\nabla\eta \\ \frac{Dh}{Dt} + h\nabla \cdot \mathbf{u} &= 0 \end{aligned} \right\}. \quad (5.164)$$

In a flat-bottomed fluid $\eta = h$, and in general $\eta = h + \eta_b$. We emphasize again that *the planetary geostrophic equations are only valid for scales of motion much larger than the deformation radius*. The height variations are then as large as the mean height field itself; that is, using (5.145), $\eta/H = \mathcal{O}(1)$.

5.8.2 Formal derivation

We assume that:

- (i) The Rossby number is small. $Ro = U/f_0L \ll 1$.
- (ii) The scale of the motion is significantly larger than the deformation scale. That is, (5.162) holds or

$$F = Bu^{-1} = \left(\frac{L}{L_d} \right)^2 \gg 1 \quad (5.165)$$

and in particular

$$FRo = \mathcal{O}(1). \quad (5.166)$$

- (iii) Time scales advectively. That is, the scaling timescale is given by $T = L/U$, so $Ro_t = 1/f_0T = U/f_0L = Ro$.

The idea is now to expand the nondimensional variables velocity and height fields in an asymptotic series with Rossby number as the small parameter, substitute into the equations of motion, and derive a simpler set of equations. It is a trivial exercise in this instance, and so it illustrates the methodology. The expansions are

$$\hat{\mathbf{u}} = \hat{\mathbf{u}}_0 + Ro\hat{\mathbf{u}}_1 + Ro^2\hat{\mathbf{u}}_1 + \cdots \quad (5.167a)$$

and

$$\hat{\eta} = \hat{\eta}_0 + Ro\hat{\eta}_1 + Ro^2\hat{\eta}_1 + \cdots \quad (5.167b)$$

Then substituting (5.167a) and (5.167b) into the momentum equation gives

$$Ro \left[\frac{\partial \hat{\mathbf{u}}_0}{\partial \hat{t}} + \hat{\mathbf{u}}_0 \cdot \nabla \hat{\mathbf{u}}_0 + \hat{\mathbf{f}} \times \hat{\mathbf{u}}_1 \right] + \hat{\mathbf{f}} \times \hat{\mathbf{u}}_0 = -\nabla \hat{\eta}_0 - Ro [\nabla_0 \hat{\eta}_1] + O(Ro^2) \quad (5.168)$$

The Rossby number is now an asymptotic ordering parameter; thus, the sum of all the terms at any particular order in Rossby number must vanish. At lowest order we obtain the simple expression

$$\hat{\mathbf{f}} \times \hat{\mathbf{u}}_0 = -\nabla \hat{\eta}_0. \quad (5.169)$$

Note that although f_0 is a representative value of f , we have made no assumptions about the constancy of f . In particular, f is allowed to vary by an order one amount, provided that it does not become so small that the Rossby number (U/f_0L) is not small.

The appropriate height (mass conservation) equation is similarly obtained by substituting (5.167a) and (5.167b) into the shallow water mass conservation equation. Because $F Ro = \mathcal{O}(1)$ at lowest order we simply retain all the terms in the equation to give

$$F Ro \left[\frac{\partial \hat{\eta}_0}{\partial \hat{t}} + \hat{\mathbf{u}}_0 \cdot \nabla \hat{\eta}_0 \right] + [1 + F Ro \hat{\eta}] \nabla \cdot \hat{\mathbf{u}}_0 = 0. \quad (5.170)$$

Equations (5.169) and (5.170) are a closed set, and constitute the nondimensional planetary geostrophic equations. The dimensional forms of these equations are just (5.164).

5.8.3 A note on the Coriolis parameter

We said that we made no assumptions about the Coriolis parameter. Suppose then that f is a constant (f_0), or nearly so. Then, from the curl of (5.169), $\nabla \cdot \mathbf{u}_0 = 0$. This means that we can define a streamfunction for the flow and, from geostrophic balance, the height field is just that streamfunction. That is,

$$\psi = \frac{1}{f_0} \eta_0, \quad \mathbf{u}_0 = -\mathbf{k} \times \nabla \psi. \quad (5.171)$$

Because we are still at very large scales $F Ro = \mathcal{O}(1)$ and (5.170) becomes

$$\frac{\partial \hat{\eta}_0}{\partial \hat{t}} + \mathbf{u}_0 \cdot \nabla \hat{\eta}_0 = 0, \quad (5.172)$$

or

$$\frac{\partial \hat{\eta}_0}{\partial \hat{t}} + J(\psi, \hat{\eta}_0) = 0. \quad (5.173)$$

But since $\hat{h}_0 \propto \psi$ the advective term is proportional to $J(\psi, \psi)$, which is zero. Thus, the flow does not evolve at this order. This seems rather odd, but there

is nothing faulty about our reasoning. The planetary geostrophic equations are *uninteresting* if the scale of the motion is such that the Coriolis parameter is not variable. On earth, the scale of motion on which this parameter regime exists is rather limited, since the planetary geostrophic equations require that the scale of motion is also larger than the deformation radius. In the earth's atmosphere, any scale that is larger than the deformation radius will be such that the Coriolis parameter varies significantly over it, and we do not encounter this parameter regime. In the earth's ocean, the deformation radius is relatively small and there exists a small parameter regime larger than the deformation radius but smaller than that on which the Coriolis parameter varies.⁸ If the scales of motion are too small for the Coriolis parameter to vary significantly, then in both atmosphere and ocean the scales of motion are quite likely to be of the same order as the deformation radius. The planetary geostrophic equations are then invalid, but we may proceed by formally separating the height field into a mean and deviation. This leads to the important quasigeostrophic equations, which we now turn to.

5.9 THE QUASI-GEOSTROPHIC EQUATIONS

The quasi-geostrophic equations are a set of equations derived from the shallow-water or stratified equations that are valid in the low Rossby number limit. As will be seen, their advantage lies in their simplicity, in particular that they have but one evolution equation; all other fields may be derived diagnostically from the single evolving variable. Explicitly, the assumptions necessary to derive the equations are

1. The Rossby number is small. $Ro = U/f_0L \ll 1$.
2. The scale of the motion is not significantly larger than the deformation scale. Specifically, we shall require that

$$Ro \left(\frac{L}{L_d} \right)^2 = \frac{Ro}{Bu} = O(Ro). \quad (5.174)$$

where $Bu = (gH/f_0^2L^2)$ is the Burger number. The assumption is a little unmotivated at the moment: from (5.146) it means that the variations in fluid depth are small compared to its total depth, but we defer explanation of its necessity.

3. Variations in the Coriolis parameter are small. That is, $|\beta L| \ll |f_0|$ where L is the length-scale of the motion.
4. Time scales advectively. That is, the scaling for time is given by $T = L/U$, so $Ro_t = 1/f_0T = U/fL = Ro$.

The idea is now to expand the variables $\hat{u}, \hat{v}, \hat{\eta}$ in an asymptotic series with Rossby number as the small parameter, substitute into the equations of motion, and derive a simpler set of equations. Thus,

$$\hat{u} = \hat{u}_0 + Ro\hat{u}_1 + Ro^2\hat{u}_2 + \dots \quad (5.175)$$

and similarly for $\hat{v}$, and

$$\hat{\eta} = \hat{\eta}_0 + Ro\hat{\eta}_1 + \dots \quad (5.176)$$

We will recognize the smallness of the beta-parameter, compared to f_0/L , by non-dimensionalizing β using U/L^2 ; that is, we let $\beta = \hat{\beta}U/L^2$, where the $\hat{\beta}$ is assumed to be a parameter of order one. Then

$$f = f_0 + \beta y \quad (5.177)$$

becomes

$$\hat{f} = f/f_0 = \hat{f}_0 + Ro\hat{\beta}\hat{y}. \quad (5.178)$$

where $\hat{f}_0$ is the nondimensional value of f_0 ; its value is unity, but it is convenient to retain it.

Substitute (5.175) into the u - and v -momentum equations, and equate powers of Ro . At lowest order we obtain

$$\hat{f}_0\hat{u}_0 = -\frac{\partial\hat{\eta}_0}{\partial\hat{y}}, \quad \hat{f}_0\hat{v}_0 = \frac{\partial\hat{\eta}_0}{\partial\hat{x}}. \quad (5.179)$$

Cross-differentiating gives

$$\nabla \cdot \hat{\mathbf{u}}_0 = 0, \quad (5.180)$$

that is to say, the velocity field is divergence-free. Note that this arises as a consequence of the momentum equation rather than the mass conservation equation.

The mass conservation equation is also, at lowest order, $\nabla \cdot \hat{\mathbf{u}}_0 = 0$. (See problem 5.16.) At next order we have

$$F\frac{\partial\hat{\eta}_0}{\partial\hat{t}} + F\hat{\mathbf{u}}_0 \cdot \nabla\hat{\eta}_0 + \nabla \cdot \hat{\mathbf{u}}_1 = 0 \quad (5.181)$$

This equation is not closed, because the evolution of the zeroth order term involves evaluation of a first order quantity. To close the equations, we go to next order in the momentum equation:

$$\frac{\partial\hat{\mathbf{u}}_0}{\partial\hat{t}} + (\hat{\mathbf{u}}_0 \cdot \nabla)\hat{\mathbf{u}}_0 - \beta y \mathbf{k} \times \hat{\mathbf{u}}_0 - \hat{f}_0 \mathbf{k} \times \hat{\mathbf{u}}_1 = -\nabla\hat{\eta}_1 \quad (5.182)$$

We obtain the vorticity equation by taking the curl of this (i.e., cross-differentiating) to give

$$\frac{\partial\zeta_0}{\partial\hat{t}} + (\hat{\mathbf{u}}_0 \cdot \nabla)(\hat{\zeta}_0 + \beta y) = -\hat{f}_0 \nabla \cdot \hat{\mathbf{u}}_1 \quad (5.183)$$

The term on the right hand side is the *vortex stretching* term. Note that only vortex stretching by the background or planetary vorticity contributes — the vortex stretching by the relative vorticity is a factor Rossby number smaller and appears only at higher order. This equation is also not closed. However, we may use (5.181) to eliminate the divergence term in (5.183) to give

$$\frac{\partial \zeta_0}{\partial t} + (\hat{\mathbf{u}}_0 \cdot \nabla)(\hat{\zeta}_0 + \hat{\beta} \hat{y}) = \hat{f}_0 \left(F \frac{\partial \hat{\eta}_0}{\partial t} + F \hat{\mathbf{u}} \cdot \nabla \hat{\eta}_0 \right) \quad (5.184)$$

or

$$\frac{\partial}{\partial t}(\zeta_0 - \hat{f}_0 F \hat{\eta}_0) + (\hat{\mathbf{u}}_0 \cdot \nabla)(\zeta_0 + \beta y - F \hat{\eta}_0) = 0 \quad (5.185)$$

The final step is to note that the lowest order vorticity and height fields are related through geostrophic balance. That is, since

$$\hat{f}_0 \mathbf{k} \times \hat{\mathbf{u}}_0 = -\nabla \hat{\eta}_0 \quad (5.186)$$

then the height field is also, apart from a constant factor, the *stream function* of the flow) and we have

$$\hat{u}_0 = -\frac{\partial \hat{\psi}}{\partial \hat{y}}, \quad \hat{v}_0 = \frac{\partial \hat{\psi}}{\partial \hat{x}}, \quad \hat{\zeta}_0 = \nabla^2 \psi_0 \quad (5.187)$$

where $\hat{\psi} = \hat{\eta}_0 / \hat{f}_0$.

Equation (5.185) can thus be written,

$$\frac{\partial}{\partial \hat{t}}(\nabla^2 \hat{\psi}_0 - \hat{f}_0^2 F \hat{\psi}_0) + (\hat{\mathbf{u}}_0 \cdot \nabla)(\zeta_0 + \hat{\beta} \hat{y} - \hat{f}_0^2 F \hat{\psi}_0) = 0, \quad (5.188)$$

or

$$\frac{D_0}{D\hat{t}}(\nabla^2 \hat{\psi}_0 + \hat{\beta} \hat{y} - \hat{f}_0^2 F \hat{\psi}_0) = 0, \quad (5.189)$$

where the subscript '0' on the material derivative indicates that the lowest order velocity, the geostrophic velocity, is the advecting velocity.

Then, restoring the dimensions, (5.189) becomes

$$\boxed{\frac{D}{Dt}(\nabla^2 \psi + \beta y - \frac{1}{L_d^2} \psi) = 0} \quad (5.190)$$

where $\psi = (g/f_0)\eta$ and the advective derivative is

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + u_g \frac{\partial}{\partial x} + v_g \frac{\partial}{\partial y} = \frac{\partial}{\partial t} - \frac{\partial \psi}{\partial y} \frac{\partial}{\partial x} + \frac{\partial \psi}{\partial x} \frac{\partial}{\partial y} = \frac{\partial}{\partial t} + J(\psi, \cdot). \quad (5.191)$$

Since $L_d = gH/f_0^2$ another form of (5.190) is

$$\frac{D}{Dt}(\zeta + \beta y - \frac{f_0}{H} \eta) = 0 \quad (5.192)$$

where $\zeta = (g/f_0)\nabla^2\eta$. Equations (5.190) and (5.192) are known as the shallow-water quasi-geostrophic potential vorticity equation. The quantity

$$q \equiv \zeta + \beta y - f_0\eta/H = \nabla^2\psi + \beta y - \frac{1}{L_d^2}\psi \quad (5.193)$$

is the (shallow-water) *quasi-geostrophic potential vorticity*.

5.9.1 More about the quasi-geostrophic potential vorticity equation

The quantity q is an approximation to the shallow water potential vorticity. This follows easily from the definition

$$Q = \frac{f + \zeta}{h}. \quad (5.194)$$

Then substitute $h = H(1 + \eta'/H)$ into (5.194), where η' is the perturbation of the free-surface height, and assume that η'/H is small to obtain

$$\begin{aligned} Q &\approx \frac{1}{H}(f + \zeta)\left(1 - \frac{\eta'}{H}\right) \\ &\approx \frac{1}{H}(f_0 + \beta y + \zeta - f_0\eta'/H) \end{aligned} \quad (5.195)$$

neglecting the smaller terms. Since f_0 is a constant it has no effect in the evolution equation and the remaining terms. The quantity given by

$$q = \beta y + \zeta - f_0\frac{\eta'}{H} \quad (5.196)$$

is thus advected by the flow. Using geostrophic balance we have $\zeta = \nabla^2\psi$ and $\eta' = f_0\psi/g$ so that this is identical to (5.193). Thus, the quasi-geostrophic potential vorticity is an approximation to the full shallow water potential vorticity, multiplied by the constant term H .

Note that the approximations needed to go from (5.194) to (5.193) are precisely the same as those used in our more long-winded derivation of the quasi-geostrophic equations. That is, we assumed that f itself is nearly constant, and that f_0 is much larger than ζ , equivalent to a low Rossby number assumption. It was also necessary to assume that $H \gg \eta'$ to enable the expansion of the height field, and since

$$\frac{\eta}{H} \sim \frac{f_0 UL}{gH} = Ro \frac{f_0^2 L^2}{gH} = RoF \quad (5.197)$$

this assumption is equivalent to assuming that F is not larger than $1/Ro$ or that the scale of motion not be significantly larger than the deformation scale. This derivation of the quasi-geostrophic system is completed by noting that the advection of the potential vorticity should be by the geostrophic velocity alone, and we recover (5.190) or (5.192).

Two interesting limits

- (i)
- Motion on scales smaller than the deformation radius.*

That is, $L \ll L_d$ and thus $Bu \gg 1$ or $F \ll 1$. Then (5.189) becomes

$$\frac{\partial \zeta}{\partial t} + \mathbf{u}_\psi \cdot \nabla \zeta = 0 \quad (5.198)$$

or

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta) = 0 \quad (5.199)$$

where $\zeta = \nabla^2 \psi$ and $J(\psi, \zeta) = \psi_x \zeta_y - \psi_y \zeta_x$. Thus, the motion obeys the two-dimensional vorticity equation. Physically, on small length scales the deviations in the height field are very small and may be neglected.

- (ii)
- Motion on scales larger than the deformation radius.*

Although scales are not allowed to become so large that $Ro(L/L_d)^2$ is of order unity, we may, *a posteriori*, still have $L \gg L_d$, whence the potential vorticity equation becomes

$$\frac{\partial \eta}{\partial t} + \mathbf{u}_\psi \cdot \nabla \eta = 0 \quad (5.200)$$

or

$$\frac{\partial \eta}{\partial t} + J(\psi, \eta) = 0 \quad (5.201)$$

but since $\psi = g\eta/f_0$ the Jacobian term vanishes. Thus, one is left with a trivial equation that implies there is no advective evolution of the height field. There is nothing wrong with our reasoning; the mathematics have indeed pointed out a limit interesting in its uninterestingness. From a physical point of view, however, such (lack of) motion is likely to be rare, because on large scales the Coriolis parameter varies considerably; the geostrophic velocity is no longer divergence free and the height field no longer equivalent to a stream function. Further analysis of large-scale motion is given in section 5.8

The most severe restriction is that variations in the height field are small: what does this have to do with geostrophy? Well, if we scale η assuming geostrophic balance then $\eta \sim fUL/g$ and $\eta/H \sim Ro(L/L_d)^2$. (These expressions assume no more than geostrophy; they are not necessarily ‘quasi-geostrophic.’) Thus, if Ro is to remain small, η/H can only be order one if $(L/L_d)^2 \gg 1$. That is, the height variations must occur on a large scale, or we are led to a scaling inconsistency. Put another way, *if there are order one height variations over a length-scale less than or order of the deformation scale, the Rossby number will not be small*. So this restriction is not so much an artificial aspect of the quasi-geostrophic equations, but a restriction demanded by initially requiring geostrophic balance. We still may indeed have large height variations if the scale of motion is large, but this contingency is described by the planetary geostrophic equations.

5.9.2 Another flow regime

Although perhaps of little terrestrial interest, we can imagine a regime in which the Coriolis parameter varies fully, but the scale of motion remains no larger than the deformation radius. This parameter regime is not quasigeostrophic, but it easily gives an interesting result. Because $\eta' / H = Ro(L / L_d)^2$ deviations of the height field are at least order Rossby number smaller than the reference height and $|\eta'| \ll H$. The dominant balance in height equation is then

$$H \nabla \cdot \mathbf{u} = 0 \quad (5.202)$$

assuming that time still scales advectively (i.e. $T \sim L/U$). This small horizontal divergence must remain consistent with geostrophic balance

$$\mathbf{f} \times \mathbf{u} = -g \nabla \eta \quad (5.203)$$

with a variable Coriolis parameter. Taking the curl of (i.e., cross-differentiating) (5.203) gives

$$\beta v + f \nabla \cdot \mathbf{u} = 0. \quad (5.204)$$

whence $v = 0$. Although not at all useful as an evolution equation, this illustrates the constraining effects that differential rotation has on meridional velocity. This may be the cause of the banded intensely zonal flow on some of the giant planets, and we will revisit this issue in our discussion of geostrophic turbulence.

5.10 THE TWO-LAYER AND MULTI-LAYER QUASI-GEOSTROPHIC SYSTEMS

Just as for the one-layer case, the two-layer shallow water equations simplify to a corresponding quasi-geostrophic system in the appropriate circumstances. In particular we again assume that:

- (i) The flow is rapidly rotating so the Rossby number is small.
- (ii) The variations in layer thickness are small compared to their mean thickness. We will see that this is equivalent to assuming that the scale of motion is comparable to the *internal deformation radius*.
- (iii) The variation in Coriolis parameter is small over the length-scale of the flow.

Although, as in the one-layer case, we can derive the quasi-geostrophic equations from the momentum and mass continuity equations (see section 5.2), here we will proceed directly from the potential vorticity equation for each layer. We will also stay in dimensional variables, foregoing a more systematic approach for the sake of informality and perhaps insight. For each layer the potential vorticity

equation is just

$$\frac{DQ}{Dt} = 0 \quad (5.205a)$$

$$Q_i = \frac{\zeta_i + f}{h_i} \quad (5.205b)$$

Let

$$h_i = H_i + h'_i \quad (5.206)$$

where $|h'_i| \ll H_i$. The potential vorticity then becomes

$$Q_i \approx \frac{1}{H_i} (\zeta_i + f) \left(1 - \frac{h'_i}{H_i} \right) \quad \text{using assumption (ii)} \quad (5.207a)$$

$$\approx \frac{1}{H_i} \left(f + \zeta_i - f \frac{h'_i}{H_i} \right) \quad \text{using assumption (i)} \quad (5.207b)$$

$$\approx \frac{1}{H_i} \left(f + \zeta_i - f_0 \frac{h'_i}{H_i} \right) \quad \text{using assumption (iii)} \quad (5.207c)$$

Further, although the Coriolis parameter is much larger than the relative vorticity, it is, by assumption (iii), nearly constant. It is only the *variations* in Coriolis parameter (e.g., βy on the beta-plane) that are important in the first term on the right-hand side of (5.207), and with this all three terms are of the same asymptotic order. Then, because mean layer thicknesses are constant, we can define the quasi-geostrophic potential vorticity in each layer by

$$q_i = \left(f + \zeta_i - f_0 \frac{h'_i}{H_i} \right). \quad (5.208)$$

As in the one-layer case this is dimensionally different from the full potential vorticity.

To obtain a closed set of equations we must relate the height field to the velocity and vorticity. Because the flow is geostrophically balanced we have, in the two-layer case,

$$f_0 \times \mathbf{u}_1 = -g \nabla \eta_0 = -g \nabla (h'_1 + h'_2 + \eta_b) \quad (5.209a)$$

$$f_0 \times \mathbf{u}_2 = -g \nabla \eta_0 + g' \nabla h'_1 = -g \nabla (h'_1 + h'_2 + \eta_b) + g' \nabla h'_1 \quad (5.209b)$$

where $g' = (\rho_2 - \rho_1) / \rho_1$ and η_b is the height of any bottom topography, and we have used assumption (iii) to replace f with f_0 on the left-hand side. This enables us to define the streamfunction for each layer by

$$\psi_1 = \frac{g}{f_0} (h'_1 + h'_2 + \eta_b) \quad (5.210a)$$

$$\psi_2 = \frac{g}{f_0} (h'_1 + h'_2 + \eta_b) - \frac{g'}{f_0} h'_1 \quad (5.210b)$$

which gives

$$h'_1 = \frac{f_0}{g'}(\psi_1 - \psi_2) \quad (5.211a)$$

$$h'_2 = \frac{f_0}{g'}(\psi_2 - \psi_1) + \frac{f_0}{g}\psi_1 - \eta_b. \quad (5.211b)$$

Using (5.208) and (5.211) the quasi-geostrophic potential vorticity for each layer becomes

$$q_1 = f + \nabla^2\psi_1 + \frac{f_0^2}{g'H_1}(\psi_2 - \psi_1) \quad (5.212a)$$

$$q_2 = f + \nabla^2\psi_2 + \frac{f_0^2}{g'H_2}(\psi_1 - \psi_2) - \frac{f_0^2}{gH_2}\psi_1 + f_0\frac{h_b}{H_2}. \quad (5.212b)$$

The penultimate term in the second equation is proportional to the free surface displacement, and this is often neglected (the rigid lid approximation), because $g \gg g'$. The potential vorticity in each layer is just advected by the geostrophic velocity, so that the evolution equation for each layer is just

$$\frac{\partial q_i}{\partial t} + J(\psi_i, q_i) = 0. \quad (5.213)$$

Multi-layer case

[not complete]

5.10.1 Non-dimensional parameters

The three assumptions in section 5.10.1 are equivalent to the following non-dimensional conditions:

- (i) Small Rossby number: $Ro = U/f_0L \ll 1$.
- (ii) Variation in layer thickness is small. From (??) we have in the flat-bottomed case

$$\frac{\Delta h_i}{H_i} \sim Ro \frac{f_0^2 L^2}{g'H} = Ro \frac{L^2}{L_d^2} \quad (5.214)$$

assuming $H_i \sim H$ where H is the total fluid depth, and where $L_d = \sqrt{g'H}/f_0$ is the *internal radius of deformation*. [A more precise definition is given in (5.239).] Often $g' \ll g$ so that the internal deformation radius is much smaller than the external deformation radius $L_e = \sqrt{gH}/f_0$. For example, in the ocean $L_d \approx 30\text{--}100$ km whereas $L_e \approx 2000$ km. The difference is less marked in the atmosphere.

- (iii) The variation in Coriolis parameter is small. On the β -plane this is formalized by setting $\beta = \mathcal{O}(Ro)f_0/L = \mathcal{O}(U^2/L)$ where U is the velocity scale and L the scale of the motion.

5.10.2 * A cautionary note

The above derivation is straightforward and physical. It seems much easier than the previous rather long-winded asymptotic derivation, and also suggests that we might be able to improve on the accuracy of the quasi-geostrophic equations. For example, a seemingly improved set of geostrophic evolution equations might be

$$\frac{\partial q_i}{\partial t} + \mathbf{u}_i \cdot \nabla q_i = 0 \quad (5.215)$$

with

$$q_i = \frac{f + \zeta_i}{h_i}. \quad (5.216)$$

where $\zeta_i = (\partial v_i / \partial x - \partial u_i / \partial y)$ and the velocities are given by geostrophic balance, namely

$$\mathbf{f} \times \mathbf{u}_1 = -g\nabla(h_1 + h_2 + h_b) \quad (5.217a)$$

$$\mathbf{f} \times \mathbf{u}_2 = -g\nabla(h_1 + h_2 + h_b) + g'\nabla h_1. \quad (5.217b)$$

Note that we now keep the full variation of the Coriolis parameter [compare (5.217) with (5.209)], and the potential vorticity is not linearized about a resting state [compare (5.216) with (5.208)]. However, the informality of the derivation hides the fact that this is not an asymptotically consistent set of equations. For example, if the Coriolis parameter and the layer thickness do vary by an $\mathcal{O}(1)$ amount, then the f term dominates the ζ_i term in (5.216) and the asymptotically consistent expression for the potential vorticity is just f/h_i . Thus, although the above set of equations does span both the planetary geostrophic and quasi-geostrophic parameter regimes, they so at the price of mixing asymptotic orders in the same equation, and good conservation properties are not assured. The set above does not, in fact, conserve energy.⁹

5.11 ROSSBY WAVES

Consider now what wave motion is possible in the quasi-geostrophic shallow water system. We will consider a flat-bottomed β -plane system, for which the unforced, inviscid quasi-geostrophic equation of motion is

$$\frac{D}{Dt}(\zeta + f - \psi/L_R^2) = 0 \quad (5.218)$$

where $\zeta = \nabla^2 \psi$ is the vorticity and ψ the streamfunction.

Infinite deformation radius

Consider first the simple case in which the scale of motion is much less than the deformation scale. For flow on the β -plane the equation of motion is then

$$\frac{D}{Dt}(\zeta + \beta y) = 0. \quad (5.219)$$

Expanding the material derivative gives

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta + \beta v = 0 \quad (5.220)$$

or equivalently

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta) + \beta \frac{\partial \psi}{\partial x} = 0 \quad (5.221)$$

The idea now is to *linearize* this equation — that is, to suppose that the flow consists of a time-independent component (the ‘mean flow’) plus a perturbation, with the perturbation being small compared to the mean flow. Such a mean flow must satisfy the time independent equation of motion, and purely zonal flow will do this, and for simplicity we choose a flow with no meridional dependence. To achieve all this we let

$$\psi = \psi_0 + \psi'(x, y, t) \quad (5.222)$$

where $\psi_0 = -Uy$ and $|\psi'| \ll |\Psi|$. (The symbol U now represents the mean zonal flow, not a magnitude for scaling purposes.) Substitute (5.222) into (5.221) and neglect the nonlinear terms involving products of ψ' to give

$$\frac{\partial \zeta'}{\partial t} + J(\psi_0, \zeta') + \beta \frac{\partial \psi'}{\partial x} = 0 \quad (5.223)$$

or

$$\frac{\partial}{\partial t} \nabla^2 \psi' + U \frac{\partial \nabla^2 \zeta'}{\partial x} + \beta \frac{\partial \psi'}{\partial x} = 0 \quad (5.224)$$

Solutions to this equation may be found in the form of a plane-wave,

$$\psi' = \text{Re } \Psi e^{i(kx + ly - \omega t)} \quad (5.225)$$

where Re indicates the real part of the function. The amplitude of the oscillation is given by Ψ and the phase by $kx + ly - \omega t$, where k and l are the x - and y -wavenumbers and ω is the frequency of the oscillation.

Substituting (5.225) into (5.224) yields

$$[(-\omega + Uk)(k^2 + l^2) + \beta k] \psi_0 = 0 \quad (5.226)$$

which, for nontrivial solutions, implies

$$\boxed{\omega = Uk - \frac{\beta k}{(k^2 + l^2)}}. \quad (5.227)$$

This is the *dispersion relation* for Rossby waves. The phase speed in the x-direction is

$$c_x = \omega / k = U - \frac{\beta}{(k^2 + l^2)} \quad (5.228)$$

The velocity U provides a uniform translation, and doppler shifts the frequency. Note that current speed required to hold waves of a particular wavenumber stationary is

$$U = \frac{\beta}{k^2 + l^2}. \quad (5.229)$$

Finite deformation radius

For finite deformation radius, $\psi_0 = -Uy$ is still a solution but the potential vorticity corresponding to this state is $q_0 = Uy / L_R^2 + \beta y$ and $\nabla q_0 = (\beta + U / L_R^2)\mathbf{j}$. The linearized equation of motion is thus,

$$\left(\frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) (\nabla^2 \psi' - \psi' / L_R^2) + (\beta + U / L_R^2) \frac{\partial \psi'}{\partial x} = 0. \quad (5.230)$$

Substituting $\psi' = \psi_0 e^{i(kx + ly - \omega t)}$ we obtain the dispersion relation,

$$\omega = Uk - k \frac{\beta + U / L_R^2}{k^2 + l^2 + 1 / L_R^2} \quad (5.231)$$

with corresponding phase speed in the x-direction

$$c_x = U - \frac{\beta + U / L_R^2}{k^2 + l^2 + 1 / L_R^2}. \quad (5.232)$$

The uniform velocity field now no longer provides just a simple Doppler shift of the frequency. Waves are stationary when

$$U(k^2 + l^2 + 1 / L_R^2) = \beta + U / L_R^2 \quad (5.233)$$

That is, the current speed required to hold waves of a particular wavenumber stationary is

$$U = \frac{\beta}{k^2 + l^2}. \quad (5.234)$$

However, this is *not* simply the magnitude of the speed of waves of that wavenumber in the absence of a current. From (5.232) this is given by

$$c_x = - \frac{\beta}{k^2 + l^2 + 1 / L_R^2} \quad (5.235)$$

Why is there a difference? It is because the current does not just provide a uniform translation, but also modifies the basic potential vorticity gradient. The basic state height field η_0 is sloping, that is $\eta_0 = -(f_0/g)\Psi y$, and the ambient potential vorticity field increases with y , that is $q_0 = (\beta + U/L_R^2)y$. Thus, the basic state defines a preferred frame of reference, and the problem is not Galilean invariant.¹⁰

5.11.1 Rossby waves in two layers

Now consider the dynamics of the two-layer model, linearized about a state of rest. The two (coupled) linear equations describing the motion in each layer are

$$\frac{\partial}{\partial t} [\nabla^2 \psi_1 + F_1(\psi_2 - \psi_1)] + \beta \frac{\partial \psi_1}{\partial x} = 0 \quad (5.236a)$$

$$\frac{\partial}{\partial t} [\nabla^2 \psi_2 + F_2(\psi_1 - \psi_2)] + \beta \frac{\partial \psi_2}{\partial x} = 0 \quad (5.236b)$$

where $F_1 = f_0^2/g'H_1$ and $F_2 = f_0^2/g'H_2$. By inspection these may be transformed into two uncoupled equations: one equation is obtained by multiplying the first by F_2 and the second by F_1 and adding, and the other is obtained by subtracting the two. Then, defining

$$\bar{\psi} = \frac{F_1\psi_2 + F_2\psi_1}{F_1 + F_2} \quad \tilde{\psi} = \frac{1}{2}(\psi_1 - \psi_2) \quad (5.237)$$

(5.236) become

$$\frac{\partial}{\partial t} \nabla^2 \bar{\psi} + \beta \frac{\partial \bar{\psi}}{\partial x} = 0 \quad (5.238a)$$

$$\frac{\partial}{\partial t} [(\nabla^2 - k_d^2)\tilde{\psi}] + \beta \frac{\partial \tilde{\psi}}{\partial x} = 0. \quad (5.238b)$$

where $k_d = (F_1 + F_2)^{1/2}$. The internal radius of deformation for this problem is the inverse of this, namely

$$\lambda_d = k_d^{-1} = \frac{1}{f_0} \left(\frac{g'H_1H_2}{H_1 + H_2} \right)^{1/2}. \quad (5.239)$$

The variables $\bar{\psi}$ and $\tilde{\psi}$ are the vertical normal co-ordinates for the two layer model; they oscillate independently of each other, and the solution in physical space is just their superposition. The equation for $\bar{\psi}$ is identical to that of the single-layer, rigid-lid model, namely (5.224) with $U = 0$, and its dispersion relation is just

$$\omega = -\frac{\beta k_x}{k_x^2 + k_y^2}. \quad (5.240)$$

The barotropic mode corresponds to synchronous, depth-independent, motion in the two layers with a flat interface — from (??) the interface slope is given by $2f_0\tilde{\psi}/g'$, proportional to the amplitude of the baroclinic mode.

The dispersion relation for the baroclinic mode is

$$\omega = -\frac{\beta k_x}{k_x^2 + k_y^2 + k_d^2} \quad (5.241)$$

The mass transport associated with the baroclinic mode is identically zero, since from (5.237) we have

$$\psi_1 = \bar{\psi} + \frac{2F_1\tilde{\psi}}{F_1 + F_2}, \quad \psi_2 = \bar{\psi} - \frac{2F_2\tilde{\psi}}{F_1 + F_2} \quad (5.242)$$

and hence

$$H_1\psi_1 + H_2\psi_2 = (H_1 + H_2)\bar{\psi}. \quad (5.243)$$

The deformation radius only affects the baroclinic mode. For scales much smaller than the deformation radius, $k_x^2, k_y^2 \gg k_d^2$, the baroclinic mode obeys the same equation as the barotropic mode so that

$$\frac{\partial}{\partial t} \nabla^2 \tilde{\psi} + \beta \frac{\partial \tilde{\psi}}{\partial x} = 0. \quad (5.244)$$

For a purely baroclinic mode $F_2\psi_1 + F_1\psi_2 = 0$ and using this in (5.244) gives

$$\frac{\partial}{\partial t} \nabla^2 \psi_i + \beta \frac{\partial \psi_i}{\partial x} = 0, \quad i = 1, 2. \quad (5.245)$$

That is to say, the two layers are uncoupled from each other. At the other extreme, for very long baroclinic waves the relative vorticity is unimportant. The phase speed is negative, westward, for both baroclinic and barotropic Rossby waves but the group velocity may be positive for sufficiently long waves — see Fig. 5.8. In the figure the frequencies, velocities and wavenumbers are all non-dimensional. With a deformation radius of 50 km, typical for the mid-latitude ocean, then a non-dimensional frequency of unity corresponds to a dimensional frequency of $5 \times 10^{-7} \text{ s}^{-1}$ or a period of about 100 days. In an atmosphere with a deformation radius of 1000 km a non-dimensional frequency of unity corresponds to $1 \times 10^{-5} \text{ s}^{-1}$ or a period of about 7 days. Nondimensional velocities of unity correspond to respective dimensional velocities of about 0.25 m s^{-1} (ocean) and 10 m s^{-1} (atmosphere).

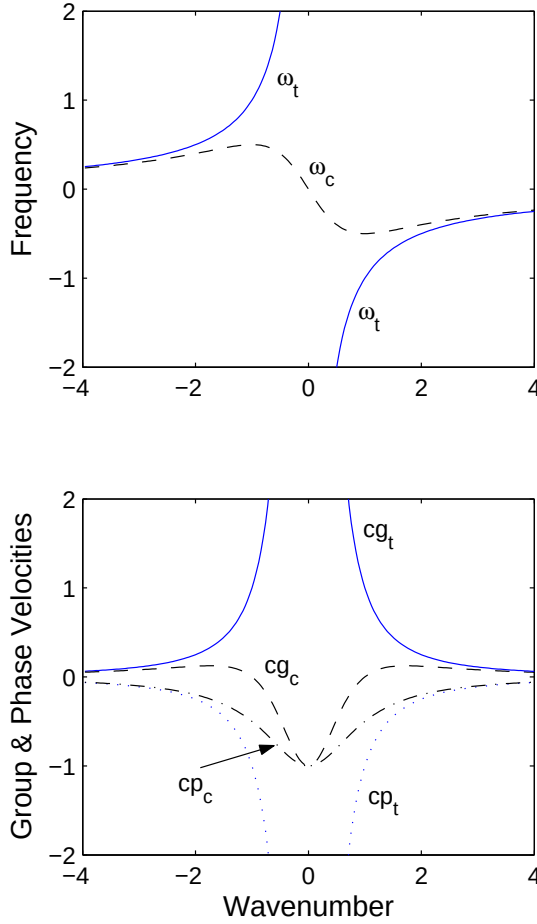


Figure 5.8 Upper panel: The dispersion relation for barotropic (ω_t , solid line) and baroclinic (ω_c , dashed line) Rossby waves in the two-layer model, calculated using (5.240) and (5.241) with $k_y = 0$, plotted here for both positive and negative zonal wavenumbers and, consequently, frequencies. The x -wavenumber is nondimensionalised by k_d , and the frequency is non-dimensionalized by β/k_d . Lower panel: the corresponding group and phase velocities, $cg = \partial\omega/\partial k_x$ and $cp = \omega/k_x$, with subscript 't' or 'c' for the barotropic or baroclinic mode. The velocities are non-dimensionalized by β/k_d^2 . The phase velocities are always negative, or westward. The group velocity of these barotropic waves is always eastward, but long baroclinic waves have a westward group velocity. For very short waves, $k^2 \gg k_d^2$, the baroclinic and barotropic velocities coincide and their phase and group velocities are equal and opposite.

5.12 GEOSTROPHIC ADJUSTMENT

The scaling of equations involves a choice: we choose scales according to the phenomena we wish to study and then simplify the equations accordingly. But the choice will be jejune unless the dynamics produces phenomena that obey that scaling. Why should the extra-equatorial circulation be in near-geostrophic balance? Why should the Rossby number be small? Arguably, the magnitude of the velocity in the atmosphere and ocean is ultimately given by the strength of the forcing, and so by the differential heating between pole and equator — although even this argument is not satisfactory, since the forcing mainly determines the energy throughput, not directly the energy itself, and the forcing is itself dependent on the atmosphere's response. But even supposing that the velocity magnitudes are given, there is no *a priori* guarantee that the dynamics will produce length-scales that are such that the Rossby number is small. The process of *geostrophic adjustment* provides part of the answer to our questions, for this is the process

whereby a fluid in an initially unbalanced state naturally evolves toward a state of geostrophic balance. We thus consider the problem of determining the free evolution of a fluid whose initial state is manifestly unbalanced, and we will do this using the linearized shallow water equations. The equations of motion are then [c.f., (5.117)]

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{f} \times \mathbf{u} = -g \nabla \eta, \quad (5.246a)$$

$$\frac{\partial \eta}{\partial t} + H \nabla \cdot \mathbf{u} = 0, \quad (5.246b)$$

where η is the free surface displacement and H is the mean fluid depth, and we omit the primes on the linearized variables.

5.12.1 Non-rotating flow

We consider first the non-rotating problem set, with little loss of generality, in one dimension. We suppose that initially the fluid is at rest but with a simple discontinuity in the height field so that

$$\eta(x, t = 0) = \begin{cases} +\eta_0 & x < 0 \\ -\eta_0 & x > 0 \end{cases} \quad (5.247)$$

and $u(x, t = 0) = 0$ everywhere. We can physically realize these initial conditions by separating two fluid masses of different depths by a thin dividing wall, and then quickly removing the wall. What is the subsequent evolution of the fluid? The general solution to the linear problem is given by (5.116) where the functional form is determined by the initial conditions so that here

$$F(x) = \eta(x, t = 0) = -\eta_0 \operatorname{sgn}(x). \quad (5.248)$$

Equation (5.116) states that this initial pattern is propagated to the right and to the left. That is, two discontinuities in fluid height simply propagate to the right and left at a speed $c = \sqrt{gH}$. Specifically, the solution is

$$\eta(x, t) = -\frac{1}{2}\eta_0 [\operatorname{sgn}(x + ct) + \operatorname{sgn}(x - ct)] \quad (5.249)$$

The initial conditions may be much more complex than a simple front, but, because the waves are dispersionless, the solution is still simply a sum of the translation of those initial conditions to the right and to the left at speed c . The velocity field in this class of problem is obtained from

$$\frac{\partial u}{\partial t} = -g \frac{\partial \eta}{\partial x} \quad (5.250)$$

giving

$$u = \frac{g}{2c} [F(x + ct) - F(x - ct)] \quad (5.251)$$

Thus, at a given location, away from the initial disturbance, the fluid remains at rest and undisturbed until the fronts arrive. After the fronts have passed, the fluid is again at rest. Ultimately, then, all the kinetic energy of the fluid is propagated away to infinity.

Rotating flow — the Rossby-Gill problem

Rotation makes a profound difference to the adjustment problem of the shallow water system, because a steady, adjusted, solution can exist with nonzero gradients in the height field, because the pressure field can be balanced by the Coriolis force. We will also find that potential vorticity conservation provides powerful constraint on the fluid evolution. In a rotating shallow fluid that conservation is represented by

$$\frac{D}{Dt} \left(\frac{\zeta + f}{h} \right) = 0 \quad (5.252)$$

which in the linear case with constant Coriolis parameter becomes

$$\frac{\partial}{\partial t} \left(\zeta - f_0 \frac{\eta}{H} \right) = 0. \quad (5.253)$$

That is, the potential vorticity field is fixed. In the example of the previous section, $f = 0$ and the fluid is initially at rest. Then $\zeta = 0$ and the fluid remains irrotational throughout the subsequent evolution of the flow. However, this is rather a weak constraint on the subsequent evolution of the fluid; it does nothing, for example, to prevent all the kinetic energy of the fluid being radiated to infinity. However, in the rotating case the potential vorticity is non-zero, and potential vorticity conservation and geostrophic balance are all we need to infer the final steady state, assuming it exists, without solving for the details of the flow evolution.

With an initial condition for the height field given by (5.247), the initial potential vorticity is given by

$$q(x, y, 0) = \begin{cases} -f_0 \eta_0 / H & x < 0 \\ f_0 \eta_0 / H & x > 0. \end{cases} \quad (5.254)$$

and this remains unchanged throughout the adjustment process. The final steady

state is then the solution of the equations:

$$\zeta - f_0 \frac{\eta}{H} = q(x, 0) \quad (5.255a)$$

$$f_0 u = -g \frac{\partial \eta}{\partial y}, \quad (5.255b)$$

$$f_0 v = g \frac{\partial h}{\partial x}. \quad (5.255c)$$

where $\zeta = \partial v / \partial x - \partial u / \partial y$. Because the Coriolis parameter is constant, the velocity field is horizontally non-divergent and we may define a streamfunction $\psi = g\eta / f_0$. Equations (5.255) then reduce to

$$\left(\nabla^2 - \frac{1}{L_d^2} \right) \psi = q(x, y, 0). \quad (5.256)$$

where $L_d = \sqrt{gH} / f_0$ is known as the *Rossby radius of deformation* or often just the ‘deformation radius’ or the ‘Rossby radius.’ It is a naturally occurring length-scale in problems involving both rotation and gravity, and arises in slightly different form in stratified fluids.

In the problem considered in the previous section, the initial conditions are a discontinuity in the height field with no variation in y . This admits of a nice analytic solution, for the flow will remain uniform in y , and (5.256) reduces to

$$\frac{\partial^2 \psi}{\partial x^2} - \frac{1}{L_d^2} \psi = -\frac{f_0 \eta_0}{H} \text{sgn}(x). \quad (5.257)$$

We may now solve this separately for $x > 0$ and $x < 0$ and then match the solutions and their first derivatives at $x = 0$, imposing also the condition that the streamfunction decay to zero as $x \rightarrow \pm\infty$. The solution is

$$\psi = \begin{cases} -(g\eta_0 / f_0)(1 - e^{-x/L_d}) & x > 0 \\ (g\eta_0 / f_0)(1 - e^{x/L_d}) & x < 0. \end{cases} \quad (5.258)$$

The velocity field associated with this is obtained from (5.255b,c), and is

$$u = 0 \quad (5.259)$$

$$v = -\frac{g\eta_0}{fL_d} e^{-|x|/L_d}. \quad (5.260)$$

The velocity is perpendicular to the slope of the free surface, and a jet forms along the initial discontinuity, as illustrated in Fig. 5.9.

The important point of this problem is that the variations in the height and field are not radiated away to infinity, as in the non-rotating problem. Rather, potential vorticity conservation constrains the influence of the adjustment to within a deformation radius (we see now why this name is appropriate) of the initial disturbance.

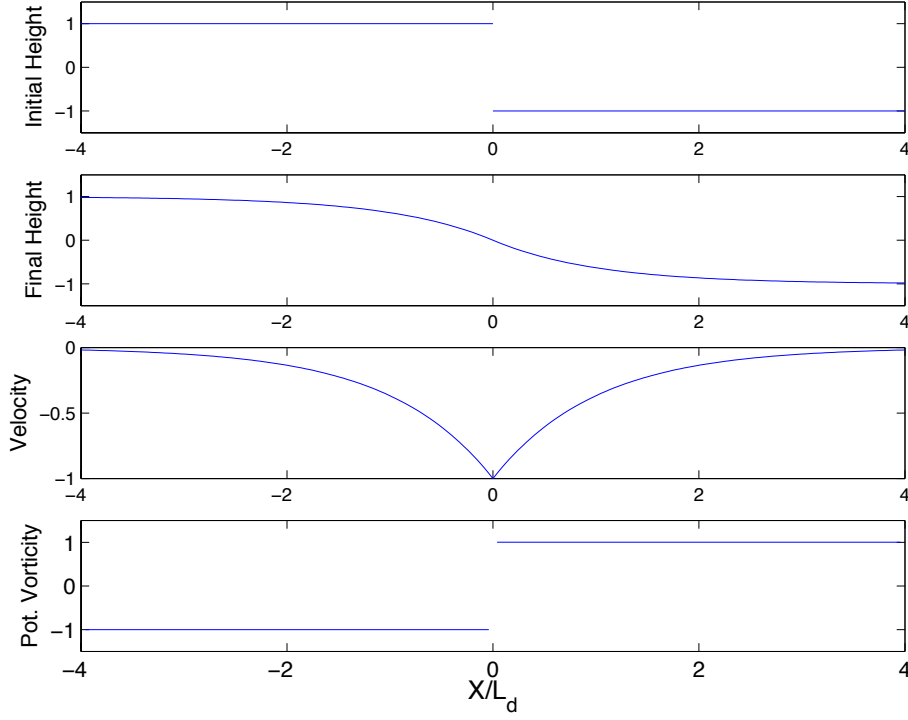


Figure 5.9 Solutions of a linear geostrophic adjustment problem. Top panel: the initial height field, given by (5.247) with $\eta_0 = 1$. Second panel: equilibrium height field, η given by (5.258) and $\eta = f_0\psi/g$. Third panel: equilibrium geostrophic velocity (normal to the gradient of height field), given by (5.259). Bottom panel: potential vorticity, given by (5.254), and this does not evolve. The distance, x is non-dimensionalized by the deformation radius $L_d = \sqrt{gH}/f_0$, and the velocity by $\eta_0(g/f_0L_d)$. Changes to the initial state occur only within $\mathcal{O}(L_d)$ of the initial discontinuity; and as $x \rightarrow \pm\infty$ the initial state is unaltered.

5.12.2 Energetics of adjustment

How much of the initial potential energy of the flow is lost to infinity by gravity wave radiation, and how much is converted to kinetic energy? The linear equations (5.246) satisfy

$$\frac{1}{2} \frac{\partial}{\partial t} (Hu^2 + g\eta^2) + gH\nabla \cdot (\mathbf{u}\eta) = 0 \quad (5.261)$$

so that energy conservation holds in the form

$$E = \frac{1}{2} \int (Hu^2 + g\eta^2) dx \quad \frac{dE}{dt} = 0. \quad (5.262)$$

The fluid has a non-zero potential energy, $(1/2) \int g\eta^2 dx$ if there are variations in fluid height. The initial potential energy is

$$PE_I = \frac{1}{2} \int_{-\infty}^{\infty} g\eta^2 dx = \int_0^{\infty} \frac{1}{2} g\eta_0^2 dx \quad (5.263)$$

which is nominally infinite if the fluid has no boundaries. In the non-rotating case the final state has no variations in free-surface height, and its potential energy is zero. That is, *all* of its initial potential energy is converted to kinetic energy which is ‘radiated away to infinity’. However, in the rotating case the final potential energy, after adjustment, is, using `sw:adjustfinal`,

$$PE_F = \frac{1}{2} g\eta_0^2 \left[\int_0^{\infty} (1 - e^{-x/L_d})^2 dx + \int_{-\infty}^0 (1 - e^{x/L_d})^2 dx \right] \quad (5.264)$$

This is also nominally infinite, but the change in potential energy is finite and is given by

$$\begin{aligned} PE_I - PE_F &= g\eta_0^2 \int_0^{\infty} (2e^{-x/L_d} - e^{-2x/L_d}) dx \\ &= \frac{3}{2} g\eta_0^2 L_d. \end{aligned} \quad (5.265)$$

The initial kinetic energy is zero, because the fluid is at rest, and its final value is, using (5.259),

$$\begin{aligned} KE_F &= \frac{1}{2} H \int u^2 dx = H \left(\frac{g\eta_0}{fL_d} \right)^2 \int_0^{\infty} e^{-2x/L_d} dx \\ &= \frac{g\eta_0^2 L_d}{2}. \end{aligned} \quad (5.266)$$

Thus, in this problem, one-third of the difference between the initial and final potential energies is converted to kinetic energy that is trapped within a distance of order a deformation radius, and the remainder, an amount $gL_d\eta_0^2$ is lost to infinity. In any finite region surrounding the initial discontinuity the final energy is less than the initial energy, but the conservation of potential vorticity prevents the complete loss of energy. In other, perhaps more realistic, adjustment problems the ratios will differ but the qualitative effect will remain. (See also problem 5.11.)

5.12.3 * General initial conditions

Because of the linearity of the (linear) adjustment problem, a spectral viewpoint is useful, in which the fields are represented as the sum or integral of

non-interacting Fourier modes. For example, suppose that the height field of the initial disturbance is a two-dimensional field given by

$$\eta = \iint \hat{\eta}_{0k,l} e^{i(k_x x + k_y y)} dk_x dk_y \quad (5.267)$$

where the Fourier coefficients $\hat{\eta}_{k,l}$ are given, and the initial velocity field is zero. Then the initial (and final) potential vorticity field is given by

$$q = -\frac{f_0}{H} \iint \hat{\eta}_{0k,l} e^{i(k_x x + k_y y)} dk_x dk_y. \quad (5.268)$$

To obtain an expression for the final height and velocity fields, we express the potential vorticity field as

$$q = \iint \hat{q}_{k_x, k_y} dk_x dk_y \quad (5.269)$$

where in the final, geostrophically balanced, state,

$$q = \frac{g}{f_0} \nabla^2 \eta - \frac{f_0}{H} \eta, \quad (5.270)$$

so that

$$\hat{q}_{k_x, k_y} = \left(-\frac{g}{f_0} k^2 - \frac{f_0}{H} \right) \hat{\eta}_{k_x, k_y}. \quad (5.271)$$

Using (5.268) and (5.271), the Fourier components of the final height field satisfy

$$\left(-\frac{g}{f_0} k^2 - \frac{f_0}{H} \right) \hat{\eta}_{k_x, k_y} = -\frac{f_0}{H} \hat{\eta}_{k,l}^{(0)} \quad (5.272)$$

or

$$\hat{\eta}_{k_x, k_y} = \frac{\hat{\eta}_{0k_x, k_y}}{k^2 L_d^2 + 1}. \quad (5.273)$$

where $k^2 = k_x^2 + k_y^2$. In physical space the final height field is just the spectral integral of this, namely

$$\eta = \iint \hat{\eta}_{k_x, k_y} dk_x dk_y = \iint \frac{\hat{\eta}_{0k_x, k_y}}{k^2 L_d^2 + 1} dk_x dk_y. \quad (5.274)$$

We see that at large scales ($k^2 L_d^2 \ll 1$) η_{k_x, k_y} is almost unchanged from its initial state; the velocity field, which is then determined by geostrophic balance, thus adjusts to the pre-existing height field. At large scales most of the energy in geostrophically balanced flow is potential energy; thus, it is energetically easier for the velocity to change to come into balance with the height field than vice

versa. At small scales, however, the final height field has much less variability than it did initially.

Conversely, at small scales the height field adjusts to the velocity field. To see this, let us suppose that the initial conditions contain vorticity but have zero height displacement. Specifically, if the initial vorticity is $\nabla^2 \psi_0$, where ψ_0 is the initial streamfunction, then it is straightforward to show that the final streamfunction is given by

$$\psi = \iint \hat{\psi}_{k_x, k_y} dk_x dk_y = \iint \frac{k_d^2 L_d^2 \hat{\psi}_{0k_x, k_y}}{k_d^2 L_d^2 + 1} dk_x dk_y. \quad (5.275)$$

The final height field then obtained from this, via geostrophic balance, by $\eta = (f_0/g)\psi$. Evidently, for *small* scales ($k_d^2 L_d^2 \gg 1$) the streamfunction, and hence the vortical component of the velocity field, are almost unaltered from their initial values. On the other hand, at large scales the final streamfunction has much less variability than it does initially, and so the height field will be largely governed by whatever variation it (and not the velocity field) had initially. In general, the final state is a superposition of the states given by (5.274) and (5.275). The divergent component of the initial velocity field does not affect the final state because it has no potential vorticity, and so all of the associated energy is lost to infinity.

Finally, we remark that just as in the problem with a discontinuous initial height profile the change in total energy during adjustment is negative — this can be seen from the form of the integrals above, although we leave the specifics as a problem to the reader. That is, some of the initial potential and kinetic energy is lost to infinity, but some is trapped by the potential vorticity constraint.

5.12.4 * A variational perspective

In the non-rotating problem, all of the initial potential energy is converted to kinetic energy. This energy is in turn radiated to infinity. In the rotating problem, the final state contains both potential and kinetic energy. Why is the energy not all radiated away to infinity? It is because potential vorticity conservation on parcels prevents all of the energy being dispersed. This suggests that it may be informative to think of the geostrophic adjustment problem as a *variational problem*: we seek to minimize the energy consistent with the conservation of potential vorticity. We stay in the linear approximation in which, because the advection of potential vorticity is neglected, potential vorticity remains constant at each point.

The energy of the flow is given by the sum of potential and kinetic energies, namely

$$\text{Energy} = \int H u^2 + g \eta^2 d\mathbf{x}, \quad (5.276)$$

(where $d\mathbf{x} \equiv dx dy$) and the potential vorticity field is

$$q = \zeta - f \frac{\eta}{H}. \quad (5.277)$$

The problem is then to extremize the energy subject to potential vorticity conservation. This is a constrained problem in the calculus of variations, sometimes called an *isoperimetric* problem because of its origins in maximizing the area of a surface for a given perimeter.¹¹ The mathematical problem is to extremize the integral

$$I = \int \{ H(u^2 + v^2) + g\eta^2 + \lambda(x, y) [(v_x - u_y) - f_0\eta/H] \} d\mathbf{x}. \quad (5.278)$$

where $\lambda(x, y)$ is a Lagrange multiplier, undetermined at this stage. It is a function of space; if it were a constant, the integral would merely extremize energy subject to a given *integral* of potential vorticity; rearrangements of potential vorticity (which are forbidden in this linear problem) would leave the integral unaltered.

Since there are three independent variables there are three Euler-Lagrange equations obtained by minimizing I . These are

$$\frac{\partial L}{\partial h} - \frac{\partial}{\partial x} \frac{\partial L}{\partial h_x} - \frac{\partial}{\partial y} \frac{\partial L}{\partial h_y} = 0 \quad (5.279a)$$

$$\frac{\partial L}{\partial u} - \frac{\partial}{\partial x} \frac{\partial L}{\partial u_x} - \frac{\partial}{\partial y} \frac{\partial L}{\partial u_y} = 0 \quad (5.279b)$$

$$\frac{\partial L}{\partial v} - \frac{\partial}{\partial x} \frac{\partial L}{\partial v_x} - \frac{\partial}{\partial y} \frac{\partial L}{\partial v_y} = 0 \quad (5.279c)$$

where L is the integrand appearing in (5.278). Equations (5.279) yield

$$2g\eta - \lambda f_0 = 0, \quad 2u + \frac{\partial \lambda}{\partial y} = 0, \quad 2v - \frac{\partial \lambda}{\partial x} = 0. \quad (5.280)$$

Eliminating λ yields the simple relationships

$$u = -\frac{g}{f_0} \frac{\partial \eta}{\partial y}, \quad v = \frac{g}{f_0} \frac{\partial \eta}{\partial x}, \quad (5.281)$$

which are the equations of geostrophic balance. These imply $\nabla \cdot \mathbf{u} = 0$ and so $\eta = f\psi/g$ and, from (5.277), $q = \nabla^2\psi - \psi/L_d^2$.

Thus, in the linear approximation, geostrophic balance is an extreme energy state for a given field of potential vorticity. It is obvious that this is a minimum energy state, since all the energy associated with gravity waves has been eliminated. This is consistent with our picture of geostrophic balance as being one in which gravity wave energy is radiated away to infinity (or dissipated on a much faster timescale than the geostrophic modes, as in section ??), leaving behind only the energy associated with geostrophic motion. Quasi-geostrophic motion is then evolution on this minimum energy manifold, determined by $Dq/Dt = 0$ and the inversion conditions (5.12.4) (see also problem 5.22).

5.13 †FRONTAL GEOSTROPHIC DYNAMICS

There is one scale of motion of physical importance that is captured by neither the quasi-geostrophic dynamics nor planetary geostrophy. This is the scale of motion larger than the deformation radius but smaller than the scale on which the Coriolis parameter varies significantly. In some sense it is between quasi-geostrophy and planetary geostrophy, and has been called 'frontal geostrophy.' We will derive the equations for single layer of shallow water.[Stay tuned]

5.14 †HIGHER ORDER QUASI-GEOSTROPHIC DYNAMICS

If something's worth doing, it's worth overdoing.

Posted on John Wilson's door at GFDL.

Notes

- 1 The algorithm to numerically solve these equations differs from that of the free-surface shallow water equations because the mass conservation equation can no longer be stepped forward in time. Rather, an elliptic equation for p_{lid} must be derived by eliminating time derivatives from (5.23) using (5.22), and this then solved at each timestep.
- 2 This effect is related to form drag as the expression is commonly used in aerodynamics, which is the force due to pressure difference between the front and rear of an object, or any other 'form', moving through a fluid. Aerodynamic form drag includes frictional effects between the wind and the surface itself, but this effect is absent in geophysical uses.
- 3 (Jules-)Henri Poincaré (1854–1912) was a prodigious French mathematician, physicist and philosopher, regarded as one the greatest mathematicians living at the turn of the 20th century. He is probably most remembered now for his original work in analysis (e.g., solvability of algebraic equations) and celestial mechanics, obtaining many results in what would be called non-linear dynamics and chaos when these fields re-emerged some 60 years later. He also independently obtained a number of the results of special relativity (he discovered, for example, that the Lorentz transformation forms a group that make the Maxwell equations invariant), as well as working on the theory of rotating fluids — hence the Poincaré waves of this chapter [also discussed by Kelvin (1879)]. Finally, he wrote extensively and successfully for the general public on the meaning, importance and philosophy of science. Among other things he discussed whether scientific knowledge was a mere convention, a notion that remains discussed and controversial to this day. (His answer: 'convention', in part, yes; 'mere', no.) He was also a proponent of the role of intuition in mathematical and scientific progress, and did not believe that mathematics could ever be wholly

reduced to pure logic, a belief that was bolstered by Gödel some 20 years after Poincaré's death. One of the last polymaths.

4 See Gill (1982) or Philander (1990) for more complete discussions

5 ?.

6 Carl-Gustav Rossby (1898-1957) played a dominant role in the development of dynamical meteorology in the early and middle parts of the 20th century. His work permeates all aspects of dynamical meteorology today. Perhaps the most fundamental non-dimensional number in rotating fluid dynamics, the Rossby number, is named for him, as is the perhaps the most fundamental wave, the Rossby wave. He also discovered the conservation of potential vorticity (later generalised by Ertel) and contributed important ideas to atmospheric turbulence and the theory of air masses. Swedish born, he studied first with V. Bjerknes before taking a position in Stockholm in 1922 with the Swedish Meteorological Hydrologic Service, also taking a 'Licentiat' at the University of Stockholm in 1925. Shortly thereafter he moved to the United States, joining the Government Weather Bureau, a precursor of NOAA's National Weather Service. In 1928 he moved to MIT, playing an important role in developing the meteorology department there, while still maintaining connections with the Weather Bureau. In 1940 he moved to the University of Chicago, where he similarly developed that department. In 1947 he became director of the newly-formed Institute of Meteorology in Stockholm, and subsequently divided his time between there and the United States. Thus, as well as his scientific contributions, he played a very influential role in the institutional development of the field.

7 (Burger 1958)

8 This is called the 'frontal geostrophic' regime (Cushman-Roisin 1994).

9 Numerical integrations of the potential vorticity equation using (5.216) and (5.217) as an inversion do indicate improved accuracy over either the quasi-geostrophic or planetary geostrophic equations (e.g., Mundt *et al.* 1997), but the equations do not conserve energy and mass is only conserved globally. Using the methods of Hamiltonian mechanics it is possible to derive equations that improve on the accuracy of the quasi-geostrophic equations and that do have good conservation properties (see for example Salmon 1983; Allen *et al.* 2002), although the resulting equations are typically rather complicated. We should also note, perhaps tritely, that within the quasi-geostrophic parameter regime the quasi-geostrophic equations are accurate and perform well.

10 This non-Doppler effect also arises quite generally, even in models in height co-ordinates. See White (1977) and problem 6.3.

11 An introduction to variational problems may be found in Weinstock (1952) and a number of other textbooks. Applications to many traditional problems in mechanics are discussed by Lanczos (1970).

Further Reading

Gill, A. E. 1982. *Atmosphere-Ocean Dynamics*.

This remains a classic reference for geostrophic adjustment and gravity waves.

The time-dependent geostrophic adjustment problem is discussed in section 7.3.

Problems

- 5.1 Derive the appropriate shallow water equations for a single moving layer of fluid of density ρ_1 above a rigid floor, and where above the moving fluid is a stationary fluid of density ρ_0 , where $\rho_0 < \rho_1$. Show that as $\rho_0 \rightarrow 0$ the usual shallow water equations emerge.
- 5.2 (a) Let us model the atmosphere as two immiscible, 'shallow water' fluids of different density stacked one above the other. Using reasonable values for the values of any needed physical parameters such as the density of the two layers, estimate the displacement of the interfacial surface associated with a pole-equator temperature gradient of 40 K.
 (b) Similarly estimate an interfacial displacement in the ocean associated with a temperature gradient of 20 K over a distance of 4000 km. (This is a crude representation of the main oceanic thermocline.)
- 5.3 $\diamond$ For a shallow water fluid the energy equation, (5.85), has the form $\partial E / \partial t + \nabla \cdot (\mathbf{v}(E + gh^2/2)) = 0$. But for a compressible fluid, the corresponding energy equation, (??), has the form $\partial E / \partial t + \nabla \cdot (\mathbf{v}(E + p)) = 0$. In a shallow water fluid, $p \neq gh^2/2$ at a point so these equations are, at least superficially, different. Explain this and reconcile the two forms. (Hint: The shallow water equations are in hydrostatic balance. What is the average pressure in a fluid column?)
- 5.4 $\diamond$ Can the shallow water equations for an incompressible fluid be derived by way of an asymptotic expansion in the aspect ratio? If so, do it. That is, without assuming hydrostasy ab initio, expand the Euler equations with a free surface in the ratio of the depth of the fluid to the horizontal scale of the motion to obtain the shallow water equations. If not, show that the equations are inherently non-asymptotic.
- 5.5 $\diamond$ The inviscid shallow water equations, rotating or not, can support gravity waves of arbitrarily short wavelengths. For sufficiently high wavenumber, the wavelength will be shorter than the depth of the fluid. Is this consistent with an asymptotic nature of the shallow water equations? Discuss.
- 5.6 Show that the appropriate generalization of (5.105) to the case with orography (i.e., a non-flat bottom) is

$$\frac{D}{Dt} \left(\frac{z - h_b}{h} \right) = 0 \quad (\text{P5.1})$$

where h_b is the height of the lower surface above a reference level, z the height above that level level, and h is the fluid thickness and. From this show that

$$w = \frac{z - h_b}{h} \frac{Dh}{Dt} + \frac{D\eta}{Dt}. \quad (\text{P5.2})$$

and interpret this result, showing that it gives sensible answers at the top and bottom of the fluid layer. Obtain an expression for the conservation of potential vorticity in this system.

- 5.7 What is the appropriate generalization of (5.116) to two-dimensions? Suppose that at time $t = 0$ the height field is given by a Gaussian distribution $h' = Ae^{-r^2/\sigma^2}$ where $r^2 = x^2 + y^2$. What is the subsequent evolution of this, in the linear approximation? Show that the distribution remains Gaussian, and that its width increases at speed $\sqrt{gH}$, where H is the mean depth of the fluid.
- 5.8 In an adiabatic shallow water fluid, it is easy to show that $D(\zeta/h)/Dt = 0$. If we are in a rotating reference frame show that the corresponding potential vorticity conservation law is

$$\frac{D}{Dt} \frac{\zeta + f}{h} = 0 \quad (\text{P5.3})$$

If the bottom of the container is not flat, show that the corresponding potential vorticity conservation law is now

$$\frac{D}{Dt} \frac{\zeta + f}{\eta - h_b} = 0 \quad (\text{P5.4})$$

where η is the height of the free surface and h_b is the height of the bottom topography, both referenced to a flat surface.

- (a) A cylindrical column of air at 30° latitude with radius 100 km expands horizontally to twice its original radius. If the air is initially at rest, what is the mean tangential velocity at the perimeter after the expansion.
- (b) An air column at 60° N with zero relative vorticity ($\zeta = 0$) stretches from the surface to the tropopause, which we assume is a rigid lid, at 10 km. The air column moves zonally onto a plateau 2.5 km high. What is its relative vorticity? Suppose it then moves southward to 30° N. What is its vorticity? (Assume density is constant.)
- 5.9 $\diamond$ In the long-wave limit of Poincaré waves, fluid parcels behave as free-agents; that is, like free solid particles moving in a rotating frame unencumbered by pressure forces. Why then, is their frequency given by $\omega = f = 2\Omega$ where Ω is the rotation rate of the co-ordinate system, and not by Ω itself? Do particles that are stationary or move in a straight line in the inertial frame of reference satisfy the dispersion relationship for Poincaré waves in this limit? Explain.
- 5.10 Consider the unforced, inviscid shallow water equations over topography obeying (??) and (??). Show that the total (kinetic plus potential) energy of the fluid is conserved in a closed domain. Obtain an expression for the energy flux and give a physical interpretation of all the terms in it.
- 5.11 Linearize the f -plane shallow-water system about a state of rest. Suppose that there is an initial disturbance that is given in the general form

$$\eta = \iint \hat{\eta}_{k,l} e^{i(kx+ly)} dk dl \quad (\text{P5.5})$$

where η is the deviation surface height and the Fourier coefficients $\hat{\eta}_{k,l}$ are given.

- (a) Obtain the geopotential field at the completion of geostrophic adjustment, and show that the deformation scale is a natural length scale in the problem.

- (b) Show that the change in total energy during the adjustment is always less than or equal to zero. Neglect any initial divergence.
N.B. Because the problem is linear, the Fourier modes do not interact.

5.12 Consider the flat-bottomed shallow water potential vorticity equation in the form

$$\frac{D}{Dt} \frac{\zeta + f}{h} = 0 \quad (\text{P5.6})$$

- (a) Suppose that deviations of the height field are small compared to the mean height field, and that the Rossby number is small (so $|\zeta| \ll f$). Further consider flow on a β -plane such that $f = f_0 + \beta y$ where $|\beta y| \ll f_0$. Show that the evolution equation becomes

$$\frac{D}{Dt} \left(\zeta + \beta y - \frac{f_0 \eta}{H} \right) = 0 \quad (\text{P5.7})$$

where $h = H + \eta$ and $|\eta| \ll H$. Using geostrophic balance in the form $f_0 u = -g \partial \eta / \partial y$, $f_0 v = g \partial \eta / \partial x$, obtain an expression for ζ in terms of η .

- (b) Linearize (P5.7) about a state of rest, and show that the resulting system supports two-dimensional Rossby waves that are similar to those of the usual two-dimensional barotropic system. Discuss the limits in which the wavelength is much shorter or much longer than the deformation radius.
(c) Linearize (P5.7) about a *geostrophically balanced state* that is translating uniformly eastwards. Note that this means that:

$$u = U + u' \quad \eta = \eta(y) + \eta' \quad (\text{P5.8})$$

where $\eta(y)$ is in geostrophic balance with U . Obtain an expression for the form of $\eta(y)$.

- (d) Obtain the dispersion relation for Rossby waves in this system. Show that their speed is different from that obtained by adding a constant U to the speed of Rossby waves in part (b), and discuss why this should be so. (That is, why is the problem not *Galilean invariant*?)

5.13 Starting with the rotating shallow water equations, nondimensionalize using f_0^{-1} as the timescale and $\sqrt{gH}/f_0$ as the length scale. Show that if we also nondimensionalize the deviation height by the mean height of the fluid, then the resulting equations have no non-dimensional parameters.

5.14 In the shallow water equations show that, if the flow is approximately geostrophically balanced, the energy at large scales is predominantly potential energy and that energy at small scales is predominantly kinetic energy. Define precisely what 'large scale' and 'small scale' mean in this context.

5.15 In the shallow-water geostrophic adjustment problem, show that at large scales the velocity adjusts to the height field, and that at small scales the height field adjusts to the velocity field.

5.16 In the derivation of the quasi-geostrophic equations, geostrophic balance leads to the lowest order velocity being divergence-free — that is, $\nabla_2 \cdot \mathbf{u}_0 = 0$. It

seems that this can also be obtained from the mass conservation equation at lowest order. Is this a coincidence? Suppose that the Coriolis parameter varied, and that the momentum equation yielded $\nabla_2 \cdot \mathbf{u}_0 \neq 0$. Would there be an inconsistency? Discuss.

- 5.17 $\diamond$ In the planetary geostrophic approximation, obtain an evolution equation and corresponding inversion conditions that conserves potential vorticity and that is accurate to one higher order in Rossby number than the usual shallow water planetary geostrophic equations.
- 5.18 The shallow-water planetary geostrophic equations are sometimes used as a model of large-scale flow in the ocean. These equations are:

$$\mathbf{f} \times \mathbf{u} = -g\nabla h, \quad \frac{Dh}{Dt} + h\nabla \cdot \mathbf{u} = 0 \quad (\text{P5.9})$$

Show that it follows from these equations that the potential vorticity equation is

$$\frac{D}{Dt} \left(\frac{f}{h} \right) = 0, \quad (\text{P5.10})$$

and comment briefly on the meaning of this. Normally, these equations apply for very large scales for which f varies. But suppose that f is constant. Show that any flow is in fact a *steady* solution to the equations. Discuss the implications of this.

- 5.19 Given the baroclinic dispersion relation

$$\omega = -\frac{\beta k_x}{k_x^2 + k_d^2}, \quad (\text{P5.11})$$

for what value of k_x is the x -component of group velocity the largest (i.e., the most positive), and what is the corresponding value of the group velocity?

- 5.20 Obtain solutions to the two-layer Rossby wave problem by seeking solutions of the form

$$\begin{aligned} \psi_1 &= \text{Re } \Psi_1 e^{i(k_x x + k_y y - \omega t)} \\ \psi_2 &= \text{Re } \Psi_2 e^{i(k_x x + k_y y - \omega t)}. \end{aligned} \quad (\text{P5.12})$$

Substitute (P5.12) directly into (5.236) to obtain the dispersion relation, and show that the ensuing two roots correspond to the baroclinic and barotropic modes. Show that the baroclinic mode has no net (vertically integrated) transport associated with it, and that the motion of the barotropic is independent of depth.

- 5.21 $\diamond$ (Not difficult, but messy.) Obtain the vertical normal modes and the dispersion relationship of the two-layer quasi-geostrophic problem with a free surface, for which the equations of motion linearized about a state of rest are

$$\frac{\partial}{\partial t} [\nabla^2 \psi_1 + F_1(\psi_2 - \psi_1)] + \beta \frac{\partial \psi_1}{\partial x} = 0 \quad (\text{P5.13a})$$

$$\frac{\partial}{\partial t} [\nabla^2 \psi_2 + F_2(\psi_1 - \psi_2) - F_{\text{ext}} \psi_1] + \beta \frac{\partial \psi_2}{\partial x} = 0. \quad (\text{P5.13b})$$

where $F_{\text{ext}} = f_0 / (gH_2)$.

- 5.22 $\diamond$ Consider the problem of minimizing the full energy [i.e., $\int (h\mathbf{u}^2) + g\eta^2 \, d\mathbf{x}$], given the potential vorticity field $q(x, y) = (\zeta + f)/h$. Show that the balance relations analogous to (5.12.4) are $uh = -\partial(Bq^{-1})/\partial y$ and $vh = \partial(Bq^{-1})/\partial x$ where B is the Bernoulli function $B = g\eta + \mathbf{u}^2/2$. Show that steady flow does not necessarily satisfy these equations. Discuss.

This extreme generality whereby the equations of motion apply to the entire spectrum of possible motions — to sound waves as well as cyclone waves — constitutes a serious defect of the equations from the meteorological point of view.

Jule Charney, *On the scale of atmospheric motions*, 1948.

CHAPTER SIX

Simplified Equations for the Ocean and Atmosphere

LARGE-SCALE FLOW IN THE OCEAN AND ATMOSPHERE is characterized by an approximate balance in the vertical between the pressure gradient and gravity (i.e., hydrostatic balance), and in the horizontal between the pressure gradient and the Coriolis force (i.e., geostrophic balance). In this chapter we exploit this to simplify the full equations for rotating stratified system and obtain various sets of ‘geostrophic’ equations. Depending on the precise nature of the assumptions we make, we are led to the *quasi-geostrophic* system for horizontal scales similar to that on which most synoptic activity takes place and, for very large-scale motion, to the *planetary-geostrophic* set of equations. There are two related reasons to consider approximate models such as these:

1. Parsimony: i.e., to simplify the equations as an aid in understanding their essential properties.
2. To eliminate unwanted and unimportant fast waves, and so allow for longer timesteps in numerical models.¹

The resulting equations properly represent, of course, only motions within the particular parameter regime we choose. That is to say, in the derivations of the simplified equations we make choices as to what phenomena to study, what the scales of motion are and what the small parameters might be.

There is no unique choice, and there indeed there is nothing guaranteeing that our choices are sensible.

6.1 ELEMENTARY SCALING IN THE MOMENTUM EQUATION

6.1.1 The Rossby Number

The Rossby number characterizes the importance of rotation in a fluid. It is a nondimensional number that is essentially the ratio of the magnitudes of the relative acceleration to that produced by the Coriolis force. It is a fundamental nondimensional number in geophysical fluid dynamics, and we introduce it briefly here, and return to it repeatedly throughout the book. We begin with the horizontal momentum equation in the form,

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{u} + \mathbf{f} \times \mathbf{u} = -\frac{1}{\rho} \nabla_z p. \quad (6.1)$$

It is clear that the approximate size of the advection term is

$$\mathbf{v} \cdot \nabla \mathbf{u} \sim \frac{U^2}{L} \quad (6.2)$$

and the approximate size of the Coriolis acceleration is

$$\mathbf{f} \times \mathbf{u} \sim fU \quad (6.3)$$

where U and L are the approximate magnitude of the horizontal velocity and a typical lengthscale over which the velocity varies, respectively. (We assume that $W/H \lesssim U/L$, so that vertical advection does not dominate the advection.) The ratio of these two terms is defined to be the Rossby number,

$$\boxed{Ro \equiv \frac{U}{fL}}. \quad (6.4)$$

If the Rossby number is small, then rotation effects are important. For large-scale flow in the earth's atmosphere, we might take $V = 10 \text{ m s}^{-1}$ and $L = 10^6 \text{ m}$. Then, with $f \approx 10^{-4} \text{ s}^{-1}$ we find that $Ro \approx 0.1$. In the ocean, a typical velocity might be 0.1 m s^{-1} , and then for a length scale of 10^5 m we have $Ro \approx 0.01$. Thus, for large-scale flows in both atmosphere and ocean we can expect that the effects of rotation indeed will be important.

Another intuitive way to think about the Rossby number is in terms of timescales. The Rossby number based on a timescale is

$$Ro_t \equiv \frac{1}{fT} \quad (6.5)$$

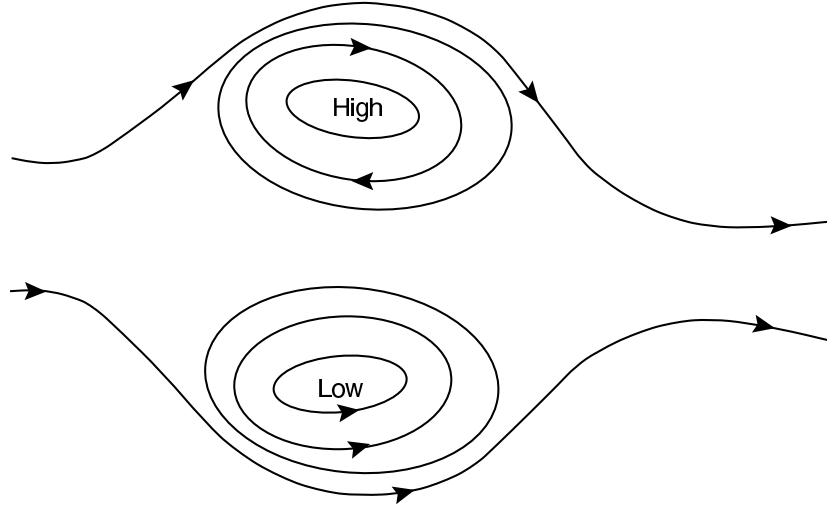


Figure 6.1 Schematic of geostrophic flow with a positive value of the Coriolis parameter f . Flow is parallel to the lines of constant pressure (isobars). Cyclonic flow is anticlockwise around a low pressure region and anticyclonic flow is clockwise around a high. If f were negative, as in the Southern hemisphere, (anti-)cyclonic flow would be (anti-)clockwise.

where T is a timescale associated with the dynamics at hand. If the timescale is an advective one, meaning that $T \sim L/U$, then this definition is equivalent to (6.4). Now, $f = 2\Omega \sin \vartheta$, where Ω is the angular velocity of the rotating frame and equal to $2\pi/T_p$ where T_p is the period of rotation (24 hours). Thus,

$$R_{ot} = \frac{T_p}{4\pi T \sin \vartheta} = \frac{T_i}{T} \quad (6.6)$$

where T_i is an inertial timescale, $1/f$, about three hours in midlatitudes. Thus, for phenomena with timescales longer than the rotation period, such as the motion of the Gulf Stream or a mid-latitude atmospheric weather system, the effects of the rotation can be expected to be important, whereas a short-lived phenomena, such as a cumulus cloud or tornado, may be oblivious to the external rotation. These are, of course, just ‘ball-park’ measures of the importance of rotation.

6.1.2 Geostrophic Balance

If the Rossby number is sufficiently small in (6.1) then the rotation term will dominate the nonlinear advection term, and if the time period of the motion scales advectively then the rotation term also dominates the local time derivative. The only term which can then balance the rotation term is the pressure term, and

therefore we must have

$$\mathbf{f} \times \mathbf{u} \approx -\frac{1}{\rho} \nabla_h p, \quad (6.7)$$

or, in Cartesian component form

$$fu \approx -\frac{1}{\rho} \frac{\partial p}{\partial y}, \quad fv \approx \frac{1}{\rho} \frac{\partial p}{\partial x} \quad (6.8)$$

This balance is known as *geostrophic balance*, and its consequences are profound, and indeed it is this balance which gives geophysical fluid dynamics a special place in the broader field of fluid dynamics. We may *define* the geostrophic flow by

$$fu_g \equiv -\frac{1}{\rho} \frac{\partial p}{\partial y}, \quad fv_g \equiv \frac{1}{\rho} \frac{\partial p}{\partial x}. \quad (6.9)$$

and for low Rossby number flow $u \approx u_g$ and $v \approx v_g$. From this simple balance we can deduce some important points:

- (i) Flow is parallel to lines of constant pressure (isobars). If $f > 0$ the flow is anti-clockwise round a region of low pressure and clockwise around a region of high pressure (see Fig. 6.1).
- (ii) If the Coriolis flow is constant then, for constant density flow, we may define the streamfunction ψ by

$$\psi \equiv \frac{p}{f_0 \rho_0} \quad (6.10)$$

whence

$$u_g = -\frac{\partial \psi}{\partial y}, \quad v_g = \frac{\partial \psi}{\partial x}. \quad (6.11)$$

The geostrophic flow is then horizontally non-divergent; that is

$$\nabla_z \cdot \mathbf{u}_g = \frac{\partial u_g}{\partial x} + \frac{\partial v_g}{\partial y} = 0. \quad (6.12)$$

The vertical component of vorticity is then

$$\omega = \mathbf{k} \cdot \nabla \times \mathbf{u} = \nabla_z^2 \psi \quad (6.13)$$

- (iii) If the Coriolis parameter is not constant, then cross-differentiating (6.9) gives, for constant density flow,

$$v_g \frac{\partial f}{\partial y} + f \nabla_z \cdot \mathbf{u}_g = 0 \quad (6.14)$$

which implies, using mass continuity,

$$\beta v_g = f \frac{\partial w}{\partial z}. \quad (6.15)$$

where $\beta \equiv \partial f / \partial y = 2\Omega \cos \vartheta / a$. This is known as *Sverdrup balance*.

6.1.3 Taylor-Proudman effect

Consider a rapidly rotating, constant density, fluid, with the rotation vector pointing in the z -direction. In the limit of zero Rossby number, we assume that the three-dimensional momentum equation may be written as

$$\mathbf{f} \times \mathbf{v} = -\nabla\phi - \nabla\chi \quad (6.16)$$

where $\mathbf{f} = 2\boldsymbol{\Omega} = 2\Omega\mathbf{k}$, $\phi = p/\rho$, and $\nabla\chi$ represents other potential forces. If $\chi = gz$ then the vertical component of this equation represents hydrostatic balance, and the horizontal components represent hydrostatic balance.

On taking the curl of this equation, the terms on the right-hand side vanish and the left-hand side becomes

$$(\mathbf{f} \cdot \nabla)\mathbf{v} - \mathbf{f}\nabla \cdot \mathbf{v} - (\mathbf{v} \cdot \nabla)\mathbf{f} + \mathbf{v}\nabla \cdot \mathbf{f} = 0 \quad (6.17)$$

But, $\nabla \cdot \mathbf{v} = 0$ by mass conservation, and, if $\mathbf{f}$ is constant, both $\nabla \cdot \mathbf{f}$ and $(\mathbf{v} \cdot \nabla)\mathbf{f}$ vanish. Thus

$$(\mathbf{f} \cdot \nabla)\mathbf{v} = 0, \quad (6.18)$$

which since $\mathbf{f} = f\mathbf{k}$, implies

$$f \frac{\partial \mathbf{v}}{\partial z} = 0. \quad (6.19)$$

In particular,

$$\frac{\partial u}{\partial z} = 0, \quad \frac{\partial v}{\partial z} = 0 \quad \text{and} \quad \frac{\partial w}{\partial z} = 0. \quad (6.20)$$

If there is a solid horizontal boundary anywhere in the fluid, for example at the surface, then $w = 0$ at that surface and thus $w = 0$ everywhere. Hence the motion occurs in planes that lie perpendicular to the axis of rotation, and the flow is ‘two-dimensional.’

A slightly different derivation proceeds as follows. If the flow is exactly in geostrophic and hydrostatic balance then

$$v = \frac{1}{f} \frac{\partial \phi}{\partial x}, \quad u = -\frac{1}{f} \frac{\partial \phi}{\partial y} \quad (6.21)$$

and

$$\frac{\partial \phi}{\partial z} = -g \quad (6.22)$$

Differentiating (6.21) with respect to z , and using (6.22) yields

$$\frac{\partial v}{\partial z} = \frac{-1}{f} \frac{\partial g}{\partial x} = 0, \quad (6.23)$$

with a similar expression for $\partial u / \partial z$. Further, noting that the geostrophic velocities are horizontally non-divergent, and using the mass conservation equation in the form

$$\frac{\partial w}{\partial z} = -\nabla_z \cdot \mathbf{u} \quad (6.24)$$

implies $\partial w / \partial z = 0$.

These manipulations imply the *Taylor-Proudman effect*,² namely that for constant density flow the vertical derivative of the horizontal and the vertical velocities vanish in the limit of zero Rossby number. If the vertical velocity is zero somewhere in the flow, it is zero everywhere in that vertical column. The horizontal flow has no vertical shear, and the fluid moves like a slab; the effects of rotation have provided a *stiffening* of the fluid in the vertical.

In neither the atmosphere nor the ocean do we observe precisely such flow, mainly because of the effects of stratification (i.e., density varying in the vertical.) However, it is typical of geophysical fluid dynamics that the assumptions underlying a derivation are rarely fully satisfied, yet there are manifestations of it in real flow. Thus, one might have naïvely expected, because $\partial w / \partial z = -\nabla_z \cdot \mathbf{u}$, that $W/H \sim U/L$, where W, U, H and L are the magnitudes of the vertical and horizontal velocities, and the vertical and horizontal scales, respectively. However, if the flow is rapidly rotating we expect that the horizontal flow will be in near geostrophic balance and therefore nearly divergence free. Thus, this scaling estimate is likely to be a gross overestimate of the magnitude of the vertical velocity, an important manifestation of the Taylor-Proudman effect.

Thermal wind balance

Thermal wind balance arises by combining the geostrophic and hydrostatic approximations, and this is most easily done in the context of the anelastic (or Boussinesq) equations, or in pressure co-ordinates.

In spherical co-ordinates, geostrophic balance may be written

$$-2\Omega \sin \vartheta v = -\frac{\partial \phi}{\partial x} = -\frac{1}{a \cos \vartheta} \frac{\partial \Phi}{\partial \lambda} \quad (6.25a)$$

$$2\Omega u \sin \vartheta = -\frac{\partial \phi}{\partial y} = -\frac{1}{a} \frac{\partial \Phi}{\partial \vartheta} \quad (6.25b)$$

Combining these with hydrostatic balance, $\partial \phi / \partial z = b$, gives

$$-2\Omega \sin \vartheta \frac{\partial v}{\partial z} = -\frac{\partial b}{\partial x} = -\frac{1}{a \cos \lambda} \frac{\partial b}{\partial \lambda} \quad (6.26a)$$

$$2\Omega \sin \vartheta \frac{\partial u}{\partial z} = -\frac{\partial b}{\partial y} = -\frac{1}{a} \frac{\partial b}{\partial \vartheta} \quad (6.26b)$$

In terms of the zonal angular momentum, (6.26b) may be written

$$\frac{\partial m}{\partial z} = -\frac{a}{2\Omega \tan \vartheta} \frac{\partial b}{\partial y} \quad (6.27)$$

Somewhat more accurate than geostrophic balance is the so-called cyclostrophic wind balance

$$2u\Omega \sin \vartheta + \frac{u^2}{a} \tan \vartheta = -\frac{\partial \phi}{\partial y} = -\frac{1}{a} \frac{\partial \phi}{\partial \vartheta}. \quad (6.28)$$

For large-scale flow this differs significantly from geostrophic balance only very close to the equator. The associated thermal wind relation takes a simple form when expressed in terms of angular momentum, namely

$$\frac{\partial m^2}{\partial z} = -\frac{a^3 \cos^3 \vartheta}{\sin \vartheta} \frac{\partial b}{\partial y} \quad (6.29)$$

where $m = (u + \Omega a \cos \vartheta) a \cos \vartheta$.

If the density or potential temperature is constant, then there is no shear, and we recover the Taylor-Proudman result. But suppose that the temperature falls in the polewards direction. Then thermal wind balance implies that the (eastwards) wind will increase with height — just as is observed in the atmosphere. In general a vertical shear of the horizontal wind is associated with horizontal temperature gradients.

6.2 EQUATIONS FOR THE VERY LARGE SCALE: THE PLANETARY GEOSTROPHIC APPROXIMATION

The first set of approximate equations we consider is appropriate for very large-scale motion in a rapidly rotating system. We will define ‘very large scale’ more precisely later on, but for now we can imagine we are dealing with near-planetary scales; however, we will demand that the Rossby number remain small, and hence equatorial dynamics are excluded. It turns out that the resulting equations of motion have more oceanographic relevance than atmospheric, and so we will begin with the (inviscid and adiabatic) Boussinesq equations of motion with the hydrostatic approximation, namely

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_z \phi, \quad (6.30)$$

$$\frac{\partial \phi}{\partial z} = b, \quad (6.31)$$

$$\frac{Db}{Dt} = 0, \quad (6.32)$$

$$\nabla \cdot \mathbf{v} = 0. \quad (6.33)$$

Anticipating that $\bar{b}$ may not scale in the same way as the deviation from it, let us separate out the contribution of the advection of the mean stratification in (6.32) by writing

$$b = \bar{b}(z) + b'(x, y, z, t), \quad (6.34)$$

Then the thermodynamic equation becomes

$$\frac{Db'}{Dt} + N^2 w = 0 \quad (6.35)$$

where $N^2 = \partial \bar{b} / \partial z$. We then let $\phi = \bar{\phi}(z) + \phi'$ where $\bar{\phi}$ is hydrostatically balanced by $\bar{b}$, and the hydrostatic equation becomes

$$\frac{\partial \phi'}{\partial z} = b'. \quad (6.36)$$

Equations (6.35) and (6.36) replace (6.32) and (6.31).

6.2.1 Non-dimensional equations

Simplification of the equations is effected by first putting the equations in non-dimensional form. We do not do this arbitrarily; rather, we choose, *a priori*, a scaling that, we believe, holds for class of motion we are interested in. If this scaling is correct, then the non-dimensional equations contain variables that are *order one* in magnitude, possibly multiplied certain non-dimensional parameters that may be large or small. The size of various terms in the equations of motion then becomes transparent.

Let us then choose typical values:

$$(x, y) \sim L, \quad (u, v) \sim U, \quad z \sim H, \quad f \sim f_0. \quad (6.37)$$

(For example, for large-scale oceanic flow we may have $U = 0.1 \text{ m/s}$, $L = 5000 \text{ km}$ and so on.) We now *assume* that the appropriate timescale is the advective one, and thus that $t \sim L/U$. The local time derivative then has the same asymptotic size as the advective term. (It is essentially this assumption that precludes wave-like motion in which the pressure term is balanced by the local time-derivative.) Further, we assume *small Rossby number* scaling, namely that $Ro = U/f_0 L \ll 1$. Then the pressure term must balance the Coriolis force,

$$\mathbf{f} \times \mathbf{u} \sim -\nabla \phi \quad (6.38)$$

and the pressure scales as

$$\phi' \sim f_0 UL. \quad (6.39)$$

where f_0 is a representative value of f . Using the hydrostatic relation, (6.39) implies that the buoyancy scales as

$$b' \sim \frac{f_0 UL}{H}. \quad (6.40)$$

Finally, we will assume that the appropriate scale of vertical motion is given by the mass conservation equation,

$$\frac{\partial w}{\partial z} = - \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right), \quad (6.41)$$

and so

$$w \sim \frac{UH}{L} \quad (6.42)$$

It is not *a priori* obvious that this scaling will always hold; in particular, we note that if the Coriolis parameter is in fact constant, the horizontal geostrophic velocity is non-divergent:

$$\left(\frac{\partial u_g}{\partial x} + \frac{\partial v_g}{\partial y} \right) = 0. \quad (6.43)$$

We should then expect that $W = \mathcal{O}(w) < UH/L$. Here we shall suppose that the scales are sufficiently large that this is not the case.

We now define the nondimensional variables by

$$\begin{aligned} (\hat{x}, \hat{y}) &= L^{-1}(x, y), & \hat{z} &= H^{-1}z \\ \hat{\phi} &= \frac{\phi'}{f_0 UL}, & \hat{b} &= \frac{H}{f_0 UL} b' \end{aligned} \quad (6.44)$$

where the hatted, nondimensional, variables are presumed to be order unity. In terms of these nondimensional variables the horizontal momentum and hydrostatic equations become

$$Ro \frac{D\hat{\mathbf{u}}}{D\hat{t}} + \hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\hat{\nabla} \hat{\phi} \quad (6.45)$$

and

$$\frac{\partial \hat{\phi}}{\partial \hat{z}} = \hat{b}. \quad (6.46)$$

The non-dimensional mass conservation equation is simply

$$\hat{\nabla} \cdot \hat{\mathbf{v}} = \left(\frac{\partial \hat{u}}{\partial \hat{x}} + \frac{\partial \hat{v}}{\partial \hat{y}} + \frac{\partial \hat{w}}{\partial \hat{z}} \right) = 0. \quad (6.47)$$

and the nondimensional thermodynamic equation is

$$\frac{f_0 U L}{H} \frac{U}{L} \frac{D\hat{b}}{D\hat{t}} + N^2 \frac{H U}{L} \hat{w} = 0, \quad (6.48)$$

or

$$Ro \frac{D\hat{b}}{D\hat{t}} + \left(\frac{L_d}{L} \right)^2 \hat{w} = 0, \quad (6.49)$$

where $L_d = NH/f_0$.

6.2.2 Simplifying the equations

The goal now is to simplify the above equations, for example by eliminating small terms, and to do so in a consistent manner. One systematic way to do this is by expanding the variables in an asymptotic expansion in powers of the Rossby number, which we have assumed small. Thus, let

$$\begin{aligned} \hat{u} &= u_0 + \epsilon u_1 + \epsilon^2 u_2 + \dots \\ \hat{b} &= b_0 + \epsilon b_1 + \epsilon^2 b_2 \dots \end{aligned} \quad (6.50)$$

and similarly for $\hat{v}$, $\hat{w}$ and $\hat{\phi}$, where $\epsilon = Ro$, the Rossby number. Substituting into the nondimensional equations of motion and equating powers of ϵ gives the lowest order momentum and mass conservation equations:

$$\hat{\mathbf{f}} \times \mathbf{u}_0 = -\hat{\nabla} \phi_0 \quad (6.51a)$$

$$\frac{\partial \phi_0}{\partial \hat{z}} = b_0 \quad (6.51b)$$

$$\hat{\nabla} \cdot \mathbf{v}_0 = 0. \quad (6.51c)$$

If we assume that $L_d/L = \mathcal{O}(1)$, then the thermodynamic equation (6.49) becomes

$$\left(\frac{L_d}{L} \right)^2 w_0 = 0. \quad (6.52)$$

Of course we have neglected any diabatic terms in this equation, which would in general provide a non-zero right-hand side. Nevertheless, in the interior of the ocean diabatic terms are believed to be very small. Evidently then, the equations we have derived, (6.51)–(6.51c), can no longer evolve — all the time derivatives have been scaled away! Thus, although instructive, these equations are not very useful.

If instead we assume that the scale of motion is much larger than the deformation scale then the other terms in the thermodynamic equation will become equally important. Thus, we suppose that $L_d \ll L^2$ or, more formally, that

$L^2 = O(Ro^{-1})L_d^2$, and then all the terms in the thermodynamic equation are retained. A closed set of equations is then given by (6.51) and the thermodynamic equation (6.49).

6.2.3 Dimensional equations

Restoring the dimensions, dropping the asymptotic subscripts, and allowing for the possibility of a source or diffusion term $Q[b]$ in the thermodynamic equation, the *planetary-geostrophic* equations of motion are:

$$\begin{array}{l} \frac{Db'}{Dt} + wN^2 = Q[b] \\ f \times \mathbf{u} = -\nabla\phi \\ \frac{\partial\phi}{\partial z} = b' \\ \nabla \cdot \mathbf{v} = 0 \end{array} \quad (6.53)$$

The advection term in the first equation is three-dimensional. Thus, in many circumstances (for example if the equations are to be solved numerically) there is little practical point in separating the advection of the mean stratification N^2 . The thermodynamic equation is then

$$\frac{Db}{Dt} = Q[b] \quad (6.54)$$

where b now represents the total stratification. The pressure, ϕ , is then the pressure that is in hydrostatic balance with b , so that $\partial\phi/\partial z = b$.

6.2.4 Applicability to the ocean and atmosphere

In the troposphere a typical buoyancy frequency, N , is about $1.2 \times 10^{-2} \text{ s}^{-1}$ (with a corresponding period, $2\pi/N$ of about 9 minutes). If a typical vertical scale, H , is 10 km, then corresponding deformation radius NH/f is a little over 1000 km. The constraint that the scale of motion be much larger than the deformation radius is thus quite hard to satisfy, since one quickly runs out of room on a planet whose equator-to-pole distance is 10000 km. Thus, only the largest planetary waves can satisfy the planetary-geostrophic scaling in the atmosphere and we would then, of course, also need to write the equations in spherical co-ordinates.

In the ocean the deformation radius is much smaller, both because the ocean is not as highly stratified (although the stratification varies considerably in the ocean) and because the vertical scale of motion is smaller. In the subtropics the upper ocean is highly stratified and the buoyancy frequency is about 10^{-2} s^{-1} , similar to its tropospheric value. In the deeper ocean, the stratification is much

smaller, with values of around 10^{-3} s^{-1} , corresponding to a period of about 100 minutes. The vertical scale of motion is nevertheless only about 1 km and using this with $N = 10^{-2} \text{ s}^{-1}$ gives a deformation scale of about 100 km. (This short discussion points, of course, to a potential weakness of scaling theory: the height scale is really part of the *solution*, and should not be imposed *a priori*.) Clearly, there is lots of ‘room’ for the planetary-geostrophic equations to hold in the ocean, and indeed much of the theory of the large-scale structure of the ocean involves the planetary-geostrophic equations.

6.3 THE QUASI-GEOSTROPHIC SYSTEM

The *quasi-geostrophic system* is perhaps the most widely used set of approximate equations of motion for low Rossby number flow in near geostrophic balance. It differs from the planetary geostrophic system primarily in the horizontal scale of the motion that we choose to describe, and now we will require that it be similar to the radius of deformation.

To simplify matters we retain a Cartesian coordinate system and assume *a priori*, that the hydrostatic approximation holds. We will also omit frictional and diffusive terms, but these are easily restored at the end of the day. The equations of motion we begin with are the anelastic or quasi-Boussinesq system discussed in chapter 2, collected here for convenience:

$$\text{The horizontal momentum equation: } \frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_z \phi. \quad (6.55)$$

$$\text{The hydrostatic equation: } \frac{\partial \phi'}{\partial z} = b'. \quad (6.56)$$

$$\text{The mass conservation equation: } \nabla \cdot \mathbf{u} + \frac{1}{\tilde{\rho}} \frac{\partial}{\partial z} (\tilde{\rho} w) = 0. \quad (6.57)$$

$$\text{The thermodynamic equation: } \frac{Db'}{Dt} + N^2 w = 0. \quad (6.58)$$

These equations hold in both an anelastic atmosphere with $b = g\delta\theta/\theta_0$ and $\tilde{\rho} = \tilde{\rho}(z)$, and in a Boussinesq ocean with $b = -g\delta\rho/\rho_0$ and $\tilde{\rho} = \rho_0$, a constant. As we did with the planetary geostrophic equations, we have separated out the mean stratification in the thermodynamic equation (6.58). A very similar derivation to the one that follows may be carried out in pressure or log-pressure co-ordinates, and we return to that later.

6.3.1 Scaling and assumptions

Scaling is intelligent nondimensionalization, the ‘intelligence’ partially resides in knowing ahead of time where one is going, and thereby making sensible nondimensionalizations that leave the magnitude of nondimensional variables order

unity. Our primary constraint, just as with the planetary-geostrophic equations, is that the motion is in near-geostrophic balance so that

$$fv \approx \frac{\partial \phi}{\partial x}, \quad fu \approx -\frac{\partial \phi}{\partial y} \quad (6.59)$$

implying

$$\phi \sim \Phi = fUL = Ro f^2 L^2. \quad (6.60)$$

Hydrostatic balance then implies that

$$b \sim B = \frac{P}{H} = \frac{fUL}{H} \quad (6.61)$$

We now make the following assumptions:

1. The Rossby number is small: $Ro = U/f_0 L \ll 1$.
2. The scale of the motion is not significantly larger than the deformation scale. That is,

$$Bu^{-1} = \left(\frac{L}{L_d} \right)^2 \sim 1. \quad (6.62)$$

where $Bu = (L_d/L)^2$ is the Burger number for stratified flow and $L_d = NH/f_0$ is the deformation radius. More precisely, we shall require that

$$Ro \left(\frac{L}{L_d} \right)^2 = \frac{Ro}{Bu} = \mathcal{O}(Ro). \quad (6.63)$$

3. Variations in the Coriolis parameter are small. That is, $|\beta L| \ll |f_0|$ where L is the length-scale of the motion.
4. Time scales advectively. That is, the scaling timescale is given by $T = L/U$.

These assumptions, and the derivation below, are similar to those involved in the analogous shallow water system of section 5.9. Similarly, we shall nondimensionalize the equations assuming small Rossby number scaling, and then expand the parameters in a series in small Rossby number.

We use the following scales for the variables:

$$(x, y) \sim L, \quad z \sim H, \quad f \sim f_0 \quad (6.64a)$$

$$(u, v) \sim U, \quad w \sim W = UH/L, \quad t \sim T = L/U, \quad (6.64b)$$

$$\phi' \sim \Phi = f_0 UL = Ro f^2 L^2, \quad b' \sim B = fUL/H. \quad (6.64c)$$

Item 2 then tells us how the perturbation stratification compares to the mean. Denoting the mean buoyancy field by $\bar{b}$, where $N^2 = \partial \bar{b} / \partial z$, we have

$$\frac{b'}{\bar{b}} \sim \frac{b'}{N^2 H} \sim \frac{f_0 U L}{N^2 H^2} = Ro \frac{f_0^2 L^2}{N^2 H^2} = Ro \frac{L^2}{L_d^2} = \frac{Ro}{Bu} = \mathcal{O}(Ro). \quad (6.65)$$

This is presumptively small and so the stratification can only change by a small amount. It is to be contrasted with the planetary geostrophic case, for which $L^2 \gg L_d^2$, and the variations in perturbation buoyancy are as large as the mean.

We now let

$$\begin{aligned} (x, y) &= L \times (\hat{x}, \hat{y}), & z &= H \hat{z} \\ \phi &= f_0 U L \hat{\phi}, & b' &= \frac{f_0 U L}{H} \hat{b}. \end{aligned} \quad (6.66)$$

which serves to define the non-dimensional, hatted, variables. Substituting into the equations of motion we easily obtain the non-dimensional the momentum and hydrostatic equations,

$$Ro \frac{D\hat{\mathbf{u}}}{D\hat{t}} + \hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\hat{\nabla}_h \hat{\phi}, \quad (6.67)$$

and

$$\frac{\partial \hat{\phi}}{\partial \hat{z}} = \hat{b}, \quad (6.68)$$

and the non-dimensional mass conservation equation ,

$$\left(\frac{\partial \hat{u}}{\partial \hat{x}} + \frac{\partial \hat{v}}{\partial \hat{y}} + \frac{1}{\hat{\rho}} \frac{\partial \hat{\rho} \hat{w}}{\partial \hat{z}} \right) = 0. \quad (6.69)$$

The thermodynamic equation is the same as (6.49), namely

$$\frac{f_0 U L}{H} \frac{U}{L} \frac{D\hat{b}}{D\hat{t}} + N^2 \frac{H U}{L} w = 0 \quad (6.70)$$

or

$$Ro \frac{D\hat{b}}{D\hat{t}} + \left(\frac{L_d}{L} \right)^2 \hat{w} = 0 \quad (6.71)$$

Now, in Cartesian co-ordinates we may express the Coriolis parameter as

$$\mathbf{f} = f_0 + \beta y \mathbf{k} \quad (6.72)$$

where $f_0 = f_0 \mathbf{k}$. The *variation* of the Coriolis parameter is however assumed small (this is a key difference between the quasi-geostrophic system and the planetary geostrophic system), and in particular we shall assume that βy is approximately the size of the relative vorticity, and so much smaller than f_0 itself.³ Thus,

$$\beta y \sim \frac{U}{L}, \quad \beta \sim \frac{U}{L^2}, \quad (6.73)$$

and so we define an $\mathcal{O}(1)$ non-dimensional beta parameter by

$$\hat{\beta} = \frac{\beta L^2}{U} = \frac{\beta L}{Ro f_0}. \quad (6.74)$$

From this it follows that if $f = f_0 + \beta y$, the corresponding nondimensional version is

$$\hat{f} = \hat{f}_0 + Ro \hat{\beta} \hat{y}. \quad (6.75)$$

where $\hat{f} = f / f_0$ and $\hat{f}_0 = f_0 / f_0 = 1$.

6.3.2 Asymptotics

We now expand the nondimensional dependent variables in an asymptotic series in Rossby number; thus, we write

$$\begin{aligned} \hat{\mathbf{u}} &= \mathbf{u}_0 + Ro \mathbf{u}_1 + \dots \\ \hat{\phi} &= \phi_0 + Ro \phi_1 + \dots \\ \hat{b} &= b_0 + Ro b_1 + \dots \end{aligned} \quad (6.76)$$

Substituting these into the equations of motion, the lowest order momentum equation is simply geostrophic balance,

$$\hat{f}_0 \times \mathbf{u}_0 = -\nabla_z \phi_0 \quad (6.77)$$

with a *constant* value of the Coriolis parameter. From (6.77) we obtain

$$\nabla_z \cdot \mathbf{u}_0 = 0, \quad (6.78)$$

that is, the horizontal flow is, to leading order, non-divergent. Note that this is a consequence of geostrophic balance, and is *not* a mass conservation equation. Using (6.78) in the mass conservation equation, (6.69) implies that

$$\frac{\partial}{\partial z}(\tilde{\rho} w_0) = 0 \quad (6.79)$$

which, like the Taylor Proudman effect, implies that if w_0 is zero somewhere (e.g., at a solid surface) then w_0 is zero everywhere. Thus, a nontrivial result of the scaling is that the lowest order vertical velocity is zero. Another, perhaps more physical, way of saying this is that the naïve scaling estimate $W = UH/L$ is an overestimate for the size of the vertical velocity because even though $\partial w / \partial z \approx \nabla \cdot \mathbf{u}$, $\nabla \cdot \mathbf{u} \ll U/L$ because the geostrophic velocity is nearly divergence free if f is nearly constant. We might even have anticipated this from the outset, and scaled w differently, perhaps using RoU/L as the scaling factor for w , but the choice

of RoU/L is by no means obvious (why not Ro^2U/L or $Ro^{1/2}U/L$ for example?) and it is better to let the asymptotics guide us.

At next order the momentum equation is

$$\frac{D_0 \mathbf{u}_0}{D\hat{t}} + \hat{\beta} \hat{\mathbf{y}} \mathbf{k} \times \mathbf{u}_0 + \hat{\mathbf{f}} \times \mathbf{u}_1 = -\nabla \phi_1, \quad (6.80)$$

where $D_0/Dt = \partial/\partial\hat{t} + (\mathbf{u}_0 \cdot \nabla)$, and the next order mass conservation equation is

$$\nabla_z \cdot (\tilde{\rho} \mathbf{u}_1) + \frac{\partial}{\partial z} (\tilde{\rho} w_1) = 0. \quad (6.81)$$

From (6.71), the lowest order thermodynamic equation is simply

$$\left(\frac{L_d}{L} \right)^2 w_0 = 0 \quad (6.82)$$

provided that, as we have assumed, the scales of motion are not sufficiently large that $Ro(L/L_d)^2 = \mathcal{O}(1)$. This is a key aspect of quasi-geostrophic motion; if we do not make this assumption, we are led down the path to the planetary geostrophic equation. However, we now must go to higher order to obtain an evolution equation for the buoyancy, and this is

$$\frac{D_0 b_0}{D\hat{t}} + w_1 \left(\frac{L_d}{L} \right)^2 = 0. \quad (6.83)$$

The potential vorticity equation

Let us now obtain an evolution equation for lowest order quantities only, and we can do this by eliminating w_1 between the thermodynamic and momentum equations. Cross differentiating the first order momentum equation (6.80) gives the vorticity equation

$$\frac{\partial \zeta_0}{\partial \hat{t}} + (\mathbf{u}_0 \cdot \nabla) \zeta_0 + \nu_0 \hat{\beta} = -\hat{f}_0 \nabla_z \cdot \mathbf{u}_1. \quad (6.84)$$

(In dimensional terms, the divergence on the right-hand side is small, but is multiplied by the large term f_0 , and their product is the same order as the terms on the left-hand side.) Using the mass conservation equation (6.81), (6.84) becomes

$$\frac{D_0}{D\hat{t}} (\zeta_0 + \hat{f}) = \frac{\hat{f}_0}{\rho_s} \frac{\partial}{\partial z} (w_1 \rho_s) \quad (6.85)$$

Combining (6.85) and (6.83) gives

$$\frac{D_0}{D\hat{t}} (\zeta_0 + \hat{f}) = -\frac{\hat{f}_0}{\rho_s} \frac{\partial}{\partial \hat{z}} \left(\frac{D_0}{D\hat{t}} (F \rho_s b_0) \right) \quad (6.86)$$

where $F \equiv (L/L_d)^2$. The right-hand side of this equation simplifies using thermal wind:

$$\frac{\partial}{\partial \hat{z}} \left(\frac{D_0 \hat{b}}{D\hat{t}} \right) = \frac{D_0}{D\hat{t}} \left(\frac{\partial \hat{b}}{\partial \hat{z}} \right) + \frac{\partial \hat{\mathbf{u}}_0}{\partial \hat{z}} \cdot \nabla_z \hat{b} \quad (6.87)$$

But

$$\mathbf{k} \times \frac{\partial \hat{\mathbf{u}}_0}{\partial \hat{z}} = -\frac{1}{\hat{f}_0} \nabla_z b_0 \quad (6.88)$$

and so the advective term $(\partial \mathbf{u}_0 / \partial \hat{z}) \cdot \nabla \hat{b}$ vanishes identically. (Because F and $\tilde{\rho}$ are only functions of z , their presence does not affect this result.) Given this, (6.86) becomes

$$\frac{D_0}{D\hat{t}} \left\{ \zeta_0 + \hat{f} + \frac{\hat{f}_0}{\rho_s} \frac{\partial}{\partial z} (\rho_s F b_0) \right\} = 0 \quad (6.89)$$

or, using the hydrostatic equation,

$$\frac{D_0}{D\hat{t}} \left[\zeta_0 + \hat{f} + \frac{\hat{f}_0}{\rho_s} \frac{\partial}{\partial z} \left(\rho_s F \frac{\partial \phi_0}{\partial z} \right) \right] = 0. \quad (6.90)$$

Since the lowest-order horizontal velocity is divergence-free, we can define a streamfunction ψ such that

$$u_0 = -\frac{\partial \hat{\psi}}{\partial y}, \quad v_0 = \frac{\partial \hat{\psi}}{\partial x} \quad (6.91)$$

where also, using (6.77), $\phi_0 = \hat{f}_0 \hat{\psi}$. The vorticity is then given by $\zeta_0 = \nabla^2 \hat{\psi}$ and (6.90) becomes a single equation in a single unknown, to wit

$$\boxed{\frac{D_0}{D\hat{t}} \left[\hat{\nabla}^2 \hat{\psi} + \hat{\beta} \hat{y} + \frac{\hat{f}_0^2}{\tilde{\rho}} \frac{\partial}{\partial \hat{z}} \left(\tilde{\rho} F \frac{\partial \hat{\psi}}{\partial \hat{z}} \right) \right]} = 0, \quad (6.92)$$

where the material derivative is evaluated using $\mathbf{u}_0 = \mathbf{k} \times \nabla \hat{\psi}$. This is the nondimensional form of the quasi-geostrophic potential vorticity equation, and is one of the most important equations in dynamical meteorology and oceanography.

Implications

Deriving the equation is a remarkable achievement, since we have reduced the Navier Stokes equations, which are six coupled nonlinear partial differential equations in six unknowns (u, v, w, T, p, ρ) to a single (albeit nonlinear) first-order partial differential equation in a single unknown, the potential vorticity.⁴ From this field, all the others — streamfunction, temperature, velocity — may be ‘inverted’, that is deduced via the solution of an elliptic equation.

The dimensional version of the equation may be written,

$$\boxed{\frac{D}{Dt} \left\{ \nabla^2 \psi + f + \frac{f_0^2}{\rho_s} \frac{\partial}{\partial z} \left(\frac{\rho_s}{N^2} \frac{\partial \psi}{\partial z} \right) \right\} = 0}. \quad (6.93)$$

The variables are related to ψ as follows:

- (i) Vorticity: $\zeta = \nabla^2 \psi$
- (ii) Perturbation pressure: $\phi = f_0 \psi$
- (iii) Velocity: $\mathbf{u} = \mathbf{k} \times \nabla \psi = -\nabla \times (\mathbf{k} \psi) = \nabla^\perp \psi$.
- (iv) Buoyancy: $b = f \partial \psi / \partial z$, where $b = g \delta \theta / \theta_0$ where θ is potential temperature, or in a Boussinesq system $b = -g \delta \rho / \rho_0$.

The equation can be written as

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + f + \frac{f_0^2}{\rho_s} \frac{\partial}{\partial z} \left(\frac{\rho_s}{N^2} \frac{\partial \psi}{\partial z} \right). \quad (6.94)$$

The quantity q is known as the *quasi-geostrophic potential vorticity*. It is analogous to the exact (Ertel) potential vorticity, although it is not in fact an approximation to that, and it is conserved when advected by the *horizontal* geostrophic flow.

Note that length-scale $L_d = NH / f_0$, emerges naturally from the QG dynamics. It is the scale at which buoyancy and relative vorticity effects contribute equally to the potential vorticity, and is called the *deformation radius*; it is analogous to the quantity $\sqrt{g'H} / f_0$ arising in shallow water theory. To see this loosely, note that

$$\frac{NH}{f_0} = \left(\frac{g}{\theta_0} \frac{\partial \theta_0}{\partial z} \right)^{1/2} \frac{H}{f_0} \sim \frac{\sqrt{g'H}}{f_0} \quad (6.95)$$

where $g' = g \Delta \theta / \theta_0$ is the reduced gravity. In the upper ocean, with $N \approx 10^{-2} \text{ s}^{-1}$, $H \approx 10^3 \text{ m}$, and $f_0 \approx 10^{-4} \text{ s}^{-1}$, then $L_d \approx 100 \text{ km}$. At high latitudes the ocean is much less stratified and $N^2 \sim 10^{-3} \text{ s}^{-1}$, and f is somewhat larger, and the deformation radius is correspondingly smaller. In the atmosphere, with $N \approx 10^{-2} \text{ s}^{-1}$, $H \approx 10^4 \text{ m}$, then $L_d \approx 1000 \text{ km}$. It is this order of magnitude difference in the deformation scales that accounts for a great deal of the quantitative difference in the dynamics of the ocean and atmosphere.

6.3.3 Boundary Conditions

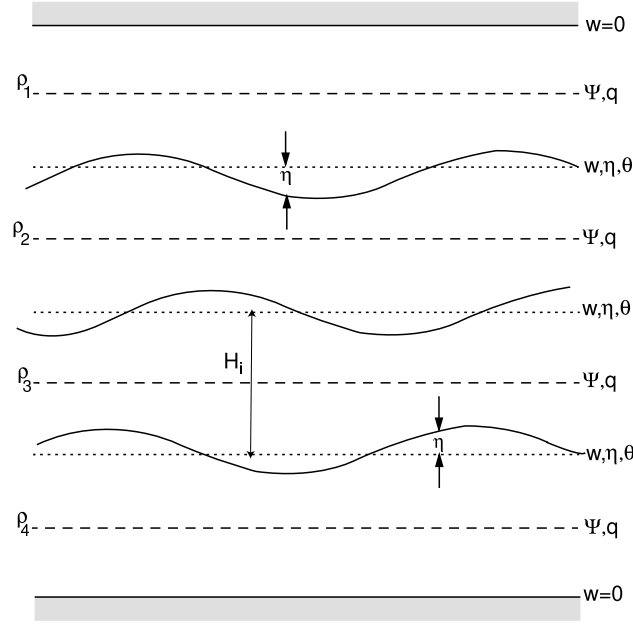


Figure 6.2 The multi-level quasi-geostrophic system.

6.4 TWO-LEVEL QUASI-GEOSTROPHIC SYSTEM

The quasi-geostrophic system derived above has continuous variation in the vertical (and horizontal). The *two-level* model is a simplification of this that allows only two-degrees of freedom in the vertical. We can regard the system either as a relatively crude finite-difference approximation to the continuous system, or as the quasi-geostrophic model appropriate for the physical model of two layers of fluid of slightly differing density, one on top of the other. We pursue the first option here, and we work with the Boussinesq approximation for simplicity.

The vertical structure of the model is a two-layer version of the one sketched in Fig. ???. We apply the quasi-geostrophic vorticity equation to levels 1 and 2, and the thermodynamic equation in the center. Then,

$$\frac{\partial \zeta_1}{\partial t} + \mathbf{u}_1 \cdot \nabla (\zeta_1 + f) = -f_0 \frac{w}{H_1} \quad (6.96)$$

and

$$\frac{\partial \zeta_2}{\partial t} + \mathbf{u}_2 \cdot \nabla (\zeta_2 + f) = f_0 \frac{w}{H_2} \quad (6.97)$$

The terms on the right-hand side are finite-difference approximations to the continuous expression $f \partial w / \partial z$, where $w = 0$ at the top and the bottom, as in the

figure.

The thermodynamic equation is

$$\frac{\partial}{\partial t} \frac{\partial \psi}{\partial z} + \mathbf{u} \cdot \nabla \frac{\partial \psi}{\partial z} + w \frac{N^2}{f_0} = 0, \quad (6.98)$$

A consistent finite-difference approximation of this is

$$\frac{\partial}{\partial t} \left(\frac{\psi_1 - \psi_2}{H/2} \right) + \bar{\mathbf{u}} \cdot \nabla \left(\frac{\psi_1 - \psi_2}{H/2} \right) + w \frac{N^2}{f_0} = 0 \quad (6.99)$$

where $\bar{\mathbf{u}} = (\mathbf{u}_1 + \mathbf{u}_2)/2$ and w is evaluated between the levels. It is then a straightforward matter to combine the vorticity equation at each level with the thermodynamic equation to give

$$\frac{Dq_1}{Dt} = 0, \quad q_1 = \left[\zeta_1 + f + \frac{2f_0^2}{N^2 H_1 H} (\psi_2 - \psi_1) \right], \quad (6.100a)$$

and

$$\frac{Dq_2}{Dt} = 0, \quad q_2 = \left[\zeta_2 + f + \frac{2f_0^2}{N^2 H_2 H} (\psi_1 - \psi_2) \right]. \quad (6.100b)$$

[To obtain this, we use the fact that the velocity associated with a streamfunction cannot itself advect that streamfunction; that is, $\mathbf{u}_i \cdot \nabla \psi_i = 0$ and so $\bar{\mathbf{u}} \cdot \nabla (\psi_1 - \psi_2) = \mathbf{u}_1 \cdot \nabla (\psi_1 - \psi_2) = \mathbf{u}_2 \cdot \nabla (\psi_1 - \psi_2)$.] The expressions in square brackets are the two-layer equivalents of the quasi-geostrophic potential vorticity. In atmospheric problems it is common to choose $H_1 = H_2$, whereas in the problems we might choose to have a thinner upper layer, representing the flow above the main thermocline. Note that the boundary conditions of $w = 0$ at the top and bottom are already taken care of in these expressions.

6.4.1 Connection to the layered system

Equations (??) have an obvious similarity with the *two-layer* equations (5.212) in the rigid lid case. Noting that $N^2 = \partial \hat{b} / \partial z$ and that $b = -g\delta\rho/\rho_0$ it is natural to let

$$N^2 = -\frac{g}{\rho_0} \frac{\rho_1 - \rho_2}{H/2} = \frac{g'}{H/2}. \quad (6.101)$$

With this identification

$$q_1 = \zeta_1 + f + \frac{f_0^2}{g' H_1} (\psi_2 - \psi_1), \quad (6.102)$$

and similarly for q_2 , the expressions being *identical* with (5.212) in the rigid lid case. This is an example of the similarity between layered models and the finite difference representation of the continuous representation, and indeed we can show that a multi-layered system, with n layers, is exactly equivalent to a finite-difference representation with n levels.⁵

6.5 ENERGETICS OF QUASI-GEOSTROPHY

If the quasi-geostrophic set of equations is to represent a real fluid system in a physically meaningful way, then it should have a consistent set of energetics. In particular, total energy should be conserved, and there should be analogs of kinetic and potential energy and conversion between the two. We now show that this is indeed the case.

Let us write the governing equations as a potential vorticity equation in the interior,

$$\frac{D}{Dt} \left[\nabla^2 \psi + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \psi}{\partial z} \right) \right] + \beta \frac{\partial \psi}{\partial x} = 0, \quad 0 < z < 1 \quad (6.103)$$

and buoyancy advection at the boundary,

$$\frac{D}{Dt} \left(\frac{\partial \psi}{\partial z} \right) = 0, \quad z = 0, 1. \quad (6.104)$$

For lateral boundary conditions we may assume that $\psi = \text{constant}$, or impose periodic conditions. If we multiply (6.103) by $-\psi$ and integrate over the domain, using the boundary conditions, we easily find

$$\frac{d\hat{E}}{dt} = 0. \quad (6.105a)$$

where

$$\hat{E} = \frac{1}{2} \int_V \left[(\nabla \psi)^2 + \frac{f_0^2}{N^2} \left(\frac{\partial \psi}{\partial z} \right)^2 \right] dV = 0 \quad (6.105b)$$

The term involving β makes no direct contribution to the energy budget. Eq. (6.105b) is the fundamental energy equation for quasi-geostrophic motion, and it states that in the absence of viscous or diabatic terms the total energy E is conserved. The two terms in (6.105b) can be identified as the kinetic and available potential energy of the flow, where

$$\begin{aligned} \text{KE} &= \frac{1}{2} \int_V (\nabla \psi)^2 dV \\ \text{APE} &= \frac{1}{2} \int_V \frac{f_0^2}{N^2} \left(\frac{\partial \psi}{\partial z} \right)^2 dV \end{aligned} \quad (6.106)$$

The available potential energy may also be written as

$$\text{APE} = \frac{1}{2} \int_V \frac{H}{L_d^2} \left(\frac{\partial \psi}{\partial z} \right)^2 dV \quad (6.107)$$

where L_d is the deformation radius NH/f_0 and we may choose H such that $z \sim H$. Thus, the ratio of the kinetic energy to the potential energy is, roughly,

$$\frac{\text{KE}}{\text{APE}} \sim \frac{L_d^2}{L^2} \quad (6.108)$$

where L is the scale of the motion. That is, for scales much larger than the deformation radius the potential energy dominates the kinetic energy.

6.5.1 Conversion between APE and KE

Let us return to the vorticity and thermodynamic equations,

$$\frac{D\zeta}{Dt} = f \frac{\partial w}{\partial z} \quad (6.109)$$

where $\zeta = \nabla^2 \psi$, and

$$\frac{Db'}{Dt} + N^2 w = 0 \quad (6.110)$$

where $b' = f_0 \partial \psi / \partial z$. From (6.109) we form a kinetic energy equation namely

$$\frac{1}{2} \frac{d}{dt} \int_V (\nabla \psi)^2 dV = - \int_V f_0 \frac{\partial w}{\partial z} \psi dV = \int_V f_0 w \frac{\partial \psi}{\partial z} dV. \quad (6.111)$$

From (6.110) we form a potential energy equation, namely

$$\frac{d}{dt} \frac{1}{2} \int_V \frac{f_0^2}{N^2} \left(\frac{\partial \psi}{\partial z} \right)^2 dV = - \int_V f_0 w \frac{\partial \psi}{\partial z} dV. \quad (6.112)$$

Thus, the *conversion* from APE to KE is represented by

$$\frac{d}{dt} \text{KE} = - \frac{d}{dt} \text{APE} = \int_V f_0 w \frac{\partial \psi}{\partial z} dV. \quad (6.113)$$

Because the buoyancy is proportional to $\partial \psi / \partial z$, when warm fluid rises there is a correlation between w and $\partial \psi / \partial z$ and available potential energy is converted to kinetic energy. Whether such a phenomenon occurs depends of course on the dynamics of the flow; however, such a conversion is in fact a common feature of geophysical flows, and in particular of baroclinic instability, as we see in chapter 7.

6.5.2 Energetics of two-layer flows

Two-layer or two-level flows are an important special case. For layers of equal thickness let us write the evolution equations as

$$\frac{D}{Dt} \left(\nabla^2 \psi_1 - \frac{1}{2} k_d^2 (\psi_1 - \psi_2) \right) + \beta \frac{\partial \psi_1}{\partial x} = 0 \quad (6.114)$$

$$\frac{D}{Dt} \left(\nabla^2 \psi_2 + \frac{1}{2} k_d^2 (\psi_1 - \psi_2) \right) + \beta \frac{\partial \psi_2}{\partial x} = 0 \quad (6.115)$$

where $k_d^2/2 = (2f_0/NH)^2$. On multiplying these two equations by $-\psi_1$ and $-\psi_2$ respectively and integrating over the horizontal domain, the advective term in the material derivatives and the beta term all vanish, and we obtain

$$\int_A \left[\frac{1}{2}(\nabla\dot{\psi}_1)^2 + \frac{1}{2}k_d^2\psi_1(\dot{\psi}_1 - \dot{\psi}_2) \right] dA, \quad (6.116)$$

and

$$\int_A \left[\frac{1}{2}(\nabla\dot{\psi}_2)^2 - \frac{1}{2}k_d^2\psi_2(\dot{\psi}_1 - \dot{\psi}_2) \right] dA. \quad (6.117)$$

where the dot indicates a partial time derivative. Adding these gives

$$\frac{d}{dt} \int_A \left[\frac{1}{2}(\nabla\psi_1)^2 + \frac{1}{2}(\nabla\psi_2)^2 + k_d^2\hat{\psi}^2 \right] dA = 0. \quad (6.118)$$

This is the energy conservation statement for the two layer model. The first two terms represent the kinetic energy and the last term the available potential energy.

Energy in the baroclinic and barotropic modes

A useful partitioning of the energy is between the energy in the barotropic and baroclinic modes. The barotropic streamfunction, $\bar{\psi}$ is the vertically averaged streamfunction and the baroclinic mode is the difference between the streamfunctions in the two layers. That is, for equal layer thicknesses,

$$\bar{\psi} \equiv \frac{1}{2}(\psi_1 + \psi_2), \quad \hat{\psi} \equiv \frac{1}{2}(\psi_1 - \psi_2) \quad (6.119)$$

The two streamfunctions are in fact (rather trivially) eigenfunctions of the vertical structure (d/dz) operator. For the continuous equations the analogous eigenfunctions are given by

$$\frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \phi}{\partial z} \right) = \lambda^2 \phi \quad (6.120)$$

The eigenfunctions and eigenvalues of this operator define the vertical modes of the continuous system.

Substituting (6.119) into (6.118) reveals that

$$\frac{d}{dt} \int_A \left[(\nabla\bar{\psi})^2 + (\nabla\hat{\psi})^2 + k_d^2\hat{\psi}^2 \right] d\mathbf{x} = 0 \quad (6.121)$$

The energy density in the barotropic mode is thus just $(\nabla\bar{\psi})^2$, and that in the baroclinic mode is $(\nabla\hat{\psi})^2 + k_d^2\hat{\psi}^2$. This partitioning will prove particularly useful when we consider baroclinic turbulence in chapter ??.

Enstrophy conservation

[]

6.6 * QUASI-GEOSTROPHY, REVISITED

Too much of a good thing is wonderful.

Mae West (1892–1990).

Rather than deriving the quasi-geostrophic approximation from the momentum equations, it is revealing (and easier!) to begin with potential vorticity, and make the corresponding approximations.⁶ We will do this in the Boussinesq approximation, although the extension to the compressible case is straightforward, and also assume hydrostasy. As before, the three main assumptions we will have to make are, written informally:

- (i) Small Rossby number;
- (ii) Basic state stratification that does not vary in the horizontal;
- (iii) Small variations in the Coriolis parameter.

We also assume that time scales advectively.

The basic idea is that we begin with the potential vorticity equation and make approximations both to the potential vorticity and to the advecting velocity. And because earlier we proceeded asymptotically, here we shall be more informal. we begin with the expression for potential vorticity in the Boussinesq hydrostatic equations [c.f., (4.115)]

$$Q = [(v_x - u_y)b_z - v_z\theta_x + u_z\theta_y + f\theta_z] \quad (6.122)$$

Without approximation, we write the stratification as:

$$b = \hat{b}(z) + b'(x, y, z, t), \quad (6.123)$$

and the potential vorticity becomes The potential vorticity is

$$\begin{aligned} Q &= (f_0 + \beta y + \zeta)(\hat{b}_z + b'_z) - (v_z b'_x - u_z b'_y) \\ &= [f_0 \hat{N}^2] + [(\beta y + \zeta)N^2 + f_0 b'_z] + [(\beta y + \zeta)b'_z - (v_z b'_x - u_z b'_y)]. \end{aligned} \quad (6.124)$$

where $N^2(z) = \hat{b}_z$. From the assumptions above, we anticipate that the terms in square brackets on the second line are in decreasing order of size. Our goal eliminate the smallest terms and to express the remaining in terms of a single variable, the streamfunction ψ .

We first invoke geostrophic and hydrostatic balance [assumption (i)]. Because $|f_0| \gg |\beta y|$ [assumption (iii)], these involve only the constant value of the Coriolis parameter and are

$$f_0 \times \mathbf{u} = -\nabla\phi \quad \text{and} \quad f_0 \times \frac{\partial \mathbf{u}}{\partial z} = -\nabla b'. \quad (6.125)$$

We thus define a streamfunction ψ by

$$\psi = \frac{\phi}{f_0} \quad (6.126)$$

and then

$$\mathbf{u}_g = \mathbf{k} \times \nabla \psi, \quad \text{and} \quad b' = f_0 \frac{\partial \psi}{\partial z}. \quad (6.127)$$

This implies that the scaling for b' is $b' \sim f_0 UL/H$, and therefore that the ratio of the size of the perturbation buoyancy to the basic state buoyancy is given by

$$\left| \frac{b'_z}{\hat{b}_z} \right| = \frac{|b'_z|}{N^2} \sim \frac{fUL}{H^2 N^2} = \frac{L^2}{L_d^2} Ro = \frac{Ro}{Bu}. \quad (6.128)$$

We will assume that the stratification varies only slightly in the horizontal or that, equivalently, $(L/L_d)^2 Ro$ is small, corresponding to assumption (ii). Thus, horizontal scales are large enough for the Rossby number to be small, but not so large as to violate this assumption.

Using (6.128), and $\beta y \sim \sim Ro f_0$ we see that the terms in square brackets in (6.124) scale like

$$\begin{aligned} [f_0 N^2] &\sim f_0 N^2 \\ [(\beta y + \zeta) N^2] &\sim Ro f_0 N^2 \\ [f_0 b'_z] &\sim Ro \frac{L^2}{L_d^2} f_0 N^2 \\ [(\beta y + \zeta) b'_z - (v_z b'_x - u_z b'_y)] &\sim Ro^2 \frac{L^2}{L_d^2} f_0 N^2 \end{aligned} \quad (6.129)$$

Using (6.124) and (6.129) we may approximate the potential vorticity by

$$Q \approx \hat{Q} + Q_h \quad (6.130)$$

where $\hat{Q} = f_0 N^2$ and

$$\begin{aligned} Q_h &= (\beta y + \zeta) N^2 + f_0 b'_z \\ &= (\beta y + \nabla^2 \psi) N^2 + f_0^2 \frac{\partial \psi}{\partial z}. \end{aligned} \quad (6.131)$$

We emphasize that if the scales of motion are too large compared to the deformation radius, then the approximation is not valid. Specifically, if $L^2/L_d^2 \sim Ro^{-1}$ then the terms in (6.129) that are putatively of $\mathcal{O}(Ro^2)$ will be as large as the terms of $\mathcal{O}(Ro)$. However, we may certainly have that $Ro(L^2/L_d^2) \ll 1$ — the stratification is then sufficiently strong that the equation of motion is the two

dimensional vorticity equation at each level. A sometimes-useful way of writing the condition that $L^2 \lesssim L_d^2$ is in terms of the Prandtl ratio:

$$L^2 \lesssim L_d^2 \quad \longrightarrow \quad \frac{f_0^2}{N^2} \lesssim \frac{H^2}{L^2} \quad (6.132)$$

Interestingly, then, the rotation cannot be too strong unless the stratification correspondingly increases. Note too that if $|\beta y| \sim f$, then the dominant dynamically active term is $\beta y N^2$ and the leading order term in the evolution equation would become $\beta v = 0$. Although this graphically illustrates the constraining effect that the β -effect has on meridional motion, it is not a very dynamically rich equation.

Potential vorticity advection

The potential vorticity equation is

$$\frac{DQ_h}{Dt} + w \frac{\partial \hat{q}}{\partial z} = 0 \quad (6.133)$$

where the advective derivative is (at this stage) fully three-dimensional. However, since f is nearly constant, the horizontal velocity is nearly non-divergent and

$$-\frac{\partial w}{\partial z} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \ll \frac{U}{L} \quad (6.134)$$

Thus,

$$w \ll \frac{UH}{L}. \quad (6.135)$$

and the vertical advection in (6.133) is important only in advecting the basic state potential vorticity $\hat{Q}$. Thus, after dividing by N^2 , (6.133) becomes

$$\frac{\partial q}{\partial t} + \mathbf{u}_g \cdot \nabla q + w \frac{\partial \hat{q}}{\partial z} = 0 \quad (6.136)$$

where

$$q = (\beta y + \zeta) + \frac{f_0}{N^2} b'_z. \quad (6.137)$$

is the approximation to (perturbation) potential vorticity in the quasi-geostrophic limit.

We can simplify this further with the help of the thermodynamic equation,

$$\frac{\partial b'}{\partial t} + \mathbf{u}_g \cdot \nabla b + w \frac{\partial \hat{b}}{\partial z} = 0 \quad (6.138)$$

Eliminating w between (6.136) and (6.138) (problem 6.??) gives the quasi-geostrophic potential vorticity equation,

$$\frac{\partial q_g}{\partial t} + \mathbf{u}_g \cdot \nabla q_g = 0, \quad (6.139)$$

where

$$q_g = (\beta y + \zeta) + \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} b' \right), \quad (6.140)$$

or, in terms of streamfunction,

$$q_g = (\beta y + \nabla^2 \psi) + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \psi}{\partial z} \right). \quad (6.141)$$

These are identical to the equations previously derived.

Summarizing some important points:

- The so-called quasi-geostrophic potential vorticity given by (6.140) or (6.141) is *not* an approximation to the Ertel potential vorticity. For that reason the QG potential vorticity is sometimes called the pseudo potential vorticity. The actual approximation to potential vorticity is given by (6.137).
- The expression (6.140) (or (6.141)) is more useful than (6.137) because it is conserved when advected by the *horizontal* geostrophic velocity. That is, the advection of the quasi-geostrophic potential vorticity by the horizontal flow is analogous to the advection of the Ertel potential vorticity along isentropic surfaces.
- The quasi-geostrophic potential vorticity, q_g , may be written as the three-dimensional divergence of a vector $\mathbf{J}$ where

$$\mathbf{J} = \nabla \psi + \mathbf{j} \beta y^2 / 2 + \mathbf{k} (f_0^2 / N^2) \partial \psi / \partial z. \quad (6.142)$$

- The streamfunction may be diagnosed from the potential vorticity by the solution of the three-dimensional elliptic equation:

$$\left(\nabla^2 + \frac{f_0^2}{N^2} \frac{\partial}{\partial z} \right) \psi = q_g - \beta y. \quad (6.143)$$

- From the streamfunction, the velocity and buoyancy are easily diagnosed as elements of the vector:

$$(u, v, b) = (-\psi_y, \psi_x, f_0 \psi_z). \quad (6.144)$$

6.7 QUASI-GEOSTROPHY IN ISENTROPIC CO-ORDINATES

[To be included] In which we show that quasi-geostrophy arises simply and naturally in isentropic co-ordinates, and that horizontal quasi-geostrophic motion is the analog of motion along isentropic surfaces.

6.8 ROSSBY WAVES IN STRATIFIED QUASI-GEOSTROPHIC FLOW

6.8.1 Preliminaries

Let us now consider the dynamics of linear waves in stratified quasi-geostrophic flow, with a resting basic state. Our emphasis will be on the basic dynamics; in chapter ?? we explore the role of Rossby waves in a more realistic setting.

The quasi-geostrophic equations, linearized about a state of rest, are the potential vorticity equation,

$$\frac{\partial}{\partial t} \left[\nabla^2 \psi + \frac{1}{\tilde{\rho}(z)} \frac{\partial}{\partial z} \left(\tilde{\rho}(z) F(z) \frac{\partial \psi}{\partial z} \right) \right] + \beta \frac{\partial \psi}{\partial x} = 0 \quad (6.145)$$

which describes flow in the interior, and the thermodynamic equation with $w = 0$,

$$\frac{\partial}{\partial t} \left(\frac{\partial \psi}{\partial z} \right) = 0. \quad (6.146)$$

which describes flow at a flat, rigid surface such as the ground (if flat), the surface and, with somewhat less justification, at the tropopause where the higher static stability of the atmosphere inhibits vertical motion. An alternative upper boundary condition is to require that perturbations decay to zero as $z \rightarrow \infty$. If the ground is not flat or if friction provides a vertical velocity by way of an Ekman layer the boundary condition must be correspondingly modified, but we will stay with the simplest case here and apply (6.146) at $z = 0$ and $z = H$. The variable $\tilde{\rho}$ is the density profile of the basic state, and $F(z) = f_0^2 / N^2$, the square of the inverse Prandtl ratio, N / f_0 . In the Boussinesq approximation $\tilde{\rho} = \rho_0$, a constant.

6.8.2 Wave motion

For simplicity we will consider motion on a horizontally doubly-periodic Cartesian β -plane. We seek solutions of the form

$$\psi = \text{Re } \Psi(z) e^{i(kx + ly - \omega t)} \quad (6.147)$$

where $\Psi(z)$ will determine the vertical structure of the waves. The case of a zonal channel is an easy step from this — assume a horizontal variation proportional of the form $\sin ly$. The case of a sphere is more complicated but introduces no truly new physical phenomena.

Substituting (6.147) into (6.145) gives

$$\omega \left[-K^2 \Psi(z) + \frac{1}{\tilde{\rho}} \frac{\partial}{\partial z} \left(\tilde{\rho} F(z) \frac{\partial \Psi}{\partial z} \right) \right] - \beta k \Psi(z) = 0. \quad (6.148)$$

where $K^2 = k^2 + l^2$. Now, if Ψ satisfies

$$\frac{1}{\tilde{\rho}} \frac{\partial}{\partial z} \left(\tilde{\rho} F(z) \frac{\partial \Psi}{\partial z} \right) = -\Gamma \Psi \quad (6.149)$$

then the equation of motion becomes

$$-\omega [K^2 + \Gamma] \Psi - \beta k \Psi = 0, \quad (6.150)$$

and the dispersion relation follows, namely

$$\omega = -\frac{\beta k}{K^2 + \Gamma}. \quad (6.151)$$

Equation (6.149), which is to be solved with boundary conditions at the vertical limits of the domain, constitutes an eigenvalue problem for the vertical structure. The resulting eigenvalues, Γ are proportional to the inverse of the squares of the deformation radii for the problem and the eigenfunctions are the vertical structure functions.

A simple example

Consider the case in which $F(z)$ and $\tilde{\rho}$ are constant, and in which the domain is confined between two rigid surfaces at $z = 0$ and $z = H$. Then the eigenvalue problem for the vertical structure is

$$F \frac{\partial^2 \Psi}{\partial z^2} = -\Gamma \Psi \quad (6.152a)$$

with boundary conditions of

$$\frac{\partial \Psi}{\partial z} = 0, \quad z = 0, z = 1. \quad (6.152b)$$

There is a sequence of solutions to this, namely

$$\Psi_n(z) = \cos(n\pi z / H), \quad n = 1, 2, \dots \quad (6.153)$$

with corresponding eigenvalues

$$\Gamma_n = n^2 \frac{F\pi^2}{H^2} = (n\pi)^2 \left(\frac{f_0}{NH} \right)^2, \quad n = 1, 2, \dots \quad (6.154)$$

This last expression may be used to define the deformation radii for this problem, the successive radii being given by

$$\lambda_n \equiv \frac{NH}{n[\pi]f_0} \quad (6.155)$$

The first deformation radius is the same as the expression obtained by dimensional analysis NH/f , except for a factor of π , which we enclose in square brackets. Definitions of the deformation radii both with and without the factor of π

are common in the literature, and neither is obviously more correct. We can only urge precision in the definition of the deformation radius when it is used in a particular context. In this book we shall, unless noted, use a definition of deformation radius that does not include the factor of π , so that the first deformation radius in a problem with uniform stratification is given by NH/f , equal to $\pi/\sqrt{F_1}$.

In addition to these baroclinic modes, it is clear that the case with $n = 0$, that is with $\Psi = 1$, is also a solution of (6.152) for any $F(z)$. Using (6.151) and (6.154) the dispersion relation becomes

$$\omega = -\frac{\beta k}{K^2 + (n\pi)^2(f_0/NH)^2}, \quad n = 0, 1, 2, \dots \quad (6.156)$$

and, of course, the horizontal wavenumbers k and l are also quantized in a finite domain. The dynamics of the barotropic mode are independent of height and independent of the stratification of the basic state, and so these Rossby waves are *identical* with the Rossby waves in a homogeneous fluid contained between two flat rigid surfaces. The structure of the baroclinic modes, which in general depends on the structure of the stratification, becomes increasingly complex as the vertical wavenumber n increases. This increasing complexity naturally leads to a certain delicacy, making it rare that they are found in nature.

Analogy with a layered model

6.8.3 More general stratification

In the more general case, for example with profiles of $N(z)$ taken from observations in the atmosphere or ocean, it is not possible to analytically calculate the vertical normal modes, but they may be calculated numerically. The results of one such calculation are given in Fig. 6.3. The case with uniform stratification reproduces cosine modes, whereas in the more realistic case the modes tend to have highest amplitude in the upper ocean, where the stratification is strongest — a result is typical of oceanic profiles.⁷

The eigenproblem for a realistic atmospheric profile is more complex because of the lack of a rigid lid at the top of the atmosphere.⁸ Nevertheless, the problem succumbs easily enough to a numerical approach, and a Matlab code and a sample problem (6.2) is provided for readers who wish to explore this further.

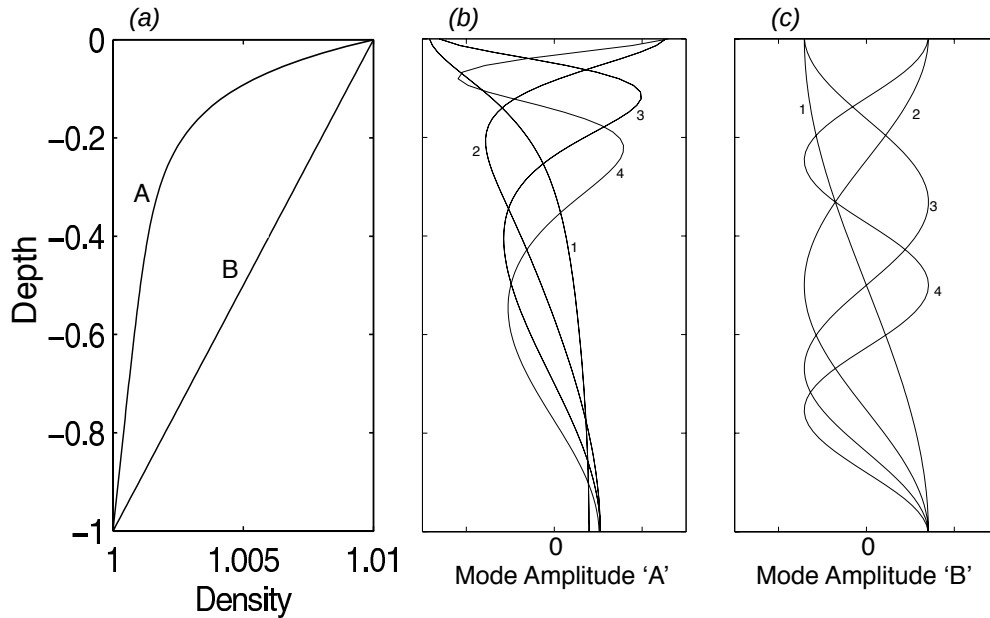


Figure 6.3 (a): two density profiles $[1 + (\rho_0 - \rho)/\rho_0]$, a fairly realistic oceanic case with enhanced stratification in the upper ocean (profile A), and uniform stratification (profile B). (b) and (c): the corresponding four baroclinic modes [eigenfunctions of $(\partial/\partial z)[N^2(z)\partial\phi/\partial z] = \lambda^2\phi$ using N^2 calculated from the profiles and with boundary conditions of $\partial\phi/\partial z = 0$ at top and bottom]. With profile B the eigenmodes are cosines, whereas in the where as in profile A they have largest amplitude in the upper ocean. The number of zero crossings is equal to the mode number.

6.9 AVAILABLE POTENTIAL ENERGY

[not yet available]

Notes

- 1 In today's world, this timestepping argument is difficult to justify. One reason is that the filtering of a wave-type is often associated with the elimination of a time-derivative in the equations of motion, which often then demands the solution of an elliptic equation at each timestep in order to solve for (i.e., diagnose) a field that was previously explicitly timestepped. It is computationally demanding to solve elliptic equations and, especially in the era of parallel computations, it may be just as efficient to take a larger number of smaller timesteps with no elliptic equation.
- 2 After G. I. Taylor and I. Proudman. Historically, the Taylor-Proudman effect has sometimes been called the Taylor-Proudman 'theorem', but it is more usefully thought of as a physical effect and we call it such.

- 3 There is a difference between the *dynamical* demands of the quasi-geostrophic system in requiring β to be small, and the *geometric* demands of the Cartesian geometry. On Earth, the use of a Cartesian system is accurate only for small variations of latitude, and so of the Coriolis parameter. But we may without dynamical inconsistency imagine a Cartesian system in which $\beta y \sim f$, and indeed this is common in idealized, planetary geostrophic, models of the large-scale ocean circulation.
- 4 Atmospheric and oceanic sciences are sometimes thought of as not being 'beautiful' in the same way as are some branches of theoretical physics. Yet surely quasi-geostrophic theory, and the quasi-geostrophic potential vorticity equation, are quite beautiful, both for their austerity of description and richness of behaviour.
- 5 The proof of equivalence in the n -layer case is given in Pedlosky (1987), section 6.18
- 6 ? clarified the relationship between quasi-geostrophic potential vorticity and Ertel potential vorticity, and derived quasi-geostrophic theory in isentropic coordinates. See also ?.
- 7 For example Kundu *et al.* (1975).
- 8 Chapman & Lindzen (1970) have argued that for a realistic atmospheric stratification profiles no baroclinic modes of a resting atmosphere exist. Of course, in reality the atmosphere is not at rest, so in that sense the specific argument is moot. For our purposes, the importance of such a Rossby wave analysis is not that any particular form of waves do or do not exist, but that it provides a foundation for more complete analyses with non-resting basic states, leading into the theory of baroclinic instability discussed in chapter 7, as well as an understanding of the propagation of Rossby waves.

Problems

- 6.1 Beginning with the vorticity and thermodynamic equations for a two layer model, obtain an expression for the conversion between available potential energy and kinetic energy in the two-layer model. Show that these expressions are consistent with the conservation of total energy as expressed by (6.118). Show also that the expression might be considered to be a simple finite-difference approximation to (6.113).
- 6.2 $\diamond$ The vertical normal modes are the eigenfunctions of

$$\frac{1}{\tilde{\rho}} \frac{\partial}{\partial z} \left(\frac{\tilde{\rho}}{N^2} \frac{\partial \Psi}{\partial z} \right) = -\Gamma \Psi \quad (\text{P6.1})$$

along with boundary conditions on $\Psi(z)$. Numerically obtain the vertical normal modes for some or all of the following profiles, or others of your choice.

- (i) $\tilde{\rho} = 1$, $N^2(z) = 1$, $\Psi_z = 0$ at $z = 0, 1$. (This is profile 'b' of Fig. 6.3. An analytic solution is possible.)
- (ii) $\tilde{\rho} = 1$ and an $N^2(z)$ profile corresponding to a density profile similar to profile 'a' of Fig. 6.3 (e.g., an exponential), and $\Psi_z = 0$ at $z = 0, 1$.

- (iii) An isothermal atmosphere, with $\Psi_z = 0$ at $z = 0, 1$. (Similar to (i), except that $\tilde{\rho}$ varies with height.)
- (iv) An isothermal atmosphere, now assuming that $\psi \rightarrow 0$ as $z \rightarrow \infty$.
- (v) An fluid with $N^2 = 1$ for $0 < z < 0.5$ and $N^2 = 4$ for $0.5 < z < 1$, with continuous b , and with $\Psi_z = 0$ at $z = 0, 1$.

In the atmospheric cases it is easiest to do the problem first with $\tilde{\rho} = 1$ (the Boussinesq case) and then extend the problem (and the code) to the compressible case. Then remove the upper boundary to larger and larger values of z .

- 6.3 $\diamond$ The non-Doppler effect arises relatively straightforwardly in the layered formulation of the quasi-geostrophic equations. Show that it also arises when the problem is formulated using geometric height as the vertical co-ordinates, using the modified quasi-geostrophic set of White (1977). In particular, obtain the dispersion relation for stratified quasi-geostrophic flow with a resting basic state. Then obtain the dispersion relation for the equations linearized about a uniformly translating state, paying attention to the lower boundary condition, and note the conditions under which the waves are stationary. Discuss.
- 6.4 Using log-pressure as a vertical co-ordinate, show that the linearized lower boundary condition equivalent to (P6.2) is

$$\frac{\partial}{\partial t} \left(\frac{\partial \psi'}{\partial z} - \frac{N^2}{g} \psi' \right) + \bar{u} \frac{\partial}{\partial x} \frac{\partial \psi'}{\partial z} - v' \frac{\partial \bar{u}}{\partial z} = - \frac{N^2}{f_0} \left(\bar{u} \frac{\partial h_b}{\partial x} + \alpha \nabla^2 \psi' \right). \quad (\text{P6.2})$$

Optional: Show that essentially the same lower boundary condition arises using the geometric-height as a vertical co-ordinate in the modified quasi-geostrophic system of White (1977).

- 6.5 (a) Obtain the dispersion relationship for free Rossby waves for the single-layer quasi-geostrophic potential vorticity equation with linear drag [i.e., for (??) with $\hat{h} = 0$].
- (b) Obtain the dispersion relation for free Rossby waves in the linearized two-layer potential vorticity equation with linear drag in the lowest layer.
- (c) $\diamond$ Obtain the dispersion relation for free waves in the continuously stratified quasi-geostrophic equations, with the effects of linear drag appearing in the thermodynamic equation for the lower boundary condition. That is, the boundary condition at $z = 0$ is $\partial(\partial\psi/\partial z)/\partial t + N^2 w = 0$ where $w = \alpha \zeta$ where α is a constant. You may make the Boussinesq approximation and assume N^2 is constant if you like.

Part II

WAVES, INSTABILITIES AND TURBULENCE

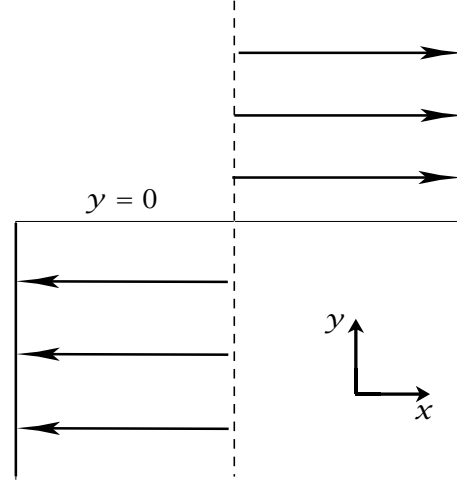
CHAPTER SEVEN

Barotropic and Baroclinic Instability

WHAT HYDRODYNAMIC STATES ACTUALLY OCCUR IN NATURE? Any flow must clearly be a solution of the equations of motion, given the initial and boundary conditions. There are, in fact, many *steady* solutions to the equations but these do not abound in nature because, in order to persist, they must be stable to those small perturbations that inevitably arise. Flows may be statistically steady, meaning that when averaged over some suitably long but finite interval the flow is unchanging, and by and large both the atmosphere and the ocean are in statistically steady states, except possibly in so far as the climate may be slowly changing. However, all the steady solutions that have been known for the large-scale flow in the earth's atmosphere and ocean have been found to be unstable. (Finding the solutions to the equations that represent statistically steady states is a very difficult problem, since it necessitates an understanding of *turbulence*, considered further in chapter ??.)

There are a myriad forms of hydrodynamic instability, but our focus in this chapter is on barotropic and baroclinic instability. The latter is at the heart of the large- and mesoscale motion in the atmosphere and ocean — it gives rise to atmospheric weather systems, for example. Barotropic instability is important to us for two reasons. First, it is important in its own right as an instability mechanisms for jets and vortices, and is a driving mechanism in both two- and three-dimensional turbulence. Second, many problems in barotropic and baroclinic instability are formally and dynamically similar, so that the solutions and

Figure 7.1 A simple basic state giving rise to Kelvin-Helmholtz instability. The velocity profile is discontinuous, the density uniform.



insight we obtain in the often simpler problems in barotropic instability are often useful in the baroclinic problem. We start with perhaps the simplest example of all, Kelvin-Helmholtz instability.

7.1 KELVIN-HELMHOLTZ INSTABILITY

To introduce the issue, we will first consider perhaps the simplest physically interesting instance of a fluid-dynamical linear instability — the *Kelvin-Helmholtz instability*. This is a shear instability, with a shear perpendicular to the fluid's mean velocity. Let us consider the simplest problem, which we pose as that of two fluids of equal density, with a common surface at $y = 0$, moving with velocities $-U$ and $+U$ in the x -direction respectively (Fig. 7.1). There is no variation in the basic flow in the z -direction (into the page), and we will assume this is also true for the instability (these restrictions are not essential). This flow is clearly a solution of the inviscid Navier Stokes equations (the Euler equations). The question to be asked is, what happens if the flow is perturbed slightly? If the perturbation is initially small then even if it grows we can, for small times after the onset of instability, neglect the nonlinear interactions in the governing equations because these are the squares of small quantities. The equations determining the evolution of the initial perturbation are then the Navier-Stokes equations (1.1), linearized about the steady solution. Thus, for $y > 0$

$$\frac{\partial \mathbf{u}'}{\partial t} + U \frac{\partial \mathbf{u}'}{\partial x} = -\nabla p', \quad (7.1a)$$

$$\nabla \cdot \mathbf{u}' = 0 \quad (7.1b)$$

where $\mathbf{u}' = u'\mathbf{i} + v'\mathbf{j}$, $\rho = 1$, and a similar equation hold for $y < 0$, but with U replaced by $-U$.

We can represent the perturbations by a Fourier expansion of the form

$$\phi(x, y, t) = \sum_k \hat{\phi}_k(y, t) \exp(ikx) \quad (7.2)$$

where ϕ is any field variable (pressure or velocity). Since (7.1a) is linear, the Fourier modes do not interact and we confine attention to just one. Taking the divergence of (7.1a), the left-hand-side vanishes and the pressure satisfies Laplace's equation

$$\nabla^2 p' = 0 \quad (7.3)$$

For $y > 0$, this has solutions in the form

$$p' = \text{Re } \hat{p}_1 e^{ikx - ky} e^{\sigma t}. \quad (7.4)$$

where Re denotes the real part, and we have assumed that the time dependence has the form $e^{\sigma t}$. In general the growth-rate σ is complex: if it has a positive real component, the amplitude of the perturbation will grow and there is an instability; if σ has a non-zero imaginary component, then there will be oscillatory motion, and there may of course be both oscillatory motion *and* an instability. To obtain the dispersion relationship, we consider the y -component of (7.1a), namely (for $y > 0$)

$$\frac{\partial w'_1}{\partial t} + U \frac{\partial v'_1}{\partial x} = -\frac{\partial p'_1}{\partial y} \quad (7.5)$$

Substituting a solution of the form $v'_1 = \hat{v}_1 \exp(ikx + \sigma t)$ yields, with (7.4),

$$(\sigma + ikU)\hat{v}_1 = k\hat{p}_1 \quad (7.6)$$

But the velocity normal to the discontinuity is, at the discontinuity, nothing but the rate of change of the discontinuity itself. That is, at the interface $y = +0$

$$v_1 = \frac{\partial \eta}{\partial t} + U \frac{\partial \eta}{\partial x}, \quad (7.7)$$

or

$$\hat{v}_1 = (\sigma + ikU)\hat{\eta}. \quad (7.8)$$

where η is the displacement of the interface from its equilibrium state. Using this in (7.6) gives

$$(\sigma + ikU)^2 \hat{\eta} = k\hat{p}_1. \quad (7.9)$$

The above few equations pertain to motion on the $y > 0$ side of the interface. Similar reasoning on the other side gives (at $y = -0$)

$$(\sigma - ikU)^2 \hat{\eta} = -k\hat{p}_2. \quad (7.10)$$

But at the interface $p_1 = p_2$ (because pressure must be continuous). The dispersion relationship then emerges from (7.9) and (7.10), giving

$$\sigma^2 = k^2 U^2. \quad (7.11)$$

This equation has two roots, one of which is positive. Thus, the amplitude of the perturbation thus grows exponentially, according to $\exp(+\sigma t)$ and the flow is *unstable*. The instability itself can be seen in the natural world when billow clouds appear wrapped up into spirals: the clouds are acting as tracers of fluid flow, indicating a shear in the atmosphere.

In this ideal example the instability appears immediately, no matter how small the shear, because of neglect of viscosity so that the Reynolds number is always infinite. If viscosity is retained the analysis becomes much more involved, but usually viscosity and surface tension will have a stabilizing effect.

7.2 INSTABILITY OF PARALLEL SHEAR FLOW

We now consider a little more systematically the instability of *parallel shear flows*, such as are illustrated in Fig. 7.2.¹ This is a classic problem in hydrodynamic stability theory, in part because one can pose and actually solve a problem of real physical interest. There are two particular reasons for our own interest:

- (i) The instability is an example of *barotropic instability*, which abounds in the ocean and atmosphere. Loosely, barotropic instability arises when a flow is unstable by virtue of its *shear*, with gravitational and buoyancy effects being secondary.
- (ii) The instability is in many ways analagous to *baroclinic instability*, which is the main instability giving rise to weather systems in the atmosphere and similar phenomena in the ocean. Baroclinic instability is an instability peculiar to rotating, stratified flows, in which potential energy is released, when, unlike normal convective instability, a single column of fluid is not gravitationally unstable.

We will define these instabilities more precisely later on. For now, we will restrict attention to two dimensional, incompressible flow. This illustrates the physical mechanisms in the most transparent way, in part because it allows for the introduction of a streamfunction and the automatic satisfaction of the mass conservation equation. In fact, for parallel two-dimensional shear flows the most unstable disturbances are two-dimensional ones,² and so for most purposes one therefore need only consider two-dimensional disturbances.

The vorticity equation for incompressible two-dimensional flow is just

$$\frac{D\zeta}{Dt} = 0. \quad (7.12)$$

We suppose the basic state to be a parallel flow in the x -direction that may vary in y -direction. That is

$$\bar{\mathbf{u}} = U(y)\mathbf{i}. \quad (7.13)$$

The linearized vorticity equation is then

$$\frac{\partial \zeta'}{\partial t} + U \frac{\partial \zeta'}{\partial x} + v' \frac{\partial \bar{\zeta}}{\partial y} = 0 \quad (7.14)$$

where $\bar{\zeta} = -U_y$. Because the mass conservation equation has the simple form

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} = 0, \quad (7.15)$$

we may introduce a streamfunction ψ such that $u' = -\partial\psi'/\partial y$, $v' = \partial\psi'/\partial x$ and $\zeta' = \nabla^2\psi'$. The linear vorticity equation is then

$$\frac{\partial \nabla^2 \psi}{\partial t} + U \frac{\partial \nabla^2 \psi'}{\partial x} + \frac{\partial \bar{\zeta}}{\partial y} \frac{\partial \psi'}{\partial x} = 0. \quad (7.16)$$

The coefficients of the x -derivatives are not themselves functions of x . Thus we may seek solutions that are harmonic functions (sines and cosines) in the x -direction, but the y -dependence must remain arbitrary at this point. Thus, anticipating our interest in wavelike or exponentially growing solutions, we seek solutions of the form,

$$\psi' = \text{Re } \hat{\psi}(y)e^{i(kx-ct)} \quad (7.17)$$

where k is the wavenumber and Re indicates the real part of the solution, a marker that we will omit where no ambiguity is caused. We envision that the solution will be a superposition of all wavenumbers, but since the problem is linear the waves do not interact and it suffices to consider them separately. The variable c determines the stability, or otherwise, of the solution. If c is purely real then c is the phase speed of the wave. If c has a positive imaginary component then the wave will grow exponentially, and any disturbance that has a projection onto wavenumber k is thus *unstable*.

From (7.17) we have

$$u' = \hat{u}(y)e^{ik(x-ct)} = -\hat{\psi}_y e^{i(kx-ct)} \quad (7.18a)$$

$$v' = \hat{v}(y)e^{ik(x-ct)} = ik\hat{\psi} e^{i(kx-ct)} \quad (7.18b)$$

$$\zeta' = \hat{\zeta}(y)e^{ik(x-ct)} = (-k^2\hat{\psi} + \hat{\psi}_{yy})e^{i(kx-ct)} \quad (7.18c)$$

Using (7.18) and (7.17) in (7.14) gives

$$(U - c)(\hat{\psi}_{yy} - k^2\hat{\psi}) - U_{yy}\hat{\psi} = 0. \quad (7.19)$$

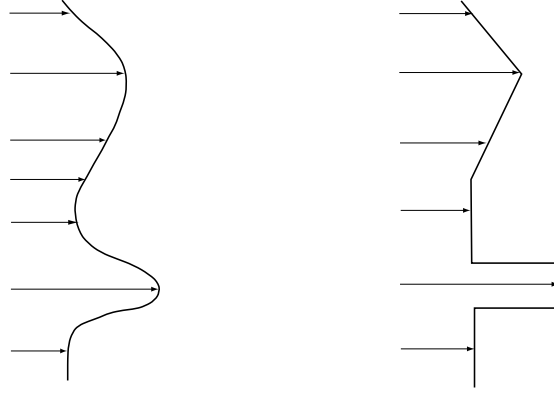


Figure 7.2 Left: example of smooth velocity profile — both the velocity and vorticity are continuous. Right: example piecewise continuous profile — the velocity and vorticity may have finite discontinuities. In general, it is impossible to analytically solve the linear eigenvalue problem (7.19) that determines stability for a continuous profile. However, solving the problem for a piecewise linear profile is often possible, for then the solution of the eigenvalue problem is effected by solving a finite number of jump conditions at the discontinuities.

This equation is sometimes known as *Rayleigh's equation*. It is the linear vorticity equation for disturbances to parallel shear flow, and in the presence of a β -effect (section ??) it is evident that the equation generalizes slightly to

$$(U - c)(\hat{\psi}_{yy} - k^2\hat{\psi}) + (\beta - U_{yy})\hat{\psi} = 0 . \quad (7.20)$$

7.2.1 Piecewise linear flows

Although Rayleigh's equation is linear and has a simple form, it is nevertheless quite difficult to analytically solve for an arbitrary smoothly varying profile. It is simpler to consider *piecewise linear* flows, in which U_y is a constant over some interval, with U or U_y changing abruptly to another value at a line of discontinuity, as illustrated in Fig. 7.2b. The curvature is accounted for through the satisfaction of matching conditions, analogous to boundary conditions, at the lines of discontinuity, and solutions in each interval are then exponential functions. The profile of Fig. 7.1 that gives rise to Kelvin-Helmholtz instability is clearly of this form — it has a discontinuity in U , and so a vortex sheet at the line of discontinuity.

Matching conditions

The idea, then, is to solve the linearized vorticity equation separately in the continuous intervals in which vorticity is constant, matching the solution with that

in the adjacent regions. The matching conditions arise from two physical conditions:

- (i) That normal stress should be continuous across the interface. For an inviscid fluid this implies that pressure be continuous.
- (ii) That the normal velocity of the fluid on either side of the interface should be consistent with the motion of the interface itself.

Continuity of pressure:

The linearized momentum equation in the direction along the interface is:

$$\frac{\partial u'}{\partial t} + U \frac{\partial u'}{\partial x} + v' \frac{\partial U}{\partial y} = -\frac{\partial p'}{\partial x}. \quad (7.21)$$

Since, for normal modes, $u' = -\hat{\psi}_y e^{ik(x-ct)}$ and $v' = ik\hat{\psi} e^{ik(x-ct)}$ this is

$$ik(U - c)\hat{\psi}_y - ik\hat{\psi}U_y = -ik\hat{p}. \quad (7.22)$$

Because pressure is continuous across the interface we have the first *matching* or *jump condition*,

$$\Delta[(U - c)\hat{\psi}_y - \hat{\psi}U_y] = 0 \quad (7.23)$$

where the operator Δ denotes the difference in the values of the argument (in square brackets) across the interface. That is, the quantity $(U - c)\hat{\psi}_y - \hat{\psi}U_y$ is continuous.

We can obtain this condition directly from Rayleigh's equation, (7.19), written in the form

$$[(U - c)\hat{\psi}_y - U_y\hat{\psi}]_y - k^2(U - c)\hat{\psi} = 0. \quad (7.24)$$

Integrating across the interface gives (7.23).

Material interface condition:

At the interface, the normal velocity v is given by the kinematic condition

$$v = \frac{D\eta}{Dt} \quad (7.25)$$

where η is the interface displacement. The linear version of this is just

$$\frac{\partial \eta}{\partial t} + U \frac{\partial \eta}{\partial x} = \frac{\partial \psi'}{\partial x}. \quad (7.26)$$

If the fluid itself is continuous (no cavitation, for instance) then this equation must hold at either side of the interface, giving two equations and their normal mode counterparts, namely,

$$\frac{\partial \eta}{\partial t} + U_1 \frac{\partial \eta}{\partial x} = \frac{\partial \psi'_1}{\partial x} \quad \Longrightarrow \quad (U_1 - c)\eta = \hat{\psi}_1, \quad (7.27)$$

$$\frac{\partial \eta}{\partial t} + U_2 \frac{\partial \eta}{\partial x} = \frac{\partial \psi'_2}{\partial x} \quad \Longrightarrow \quad (U_2 - c)\eta = \hat{\psi}_2. \quad (7.28)$$

Material continuity at the interface thus gives the second jump condition

$$\Delta \left[\frac{\hat{\psi}}{U - c} \right] = 0. \quad (7.29)$$

Thus, $\hat{\psi}/(U - c)$ is continuous at the interface. Note that if U is continuous across the interface the condition becomes one of continuity of the normal velocity.

7.2.2 Kevin-Helmholtz instability, revisited

We now use Rayleigh's equation and the jump conditions to consider the situation illustrated in Fig. 7.1; that is, vorticity is everywhere zero except in a thin sheet at $y = 0$. We suppose a solution to the linearized vorticity equation

$$\psi'_i = \hat{\psi}_i(y)e^{ik(x-ct)} \quad i = 1, 2 \quad (7.30)$$

and then, on either side of the interface, Rayleigh's equation is simply

$$(U - c)(\hat{\psi}_{iyy} - k^2\hat{\psi}_i) = 0 \quad i = 1, 2 \quad (7.31)$$

or, assuming that $U \neq c$, $\hat{\psi}_{iyy} - k^2\hat{\psi}_i = 0$. (This is just Laplace's equation, coming from $\nabla^2\psi' = \zeta'$, with $\zeta' = 0$ everywhere except at the interface.) Solutions that decay away on either side of the interface are

$$y > 0 : \quad \hat{\psi}_1 = \Psi_1 e^{-ky}, \quad (7.32a)$$

$$y < 0 : \quad \hat{\psi}_2 = \Psi_2 e^{ky}, \quad (7.32b)$$

where Ψ_1 and Ψ_2 are constants. The boundary condition (7.23) gives

$$(U_1 - c)(-k)\Psi_1 = (U_2 - c)(k)\Psi_2, \quad (7.33)$$

and (7.29) gives

$$\frac{\Psi_1}{(U_1 - c)} = \frac{\Psi_2}{(U_2 - c)}. \quad (7.34)$$

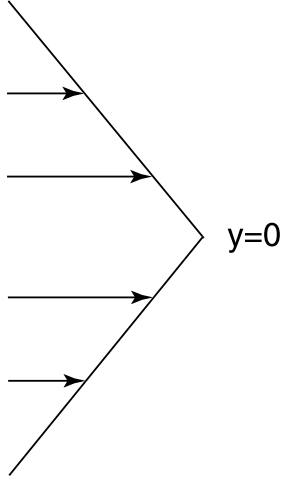


Figure 7.3 Velocity profile of a stable jet. Although the vorticity is discontinuous, a small perturbation gives rise only to *edge waves* centered at $y = 0$.

The last two equations combine to give

$$(U_1 - c)^2 = -(U_2 - c)^2 \quad (7.35)$$

which, supposing (without loss of generality, because of the Galilean invariance of the problem) that $U = U_1 = -U_2$ gives

$$c^2 = -U^2. \quad (7.36)$$

Thus, since U is purely real, $c = \pm iU$. This is the same result as we obtained in section 7.1 and indicates that the disturbance grows exponentially, at a rate $\exp(kU_1 t)$. All wavelengths are unstable, and indeed the shorter the wavelength the greater the instability. In reality of course viscosity will damp the smallest waves, but of course in reality the presence of viscosity means that initial profile is also not a steady solution of the equations of motion.

7.2.3 Edge Waves

We now consider a case sketched in Fig. 7.3 in which the velocity is continuous, but the vorticity is discontinuous. Since on either side of the interface $U_{yy} = 0$, Rayleigh's equation is just

$$(U - c)(\hat{\psi}_{yy} - k^2 \hat{\psi}) = 0. \quad (7.37)$$

Provided $c \neq U$ this has solutions,

$$\hat{\psi} = \begin{cases} \Psi_1 e^{-ky} e^{ik(x-ct)} & y > 0 \\ \Psi_2 e^{ky} e^{ik(x-ct)} & y < 0. \end{cases} \quad (7.38)$$

The value of c is found by applying the jump conditions (7.23) and (7.29) at $y = 0$. These give

$$-k(U_0 - c)\Psi_1 - \Psi_1 U_{1y} = k(U_0 - c)\Psi_2 - \Psi_2 U_{2y} \quad (7.39a)$$

$$\Psi_1 = \Psi_2 \quad (7.39b)$$

where $U_1 = U_2 = U_0$ at the interface. After a line of algebra these equations give

$$c = U_0 + \frac{U_{1y} - U_{2y}}{2k}. \quad (7.40)$$

This is the dispersion relationship for *edge waves* that propagate along the interface with the fluid speed plus a factor proportional to the difference in the vorticity between the two layers. No matter what the shear is on either side of the interface, the phase speed is purely real and there is no instability. If we write (7.40) in the form

$$c = U_0 - \frac{\bar{\zeta}_1 - \bar{\zeta}_2}{2k}. \quad (7.41)$$

where $\bar{\zeta}_1, \bar{\zeta}_2$ is the vorticity of the mean state, then we see an analogy with the Rossby wave dispersion relation $c = U_0 - \beta/k^2$. The analogy reflects a similarity in the physics — β is a planetary vorticity gradient, which in (7.41) is collapsed to a front and represented by the difference $(\bar{\zeta}_1 - \bar{\zeta}_2)$.

7.2.4 Interacting edge waves giving barotropic instability

Now we consider a slightly more complicated case in which edge waves may interact giving rise, as we shall see, to an instability. The physical situation is illustrated in Fig. 7.4. We consider the simplest case, that of a shear layer (which we denote region 2) sandwiched between two semi-infinite layers, regions 1 and 3, as in the left panel of the figure. Thus, the basic state is:

$$y > a : \quad U = U_1 = U_0 \text{ (a constant)}, \quad (7.42a)$$

$$-a < y < a : \quad U = U_2 = \frac{U_0}{a}y, \quad (7.42b)$$

$$y < -a : \quad U = U_3 = -U_0 \quad (7.42c)$$

We assume a solution of Rayleigh's equation of the form:

$$y > a : \quad \psi_1 = Ae^{-k(y-a)} \quad (7.43a)$$

$$-a < y < a : \quad \psi_2 = Be^{-k(y-a)} + Ce^{k(y+a)} \quad (7.43b)$$

$$y < -a : \quad \psi_3 = De^{k(y+a)} \quad (7.43c)$$

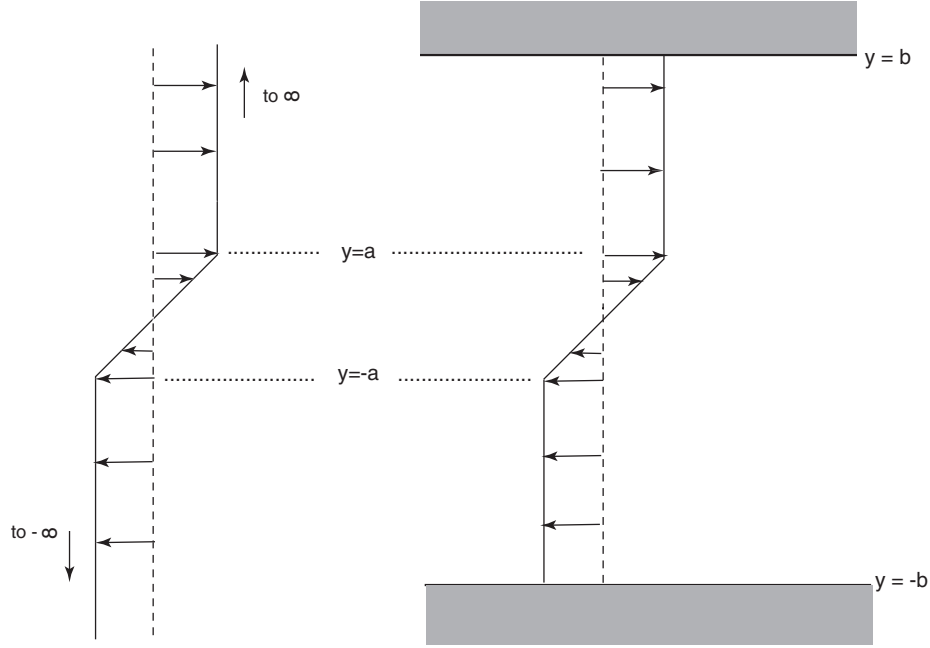


Figure 7.4 Barotropically unstable velocity profiles. In the simplest case, on the left, a region of shear is sandwiched between two infinite regions of constant velocity. The edge waves at $y = \pm a$ interact to produce an instability. If $a = 0$, then the situation corresponds to that of Fig. 7.1, giving Kelvin-Helmholtz instability. In the case on the right, the flow is bounded at $y = \pm b$. The flow is still unstable, provided that b is sufficiently larger than a . If $b = a$, the situation corresponds to plane Couette flow, which is stable to infinitesimal disturbances.

Applying the jump conditions (7.23) and (7.29) at the interfaces at $y = a$ and $y = -a$ gives the following relations between the coefficients:

$$A[(U - c)k] = B \left[(U - c)k + \frac{U}{a} \right] + Ce^{2ka} \left[\frac{U}{a} - (U - c)k \right] \quad (7.44a)$$

$$A = B + Ce^{2ka} \quad (7.44b)$$

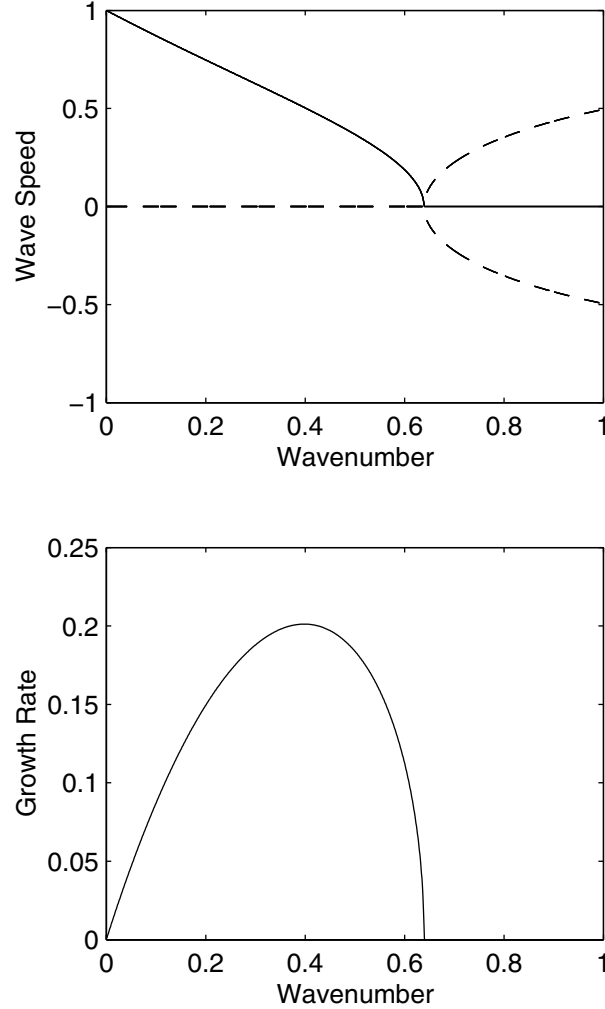
$$(7.44c)$$

and

$$D[(U + c)k] = Be^{2ka} \left[-(U + c)k + \frac{U}{a} \right] + C \left[\frac{U}{a} + (U + c)k \right] \quad (7.44d)$$

$$D = Be^{2ka} + C \quad (7.44e)$$

Figure 7.5 Upper panel: Real (c_r , dashed) and imaginary (c_i , solid) wave speeds calculated from (7.46) with c nondimensionalized by U and k nondimensionalized by $1/a$ (equivalent to setting $a = U = 1$). Lower panel: corresponding growth rate ($\sigma = kc_i$). The flow is unstable for $k < 0.63$, with the maximum instability occurring at $k = 0.39$.



These are a set of four homogeneous equations, with the unknown parameters A , B , C and D , which may be written in the form of a matrix equation,

$$\begin{pmatrix} k(U - c) & -k(U - c) - U/a & e^{2ka}[k(U - c) - (U/a)] & 0 \\ 1 & -1 & -e^{2ka} & 0 \\ 0 & e^{2ka}[k(U + c) - (U/a)] & -k(U + c) - (U/a) & k(U + c) \\ 0 & e^{2ka} & 1 & -1 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} = 0. \quad (7.45)$$

For non-trivial solutions the determinant of the matrix of coefficients must be zero, and solving the ensuing equation finally gives the dispersion relationship,

first obtained by Rayleigh,

$$c^2 = \left(\frac{U}{2ka} \right)^2 [(1 - 2ka)^2 - e^{-4ka}]. \quad (7.46)$$

The flow is unstable for sufficiently long wavelengths, for then the right-hand side of (7.46) is negative. The critical wavenumber below which instability occurs is found by solving $(1 - 2ka)^2 = e^{-4ka}$, which gives instability for $ka < 0.63293$ — see Fig. 7.5. From a physical standpoint, instability arises when the edge waves are of sufficient cross-stream extent that they interact with each other, and this occurs for sufficiently long wavelengths because the cross-stream decay scale is proportional to the along-stream wavelength. A numerical solution of the initial value problem is illustrated in Fig. 7.6 and Fig. 7.7.³ Here, the initial perturbation is small and random, containing components at all wavenumbers. All the modes in the unstable range grow exponentially, and the pattern is soon dominated by the mode that grows fastest — a horizontal wavenumber three in this problem. Eventually, the perturbation grows sufficiently that the linear equations are no longer valid and, as is seen in the lower three panels of Fig. 7.6, vortices form and pinch off. Eventually, the vortices interact and merge, and the flow develops into two-dimensional turbulence, considered in chapter ??.

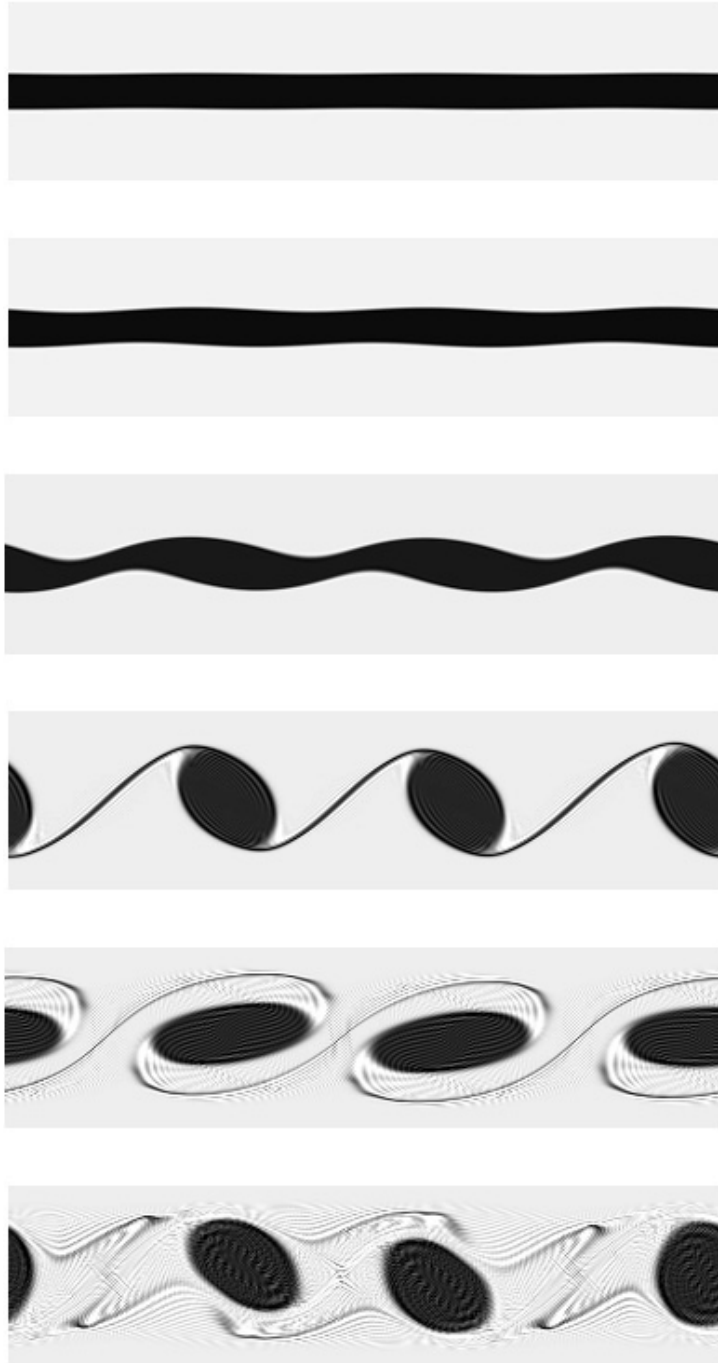


Figure 7.6 A sequence of plots of the vorticity, at equal time intervals, from a numerical solution of the nonlinear vorticity equation (7.12), with initial conditions as in Fig. 7.4, plus a very small random perturbation. The solution is obtained in a rectangular (4×1) domain, with periodic conditions in the x -direction and slippery walls at $y = (0, 1)$. The initial shear is confined to a region $0.4 < y < 0.6$ so that $a = 0.1$, and the maximum instability thus occurs for a wavelength of 1.57. For a domain of length 4, this corresponds to a nondimensional wavenumber of 2.55. Since the periodic domain quantizes the allowable wavenumbers, the maximum instability is at wavenumber 3, and this is what emerges. Only in the first two or three frames is the linear approximation valid.

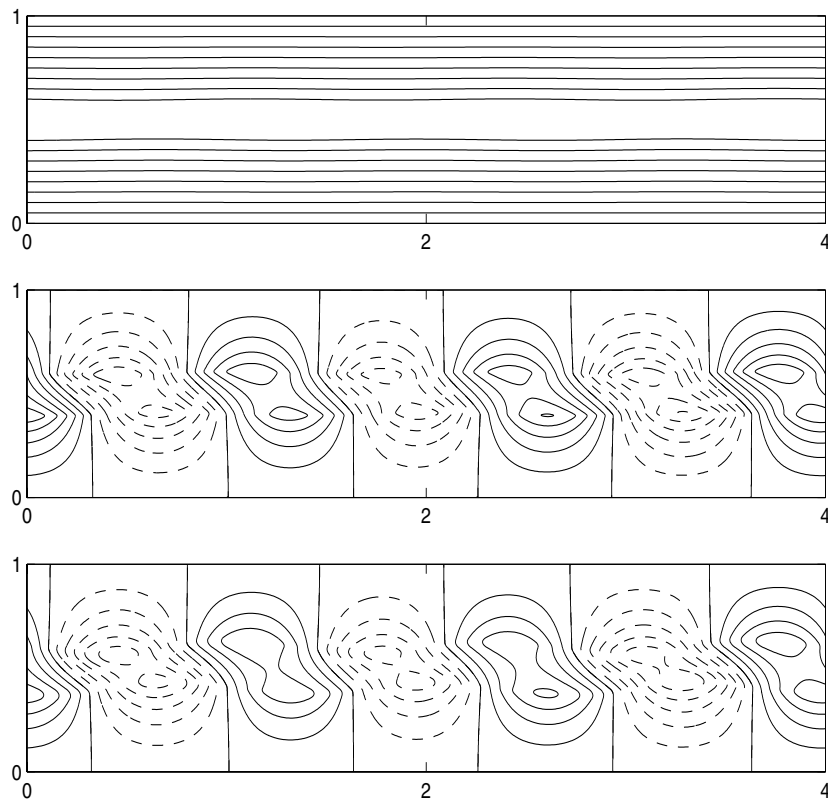


Figure 7.7 Stream function in the same numerical calculation as in Fig. 7.6. Top two frames: total streamfunction and perturbation streamfunction at at time corresponding to the second frame of Fig. 7.6. Bottom frame: perturbation streamfunction at the time of the third frame of Fig. 7.6. Positive values (a clockwise circulation) are solid lines, and negative values are dashed. The pattern grows exponentially (the contour interval in the third frame is much larger than in the second) but is locked in place.

7.3 NECESSARY CONDITIONS FOR INSTABILITY

Let's take a taxi to the finish line.

Chris Garrett, Ocean Science Meeting, Hawaii 2002.

7.3.1 Rayleigh's criterion

Why should a discontinuous velocity profile be unstable, as in Kelvin-Helmholtz instability? And why should a single discontinuity of vorticity be stable, giving edge waves, but if two discontinuities exist, as in the last section, why should the flow be unstable? For simple profiles it may be possible to calculate, or even intuit, the instability properties, but for continuous profiles of $U(y)$ this is usually

impossible and it would be nice to have some general guidelines as to when a profile might be unstable. To this end, we will derive a *necessary* condition for instability, or a *sufficient* criterion for stability that will at least tell us if a flow *might* be unstable.

Write Rayleigh's equation, including a β -effect as

$$\hat{\psi}_{yy} - k^2 \hat{\psi} + \frac{\beta - U_{yy}}{U - c} \hat{\psi} = 0 \quad (7.47)$$

Multiply by $\hat{\psi}^*$ (the complex conjugate of $\hat{\psi}$) and integrate over the domain of interest. After integrating the first term by parts, this gives

$$\int_{y_1}^{y_2} \left[\left| \frac{\partial \hat{\psi}}{\partial y} \right|^2 + k^2 |\hat{\psi}|^2 \right] dy - \int_{y_1}^{y_2} \frac{\beta - U_{yy}}{U - c} |\hat{\psi}|^2 dy = 0. \quad (7.48)$$

assuming that $\hat{\psi}$ vanishes at the boundaries. (The limits to the integral may be infinite, in which case it is assumed that $\hat{\psi}$ decays to zero as $|y|$ approaches ∞ .) The only variable in this expression that is complex is c , and thus the first integral is real. The imaginary component of the second integral is

$$c_i \int \frac{\beta - U_{yy}}{|U - c|^2} |\hat{\psi}|^2 dy = 0. \quad (7.49)$$

Thus, *either* c_i vanishes or the integral does. For there to be an instability, c_i must be nonzero and because the eigenvalues of Rayleigh's equation come in pairs (because it is a second order ODE), for each decaying mode (negative c_i) there is a corresponding growing mode (positive c_i). Therefore:

A necessary condition for instability is that the expression

$$\beta - U_{yy}$$

change sign somewhere in the domain (Rayleigh, Kuo).

That is, there must be an inflexion point in the absolute vorticity of the basic state. Equivalently, a sufficient criterion for stability is that $\beta - U_{yy}$ not vanish in the domain interior; it may vanish at the boundaries, or tend to zero for large $|y|$.

This condition is known as Rayleigh's inflexion point criterion. (Rayleigh himself did not consider the case with β ; this was added by Kuo in 1949.)

An alternate, more general, derivation

Consider again the vorticity equation, linearized about a parallel shear flow,

$$\frac{\partial \zeta}{\partial t} + U \frac{\partial \zeta}{\partial x} + \nu \left(\frac{\partial \bar{\zeta}}{\partial y} + \beta \right) = 0. \quad (7.50)$$

(dropping the primes on the perturbation quantities). Multiply by ζ and divide by $\beta + \bar{\zeta}_y$ to obtain

$$\frac{\partial}{\partial t} \left(\frac{\zeta^2}{\beta + \bar{\zeta}_y} \right) + \frac{U}{\beta + \bar{\zeta}_y} \frac{\partial \zeta^2}{\partial x} + v\zeta = 0, \quad (7.51)$$

and then integrate with respect to x to give

$$\frac{d}{dt} \int \left(\frac{\zeta^2}{\beta + \bar{\zeta}_y} \right) dx = - \int v\zeta dx. \quad (7.52)$$

Now,

$$v\zeta = -\frac{\partial}{\partial y}(uv) + \frac{1}{2} \frac{\partial}{\partial x}(u^2 + v^2). \quad (7.53)$$

That is, *the flux of vorticity is the divergence of some quantity*. Its integral therefore vanishes provided there are no contributions from the boundary, and integrating (7.52) with respect to y gives

$$\frac{d}{dt} \int \left(\frac{\zeta^2}{\beta + \bar{\zeta}_y} \right) dx dy = 0. \quad (7.54)$$

If there is to be an instability ζ must grow, but the integral is identically zero. These two conditions can only be simultaneously satisfied if $\beta + \bar{\zeta}_y$, or equivalently $\beta - U_{yy}$, is zero somewhere in the domain.

This derivation clearly shows that the Rayleigh-Kuo criterion applies even if disturbances are not of normal-mode form. The quantity $\zeta^2 / (\beta + \bar{\zeta}_y)$ is an example of a *wave-activity density* — a wave activity being a conserved quantity, quadratic in the amplitude of the wave. Such quantities play an important role in instabilities, and we consider then further in chapter 8.

7.3.2 Fjortoft's criterion

A slightly stronger condition was obtained by Fjortoft. Consider the real part of (7.47) namely

$$\int_{y_1}^{y_2} (\beta - U_{yy}) \frac{(U - c_r)}{|U - c|^2} |\hat{\psi}|^2 dy = \int_{y_1}^{y_2} \left| \frac{\partial \hat{\psi}}{\partial y} \right|^2 + k^2 |\hat{\psi}|^2 dy \geq 0 \quad (7.55)$$

where the inequality holds if there is an instability. Now, for an instability we also know that

$$\int_{y_1}^{y_2} \frac{\beta - U_{yy}}{|U - c|^2} |\hat{\psi}|^2 dy = 0. \quad (7.56)$$

Using this and (7.55) it is clear that, for an instability,

$$\int_{y_1}^{y_2} (\beta - U_{yy}) \frac{(U - U_s)}{|U - c|^2} |\hat{\psi}|^2 dy > 0 \quad (7.57)$$

where U_s is any real constant. It is most useful to choose this constant to be the value of $U(y)$ at which $\beta - U_{yy}$ vanishes. This leads directly to Fjortoft's criterion:

A necessary condition for stability is that the expression

$$(\beta - U_{yy})(U - U_s)$$

where U_s is the value of $U(y)$ at which $\beta - U_{yy}$ vanishes, be positive somewhere in the domain (Fjortoft).

This is equivalent to saying that the magnitude of the vorticity must have a maximum inside the domain, and not at the boundary or at infinity, as can be seen by perusing Fig. 7.8. Why choose U_s in the manner we did? Suppose we chose U_s to have a very large negative or large positive value, so that $U - U_s$ is of one sign everywhere. Then (7.57) just implies that $\beta - U_{yy}$ must be negative somewhere, or must be positive somewhere, which is already known from Rayleigh's criterion. The most stringent criterion is obtained by choosing U_s to be the value of $U(y)$ at which $\beta - U_{yy}$ vanishes. Again it should be noted that both Fjortoft's and Rayleigh's criteria are necessary conditions for instability, and examples may be constructed which do satisfy their criterion, yet which are stable to infinitesimal perturbations. Note that the β -effect can stabilize the middle two profiles of Fig. 7.8, because if it is large enough $\beta - U_{yy}$ will be one-signed. However, the β -effect can destabilize a westward point jet, $U(y) = -(1 - |y|)$ (the negative of the jet in Fig. 7.3), because $\beta - U_{yy}$ is negative at $y = 0$ and positive elsewhere. An eastward point jet is stable, with or without β .

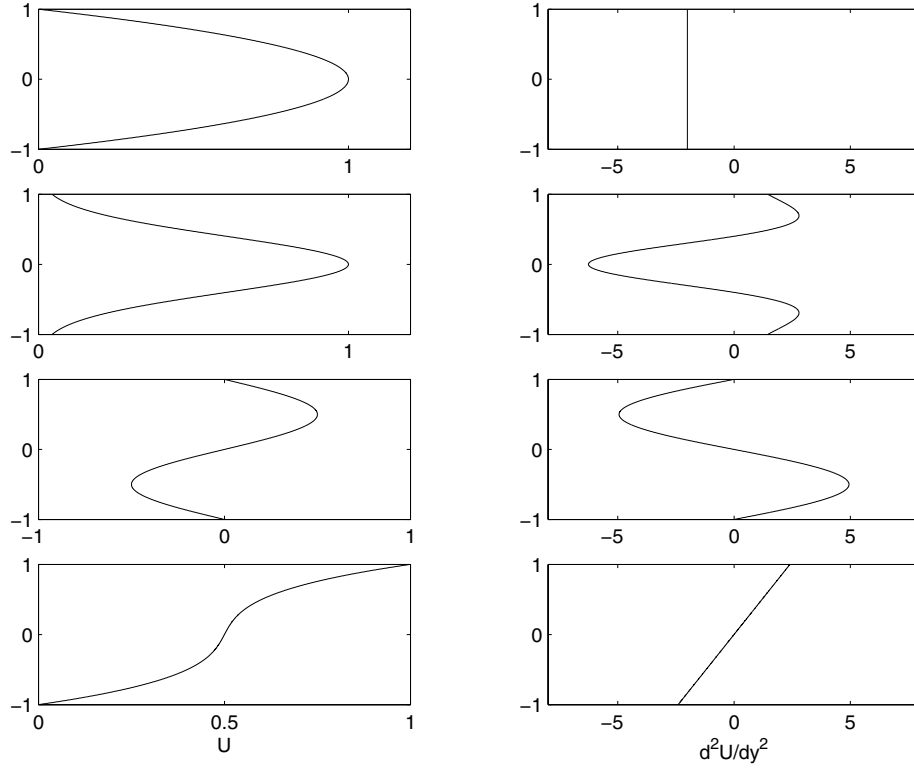


Figure 7.8 Example parallel velocity profiles (left column) and their second derivatives (right column). From the top: Poiseuille flow ($u = 1 - y^2$); a Gaussian jet; a sinusoidal profile; a polynomial profile. By Rayleigh's criterion, the top profile is stable, whereas the lower three are potentially unstable. However, the bottom profile is stable by Fjortoft's criterion — note that the vorticity maxima are at the boundaries. If the β -effect were present and large enough it would stabilize the middle two profiles.

7.4 BOUNDS ON THE GROWTH RATE

7.4.1 Howard's semicircle theorem

If a flow is unstable, how fast will the instability grow? We cannot answer this question precisely without calculating the growth rate, and this is impossible for an arbitrary shear flow. However, it is possible to find bounds on the growth rates, and wave speeds, of a given profile.⁴ Although the calculation itself appears rather contrived, it gives a very useful result, illustrated in Fig. 7.9.

Consider the y -displacement in of a fluid parcel, η , obeying

$$\frac{D\eta}{Dt} = v, \quad (7.58)$$

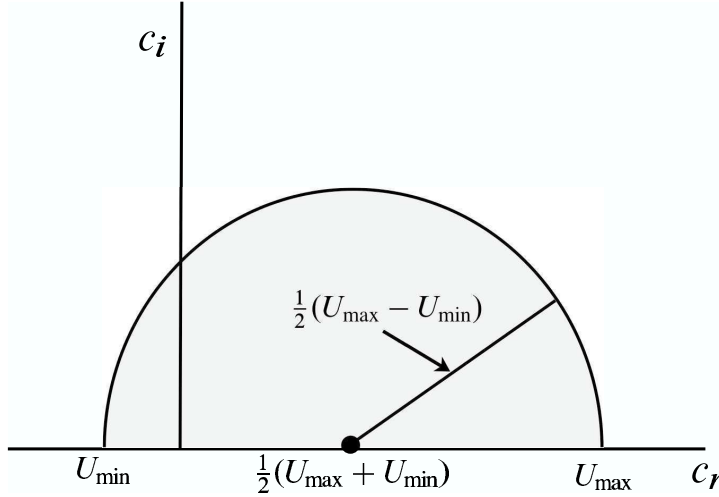


Figure 7.9 Howard's semi-circle theorem. The complex eigenvalue of a shear flow must lie within the shaded semi-circle.

which, after linearization, is

$$\frac{\partial \eta}{\partial t} + U \frac{\partial \eta}{\partial x} = \frac{\partial \psi}{\partial x}, \quad (7.59)$$

or, for normal modes,

$$(U - c)\hat{\eta} = \hat{\psi}. \quad (7.60)$$

Now recall Rayleigh's equation, with $\beta = 0$,

$$\frac{\partial^2 \hat{\psi}}{\partial y^2} - k^2 \hat{\psi} - \frac{U_{yy}}{U - c} \hat{\psi} = 0 \quad (7.61)$$

and substitute (7.60) into this to give

$$(U - c) \left[\left(\frac{\partial^2}{\partial y^2} - k^2 \right) (U - c) \hat{\eta} \right] - U_{yy} (U - c) \hat{\eta} = 0. \quad (7.62)$$

This can be written

$$-k^2 (U - c)^2 \hat{\eta} + (U - c) \hat{\eta} \left(\frac{\partial^2}{\partial y^2} (U - c) \hat{\eta} \right) - U_{yy} (U - c) \hat{\eta} = 0 \quad (7.63)$$

or

$$\frac{\partial}{\partial y} \left((U - c)^2 \frac{\partial \hat{\eta}}{\partial y} \right) - k^2 (U - c)^2 \hat{\eta} = 0. \quad (7.64)$$

Multiply by the complex conjugate of $\hat{\eta}$ and integrate with respect to y over the domain $\mathcal{Y}$ to give

$$\int_{\mathcal{Y}} (U - c)^2 (|\hat{\eta}_y|^2 + k^2 |\hat{\eta}|^2) dy = 0. \quad (7.65)$$

The real and imaginary components of this are:

$$\text{Real:} \quad \int_{\mathcal{Y}} [(U - c_r)^2 - c_i^2] E \, dy = 0, \quad (7.66a)$$

$$\text{Imaginary:} \quad 2c_i \int_{\mathcal{Y}} (U - c_r) E \, dy = 0. \quad (7.66b)$$

where $E = |\hat{\eta}_y|^2 + k^2 \hat{\eta}^2 > 0$. If $c_i \neq 0$ then the second integral implies that $U - c_r$ must be both positive and negative, and so that c_r must lie in the range of U :

$$U_{\min} < c_r < U_{\max}. \quad (7.67)$$

This result, first derived by Rayleigh, states that somewhere in the fluid the wavespeed equals the fluid speed, and so that the instability can grow in place, without moving relative to the fluid. The position where $c_r = U$ is called a *critical level*; such a level must exist for instability of parallel shear flow.

Now, note the inequality

$$\int_{\mathcal{Y}} (U - U_{\min})(U - U_{\max}) E \, dy \leq 0, \quad (7.68)$$

and note also that, using (7.66b), (7.66a) is equivalent to

$$\int_{\mathcal{Y}} (U^2 - c_r^2 - c_i^2) E \, dy = 0. \quad (7.69)$$

The last two equations give

$$\int_{\mathcal{Y}} [U_{\max} U_{\min} + c_r^2 + c_i^2 - U(U_{\max} + U_{\min})] E \, dy < 0. \quad (7.70)$$

and using (7.66b) again

$$\int_{\mathcal{Y}} [U_{\max} U_{\min} + c_i^2 + c_r^2 - c_r(U_{\max} + U_{\min})] E \, dy < 0. \quad (7.71)$$

Thus, since E is positive definite,

$$[U_{\max} U_{\min} + c_r^2 + c_i^2 - c_r(U_{\max} + U_{\min})] \leq 0, \quad (7.72)$$

or

$$\left(c_r - \frac{U_{\max} + U_{\min}}{2} \right)^2 + c_i^2 \leq \left(\frac{U_{\max} - U_{\min}}{2} \right)^2. \quad (7.73)$$

Eq. (7.73) implies that in the complex c -plane, c_i, c_r must lie inside a circle centered at $[(U_{\max} + U_{\min})/2, 0]$ with a radius of $(U_{\max} - U_{\min})/2$. For growing

disturbances c_i is positive and the imaginary wave-speed must lie in upper semi-circle — thus Howard's *semi-circle* theorem (Fig. 7.9). Note that on the real axis, we have again $U_{\min} < c_r < U_{\max}$, recovering Rayleigh's result. The imaginary wave speed is bounded by

$$c_i \leq \frac{U_{\min} - U_{\max}}{2}. \quad (7.74)$$

This gives an upper bound on the growth rate, kc_i , of the instability, although it can only be more precisely determined if the wavelength of the instability is known or can be bound.

7.5 BAROCLINIC INSTABILITY

Baroclinic instability is a hydrodynamic instability that occurs in stably stratified, rotating fluids, and it is ubiquitous in the planetary atmospheres and oceans. It gives rise to weather, and thus is the form of hydrodynamic instability that most affects the human condition.

7.5.1 A physical picture

We will first draw a picture of baroclinic instability as a form of sloping convection in which the fluid, although statically stable, is able to release available potential energy when parcels move along a sloping path. Although it will illustrate a fundamental mechanism whereby mean energy can be transferred to that of the disturbance, the picture pays no attention to the *dynamical* constraints imposed by the equations of motion, and we will come back to that later on.

What is the basic state that is baroclinically unstable? In a stably stratified fluid potential density (potential temperature) decreases (increases) with height; we can also easily imagine a state in which the basic state temperature decreases polewards. Can we construct a steady solution from these two conditions? The answer is yes, provided the fluid is also rotating; rotation is necessary because the meridional temperature gradient generally implies a meridional pressure gradient; there is nothing to balance this in the absence of rotation, and a fluid parcel would therefore accelerate. In a rotating fluid this pressure gradient can be balanced by the Coriolis force and a steady solution (namely thermal wind balance) maintained even in the absence of viscosity. Consider a stably-stratified Boussinesq fluid in geostrophic and hydrostatic balance on an f -plane, with buoyancy decreasing uniformly polewards (as in Fig. 7.10). Then

$$fu = -\frac{\partial\phi}{\partial y} \quad (7.75)$$

and

$$\frac{\partial\phi}{\partial z} = b \quad (7.76)$$

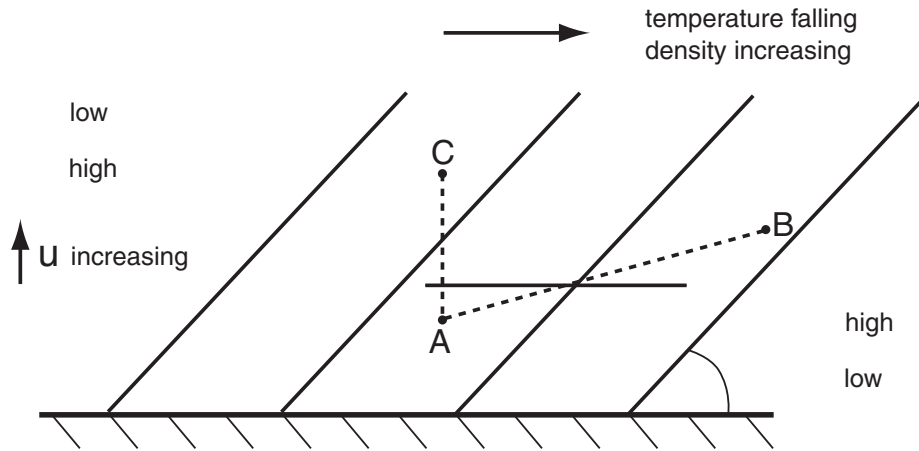


Figure 7.10 Basic state giving rise to baroclinic instability. Potential temperature (θ) increases upwards and equatorwards, or equivalently potential density (ρ) decreases upwards and polewards. The horizontal temperature gradient implies a vertical shear of the horizontal flow, by the thermal wind relationship. The horizontal pressure gradient is balanced by the Coriolis force; thus, the stationary basic state can only exist in the presence of rotation.

where $\phi = p/\rho_0$ and $b = -g\rho_1/\rho_0$. These together give the thermal wind relation,

$$\frac{\partial u}{\partial z} = g \frac{\partial \rho}{\partial y} \quad (7.77)$$

If there is no variation of these fields in the zonal direction, then, for any variation of ρ with y , this is a steady solution to the primitive equations of motion: for $v = w = 0$ and $Du/Dt = Dv/Dt = 0$. The density structure corresponding to a uniform decrease of temperature in the meridional direction is illustrated in Fig. 7.10. Is this structure stable to perturbations? The answer is no, although the perturbations must be a little special. Suppose the particle at 'A' is displaced vertically; then, since the fluid is (by assumption) stably stratified it will, if displaced upwards (downwards) be denser (less dense) than its surroundings and hence experience a restoring force. Since the perturbation involved moving a relatively heavy parcel upwards, or a light parcel downwards, the total potential energy of the fluid increases and therefore energy must be provided to the fluid for the perturbation to occur. Thus, it is not convectively unstable or statically unstable.

Suppose, however, we interchange the two parcels at positions 'A' and 'B'. Parcel 'A' finds itself surrounded by parcels of higher density than itself, and it is therefore buoyant. Plus, it is higher than where it started. Parcel 'B' is negatively buoyant, and at a lower altitude than where it started. Thus, overall, the centre of gravity of the fluid has been lowered, and therefore its overall potential energy

lowered. This loss in potential energy of the basic state must be accompanied by a gain in kinetic energy of the perturbation. That is, the perturbation can amplify. The basic state is *unstable* to ‘slantwise’ perturbations.

The change in potential energy is easily calculated. Since

$$\text{P.E.} = \int \rho g dz \quad (7.78)$$

the change in potential energy due to the interchange is

$$\begin{aligned} \Delta \text{P.E.} &= g(\rho_A z_A + \rho_B z_B - \rho_A z_B - \rho_B z_A) \\ &= g(z_A - z_B)(\rho_A - \rho_B) = g\Delta\rho\Delta z \end{aligned} \quad (7.79)$$

If both $\rho_B > \rho_A$ and $z_B > z_A$ then the initial potential energy is larger than the final, energy is released and the state is unstable. If the slope of the isopycnals is ϕ [so that $\phi = -(\partial\rho/\partial y)/(\partial\rho/\partial z)$] and the slope of the displacements is α , then for a displacement of horizontal distance l the change in potential energy is given by

$$\begin{aligned} \Delta \text{PE} &= g\Delta\rho\Delta z = g(l\frac{\partial\rho}{\partial y} + l\alpha\frac{\partial\rho}{\partial z})\alpha l \\ &= gl^2\alpha\frac{\partial\rho}{\partial y}(1 - \frac{\alpha}{\phi}) \end{aligned} \quad (7.80)$$

if α and ϕ are small. If $0 < \alpha < \phi$ then energy is released by the perturbation, and it is maximized when $\alpha = \phi/2$. For the atmosphere the actual slope of the isotherms is about 10^{-3} , so that the slope and potential parcel trajectories are indeed shallow.

Although intuitively appealing, the thermodynamic arguments presented in this section pay no attention to satisfying the dynamical constraints of the equations of motion, and we now turn our attention to that,

7.6 LINEARIZED QUASI-GEOSTROPHIC EQUATIONS

We will explore the baroclinic instability problem using the quasi-geostrophic equations. Thus, *a priori*, we are assuming that the scale of motion of the resulting instabilities will be roughly the deformation scale — strictly it means that the squared ratio of the scale of motion to the first deformation scale, multiplied by the Rossby number, is of order the Rossby number itself. That is, $Ro(L/L_R)^2 = O(Ro)$. Although this cannot formally be violated, in an asymptotic sense, by the results analysis that itself uses the quasi-geostrophic equations, we should check that the scale of the instabilities is roughly the deformation scale when the analysis is all done. Further, the scale of the instabilities should not be so large that $\beta y \sim f_0$, and again an *a posteriori* check will be in order.

Recall that the quasi-geostrophic equations comprise a potential vorticity equation for the fluid interior, and the buoyancy or temperature equation for the boundaries that satisfies $w = 0$. (We could also consider the case of a semi-infinite fluid with no upper boundary; the upper boundary condition is then that the perturbation decays away as $z \rightarrow \infty$. We defer consideration of that case.) For a Boussinesq fluid the equations are:

$$\begin{aligned} \frac{\partial q}{\partial t} + \mathbf{u} \cdot \nabla q &= 0, & 0 < z < H \\ q &= \nabla^2 \psi + \beta y + \frac{\partial}{\partial z} \left(F \frac{\partial \psi}{\partial z} \right), \end{aligned} \quad (7.81)$$

and

$$\begin{aligned} \frac{\partial b}{\partial t} + \mathbf{u} \cdot \nabla b &= 0, & z = 0, z = H \\ b &= f_0 \frac{\partial \psi}{\partial z}, \end{aligned} \quad (7.82)$$

where $F = F(z) = f_0^2 / N^2$.

A solution of these equations is a purely zonal flow, $\mathbf{u} = U(y, z)\mathbf{i}$ with a corresponding temperature field given by thermal wind balance. The potential vorticity of this basic state is

$$\bar{q} = \beta y - U_y + \frac{\partial}{\partial z} F \frac{\partial \bar{\psi}}{\partial z} = \beta y + \bar{\psi}_{yy} + \frac{\partial}{\partial z} F \frac{\partial \bar{\psi}}{\partial z} \quad (7.83)$$

where $\bar{\psi}$ is the streamfunction of the basic state, related to U by $U = -\partial \bar{\psi} / \partial y$. Linearizing (7.81) and (7.82) about a zonal flow gives the potential vorticity equation for the interior,

$$\begin{aligned} \frac{\partial q'}{\partial t} + U \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} &= 0, & 0 < z < H \\ q' &= \nabla^2 \psi' + \frac{\partial}{\partial z} \left(F \frac{\partial \psi'}{\partial z} \right), & v' = \frac{\partial \psi'}{\partial x}, \end{aligned} \quad (7.84)$$

and the buoyancy equation

$$\frac{\partial b'}{\partial t} + U \frac{\partial b'}{\partial x} + v' \frac{\partial \bar{b}}{\partial y} = 0, \quad z = 0, H \quad (7.85)$$

or

$$\frac{\partial}{\partial t} \frac{\partial \psi'}{\partial z} + U \frac{\partial}{\partial x} \frac{\partial \psi'}{\partial z} - U_z \frac{\partial \psi'}{\partial x} = 0, \quad z = 0, H \quad (7.86)$$

using $\partial \bar{b} / \partial y = f_0 \partial (\partial \bar{\psi} / \partial z) / \partial y = -f_0 \partial U / \partial z$.

Just as for the barotropic problem, a standard way to proceed is to seek normal-mode solutions of these equations. Since the coefficients of the equations are functions of y and z , but not of x , we may solutions of the form

$$\psi'(x, y, z, t) = \hat{\psi}(y, z)e^{ik(x-ct)} \quad (7.87)$$

and similarly for the derived quantities u' , v' and q' . In particular

$$q'(x, y, z, t) = \hat{q}(y, z)e^{ik_x(x-ct)} \quad (7.88)$$

where

$$\hat{q} = \hat{\psi}_{yy} + \frac{\partial}{\partial z} F \frac{\partial \hat{\psi}}{\partial z} - k_x^2 \hat{\psi}. \quad (7.89)$$

Substituting (7.87) and (7.88) into (7.81) and (7.82) gives

$$(U - c) (\hat{\psi}_{yy} + (F\hat{\psi}_z)_z - k_x^2 \hat{\psi}) + \bar{q}_y \hat{\psi} = 0 \quad (7.90)$$

and

$$(U - c)\hat{\psi}_z - U_z \hat{\psi} = 0 \quad \text{at } z = 0, H. \quad (7.91)$$

These equations, similar to Rayleigh's equation for parallel shear flow, are the starting point for our studies of linear baroclinic instability.

7.6.1 Necessary conditions for instability

Necessary conditions for instability may be obtained following a procedure analogous to that used for parallel shear flows. First we note the identities

$$\int_{y_1}^{y_2} \hat{\psi}^* \hat{\psi}_{yy} dy = [\hat{\psi}^* \hat{\psi}_y]_{y_1}^{y_2} - \int |\hat{\psi}_y|^2 dy \quad (7.92)$$

If the integral is performed between two quiescent latitudes, or the domain is a channel with $\psi = 0$ at the boundaries, then the first term on the right-hand-side vanishes. Similarly,

$$\begin{aligned} \int_0^H \hat{\psi}^* (F\hat{\psi}_z)_z dz &= [F\hat{\psi}^* \hat{\psi}_z]_0^H - \int F |\hat{\psi}_z|^2 dz \\ &= \left[\frac{FU_z |\hat{\psi}|^2}{(U - c)} \right]_0^H - \int F |\hat{\psi}_z|^2 dz, \end{aligned} \quad (7.93)$$

using (7.91). Multiply (7.90) by $\hat{\psi}^*$ and integrate over y and z , and use (7.92) and (7.93) to obtain

$$\begin{aligned} &\int |\hat{\psi}_{yy}|^2 + F |\hat{\psi}_z|^2 + k_x^2 |\hat{\psi}|^2 dy dz \\ &- \int \left\{ \int \frac{\bar{q}_y}{U - c} |\hat{\psi}|^2 dz + \left[\frac{FU_z |\hat{\psi}|^2}{U - c} \right]_0^H \right\} dy = 0. \end{aligned} \quad (7.94)$$

The first term is purely real. The second term is complex, and its imaginary component is given by

$$-c_i \int \left\{ \int \frac{\bar{q}_y}{|U - c|^2} |\hat{\psi}|^2 dz + \left[\frac{FU_z |\hat{\psi}|^2}{|U - c|^2} \right]_0^H \right\} dy = 0. \quad (7.95)$$

If there is to be instability, $c_i \neq 0$ and the integrand must therefore vanish, giving the *Charney-Stern-Pedlosky* criterion for instability. Necessary conditions for instability are then that one of the following conditions is satisfied:

1. $\bar{q}_y$ changes sign in the interior.
2. $\bar{q}_y$ is the opposite sign to U_z at the upper boundary, $z = H$.
3. $\bar{q}_y$ is the same sign as U_z at the lower boundary, $z = 0$.
4. U_z is the same sign at the upper and lower boundaries, a condition which differs from (2) or (3) if $\bar{q}_y = 0$.

In the earth's atmosphere, q_y is often dominated by β , and is positive everywhere, as, frequently, is the shear. The instability criterion is normally satisfied through condition (3): both $\bar{q}_y$ and $U_z(0)$ are positive.

7.7 THE EADY PROBLEM

Historically, the Eady problem (?) was one of the first two mathematical descriptions of baroclinic instability, the other being the 'Charney problem' (?).⁵ The two were formulated independently, each being the (largely unsupervised) Ph.D. thesis of its respective author, and although the Charney problem is in some respects more complete (for example in allowing a β -effect) the Eady problem displays the instability in a more austere and basic form. The Charney problem in its entirety is also quite mathematically opaque,⁶ and for these reasons we will first consider the Eady problem. The β -effect can be incorporated relatively simply in the two-layer model (the 'Phillips problem') considered in the next section, and in section 7.9.1 we look at some aspects of the Charney problem approximately. These models were all initially envisioned as models for instabilities in the atmosphere, but the process of baroclinic instability is also ubiquitous in the ocean. To begin, let us make the following simplifying assumptions:

- (i) The instability will be described by the quasi-geostrophic equations. This implies that we are restricted to instabilities of a scale of the same order as the deformation radius.
- (ii) The motion is on the f -plane ($\beta = 0$). This is not a particularly realistic assumption for flow in the atmosphere at the deformation scale, but it greatly simplifies the analysis.

- (iii) The fluid is uniformly stratified. That is, N^2 is a constant. This is a reasonable approximation for the atmosphere below the tropopause, but less so for the ocean where the stratification is quite non-uniform, being much larger in the upper ocean (in the main thermocline).
- (iv) The basic state is of uniform shear. That is $u_o = Uz/H$ where U is a constant and z the height co-ordinate and H the domain depth. Again, this is more appropriate for the atmosphere than the ocean — below the thermocline the ocean is relatively quiescent and the shear small.
- (v) The motion is contained between two rigid, flat horizontal surfaces. In the atmosphere this straightforwardly corresponds to the ground and a 'lid' at a constant-height tropopause. In the ocean the surface and floor of the ocean are perhaps the obvious analogues, although since the stratification is largely confined to the upper ocean, taking the lower surface as the ocean floor is not *a priori* appropriate.

Assumptions (iii)–(v) are rather inappropriate for the ocean, and will preclude us from drawing any quantitative conclusions about that system from our analysis. The most restrictive assumption vis-a-vis the atmosphere is (ii), and we explore the role of the β -effect in baroclinic instability later on.

The quasi-geostrophic potential vorticity is, in general,

$$q = \nabla^2 \psi + \beta y + f_0^2 \frac{\partial}{\partial z} \left(\frac{1}{N^2} \frac{\partial \psi}{\partial z} \right) \quad (7.96)$$

where ψ is the streamfunction. It is convenient to rescale the vertical co-ordinate, letting $z \rightarrow zH$ where H is the depth of the domain. Then, since f and N are both constants, the expression for potential vorticity becomes

$$q = \nabla^2 \psi + \frac{1}{L_d^2} \frac{\partial}{\partial z} \left(\frac{\partial \psi}{\partial z} \right) \quad (7.97)$$

where $L_d = NH/f$ is the radius of deformation and the fluid motion is now contained within $z \in \{0, 1\}$. The basic state potential vorticity is given by

$$q_0 = \nabla^2 \psi_0 + \frac{1}{L_d^2} \frac{\partial}{\partial z} \left(\frac{\partial \psi_0}{\partial z} \right) \quad (7.98)$$

where $\psi_0 = -Uzy$. Both the terms in (7.96) are individually zero, and the basic state potential vorticity is therefore zero. As we will see, this makes the Eady problem a special case in some regards.

7.7.1 Linearization and solution

The linearized potential vorticity equation is

$$\frac{\partial q'}{\partial t} + \mathbf{u}_0 \cdot \nabla q' + \mathbf{u}' \cdot \nabla q_0 = 0 \quad (7.99)$$

which, using (7.97) and $q_0 = 0$, becomes

$$\frac{\partial q'}{\partial t} + Uz \frac{\partial}{\partial x} \left(\nabla^2 \psi' + \frac{\partial^2 \psi}{\partial z^2} \right) = 0 \quad (7.100)$$

We shall seek solutions to this equation in the form

$$\psi'(x, y, z, t) = \Psi(y, z) e^{ik_x(x-ct)}. \quad (7.101)$$

Why do we choose this form? It is because we don't *a priori* know the y and z structure of solutions to (7.100) because of the presence of variable coefficients in the equation (i.e. z), and because lateral boundary conditions must be satisfied in the y -direction. In the x -direction, however, we can expect travelling wave solutions.

Substituting (7.101) into (7.100) yields

$$(Uz - c) \left(\frac{\partial^2 \psi'}{\partial y^2} + \frac{1}{L_d^2} \frac{\partial^2 \psi'}{\partial z^2} - k^2 \psi' \right) = 0 \quad (7.102)$$

Boundary Conditions

There are two sets of boundary conditions to satisfy, the vertical boundary conditions at $z = 0$ and $z = 1$ and the lateral boundary conditions. In the vertical

$$w = 0 \quad \text{at} \quad z = 0, 1. \quad (7.103)$$

In the horizontal plane we may either consider the flow to be confined to a channel or, with a slightly greater degree of idealization, suppose that the disturbance is periodic in the y -direction. Either case is dealt with easily enough by the choice of geometric basis function. For a channel of width L we impose

$$\psi = 0 \quad \text{at} \quad y = +L/2, -L/2 \quad (7.104)$$

and seek solutions of the form

$$\Psi = \Phi(z) \cos k_y y \quad (7.105)$$

where $k_y = (2n + 1)\pi / L$. For periodic case we would choose

$$\Psi = \text{Re } \Phi(z) \exp(k_y y) \quad (7.106)$$

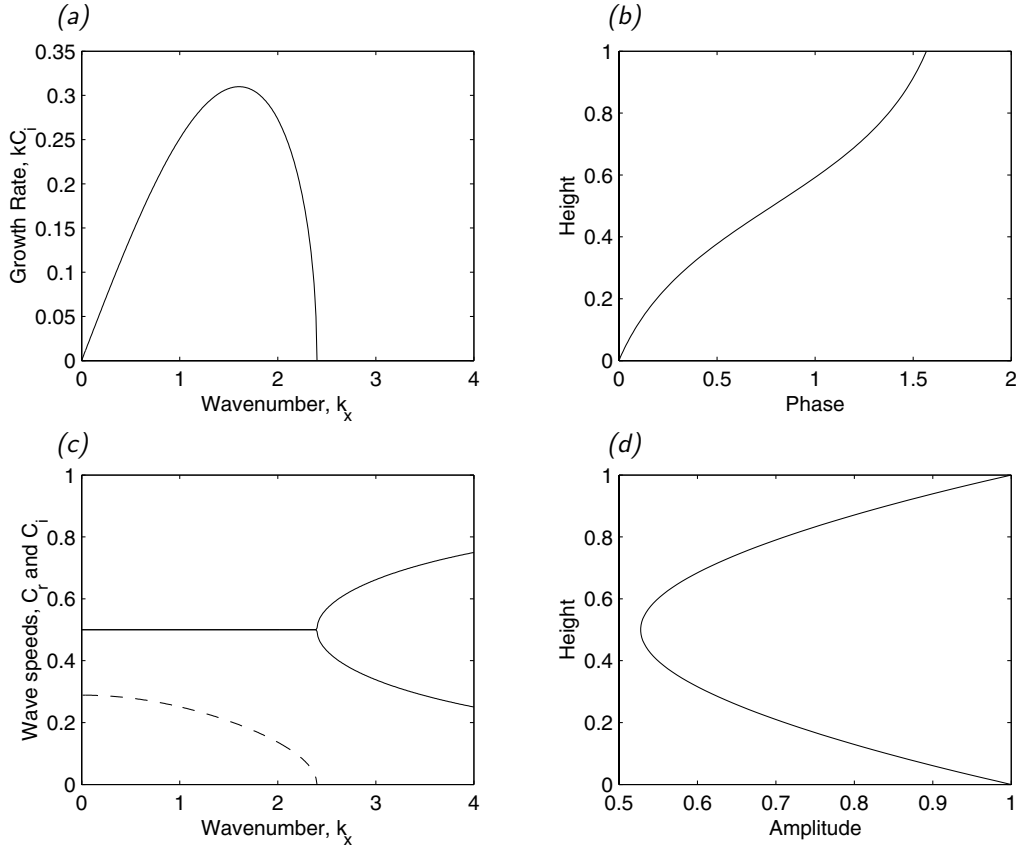


Figure 7.11 Solution of the Eady problem, in non-dimensional units. (a): nondimensional growth rate, $k_x c_i$, of the most unstable Eady modes (i.e., those with the gravest meridional scale) as a function of scaled wavenumber μ , from (7.123) with $U = 1$. The scaled x-wavenumber k_x is related to μ by (7.124), where we choose $k_y = 0$. (c): the real (solid) and imaginary (dashed) wave speeds of those modes, again as a function of horizontal wavenumber. (b): the phase of the single most unstable mode as a function of height. (d): the amplitude of that mode as a function of height. To obtain dimensional values, multiply the growth rate by U/L_d and the wavenumber by $1/L_d$, where L_d is the deformation radius and U is such that $u(z) = Uz/H$. If the shear is positive — i.e., increasing toward the east with height, then the increase of the phase with height implies the tilting westward of the wave with height, which in turn allows for the release of potential energy of the disturbance.

where $k_y = 2\pi n/L$ and L is again the meridional extent of the domain. (See problem 7.5.) The meridional boundary conditions, unlike the vertical boundary conditions, are not an essential aspect of the Eady problem.

To satisfy the vertical boundary conditions we look to the thermodynamic equation,

$$\frac{Db}{Dt} + wN^2 = 0. \quad (7.107)$$

or, in terms of streamfunction,

$$\frac{D}{Dt} \left(\frac{\partial \psi}{\partial z} \right) + w \frac{HN^2}{f_0} = 0. \quad (7.108)$$

where again height is scaled by H . Linearizing this gives,

$$\frac{\partial}{\partial t} \left(\frac{\partial \psi'}{\partial z} \right) + \mathbf{u}_0 \cdot \nabla \left(\frac{\partial \psi'}{\partial z} \right) + \mathbf{u}' \cdot \nabla \left(\frac{\partial \psi_0}{\partial z} \right) + w' \frac{N^2}{f_0} = 0 \quad (7.109)$$

which, since $\psi_0 = -Uyz$, is

$$\frac{\partial}{\partial t} \left(\frac{\partial \psi'}{\partial z} \right) + Uz \cdot \nabla \left(\frac{\partial \psi'}{\partial z} \right) - U \frac{\partial \psi'}{\partial x} + w' \frac{N^2 H}{f_0} = 0. \quad (7.110)$$

Applying the boundary conditions at $z = 0$ and $z = 1$ yields

$$\text{at } z = 0 : \quad \frac{\partial}{\partial t} \left(\frac{\partial \psi'}{\partial z} \right) - U \frac{\partial \psi'}{\partial x} = 0 \quad (7.111a)$$

$$\text{at } z = 1 : \quad \frac{\partial}{\partial t} \left(\frac{\partial \psi'}{\partial z} \right) + U \frac{\partial}{\partial x} \left(\frac{\partial \psi'}{\partial z} \right) - U \frac{\partial \psi'}{\partial x} = 0. \quad (7.111b)$$

Substituting (7.101) into these equations finally gives

$$\text{at } z = 0 : \quad -c \frac{\partial \Psi}{\partial z} - U \Psi = 0 \quad (7.112a)$$

$$\text{at } z = 1 : \quad (U - c) \frac{\partial \Psi}{\partial z} - U \Psi = 0. \quad (7.112b)$$

Solutions

Substituting (7.105) in to the linearized equations (7.102) and (7.112) gives

$$(Uz - c) \left[\frac{1}{L_d^2} \frac{\partial^2 \Phi}{\partial z^2} - (k_x^2 + k_y^2) \Phi \right] = 0 \quad (7.113)$$

along with

$$c \frac{d\Phi}{dz} + U \Phi = 0 \quad (7.114a)$$

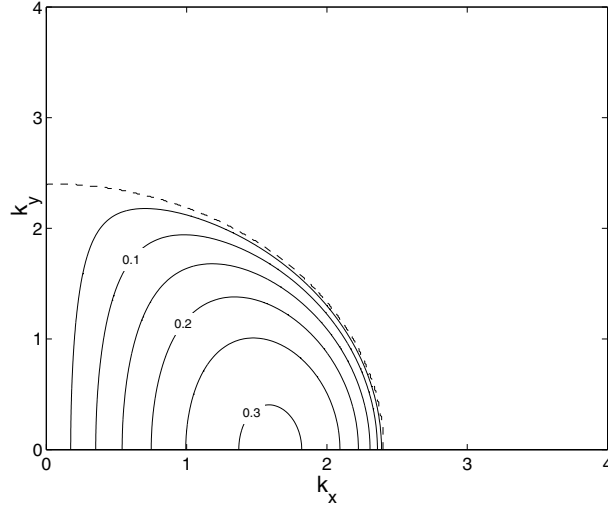
and

$$(c - U) \frac{d\Phi}{dz} + U \Phi = 0 \quad (7.114b)$$

If $Uz \neq c$ then (7.113) becomes

$$\frac{d^2 \Phi}{dz^2} - \mu^2 \Phi = 0 \quad (7.115)$$

Figure 7.12 Contours of growth rate, σ , in the Eady problem, in the k_x - k_y plane using (7.123) and (7.124). Both the wavenumber and growth rate are in the same nondimensional units as in Fig. 7.11. The dimensional wavenumbers are k_x/L_d and k_y/L_d , and the dimensional growth rate is $\sigma U/L_d$. The growth rate peaks near the deformation scale, and for any given zonal wavenumber the most unstable wavenumber is that with the gravest meridional scale.



where

$$\mu^2 = L_d^2(k_x^2 + k_y^2). \quad (7.116)$$

The parameter μ is a nondimensional horizontal wavenumber, scaled by the inverse of Rossby radius of deformation. Solutions of (7.115) are

$$\Phi(z) = A \cosh \mu z + B \sinh \mu z, \quad (7.117)$$

so we see that μ determines the vertical structure of the solution. The boundary conditions (7.114) are satisfied if

$$\begin{aligned} A \times [U] + B \times [\mu c] &= 0, \\ A \times [(c - U)\mu \sinh \mu + U \cosh \mu] + B \times [(c - U)\mu \cosh \mu + U \sinh \mu] &= 0. \end{aligned} \quad (7.118)$$

Equations (7.118) are two coupled homogeneous equations in the two unknowns A and B . Non-trivial solutions will only exist if the determinant of their coefficients (the terms in square brackets) vanish. That is, provided

$$c^2 - Uc + U^2(\mu^{-1} \coth \mu - \mu^{-2}) = 0 \quad (7.119)$$

(where $\coth \mu = \cosh \mu / \sinh \mu$), the solution of which is

$$c = \frac{U}{2} \pm \frac{U}{\mu} \left[\left(\frac{\mu}{2} - \coth \frac{\mu}{2} \right) \left(\frac{\mu}{2} - \tanh \frac{\mu}{2} \right) \right]^{1/2} \quad (7.120)$$

The solution, being proportional to $\exp(-ik_x ct)$, will grow exponentially if c has an imaginary part. Since $\mu/2 > \tanh(\mu/2)$ for all μ , we require that

$$\frac{\mu}{2} < \coth \frac{\mu}{2} \quad (7.121)$$

which is satisfied when

$$\mu < \mu_c \quad (7.122)$$

where $\mu_c = 2.399$.

The growth rates of the instabilities themselves are given by the imaginary part of (7.120), multiplied by the x-wavenumber. That is

$$\sigma = kc_i = k \frac{U}{\mu} + k \frac{U}{\mu} \left[\left(\frac{\mu}{2} - \coth \frac{\mu}{2} \right) \left(\frac{\mu}{2} - \tanh \frac{\mu}{2} \right) \right]^{1/2} \quad (7.123)$$

The maximum growth rate occurs when $\mu = 1.61$ and the associated the (nondimensional) growth rate is 0.31. See Fig. 7.11 and Fig. 7.12.

7.7.2 Scales and dimensionalization

The scaled x- and y- wavenumbers of Fig. 7.12 are such that

$$k_x^2 + k_y^2 = \mu^2. \quad (7.124)$$

In Fig. 7.11 we plot the most unstable wave in a periodic box, with $k_y = 0$, so that $k_x = \mu$.

The associated *dimensional* wavenumbers are given by

$$\begin{aligned} k_x^* &= k_x / L_d \\ k_y^* &= k_y / L_d \end{aligned} \quad (7.125)$$

so that the unstable wavenumbers occur for

$$k^* < \mu_c / L_d = 2.4 / L_d \quad (7.126)$$

The corresponding unstable wavelengths occur for

$$\lambda > \lambda_c = \frac{2\pi L_d}{\mu_c} = 2.6 L_d \quad (7.127)$$

The wavelength at which the instability is greatest is

$$\lambda_U = \frac{2\pi L_d}{\mu_u} = 3.9 L_d \quad (7.128)$$

where $\mu_u = 1.61$ is the scaled wavenumber at which the instability is greatest. In the Eady problem the wave-speeds are scaled by U , and the growth rates are scaled by U / L_d . That is, a natural timescale in the Eady problem is given by

$$T_e = \frac{L_d}{U} = \frac{NH}{f_0 U} = \frac{1}{Fr f_0} = \frac{\sqrt{Ri}}{f_0}, \quad (7.129)$$

(where Fr and Ri are the Froude and Richardson numbers, respectively), sometimes known as the Eady timescale; its inverse is the Eady growth rate.

Atmospheric parameters

We choose

$$H \sim 10 \text{ km} \quad U \sim 10 \text{ m s}^{-1} \quad N \sim 10^{-2} \text{ s}^{-1}. \quad (7.130)$$

We then obtain the following:

Deformation Radius:

$$L_d = \frac{NH}{f} \approx \frac{10^{-2} 10^4}{10^{-4}} \approx 1000 \text{ km} \quad (7.131)$$

Scale of maximum instability:

$$L_{\max} \approx 3.9 L_d \approx 4000 \text{ km} \quad (7.132)$$

Growth Rate:

$$\sigma \sim 0.31 \frac{U}{L_d} \approx \frac{0.31 \times 10}{10^6} \text{ s}^{-1} \approx 0.26 \text{ day}^{-1} \quad (7.133)$$

Oceanic parameters (in main thermocline)

We choose

$$H \sim 1 \text{ km} \quad U \approx 0.1 \text{ m s}^{-1} \quad N \sim 10^{-2} \text{ s}^{-1} \quad (7.134)$$

We then obtain the following:

Deformation Radius:

$$L_d = \frac{NH}{f} \approx \frac{10^{-2} 10^3}{10^{-4}} = 100 \text{ km} \quad (7.135)$$

Scale of maximum instability:

$$L_{\max} \approx 3.9 L_d \approx 400 \text{ km} \quad (7.136)$$

Growth Rate:

$$\sigma \sim \frac{0.3U}{L_d} \approx \frac{0.3 \times 0.1}{10^5} \text{ s}^{-1} \approx 0.026 \text{ day}^{-1} \quad (7.137)$$

In the ocean, the Eady problem is not really quantitatively applicable because of the non-uniformity of the stratification, and we cannot simply impose a lid at the base of the thermocline. Nevertheless, the above estimates give a qualitative sense of the scale and growth rate of the instability relative to the corresponding values in the atmosphere.

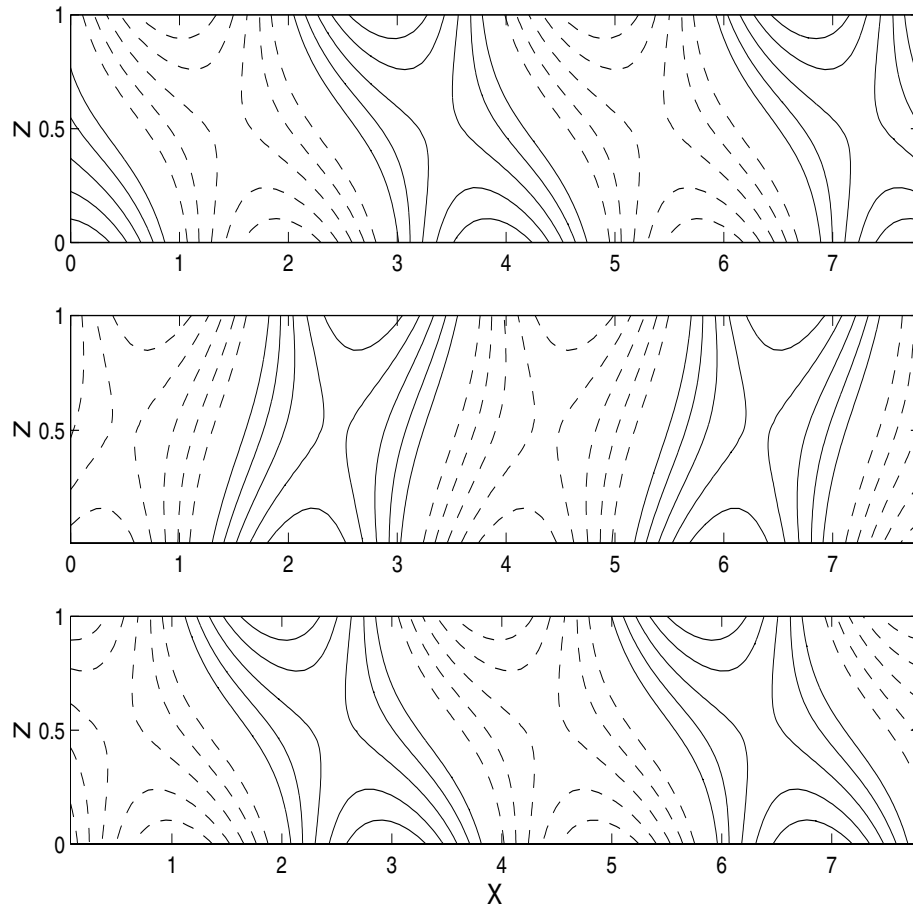


Figure 7.13 Vertical structure of the most unstable Eady mode. Top: contours of streamfunction. Middle: temperature, proportional to $\partial\psi/\partial z$. Bottom: meridional velocity, proportional to $\partial\psi/\partial y$. Negative contours are dashed, and two complete wavelengths are present in the horizontal. Poleward flowing (positive v) air is generally warmer than equatorward flowing air.

7.7.3 Summary

Summarizing, important properties of the Eady model are:

- (i) The length-scale of the instability is similar to or a few times larger than the deformation scale.
- (ii) The growth rate of the instability is

$$\sigma \sim \frac{U}{L_d} = \frac{U}{H} \frac{f_0}{N} \quad (7.138)$$

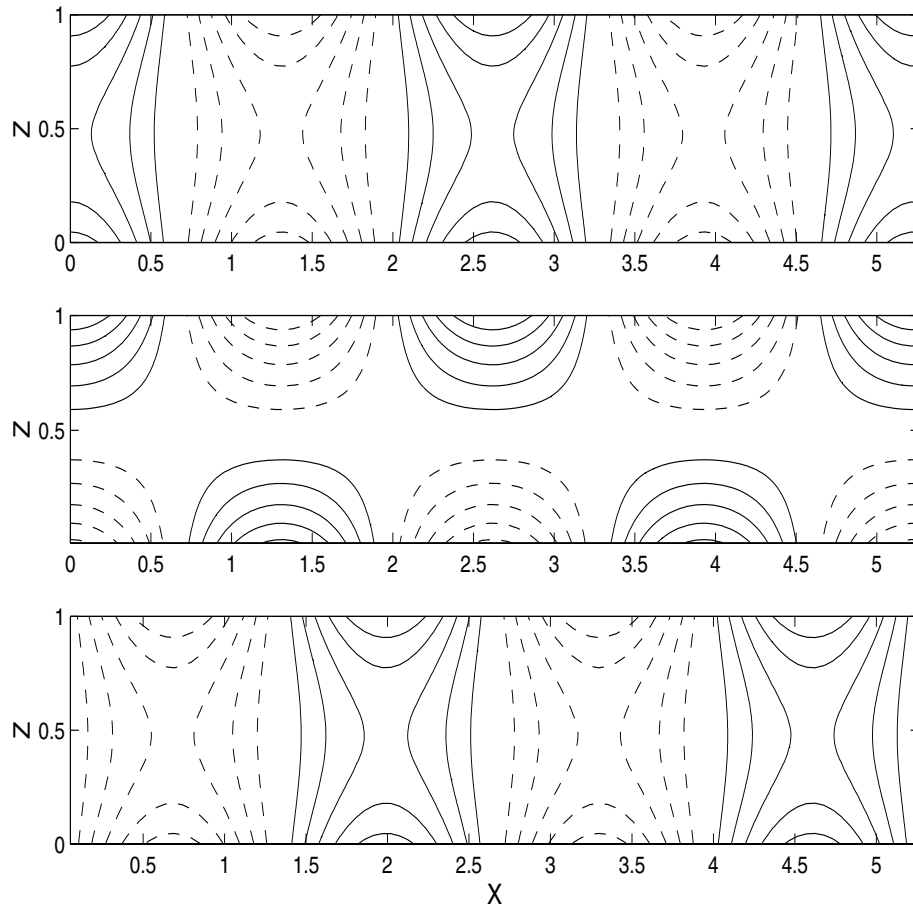


Figure 7.14 As for Fig. 7.13, but now for a wave just beyond the short-wave cut-off. There is no phase-tilt in the vertical, and the temperature perturbations at the upper and lower boundaries are no longer able to interact. At still higher wavenumbers, the perturbations become more confined to the upper or lower boundaries.

That is, it is proportional to the shear, and scaled by the Prandtl ratio f_0/N .

- (iii) There is a *short-wave cutoff* beyond which (i.e., at higher wavenumber than) there is no instability. This occurs near the deformation radius.
- (iv) The most unstable waves for a given zonal scale are those with the gravest meridional scale.
- (v) The Eady waves grow by way of an interference between waves at the upper and lower boundaries. The instabilities must therefore have a fairly simple vertical structure. If either boundary is removed, the instability dies.

7.8 TWO-LAYER BAROCLINIC INSTABILITY

We saw that the eigenfunctions displaying the largest growth rates in the Eady problem have a relatively simple vertical structure. This suggests that an even simpler mathematical model of baroclinic instability might be constructed in which the vertical structure is *a priori* restricted to a very simple form. A single-layer quasi-geostrophic is too simple, since the complete absence of vertical structure will eliminate the mechanism of baroclinic instability. However, the two-layer quasi-geostrophic equations do have sufficient structure, and are probably the simplest model of baroclinic instability. We will find that an advantage over the Eady model is that it will be possible to include the β -effect in a simple way.

We begin with the two-layer potential vorticity equations, which we write in the dimensional form,

$$\frac{D}{Dt} \left\{ \zeta_i + \beta y + \frac{k_d^2}{2} (\psi_j - \psi_i) \right\} = 0 \quad (7.139)$$

where $i = 1, 2$ and $j = 3 - i$ and

$$\frac{k_d^2}{2} = \left(\frac{2f_0}{NH} \right)^2 \implies k_d = \frac{\sqrt{8}}{L_d}, \quad (7.140)$$

where H is the total depth of the domain, as in the Eady problem.

The basic state we choose is:

$$\begin{aligned} \bar{\psi}_1 &= -U_1 y \\ \bar{\psi}_2 &= -U_2 y = +U_1 y \end{aligned} \quad (7.141)$$

Because there is no topography the system is Galilean invariant, and we may choose $U_2 = -U_1$. The basic state does imply a non-zero basic state potential vorticity gradient, given by

$$\begin{aligned} \bar{q}_1 &= \beta y + k_d^2 U_1 y \\ \bar{q}_2 &= \beta y - k_d^2 U_1 y \end{aligned} \quad (7.142)$$

Note though, that in the absence of β there is a non-zero potential vorticity gradient. Why should this be different from the Eady problem? After all, the shear is uniform in both problems. The difference arises from the vertical boundary conditions. In the standard layered formulation the temperature gradient at the boundary conditions is absorbed into the definition of the potential vorticity in the interior. This results in a nonzero interior potential vorticity gradient at the two levels adjacent to the boundary (the only layers in the two-layer problem), but isothermal boundary conditions $D/Dt(\partial\psi/\partial z) = 0$. In the Eady problem we have a zero interior gradient of potential vorticity but a temperature gradient at

the boundary. The two formulations are equivalent — a simple finite difference example of the ‘Bretherton’s boundary layer’.

The linearized potential vorticity equation is, for each layer,

$$\frac{\partial q'_i}{\partial t} + U_i \frac{\partial q'_i}{\partial x} + v'_i \frac{\partial \bar{q}_i}{\partial y} = 0 \quad (7.143)$$

which explicitly is

$$\frac{\partial q'_1}{\partial t} + U_1 \frac{\partial}{\partial x} \left\{ \nabla^2 \psi'_1 + \frac{k_d^2}{2} (\psi'_2 - \psi'_1) \right\} + \frac{\partial \psi'_1}{\partial x} (\beta + k_d^2 U_1) = 0 \quad (7.144a)$$

$$\frac{\partial q'_2}{\partial t} + U_2 \frac{\partial}{\partial x} \left\{ \nabla^2 \psi'_2 + \frac{k_d^2}{2} (\psi'_1 - \psi'_2) \right\} + \frac{\partial \psi'_2}{\partial x} (\beta - k_d^2 U_1) = 0 \quad (7.144b)$$

For simplicity we will set the problem in a square doubly-periodic domain, and so seek solutions in the form,

$$\psi'_i = \text{Re } \Psi_i e^{i(k_x x + k_y y - \omega t)} = \text{Re } \Psi_i e^{ik_x(x-ct)} e^{ik_y y}, \quad i = 1, 2 \quad (7.145)$$

Here, k_x and k_y are the x - and y -wavenumbers, and $(k_x, k_y) = (2\pi/L)(m, n)$ where L is the size of the domain and m and n are integers. The constant Ψ_i is the complex amplitude.

The solution

Substituting (7.145) into (7.144) we obtain

$$[ik_x(U_1 - c)] \left[-k^2 \Psi_1 + \frac{k_d^2}{2} (\Psi_2 - \Psi_1) \right] + ik_x \Psi_1 (\beta + k_d^2 U_1) = 0 \quad (7.146)$$

and

$$[-ik_x(U_1 + c)] \left[-k^2 \Psi_2 + \frac{k_d^2}{2} (\Psi_1 - \Psi_2) \right] + ik_x \Psi_1 (\beta - k_d^2 U_1) = 0 \quad (7.147)$$

where $k^2 = k_x^2 + k_y^2$. Adding and subtracting these two equations gives

$$\left[(U_1 - c) \left(\frac{k_d^2}{2} + k^2 \right) - (\beta + k_d^2 U_1) \right] \Psi_1 - \left[\frac{k_d^2}{2} (U_1 - c) \right] \Psi_2 = 0 \quad (7.148a)$$

and

$$- \left[\frac{k_d^2}{2} (U_1 + c) \right] \Psi_1 + \left[(U_1 + c) \left(\frac{k_d^2}{2} + k^2 \right) + (\beta - k_d^2 U_1) \right] \Psi_2 = 0 \quad (7.148b)$$

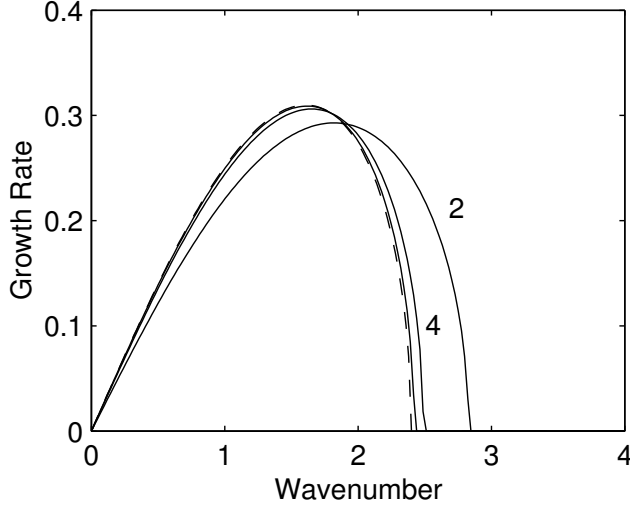


Figure 7.15 Growth rates for models with varying numbers of vertical layers, all with $\beta = 0$ and the same uniform stratification and shear. The dashed line is the analytic solution to the continuous (Eady) problem, and the solid lines are results obtained using two layers (obtained analytically or numerically), and four and eight layers (obtained numerically). The two and four layer results are labelled, and the eight layer result is almost coincident with the dashed line. Results with 16 layers (not shown) are indistinguishable from the Eady result.

Equations (7.148a) are of the form

$$\begin{aligned} \mathbf{A}\Psi_1 + \mathbf{B}\Psi_2 &= 0 \\ \mathbf{C}\Psi_1 + \mathbf{D}\Psi_2 &= 0 \end{aligned} \quad (7.149)$$

where $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ correspond to the terms in square brackets. For nontrivial solutions the determinant of coefficients must be zero. That is,

$$\mathbf{AD} - \mathbf{BC} = 0 \quad (7.150)$$

This gives a quadratic equation in c and solving this we obtain

$$c = -\frac{\beta}{k^2 + k_d^2} \left\{ 1 + \frac{k_d^2}{2k^2} \pm \frac{k_d^2}{2k^2} \left[1 + \frac{4k^4(k^4 - k_d^4)}{k_\beta^4 k_d^4} \right]^{1/2} \right\}, \quad (7.151)$$

where $k_\beta = \sqrt{\beta/U}$; its inverse is known as the Kuo scale. We may nondimensionalize this equation using the deformation radius L_d as the length scale and the shear velocity U as the velocity scale.⁷ Then, denoting non-dimensional parameters with hats, we have

$$k = \frac{\hat{k}}{L_d}, \quad c = \hat{c}U, \quad t = \frac{L_d}{U} \hat{t} \quad (7.152)$$

and the nondimensional form of (7.153) is just

$$\hat{c} = -\frac{\hat{k}_\beta^2}{\hat{k}^2 + \hat{k}_d^2} \left\{ 1 + \frac{\hat{k}_d^2}{2\hat{k}^2} \pm \frac{\hat{k}_d^2}{2\hat{k}^2} \left[1 + \frac{4\hat{k}^4(\hat{k}^4 - \hat{k}_d^4)}{\hat{k}_\beta^4 \hat{k}_d^4} \right]^{1/2} \right\} \quad (7.153)$$

where $\hat{k}_\beta = k_\beta L_d$ and $\hat{k}_d = \sqrt{8}$, as before. The nondimensional parameter

$$\gamma = \frac{1}{4} \hat{k}_\beta^2 = \frac{\beta L_d^2}{4U}, \quad (7.154)$$

is often useful as a measure of the importance of β ; it is proportional to the square of the ratio of the deformation radius to the Kuo scale $\sqrt{U/\beta}$. Generalizing to the continuous case, if the basic-state shear is given by $\Lambda = \partial U / \partial z$ then the corresponding nondimensional number, the Charney-Green number,⁸ is

$$\gamma = \frac{\beta N^2 H}{f_0^2 \Lambda}. \quad (7.155)$$

where H is an appropriate vertical domain scale (see also section 7.9.1). Typically, the Charney-Green number is somewhat less than one for baroclinic instability problems in the earth's atmosphere, and considerably less than one for such problems in the ocean. Let us look at two special cases first, before considering the general solution to these equations.

(i) *Zero Shear*

If there is no shear (i.e., $U = 0$) then (7.148a) and (7.148b) are identical and two roots of the equation give the purely real phase speeds c

$$c = -\frac{\beta}{k^2}, \quad c = -\frac{\beta}{k^2 + k_d^2} \quad (7.156)$$

The first of these is the dispersion relationship for Rossby waves in a purely barotropic flow, and corresponds to the eigenfunction $\Psi_1 = \Psi_2$. The second solution corresponds to the baroclinic eigenfunction $\Psi_1 + \Psi_2 = 0$.

(ii) *Zero β*

If $\beta = 0$, then (7.148a) and (7.148b) easily yield

$$c = \pm U_1 \left[\frac{k_d^2 - k^2}{k_d^2 + k^2} \right]^{1/2} \quad (7.157)$$

or, defining the growth rate σ by $\sigma = -i\omega$,

$$\sigma = U_1 k_x \left[\frac{k_d^2 - k^2}{k^2 + k_d^2} \right]^{1/2} \quad (7.158)$$

These expressions have similar properties to those for the Eady problem, as illustrated in Fig. 7.15. We note that:

- There is an instability for *all* values of U , for $|\mathbf{k}| < \hat{k}_d$.

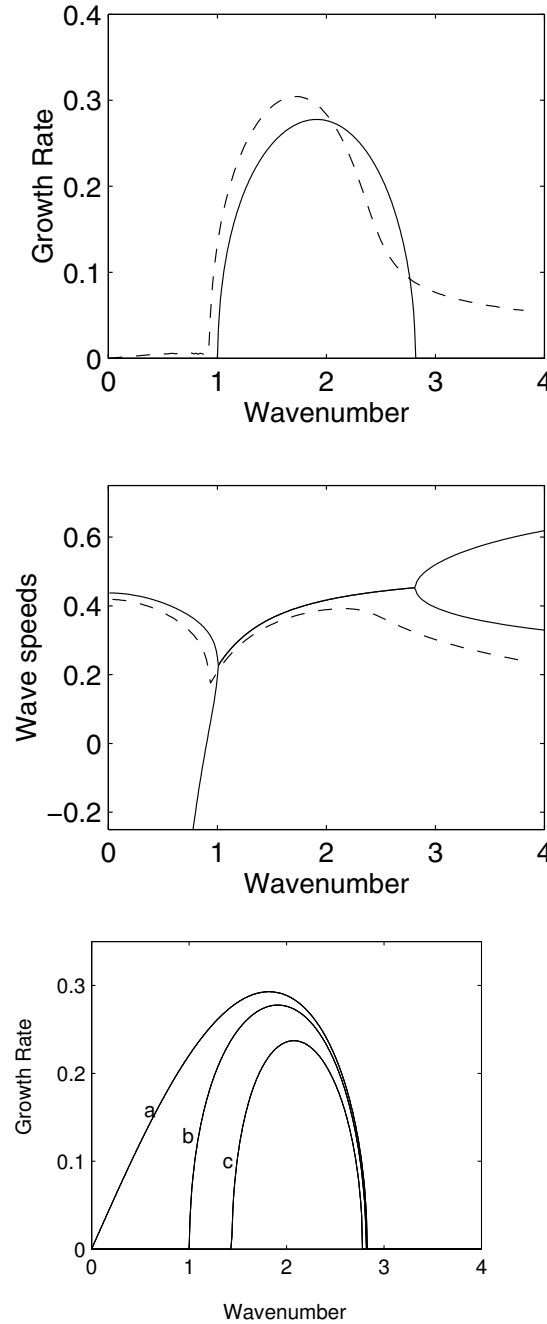


Figure 7.16 Growth rates and wave speeds for the two-layer (solid) and continuous (dashed) models, with the same (nonzero) β and the same uniform shear and stratification. [The two-layer solutions, from (7.153) have $\hat{k}_d = \sqrt{8}$, $k_\beta = \sqrt{2}$ and Charney-Green number $\gamma = \beta L_d^2 / [2(U_1 - U_2)] = 0.5$.] In the continuous case only the (real) wave speed associated with the unstable mode is shown. In the two-layer case there are two real wave speeds which coalesce in the unstable region. The two-layer model has an abrupt short-wave and long-wave cut-off, whereas the growth rate of the continuous model tails off gradually at small wavelengths, and has a weak instability (the ‘Green modes’) at large wavelengths.

Figure 7.17 Growth rates and wave speeds for the two-layer baroclinic instability problem, from (7.153), with three (nondimensional) values of β : a, $\gamma = 0$ ($k_\beta = 0$); b, $\gamma = 0.5$ ($k_\beta = \sqrt{2}$); c, $\gamma = 1$ ($k_\beta = 2$). As β increases, so does the low-wavenumber cut-off to instability, but the high-wavenumber cut-off is little changed. (The solutions are obtained from (7.153), with $\hat{k}_d = \sqrt{8}$ and $U_1 = -U_2 = 1/4$.)

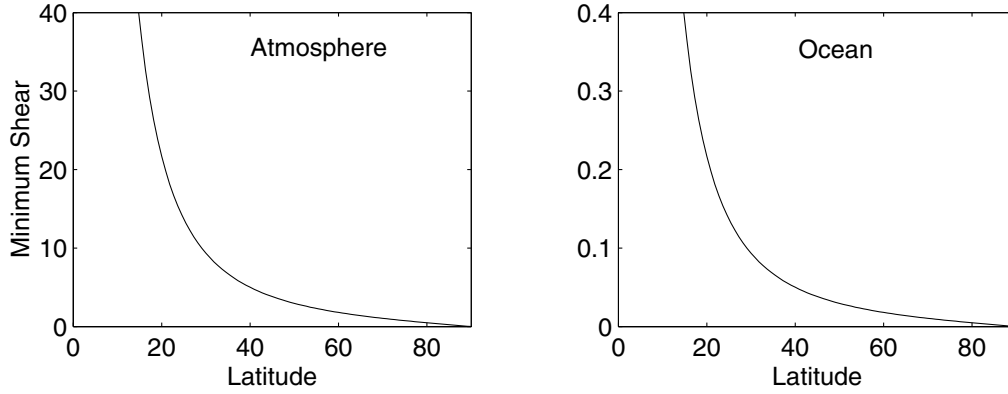


Figure 7.18 The minimum shear ($U_s = U_1 - U_2$, in m/s) required for baroclinic instability in a two-layer model, calculated using (7.167), i.e. $U_s = \beta L_d^2 / 4$ where $\beta = 2\Omega a^{-1} \cos \vartheta$ and $L_d = NH/f$, where $f = 2\Omega \sin \vartheta$. The left panel uses atmospheric parameters of $H = 10$ km and $N = 10^{-2} \text{ s}^{-1}$, the right panel uses parameters representative of the main thermocline in the ocean, $H = 1$ km and $N = 10^{-2} \text{ s}^{-1}$. Because of the nonuniform stratification and nonzonal flow in the ocean, these values are likely to be even less quantitatively accurate than those of the atmosphere. However, the implication that the minimum shear is much less for the ocean, and that both in the ocean and atmosphere the minimum shear increases rapidly at low latitudes, is quite robust.

- There is a high-wavenumber cut-off, at the radius of deformation. That is, if $k > k_d = 2.82/L_d$ there is no growth. (For the Eady problem, the high wavenumber cut-off occurs at $2.4/L_d$.)
- There is no low wavenumber cut-off.
- For any given k , the highest growth rate occurs for $l = 0$. From (7.158), for $l = 0$ the maximum growth rate occurs when $k = 0.634\hat{k}_d = 1.79/L_d$. (For the Eady problem, the maximum growth rate occurs at $1.61/L_d$.)

The general case

The growth rate and wave speeds as function of wavenumber are plotted in Fig. 7.16 and Fig. 7.17. We observe that there still appears to be a high wave number cut-off and, interestingly, there is a low-wavenumber cut-off. A little elementary analysis elucidates these features.

The neutral curve

For instability, there must be an imaginary component to the phase speed in (7.153). That is, we require

$$k_p^4 k_d^4 + 4k^4(k^4 - k_d^4) < 0 \quad (7.159)$$

This may be considered a quadratic equation in k^4 for the value of k , k_c say, at which the growth rate is zero. Solving, we find

$$k_c^4 = \frac{1}{2}k_d^4 \left(1 \pm \sqrt{1 - k_\beta^4/k_d^4} \right). \quad (7.160)$$

If $k_\beta \ll k_d$ then we can expand the term under the square-root sign. Taking the negative values of the square root, we find

$$k_c^2 \approx \frac{\beta}{2U} \quad (7.161)$$

or, equivalently, an approximate expression for the critical shear U_c in terms of wavenumber,

$$U_c \approx \frac{\beta}{2k^2}. \quad (7.162)$$

This is valid for wavenumbers to the left of the minimum shear (see Fig. 7.19). Taking the positive value of the square root, and for $k \approx k_d$, we find that the neutral curve is approximately given by

$$k_d^4 - k_c^4 \approx \frac{\beta^2}{4U^2} \quad (7.163)$$

and that, equivalently, the critical shear for instability is then

$$U_c^2 \approx \frac{\beta^2}{4(k_d^4 - k^4)}. \quad (7.164)$$

This expression holds for wavenumbers higher than that at which the critical shear is a minimum.

Minimum shear for instability

From (7.159), instability arises when $\beta^2 \hat{k}_d^4 / U^2 < 4k^4(\hat{k}_d^4 - k^4) < 0$. The maximum value of the right-hand-side of this expression arises when $k^4 = \hat{k}_d^4/2$; thus, instability arises only when

$$\frac{\beta^2 k_d^4}{U^2} < 4 \frac{k_d^4}{2} \frac{k_d^4}{2} \quad (7.165)$$

or

$$\boxed{U_s > \frac{2\beta}{k_d^2}} \quad (7.166)$$

where $U_s = U_1 - U_2 = 2U$. In terms of the deformation radius itself the minimum shear for instability is

$$U_s > \frac{1}{4}\beta L_d^2 \quad (7.167)$$

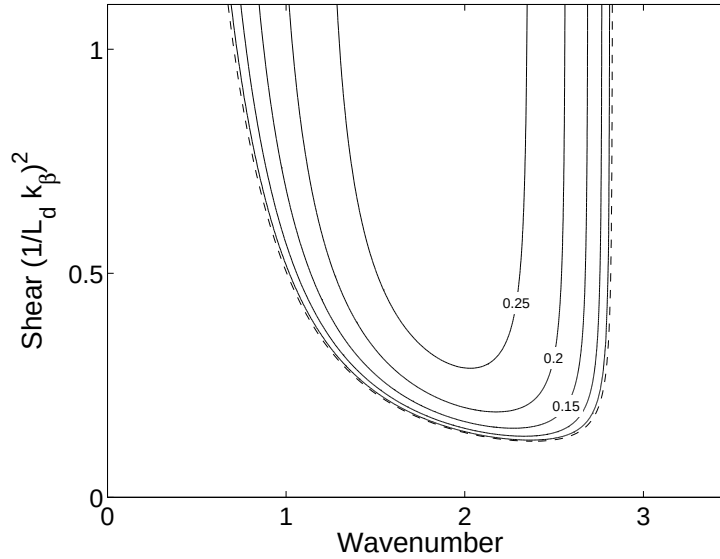


Figure 7.19 Contours of growth rate in the two-layer baroclinic instability problem. The dashed line is the neutral stability curve obtained from (7.161), and the other curves are contours of growth rates obtained from (7.153). Outside of the dashed line, the flow is stable. The deformation radius is $\sqrt{8}$. The growth rates are scaled by the inverse of the Eady timescale, i.e., by U/L_d . Thus, for $L_d = 1000$ km and $U = 10$ m s $^{-1}$, a nondimensional growth rate of 0.25 corresponds to a dimensional growth rate of $0.25 \times 10^{-5} \text{ s}^{-1} = 0.216 \text{ day}^{-1}$.

(In this formula, the shear is the difference in the velocity between level 1 and level 2, whereas the deformation radius, NH/f_0 , is based on the total height of the fluid. If we were to use half the depth of the fluid in the definition of the deformation radius, the factor of 4 would cancel.) If the shear is just this critical value, the instability occurs at $k = 2^{-1/4}k_d = 0.84k_d = 2.37/L_d$. As the shear increases, the wavenumber at which the growth rate is maximum decreases slightly (see Fig. 7.19), and for a sufficiently large shear the β -effect is negligible and the wavenumber of maximum instability is, as we saw earlier, $0.634 k_d$ or $1.79/L_d$.

Note the relationship of the minimum shear to the basic state potential vorticity gradient in the respective layers. In the upper layer the potential vorticity gradient is given by

$$Q_y = \beta + k_d^2 U \quad (7.168)$$

and in the lower layer,

$$Q_y = \beta - k_d^2 U \quad (7.169)$$

Thus, the requirement for instability is exactly that which causes the potential vorticity gradient to change sign somewhere in the domain, in this case in

the lower layer. This is an example of the general rule that potential vorticity (suitably generalized to include the surface boundary conditions) must change sign somewhere in the domain in order for there to be an instability.

High-wavenumber cut-off

Instability can only arise when, from (7.159),

$$4k^4(k_d^4 - k^4) > k_\beta^4 k_d^4, \quad (7.170)$$

so that a necessary condition for instability is

$$k_d^2 > k^2. \quad (7.171)$$

Thus, waves shorter than the deformation radius are always stable, no matter the value of β . We also see from Fig. 7.17 and Fig. 7.19 that the high wavenumber cut-off in fact varies little with β if $k_d \gg k_\beta$. Note that the critical shear required for instability approaches infinity as k approaches k_d .

Low-wavenumber cut-off

Suppose that $k \ll k_d$. Then (7.159) simplifies to $k_\beta^4 \hat{k}_d^4 < 4k^4 \hat{k}_d^4$. That is, for instability we require

$$k^2 > \frac{1}{2} k_\beta^2 = \frac{\beta}{2U}. \quad (7.172)$$

Thus, using (7.172) and (7.172) the unstable waves lie approximately in the interval $\beta / (\sqrt{2}U) < k < k_d$.

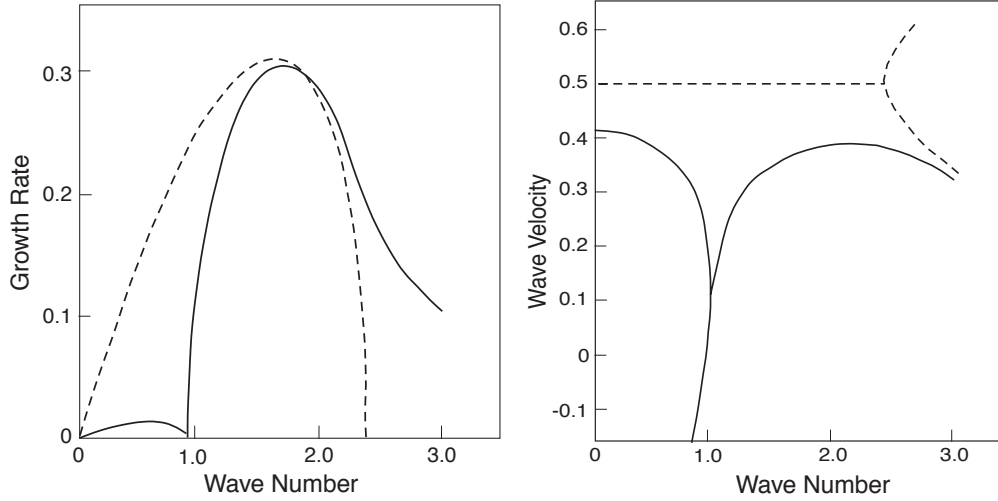


Figure 7.20 Growth rate (left) and wave speed (right) in a fluid with a uniform shear ($u = z$) and uniform stratification ($N^2 = 1$). Dashed line: $\beta = 0$, the Eady problem; solid line, $\gamma = 0.5$. See also Fig. 7.16.

7.9 * BETA, SHEAR AND STRATIFICATION IN A CONTINUOUS MODEL

The two-layer model illustrates many of the qualitative effects of β on baroclinic instability. Do these carry over to the continuously stratified case? The answer by-and-large is yes, but with some important qualifications that generally concern weak or shallow instabilities. In particular, we will find that there is no short-wave cut-off in the continuous model with non-zero beta, and that the instability determines its own depth scale. We will illustrate these properties first by way of scaling arguments, and then by way of numerical calculations of the instability properties.⁹

7.9.1 Scaling arguments for growth rates, scales and depth

With finite density scale height and non-zero β , the potential vorticity equation, linearized about a mean zonal velocity $\bar{u}(z)$, is

$$\left(\frac{\partial}{\partial t} + \bar{u} \frac{\partial}{\partial x} \right) q' + \frac{\partial \psi'}{\partial x} \frac{\partial \bar{q}}{\partial y} = 0 \quad (7.173)$$

where

$$q' = \nabla^2 \psi' + \frac{f_0^2}{\rho} \frac{\partial}{\partial z} \left(\frac{\rho}{N^2} \frac{\partial \psi'}{\partial z} \right) \quad (7.174)$$

ρ is a specified density profile and

$$\frac{\partial \bar{q}}{\partial y} = \beta - \frac{f_0^2}{\rho} \frac{\partial}{\partial z} \left(\frac{\rho}{N^2} \frac{\partial \bar{u}}{\partial z} \right). \quad (7.175)$$

If we assume that $\bar{u} = \Lambda z$ where Λ is constant and that N is constant, and let $H^{-1} = -\rho^{-1} \partial \rho / \partial z$, then

$$\frac{\partial \bar{q}}{\partial y} = \beta + \frac{f_0^2 \Lambda}{N^2 H} = \beta(1 + \alpha) \quad (7.176)$$

where

$$\alpha = \left(\frac{f_0^2 \partial \bar{u} / \partial z}{\beta N^2 H} \right), \quad (7.177)$$

which is a nondimensional quantity of fundamental importance. The boundary conditions on (7.173) are

$$\left(\frac{\partial}{\partial t} + \bar{u} \frac{\partial}{\partial x} \right) \frac{\partial \psi'}{\partial z} - \frac{\partial \psi'}{\partial x} \frac{\partial \bar{u}}{\partial z} = 0, \quad \text{at } z = 0 \quad (7.178)$$

and that $\psi \rightarrow 0$ as $z \rightarrow \infty$. The problem we have defined essentially constitutes the Charney problem. We can reduce this to the Eady problem by setting $\beta = 0$ and $H = \infty$, and providing a lid some finite height above the ground.

As in the Eady problem, we seek solutions of the form

$$\psi = \text{Re } \Psi(z) e^{i(kx + ly - kct)} \quad (7.179)$$

and substituting into (7.173) gives

$$\left(\frac{f_0^2}{H^2 N^2} \right) \left(H^2 \frac{d^2 \Psi}{dz^2} - H \frac{d\Psi}{dz} \right) - \left(K^2 - \frac{\beta + \Lambda f_0^2 / (N^2 H)}{\Lambda z - c} \right) \Psi = 0 \quad (7.180)$$

The Boussinesq version of this expression for a fluid contained between two horizontal surfaces is just

$$\left(\frac{f_0^2}{N^2} \right) \frac{d^2 \Psi}{dz^2} - \left(K^2 - \frac{\beta}{\Lambda z - c} \right) \Psi = 0 \quad (7.181)$$

It seems natural to nondimensionalize (7.180) using:

$$z = H \hat{z}, \quad c = \Lambda H \hat{c}, \quad K = \left(\frac{f_0}{NH} \right) \hat{K}, \quad (7.182)$$

whence it becomes

$$\frac{d^2 \Psi}{d\hat{z}^2} - \frac{d\Psi}{d\hat{z}} - \left(\hat{K}^2 - \frac{\gamma + 1}{\hat{z} - \hat{c}} \right) \Psi = 0 \quad (7.183)$$

where

$$\gamma = \alpha^{-1} = \frac{\beta N^2 H}{f_0^2 \Lambda} = \frac{H}{h}. \quad (7.184)$$

where $h \equiv \Lambda f_0^2 / (\beta N^2)$. The Boussinesq version, (7.181), may be non-dimensionalized using H_l in place of H , where H_l is the depth of the fluid between two rigid surfaces. In that case

$$\frac{d^2 \Psi}{d\hat{z}^2} - \left(\hat{K}^2 - \frac{\gamma}{\hat{z} - \hat{c}} \right) \Psi = 0, \quad (7.185)$$

where here the non-dimensional variables are scaled with H_l .

Now, suppose that γ is large, meaning that either β or the static stability are large or the shear is weak. Eq. (7.183) admits of no non-trivial balance, suggesting that we rescale the variables using h instead of H as the vertical scale in (7.182). The rescaled version of (7.183) is then

$$\frac{d^2 \Psi}{d\hat{z}^2} - \frac{1}{\gamma} \frac{d\Psi}{d\hat{z}} - \left(\hat{K}^2 - \frac{1 + \gamma^{-1}}{\hat{z} - \hat{c}} \right) \Psi = 0, \quad (7.186)$$

or, approximately,

$$\frac{d^2 \Psi}{d\hat{z}^2} - \left(\hat{K}^2 - \frac{1}{\hat{z} - \hat{c}} \right) \Psi = 0. \quad (7.187)$$

This is exactly the same equation as results from a similar rescaling of the Boussinesq system, (7.185), as we might have expected because now the dynamical vertical scale is much smaller than the scale height and the system is essentially Boussinesq. Thus, noting that (7.187) has the same nondimensional form as (7.185) save that γ is replaced by unity, and that (7.187) with $\gamma = 1$ must produce the same scales and growth rates as in the Eady problem, we may deduce that:

- (i) The wavelength of the instability is $\mathcal{O}(Nh/f_0)$.
- (ii) The growth rate of the instability is $\mathcal{O}(Kc) = \mathcal{O}(f_0 \Lambda / N)$.
- (iii) The vertical scale of the instability is $\mathcal{O}(h) = \mathcal{O}(f_0^2 \Lambda / (\beta N^2))$.

These are the same values as for the Eady problem, except with the dynamical height h replacing the geometric or scale height H .

In the limit $\gamma \ll 1$, the Boussinesq and compressible problems differ. The Boussinesq problem reduces to the Eady problem, considered previously, whereas (7.183) becomes, approximately,

$$\frac{d^2 \Psi}{d\hat{z}^2} - \frac{d\Psi}{d\hat{z}} - \left(\hat{K}^2 - \frac{1}{\hat{z} - \hat{c}} \right) \Psi = 0, \quad (7.188)$$

and in this limit the appropriate vertical scale is H .

The scale h does not arise in the two-level model, but there is a connection between it and the critical shear for instability in the two-level model. From the definition of h we have

$$h\Lambda = \beta \left(\frac{Nh}{f_0} \right)^2, \quad (7.189)$$

and in a two-level model we previously derived the necessary condition for instability, (7.167), that

$$(U_1 - U_2) > \beta \left(\frac{NH'}{f_0} \right)^2 \quad (7.190)$$

where H' is the vertical distance between the two levels. Thus, when the shear is strong (or when β is small), the two-level model is unstable and the continuous model produces deep modes. When the shear is weak or when β is large, the two-level model produces no instability, and the continuous model produces shallow unstable modes of depth h . In the two-level model, all modes are 'deep', and the model fails (as it should) to capture the shallow modes of the continuous system. For similar reasons, there is a high-wavenumber cut-off in the two-level model: In the continuous model these modes are shallow, and cannot be captured by two-level dynamics that demand that modes are deep. It is left as an exercise for the reader to demonstrate how the characteristic vertical scale of the waves decreases as wavenumber increases. Interestingly, and somewhat counter-intuitively, for these modes the β -effect must be important, even though the modes are small scale, because when $\beta = 0$ the instability arises via an interaction between edge waves at the top and bottom of the domain and, because high-wavenumber modes are shallow, there is a high-wavenumber cut-off.

7.9.2 Some numerical calculations

Our first step is to add the β -effect to the Eady problem.¹⁰ That is, we suppose a Boussinesq fluid with uniform stratification, that the shear is zonal and constant, and that the entire problem is sandwiched between two rigid surfaces. Growth rates and phase speeds of such an instability calculation are illustrated in Fig. 7.20. As in the two-layer problem, there is a low-wavenumber cut-off to the instability although there is a weak instability at very large-scales. These so-called *Green modes* have no counterpart in the two-layer model — they are deep, slowly growing modes that may or may not be important in the real atmosphere, the latter being an option because the much faster growing instabilities closer to the deformation scale will soon dominate in any real situation. Furthermore, the fact that the Green modes have a scale much larger than the deformation scale suggest that a degree of caution in the accuracy of the quasi-geostrophic calculation is warranted.

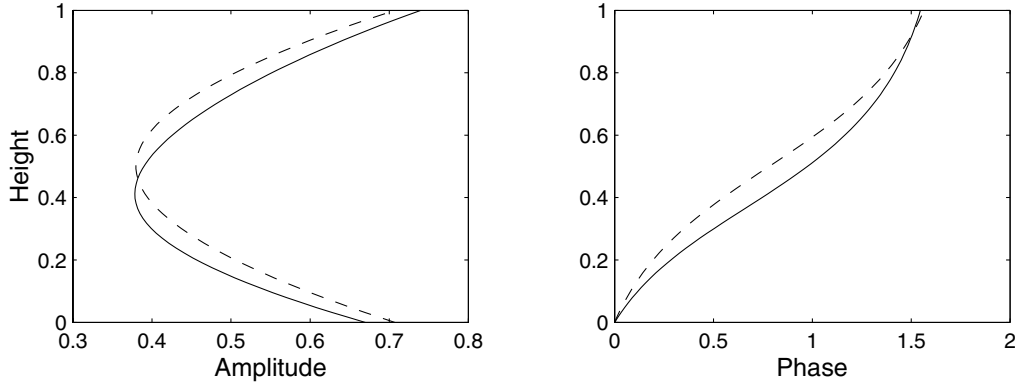


Figure 7.21 Vertical structure of the most unstable modes in instability calculation with $\beta = 0$ (dashed lines, the Eady problem) and $\beta \neq 0$ (solid lines), as in Fig. 7.20. The effect of beta is to depress the height of maximum amplitude of the instability.

7.9.3 Effects of nonuniform shear and stratification

If the shear or stratification is non-uniform an analytic treatment is, even in problems without β , usually impossible and the resulting equations must be solved numerically. However, if we restrict attention to a *discontinuity* in the shear or the stratification, then the resulting problem is very similar to the problem with rigid boundaries, and this property provides much of the justification for using the Eady problem to model instabilities in the earth's atmosphere: in the troposphere the stratification is (approximately) constant, and the rapid increase in stratification in the stratosphere can be approximated by a lid at tropopause. Heuristically, we can see this from the form of the thermodynamic equation, namely

$$\frac{Db}{Dt} + N^2 w = 0. \quad (7.191)$$

If N^2 is high this suggests w will be small, and a lid is the limiting case of this. The oceanic problem is rather more difficult, because although both the stratification and shear are concentrated in the upper ocean, they vary relatively smoothly. Furthermore, the shear is high where the stratification is high.

As usual we start with the potential vorticity equation (for the interior) and the surface temperature equation, both linearized about a zonally uniform state $\psi_0(y, z)$. The potential vorticity equation is

$$\frac{\partial q'}{\partial t} + u_0(y, z) \frac{\partial q'}{\partial x} + v' \frac{\partial q_{0y}}{\partial y} = 0 \quad (7.192a)$$

where $u_0 = -\partial\psi_0/\partial y$ and

$$q' = \nabla^2 \psi' + \frac{\partial}{\partial z} \left(S(z) \frac{\partial \psi'}{\partial z} \right), \quad (7.192b)$$

$$q_{0y} = \beta - \frac{\partial^2 u_0}{\partial y^2} - \frac{\partial}{\partial z} \left(S(z) \frac{\partial u_0}{\partial z} \right). \quad (7.192c)$$

The surface boundary condition ($w = 0$ at $z = 0$) is effected by

$$\frac{\partial b'}{\partial t} + u_0(y, 0) \frac{\partial b'}{\partial x} + v' \frac{\partial b_{0y}}{\partial y} = 0, \quad \text{at } z = 0, \quad (7.193a)$$

where $b' = f_0 \partial \psi' / \partial z$ and

$$b_{0y} = -f_0 \frac{\partial u_0(y, 0)}{\partial z}. \quad (7.193b)$$

Consider a normal mode solution of the form

$$\psi'(x, y, z, t) = \hat{\psi}(y, z) e^{ik_x(x-ct)} \quad (7.194)$$

Substituting into (7.192) and (7.193) results in

$$(u_0 - c) \left[\frac{\partial^2}{\partial y^2} - k_x^2 + \frac{\partial}{\partial z} \left(S(z) \frac{\partial}{\partial z} \right) \right] \hat{\psi} + q_{0y} \hat{\psi} = 0 \quad (7.195)$$

and

$$(u_0 - c) \frac{\partial \hat{\psi}}{\partial z} - \frac{\partial u_0}{\partial z} \hat{\psi} = 0. \quad \text{at } z = 0. \quad (7.196)$$

Now suppose that there is a discontinuity in the shear and/or the stratification in the interior of the fluid, at some level $z = z_c$. Integrating (7.195) across the discontinuity gives

$$(u_0 - c) \left[S(z) \frac{\partial \hat{\psi}}{\partial z} \right]_{z_c^-}^{z_c^+} - \hat{\psi}(y, z_c) \left[S(z) \frac{\partial u_0}{\partial z} \right]_{z_c^-}^{z_c^+} = 0. \quad (7.197)$$

which has similar form to (7.196). This construction is evocative of that leading to Bretherton's boundary layer, except that now a discontinuity in the potential vorticity in *interior* has a near-equivalence with a rigid boundary.

We can illustrate this by numerically solving the linear eigenvalue problem. For simplicity, we pose the problem on the f -plane, in a horizontally doubly-periodic domain, with no horizontal variation of shear, and between two horizontal rigid lids. The eigenvalue problem is then defined by

$$(u_0 - c) \left[k^2 - \frac{\partial}{\partial z} \left(S(z) \frac{\partial}{\partial z} \right) \right] \hat{\psi} - q_{0y} \hat{\psi} = 0 \quad (7.198)$$

where

$$q_{0y} = \frac{\partial}{\partial z} \left(S(z) \frac{\partial u_0}{\partial z} \right), \quad (7.199)$$

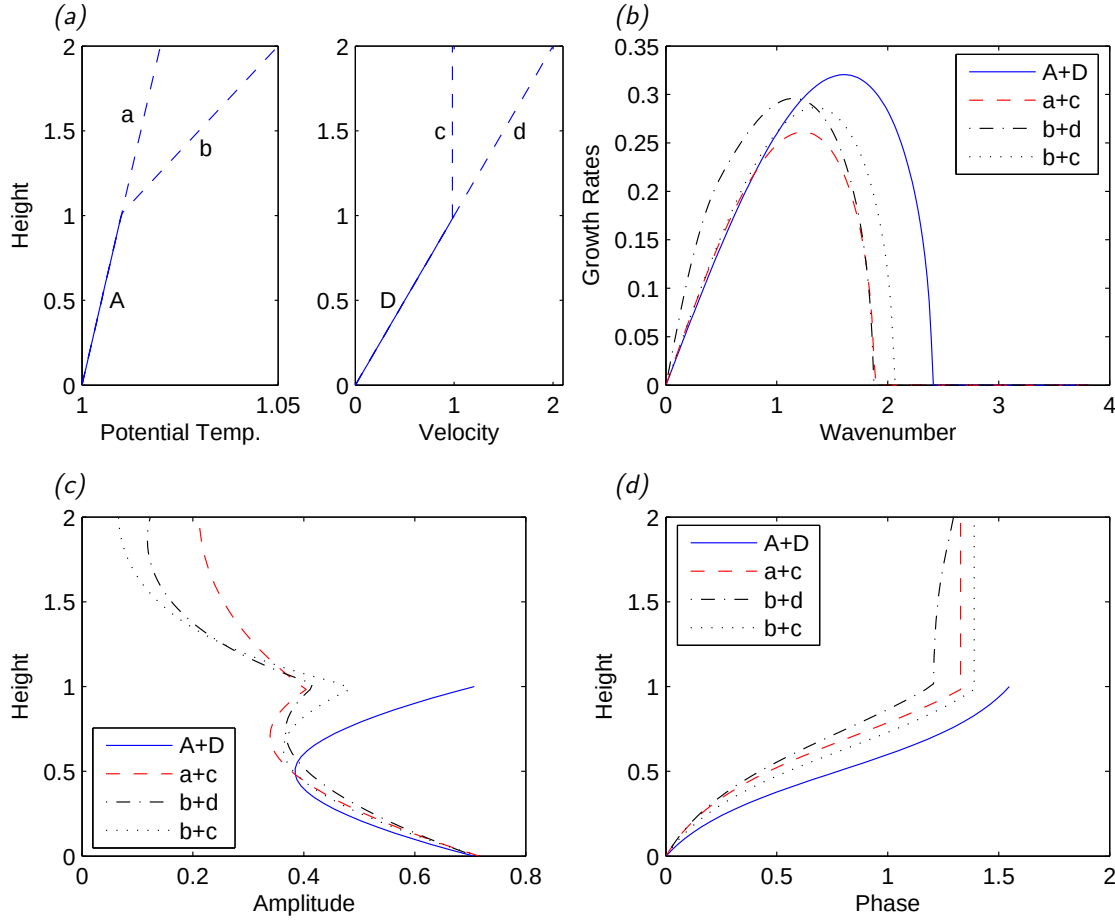


Figure 7.22 The effect of a stratosphere on baroclinic instability. (a) the given profiles of shear and stratification; (b) The growth rate of the instabilities; (c) amplitude of the most unstable mode as a function of height; (d) phase of the most unstable mode. The instability problem is numerically solved with various profiles of stratification and shear. In each profile, in the idealized troposphere ($z < 1$) the shear and stratification are uniform and the same in each case. We consider four idealized stratospheres ($z \geq 1$): 1, A lid at $z = 1$, i.e., no stratosphere, so Eady problem itself (profiles A+D, solid lines); 2, Stratospheric stratification same as the troposphere, but zero shear (profiles a+c, dashed); 3, Stratospheric shear same as troposphere, but stratification (N^2) four times the tropospheric value (b+d, dot-dashed); 4, Zero shear and high stratification in the stratosphere (b+c, dotted). In the troposphere the amplitude and structure of the instability is similar in all cases, illustrating the similarity of a rigid-lid and abrupt changes in shear or stratification. Either a high stratification or a low shear (or both) will result in weak stratospheric instability.

and, at the top and bottom,

$$(u_0 - c) \frac{\partial \hat{\psi}}{\partial z} - \frac{\partial u_0}{\partial z} \hat{\psi} = 0. \quad (7.200)$$

The numerical procedure then solves for the eigenvalue c and the complex eigen-

function $\hat{\psi}(z)$, and various results are illustrated in Fig. 7.22. To parse this rather complex figure, first look at the solid curves in all the panels. These arise when the problem is solved with a uniform shear and a uniform stratification [panel (b)], with a lid at $z = 0$ and $z = 1$, so simply giving the Eady problem. The familiar growth rates and vertical structure of the solution are given by the solid curves in panels (a), (c) and (d), and these are just the same as in Fig. 7.11. The various dotted and dashed curves show the results when the lid at $z = 1$ is replaced by stratosphere stretching from $1 < z < 2$ either with high stratification, zero shear, or both. In all of these cases the stratosphere acts in the same qualitative way as a rigid lid; in all three cases the vertical structure of the solution in the troposphere is quite similar, and the amplitude decaying rapidly above the idealized tropopause, consistent with the almost uniform phase of the disturbance illustrated in Fig. 7.22d (recall that a tilting of the disturbance with height is necessary for instability). It is these properties that make the Eady problem, or more generally any baroclinic instability problem that is posed between two rigid lids, of more general applicability to the earth's atmosphere than might be first thought: the high stratification above the tropopause and consequent decay of the instability is mimicked by the imposition of a rigid lid.

In the ocean, the stratification is highest in the upper ocean where the shear is also strongest. Calculations of the structure and growth rate are illustrated in Fig. 7.23. Like the calculation shown in Fig. 7.22 this numerical, quasi-geostrophic calculation of the instability of a zonal flow on the f -plane, but now with more oceanic profiles of the shear and stratification. Much of the ocean is characterized by having both a higher shear and a higher stratification in the upper 1 km or so, and this case is the one with the dotted line in Fig. 7.23. In this case the amplitude of the instability is also largely confined to the upper ocean, and unlike the Eady problem it does not arise through the interaction of edge waves at the top and bottom: the potential vorticity changes sign because of the interior variations due to the nonuniform shear. The ocean itself is still more complicated, because the most unstable regions near intense western boundary currents are often also barotropically unstable, and the mean flow itself may be meridionally directed. Nevertheless, the result that linear baroclinic instability is primarily an *upper ocean* phenomenon is quite robust.¹¹

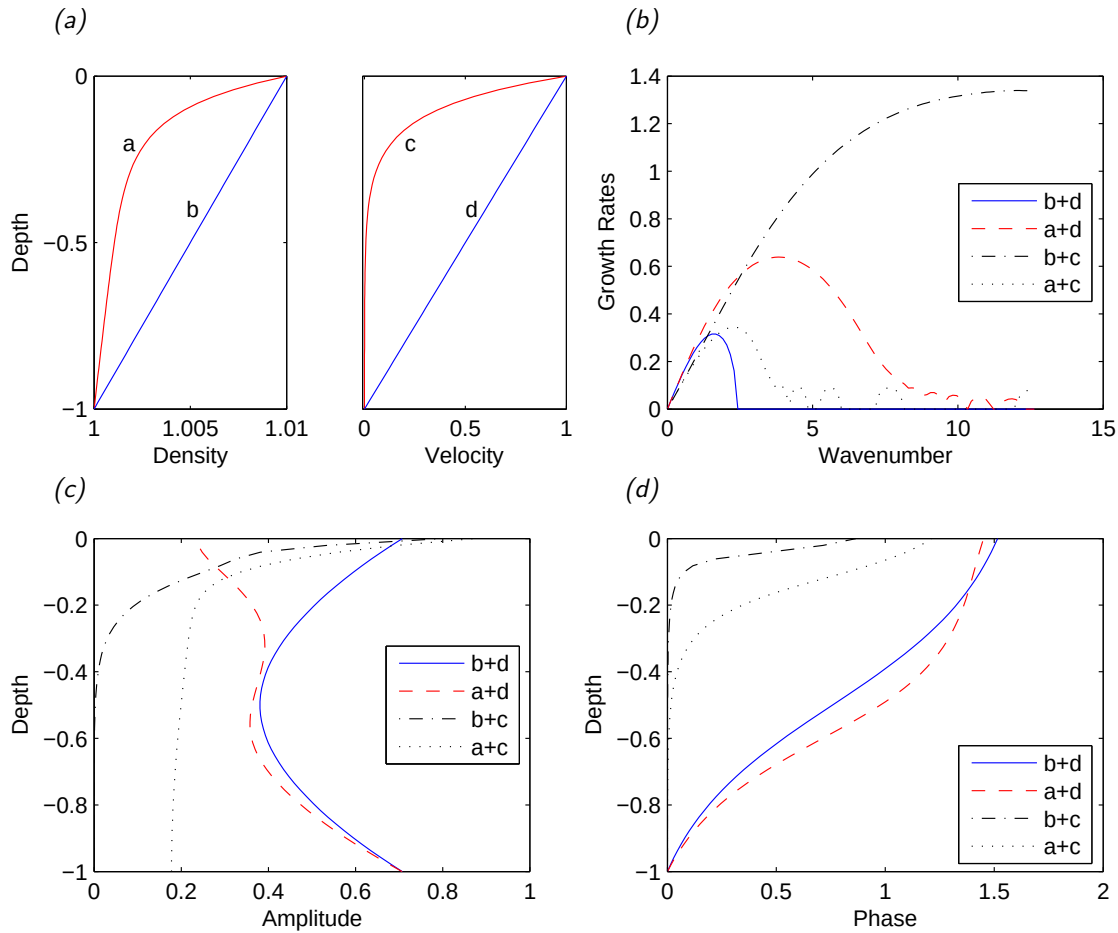


Figure 7.23 The baroclinic instability in an idealized ocean, with four different profiles of shear or stratification. The panels are: (a) The profiles of velocity and density (and so N^2) used; (b) the growth rates of the various cases; (c) The vertical structure of the amplitude of the most unstable modes; (d) the phase in the vertical of the most unstable modes. The instability is numerically calculated with four combinations of shear and stratification: 1, Uniform stratification and shear i.e. the Eady problem, (profiles b+d, solid lines). 2, Uniform shear, upper-ocean enhanced stratification (a+d, dashed); 3, Uniform stratification, upper ocean enhanced shear (b+c, dot-dashed); 4, Both stratification and shear enhanced in upper ocean (a+c, dotted). Case 2 is really more like to an atmosphere with a stratosphere (see Fig. 7.22), and the amplitude of the disturbance falls off, rather unrealistically, in the upper ocean. Case 4 (a+c, dotted) is the most oceanically relevant.

7.10 ENERGETICS AND DIAGNOSTICS OF BAROCLINIC INSTABILITY

In this section we look at how heat is transferred and energy is transformed by baroclinic instability. Insofar as the physical picture of section 7.5 represents the parcel trajectories, we expect warm parcels to be transferred to a location at higher latitudes where the potential temperature is smaller. That is, warm

parcels move poleward and, similarly, cold parcels move equatorward, and there is a net poleward heat transport. We also expect that the instability draw on the available potential energy of the mean state, and it does this because warm light parcels move upward, and cold dense parcels downward and the height of the mean center of gravity of the fluid falls. Because the instability is growing, the energy of the perturbation is of course not conserved, and both the kinetic energy and the available potential energy of the perturbation will grow. However, we will see that there is a meaningful way in which baroclinic instability converts potential energy to kinetic energy. We will discuss this in the simplest model of baroclinic instability — the two-layer model with $\beta = 0$.

7.10.1 Heat transport

It is useful to first re-derive the instability *ab initio* from the equations of motion, which we now write in terms of the baroclinic streamfunction $\hat{\psi}$ and the barotropic streamfunction $\bar{\psi}$ defined as

$$\hat{\psi} = \frac{1}{2}(\psi_1 - \psi_2) \quad \bar{\psi} = \frac{1}{2}(\psi_1 + \psi_2). \quad (7.201)$$

The nonlinear equations of motion, equivalent to (7.139), are

$$\frac{\partial}{\partial t} \nabla^2 \bar{\psi} + J(\bar{\psi}, \nabla^2 \bar{\psi}) + J(\hat{\psi}, (\nabla^2 - k_d^2) \hat{\psi}) = 0 \quad (7.202a)$$

$$\frac{\partial}{\partial t} (\nabla^2 - k_d^2) \hat{\psi} + J(\hat{\psi}, \nabla^2 \bar{\psi}) + J(\bar{\psi}, (\nabla^2 - k_d^2) \hat{\psi}) = 0 \quad (7.202b)$$

Linearizing, as before, around a sheared basic state with $\bar{\Psi} = 0$ and $\hat{\Psi} = -Uy$ we obtain the simple equations

$$\frac{\partial}{\partial t} \nabla^2 \bar{\psi} = -U \frac{\partial}{\partial x} \nabla^2 \hat{\psi} \quad (7.203)$$

and

$$\frac{\partial}{\partial t} (\nabla^2 - k_d^2) \hat{\psi} = -U \frac{\partial}{\partial x} (\nabla^2 + k_d^2) \bar{\psi}. \quad (7.204)$$

Seeking solutions of the form $\psi = \text{Re } \Psi \exp[ik(x - ct)]$ gives

$$c \bar{\Psi} - U \hat{\Psi} = 0 \quad (7.205)$$

$$c(k^2 + k_d^2) \hat{\Psi} - U(k^2 - k_d^2) \bar{\Psi} = 0 \quad (7.206)$$

These equations only have nontrivial solutions if the determinant of the matrix of coefficients is zero. That is, we require

$$c^2(k^2 + k_d^2) - U^2(k^2 - k_d^2) = 0 \quad (7.207)$$

or

$$c = U \left(\frac{k^2 - k_d^2}{k^2 + k_d^2} \right)^{1/2} \quad (7.208)$$

This is the dispersion relationship for the two-layer model. Instabilities occur for $k^2 < k_d^2$ for which c is purely imaginary, equal to ic_i where $c_i = U [(k_d^2 - k^2)/(k^2 + k_d^2)]^{1/2}$. Growing waves have $c_i > 0$.

From (7.205) unstable modes have

$$\hat{\Psi} = i \frac{c_i}{U} \bar{\Psi} = e^{i\pi/2} \frac{c_i}{U} \bar{\Psi} \quad (7.209)$$

That is, $\hat{\psi}$ lags $\bar{\psi}$ by 90° for a growing wave. Similarly, $\hat{\psi}$ leads $\bar{\psi}$ by 90° for a decaying wave, for which $c_i < 0$. Now, the temperature is proportional to $\hat{\psi}$, and the vertically averaged perturbation meridional velocity, $\bar{v}$, (which advects the temperature) is given by $\bar{v} = \partial \bar{\psi} / \partial x = ik \bar{\Psi}$. Thus, for growing or decaying waves,

$$\bar{v} = \hat{\Psi} \frac{kU}{c_i} \quad (7.210)$$

and the meridional velocity is exactly *in phase* with the temperature for growing modes, and is *out of phase* with the temperature for decaying modes. That is, for unstable (decaying) modes, polewards flow is correlated with high (low) temperatures. For neutral waves, $\hat{\psi} = c_r \bar{\Psi} / U$ and so $\bar{v} = ik \hat{\Psi} U / c_r$ and the meridional velocity and temperature are $\pi/2$ out of phase. Thus, to summarize:

- Growing waves transport heat (or buoyancy) polewards.
- Decaying waves transport heat equatorward.
- Neutral waves do not transport heat.

7.10.2 Energetics

As in section 6.5, the energy may be partitioned into kinetic energy and available potential energy. In a three-dimensional quasi-geostrophic flow the kinetic energy is given by, in general,

$$\text{KE} = \frac{1}{2} \int (\nabla \psi)^2 dV \quad (7.211)$$

which, in the case of the two-layer model becomes

$$\begin{aligned} \text{KE} &= \frac{1}{2} \int (\nabla \psi_1)^2 + (\nabla \psi_2)^2 dA \\ &= \int (\nabla \bar{\psi})^2 + (\nabla \hat{\psi})^2 dA. \end{aligned} \quad (7.212)$$

Restricting attention to a single Fourier mode, $\psi = \Psi \exp[i(kx - ct)] + \text{c.c.}$, this becomes

$$\text{KE} = k^2 \overline{\Psi}^2 + k^2 \hat{\Psi}^2. \quad (7.213)$$

The available potential energy is given by

$$\text{APE} = \frac{1}{2} \int \left(\frac{f_0}{N} \right)^2 \left(\frac{\partial \psi}{\partial z} \right)^2 dV. \quad (7.214)$$

For a single Fourier mode in a two layer model this becomes

$$\text{APE} = k_d^2 \hat{\Psi}^2. \quad (7.215)$$

The evolution equations can be written as a vorticity equation and a thermodynamic equation. The nonlinear vorticity equations for each level is

$$\frac{\partial}{\partial t} \nabla^2 \psi_1 + J(\psi_1, \nabla^2 \psi_1) = -2 \frac{f_0 w}{H} \quad (7.216a)$$

$$\frac{\partial}{\partial t} \nabla^2 \psi_2 + J(\psi_2, \nabla^2 \psi_2) = 2 \frac{f_0 w}{H} \quad (7.216b)$$

where w is the vertical velocity between the levels. Multiplying these equation by $-\psi_1$ and $-\psi_2$ and adding we find

$$\frac{d}{dt} \text{KE} = \frac{4f_0}{H} \int w \hat{\psi} dA \quad (7.217a)$$

For a single Fourier mode this becomes

$$\frac{d}{dt} \text{KE} = \text{Re} \frac{4f_0}{H} W \hat{\Psi}^* \quad (7.217b)$$

where $w = W \exp[i(kx - ct)] + \text{c.c.}$, and the asterisk denotes complex conjugacy.

The continuous thermodynamic equation is

$$\frac{Db}{Dt} + wN^2 = 0 \quad (7.218)$$

which, using $b = f_0 \partial \psi / \partial z$ and finite-differencing, becomes in the two-level model

$$\frac{\partial \hat{\psi}}{\partial t} + J(\overline{\psi}, \hat{\psi}) + \frac{wN^2 H}{4f_0} = 0. \quad (7.219)$$

The change of available potential energy is obtained from this by multiplying by $k_d^2 \hat{\psi}$ and integrating, giving

$$\int \left[\frac{1}{2} \frac{d}{dt} k_d^2 \hat{\psi}^2 + \hat{\psi} w \frac{2f_0}{H} \right] dA = 0 \quad (7.220)$$

or

$$\frac{d}{dt} \text{APE} = -\frac{4f_0}{H} \int w \hat{\psi} dA \quad (7.221a)$$

or, for a single Fourier mode,

$$\frac{d}{dt} \text{APE} = -\text{Re} \frac{4f_0}{H} W \hat{\Psi}^*. \quad (7.221b)$$

From (7.217) and (7.221) it is clear that in the nonlinear equations the sum of the kinetic energy and the available potential energy is conserved.

We now specialize by obtaining w from the linear baroclinic instability problem. Using this in (7.217) and (7.221) will give us the conversion between kinetic energy and potential energy in the growing baroclinic wave. It is important to realize that the total energy of the disturbance is not conserved — both the potential and kinetic energy are growing, exponentially in this problem, because they are extracting energy from the mean state.

To calculate w we use the linearized thermodynamic equation. From (7.219) this is

$$\frac{\partial \hat{\psi}}{\partial t} - U \frac{\partial \bar{\psi}}{\partial x} + \frac{HwN^2}{4f_0} = 0, \quad (7.222)$$

omitting the primes on perturbation quantities. For a Fourier mode,

$$-ikc\hat{\Psi} - ikU\bar{\Psi} + \frac{HwN^2}{4f_0} = 0. \quad (7.223)$$

Thus,

$$\frac{HN^2}{4f_0} W = ik(c\hat{\Psi} + U\bar{\Psi}). \quad (7.224)$$

But, from (7.205), $c\bar{\Psi} = U\hat{\Psi}$ in two-layer f -plane baroclinic instability and so

$$\frac{HN^2}{4f_0} W = ikc\hat{\Psi} \left(1 + \frac{U^2}{c^2} \right) = ikc\hat{\Psi} \left(\frac{2k^2}{k^2 - k_d^2} \right). \quad (7.225)$$

using (7.208). For stable waves, $k^2 > k_d^2$ and $c = c_r$ and in that case the vertical velocity is $\pi/2$ out of phase with the temperature, and there is no conversion of APE to KE. For unstable waves $c = ic_i$ and $k^2 < k_d^2$, and the vertical velocity is in phase with the temperature. That is, warm air is rising and so there is a conversion of APE to KE. To see this more formally, recall that the conversion from APE to KE is given by $4W\hat{\Psi}^* f_0 / H$. Thus, using (7.225),

$$\frac{d}{dt} (\text{APE} \rightarrow \text{KE}) = \text{Re} 2ikck_d^2 \left(\frac{2k^2}{k^2 - k_d^2} \right) \hat{\Psi}^2. \quad (7.226)$$

If the wave is growing, then $k^2 < k_d^2$ and $c = ic_i$ and the right-hand side is real and positive. For neutral waves, If $c = c_r$ the right-hand side of (7.226) is pure imaginary, and so the conversion is zero. This completes our demonstration that baroclinic instability converts potential energy into kinetic energy.

7.11 PHYSICAL INTERPRETATION OF BAROCLINIC INSTABILITY

7.11.1 A form drag instability

[To be included]

7.11.2 The interference of travelling waves

[To be included]

Interfering waves on upper and lower boundaries. At large wavenumbers there is no interference.

For small scales (large μ),

$$\Phi(z) = e^{-\mu z} \qquad c \rightarrow 0 \qquad (7.227a)$$

$$\Phi(z) = e^{-\mu(z-1)} \qquad c \rightarrow 1 \qquad (7.227b)$$

Neutral waves are trapped near an upper and lower boundary.

Notes

- 1 Our analytic description of shear instabilities is fairly standard. See ? and ? for more detail.
- 2 Squire's theorem states that for every three-dimensional disturbance to a plane-parallel flow, there corresponds a more unstable two-dimensional one. This means that the most unstable disturbances are two-dimensional, and there is no need to consider three dimensional effects to determine whether such a flow is unstable. See ?.
- 3 The solution of Fig. 7.6 is obtained with a gridpoint code, here with 400×400 equally spaced gridpoints. For this kind of problem, which has a small number of possible values of vorticity, a contour dynamics code is very well suited, as in ?.
- 4 Following ?.
- 5 Eric Eady (1915–1966) is best remembered today as the author of the 'Eady model' of baroclinic instability which describes the fundamental hydrodynamic instability mechanism that gives rise to weather systems. After an undergraduate education in mathematics he joined the U. K. Meteorological Office in 1937, becoming a forecaster and upper air analyst, in which capacity he served throughout the war. In 1946 he joined the Department of Mathematics at Imperial College, presenting his Ph. D. thesis in 1948 on 'The theory of development in dynamical meteorology', subsequently summarized in *Tellus* (?). This work, masterly in its combination of austerity and relevance, provides a mathematical description of the essential aspects of cyclone development that stands to this day as a canonical model in the field. It also includes, rather obliquely, a derivation of the stratified quasi-geostrophic equations, albeit in a special form. The impact of the work was immediate and it led to invitations and visits to Bergen (in 1947 with J. Bjerknes), Stockholm (in 1952 with C.-G. Rossby) and Princeton (in 1953 with J. von Neumann and Charney). Eady followed his baroclinic instability work with a prescient discussions of the general circulation of the atmosphere (???). A perfectionist who sought to understand it all, Eady's subsequent published output was small and he later turned his attention to fundamental problems in other areas of fluid mechanics, the dynamics of the sun and the earth's interior, and biochemistry. He finally took his own life. There is little published about him, save for the obituary by ?.

Jule Charney (1917–1981) played a defining role in dynamical meteorology in the second half of the 20th century. He made seminal contributions in many areas including: the theory of baroclinic instability (?); a systematic scaling theory for large-scale atmospheric motions and the derivation of the quasi-geostrophic equations (Charney 1948); a theory of stationary waves in the atmosphere (?); the demonstration of the feasibility of numerical weather forecasts (Charney *et al.* 1950); planetary wave propagation into the stratosphere (?); a criterion for baroclinic instability (?); a theory for hurricane growth (?); the concept of geostrophic turbulence (?). His Ph.D. is from UCLA in 1946

and this, entitled 'Dynamics of long waves in a baroclinic westerly current', became his well-known 1947 paper. After this he spent a year at Chicago and another at Oslo, and in 1948 joined the Institute of Advanced Study in Princeton where he stayed until 1956 (and where he briefly overlapped with Eady). He spent most of his subsequent career at MIT, interspersed with many visits to Europe, especially Norway. For a more complete picture of Charney, see Lindzen *et al.* (1990) and a brief biography by N. Phillips available at <http://www.nap.edu/readingroom/books/biomems/jcharney.html>

- 6 At least I find it so.
- 7 Our nondimensionalization of the two-layer system is such as to be in correspondence with that for the continuous system. Thus we choose H to be the total depth of the domain, rather than the distance between the two levels. This choice produces growth rates and wavenumbers that are equivalent to those in the Eady problem.
- 8 After ?, in whose problem it appears, and ?, who appreciated its importance.
- 9 See ? and ?. ? also provide an approximate calculation of growth rates in the Charney problem.
- 10 Our numerical procedure is to assume a wavelike solution in the horizontal of the form $\Psi \exp[i(kx + ly - \omega t)]$, and to finite difference the equations in the vertical. The resulting eigenvalue equations are solved by standard matrix methods, for each horizontal wavenumber. See ?.
- 11 ? and ? were among the first to look at baroclinic instability in the ocean. Since then there have been many studies in various configurations.

Further Reading

Drazin P. and W. H. Reid 1981. *Hydrodynamic Stability*.

A standard text on hydrodynamic instability theory. It discusses nearly all the classic cases in a straightforward and clear fashion. It includes a more extensive discussion of the linear instability of parallel shear flow than is contained here, although the treatment of baroclinic instability is rather brief.

Chandrasekhar, S. 1961. *Hydrodynamic and Hydromagnetic Stability*.

A classic text discussing many forms of instability but not, alas, baroclinic instability.

Pierrehumbert and Swanson (1995) review many aspects of baroclinic instability from a fairly modern point of view.

Problems

- 7.1 Derive the jump condition (7.29) without directly considering the motion of the interface. In particular, from the momentum equation along the interface show that

$$\frac{\partial}{\partial y} \left(\frac{\hat{\psi}}{U - c} \right) = - \frac{\hat{\psi}}{(U - c)^2} \quad (\text{P7.1})$$

and show that (7.29) follows. Be explicit about the conditions under which the right-hand-side vanishes when integrated across the interface. (See ?, for help).

- 7.2 By applying the matching conditions (7.23) and (7.29) at $y = \pm a$ to Rayleigh's equation, explicitly derive the dispersion relationship (7.46).
- 7.3 Show that for very long waves, or as the shear layer becomes thinner, the growth rate given by (7.46) reduces to that of Kelvin-Helmholz instability of a vortex sheet.
- 7.4 Obtain the stability properties of the triangular jet, with a basic state velocity given by

$$U(y) = \begin{cases} 0 & \text{for } z \geq 1 \\ 1 - |y| & \text{for } -1 \leq y \leq 1 \\ 0 & \text{for } z \leq -1 \end{cases} \quad (\text{P7.2})$$

In particular, obtain the eigenfunctions and eigenvalues of the problem, and show that each eigenfunction is either even or odd. Perturbations with even ψ' are known as 'sinuous modes' and those with odd ψ' are 'varicose modes'. Show that sinuous waves are unstable for sufficiently long wavelengths in the z -direction, but that all varicose modes are stable.

- 7.5 In a doubly-periodic domain on the β -plane, how can a parcel conserve its potential vorticity as it (apparently) leaves the northern part of the domain and re-enters at the southern boundary with a different value of β , without there being a discontinuity in the vorticity field? (See also chapter ??.)
- 7.6 Show numerically or analytically that, in the Eady problem:
- Instability occurs for $\mu < 2.399$.
 - The wavenumber at which the instability is greatest is $\mu = 1.61$.
 - The nondimensional growth rate at that wavenumber is 0.31.
- 7.7 $\diamond$ In which we consider the vertical modes of the Eady and other stratified problems.
- When solving the continuous form of the eigenvalue stability problem (as in the Eady problem, for example) the differential equation typically has just one pair of eigenvalues. However, if the equation is solved on a vertical grid with N levels, the resulting difference equation has N roots. Does this mean that $N - 2$ roots are spurious? How might they be eliminated, or how might the 'correct' eigenvalue be identified, in a numerical procedure?
 - The N -level problem is equivalent to a physically realizable N -layer system, in which there are presumably N physically meaningful eigenvalues. As N becomes large, with the density differences and thicknesses of each layer chosen to become smaller in a consistent way, the equations describing the layered system presumably converge to those describing the continuous system, yet there are still N eigenvalues in the former. How may this paradox be resolved? What is the physical nature of the extra eigenvalues in the layered system?
- 7.8 Show, using the two-layer model (or otherwise) that the presence of β reduces the efficiency of baroclinic instability. For example, show that it makes the meridional velocity slightly out of phase with the temperature.

- 7.9 $\diamond$ Consider the baroclinic instability problem with a discontinuity in the stratification, but a uniform shear. For example, suppose the shear is uniform for $z \in (0, 1)$ with an abrupt change in stratification at $z = 0.5$. How does the amplitude of the instability vary on either side of the discontinuity? Your answer may be an analytical or a numerical calculation, or both.

CHAPTER EIGHT

Wave-Mean Flow Interaction

WAVE-MEAN FLOW INTERACTION is concerned with how some mean flow, perhaps a time or zonal average, interacts with a ‘wave’, meaning a departure from that mean, and this chapter provides an elementary introduction to a number of topics in that area. It is ‘elementary’ because our derivations and discussion are obtained by direct and straightforward manipulations of the equations of motion, often in the simplest case that will illustrate the relevant principle. It is implicit in what we do that it is a sensible thing to decompose the fields into a mean plus an eddy, and one case when this is so is when the eddies are small amplitude. Eddies are of course not always small; for example, in the mid-latitude troposphere the eddies are often of similar amplitude to the mean flow, and chapter ?? and ?? explore this from the standpoint of turbulence. However, here we will generally assume that eddies are of small amplitude, and the fact that Rossby wave dynamics often then provide a leading order description of the perturbation dynamics means that the appellation ‘wave-mean flow’ is indeed appropriate. However, the usefulness of the methods and results is not limited to cases in which the eddies have an infinitesimal amplitude, and wave-mean flow interaction theory provides many useful insights into the finite-amplitude problem. Indeed, the goal of this chapter is to provide an introduction to a way of understanding more realistic situations, and to suggest diagnostics that might be used to analyze both observations and numerical solutions of the fully nonlinear equations. We will almost exclusively concern

ourselves with a *zonal* mean, for this is the simplest and often most useful case, although with care many of our results can be extended to the case of a temporal mean. We will also be mainly concerned with quasi-geostrophic dynamics on a β -plane.

8.1 QUASI-GEOSTROPHIC PRELIMINARIES

To fix our dynamical system and our notation, we write down the quasi-geostrophic potential vorticity equation

$$\frac{\partial q}{\partial t} + J(\psi, q) = D \quad (8.1)$$

where D represents any nonconservative terms and the potential vorticity in a Boussinesq system is

$$q = \beta y + \zeta + \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} b \right), \quad (8.2)$$

where, as usual, ζ is the relative vorticity and b is the buoyancy $-g\delta\rho/\rho_0$. In an ideal gas we have very similar equations, but with $b = g\theta/\theta_0$ where θ is the potential temperature, and $q = \beta y + \zeta + (f_0/\rho_s)\partial/\partial z (\rho_s b/N^2)$. We will refer to lines of constant b as isentropes, and the parameter N^2 is the buoyancy frequency, equal to db_0/dz where $b_0(z)$ is a reference profile. In terms of streamfunction the variables are

$$\begin{aligned} \zeta &= \nabla^2 \psi, & b &= f_0 \frac{\partial \psi}{\partial z}, \\ q &= \beta y + \left[\nabla^2 + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial}{\partial z} \right) \right] \psi. \end{aligned} \quad (8.3)$$

where $\nabla^2 \equiv (\partial_x^2 + \partial_y^2)$. The potential vorticity equation holds in the fluid interior; the boundary conditions on (8.3) are provided by the thermodynamic equation

$$\frac{\partial b}{\partial t} + J(\psi, b) + wN^2 = J, \quad (8.4)$$

where J represents heating terms. The vertical velocity at the boundary, w , is zero in the absence of topography and Ekman friction, and if J is also zero the boundary condition is just:

$$\frac{\partial b}{\partial t} + J(\psi, b) = 0. \quad (8.5)$$

If both D and J are zero the above equations conserve both the total energy and the total enstrophy:

$$\begin{aligned}\frac{dE}{dt} &= 0, & E &= \int_V (\nabla\psi)^2 + \frac{f_0^2}{N^2} \left(\frac{\partial\psi}{\partial z} \right)^2 dV \\ \frac{dZ}{dt} &= 0, & Z &= \int_V q^2 dV.\end{aligned}\tag{8.6}$$

where V is a volume bounded by surfaces at which the normal velocity is zero, or that has periodic boundary conditions. The enstrophy is also conserved layerwise — that is, the horizontal integral of q^2 is conserved at every level.

8.1.1 Potential vorticity and buoyancy fluxes in the linear equations

Let us decompose the fields into a mean (to be denoted with an overbar) plus a perturbation (denoted with a prime), and let us suppose the perturbation fields are of small amplitude. The linearised quasi-geostrophic potential vorticity equation is

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + u' \frac{\partial \bar{q}}{\partial x} + \bar{v} \frac{\partial q'}{\partial y} + v' \frac{\partial \bar{q}}{\partial y} = D'.\tag{8.7}$$

If the mean is a zonal mean, or just if the basic state is not a function of x , then $\bar{v} = 0$ and this simplifies to

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = D'.\tag{8.8}$$

where

$$\frac{\partial \bar{q}}{\partial y} = \beta - \frac{\partial^2 \bar{u}}{\partial y^2} - \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \bar{u}}{\partial z} \right).\tag{8.9}$$

Multiplying by q' and zonally averaging gives the perturbation enstrophy equation:

$$\frac{1}{2} \frac{\partial \overline{q'^2}}{\partial t} = -\overline{v'q'} \frac{\partial \bar{q}}{\partial y} + \overline{D'q'}.\tag{8.10}$$

The quantity $\overline{v'q'}$ is the meridional flux of potential vorticity; this is downgradient when first term on the right-hand side is positive, and it then acts to increase the variance of the perturbation. (This occurs, for example, when the flux is diffusive so that $\overline{v'q'} = -\kappa \partial \bar{q} / \partial y$ where κ may vary but is everywhere positive.) This argument may be inverted: for inviscid flow, if the waves are growing, as for example in the canonical models of baroclinic instability discussed in chapter 7, then *the potential vorticity flux is downgradient*. If the second term on the right-hand side is negative, as it will be if D is a dissipative process (for example if $D = \kappa' \nabla^2 q$ or if $D = -rq$, where κ' and r are positive) then a statistical balance can be achieved between enstrophy production via downgradient transport,

and dissipation. If the waves are steady (by which we mean statistically steady, neither growing nor decaying in amplitude) and conservative (i.e., $D' = 0$) then we must have:

$$\overline{v'q'} = 0. \quad (8.11)$$

Similar results follow for the potential temperature at the boundary. Thus, linearizing the thermodynamic equation (8.5) gives

$$\frac{\partial b'}{\partial t} + \bar{u} \frac{\partial b'}{\partial x} + v' \frac{\partial \bar{b}}{\partial y} = J'. \quad (8.12)$$

and multiplying by b' and averaging gives

$$\frac{1}{2} \frac{\partial \overline{b'^2}}{\partial t} = -\overline{v'b'} \frac{\partial \bar{b}}{\partial y} + \overline{J'b'}. \quad (8.13)$$

Thus growing adiabatic waves have a downgradient flux of temperature at the boundary. In the Eady problem there is no interior gradient of basic-state potential vorticity and all the terms in (8.10) are zero, but the perturbation grows by way of the boundary condition. If the waves are steady and adiabatic then, analogously to (8.11),

$$\overline{v'b'} = 0. \quad (8.14)$$

The boundary conditions and fluxes may be absorbed into the interior definition of potential vorticity and its fluxes by way of Bretherton's boundary layer construction, described in chapter 6. This can provide notational and conceptual advantages over dealing with boundary fluxes explicitly, but if an actual calculation is to be performed there is often little to be gained, for the boundary terms have to be dealt with one way or the other.

8.2 THE ELIASSEN-PALM FLUX

In terms of the flux of vorticity and buoyancy the eddy flux of potential vorticity is

$$v'q' = v'\zeta' + f_0 v' \frac{\partial}{\partial z} \left(\frac{b'}{N^2} \right) \quad (8.15)$$

The second term on the right-hand side can be written

$$\begin{aligned} f_0 v' \frac{\partial}{\partial z} \left(\frac{b'}{N^2} \right) &= f_0 \frac{\partial}{\partial z} \left(\frac{v'b'}{N^2} \right) - f_0 \frac{\partial v'}{\partial z} \frac{b'}{N^2} \\ &= f_0 \frac{\partial}{\partial z} \left(\frac{v'b'}{N^2} \right) - f_0 \frac{\partial}{\partial x} \left(\frac{\partial \psi'}{\partial z} \right) \frac{b'}{N^2} \\ &= f_0 \frac{\partial}{\partial z} \left(\frac{v'b'}{N^2} \right) - \frac{f_0^2}{N^2} \frac{\partial}{\partial x} \left(\frac{1}{2} \frac{\partial \psi'}{\partial z} \right)^2 \end{aligned} \quad (8.16)$$

using $b' = f_0 \partial \psi' / \partial z$.

Similarly, the flux of relative vorticity can be written

$$v' \zeta' = -\frac{\partial}{\partial y} u' v' + \frac{1}{2} \frac{\partial}{\partial x} (v'^2 - u'^2) \quad (8.17)$$

Using (8.16) and (8.17), (8.15) becomes

$$v' q' = -\frac{\partial}{\partial y} (u' v') + \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} v' b' \right) + \frac{\partial}{\partial x} \left(\frac{1}{2} (v'^2 - u'^2) - \frac{b'^2}{N^2} \right) \quad (8.18)$$

Thus the potential vorticity flux, in the quasi-geostrophic approximation, can be written as the divergence of a vector: $v' q' = \nabla \cdot \mathcal{E}$ where

$$\mathcal{E} \equiv \left(\frac{1}{2} (v'^2 - u'^2) - \frac{b'^2}{N^2} \right) \mathbf{i} - (u' v') \mathbf{j} + \left(\frac{f_0}{N^2} v' b' \right) \mathbf{k}. \quad (8.19)$$

A particularly useful form of this arises after zonally averaging, after which (8.18) becomes

$$\overline{v' q'} = -\frac{\partial}{\partial y} \overline{u' v'} + \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} \overline{v' b'} \right). \quad (8.20)$$

The vector defined by

$$\mathcal{F} \equiv -\overline{u' v'} \mathbf{j} + \frac{f_0}{N^2} \overline{v' b'} \mathbf{k} \quad (8.21)$$

is called the *Eliassen-Palm flux*,¹ and its divergence gives the polewards flux of potential vorticity:

$$\overline{v' q'} = \nabla_m \cdot \mathcal{F}, \quad (8.22)$$

where $\nabla_m \cdot \equiv (\partial / \partial y, \partial / \partial z)$ is the divergence in the meridional plane, with a straightforward extension to spherical co-ordinates.

8.2.1 The Eliassen-Palm relation

On dividing by $\partial \bar{q} / \partial y$ and using (8.22), the perturbation enstrophy equation (8.10) can be written

$$\frac{\partial \mathcal{A}}{\partial t} + \nabla \cdot \mathcal{F} = \mathcal{D}, \quad (8.23a)$$

where

$$\mathcal{A} = \frac{\overline{q'^2}}{2 \partial \bar{q} / \partial y}, \quad \mathcal{D} = \frac{\overline{D' q'}}{\partial \bar{q} / \partial y} \quad (8.23b)$$

This is known as the *Eliassen-Palm relation*, and it is a conservation law for the the *wave activity density* $\mathcal{A}$, for if we integrate this expression over a meridional

area A bounded by walls where the eddy activity vanishes, and if $\mathcal{D} = 0$, we obtain

$$\frac{d}{dt} \int_A \mathcal{A} dA = 0. \quad (8.24)$$

In general, a wave activity is a quantity that is quadratic in the amplitude of the perturbation and that is conserved in the absence of forcing and dissipation. (Although by this definition $\mathcal{A}$ is a wave activity density it is sometimes referred to simply as a wave activity, just as ρv is sometimes referred to as momentum.) More specifically, $\mathcal{A}$ is the negative of the *pseudomomentum*, for reasons we will encounter later. Note that neither perturbation energy nor perturbation enstrophy are wave activities of the linearized equations, because there can be an exchange of energy or enstrophy between mean and perturbation — indeed, this is how a perturbation grows in baroclinic or barotropic instability! This is already evident from (8.10), or in general take (8.7) with $D' = 0$ and multiply by q' to give the enstrophy equation

$$\frac{1}{2} \frac{\partial q'^2}{\partial t} + \frac{1}{2} \bar{\mathbf{u}} \cdot \nabla q'^2 + \mathbf{u}' q' \cdot \nabla \bar{q} = 0 \quad (8.25)$$

where here the overbar is an average, but need not be a zonal average. Integrating this over a volume V gives

$$\frac{dZ'}{dt} = \frac{d}{dt} \int \frac{1}{2} q'^2 dV = - \int_V \mathbf{u}' q' \cdot \nabla \bar{q} dV. \quad (8.26)$$

The right-hand side does not in general vanish and so Z' is not in general conserved. It is thus a remarkable property of $\mathcal{A}$ that it is both a measure of the amplitude of a wave and that it is a conserved quantity, in the sense of (8.23a).

The vector $\mathcal{F}$ describes how the wave activity propagates. In general, we cannot express it simply in terms of $\mathcal{A}$, but in the case in which the disturbance is composed of plane or almost plane waves that permit a group velocity to be defined then $\mathcal{F} = \mathbf{c}_g \mathcal{A}$, where $\mathbf{c}_g$ is the group velocity and (8.23a) becomes

$$\frac{\partial \mathcal{A}}{\partial t} + \nabla \cdot (\mathcal{A} \mathbf{c}_g) = 0. \quad (8.27)$$

We shall now show this explicitly when the waves in question are plane Rossby waves, although the property holds for waves of other types when the group velocity itself can be defined.²

8.2.2 The group velocity property and Rossby waves

The Boussinesq quasi-geostrophic equation on the β -plane, linearized around a uniform zonal flow and with constant static stability, is

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0 \quad (8.28)$$

where $q' = [\nabla^2 + (f_0^2/N^2)\partial^2/\partial z^2]\psi'$ and, if $\bar{u}$ is constant, $\partial\bar{q}/\partial y = \beta$. Thus we have

$$\left(\frac{\partial}{\partial t} + \bar{u}\frac{\partial}{\partial x}\right) \left[\nabla^2\psi' + \frac{\partial}{\partial z}\left(\frac{f_0^2}{N^2}\frac{\partial\psi'}{\partial z}\right)\right] + \beta\frac{\partial\psi'}{\partial x} = 0. \quad (8.29)$$

Seeking solutions of the form

$$\psi' = \text{Re } \Psi e^{i(kx+ly+mz-\omega t)}, \quad (8.30)$$

we find the dispersion relation,

$$\omega = \bar{u}k - \frac{\beta k}{\kappa^2}. \quad (8.31)$$

with group velocities,

$$c_{gy} = \frac{2\beta kl}{\kappa^2}, \quad c_{gz} = \frac{2\beta km f_0^2/N^2}{\kappa^2}, \quad (8.32)$$

where $\kappa^2 = (k^2 + l^2 + m^2 f_0^2/N^2)$. Also, if $u' = \text{Re } \hat{u} \exp[i(kx + ly + mz - \omega t)]$, and similarly for the other fields, then

$$\begin{aligned} \hat{u} &= -\text{Re } ik\Psi, & \hat{v} &= \text{Re } il\Psi \\ \hat{b} &= \text{Re } imf_0\Psi, & \hat{q} &= -\text{Re } \kappa^2\Psi. \end{aligned} \quad (8.33)$$

The wave activity is then

$$\mathcal{A} = \frac{1}{2} \frac{\overline{q'^2}}{\beta} = \frac{\kappa^2}{4\beta} |\Psi^2| \quad (8.34)$$

where the additional factor of 2 in the denominator arises from the averaging. Using (8.33) the EP flux is

$$\begin{aligned} \mathcal{F}_y &= -\overline{u'v'} = \frac{1}{2} kl |\Psi^2| \\ \mathcal{F}_z &= \frac{f_0}{N^2} \overline{v'b'} = \frac{f_0^2}{2N^2} km |\Psi^2|. \end{aligned} \quad (8.35)$$

Using this (8.32) and (8.34) gives

$$\boxed{\mathcal{F} = (\mathcal{F}_y, \mathcal{F}_z) = \mathbf{c}_g \mathcal{A}}. \quad (8.36)$$

If the properties of the medium are varying, but only slowly so that we can still define a group velocity, then this is a useful expression to estimate how the wave activity propagates and to interpret the results of numerical simulations, as we will see in chapter ??.

8.3 PROPERTIES AND CONSEQUENCES OF WAVE ACTIVITY

8.3.1 Orthogonality of modes

It is a direct consequence of the conservation of wave activity that disturbance modes are orthogonal in the wave activity norm, and thus are a useful measure of the amplitude of a particular mode. Consider again the linearized potential vorticity equation,

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0 \quad (8.37)$$

Let us seek solutions of the form $\psi' = \text{Re } \Psi \exp(ikx)$ where Ψ is the sum of modes

$$\Psi = \sum_n \psi_n(y, z) e^{-ikc_n t}. \quad (8.38)$$

that satisfy

$$(\bar{u} \Delta_k^2 + \bar{q}_y) \psi_n = c_n \Delta_k^2 \psi_n, \quad (8.39)$$

where

$$\Delta_k^2 = \frac{\partial^2}{\partial y^2} + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial}{\partial z} \right) - k^2. \quad (8.40)$$

The upper and lower boundary conditions (at $z = 0, H$) are given by the thermodynamic equation

$$\frac{\partial b'}{\partial t} + \bar{u} \frac{\partial b'}{\partial x} + v' \frac{\partial \bar{b}}{\partial y} = 0, \quad (8.41)$$

and if we simplify further by supposing $\partial \bar{u} / \partial z = 0$ then the boundary condition becomes

$$\frac{\partial \psi'_z}{\partial t} + \bar{u} \frac{\partial \psi'_z}{\partial x} = 0. \quad (8.42)$$

and there are no meridional buoyancy fluxes at the boundary. If N^2 is a constant then we can let $\psi_n(y, z) = \psi_n(y) \cos mz$, with $m = j\pi/H$ where j is an integer, so that $q_n = \Delta_{k,m}^2 \psi_n$ where

$$\Delta_{k,m}^2 = \frac{\partial^2}{\partial y^2} - (f_0^2 / N^2) m^2 - k^2, \quad (8.43)$$

and the boundary conditions are then built-in to any solution we construct from (8.39) and (8.43).³ We may then consider a single zonal and a single vertical wavenumber, with the mode n characterizing the meridional variation. (Of course if there is no horizontal variation of the shear, then these too are harmonic functions, with $\psi_n \propto \sin(n\pi y / L)$ for a channel of width L .)

For a given basic state we may imagine solving (8.39), numerically or analytically, and determining the modes. However, these modes are not orthogonal in the sense of either energy or enstrophy. That is, denoting the inner product by

$$\langle a, b \rangle \equiv L^{-1} \int_L ab \, dy, \quad (8.44)$$

then it is *not* the case that

$$\begin{aligned} I_E &= \langle \psi_n, q_m \rangle = 0, \\ I_Z &= \langle q_n, q_m \rangle = 0 \end{aligned} \quad (8.45)$$

for $n \neq m$, where $q_n = \Delta_{k,m}^2 \psi_n$. Perturbation energy and enstrophy are thus not wave activities of the linearized equations, and it is not meaningful to talk about the energy or enstrophy of a particular mode. However, by the same token we may expect orthogonality in the wave activity norm. To prove this and understand what it means, suppose that at $t = 0$ the disturbance consists of two modes, n and m , so that at a later time $q = (q_n e^{-ikc_n t} + q_m e^{-ikc_m t} + \text{c.c.})$, where $c_r \neq c_s$ and we assume that both are real. The wave activity is

$$P \equiv \int \mathcal{A} \, dy \, dz = \langle q_n, q_m^* / \bar{q}_y \rangle e^{ik(c_n - c_m)t} + \langle q_m, q_n^* / \bar{q}_y \rangle + \langle q_n, q_n^* / \bar{q}_y \rangle + \text{c.c.} \quad (8.46)$$

The second and third terms of this are the wave activities of each mode, and these are constants (to see this, consider the case when the disturbance is just a single mode). Now, because $dP/dt = 0$ the first term must vanish if $c_n \neq c_m$, implying the modes are orthogonal and in particular

$$\text{Re} \int \frac{1}{\bar{q}_y} q_n q_m^* \, dy = 0 \quad (8.47)$$

for $n \neq m$. Readers who would prefer a more direct derivation of the orthogonality condition directly from the eigenvalue equation (8.39) should see problem 8.6. Orthogonality is a useful result, for what it means is that the wave activity is a proper measure of the amplitude of a given mode. One may then sensibly distinguish between (for example) the amplitude of continuous modes and discrete modes. Related reasoning will provide us with a necessary condition for stability, in section 8.7.

8.3.2 The nonacceleration theorem

Let us now consider how the potential vorticity fluxes affect the mean fields. First note that if the waves are steady and conservative then, from (8.11) or (8.23a),

the EP flux is non-divergent and the meridional potential vorticity flux is zero. Now, the zonally-averaged potential vorticity equation is

$$\frac{\partial \bar{q}}{\partial t} + \frac{\partial \overline{v'q'}}{\partial y} = \bar{D} \quad (8.48)$$

Thus, if the waves are steady and conservative and if $\bar{D} = 0$ we have that $\partial \bar{q} / \partial t = 0$. Now, in quasi-geostrophic theory the buoyancy and geostrophically balanced velocity can be determined from the potential vorticity via an elliptic equation, and in particular

$$\bar{q} - \beta y = \frac{\partial^2 \bar{\psi}}{\partial y^2} + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \bar{\psi}}{\partial z} \right). \quad (8.49)$$

and so (8.48) becomes

$$\left[\frac{\partial^2}{\partial y^2} + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial}{\partial z} \right) \right] \bar{u}_t = (\nabla \cdot \mathcal{F})_{yy} - \bar{D}_y. \quad (8.50)$$

Thus, if the terms on the right-hand side are zero then a solution is

$$\frac{\partial \bar{u}}{\partial t} = 0. \quad (8.51)$$

and this is the unique solution if there is no acceleration at the boundaries. This is a *nonacceleration result*.⁴ That is to say, under the non-acceleration conditions the tendency of the mean balanced fields, and in particular the zonally-averaged zonal flow, are independent of the waves and, in the absence of other terms, are zero. Explicitly, the nonacceleration conditions are that the waves:

- (i) are steady (so that $\mathcal{A}$ does not vary),
- (ii) are conservative (i.e., $\mathcal{D} = 0$, so that it is not the presence of dissipative terms that allow there to be a non-zero potential vorticity flux — see (8.23a)),
- (iii) are of small amplitude (all of our analysis has neglected terms that are cubic in perturbation amplitude),
- (iv) do not affect the boundary conditions (so there is no boundary contributions to the acceleration).

The result applies to the buoyancy and velocity fields that are directly invertible from potential vorticity, and not to the ageostrophic velocities. Given the way we have derived it it is not a surprising result, but we will see that it can be a powerful and counter-intuitive, for it means that steady waves (i.e., nonamplifying and nondecaying) do not affect the zonal flow. However, they do affect the meridional overturning circulation, and the vorticity flux also may be nonzero. In fact, the nonacceleration theorem is telling us that the changes in the vorticity flux are exactly compensated for by changes in the meridional circulation, and there is no

net effect on the zonally-averaged zonal flow. It is *irreversibility*, often manifested by the breaking of waves, that leads to permanent changes in the mean flow.

The derivation of this result by way of the momentum equation, which one might expect to be more natural, is in fact rather awkward because one must consider momentum and buoyancy fluxes separately. Furthermore, the zonally averaged meridional circulation comes into play: for example, meridional velocity, $\bar{v}$, is, although small because it is purely ageostrophic, not zero and we cannot neglect it because it is multiplied by Coriolis parameter, which is large. Thus, the eddy vorticity fluxes can affect both the meridional circulation and the acceleration of the zonal mean flow, and it might seem impossible to disentangle the two effects without completely solving the equations of motion. Fortunately, the situation is simplified and clarified via the use of the ‘transformed Eulerian mean’, which we now introduce.

8.4 THE TRANSFORMED EULERIAN MEAN

The so-called *transformed Eulerian mean*, or TEM, provides a useful framework for discussing eddy effects under a wide range of conditions.⁵ It is useful because, as we shall see, it is equivalent to a very natural form of averaging the equations that serves to eliminate eddy fluxes in the thermodynamic equation and collect them together, in a simple form, in the momentum equation, highlighting the role of potential vorticity fluxes. It also provides a natural separation between diabatic and adiabatic effects or between advective and diffusive fluxes and, in the case in which the flow is adiabatic, a nice simplification of the equations. In later chapters we will use the TEM to better understand the mid-latitude troposphere and the dynamics of the Antarctic Circumpolar current, and as a framework for the parameterization of eddy fluxes. Of course, there being no free lunch, the TEM brings with it its own difficulties, and in particular the implementation of boundary conditions can cause difficulties, especially in the actual numerical integration of the TEM equations.

8.4.1 Quasi-geostrophic form

Recall the conventional zonally averaged Eulerian mean equations for the zonally averaged zonal velocity and the buoyancy on the beta-plane:

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v} = -\frac{\partial}{\partial y} \overline{u'v'} - \frac{\partial}{\partial z} \overline{u'w'} + \bar{F} \quad (8.52)$$

$$\frac{\partial \bar{b}}{\partial t} + N^2 \bar{w} = -\frac{\partial}{\partial y} \overline{v'b'} - \frac{\partial}{\partial z} \overline{w'b'} + \bar{J} \quad (8.53)$$

where $\bar{F}$ and $\bar{J}$ represent frictional and heating terms. Note that the only contribution to $\bar{v}$ is from the ageostrophic meridional velocity. Using quasi-geostrophic scaling we neglect the vertical eddy flux divergences and all ageostrophic velocities except when multiplied by f_0 or N^2 , and assume that b is in thermal wind balance with $\bar{u}$, $f_0 \partial u / \partial z = -\partial b / \partial y$ (in the Boussinesq approximation). The equations then become

$$\frac{\partial \bar{u}}{\partial t} = f_0 \bar{v} - \frac{\partial}{\partial y} \overline{u'v'} + \bar{F}_u \quad (8.54)$$

$$\frac{\partial \bar{b}}{\partial t} = -N^2 \bar{w} - \frac{\partial}{\partial y} \overline{v'b'} + J \quad (8.55)$$

One ‘problem’ with these equations is that in the extratropics the dominant balance in these equations is usually between the first two terms on the right-hand sides of each equation, even in time-dependent cases. Thus, the Coriolis force closely balances the divergence of the eddy momentum fluxes, and the advection of the mean stratification ($N^2 \bar{w}$, or ‘adiabatic cooling’) often balances the convergence of eddy heat flux, with heating being a small residual. This can underestimate its importance, since diabatic processes are ultimately responsible for the mean meridional circulation. Thus, in the thermodynamic equation we might seek to combine the terms $N^2 \bar{w}$ and the eddy flux into a single total (or ‘residual’) heat transport term that in a steady state is balanced by the diabatic term J . The TEM provides this reformulation, and in doing so the eddy terms in the momentum equation also take a different form.

To begin, note that because $\bar{v}$ and $\bar{w}$ are related by mass conservation, we can define a mean meridional streamfunction ψ_m such that

$$(\bar{v}, \bar{w}) = \left(-\frac{\partial \psi_m}{\partial z}, \frac{\partial \psi_m}{\partial y} \right). \quad (8.56)$$

Then, defining a ‘residual’ streamfunction by

$$\psi^* \equiv \psi_m + \frac{1}{N^2} \overline{v'b'} \quad (8.57)$$

the components of the *residual mean meridional circulation* are given by

$$(\bar{v}^*, \bar{w}^*) = \left(-\frac{\partial \psi^*}{\partial z}, \frac{\partial \psi^*}{\partial y} \right), \quad (8.58)$$

and

$$\bar{v}^* = \bar{v} - \frac{\partial}{\partial z} \left(\frac{1}{N^2} \overline{v'b'} \right), \quad \bar{w}^* = \bar{w} + \frac{\partial}{\partial y} \left(\frac{1}{N^2} \overline{v'b'} \right). \quad (8.59)$$

Note that by construction, the residual overturning circulation satisfies

$$\frac{\partial \bar{v}^*}{\partial y} + \frac{\partial \bar{w}^*}{\partial z} = 0. \quad (8.60)$$

Substituting (8.59) into (8.54) and (8.55) the zonal momentum and buoyancy equations then take the simple forms

$$\left. \begin{aligned} \frac{\partial \bar{u}}{\partial t} &= f_0 \bar{v}^* + \overline{v'q'} + \bar{F}_u \\ \frac{\partial \bar{b}}{\partial t} &= -N^2 \bar{w}^* + \bar{J} \end{aligned} \right\}, \quad (8.61)$$

which are known as the (quasi-geostrophic) *transformed Eulerian mean equations*. The potential vorticity flux is given in terms of the heat and vorticity fluxes by (8.15) or (8.20). (The quasi-geostrophic TEM easily generalizes to a non-Boussinesq form, and can be couched in spherical co-ordinates, and these are left as exercises — see problem 8.??)

The TEM equations make it apparent that it is the potential vorticity fluxes, rather than the separate contributions of the vorticity and heat fluxes, that drive the circulation. If we know the potential vorticity flux, and $\bar{F}$ and $\bar{J}$, then (8.60) and (8.61), along with thermal wind balance

$$f_0 \frac{\partial \bar{u}}{\partial z} = -\frac{\partial \bar{b}}{\partial y} \quad (8.62)$$

form a complete set. The meridional overturning circulation is obtained by eliminating time derivatives from (8.61) using (8.62) to give

$$f_0^2 \frac{\partial^2 \psi^*}{\partial z^2} + N^2 \frac{\partial^2 \psi^*}{\partial y^2} = f_0 \frac{\partial}{\partial z} \overline{v'q'} + f_0 \bar{F}_{uz} + \bar{J}_y. \quad (8.63)$$

Thus, the residual or net overturning circulation too is driven by the (vertical derivative of the) potential vorticity fluxes and the diabatic terms — driven in the sense that if we know those terms we can calculate the overturning circulation. Note that this equation applies at every instant, even if the equations are not in a steady state.

8.4.2 The TEM in isentropic co-ordinates

The residual circulation has an illuminating interpretation if we think of the fluid as comprising multiple layers of shallow water, or equivalently if we cast the

problem in isentropic co-ordinates (section 2.6.2). The momentum and mass conservation equation can then be written as

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} - f\mathbf{v} = F \quad (8.64a)$$

$$\frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{u}) = J[h] \quad (8.64b)$$

In isentropic co-ordinates h is the thickness between two isentropic surfaces and $H = H(b)$ is its mean thickness, and the layer thickness is a measure of the temperature of the layer. With quasi-geostrophic scaling (so that β and variations in layer thickness are small) zonally averaging in a conventional way gives

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v} = \overline{v' \zeta'} + \bar{F} \quad (8.65a)$$

$$\frac{\partial \bar{h}}{\partial t} + H \frac{\partial \bar{v}}{\partial y} = -\frac{\partial}{\partial y} \overline{v' h'} + \bar{J}[h] \quad (8.65b)$$

The overbars in these equations denote averages taken along isentropes, but are otherwise conventional, and the meridional velocity is purely ageostrophic. We now choose to define the residual circulation by

$$\bar{v}^* = \bar{v} + \frac{1}{H} \overline{v' h'} \quad (8.66)$$

which is analogous to (8.59). Using (8.66) in (8.65) gives

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \overline{v' q'} + \bar{F} \quad (8.67a)$$

$$\frac{\partial \bar{h}}{\partial t} + H \frac{\partial \bar{v}^*}{\partial y} = \bar{J}[h]. \quad (8.67b)$$

where

$$\overline{v' q'} = \overline{v' \zeta'} + \frac{f_0}{H} \overline{v' h'}. \quad (8.68)$$

From (8.66) we see that the residual velocity is a measure of the *total meridional mass flux*, eddy plus mean, in an isentropic layer. This is often a more useful quantity than the Eulerian velocity $\bar{v}$ because it is generally the former, not the latter, that is constrained by the external forcing, and we will exploit this further in chapter ???. What we have done, of course, is to effectively use a mass-weighted mean in (8.64b); that is, if we define the mass-weighted mean by

$$\bar{v}_* = \frac{\overline{h v}}{\bar{h}} \quad (8.69)$$

so that

$$\bar{v}_* = \bar{v} + \frac{1}{h} \overline{v'h'}, \quad (8.70)$$

the zonal average of (8.64b) is just

$$\frac{\partial \bar{h}}{\partial t} + \frac{\partial}{\partial y} (\bar{h} \bar{v}_*) = \bar{J}[h]. \quad (8.71)$$

which is the same as (8.67b) if we replace $\bar{h}$ by H using quasi-geostrophic scaling. Then, using (8.70) in (8.65a) we obtain (8.67a). Evidently if the averages are taken in this way, we do not need to explicitly worry about eddy heat fluxes. Thus, in isentropic co-ordinates it is clear that the equations in TEM form are equivalent to the equations that arise from a particular form of averaging — mass weighted averaging — rather than the conventional Eulerian averaging. Is there a similar relationship in height co-ordinates? The answer is yes, as we now see.

8.4.3 Connection between the residual and mass-weighted circulation

We now explicitly show that averaging the total transport in isentropic layers is equivalent to the mass transport evaluated by the TEM formalism in height co-ordinates. The two derivations following are essentially equivalent, but differ in detail.⁶

A geometric derivation

Consider two isentropic surfaces, η_1 and η_2 with mean positions $\bar{\eta}_1$ and $\bar{\eta}_2$. (We use z to denote the vertical co-ordinate, and η to denote the location of isentropic surfaces.) The meridional transport between these surfaces is given by

$$T = \int_{\eta_2}^{\eta_1} v \, dz. \quad (8.72)$$

If the velocity does not vary with height within the layer (and later we take the layer thickness to zero so this is not a real restriction) then $T = v h$ where $h = \eta_1 - \eta_2$ is the thickness of the isentropic layer. The zonally-averaged transport is then given by

$$\begin{aligned} \bar{T} &= \frac{1}{L} \int_L T \, dx = \frac{1}{L} \int_L \left(\int_{\eta_2}^{\eta_1} v \, dz \right) dx = \overline{\int_{\eta_2}^{\eta_1} v \, dz} \\ &= \overline{v h} = \bar{v} \bar{h} + \overline{v'h'} \end{aligned} \quad (8.73)$$

with obvious notation, and an overbar denoting a zonal average. Note that the averaged quantity $\bar{T}/\bar{h}$ is not simply proportional to the average of the velocity

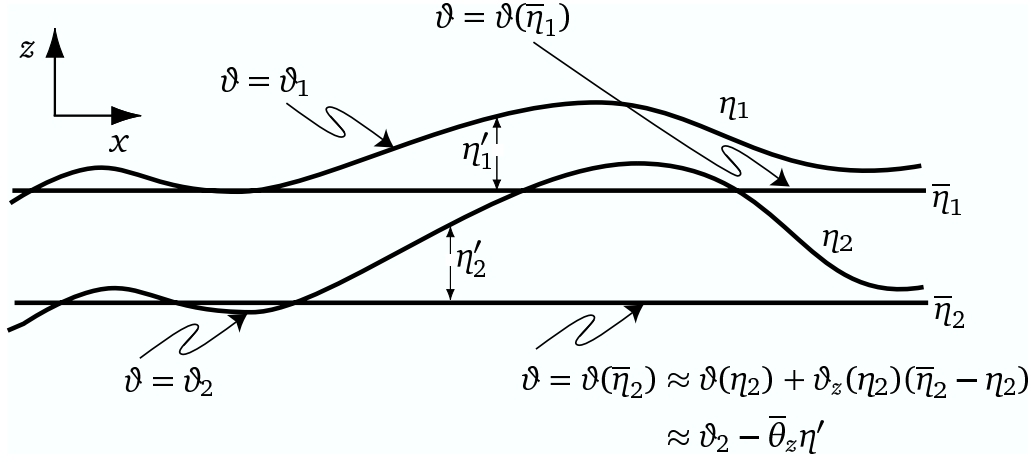


Figure 8.1 Two isentropic surfaces, η_1 and η_2 , and their mean positions, $\bar{\eta}_1$ and $\bar{\eta}_2$. The departure of the isentrope from its mean position is proportional to the temperature perturbation on the mean isentrope. The residual meridional velocity is proportional to the thickness-weighted average velocity between the isentropes, and is a measure of the total (mean plus eddy) meridional transport.

at constant height, or even to the average along an isentrope; rather, it is the *thickness-weighted* zonal average of the velocity *between* two isentropic surfaces. The mean velocity $\bar{v} = \bar{v}(\eta)$ is the average velocity within the isentropic layer, and $\bar{h} = \bar{h}(\eta)$ its mean thickness. [Other kinds of average (e.g., time) may similarly be defined, and the derivations proceed in a similar way.]

Now, if $\bar{\eta}_i$ is the mean position of the interface, so that $\eta_i = \bar{\eta}_i + \eta'_i$, then the transport may be written as

$$T = \int_{\bar{\eta}_2 + \eta'_2}^{\bar{\eta}_1 + \eta'_1} v \, dz = \int_{\bar{\eta}_2}^{\bar{\eta}_1} v \, dz + \int_{\bar{\eta}_1}^{\bar{\eta}_1 + \eta'_1} v \, dz - \int_{\bar{\eta}_2}^{\bar{\eta}_2 + \eta'_2} v \, dz. \quad (8.74)$$

This is helpful because, if the amplitudes of the perturbation are small, we can usefully expand the velocities in a Taylor series about the fixed heights. Thus, $v(z) = v(\bar{\eta}_1) + v_z(\bar{\eta}_1)(z - \bar{\eta}_1) + \dots$ and we obtain

$$\int_{\bar{\eta}_1}^{\bar{\eta}_1 + \eta'_1} v \, dz = v(\bar{\eta}_1)\eta'_1 + \frac{1}{2}v_z(\bar{\eta}_1)\eta'^2_1 + \mathcal{O}(\alpha^3), \quad (8.75)$$

where the perturbation amplitude is $\mathcal{O}(\alpha)$, and the averaged expression

$$\overline{\int_{\bar{\eta}_1}^{\bar{\eta}_1 + \eta'_1} v \, dz} = \overline{v'(\bar{\eta}_1)\eta'_1} + \mathcal{O}(\alpha^3). \quad (8.76)$$

To obtain this expression note that, after averaging, the second term on the right-hand side of (8.75) is $\mathcal{O}(\alpha^3)$, because using thermal wind balance there is no

geostrophic zonally-averaged vertical shear unless the zonal integration path is blocked by orography, so that both v_z and η' are perturbation quantities and so are $\mathcal{O}(\alpha)$.

The averages in (8.76) are still along isentropic layers. Now, we can express the geometric deviation of the isentropic surface from its mean position in terms of the temperature perturbation at the location of the mean isentrope by Taylor expanding b around its value on that mean isentrope. That is,

$$b(\eta_1) = b(\bar{\eta}_1) + \left(\frac{\partial b}{\partial z} \right)_{z=\bar{\eta}_1} (\eta_1 - \bar{\eta}_1) + \cdots, \quad (8.77)$$

giving

$$\eta'_1 \approx \frac{-b'(\bar{\eta}_1)}{\partial b(\bar{\eta}_1) / \partial z}. \quad (8.78)$$

where $\eta'_1 = \eta_1 - \bar{\eta}_1$ and $b' = b(\bar{\eta}) - b(\eta)$. The reader should be convinced of this expression, and in particular the minus sign, before proceeding, for it is this which enables the connection between isentropic and height co-ordinates to be made. Using (8.78) in (8.76) gives

$$\overline{\int_{\bar{\eta}_1}^{\bar{\eta}_1 + \eta'_1} v \, dz} = - \frac{\overline{v'(\bar{\eta}_1) b'(\bar{\eta}_1)}}{\bar{b}_z(\bar{\eta}_1)}. \quad (8.79)$$

Note that the various terms on the right-hand side are functions only of the position of the mean isentrope, and so the overbar represents an average at a constant height. Substituting (8.79), and a similar expression involving $\bar{\eta}_2$, into (8.74) results in

$$\bar{T} = \int_{\bar{\eta}_2}^{\bar{\eta}_1} v \, dz - \frac{\overline{v'(\bar{\eta}_1) b'(\bar{\eta}_1)}}{\bar{b}_z(\bar{\eta}_1)} + \frac{\overline{v'(\bar{\eta}_2) b'(\bar{\eta}_2)}}{\bar{b}_z(\bar{\eta}_2)} \quad (8.80)$$

In the limit as η_1 and η_2 approach each other we therefore obtain, on dividing by $\bar{h} = \bar{\eta}_1 - \bar{\eta}_2$,

$$\boxed{\frac{\overline{vh}}{\bar{h}} \equiv \bar{v}(\eta) + \frac{\overline{v'h'}}{\bar{h}} = \bar{v}(z) - \left(\frac{\overline{v'b'}}{\bar{b}_z} \right)_z}. \quad (8.81)$$

where $h = \eta_1 - \eta_2$ is the infinitesimal layer thickness and $z = \bar{\eta}$.

This equation requires some interpretation. The leftmost term is the thickness-weighted mean isentropic velocity, and the average is taken along an infinitesimal isentropic layer of thickness h . In a Boussinesq fluid the layer thickness is related to the continuous variables by $h \propto \partial z / \partial b$, with a similar expression for an ideal gas. Just like any average, it may be decomposed into two terms, a mean term $\bar{v}(\eta)$ and an eddy term $\overline{v'h'}$, and both of these are averages along isentropes. Thus, for example, $\bar{v}(\eta) = L^{-1} \int_L v(\eta) \, dx$. The averages in the expressions on the

right-hand side are of quantities evaluated at constant height, that height being the mean position of the isentrope $\eta(b)$, so that $z = \bar{\eta}(b)$. The two quantities $\bar{v}(\eta)$ and $\bar{v}(z)$ do not in general equal each other, although if the isentropic slopes are small, as assumed in quasi-geostrophic dynamics, they will be very similar. Evidently, the right-hand side is the residual mean circulation that appears in the TEM formalism discussed in the previous section, and the corresponding residual mean streamfunction for the zonally averaged residual circulation is just

$$\psi^* = \psi + \left(\frac{\overline{v'b'}}{\bar{b}_z} \right), \quad (8.82)$$

as in (8.57).

An analytic derivation

Let us first connect an average along an isentrope of some variable χ to its average at constant height by writing, for small isentropic displacements,

$$\bar{\chi}^b = \overline{\chi(z + \eta')^z} \approx \overline{\chi(z) + \eta' \partial \chi / \partial z}^z \quad (8.83)$$

where the superscript explicitly denotes how the zonal average is taken, and η' is the displacement of the isotherm from its mean position. This departure is related to the temperature perturbation on the mean isotherm by

$$\eta' \approx - \frac{b'}{\partial \bar{b} / \partial z} \quad (8.84)$$

so that

$$\bar{\chi}^b = \overline{\chi(z)}^z - \frac{\overline{b' \partial \chi / \partial z}^z}{(\partial \bar{b} / \partial z)} \quad (8.85)$$

Now, if $\chi = v = \partial \psi / \partial x$ then, because $b' = f_0 \partial \psi / \partial z$, the second term vanishes and we have

$$\bar{v}^b \approx \bar{v}^z. \quad (8.86)$$

Also, neglecting terms that are third order in perturbation amplitude

$$\overline{v'h'}^b \approx \overline{v'h'}^z. \quad (8.87)$$

so that, using (8.70)

$$\bar{v}_* \approx \bar{v}^z + \frac{1}{h} \overline{v'h'}^z. \quad (8.88)$$

Now, using (8.84), the thickness of an isothermal layer is given by

$$h' \approx -\bar{h} \frac{\partial}{\partial z} \left(\frac{b'}{\partial \bar{b} / \partial z} \right). \quad (8.89)$$

and therefore, for fields in thermal wind balance,

$$\overline{v'h'} \approx -\bar{h} \frac{\partial}{\partial z} \left(\frac{\overline{b'v'^z}}{\partial \bar{b} / \partial z} \right) \quad (8.90)$$

Thus, using (8.86), (8.88) and (8.90) we have

$$\bar{v}_* = \frac{\overline{vh}}{\bar{h}} = \bar{v}^b + \frac{\overline{v'h'^b}}{\bar{h}} \approx \bar{v}^z - \left(\frac{\overline{v'b'^z}}{\bar{b}_z} \right)_z = \bar{v}^* \quad (8.91)$$

With $N^2 = \bar{b}_z$ and $H = \bar{h}$ we see the equivalence of the thickness weighted mean velocity on the left-hand side and the residual velocity on the right-hand side.

8.4.4 * The TEM in the primitive equations

We now look at the TEM equations in a slightly more general way, although we will largely restrict our attention to the thermodynamic budget. The basic idea, as before, is that we seek to replace the advecting velocity in the equations of motion by the residual velocity $\bar{v}^*$ where

$$\bar{\mathbf{v}}^* = \mathbf{v} - \nabla \times \boldsymbol{\psi} \quad (8.92)$$

where we choose the vector streamfunction $\boldsymbol{\psi}$ to simplify the resulting equations as far as possible, and in particular we will now try to distinguish clearly between diffusive and advective eddy fluxes. The meaning of this last statement will become clearer below, and at this stage the transformation (8.92) is purely formal. (The separation will be particularly useful in the context of diffusive parameterizations of turbulent eddy fluxes. See section ?? and the ones following.)

We begin with the standard, incompressible, Boussinesq primitive equations. If we decompose these into Eulerian mean and eddy terms, denoted with an overbar and a prime, then the momentum and thermodynamic equations may be written

$$\frac{\partial \bar{\mathbf{u}}}{\partial t} + \bar{\mathbf{v}} \cdot \nabla \bar{\mathbf{u}} + \mathbf{f} \times \bar{\mathbf{u}} = -\nabla \bar{\phi} + \mathbf{R}, \quad (8.93a)$$

$$\frac{\partial \bar{b}}{\partial t} + \bar{\mathbf{u}} \cdot \nabla \bar{b} = J[b] - \nabla \cdot \mathbf{F}[b], \quad (8.93b)$$

and a tracer, ϕ , obeys

$$\frac{\partial \bar{\phi}}{\partial t} + \bar{\mathbf{u}} \cdot \nabla \bar{\phi} = J[\phi] - \nabla \cdot \mathbf{F}[\phi]. \quad (8.93c)$$

The equations are completed with the hydrostatic ($\partial \phi / \partial z = \bar{b}$) and mass conservation ($\nabla \cdot \bar{\mathbf{v}} = 0$) equations. The term $\mathbf{R}$ represents the various Reynolds stress

Aspects of the TEM formulation

Properties and features

- ★ The residual mean circulation is equivalent to the total mass-weighted (eddy plus Eulerian mean) circulation, a useful and it is this circulation that is driven by the diabatic forcing.
- ★ There are no explicit eddy fluxes in the buoyancy budget; the only eddy term is the flux of potential vorticity, and this is divergence of the Eliassen-Palm flux; that is $\overline{v'q'} = \nabla_m \cdot \mathbf{E}$.
- ★ The residual circulation, $\bar{v}^*$, becomes part of the solution, just as $\bar{v}$ is part of the solution in an Eulerian mean formulation.

But note

- ★ The TEM formulation does not solve the parameterization problem, and eddy fluxes are still present in the equations.
- ★ The theory and practice are well developed for a zonal average, less so for three-dimensional, non-zonal flow.⁷
- ★ The boundary conditions on the residual circulation are neither necessarily simple nor easily determined; for example, at a horizontal boundary $\bar{w}^*$ is not zero if there are horizontal buoyancy fluxes.

terms, the J terms are sources such as diffusion and heating, and the $\nabla \cdot \mathbf{F}$ terms are eddy fluxes of the tracer indicated in the square brackets. Thus,

$$\nabla \cdot \mathbf{F}[\varphi] = \frac{\partial \overline{u'\varphi'}}{\partial x} + \frac{\partial \overline{v'\varphi'}}{\partial y} + \frac{\partial \overline{w'\varphi'}}{\partial z} = \nabla \cdot \overline{\mathbf{v}'\varphi'}. \quad (8.94)$$

and similarly for $\nabla \cdot \mathbf{F}[b]$. (We will omit the identifying label in square brackets if there is no ambiguity. Also, in this section, the divergence and related operators are three dimensional unless noted.) The fluxes may be divided into components along and across the tracer gradient,

$$\mathbf{F} = \mathbf{F}_\perp + \mathbf{F}_\parallel = (\mathbf{n} \cdot \mathbf{F})\mathbf{n} + (\mathbf{n} \times \mathbf{F}) \times \mathbf{n} \quad (8.95)$$

where $\mathbf{n} = \nabla\bar{\phi}/|\nabla\bar{\phi}|$. Thus,

$$\mathbf{F}_\perp = \frac{\mathbf{F} \cdot \nabla\bar{\phi}}{|\nabla\bar{\phi}|^2} \nabla\bar{\phi}, \quad \mathbf{F}_\parallel = \frac{\nabla\bar{\phi} \times \mathbf{F}}{|\nabla\bar{\phi}|^2} \times \nabla\bar{\phi} \quad (8.96)$$

so that $\mathbf{F}_\perp$ ($\mathbf{F}_\parallel$) is perpendicular to (along) iso-surfaces of tracer. Such a decomposition is possible for any vector, of course. The component $\mathbf{F}_\parallel$ is known as a *skew flux*, and this may be written as if it were an advection by some divergence-free velocity $\tilde{\mathbf{v}}$. That is,

$$\nabla \cdot \mathbf{F}_\parallel = \nabla \cdot \left(\frac{\nabla\bar{\phi} \times \mathbf{F}}{|\nabla\bar{\phi}|^2} \times \nabla\bar{\phi} \right) = \left(\nabla \times \frac{\nabla\bar{\phi} \times \mathbf{F}}{|\nabla\bar{\phi}|^2} \right) \cdot \nabla\bar{\phi} \equiv \tilde{\mathbf{v}} \cdot \nabla\bar{\phi}, \quad (8.97)$$

using the vector identity $\nabla \cdot (\mathbf{A} \times \nabla\bar{\phi}) = \nabla\bar{\phi} \cdot \nabla \times \mathbf{A}$. Furthermore, $\tilde{\mathbf{v}} \cdot \nabla\bar{\phi} = \nabla \cdot (\tilde{\mathbf{v}}\bar{\phi})$ so that the skew flux is adiabatic: that is, it serves to redistribute $\bar{\phi}$, but, just like advection by a true incompressible velocity, it does not of itself change the distribution, or the census, of $\bar{\phi}$. (It may create a situation in which true dissipation is more effective, by the production of small scales and enhanced tracer gradients.)

If we use (8.92) in (8.93c) then we obtain

$$\frac{\partial\bar{\phi}}{\partial t} + \bar{\mathbf{v}}^* \cdot \nabla\bar{\phi} = J[\bar{\phi}] - \nabla \cdot \mathbf{F}^*[\bar{\phi}]. \quad (8.98)$$

where

$$\nabla \cdot \mathbf{F}^*[\bar{\phi}] = \nabla \cdot (\mathbf{F}[\bar{\phi}] + \bar{\phi} \nabla \times \boldsymbol{\psi}) = \nabla \cdot (\mathbf{F}[\bar{\phi}] - \nabla\bar{\phi} \times \boldsymbol{\psi}), \quad (8.99)$$

using the vector identity $\nabla \cdot (\bar{\phi} \nabla \times \boldsymbol{\psi}) = \nabla \cdot (\boldsymbol{\psi} \times \nabla\bar{\phi})$. Thus, we may define the *residual flux* with respect to $\bar{\phi}$ by

$$\mathbf{F}^*[\bar{\phi}] \equiv \mathbf{F}[\bar{\phi}] - \nabla\bar{\phi} \times \boldsymbol{\psi}, \quad (8.100)$$

remembering that the flux is only relevant up to a non-divergent additive factor. From (8.100) we see that, no matter how we choose $\boldsymbol{\psi}$, the components of the residual and raw flux along the mean tracer gradient, or across isosurfaces of tracer, are the same:

$$\mathbf{F}^*[\bar{\phi}] \cdot \nabla\bar{\phi} = \mathbf{F}[\bar{\phi}] \cdot \nabla\bar{\phi}. \quad (8.101)$$

However, the two fluxes along the tracer contours, the skew fluxes, differ and we may choose $\boldsymbol{\psi}$ in such a way that this is zero (for example) for the residual flux. Why is this a useful choice? It is because a skew flux is already present in the advective term, $\bar{\mathbf{v}}^* \cdot \nabla\bar{\phi}$ on the left-hand side and so it is useful for the right-hand side to contain only along-gradient fluxes.

To remove the residual skew flux, we simply make the choice

$$\boldsymbol{\psi} = \frac{\mathbf{F}[\bar{\phi}] \times \nabla\bar{\phi}}{|\nabla\bar{\phi}|^2}. \quad (8.102)$$

The flux associated with this is just the negative of the raw skew flux, because from (8.100) this is

$$-\nabla\bar{\varphi} \times \boldsymbol{\psi} = \frac{\nabla\bar{\varphi} \times \mathbf{F}[\varphi]}{|\nabla\bar{\varphi}|^2} \times \nabla\bar{\varphi} = -\mathbf{F}_{\parallel}. \quad (8.103)$$

Making this choice, the tracer evolution equation may be written

$$\frac{\partial\bar{\varphi}}{\partial t} + \bar{\mathbf{v}}^* \cdot \nabla\bar{\varphi} = -\nabla \cdot \mathbf{F}^*[\varphi] \quad (8.104a)$$

where

$$\bar{\mathbf{v}}^* = \bar{\mathbf{v}} - \nabla \times \boldsymbol{\psi} = \bar{\mathbf{v}} - \nabla \times \frac{\mathbf{F}[\varphi] \times \nabla\bar{\varphi}}{|\nabla\bar{\varphi}|^2} \quad (8.104b)$$

and

$$\mathbf{F}^*[\varphi] = \frac{\mathbf{F}[\varphi] \cdot \nabla\bar{\varphi}}{|\nabla\bar{\varphi}|^2} \nabla\bar{\varphi}. \quad (8.104c)$$

Evidently there is a different residual flux and a TEM formulation for any advected tracer, be it a passive tracer or an active one like potential temperature or potential vorticity. We must choose just one, and in doing so the skew flux of the other tracers will not be removed. The one most commonly chosen, for both oceanographic and atmospheric applications, is potential temperature, a sensible choice for two reasons. First, the potential temperature is a particularly important tracer. Second, the various eddy fluxes are in reality caused by advection by small-scale turbulent motion, and such motion acts similarly on all conserved tracers. Thus, removing the skew flux in potential temperature is likely to nearly remove it for all nearly-conserved tracers. Potential vorticity might be a suitable choice in some circumstances.

Suppose that the eddying flow is adiabatic, which is a particularly good approximation in the ocean interior and, if not to quite the same extent, in the stratosphere. This means that there is no eddy buoyancy flux across isopycnals (or heat flux across isotherms) and the residual flux with respect to buoyancy is zero. The thermodynamic equation is simply

$$\frac{\partial b}{\partial t} + \bar{\mathbf{v}}^* \cdot \nabla b = J[b]. \quad (8.105)$$

Thus, in much the same way as arose in the quasi-geostrophic equations, the transformed thermodynamic equation appears devoid of explicit eddy terms. However, the (usually numerical) solution of the transformed equations is not necessarily straightforward, because the residual velocity does not vanish at the boundary. Thus, for example, in $\bar{\mathbf{w}}^* \neq 0$ at a surface where there are nonzero heat fluxes.

A zonal average

Let the averages in the above expression (as denoted by the overbar) be a zonal average; the residual streamfunction is then a scalar, ψ , where $\boldsymbol{\psi} = \mathbf{i}\psi$ and

$$\bar{\mathbf{v}}^* = -\frac{\partial\psi}{\partial z}, \quad \bar{\mathbf{w}}^* = \frac{\partial\psi}{\partial y}. \quad (8.106)$$

The residual streamfunction of (8.102) becomes

$$\psi = \frac{\overline{v'b' \bar{b}_z} - \overline{w'b' \bar{b}_y}}{|\nabla_m b|^2} \quad (8.107)$$

and the residual flux is

$$\mathbf{F}^*[b] = \frac{\overline{v'b' \bar{b}_y} + \overline{w'b' \bar{b}_z}}{|\nabla_m b|^2} \nabla b \quad (8.108)$$

where $\nabla_m b = \mathbf{j} \partial b / \partial y + \mathbf{k} \partial b / \partial z$. Given a residual streamfunction, the zonally averaged zonal momentum equation can then be cast in TEM form, and a form of the EP flux naturally appears on the right-hand side.⁸

Relation to quasi-geostrophy

If the isotherms are nearly flat then the above forms reduce to the quasi-geostrophic TEM equations. Specifically, (i) $|\nabla \bar{b}|$ is approximated by $\bar{b}_z = b_{0z}$, and (ii) vertical eddy thermal fluxes are neglected. The streamfunction (8.107) then reduces to

$$\psi = \frac{\overline{v'b'}}{\bar{b}_z} \quad (8.109)$$

which is the same as the eddy part of (8.57).

** More general forms*

Transformations more general than that of (8.102) are possible that still lead to a transformed thermodynamic budget with the same general form, and in particular still partition the flux into a skew component and a digradient component, and that may be particularly useful as boundaries are approached.⁹ In particular, we may choose

$$\boldsymbol{\psi} = \frac{\mathbf{F}[\bar{\varphi}] \times \nabla \bar{\varphi}}{|\nabla \bar{\varphi}|^2} + \frac{\mathbf{F}[\varphi] \cdot \nabla \bar{\varphi}}{|\nabla \bar{\varphi}|^2} \boldsymbol{\alpha}. \quad (8.110)$$

where $\boldsymbol{\alpha}$ is an arbitrary vector. The transformed evolution equation for $\bar{\varphi}$ is then

$$\frac{\partial \bar{\varphi}}{\partial t} + \bar{\mathbf{v}}^* \cdot \nabla \bar{\varphi} = -\nabla \cdot \mathbf{F}^*[\varphi]. \quad (8.111)$$

where

$$\mathbf{F}^*[\varphi] = \frac{\mathbf{F}[\varphi] \cdot \nabla \bar{\varphi}}{|\nabla \bar{\varphi}|^2} (\nabla \bar{\varphi} + \boldsymbol{\alpha} \times \nabla \bar{\varphi}) \quad (8.112)$$

and $\bar{\mathbf{v}}^* = \mathbf{v} - \nabla \times \boldsymbol{\psi}$. The residual flux is still proportional to the diagraphic flux, and will be zero if the eddies are adiabatic. However, unlike (8.104c) the flux is not directed parallel to the gradient.

A particularly common choice is to choose $\boldsymbol{\alpha}$ to be proportional to the isothermal slope. Then, with $\varphi = b$, we have

$$\boldsymbol{\alpha} = -\frac{\mathbf{k} \times \nabla \bar{b}}{\bar{b}_z} \quad (8.113)$$

or, in the zonally averaged case $\boldsymbol{\alpha} = \mathbf{i}\alpha = -\mathbf{i}\bar{b}_y/\bar{b}_z$. After some algebra, we then find

$$\boldsymbol{\psi} = \frac{\mathbf{F}[b] \times \mathbf{k}}{\bar{b}_z} = \frac{1}{\bar{b}_z} (\mathbf{i} \overline{v'b'} - \mathbf{j} \overline{u'b'},) \quad (8.114)$$

and

$$\mathbf{F}^*[b] = \frac{\mathbf{F}[b] \cdot \nabla \bar{b}}{\bar{b}_z} \mathbf{k}. \quad (8.115)$$

In full, the averaged thermodynamic equation may be written as

$$\frac{\partial \bar{b}}{\partial t} + (\bar{\mathbf{u}} + \tilde{\mathbf{u}}) \cdot \nabla \bar{b} + (\bar{w} + \tilde{w}) \frac{\partial \bar{b}}{\partial z} = \frac{\partial G}{\partial z}. \quad (8.116)$$

where

$$G = \frac{\overline{v'b'} \cdot \nabla \bar{b}}{\bar{b}_z} = \frac{1}{\bar{b}_z} \left(\overline{u'b'} \cdot \nabla_z \bar{b} + \overline{w'b'} \frac{\partial \bar{b}}{\partial z} \right). \quad (8.117)$$

It may also be readily verified by direct manipulation that this is exactly equivalent to the usual, Eulerian averaged, thermodynamic equation. The eddy induced velocities $\tilde{\mathbf{u}}$ and $\tilde{w}$ may also be conveniently written in terms of a streamfunction as

$$\tilde{\mathbf{u}} = -\frac{\partial \tilde{\psi}}{\partial z}, \quad \tilde{w} = \nabla \cdot \tilde{\boldsymbol{\psi}} \quad (8.118a)$$

where

$$\tilde{\boldsymbol{\psi}} = \frac{\mathbf{F}[b]}{\bar{b}_z} = \frac{\overline{\mathbf{u}'b'}}{\bar{b}_z}. \quad (8.118b)$$

[Note $\tilde{\boldsymbol{\psi}}$ is different from the $\boldsymbol{\psi}$ of (8.114).] The quasi-geostrophic form follows particularly easily from this by setting $G = 0$ and $\bar{b} = b_0$ in (8.118).

This can be a particularly useful form in the diagnosis of numerical experiments and observations in the interior of the ocean or atmosphere where isopycnal slopes are small, and here it differs little from (8.104) but is easier to calculate and interpret. It is also useful for eddy flux parameterization schemes,

considered in chapter ???. The various forms of TEM do differ when isopycnals are steeply sloping and, relatedly, near boundaries. Note, for example, that the vertical residual velocity does not vanish at a horizontal boundary if there is a non-zero horizontal buoyancy flux.¹⁰

8.5 THE TEM, MOMENTUM FLUXES, AND FORM DRAG

For the rest of this chapter we return to quasi-geostrophic dynamics, and consider further the interpretation and application of potential vorticity fluxes and its relatives.

8.5.1 The nonacceleration theorem, revisited

Using the momentum equation: a two-dimensional case

Consider two-dimensional incompressible flow on the β -plane, for which there is no buoyancy flux. The linearized vorticity equation is

$$\frac{\partial \zeta'}{\partial t} + \bar{u} \frac{\partial \zeta'}{\partial x} + v' \frac{\partial \bar{\zeta}}{\partial y} = D \quad (8.119)$$

from which we derive, analogously to (8.23a), the Eliassen-Palm relation

$$\frac{\partial \mathcal{A}}{\partial t} + \frac{\partial \mathcal{F}}{\partial y} = \mathcal{D} \quad (8.120)$$

where $\mathcal{F} = -\overline{u'v'}$, $\mathcal{D}$ represents nonconservative forces, and

$$\mathcal{A} = \frac{\overline{\zeta'^2}}{2\partial \bar{\zeta} / \partial y} = \frac{1}{2} \overline{\eta'^2} \frac{\partial \bar{\zeta}}{\partial y} \quad (8.121)$$

The quantity $\eta' \equiv -\zeta' / \partial \bar{\zeta} / \partial y$ is proportional to the meridional particle displacement in a disturbance. Now consider the x-momentum equation

$$\frac{\partial u}{\partial t} = -\frac{\partial u^2}{\partial x} - \frac{\partial uv}{\partial y} - \frac{\partial \phi}{\partial x} + fv \quad (8.122)$$

Zonally averaging, noting that $\bar{v} = 0$, gives

$$\frac{\partial \bar{u}}{\partial t} = -\frac{\partial \overline{uv}}{\partial y} = \overline{v'\zeta'} = \frac{\partial \mathcal{F}}{\partial y}. \quad (8.123)$$

Finally, combining (8.120) and (8.123) gives

$$\frac{\partial}{\partial t} (\bar{u} + \mathcal{A}) = \mathcal{D} \quad (8.124)$$

In the absence of nonconservative terms (i.e., if $\mathcal{D} = 0$) the quantity $\bar{u} + \mathcal{A}$ is constant.¹¹ Further, if the waves are steady and conservative then $\mathcal{A}$, and hence $\bar{u}$, are constant.

The stratified case

In the stratified case we can use the TEM form of the momentum equation to derive a similar result. The unforced zonally averaged zonal momentum equation can be written

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \nabla \cdot \mathcal{F}, \quad (8.125)$$

and using the Eliassen-Palm relation this may be written,

$$\frac{\partial}{\partial t}(\bar{u} + \mathcal{A}) - f_0 \bar{v}^* = \mathcal{D} \quad (8.126)$$

and so again $\mathcal{A}$ is related to the momentum of the flow. If, furthermore, the waves are steady and conservative then $\partial \bar{u} / \partial t - f_0 \bar{v}^* = 0$. However, under these same conditions the residual circulation will also be zero. This is because the residual meridional circulation $(\bar{v}^*, \bar{w}^*)$ arises via the necessity to keep the temperature and velocity fields in thermal wind balance, and is thus determined by an elliptic equation, namely (8.63). If the waves are steady and adiabatic then the right-hand side of this equation is zero and

$$f_0^2 \frac{\partial^2 \psi^*}{\partial z^2} + N^2 \frac{\partial^2 \psi^*}{\partial y^2} = 0. \quad (8.127)$$

If $\psi^* = 0$ at the boundaries, then the unique solution of this is $\psi^* = 0$ everywhere. At the meridional boundaries we may certainly suppose that ψ^* vanishes if these are quiescent latitudes, and at the horizontal boundaries the buoyancy flux will vanish if the waves there are steady, because from (8.10)

$$\overline{v' b'} \frac{\partial \bar{b}}{\partial y} = -\frac{1}{2} \frac{\partial}{\partial t} \overline{b'^2} = 0. \quad (8.128)$$

Under these circumstances, then, the residual meridional circulation vanishes in the interior, and from (8.125), the mean flow is steady.

Compare (8.125) with the momentum equation in conventional Eulerian form, namely

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v} = \overline{v' \zeta'} \quad (8.129)$$

There is no reason that the vorticity flux should vanish when waves are present, even if they are steady. However, such a flux is (under nonacceleration conditions) precisely compensated by the meridional circulation $f_0 \bar{v}$, something that is almost impossible to infer or intuit directly from (8.129). Even when nonacceleration conditions do not apply there will be a significant cancellation between the two terms, and resulting acceleration of the mean flow will be a result mainly of any additional terms in the equation.

Unlike the proof of the nonacceleration result given in section 8.3.2 the above argument does not use the invertibility property of potential vorticity so directly, suggesting an extension to the primitive equations. However, the primitive-equation form of the wave activity density is complicated and we do not give the proof here.¹²

8.5.2 The EP flux and form drag

It may seem a little magical that the zonal flow is driven by the Eliassen-Palm flux via (8.125). The polewards vorticity flux is clearly related to the momentum flux convergence, but why should a polewards buoyancy flux affect the momentum? Omitting the Coriolis and vorticity flux terms the TEM momentum equation is

$$\frac{\partial \bar{u}}{\partial t} = \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} \overline{v'b'} \right). \quad (8.130)$$

and the right-hand side certainly does not look like a force. However, it turns out to be directly proportional to the *form drag* between isentropic layers. Recall from section 5.2.3 that the form drag, τ_d at an interface between two layers of shallow water is

$$\tau_d = -\eta' \frac{\partial p'}{\partial x} \quad (8.131)$$

where η is the interfacial displacement. But from (8.84) $\eta = -b'/N^2$ and with this and geostrophic balance we have

$$\tau_d = \frac{f_0}{N^2} \overline{v'b'} \quad (8.132)$$

Thus, the vertical component of the EP flux (i.e., the meridional buoyancy flux) is in fact a real stress acting on a fluid layer and equal to the momentum flux caused by the wavy interface. The net momentum convergence into an infinitesimal layer of mean thickness $\bar{h}$ is then [c.f., (5.65)],

$$F_d = \bar{h} \frac{\partial \tau_d}{\partial z} = \bar{h} \frac{\partial}{\partial z} \left(\frac{\overline{v'b'}}{N^2} \right). \quad (8.133)$$

and a layer of mean thickness $\bar{h}$ is accelerated according to

$$\frac{\partial \bar{u}}{\partial t} = f_0 \frac{\partial}{\partial z} \left(\frac{\overline{v'b'}}{\bar{b}_z} \right) + F_m, \quad (8.134)$$

where $F_m = \overline{v'\zeta'} + f_0 \bar{v}^*$ represents forces from the momentum flux and Coriolis force. The appearance of the buoyancy flux is really a consequence of the way

we have chosen to average the equations: obtaining (8.134) involved averaging the forces over an isentropic layer, and given this it can only be the residual circulation that contributes to the Coriolis force. Thus, so to speak, the vertical component of the EP flux is a force in drag, a form masquerading as a bouyancy flux.

8.6 EXAMPLES OF WAVE-MEANFLOW INTERACTION

[This section is not yet written. Suggestions welcome.]

8.7 NECESSARY CONDITIONS FOR INSTABILITY

8.7.1 Stability conditions from wave activity conservation

In sections 7.3 and 7.6.1 we derived necessary conditions for instability, assuming that the instability is of normal-mode form. Here we show that such an assumption is not necessary, and give a derivation of the necessary conditions that is both more general and, in some regards, simpler. Our results are based on the conservation properties of the equations, and the fact that the potential vorticity flux may be written as a divergence, and hence when integrated over a domain vanishes, apart possibly from boundary contributions.

Consider again the perturbation enstrophy equation

$$\frac{1}{2} \frac{\partial \overline{q'^2}}{\partial t} = - \frac{\partial \bar{q}}{\partial y} \nabla_m \cdot \mathcal{F}. \quad (8.135)$$

where $\mathcal{F}$ is the Eliassen-Palm flux given by (8.21)

Dividing by $\partial \bar{q} / \partial y$ and integrating over a domain $\mathcal{D}$ which is such that the Eliassen-Palm flux vanishes at the boundaries (assuming for now that this is possible) gives

$$\int_{\mathcal{D}} \frac{\partial \overline{q'^2} / \partial t}{\partial \bar{q} / \partial y} dx dy = 0 \quad (8.136)$$

The quantity $\overline{q'^2} / \partial \bar{q} / \partial y$ is the pseudo-momentum for this problem. Eq. (8.136) implies that, in the *norm* $[\overline{q'^2} / \partial \bar{q} / \partial y]$, the perturbation cannot grow unless $\partial \bar{q} / \partial y$ changes sign somewhere in the domain, or at the boundaries. This result does not depend on there being a normal mode instability. The simplest result of all occurs in a barotropic problem with no vertical variation. Then $\partial \bar{q} / \partial y = \partial \bar{\zeta}_a / \partial y = \beta - \bar{u}_{yy}$, and demanding that this must change sign for an instability is the Rayleigh's condition for instability, or the Rayleigh-Kuo condition if β is present. In the more general case, if $\partial \bar{q} / \partial y$ changes sign along a vertical line then the instability is called a baroclinic instability, and if it changes sign along a horizontal the instability is barotropic — these may be taken as the definitions

of those terms. A mixed instability has a change of sign along both a horizontal and vertical.

8.7.2 Inclusion of boundary terms

Suppose now the flow is contained between two flat boundaries, at $z = 0$ and $z = H$, and the relevant equations of motion are now the potential vorticity evolution in the interior, supplemented by the thermodynamic equation at the boundary. These give, in the usual way, [c.f. (8.10) and (8.13)]

$$\frac{\partial}{\partial t} \left(\frac{1}{2} \frac{\overline{q'^2}}{\partial \bar{q} / \partial y} \right) = -\overline{v'q'} \quad 0 < z < H \quad (8.137)$$

and

$$\frac{\partial}{\partial t} \left(\frac{1}{2} \frac{\overline{b'^2}}{\partial \bar{b} / \partial y} \right) = -\overline{v'b'} \quad z = 0, H \quad (8.138)$$

Now polewards flux of potential vorticity is

$$\overline{v'q'} = -\frac{\partial}{\partial y} \overline{u'v'} + \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} \overline{v'b'} \right), \quad (8.139)$$

and integrating this expression with respect to both y and z gives

$$\int_D \overline{v'q'} dy dz = \left[\frac{f_0}{N^2} \overline{v'b'} \right]_0^H, \quad (8.140)$$

assuming that the meridional boundaries are at quiescent latitudes.

Integrating (8.137) over y and z , and using (8.140) gives

$$\frac{\partial}{\partial t} \iint \frac{1}{2} \frac{\overline{q'^2}}{\partial \bar{q} / \partial y} dy dz = -\frac{f_0}{N^2} \left[\frac{f_0}{N^2} \overline{v'b'} \right]_0^H. \quad (8.141)$$

Using (8.138) to eliminate $\overline{v'b'}$ finally gives

$$\frac{\partial}{\partial t} \left\{ \iint \frac{1}{2} \frac{\overline{q'^2}}{\partial \bar{q} / \partial y} dy dz - \int \left[\frac{1}{2} \frac{f_0}{N^2} \frac{\overline{b'^2}}{\partial \bar{b} / \partial y} \right]_0^H dy \right\} = 0. \quad (8.142)$$

If this expression is positive or negative definite the perturbation cannot grow and therefore the basic state is stable. Stability thus depends on the meridional gradient of potential vorticity in the interior, and the meridional gradient of buoyancy at the boundary. If either $\partial \bar{q} / \partial y$ or $\partial \bar{b} / \partial y$ changes sign in the interior or at the boundary, respectively, we have the potential for instability. If these are both one signed, then various possibilities exist, for example: (with $f_0 > 0$):

I *A stable case:*

$$\frac{\partial \bar{q}}{\partial y} > 0 \text{ and } \frac{\partial \bar{b}}{\partial y} \Big|_{z=0} > 0 \text{ and } \frac{\partial \bar{b}}{\partial y} \Big|_{z=H} < 0 \implies \text{stability.} \quad (8.143)$$

Using the thermal wind relationship, $f_0 \partial \bar{u} / \partial z - \partial \bar{b} / \partial y$ this is equivalent to:

$$\frac{\partial \bar{q}}{\partial y} > 0 \text{ and } \frac{\partial u}{\partial z} \Big|_{z=0} < 0 \text{ or } \frac{\partial u}{\partial z} \Big|_{z=H} > 0 \implies \text{stability.} \quad (8.144)$$

Stability also ensues if all inequalities are switched.

II *Instability via interior-surface interactions:*

$$\frac{\partial \bar{q}}{\partial y} > 0 \text{ and } \frac{\partial \bar{b}}{\partial y} \Big|_{z=0} < 0 \text{ or } \frac{\partial \bar{b}}{\partial y} \Big|_{z=H} > 0 \implies \text{potential instability.} \quad (8.145)$$

Using thermal wind this may be written:

$$\frac{\partial \bar{q}}{\partial y} > 0 \text{ and } \frac{\partial u}{\partial z} \Big|_{z=0} > 0 \text{ or } \frac{\partial u}{\partial z} \Big|_{z=H} < 0 \implies \text{potential instability.} \quad (8.146)$$

The condition $\partial \bar{q} / \partial y > 0$ and $(\partial u / \partial z)_{z=0} > 0$ is the most common criterion for instability that is met in the atmosphere. In the troposphere we can sometimes ignore contributions of the buoyancy fluxes at the tropopause ($z = H$), and stability is then determined by the interior potential vorticity gradient and the surface buoyancy gradient. Similarly, in the ocean contributions from the ocean floor are normally very small.

III *Instability via edge wave interaction:*

$$\frac{\partial \bar{b}}{\partial y} \Big|_{z=0} < 0 \text{ and } \frac{\partial \bar{b}}{\partial y} \Big|_{z=H} < 0 \implies \text{potential instability.} \quad (8.147)$$

And similarly, with both inequalities switched. Using thermal wind this may be written:

$$\frac{\partial u}{\partial z} \Big|_{z=0} > 0 \text{ or } \frac{\partial u}{\partial z} \Big|_{z=H} > 0 \implies \text{stability.} \quad (8.148)$$

This occurs in the Eady problem, and in the real atmosphere.

Typically, in the atmosphere, the zonally averaged potential vorticity gradient in the interior is positive almost everywhere, and baroclinic instability arises because of temperature gradient at the lower boundary. Consider, for example, the

‘Eady problem’ in which the basic state is $\bar{u} = \Gamma z$, N^2 is a constant and $\beta = 0$. Then the basic state has $\bar{q} = 0$ in the interior and the instability arises because $\partial \bar{b} / \partial y$ has the same sign at the upper and lower boundaries, as in case III above. If there is no lid at $z = H$, then the instability in the Eady problem disappears. However, instability will arise if there is a positive shear at the lower boundary and a positive potential vorticity gradient in the interior, with no necessary condition at the upper boundary, and this occurs in the Charney problem.

One consequence of the upper boundary condition is that it provides a condition on depth of the disturbance. In the Eady problem the evolution of the system is determined by temperature evolution at the surface,

$$\frac{Db}{Dt} = 0 \quad \text{at } z = 0, H \quad (8.149)$$

(where $b = f_0 \partial \psi / \partial z$) and zero potential vorticity in the interior,

$$\nabla^2 \psi + k_d^2 H^2 \frac{\partial^2 \psi}{\partial z^2} = 0 \quad 0 < z < H \quad (8.150)$$

where $k_d = f_0 / (HN)$. Assuming a solution of the form $b \sim \sin kx$ then the Poisson equation (8.150) becomes

$$H^2 k_d^2 \frac{\partial^2 \psi}{\partial z^2} = -k^2 \psi \quad (8.151)$$

with solutions $\psi = A \exp(-\alpha z) + B \exp(\alpha z)$ where $\alpha^2 = k^2 N^2 / f_0^2$. The scale height of the disturbance is thus

$$h \sim \frac{f_0 L}{2\pi N} \quad (8.152)$$

where $L \sim 2\pi / k$ is the horizontal scale of the disturbance. If the upper boundary is higher than this, it cannot interact strongly with the surface, because the disturbances at either boundary decay too much before reaching the other. Put another way, if the structure of the disturbance is such that it is shallower than H , the presence of the upper boundary is not felt. In the Eady problem, we know that the upper boundary must be important, because it is only by its presence that the flow can be unstable. Thus, all unstable modes in the Eady problem must be ‘deep’ in this sense, which can be verified by direct calculation. This condition gives rise to a physical interpretation of the high-wavenumber cut-off: if L is too small, the modes are too shallow to span the full depth of the fluid, and from (8.152) the condition for this is

$$L_c = 2\pi \frac{NH}{f_0} \quad (8.153)$$

or

$$k_c = \frac{f_0}{NH} = L_d. \quad (8.154)$$

Wavenumbers larger than the deformation radius are stable in the Eady problem. If β is non-zero, this condition does not apply, because the necessary condition for instability can be satisfied by a combination of a surface temperature gradient and an interior gradient of potential vorticity provided by β , as in condition (II) in section 8.7.2. Thus, we may expect that, if $\beta \neq 0$, higher wavenumbers ($k > k_d$) may be unstable but if so they will be shallow, and this may be confirmed by explicit calculation (see figures 7.14, 7.16 and 7.20.) In the two-level model shallow modes are, by construction, not allowed so that high wavenumbers will be stable, with or without beta.

8.8 * NECESSARY CONDITIONS FOR INSTABILITY: PSEUDOENERGY

In this section we derive another necessary condition for instability, based on the conservation properties of energy and enstrophy.¹³

8.8.1 Two-dimensional flow

Let us first consider inviscid, incompressible two-dimensional flow governed by the equation of motion

$$\frac{\partial q}{\partial t} + J(\psi, q) = 0 \quad (8.155)$$

where $q = \zeta + f = \nabla^2 \psi + f$ is the absolute vorticity and ψ is the streamfunction. A steady solution of this equation is given by

$$q = Q(\psi). \quad (8.156)$$

where Q is a differentiable but otherwise arbitrary function of its argument. To see this, substitute it into (8.155) to give

$$\frac{\partial q}{\partial t} = -\frac{dQ}{d\psi} J(\psi, \psi) = 0. \quad (8.157)$$

We will consider the special case $Q = \gamma\psi$, and prove that if $\gamma > 0$ then the flow is stable. To this end, we consider the evolution of perturbations about such a steady state. Let

$$q = Q + q' \quad (8.158)$$

$$\psi = \Psi + \psi' \quad (8.159)$$

where $Q = \gamma\Psi$. Then, using (8.155), the equation of motion for the perturbation is

$$\frac{\partial q'}{\partial t} + J(\psi', \gamma\Psi) + J(\Psi, q') + J(\psi', q') = 0 \quad (8.160)$$

or

$$\frac{\partial q'}{\partial t} + J(\Psi, q' - \gamma\psi') + J(\psi', q') = 0. \quad (8.161)$$

We emphasize that the perturbation quantities need *not* be assumed small compared to the steady-state values, and henceforth we will drop the primes on the perturbation quantities.

We form the perturbation enstrophy and energy equations by multiplying by q and $-\psi$ respectively to give

$$\frac{dZ}{dt} = \frac{1}{2} \frac{d\langle q^2 \rangle}{dt} = -\langle qJ(\Psi, -\gamma\psi) \rangle \quad (8.162a)$$

$$\frac{dE}{dt} = \frac{1}{2} \frac{d\langle (\nabla\psi)^2 \rangle}{dt} = \langle \psi J(\Psi, q) \rangle. \quad (8.162b)$$

The angle brackets denote horizontal integrations over the domain, and we have assumed that there is no contribution from the boundaries, which holds with periodic or no-normal-flow conditions.

Now, a property of Jacobians that may simply be checked by integrating by parts is that

$$\langle aJ(b, c) \rangle = \langle bJ(c, a) \rangle = \langle cJ(a, b) \rangle = -\langle cJ(b, a) \rangle \quad (8.163)$$

again assuming appropriate boundary conditions (problem 8.1). Thus, (8.162) becomes

$$\frac{1}{\gamma} \frac{dZ}{dt} = -\langle \psi J(\Psi, q) \rangle \quad (8.164a)$$

$$\frac{dE}{dt} = +\langle \psi J(\Psi, q) \rangle \quad (8.164b)$$

and so

$$\frac{d}{dt} \left[E + \frac{Z}{\gamma} \right] = 0. \quad (8.165)$$

Now, E and Z are both positive definite quantities. Thus, if γ is also positive everywhere $(E + Z/\gamma)$ is a strictly positive quantity, and the growth of the perturbation energy and the perturbation enstrophy is strictly limited. Therefore the basic state is *stable*, or in particular it is said to be *stable in the sense of Liapunov*. This means that the magnitude of the perturbation, as measured by some norm,

is bounded by its initial magnitude or a function thereof. In this case we define the norm

$$\|\psi\|^2 \equiv \iint \left[(\nabla\psi)^2 + \frac{1}{\gamma} (\nabla^2\psi)^2 \right] dx dy. \quad (8.166)$$

Then, clearly,

$$\|\psi(t)\|^2 = \|\psi(0)\|^2 \quad (8.167)$$

We have thus established a sufficient condition for nonlinear stability. Even though the stability is defined for a particular norm, stability in this sense provides bounds on the growth of either the disturbance energy or enstrophy. For example, if perturbation energy is to grow perturbation enstrophy must shrink, but it cannot shrink past zero. The stability condition also precludes normal mode instability in which all modes grow exponentially.

The norm (8.166) defines the *pseudo-energy* of the disturbance. The 'energy' part arises because it is clearly an energy-like quantity. It is not the energy of the disturbance (hence the 'pseudo'), but it is a measure of the amplitude of the disturbance that is quadratic in the disturbance fields. When conserved, such quantities are generically called *wave activities*. Thus, pseudo-energy is an example of a wave activity. Often, wave activities can only be obtained in the limit of small amplitude disturbances, but here we have obtained a wave activity that is valid at finite amplitude.

8.8.2 Quasi-Geostrophic flow

The extension to quasi-geostrophic flow is mostly straightforward, but with a complication from the vertical boundary conditions at the surface and at an upper boundary.¹⁴ The reader willing to accept this may skip straight to the result, equation (8.176).

The equations of motion are

$$\frac{\partial q}{\partial t} + J(\psi, q) = 0 \quad 0 < z < H \quad (8.168a)$$

$$\frac{\partial b}{\partial t} + J(\psi, b) = 0 \quad z = 0, H \quad (8.168b)$$

where

$$q = \nabla^2\psi + \frac{\partial}{\partial z} \left(S(z) \frac{\partial\psi}{\partial z} \right) \quad (8.169)$$

$$b = f_0 \frac{\partial\psi}{\partial z} \quad z = 0, H \quad (8.170)$$

where $S(z) = f_0^2 / N^2$ is positive. We suppose that the basic state is given by

$$Q = \gamma(z)\Psi \quad \text{and} \quad b(0) = \gamma_0\psi, \quad b(H) = \gamma_H\psi. \quad (8.171a)$$

Analogous to (8.161) we obtain the equations of motion for the perturbation,

$$\frac{\partial q}{\partial t} + J(\Psi, q - \gamma(z)\psi) + J(\psi, q) = 0 \quad z < 0 < H, \quad (8.172a)$$

$$\frac{\partial b}{\partial t} + J(\Psi, b - \gamma_0\psi) + J(\psi, b) = 0 \quad z = 0, \quad (8.172b)$$

$$\frac{\partial b}{\partial t} + J(\Psi, b - \gamma_H\psi) + J(\psi, b) = 0 \quad z = H. \quad (8.172c)$$

Now form the energy and enstrophy equations by multiplying (8.172a) by q and $-\psi$ and integrating over the three-dimensional domain. In the interior we obtain

$$\frac{1}{2} \frac{d \langle q^2 \rangle}{dt} + \langle qJ(\Psi, -\gamma\psi) \rangle = 0 \quad (8.173a)$$

$$\frac{1}{2} \frac{d}{dt} \overline{\left\langle (\nabla\psi)^2 + S \left(\frac{\partial\psi}{\partial z} \right)^2 \right\rangle} - \overline{\langle \psi J(\Psi, q) \rangle} = \left[\left\langle S\psi \frac{\partial}{\partial t} \left(\frac{\partial\psi}{\partial z} \right) \right\rangle \right]_0^H \quad (8.173b)$$

where the overbar here denotes a vertical integration. At the lower boundary we obtain, from (8.172b):

$$\frac{1}{2} \frac{d \langle b^2 \rangle}{dt} + \langle bJ(\Psi, -\gamma_0\psi) \rangle = 0 \quad (8.174a)$$

$$\left\langle \psi \frac{\partial b}{\partial t} \right\rangle + \langle \psi J(\Psi, b) \rangle = 0 \quad (8.174b)$$

with a similar equation at the upper boundary. Using (8.173b) and (8.174b) (and its upper boundary equivalent) gives

$$\begin{aligned} \frac{dE}{dt} &= \langle \psi J(\Psi, q) \rangle + S \langle \psi J(\Psi, b) \rangle \big|_{z=0} - S \langle \psi J(\Psi, b) \rangle \big|_{z=H} \quad (8.175) \\ &= -\frac{1}{\gamma} \frac{dZ}{dt} + S \langle \psi J(\Psi, b) \rangle \big|_{z=0} - S \langle \psi J(\Psi, b) \rangle \big|_{z=H} \quad [\text{using (8.173a)}] \\ &= -\frac{1}{\gamma} \frac{dZ}{dt} - \frac{S}{\gamma_0} \frac{dB_0}{dt} + \frac{S}{\gamma_H} \frac{dB_H}{dt} \quad [\text{using (8.174a)}] \end{aligned}$$

In these equations, $E = (1/2) \overline{\langle (\nabla\psi)^2 + S(\partial\psi/\partial z)^2 \rangle}$, $Z = (1/2) \overline{\langle q^2 \rangle}$ and $B_H = (1/2) \langle b(H)^2 \rangle \big|_H$, $B_0 = (1/2) \langle b(0)^2 \rangle$ and each are positive definite. Thus, just as in the two-dimensional case we have constructed a pseudoenergy that is conserved. That is,

$$\frac{dA}{dt} = \frac{d}{dt} \left[E + \frac{Z}{\gamma} + \frac{SB_0}{\gamma_0} - \frac{SB_H}{\gamma_H} \right] = 0 \quad (8.176)$$

If the quantity in square brackets is positive definite, the perturbation cannot grow in the norm defined by A , and the flow is stable; evidently positive definiteness is assured if $\gamma > 0$, $\gamma_0 > 0$ and $\gamma_H < 0$.¹⁵

8.8.3 * Necessary Conditions for Baroclinic Instability

What good are the above manipulations? We show that the conditions derived in the previous section can be used to provide necessary conditions for baroclinic instability.¹⁶

Minimum shear in two-layer flow

Pseudo-energy

In two-level quasi-geostrophic flow, as conventionally written, the boundary conditions are ‘built-in’ to the definitions of potential vorticity — Bretherton’s boundary layer is an implicit part of the two-level equations of motion. The two-layer formulation is formally identical. We consider two-layers of equal depth, on a flat-bottomed Cartesian β -plane. The potential vorticity in each layer is then

$$q_i = \nabla^2 \psi + \frac{k_d^2}{2}(\psi_j - \psi_i) \quad i = 1, 2 \quad j = 3 - i. \quad (8.177)$$

Let the basic state be one of constant shear with

$$\Psi_1 = -U_1 y \quad \Psi_2 = -U_2 y \quad (8.178)$$

and therefore the basic state potential vorticity is

$$\begin{aligned} Q_1 &= \beta y - \frac{k_d^2}{2}(U_2 - U_1)y \\ Q_2 &= \beta y - \frac{k_d^2}{2}(U_1 - U_2)y \end{aligned} \quad (8.179)$$

Thus the basic state is characterized by

$$Q_i = \gamma_i \Psi_i, \quad i = 1, 2 \quad (8.180)$$

where

$$\gamma_1 = -\frac{(\beta + \lambda^2 U)}{(U_g + U)}, \quad \gamma_2 = -\frac{(\beta - \lambda^2 U)}{(U_g - U)} \quad (8.181)$$

and $U_1 = U_g + U$ and $U_2 = U_g - U$. That is, U_g is the barotropic flow, and U is the shear.

The conserved pseudo-energy corresponding to (8.176) obeys

$$\frac{dA}{dt} = \frac{d}{dt} \left\langle (\nabla \psi_1)^2 + (\nabla \psi_2)^2 + \frac{1}{2}k_d^2(\psi_1 - \psi_2) + \frac{q_1^2}{\gamma_1} + \frac{q_2^2}{\gamma_2} \right\rangle = 0 \quad (8.182)$$

Evidently, this is not positive-definite. However, the barotropic flow does not affect the stability properties, which is equivalent to saying that the system is

Galilean invariant. Thus, we choose $U_g < -U$, and this makes $\gamma_1 > 0$. Then γ_2 is also positive if $\beta > k_d^2 U / 2$ or

$$U < \frac{\beta}{k_d^2} \quad (8.183)$$

Pseudo-momentum

The flow will also be stable if $\partial Q / \partial y > 0$, for then the conserved pseudo-momentum will be positive definite. From (8.179), we have

$$\begin{aligned} \frac{dQ_1}{dy} &= \beta - \frac{k_d^2}{2}(U_2 - U_1) \\ \frac{dQ_2}{dy} &= \beta - \frac{k_d^2}{2}(U_1 - U_2) \end{aligned} \quad (8.184)$$

Clearly $dQ_1 / dy > 0$ and dQ_2 / dy will be positive if

$$U = (U_1 - U_2) < \frac{\beta}{k_d^2} \quad (8.185)$$

as in (8.183). Thus, the minimum shear for instability is

$$\boxed{U_{min} = \frac{\beta}{k_d^2}}. \quad (8.186)$$

The importance of these results is that they establish *nonlinear* stability — that is stability to finite-size perturbations. Normal mode instabilities are certainly precluded, but the result is much stronger than that. If the shear is below the critical level the energy that can be extracted from the mean flow is constrained by the size of the initial perturbation, and indeed in the pseudo-energy norm the perturbation cannot grow at all, no matter its initial size.

Multi-layer flow

We saw earlier that if the shear in the interior is uniform, as in the Eady problem for example, there is no critical shear for instability. In general, the instabilities occur for small shears are *shallow*, becoming deep only as the shear increases past the critical value in the two-layer model. If the vertical shear is not uniform, but instead is of the form

$$U = U_0 \cos(\pi p z / H) \quad (8.187)$$

then there is in fact a critical shear. The parameter p is the vertical wavenumber of the basic state, the fluid is confined between rigid surfaces at $z = 0$ and H . The basic state has no temperature gradient at the boundary, and so stability is assured if $\partial Q / \partial y$ is one-signed. As in the Eady problem we assume that the stratification is uniform, and so the basic state potential vorticity is

$$Q = \nabla^2 \psi + \frac{f_0^2}{N^2} \frac{\partial^2 \Psi}{\partial z^2} + \beta y. \quad (8.188)$$

Thus

$$\frac{dQ}{dy} = \beta - k_d^2 p^2 U_0 \quad (8.189)$$

where here $K_d^2 = f_0^2 H^2 / N^2$ is the inverse deformation radius for the problem. Stability is assured if

$$U_0 < \frac{\beta}{K_d^2 p^2} \quad (8.190)$$

which is analogous to (8.185). If $p = 1$ then the basic state has a very simple structure, as in the two-layer problem, and a similar stability result ensues. The more complex the basic state (i.e., the higher the Fourier mode representing the basic shear) the lower the critical shear for instability. If the interior shear is uniform, then the basic state is a sum of all Fourier modes, a consequence of which is that there is no minimum shear. We defer further physical interpretation to section ??.

8.8.4 High Wavenumber Cut-off

** From the equations of motion*

(This subsection will probably be omitted in the final version!) The basic state of each layer is a function of y alone. Without loss of generality we choose the barotropic part to be zero, so that

$$\Psi_1 = -Uy, \quad \Psi_2 = +Uy. \quad (8.191)$$

We also choose $\beta = 0$ so that

$$Q_1 = k_d^2 Uy, \quad Q_2 = -k_d^2 Uy = -Q_1 \quad (8.192)$$

The perturbation potential vorticity as

$$\begin{aligned} q_1 &= \nabla^2 \psi_1 + \frac{k_d^2}{2} (\psi_2 - \psi_1) \\ q_2 &= \nabla^2 \psi_2 + \frac{k_d^2}{2} (\psi_1 - \psi_2) \end{aligned} \quad (8.193)$$

and the linearized perturbation equation for each layer becomes

$$\begin{aligned}\frac{\partial q_1}{\partial t} + U \frac{\partial q_1}{\partial x} + \frac{\partial Q_1}{\partial y} \frac{\partial \psi_1}{\partial x} &= 0 \\ \frac{\partial q_2}{\partial t} - U \frac{\partial q_2}{\partial x} - \frac{\partial Q_1}{\partial y} \frac{\partial \psi_2}{\partial x} &= 0\end{aligned}\quad (8.194)$$

Adding and subtracting these equations yields

$$\begin{aligned}\frac{\partial}{\partial t}(q_1 + q_2) + \left[U - \frac{\partial Q_1 / \partial y}{k^2 + k_d^2} \frac{\partial}{\partial x}(q_1 - q_2) \right] &= 0 \\ \frac{\partial}{\partial t}(q_1 - q_2) + \left[U - \frac{\partial Q_1 / \partial y}{k^2} \frac{\partial}{\partial x}(q_1 + q_2) \right] &= 0\end{aligned}\quad (8.195)$$

Combining these, and assuming a variation in x of the form $\exp(ikx)$, gives

$$\frac{\partial^2}{\partial t^2}(q_1 - q_2) + k_x^2 \left\{ \left[U - \frac{\partial Q_1 / \partial y}{k^2 + k_d^2} \right] \left[U - \frac{\partial Q_1 / \partial y}{k^2} \right] \right\} (q_1 - q_2) = 0. \quad (8.196)$$

It is clear that the perturbation can grow exponentially only if the term in curly bracket is negative, and this in turn can only happen if

$$\frac{\partial Q_1 / \partial y}{U} - k_d^2 < k^2 < \frac{\partial Q_1 / \partial y}{U}. \quad (8.197)$$

Thus, there is a short-wave cut-off to instability at

$$k_{max}^2 = \frac{\partial Q_1 / \partial y}{U} = k_d^2. \quad (8.198)$$

Using pseudo-energy

Turning back to the two-layer case, let us choose the barotropic flow to be $U_g = \beta / k_d^2$. Then the pseudo-energy (8.182) takes the form

$$\frac{dA}{dt} = \frac{d}{dt} \sum_k \{ k^2 \psi_k^2 (k^2 - k_d^2) + k'^2 \tau_k^2 (k'^2 - k_d^2) \} = 0 \quad (8.199)$$

where $\psi = (\psi_1 + \psi_2) / 2$, $\tau = (\psi_1 - \psi_2) / 2$ and $k'^2 = k^2 + k_d^2$. We have replaced the integral over physical space by a sum over Fourier space, using Parseval's theorem.

If the sum (8.199) is restricted to modes for which $k > k_d$, then the flow is stable no matter what the shear may be. In the linear problem, in which perturbation modes do not interact, there is thus a *high-wavenumber* cut-off to

instability, as we previously noted. If geometric reasons dictate that the smallest allowable wavenumber is greater than k_d then the flow is always nonlinearly stable. Such a situation might arise on a planet whose circumference were less than the deformation radius, or in a small ocean basin.

No such high wavenumber cut-off can be derived in general in the continuously stratified case. However, if the vertical shear takes the form (8.187) then a cut-off does arise. To derive this, we choose a barotropic flow equal to $-\beta/(k_d^2 p^2)$ so that the mean flow is $U = U_0 \cos(\pi p z / H) - \beta/(k_d^2 p^2)$ and $\gamma(z) = Q/\Psi = -k_d^2 p^2$. If the perturbation streamfunction is expanded as

$$\psi = \sum_{\mathbf{k}, \alpha} \psi_{\mathbf{k}, \alpha} e^{i\mathbf{k} \cdot \mathbf{x}} \cos \alpha z \quad (8.200)$$

then the pseudo-energy constraint is

$$\frac{dA}{dt} = \frac{d}{dt} \sum_{\mathbf{k}, \alpha} \{ (k^2 + k_d^2 \alpha^2) [(k^2 + k_d^2 \alpha^2) - k_d^2 p^2] \} = 0 \quad (8.201)$$

In the linear problem, the sum is zero for each individual horizontal wavenumber $\mathbf{k}$. Then, for $p = 1$ this sum is positive definite if $k^2 > k_d^2$ so that, for this problem, there is again a high-wavenumber cut-off for instability, just as in the two-layer case, even when $\beta \neq 0$. Again, if geometric constraints assure that $k > k_d, \forall k$, then the basic state is nonlinearly stable.

The power of this derivation over the one directly from the equations of motion is that

- (i) It includes the β effect.
- (ii) It is potentially a nonlinear result.

Notes

- 1 After ?.
- 2 A more general demonstration of the group velocity property follows directly from (8.23a) by way of a result given in ?.
- 3 These restrictions on the basic state are not necessary to prove orthogonality, but they make the algebra simpler. Also, we don't pay attention here to the nature of the eigenvalues of (8.39), which in general consist of both a discrete and continuous spectrum. See ?, ? and ?.
- 4 Nonacceleration arguments have a long history, including contributions from ?, ?, ? and, particularly, ?, and the final result may be referred to as the 'Charney-Drazin-Boyd theorem'. ? put these results in the context of the EP flux and the TEM formalism.

- 5 The TEM formulation stems from ?, ? and ?. Later on we will also draw on ?.
- 6 The first derivation is based on ?. See also ?, and ? and ? for generalizations. The second 'analytic' derivation is based on one by A. Plumb. In the final version of this book the two derivations may be merged as they are not independent.
- 7 This problem can be worked around in some cases, especially if one's primary interest is in the buoyancy budget. See (among others) ? or ? for various ways to deal with three-dimensional flows.
- 8 ?.
- 9 ?. Also see ?.
- 10 The issues involved are not settled and progress is being made apace, so it is hard to say more without soon being out of date.
- 11 Conservation laws of this ilk, their connection to the underlying symmetries of the basic state and (relatedly) their finite amplitude extension, are discussed by ? and ?. Conservation of momentum is related to the translational invariance of the medium; conservation of $\mathcal{A}$ is related to the translational invariance of the mean flow, and for this reason $\mathcal{A}$ is called pseudomomentum.
- 12 See ?.
- 13 It is an example of a so-called Arnol'd condition. Such conditions can be derived most generally through the machinery of Hamiltonian fluid mechanics, but our derivation will be elementary and direct. The original papers are ??, and many results were developed by ?. See ? for a nice review.
- 14 See ?.
- 15 This derivation may proceed by the use of Bretherton's δ -function construction, in which the evolution of the fluid system may be regarded as being determined purely by potential vorticity evolution in the interior, and with isothermal boundary conditions, plus a thin sheet or δ -function of potential vorticity at the boundary. In that case, the boundary conditions play no role and the derivation proceeds from (8.172a) alone, just as in the two-dimensional derivation from (8.161). However, some manipulations are then required to obtain a final form with the surface conditions represented explicitly.
- 16 Following ? and ?.

Further Reading

Andrews, D. G., Holton, J. R. and Leovy, C.B., 1987. *Middle Atmosphere Dynamics*.

Provides a discussion of a number of topics in wave dynamics and wave-meanflow interaction, including the TEM, mainly in the context of stratospheric dynamics.

Problems

- 8.1 Prove that

$$\langle aJ(b, c) \rangle = \langle bJ(c, a) \rangle = \langle cJ(a, b) \rangle = -\langle cJ(b, a) \rangle \quad (\text{P8.1})$$

where the angle brackets denote a horizontal integration, and the boundary conditions are either no-normal-flow or periodic.

- 8.2 Consider an axi-symmetric barotropic shear flow given by

$$\zeta = \begin{cases} 2\Omega & r \leq R \\ 0 & r > R \end{cases} \quad (\text{P8.2})$$

where Ω and R are constants. Thus, the inner region is in solid body rotation and the outer region is irrotational, and we suppose that the velocity is continuous. (Is this implied by (P8.2) or is it an extra condition?) Suppose that the boundary between the two regions is perturbed. Find the phase speed of this disturbance in terms of its azimuthal wavenumber. Show that for large wavenumbers this reduces to the dispersion relation for a point jet [i.e., , or $c = U_0 + (\zeta_1 - \zeta_2)/2k$ — see (7.41)], and define what ‘large’ means in this context.

- 8.3 Demonstrate the group velocity property for the Eliassen-Palm flux for barotropic Rossby waves on the β -plane when the basic state is horizontally sheared.
- 8.4 Show that on the sphere and in a compressible, ideal gas atmosphere the EP flux is given by

$$\mathbf{F} = [F_\vartheta, F_z] = [-\rho_s \cos \vartheta \overline{u'v'}, \rho_s f \cos \vartheta \overline{v'\theta'} / (\partial \bar{\theta} / \partial z)]. \quad (\text{P8.3})$$

where ϑ is latitude, θ is potential temperature and ρ_s is a reference profile. You may use either height co-ordinates with the anelastic approximation, or log-pressure co-ordinates. In this expression, should f vary or should it be equal to a constant, f_0 ?

- 8.5 Show that on the sphere and in a compressible, ideal gas atmosphere the residual streamfunction corresponding to (8.57) is given by

$$\psi^* = \psi + a \rho_s \cos \vartheta \frac{\overline{v'\theta'}}{\bar{\theta}_z}. \quad (\text{P8.4})$$

where θ is potential temperature and ϑ is latitude.

- 8.6 $\diamond$ Show that perturbations to a horizontally sheared flow are orthogonal with respect to the wave activity norm. You may restrict attention to two-dimensional (barotropic) flow on the β -plane.

Partial solution: The eigenvalue equation is

$$(\bar{u} \nabla_k^2 + \bar{q}_y) \psi = c \nabla_k^2 \psi \quad (\text{P8.5})$$

where $\nabla_k^2 = \partial_{yy} - k^2$. If $q = \nabla_k^2 \psi$ then the eigenvalue equation may be written

$$Mq \equiv (\bar{u} + \bar{q}_y \nabla_k^{-2}) q = cq \quad (\text{P8.6})$$

or

$$Nq \equiv \left(\frac{\bar{u}}{\bar{q}_y} + \nabla_k^{-2} \right) q = c \frac{q}{\bar{q}_y} \quad (\text{P8.7})$$

The operator N on the left-hand side of (P8.7) is self-adjoint (show this) so that the eigenfunctions associated with two different eigenvalues are orthogonal with respect to $\bar{q}_y$; that is, for $n \neq m$,

$$\int \int \frac{q_n q_m}{\bar{q}_y} dy = 0. \quad (\text{P8.8})$$

If this derivation holds, why is not simply the case that, from (P8.6), that

$$\int q_n q_m dy = 0. \quad (\text{P8.9})$$

for $n \neq m$? (Is M self-adjoint?)

- 8.7 Obtain an expression for the EP flux due to equatorial Kelvin waves. What is the sign of the wave drag in a region of Kelvin wave generation?

References

- ALLEN, J. S., HOLM, D. D. & NEWBERGER, P. A. 2002 Extended-geostrophic Euler-Poincaré models for mesoscale oceanographic flow. In *Large-Scale Atmosphere-Ocean Dynamics I* (ed. J. Norbury & I. Roulstone). Cambridge University Press.
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